

Mathematics for Homeschooling

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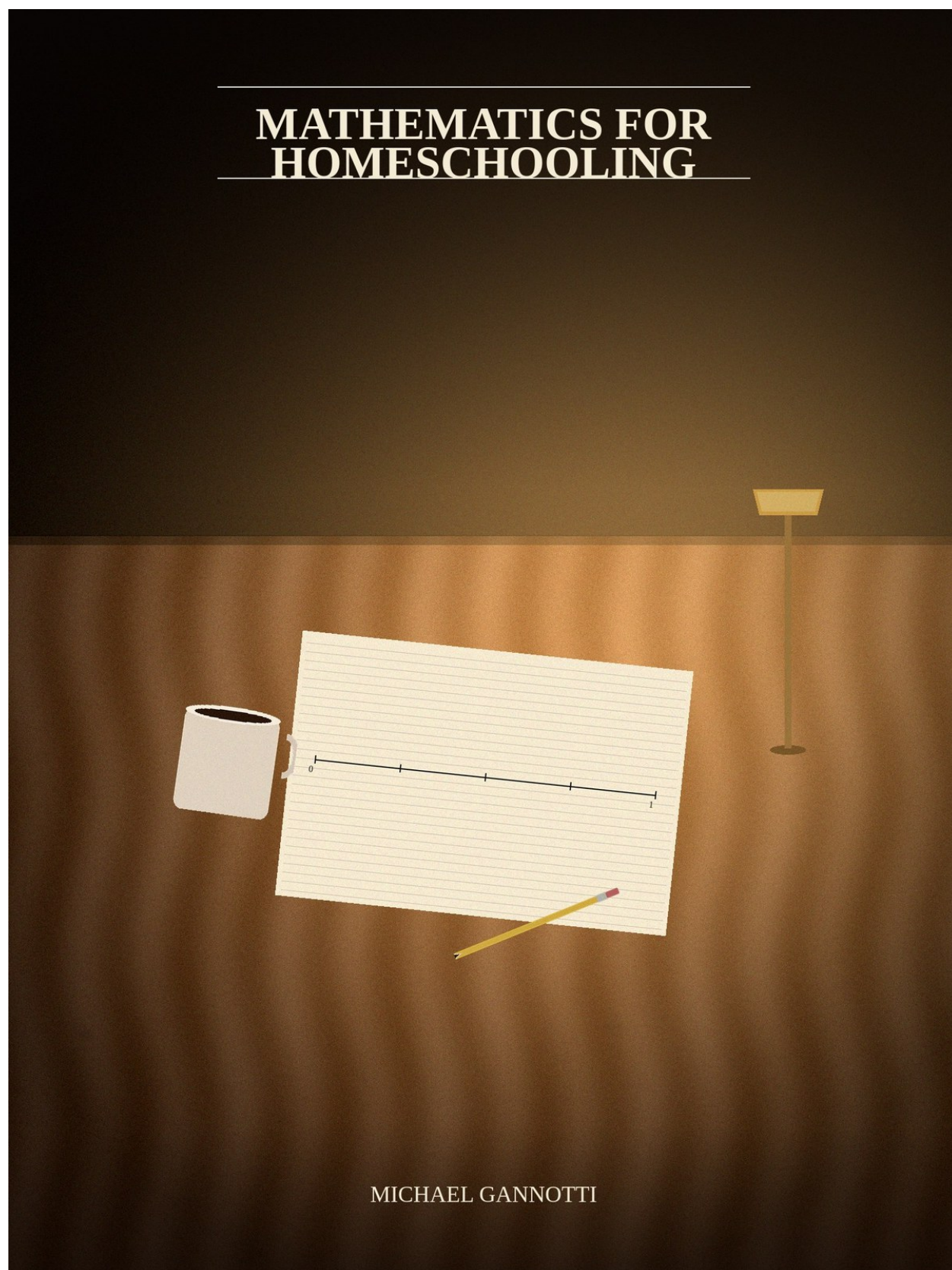


Figure 1: Cover

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This illustrated edition merges the documented final draft with its Chicago bibliography. Endnotes run in one series. Chapter-opening images are original still-lives, not portraits of persons or children.

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Preface

Hear a Wrong Answer

Two facts sit on the table this book is written from. Neither is a kitchen scene. Neither is a chatbot. Neither is a percentile.

The first is a federal count. The National Center for Education Statistics, in *Parent and Family Involvement in Education: 2023* (NCES 2024-113), reports that among students ages 5–17 with a K–12 grade equivalent, 3.4 percent were homeschooled in 2022–23 — approximately 1,765,000 students.¹ That is the latest national homeschool headcount this manuscript opened. It is not a 2026 total. It is not the experimental Household Pulse series, which is a different instrument, low-response, and not comparable to the National Household Education Surveys. It is not the National Home Education Research Institute’s advocacy synthesis of Pulse plus state registers plus under-count adjustments. Those other figures exist. They are labelled where they appear. They are not this opening.

The second fact is a constraint, not a guarantee. A parent who is not a math specialist can take a child from first-grade number sense through high-school calculus *if* three things hold: the parent understands the concept well enough to hear a wrong answer; the child does the struggle before the tool; and any agent is a supplement hosted and constrained by the parent — not a substitute for the child’s work and not a school-safe product by default.² The sources this book stands on do not prove that thesis as a causal claim about homeschool mathematics. They do not contradict it. They constrain how it can be attempted. The preface names the constraint. The introduction takes apart the popular story that would replace it. The chapters show which conjuncts survive.

Inside that count, the teacher of record is usually one adult. In the last Parent and Family Involvement tables that published the teaching-style and subject rows — the 2016 survey, reported as NCES 2020-001 — the mother was the main instructional provider for 78 percent of homeschooled

¹Ellena Sempeles and Jiashan Cui, *Parent and Family Involvement in Education: 2023* (NCES 2024-113) (Washington, DC: National Center for Education Statistics, September 2024), Table A-6 and footnote: 3.4 percent homeschooled, approximately 1,765,000 students, ages 5–17 with a K–12 grade equivalent, 2022–23; 5.2 percent instruction at home. <https://nces.ed.gov/pubs2024/2024113.pdf>. Access date for all URLs in these notes: 30 August 2026. Latest federal PFI opened; not a 2026 national total. Do not treat the National Home Education Research Institute’s 2024–25 estimate of 3.408 million as NCES.

²Working thesis, locked in FRAME.md (30 August 2026) and restated in *Mathematics for Homeschooling: Book Brief* (30 August 2026). The sources constrain the attempt; they do not prove a causal claim about homeschool mathematics.

students, and 77 percent of families used a mostly or strictly formal curriculum.³ Homeschooling is legal in all fifty states and the District of Columbia. Most of the state cards this research opened do not require the parent to hold a mathematics degree; several require no teacher qualification at all. Those cards are advocacy summaries, attorney-reviewed for members, not statutes. The parent still has to read the code of the state they live in. The teaching chapters cannot assume a single paperwork load, and they cannot assume a specialist at the table. One adult, usually the mother, usually a purchased book, legally allowed in most of the country to teach Algebra II without a mathematics degree: that is the kitchen this book is written for. It is not a district.

What this book is not, in one page. It is not a franchise. It names and compares Saxon, Singapore / Dimensions, Math-U-See, Beast Academy and Art of Problem Solving, Teaching Textbooks, CTC-Math, and Khan Academy. It does not sell any of them. A “grade 6” book is a publisher’s scope, not a legal grade; the comparison is grain, parent load, and how they place, not a phantom homeschool meta-analysis. It is not a 180-day plan and not a worksheet dump: twelve fat chapters, not twelve thin “Grade *N*” repeats, and not 180 days times twelve grades of photocopies. It is not a Common Core advocacy tract and not an anti-Common Core tract. The Common Core State Standards for Mathematics are treated as a map a parent can steal clusters from, not as a homeschool mandate and not as a teaching method. It is not a sequel to *Autonomous AI and Education*. That earlier book already established that agents are not chatbots, that Hermes 4 is not Hermes Agent, that there are three Groks, that unguarded solvers raise practice scores and cut unaided exams, that detectors fail, and that no K–12 student-facing autonomous open-web school agent was documented on the pages opened for that work. This book gets one chapter on the kitchen condition — the agent does not take the test — and then an AI lane in every later mathematics chapter. It does not duplicate classroom-tutor design at monograph length. It is not a What Works Clearinghouse-grade claim where none exists: no homeschool-curriculum RCT was found, and none is invented here.

The popular story collapses several claims that do not survive together. Homeschoolers score higher in math. They’ll learn fractions by cooking. Just buy the video curriculum. The chatbot will tutor. I stopped at algebra, so I will help less. No serious source in the research this manuscript stands on endorses all of those conjuncts. The collapse is the introduction’s job to take apart. The preface only names it, so the reader is not asked to treat a convenience-sample percentile, a publisher grade, a cooking afternoon, and a path through calculus as the same object.

The three conditions, in miniature, are the book’s spine. Hearing a wrong answer is not a personality trait. It is the parent knowing the day’s concept well enough to stay out of the child’s working memory, to recognize $3 + 4 + 5 = 3 + \square$ written as 15 as an operational equal sign, and to recognize $1/2 + 1/3 = 2/5$ as a magnitude failure rather than a cute arithmetic slip. The child doing the struggle before the tool is not a slogan about grit. Current performance is often an unreliable index of learning; a practice score that rose because a solver finished the page is not the unaided exam. The agent as parent-hosted supplement is not a product pitch. It is a ban on paste-the-worksheet and a set of parent workflows — explain the concept to the adult first; generate isomorphic practice with the key held back; diagnose work the child already produced — grounded in a field experiment

³Jiashan Cui and Rachel Hanson, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey*, NCES 2020-001 (Washington, DC: NCES, 2019), Table 3 (mother 78 percent main provider) and Table 6 (77 percent mostly or strictly formal curriculum; 12 percent mostly or strictly informal). <https://nces.ed.gov/pubs2020/2020001.pdf>. The 2023 First Look did not republish the teaching-style or subject tables.

and a learning-versus-performance review, not in a homeschool RCT. Where the sources constrain those three conditions, the chapters say so. This preface does not dump Chapter 2's method lock, Chapter 3's ban-versus-allow list, or Chapter 5's fractions gate. The introduction maps the twelve chapters. The chapters do the work.

The contract with the reader is documentary. Documents do the work; no invented kitchen is presented as a reported case. Composite hypotheticals, if a later chapter needs one to show a move — a multi-age table, a parent-first prompt, an unaided exit ticket — are labelled *hypothetical* in the sentence that introduces them. No portraits of living children. Grade labels in this book are sequence anchors, not prison bars: the 2016 tables already showed Algebra I taught that year to 41 percent of the homeschooled 6–8 band, and arithmetic still taught to 31 percent of 9–12.⁴ Every mathematics chapter of the finished book (Chapters 4–11) owes the parent four lanes: explain the mathematics to the parent; teaching guidance; tips that survive evidence; parent-hosted AI, with the child doing the work and the parent holding the key. Those lanes are not four identical boilerplate headings. Chicago notes; load-bearing numbers cited on the page; longer source fights — **UNVERIFIED** flags, vendor-claim labels, preprint labels, advocacy labels, journalism labels — in the notes. Items the research marked unverified stay flagged until a primary is read. Chicago notes run in one series from the preface through the coda. Access date for all fetches remains 30 August 2026. Open questions are left standing rather than filled with analogy. Promise a path, not a percentile.

If the popular story is a collapse, the reader needs the corrections before any grade band is explained at length. That is the introduction's job.

⁴Cui and Hanson, NCES 2020-001, Table 9: Algebra I taught that year to 41 percent of the 6–8 homeschool band; arithmetic still taught to 31 percent of 9–12. Grade labels in this book are sequence anchors, not prison bars.

Introduction

The Conjunction That Does Not Survive

The popular story about homeschool mathematics is a conjunction, and it is how the advice functions in the wild. Homeschoolers score higher in math. A “grade 6” book is a legal grade. Fractions are a cute side topic they will pick up by cooking. Understand first, *or* compute first — pick a religion. Timed tests are either the identity of being good at mathematics or a settled cause of anxiety. The chatbot will tutor. I stopped at algebra, so I will help less. Homeschoolers go to college at higher rates. Buy the video curriculum and the parent can stop hearing wrong answers.⁵

A reader has met this story in a curriculum catalog, a co-op hallway, a Facebook thread, a vendor FAQ, and a well-meant kitchen rule, often in the same week. The story is congratulatory when the speaker is “we” and catastrophic when the speaker is a district, a test, or “kids today.” Both tones need the same object: one path, already delivered by the act of homeschooling, certifiable by a publisher grade and a percentile, and lately outsourced to a fluent machine. Chapter 1 names the kitchen so later chapters can refuse the district analogy without sounding pedantic. This introduction does the prior job. It puts eight corrections in the way, so that a reader who still wants the popular story has to go through the documents. Almost none of the popular story is how the record works.

Almost none of that is how the record works. CCSSM is a map. Fractions are the gate. A “grade 6” book is a publisher’s scope. Pathways are models, not mandates, and AP Statistics does not substitute for calculus in STEM. Concepts and procedures iterate; ranking the strands is misguided. Effect sizes are not interchangeable. The only federal headcount this book may crown is 3.4 percent / approximately 1.765 million in 2022–23; “homeschoolers score higher in math” is a fight. Unguarded GPT raises practice scores and cuts the unaided exam. This introduction’s job is to make the reader unable to treat a vendor grade, a convenience-sample percentile, a cooking afternoon, a chatbot, and a path through calculus as the same object. Eight corrections, then a map, then the questions the chapters will not fill. Documents do the work. No invented kitchen. The popular story is already public. The corrections are already sourced. The map is already the book.

⁵The conjunction is assembled from the popular-story items in *Mathematics for Homeschooling: Book Brief* (30 August 2026), section “What a book must get right.” No single source endorses all conjuncts. Teaching Textbooks “does 100% of the teaching” is a **vendor** claim, not a finding. Access date for all URLs in these notes: 30 August 2026.

The popular story

Name the conjunction so it can be taken apart. Franchise: there is a right book, usually the one already on the shelf or the one a catalog ranks, and a homeschool-effectiveness trial somewhere behind it. Percentile: homeschoolers, as a class, sit in the eightieth-percentile range in mathematics, so the kitchen has already succeeded. Kitchen-math: cooking, money, and making *are* fractions and algebra; the formal page is optional. Chatbot-tutor: paste the Saxon lesson, get the key, feel fine; or buy a Socratic tutor and walk away. “I stopped at algebra, so I will help less”: the school-homework finding that anxious frequent help predicts worse child math, misread as a daily strategy for the person who is the teacher. Video curriculum as the release valve that lets the parent stop hearing wrong answers, because the product “does 100 percent of the teaching.” Timed tests as either the whole of fluency or a settled mechanism of anxiety. Statistics as stealth calculus, or calculus as the only college-equivalent that matters. A calendar grade as a legal object. A satisfied state statute as a transcript a registrar can read.

No serious source in the research this manuscript stands on endorses all of those conjuncts. Some of them are vendor sentences, labelled vendor where they appear. Some are advocacy syntheses. Some are real findings pointed at the wrong object. Some are open questions filled with analogy. The eight corrections that follow are not decorations. A chapter that still told the popular story after reading them would be the wrong book. The design user inside the corrections is the parent who stopped at algebra. Most hours of homeschool mathematics are still parent-supervised. Not every reader is a stay-at-home parent. The book is not written to the child.

Correction 1 — CCSSM is a map, not a homeschool mandate

The Common Core State Standards for Mathematics, issued by the National Governors Association Center for Best Practices and the Council of Chief State School Officers in 2010, are a grade-by-grade map in kindergarten through 8 and a conceptual-category map in high school: Number and Quantity, Algebra, Functions, Modeling, Geometry, Statistics and Probability. They “do not dictate curriculum or teaching methods,” and the order of topics inside a grade is not a required teaching order.⁶ Many homeschoolers do not use CCSSM. Many U.S. states have adopted, adapted, or replaced it. A 2026 state-by-state census of who still uses the standards is **UNVERIFIED** in the research this manuscript stands on. This book will not say “most states” without a current count.

Treat CCSSM as the most fully specified public map a parent can steal clusters from, not as a law and not as a teaching method. The eight Standards for Mathematical Practice run through every grade: make sense of problems and persevere; reason abstractly and quantitatively; construct viable arguments and critique the reasoning of others; model with mathematics; use appropriate tools strategically; attend to precision; look for and make use of structure; look for and express regularity in repeated reasoning.⁷ They rest on the National Council of Teachers of Mathematics process

⁶National Governors Association Center for Best Practices and Council of Chief State School Officers, *Common Core State Standards for Mathematics* (2010), “How to read the grade level standards” and the sentence that the standards “do not dictate curriculum or teaching methods.” https://www.corestandards.org/assets/CCSSI_Math%20Standards.pdf.

⁷CCSSM, pp. 6–8, Standards for Mathematical Practice. Rest on NCTM process standards and on National Research Council, *Adding It Up: Helping Children Learn Mathematics*, ed. Jeremy Kilpatrick, Jane Swafford, and Brad-

standards — problem solving, reasoning and proof, communication, connections, representation — and on the National Research Council’s *Adding It Up* strands. NCTM *Principles and Standards for School Mathematics* (2000) organizes school mathematics into five content standards and five process standards across four grade bands, not thirteen separate grade lists. The 2006 *Curriculum Focal Points* then named three “most important mathematical topics” per grade, prekindergarten through 8. Those documents predate CCSSM. They are not “the current U.S. standard.” They remain useful as professional consensus on strands and per-grade focus. Grade lists from Focal Points in this manuscript were read from a third-party posting of NCTM text; the official book is paywalled and stays flagged. This book is not a Common Core advocacy tract and not an anti-Common Core tract. Later chapters use the map the way a traveler uses a map: to name the terrain, not to pretend the terrain is a statute.

Correction 2 — Fractions are the gate

The National Mathematics Advisory Panel, in *Foundations for Success* (2008), named three Critical Foundations of Algebra: fluency with whole numbers; fluency with fractions, including decimals, percents, and negative fractions; and particular aspects of geometry and measurement. Of these, “knowledge of fractions is the most important foundational skill not currently developed among American students.”⁸ A nationally representative survey of Algebra I teachers commissioned by the Panel — 743 teachers, 2007 — rated students as having very poor preparation in “rational numbers and operations involving fractions and decimals.” Those are 2008 data. They are still the best nationally representative teacher-voice primary on *incoming* Algebra I gaps this research opened. They are not a 2026 replication. Do not write “teachers today say.”

Robert Siegler, Greg Duncan, Pamela Davis-Kean, Kathryn Duckworth, Amy Claessens, Mimi Engel, Maria Ines Susperreguy, and Meichu Chen (2012), analyzing the British Cohort Study ($N = 3,677$) and the U.S. Panel Study of Income Dynamics Child Development Supplement ($N = 599$), found that elementary students’ knowledge of fractions and of division uniquely predicted algebra knowledge and overall mathematics achievement five or six years later, after controlling for other mathematical knowledge, general intellectual ability, working memory, and family income and education. A 1-SD increase in age-10 fractions knowledge was uniquely associated with a 0.15-SD (UK) / 0.17-SD (US) increase in high-school algebra and a 0.16-SD / 0.18-SD increase in overall high-school math. Concurrent correlations of high-school fraction knowledge with overall math were $r = .81$ (UK) and $r = .87$ (US).⁹ This is a *prediction* finding, not a classroom experiment proving that teaching fractions a certain way *causes* algebra success. Do not upgrade it. Chapter 5 is the teaching chapter. This correction only installs the gate.

ford Findell (Washington, DC: National Academy Press, 2001).

⁸National Mathematics Advisory Panel, *Foundations for Success: The Final Report of the National Mathematics Advisory Panel* (Washington, DC: U.S. Department of Education, 2008), Executive Summary findings 4 and 12, and Chapter 4. ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>. Prefer the Panel’s sentence over a journalism paraphrase. NMAP is the still-cited algebra-foundations synthesis, not the science of 2026 classrooms.

⁹Robert S. Siegler, Greg J. Duncan, Pamela E. Davis-Kean, Kathryn Duckworth, Amy Claessens, Mimi Engel, Maria Ines Susperreguy, and Meichu Chen, “Early Predictors of High School Mathematics Achievement,” *Psychological Science* 23, no. 7 (2012): 691–97. Author PDF, <https://sieglertc.columbia.edu/wp-content/uploads/2019/02/Siegler-et-al-PsySci12.pdf>. British Cohort Study $N = 3,677$; U.S. PSID-CDS $N = 599$. Prediction after controls, not a curriculum RCT.

Do not treat “multiply makes bigger / divide makes smaller” as cute folklore. It is a whole-number property that fails for multipliers and divisors less than 1 — a central conceptual-change problem in the whole-number-bias literature (Ni and Zhou 2005; Siegler and colleagues 2012). CCSSM 5.NF.5 exists specifically to break it. Adding numerators and denominators — $1/2 + 1/3 = 2/5$ — is a magnitude failure the IES fractions practice guide (Siegler and colleagues, 2010, Recommendation 2) names as often rooted in never treating fractions as numbers on a line. Pies introduce part-whole. They do not by themselves teach that $3/4$ is a number with a magnitude. The Panel: representing fractions on a number line is “one key mechanism linking conceptual and procedural knowledge.” Fuchs and colleagues 2013 and after are the experimental half of that research program; numerical effect sizes from those papers were **not extracted** in the research this manuscript stands on — do not invent them. Greer 1994 is **UNVERIFIED** as to that paper’s data this pass; cite Ni and Zhou, Siegler, and 5.NF.5 instead. NMAP 2008 is not the science of 2026 classrooms. It is the still-cited algebra-foundations synthesis. Diagnose fractions, signed numbers, and the equal sign before buying Algebra I.

Correction 3 — A “grade 6” book is a publisher’s scope

Singapore Math Inc., labelled **vendor**, states that Dimensions Math 6 “includes some Pre-algebra topics, but is not a Pre-algebra text in the traditional sense; it is a Grade 6 text,” and that Dimensions 7 and 8 together cover Pre-algebra and Algebra 1 topics with some Geometry.¹⁰ Saxon’s homeschool sequence uses split names — Math 5/4, 6/5, 7/6, 8/7 — that do not match calendar grade; Houghton Mifflin Harcourt placement tests set the starting book by score and treat Saxon as skill-level rather than grade-level. Math-U-See uses Greek letters uncoupled from calendar grade: Epsilon is a fractions *level*, not “fifth grade.” Beast Academy, labelled **vendor** / Art of Problem Solving, “generally recommend[s] that students start about a level ‘behind’ their current grade level” because of problem difficulty, and states that levels 4–5 cover many middle-grades standards. A child of ten may place into Singapore 3A, Saxon 6/5, and Math-U-See Gamma simultaneously. That is not a contradiction. It is three different grain-sizes of “grade.”

The population already shows the stretch. NCES Parent and Family Involvement 2016: Algebra I taught that year to 41 percent of the 6–8 homeschool band; arithmetic still taught to 31 percent of 9–12.¹¹ Grade labels in this book are sequence anchors, not prison bars. Placement tests exist because the labels lie. The National Mathematics Advisory Panel’s Finding 15: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; claims that children of a given age cannot learn certain content because they are “too young,” “not in the appropriate stage,” or “not ready” have “consistently been shown to be wrong.” Accelerate a child who has the prerequisites. Do not skip fractions because a child is “good at math.” Cathy Duffy Reviews and Rainbow Resource are catalogue and reviewer, not independent effectiveness trials. HMH’s school-side Saxon ESSA ratings are **vendor** evidence about school implementations, not home-

¹⁰Singapore Math Inc., Dimensions Math FAQs. **Vendor.** Dimensions 6 “includes some Pre-algebra topics, but is not a Pre-algebra text in the traditional sense; it is a Grade 6 text.” Art of Problem Solving, Beast Academy FAQ, <https://beastacademy.com/faq/books>. **Vendor.** “We generally recommend that students start about a level ‘behind’ their current grade level.” Math-U-See / Demme Learning: Epsilon is a fractions *level*. **Vendor.** Houghton Mifflin Harcourt / Saxon Homeschool middle-grades placement test: skill-level score bands. **Vendor.**

¹¹Jiashan Cui and Rachel Hanson, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey*, NCES 2020-001 (Washington, DC: NCES, 2019), Table 9.

school RCTs. Kunzman and Gaither (2020) note that curriculum research on homeschooling is thin relative to demographics and law. No homeschool-curriculum RCT was found. Choose by fit — parent math confidence, student stance, special needs, budget, college target — not by a phantom homeschool meta-analysis. Teaching Textbooks’ sentence that the program “does 100% of the teaching” is a **vendor** claim. It is relevant to the parent who stopped at algebra. It is not a finding that the parent can stop hearing wrong answers.

Correction 4 — Typical U.S. high school is Algebra I → Geometry → Algebra II

CCSSM Appendix A, published by Achieve in partnership with the Common Core writing team, prints four model pathways: Traditional (Algebra I, Geometry, Algebra II); Integrated (Mathematics I, II, III); and compacted versions of each that complete Grade 7, Grade 8, and the first high-school course in Grades 7–8 so that a student can reach Calculus or another college-level course by senior year. “The pathways and courses are models, not mandates.”¹² A non-accelerated student who starts Algebra I in Grade 9 finishes Algebra II in Grade 11 and has senior year for precalculus, statistics, or a first calculus course — not both precalculus and BC unless the fourth course is overloaded. The Panel recommended that all *prepared* students have access to an authentic algebra course by Grade 8; “authentic” means the Major Topics of School Algebra, not a course that is algebra in name only. “Grade 8 algebra” is ambiguous: it can mean CCSSM Grade 8 expressions, equations, and functions — linear equations, systems, the function concept — or a full Algebra I taken in Grade 8. Those are not the same course. Singapore Secondary 2 (U.S. Grade 8) G3 and England’s Key Stage 3 already expect quadratic graphs that U.S. CCSSM places in Algebra I. Do not say Singapore or England is “two years ahead” as a single number. Say: if the non-U.S. book is doing quadratic graphs in what it calls Grade 8, the family is on a different train, often earlier on the algebra car. Integrated I–III topic-by-topic from Appendix A was only partly extracted; if a later chapter needs a full integrated-path box, that re-open stays an open question.

Advanced Placement Calculus AB is eight units; BC is those eight plus parametric, polar, and vector functions and infinite sequences and series. Students who take BC receive an AB subscore. CLEP Calculus is a different object: a one-semester college-calculus exam, about 60 percent limits and differential calculus and 40 percent integral calculus, 44 questions, about 90 minutes; ACE recommended credit-granting score 50 for 4 semester hours; each institution sets its own policy. AP Statistics, effective Fall 2026 Course and Exam Description, is “equivalent to a one-semester, introductory, **non calculus-based** college course in statistics.” The only published prerequisite is “successful completion of a first-year algebra course.” Five units, not the former nine. College Board’s revised framework encourages students with appropriate background to take **both** AP Statistics and AP Calculus. It is not published as a substitute for calculus in STEM calculus sequences.¹³ A parent choosing AP Statistics instead of precalculus or calculus is choosing a differ-

¹²Achieve in partnership with the CCSS writing team, *Appendix A: Designing High School Mathematics Courses Based on the Common Core State Standards*, Oregon ODE copy. “The pathways and courses are models, not mandates.” <https://www.oregon.gov/ode/educator-resources/standards/mathematics/Documents/math-appendix-a-model-course-pathways.pdf>. Integrated I–III topic-by-topic only partly extracted (open question 14).

¹³College Board, AP Calculus AB and BC course pages and *AP Calculus AB and BC Course and Exam Description Clarifications and Corrections, Effective Fall 2026*; *AP Statistics Course and Exam Description, Effective Fall 2026*;

ent college-equivalent, legitimate for a non-STEM path and a weak stealth substitute for Calculus I. NCAA mathematics core courses are Algebra I or higher; CLEP is not an NCAA core course. Named colleges this research opened want, as a pattern and not a national rule, about three years through Algebra II as a floor. State compliance is not college evidence. There is no national home-school diploma. Chapter 10 keeps AB, BC, and CLEP from collapsing. Chapter 11 keeps data and chance from being stuffed into calculus. Chapter 12 is the export.

Correction 5 — Conceptual and procedural iterate; five strands; ranking them is misguided

Bethany Rittle-Johnson, Michael Schneider, and Jon Star, reviewing longitudinal and experimental evidence in 2015, wrote: “It is a myth that it is a one-way street from conceptual knowledge to procedural knowledge.” Alternative orderings of instruction have “rarely been compared.”¹⁴ Studies often cited for concepts-first compared conceptually rich teaching to typical procedural classrooms that gave little or no conceptual instruction. They do not compare concepts-then-procedures to procedures-then-concepts. The classroom experiment that held content constant and varied only order — Rittle-Johnson and Kenneth Koedinger, 2009, sixth-grade decimals — favored iteration, not a long conceptual foundation: more procedural learning including transfer; conceptual knowledge fairly comparable. NCTM’s 2014 sentence that procedural fluency follows conceptual understanding is a policy stance, not a finding tested against a procedures-then-concepts alternative.

Adding It Up (National Research Council / Kilpatrick, Swafford, and Findell, 2001, p. 116) defines mathematical proficiency as five interwoven strands: conceptual understanding, procedural fluency, strategic competence, adaptive reasoning, and productive disposition. The strands “are interwoven and interdependent.” A later recommendation (p. 409) adds that instruction “should not be based on extreme positions that students learn, on one hand, solely by internalizing what a teacher or book says or, on the other hand, solely by inventing mathematics on their own.”¹⁵ NMAP Finding 10: debates about the relative importance of conceptual understanding, computational fluency, and problem solving “are misguided”; they are mutually supportive. Any homeschool program that advertises only one of these is selling a fragment. Do not wait for full understanding before any practice, and do not drill a procedure with zero talk about why.

NMAP Finding 23: high-quality research does not support exclusive use of either all-encompassing

CLEP Calculus exam page. Accessed 30 August 2026. AP Statistics is non-calculus-based; prerequisite is first-year algebra; not published as a STEM-calculus substitute. AB eight units; BC those eight plus Units 9–10. CLEP Calculus is a different object.

¹⁴Bethany Rittle-Johnson, Michael Schneider, and Jon R. Star, “Not a One-Way Street: Bidirectional Relations Between Procedural and Conceptual Knowledge of Mathematics,” *Educational Psychology Review* 27, no. 4 (2015): 587–97. Bethany Rittle-Johnson and Kenneth R. Koedinger, “Iterating between Lessons on Concepts and Procedures Can Improve Mathematics Knowledge,” *British Journal of Educational Psychology* 79 (2009): 483–500. NCTM, *Principles to Actions: Ensuring Mathematical Success for All* (Reston, VA: NCTM, 2014), 42: policy stance, not an experiment.

¹⁵National Research Council, *Adding It Up*, 116 (five strands “interwoven and interdependent”); 409 (instruction “should not be based on extreme positions”). NMAP Finding 10 (ranking misguided; capabilities are “mutually supportive”); Finding 23 (no exclusive teacher-directed or student-centered prescription); Finding 27 (explicit instruction for students already struggling — not all instruction). Manu Kapur, “Productive Failure in Learning Math,” *Cognitive Science* 38, no. 5 (2014): 1008–22, is a different comparison; do not use it to cancel Finding 27.

teacher-directed or student-centered prescriptions. Finding 27: explicit instruction — clear models, extensive practice, think-alouds, extensive feedback — has consistently positive effects for students with mathematical difficulties and for the lowest third of a typical class, and this “does not mean that all of a student’s mathematics instruction should be delivered in an explicit fashion.” Manu Kapur (2014): productive failure — problem-solving first, then instruction — produced comparable procedural knowledge to direct instruction first, but greater conceptual understanding and transfer. These two literatures are not the same comparison. Do not use Kapur to cancel NMAP 27. Stockard and colleagues’ 0.60 is a meta-analysis of a named Direct Instruction curriculum family, not “lecture works.” Kirschner, Sweller, and Clark 2006 is a position paper. Confrey, Maloney, and Nguyen’s 2008 methodological critique of the Panel is real; NMAP is still the synthesis later documents cite. Report both. Chapter 2 is the method lock. This correction only kills the one-way street.

Correction 6 — Do not mash effect sizes or timed-test objects

Doug Rohrer, Robert Dedrick, and Kaleena Burgess (2014): grade-7 classroom crossover, $n = 140$, unannounced test two weeks later, interleaved 72 percent versus blocked 38 percent, $d = 1.05$, 95 percent CI [0.80, 1.30]. Four superficially dissimilar kinds; pretest 0.7 percent for full solutions. Rohrer, Dedrick, Marissa Hartwig, and Chi-Ngai Cheung (2020): preregistered cluster RCT, 54 seventh-grade classes, four months of assignments, unannounced test one month later, interleaved 61 percent versus blocked 38 percent, $d = 0.83$. The authors note that interleaving also spaces same-kind problems, so the d values should not be treated as “pure interleaving.” Almost all clean classroom RCTs of interleaving in mathematics are grade 7. John Dunlosky, Katherine Rawson, Elizabeth Marsh, Mitchell Nathan, and Daniel Willingham (2013) rate practice testing and distributed practice as high-utility techniques and interleaved practice as moderate-utility — less classroom generality at the time of that review. Do not mash a $d = 1.05$ from a nine-week classroom interleaving study with Dunlosky’s “high utility” rating for spacing, and do not mash 2014 with 2020. Brunmair and Richter’s later meta-analysis of interleaving, sometimes cited as $g = 0.34$ for mathematical tasks, is **UNVERIFIED as to exact coefficient** this pass.¹⁶

¹⁶Doug Rohrer, Robert F. Dedrick, and Kaleena Burgess, “The Benefit of Interleaved Mathematics Practice Is Not Limited to Superficially Similar Kinds of Problems,” *Psychonomic Bulletin & Review* 21 (2014): 1323–30, $d = 1.05$, 72% vs. 38%, $n = 140$, two-week delay. Doug Rohrer, Robert F. Dedrick, Marissa K. Hartwig, and Chi-Ngai Cheung, “A Randomized Controlled Trial of Interleaved Mathematics Practice,” *Journal of Educational Psychology* 112, no. 1 (2020): 40–52, 61% vs. 38%, $d = 0.83$, 54 classes, one-month delay, joint with spacing. John Dunlosky, Katherine A. Rawson, Elizabeth J. Marsh, Mitchell J. Nathan, and Daniel T. Willingham, “Improving Students’ Learning with Effective Learning Techniques,” *Psychological Science in the Public Interest* 14, no. 1 (2013): 4–58. Asha K. Jitendra, Jon R. Star, Danielle Dupuis, and Michael C. Rodriguez, “Effectiveness of Schema-Based Instruction for Improving Seventh-Grade Students’ Proportional Reasoning,” *Journal of Research on Educational Effectiveness* 5 (2012): immediate $g = 1.24$, six-week $g = 1.27$, null transfer. Asha K. Jitendra et al., “A Randomized Controlled Trial of the Impact of Schema-Based Instruction on Mathematical Outcomes for Third-Grade Students with Mathematics Difficulties,” *The Elementary School Journal* 114, no. 2 (2013): 252–76. Corey Peltier and Kimberly J. Vannest, “A Meta-Analysis of Schema Instruction on the Problem-Solving Performance of Elementary School Students,” *Review of Educational Research* (2017): $g = 1.57 / 1.09$ on mostly proximal measures. Lynn S. Fuchs et al., *Assisting Students Struggling with Mathematics: Intervention in the Elementary Grades* (WWC 2021006) (IES, 2021), Recommendation 6. Jo Boaler, *Fluency Without Fear* (Youcubed working paper, 2015), labelled working paper / commentary. Brunmair and Richter exact math-task g : **UNVERIFIED**.

Word problems are schemas, not keywords. Asha Jitendra, Jon Star, Danielle Dupuis, and Michael Rodriguez (2012): seventh-grade RCT, 1,163 students in 42 classrooms, schema-based instruction versus regular instruction, proportion problems, immediate $g = 1.24$, six-week retention $g = 1.27$, no significant transfer. Jitendra and colleagues (2013): third-graders with mathematics difficulties, small-group SBI versus control, word-problem posttest $g = 0.46$; district achievement $g = 0.34$; no significant retention at 8 weeks. Corey Peltier and Kimberly Vannest (2017) meta-analysis of schema instruction in elementary school: Hedge's $g = 1.57$ immediate and $g = 1.09$ transfer, 21 studies, 3,408 students — a large number that includes many researcher-developed proximal measures. Do not treat it as a general-achievement effect, and do not mash it with Jitendra 2012. IES 2021 Recommendation 5 (strong evidence, 18 studies) and the 2009 RtI guide both reject keyword strategies. Teach problem types — change, compare, equal groups, proportion — not magic words.

Timed facts are two different objects. Fluency means flexible, accurate, efficient retrieval — not a race and not optional (NMAP Finding 11; NRC 2001). Lynn Fuchs and colleagues (2021) Recommendation 6, strong evidence, 27 studies: regularly include timed activities as one way to build fluency in intervention. The how-to is brief, mixed, already-taught material, often with the child graphing their own score — not a high-stakes three-minute 50-item test in grade 1. Russell Gersten and colleagues (2009): about 10 minutes per intervention session on fact fluency. Jo Boaler (2012, 2015 *Fluency Without Fear* working paper / Youcubed) argues that timed tests are a “direct cause” of early math anxiety and that fluency has been misread as speed. Labelled working paper and commentary, not a journal experiment. The two sides are not measuring the same practice. A 2025 preprint meta-analysis (Peltier and colleagues, OSF, labelled **preprint**) notes that the 27 IES studies generally bundled timing with other intervention components, so timing is not isolated. Parent-facing sentence: retrieve known facts often, sometimes against a clock, after the child can already get them right untimed; do not make the kitchen timer the identity of being good at math.

Kitchen-math transfer is hard. NMAP Finding 26: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving. Cooking can motivate and can provide a second representation of a fraction the child is already being taught as a number. It does not replace the number-line work, the schema work, or the retrieval. Money is a decimal-and-place-value context, useful, also not a curriculum. Building and making are spatial; they do not automatically produce proof. A child who can double a recipe and cannot place $3/2$ on a line has performed, not learned. Nicholas Soderstrom and Robert Bjork (2015): current performance is “often an unreliable index” of learning.¹⁷ Heterogeneity is the story. Do not average Rohrer 1.05 with Rohrer 0.83 with Jitendra 1.24 with Peltier and Vannest 1.57 with Carbonneau's manipulatives d of approximately 0.37 — itself larger on retention than on transfer or justification — with Stockard 0.60 with Bastani's plus 48 / minus 17 percents of control grades with Oreopoulos and Low's 0.040 SD per term. Chapter 2 prints the hygiene. This correction only forbids the mash.

¹⁷Nicholas C. Soderstrom and Robert A. Bjork, “Learning Versus Performance: An Integrative Review,” *Perspectives on Psychological Science* 10, no. 2 (2015): 176–99: current performance is “often an unreliable index” of learning. NMAP Finding 26 on real-world contexts. Kira J. Carbonneau, Scott C. Marley, and James P. Selig, “A Meta-Analysis of the Efficacy of Teaching Mathematics with Concrete Manipulatives,” *Journal of Educational Psychology* 105, no. 2 (2013): 380–400, overall d approximately 0.37 with moderators.

Correction 7 — Quote NCES; do not crown 2026

NCES *Parent and Family Involvement in Education: 2023* (NCES 2024-113), from NHES:2023: 3.4 percent homeschooled, approximately 1,765,000 students, 2022–23, ages 5–17 with a K–12 grade equivalent. “Instruction at home” — homeschool or full-time virtual — was 5.2 percent. Weighted PFI sample 19,562 representing about 53 million K–12 students. This is the latest NCES national homeschool count this manuscript opened. **Do not treat any 2026 national total as official.** NHES 2018–19: 2.8 percent / about 1,457,000. The COVID bump is real in the experimental Household Pulse series (5.4 percent in 2020–21; NCES *Condition of Education*, May 2022) and in NCES’s “instruction at home” series (3.7 percent in 2018–19 to 5.2 percent in 2022–23). Pulse is experimental, low-response — 6.6 percent in the cited wave — and **not comparable** to NHES. NHERI, labelled **advocacy**, estimates 3.408 million for 2024–25 from Pulse plus 23 state registers plus undercount adjustments. Advocacy synthesis, not NCES. Pew, 20 February 2025, journalism restating NCES, notes an additional 1.8 percent full-time virtual public or private whose parents did *not* consider them homeschooled. Do not collapse full-time public virtual into parent-directed homeschool.¹⁸

Homeschooling is legal in all fifty states and the District of Columbia. Regulation ranges from no-notice — Texas, Illinois, Idaho as HSLDA-labelled examples — through notification to local approval. Most opened states do **not** require the parent to hold a mathematics degree. HSLDA summaries are **advocacy**, attorney-reviewed for members, not statutes. Parents must read their own state’s code. NCES PFI 2016: mother 78 percent main provider; 77 percent mostly or strictly formal curriculum; 12 percent mostly or strictly informal; 23 percent any tutor; 31 percent any local homeschool group or co-op. High-school-level homeschooled students in 2016 had ever been taught Algebra I (79 percent), geometry (52 percent), Algebra II (49 percent), calculus (15 percent, interpret with caution). The 2012 report found 88 percent of high-school-level homeschoolers had been taught basic algebra. Do not narrate 88 → 79 as decline without checking standard errors and questionnaire differences. Both figures say: Algebra I is common, Algebra II and geometry are not universal, calculus at home is a minority. The 2023 First Look did not republish the teaching-style or subject tables. NHES 2023 labor-force split: two-parent, one full-time and one not in labor force, 8.8 percent homeschooled versus both full-time 1.1 percent. Most *hours* are still parent-supervised. Not every reader is stay-at-home. Unschooling is a named 12 percent, not the market majority. The thesis is closer to structured and mastery. Say so without scorning the 12 percent. Academic dissatisfaction is common among reasons named important; it is not the modal *most* important reason in NHES 2023 Table A-7.

“Homeschoolers score higher in math” is a fight, not a finding. Brian Ray / NHERI convenience samples, often recruited through testing services and homeschool networks, repeatedly report eightieth-percentile-range composites. The 2009 *Homeschool Progress Report* specimen: about the 86th percentile overall, **84th in math**, 89th in reading; sample disproportionately White, Christian, two-parent, higher-income, higher parental education. Critics — Lubienski, Puckett, and Brewer 2013; McCracken / ICHER; Kunzman and Gaither 2020; the *Washington Post*, 11

¹⁸Ellena Sempeles and Jiashan Cui, NCES 2024-113. NCES, *The Condition of Education* (May 2022), on Pulse 5.4 percent as experimental and not comparable to NHES. NHERI 2024–25 estimate 3.408 million: **advocacy** synthesis, not NCES. Pew Research Center, 20 February 2025: journalism restating NCES 3.4 percent and the additional 1.8 percent full-time virtual not considered homeschooled by their parents.

December 2023, quoting Kunzman — argue selection, nonresponse, and demographic confounding. Cardus Education Survey 2023 (Cheng and Watson, 2024; Ipsos KnowledgePanel; 2,350 U.S. adults ages 24–39; about 8 percent / $n = 181$ ever homeschooled) finds **lower** bachelor’s attainment among medium-term (29 percent) and long-term (27 percent) homeschoolers than among never-homeschooled (46 percent), after demographic controls, and does not claim a math-score advantage. Only 17 percent of ever-homeschooled adults were homeschooled all K–12. The authors explicitly refuse a causal interpretation. NCES does not publish a nationally representative homeschool NAEP mathematics score. **Do not crown a winner with a convenience sample.** The teaching chapters promise a path, not a percentile. No named nationally representative source opened for this research gives the share of homeschool graduates who complete a college calculus course or who major in STEM.

The design user is “I stopped at algebra.” Erin Maloney, Gerardo Ramirez, Elizabeth Gunderson, Susan Levine, and Sian Beilock (*Psychological Science*, 2015) found that first- and second-graders of more math-anxious parents learned less math over the school year and had more math anxiety at year’s end — **but only if those parents reported frequent homework help.** When help was infrequent, parent anxiety was unrelated to child outcomes. The effect was specific to math, not reading.¹⁹ A 2021 follow-up (Retanal, Johnston, Di Lonardo Burr, Storozuk, DiStefano, and Maloney) found that a controlling-supportive homework-helping style partially mediated the parent-anxiety / child-achievement relation in a survey of parents of 11- to 14-year-olds. Mark Ashcraft and Elizabeth Kirk (2001): high math-anxious adults showed smaller working-memory spans on computation-based span tasks (full-scale $r = -.40$); the disruption is transitory occupation of the central executive by worry. A homeschool parent who is the daily instructor cannot “help less often” the way a school parent can. The risk is therefore higher, not lower. The mechanism to watch is not “parent is bad at math.” It is parent anxiety leaking as controlling help, speed-pressure, or “I’m not a math person, and you take after me.” The constructive response the literature actually supports: the parent learns the day’s concept well enough to stay out of the child’s working memory; the child generates; the parent does not narrate worry. Do not treat the finding as “anxious parents should not teach.” Replication status of Maloney and colleagues 2015 is an open question; some later work mixed on the exact interaction; no registered replication was opened. Flag as a single large, well-cited field study plus a 2021 mediation follow-up. It is a school-homework study, not a parent-as-sole-teacher RCT.

IDEA at home is not an IEP with FAPE. Under the Individuals with Disabilities Education Act, children parentally placed in private schools have no individual right to a free appropriate public education (34 C.F.R. § 300.137(a)). If a state treats homeschools as private schools, they may be in the child-find and equitable-services bucket; if it does not, even that may not apply. OSEP

¹⁹Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, “Intergenerational Effects of Parents’ Math Anxiety on Children’s Math Achievement and Anxiety,” *Psychological Science* 26 (2015): 1480–88. Interaction with frequent homework help; math-specific; school-homework study, not a parent-as-sole-teacher RCT. Mark H. Ashcraft and Elizabeth P. Kirk, “The Relationships among Working Memory, Math Anxiety, and Performance,” *Journal of Experimental Psychology: General* 130, no. 2 (2001): 224–37, full-scale $r = -.40$. Cardus / Albert Cheng and Angela R. Watson, 2024, Ipsos KnowledgePanel: medium-term 29 percent / long-term 27 percent bachelor’s versus 46 percent never-homeschooled, after demographic controls, noncausal. Brian D. Ray / NHERI 2009 specimen (~84th percentile in math): convenience sample, **advocacy**. IDEA: 34 C.F.R. § 300.137(a); OSEP Q&A 22-01 (February 2022). No individual FAPE at home.

Q&A 22-01 (February 2022) is the federal guidance. Pennsylvania additionally requires that an IDEA-identified student's home program be approved by a certified special educator or licensed psychologist (24 P.S. § 13-1327(d) — HSLDA **advocacy** citing the statute). Do not promise an "IEP at home." Do promise: request an evaluation; know whether the state treats homeschool as a private school; if the parent is the ceiling for a dyscalculic learner, buy expertise rather than an AI workaround. DSM-5 estimates specific learning disorders as a class — reading, writing, and mathematics — at 5–15 percent of school-age children. That is not a dyscalculia base rate. One primary-school sample applying DSM-5 SLDM criteria reported 6 percent; do not treat 6 percent as a universal prevalence. No national study of typical daily math minutes in U.S. homeschools was found. Do not write "the average homeschooler does 45 minutes of math."

Correction 8 — The agent does not take the test

Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman (*Proceedings of the National Academy of Sciences*, 2025): RCT, nearly 1,000 Turkish high-school mathematics students, grades 9–11, four 90-minute sessions. GPT Base, ChatGPT-like, improved assisted practice grades **48 percent** relative to control, then **cut unaided exam grades 17 percent**. GPT Tutor — hints, no full solution, teacher-supplied answers in the hidden prompt — improved practice **127 percent** and was statistically indistinguishable from control on the unaided exam (point estimate -0.004). GPT Base returned a correct answer only **51 percent** of the time on the practice set (42 percent logical errors, 8 percent arithmetic errors). Students used GPT Base as a crutch and did not perceive the harm.²⁰ This is not a homeschool study. It is not GPT-5. It is not Khanmigo. It is the cleanest field experiment that the kitchen-table pattern — "paste the Saxon lesson, get the key, feel fine" — predicts a worse unaided test. GPT Tutor's hidden trick, which a parent can copy and a child-facing chatbot usually cannot: the prompt contained the **correct solution** and **common mistakes**. Ban: paste the worksheet, "what is the answer," photo-to-key, companion persona, open-web agent in the child's hands. Allow, parent-hosted: explain the concept to the *parent* first; generate a lesson outline; generate extra isomorphic practice with the answer key held by the parent; generate a hint script; make a hands-on activity; diagnose a wrong answer the child already produced; explain a worked example *after* the child attempts. These "allow" items are practice suggestions grounded in Bastani plus Soderstrom and Bjork plus the Khanmigo engagement finding, **not a new RCT of homeschool AI**. Photomath, Mathway, and Symbolab are solvers unless the parent constrains them. No Photomath-specific RCT of unaided transfer was opened (**UNVERIFIED** as a named Photomath harm trial). The transferable evidence is Bastani's crutch finding. One AI chapter, not five. This book is not a sequel to *Autonomous AI and Education*.

Philip Oreopoulos and Nina Low, NBER working paper 35620 (August 2026), two-year cluster RCT in 18 Tennessee middle schools: assignment to Khan Academy plus Khanmigo raised MAP math by **1.26 national percentile ranks per term** (SE 0.60), about **0.040 population SD per term**

²⁰Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman, "Generative AI without guardrails can harm learning: Evidence from high school mathematics," *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122. Author PDF, https://hamsabastani.github.io/education_llm.pdf. GPT Base +48 percent practice / -17 percent unaided exam; GPT Tutor +127 percent / -0.004 NS vs. control; GPT Base fully correct 51 percent of the time on the practice set. Percentages of control grades, not Cohen's *d*. Not a homeschool study.

and **about 0.06–0.08 SD over a school year**; implied effect of a full year of active participation in year 2 is **0.14 SD**. Authors: “These gains resemble those from Khan Academy practice without AI assistance.” Ninety-six percent of students tried Khanmigo; the median student messaged it on only a third of practice days and in **17 percent** of mistake sessions; **14.5 percent** of messages contained a mathematical question or a step of reasoning.²¹ Do not invent standalone 0.06 as the ITT. Khanmigo-the-product is designed like GPT Tutor (**vendor**: “never gives you the answer”). Khanmigo-as-used, even in a mandatory daily block with a teacher present, was mostly unused as a tutor. A homeschool parent who buys Khanmigo and walks away should expect the Tennessee pattern, not a college physics custom-tutor pattern.

Hermes 4 is not Hermes Agent. There are three Groks. Adaptive is not an agent. Hermes Agent is a parent-machine tool, not a child-safety school product. Grok Bot requires cloud data storage; no education SKU on the pages opened. El Salvador’s Grok deployment is the assistant in a ministry program, announcement not RCT, not Grok Bot. ChatGPT for Teachers and Claude for Teachers are teacher-facing SKUs; homeschool parents are not verified U.S. K–12 educators by default. ALEKS, Khan Academy, IXL, DreamBox, and MATHia sequence practice; they are not agents. **No page opened on 30 August 2026 documented a district-sold, student-facing, autonomous open-web agent.** Chapter 3 prints the landscape. This correction only forbids the collapse.

COPPA amended Rule: published 22 April 2025, general compliance **22 April 2026**. The Commission **did not finalize** the proposed ed-tech / school-authorization amendments. California A.B. 1159 (Addis), as of 30 August 2026: Active Bill — In Floor Process, Senate third reading. Last amended 21 August 2026. **Not law.** Weixin Liang, Mert Yuksekogul, Yining Mao, Eric Wu, and James Zou (2023): average false-positive rate on human TOEFL essays **61.22 percent** (arXiv) / **61.3 percent** (*Patterns*). Do not use detectors to police a homeschooler’s written math explanations. Companion cases — Garcia / Setzer v. Character Technologies; Raine v. OpenAI — are **allegations**, not verdicts. They are why a math agent should not be a companion. They are not a literature on math learning. GSM8K saturation is not geometry proof. Facts are not an LLM job. AP free response is unaided. Wolfram is a CAS check, not an explainer. Frontier models have long since scored very high on grade-school word problems; that number is not a proof-writing score. AlphaGeometry2 (2025) solved 84 percent of IMO 2000–2024 geometry problems as a **specialized neuro-symbolic system**; DeepMind-adjacent reporting says flagship chat LLMs solved **none** of that IMO-geometry set. A kitchen-table chatbot is not AlphaGeometry. Named 2026 eval of consumer ChatGPT on high-school two-column proofs: **UNVERIFIED**. Treat model-written proofs as untrusted drafts; the child writes the proof unaided. If the family remembers only one sentence from Chapter 3: the model may prepare the adult and the next problem; it may not do the child’s problem.

²¹Philip Oreopoulos and Nina Low, NBER Working Paper 35620 (August 2026): Khanmigo ITT 0.040 population SD per term; about 0.06–0.08 SD over a school year; gains resemble Khan without AI. Do not invent standalone 0.06 as ITT. Weixin Liang, Mert Yuksekogul, Yining Mao, Eric Wu, and James Zou, “GPT detectors are biased against non-native English writers,” *Patterns* 4, no. 7 (2023): 61.22 percent arXiv / 61.3 percent *Patterns* false-positive on human TOEFL essays. COPPA: 90 FR 16918, general compliance 22 April 2026. California A.B. 1159: Senate third reading as of 30 August 2026, **not law**.

Map of the book

Twelve chapters, one paragraph, then the list this book will not fill. Chapter 1 names the kitchen so later chapters can refuse the district analogy: one adult, no district scope, the parent who stopped at algebra, counts quoted from NCES and not crowned, curriculum as a tool not a franchise, a path not a percentile. Chapter 2 locks method: five strands, iteration, the fractions gate as prediction, the number line, teaching moves that survive with their actual objects, fluency as two objects, NMAP 27 not mashed with Kapur, parent-as-teacher as hear-a-wrong-answer. Chapter 3 locks the AI lane: performance is not learning, Bastani, product distinctions, ban versus allow, COPPA, detectors, one sentence. Chapter 4 is grades 1–2: number, operations, place value, and the equal sign that later algebra will need; equal shares as the on-ramp to fractions. Chapter 5 is grades 3–5 and the heaviest elementary chapter: multiplicative reasoning and fractions as numbers on a line; the gate the Panel named. Chapter 6 is grades 6–8: ratio, signed numbers, expressions, early functions; a “grade 6” book still a publisher’s scope; diagnose fractions, integers, and the equal sign before Algebra I. Chapter 7 is Algebra I: generalized arithmetic, functions, the NMAP Major Topics as far as this course carries them, incorrect examples, NCAA titles. Chapter 8 is Geometry: congruence, similarity, proof, right-triangle trigonometry that is not yet the unit circle. Chapter 9 is Algebra II, functions, and precalculus: the rest of the Major Topics, the unit circle, the “have we finished Algebra II?” checklist. Chapter 10 is calculus: AB’s eight units, BC’s two more, CLEP as a different object, the function concept the derivative attaches to. Chapter 11 is data and chance across the years: statistics is a different college-equivalent; CCSSM placed it from Grade 6; AP Statistics does not substitute for calculus in STEM. Chapter 12 is records, transcripts, college-ready: no national diploma; compliance is not college evidence; the record still has to leave the house. The coda is what an honest book can promise versus what it cannot. Every mathematics chapter (4–11) weaves four lanes: explain the mathematics to the parent; teaching guidance; tips that survive evidence; parent-hosted AI, child does the work, parent holds the key. Those lanes are book architecture, not a finding.

What an honest book will not fill, named here so a later chapter cannot fill them with analogy:

1. No 2026 national homeschool headcount. NHES:2023 is the latest federal PFI opened.
2. NHES:2023 microdata on teaching style, curriculum sources, and math subjects taught — the First Look did not republish the 2016 subject tables.
3. No homeschool-curriculum effectiveness RCT.
4. No national study of typical daily math minutes (**UNVERIFIED / no study found**).
5. STEM-major and college-calculus completion rates for homeschool graduates in a probability sample: no named source opened.
6. Parent-as-sole-teacher: Maloney 2015 is a school-homework study; no fetched experiment on math-anxious parents as full-time instructors.
7. No year-plus homeschool RCT of parent-constrained GPT-Tutor workflows.
8. No Photomath- or Chegg-specific RCT of unaided transfer opened; a 2026 Chinese staggered-adoption study was not opened as a PDF.
9. Khanmigo durability past the term, and whether a parent sitting next to the child is a different intervention than the Tennessee ITT: design hypotheses, not fitted results.
10. AP Calculus CED full PDF not fully extracted; AP Precalculus full CED not opened.
11. NMAP Table 2 geometry/measurement benchmarks truncated in the ERIC fetch.

12. NCTM Focal Points official book paywalled; third-party posting used.
13. Named vendor primaries still closed (Teaching Textbooks official scope; RightStart; Math Mammoth; Life of Fred; live Khan course list; some 30 August 2026 list prices).
14. Integrated Math I–III topic-by-topic from Appendix A only partly extracted.
15. State-by-state status of CCSSM in 2026 **UNVERIFIED**.
16. Homeschool-specific dual-enrollment and AP exam participation rates not in these sources.
17. Fuchs 2013/2014 fractions-intervention numerical effect sizes not copied into the memos.
18. Isolated effect of the timer; interleaving in grades 1–5 and in proof/calculus.
19. Optimal dosage and order of concepts versus procedures for whole topics — still thin.
20. Transfer from informal activity (cooking, money, games) to formal items, measured, not assumed.
21. Proof, modeling, and the limit concept — thin experimental base; named 2026 eval of consumer ChatGPT on two-column proofs **UNVERIFIED**.
22. A.B. 1159 final enactment; El Salvador evaluation; Setzer/Garcia docket terms; whether any U.S. district now sells a student-facing open-web autonomous agent (negative finding is pages opened, not an exhaustive census).
23. College-credit comparison study of AP versus dual enrollment versus CLEP math outcomes at scale: not found; IES dual-enrollment outcomes report expected 2027.
24. NACAC current official guidance specific to homeschool math: not opened as a 2026 PDF.
25. Unschooling math outcomes beyond the 12 percent informal share; 2026 AI-usage surveys **UNVERIFIED**; Massachusetts 2026 legislative outcome on approval versus notice; and the remaining closed primaries the brief already listed (Levine/Mix, Beilock 2010, Greer 1994, Brunmair and Richter *g*, and others).

Hold the thesis unless a primary contradicts it. Chapter 1 does not tour standards. It names the kitchen so the method lock and the mathematics can begin.

Chapter 1

The Kitchen Table Is Not a District



Figure 2: An empty wooden chair, a hardcover book with a blank spine, and a manila envelope on a kitchen table.

Three documents sit on the table this book is written from. None of them is a kitchen scene. None of them is a chatbot. The first is a federal First Look table. Ellena Sempeles and Jiashan Cui, writing for the National Center for Education Statistics as NCES 2024-113, report that among students ages

5–17 with a K–12 grade equivalent, 3.4 percent were homeschooled in the 2022–23 school year — approximately 1,765,000 students — and that 5.2 percent received instruction at home, a broader object that includes full-time virtual school.²² The weighted Parent and Family Involvement sample of 19,562 students represents about 53 million K–12 students. That is the latest national homeschool count this book will treat as official. It is not a 2026 total. There is no 2026 national total in any federal PFI this research opened. Anyone who hands you one is selling a different instrument.

The second document is a state card from the Home School Legal Defense Association. Texas, on the card this research opened: notification, no; assessment, no; subjects, yes; teacher qualifications, none. Mathematics is a named subject. A written curriculum is required under the *Leeper* private-school reading the card cites. A mathematics degree is not.²³ The card is advocacy, attorney-reviewed for members. It is not the Texas Education Code and it is not the *Leeper* opinion. Illinois and Idaho sit next to Texas on the same advocacy map as other no-notice examples. New York, on a different HSLDA how-to, is notification plus an Individualized Home Instruction Plan, hours, quarterly reports, and a standardized battery that must include mathematics in high school. The legal object is not “a math program.” The teaching chapters of this book cannot assume a single paperwork load.

The third document is a publisher placement test. Houghton Mifflin Harcourt’s Saxon middle-grades PDF places by score band into Math 5/4, 6/5, 7/6, 8/7, Algebra 1/2, or Algebra 1, and the publisher says the test is not infallible and that Saxon is skill-level rather than grade-level.²⁴ Singapore Math Inc. ships untimed PDFs and tells parents that starting at an earlier level is common and acceptable. Math-U-See’s placement tool is a yes/no mastery questionnaire; Epsilon is a fractions *level*, not “fifth grade.” Beast Academy, on the Art of Problem Solving pages opened this research day, generally recommends that students start about a level behind their current grade because of problem difficulty.²⁵ Those are vendor documents. They disagree with one another about what “grade 6” names. A child of ten can place into Singapore 3A, Saxon 6/5, and Math-U-See Gamma on the same afternoon. That is not a contradiction. It is three different grain-sizes of “grade.” The National Center for Education Statistics already knew the sequence was stretched. In the 2016 Parent and Family Involvement tables, Algebra I was taught that year to 41 percent of the homeschooled 6–8 band; arithmetic was still taught to 31 percent of the 9–12 band.²⁶ Grade labels in

²²Ellena Sempeles and Jiashan Cui, *Parent and Family Involvement in Education: 2023* (NCES 2024-113) (Washington, DC: National Center for Education Statistics, September 2024), Table A-6 and footnote: 3.4 percent homeschooled, approximately 1,765,000 students; 5.2 percent instruction at home; weighted $n = 19,562$ representing about 53 million K–12 students. <https://nces.ed.gov/pubs2024/2024113.pdf>. Access date for all URLs in these notes: 30 August 2026. Latest federal PFI opened; not a 2026 national total.

²³Home School Legal Defense Association, “Texas Homeschool Laws (at a glance),” <https://hsllda.org/legal/texas>. **Advocacy**, attorney-reviewed for members, not the Texas Education Code and not the *Leeper* opinion. Teacher qualifications: none. Subjects required, including mathematics. Notification: no. Assessment: no.

²⁴Houghton Mifflin Harcourt / Saxon Homeschool, *Middle Grades Math Placement Test* (PDF), <https://images.salsify.com/image/upload/s-hxkiZwTr-/60b1c78e60f3fc141b8ed703c0c62f98d9b68153.pdf>. **Vendor**. Score bands into Math 5/4 through Algebra 1. Publisher: the test is not infallible; Saxon is skill-level rather than grade-level.

²⁵Art of Problem Solving, Beast Academy books and FAQ, <https://beastacademy.com/books> and <https://beastacademy.com/faq/books>. **Vendor**. “We generally recommend that students start about a level ‘behind’ their current grade level.” Levels 4–5 cover many middle-grades standards.

²⁶Jiashan Cui and Rachel Hanson, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey*, NCES 2020-001 (Washington, DC: NCES, 2019), Table 9: Algebra I taught that year to

this book are sequence anchors, not prison bars.

Those three documents are the kitchen this chapter has to name. One adult, usually the mother, usually a purchased formal curriculum, legally allowed in most of the country to teach Algebra II without a mathematics degree. The design user stopped at algebra. The legal object is not a math program. Curriculum is a tool, not a franchise. Homeschool math-superiority is a fight, not a finding. This chapter prints those sentences so later chapters can refuse the district analogy without sounding pedantic. Chapter 2 will lock method — five strands, iteration, the fractions gate, the moves that survive. Chapter 3 will lock the AI lane — the agent does not take the test. This chapter does not rewrite either. It names the room those chapters inherit.

A district has a scope-and-sequence committee, a teacher of record with a credential, a special-education team that writes an Individualized Education Program under a Free Appropriate Public Education duty, a transcript office, and a testing calendar. A kitchen has one adult, a book the adult chose, a state statute or a regulation that may require a notice or a test or nothing, and a college or an athletic association that will later want a document the state never asked for. Confusing those two rooms is how a parent buys a “grade 6” book as if it were a legal grade, waits for an IEP that will not arrive, treats an advocacy percentile as a finding, or stops at Algebra I because “the state doesn’t require it” and then meets a university page that wants three years through Algebra II. The documents do not let that confusion stand.

What this chapter is not. It is not a 50-state legal manual. It is not a franchise catalog. It is not a 180-day plan. It is not a sequel to *Autonomous AI and Education*. It does not open with a percentile and it does not open with a product. It does not invent a kitchen scene and present it as reported. If a later paragraph needs a composite to show a multi-age table, that composite will be labelled *hypothetical* in the sentence that introduces it. Access date for the fetches under these claims remains 30 August 2026.

The popular story about this kitchen collapses several claims that do not survive together. Homeschoolers score higher in math. A purchased grade label is a legal grade. The parent who stopped at algebra should help less, or should buy a video that “does 100 percent of the teaching,” or should hand the worksheet to a chatbot. Homeschoolers go to college at higher rates. The first of those is a fight between convenience samples and their critics, not a nationally representative mathematics score. The second is a publisher’s scope. The third is a misreading of a school-homework study and a vendor slogan. The fourth is contradicted, as a rate, by a nationally representative adult survey that does not claim a math cause. This chapter prints the corrections that belong here. The rest of the book inherits them.

The thesis this book will keep defending is already in the frame. A parent who is not a math specialist can take a child from first-grade number sense through high-school calculus if three things hold: the parent understands the concept well enough to hear a wrong answer; the child does the struggle before the tool; and any agent is a supplement hosted and constrained by the parent — not a substitute for the child’s work and not a school-safe product by default. The sources do not prove that thesis as a causal claim about homeschool mathematics. They do not contradict it. They constrain how it can be attempted. No homeschool-curriculum randomized trial was opened for

41 percent of the 6–8 band; arithmetic still taught to 31 percent of 9–12. <https://nces.ed.gov/pubs2020/2020001.pdf>.

this project. No nationally representative study of daily homeschool math minutes was opened. No 2026 national enrollment total exists to crown a headcount. Promise a path, not a percentile.

What follows is the kitchen, named from documents. Section 1.1 installs the actual teacher — one adult, usually the mother, usually a purchased book — before any standard is explained. Section 1.2 splits compliance from teaching so later chapters do not assume New York hours or Texas silence. Section 1.3 prints the only federal headcount this book is allowed to use, labelled against Pulse and against an advocacy synthesis. Section 1.4 makes “I stopped at algebra” a design constraint, not a marketing persona, and points at Chapter 2 for the method that constraint requires. Section 1.5 kills the calendar-grade as a legal object. Section 1.6 names the tools the parent will actually meet without selling any of them. Section 1.7 restates the thesis, prints the live fight about outcomes, and maps the twelve chapters in short. College-ready documentation is Chapter 12’s job. This chapter plants it. It does not dump an NCAA worksheet or an AP order deadline here.

1.1 One adult, usually the mother, usually a purchased book

The teacher this book is written for is already in a federal table. The 2016 Parent and Family Involvement Survey, published as NCES 2020-001, asked who provided the instruction. The mother was the main provider for 78 percent of homeschooled students. The father was the main provider for 13 percent. Another relative, 3 percent. “Other person,” 4 percent. Virtual school as the main provider, 2 percent, a cell the agency flags to interpret with caution.²⁷ That is not a composite family invented for warmth. It is the last federal table this research opened that asked the question. The 2023 First Look did not republish the teaching-style table or the subject-taught tables. Whether the 2023 microdata, if later analyzed, still show the same shares is an open question this chapter will not fill.

The same 2016 tables describe how formal the work was. Seventy-seven percent of homeschooled students were taught with a “mostly or strictly formal curriculum.” Twelve percent were taught with “mostly or strictly informal learning.” The informal share was higher among students whose parent reported a disability — 20 percent informal, 62 percent formal — which is a fact about a minority practice, not a finding that informal is the special-education method.²⁸ The unschooling tradition associated with John Holt is real. It is named. It is not the modal United States homeschool mathematics practice in the last federal table that asked. This book’s thesis — a parent who can hear a wrong answer, a child who struggles before the tool, an agent that does not take the test — is closer to structured or mastery work than to pure unschooling. That is a design choice, not a scorn. The twelve percent are not a failure of the kitchen. They are a poor match for a New York Individualized Home Instruction Plan, a Pennsylvania portfolio, and a National Collegiate Athletic Association core-course worksheet unless the parent translates lived mathematics into a course title after the fact. Chapter 12 will say so again. This chapter says it once so the teaching chapters do not have to pretend the market is Holt.

²⁷Cui and Hanson, NCES 2020-001, Table 3: mother 78 percent main provider; father 13; another relative 3; “other person” 4; virtual as main provider 2 (interpret with caution).

²⁸Cui and Hanson, NCES 2020-001, Table 6: 77 percent “mostly or strictly formal curriculum”; 12 percent “mostly or strictly informal learning.” Informal share higher among students whose parent reported a disability (20 percent informal / 62 percent formal). The 2023 First Look did not republish the teaching-style table. **UNVERIFIED** whether 2023 microdata still show ~12 percent informal.

Any instruction by a tutor or private teacher: 23 percent. Any instruction from a local homeschooling group or co-op: 31 percent. Any online course: 23 percent of homeschoolers, 30 percent of the 9–12 band.²⁹ Those are supplements, not replacements for the 78 percent. The kitchen is one adult, usually the mother, sometimes supplemented. A later chapter that writes as if a certified mathematics teacher is in the room has left the table. A later chapter that writes as if no adult is supervising has left it the other way.

Where the books come from, in the same 2016 table, is a consumer market plus the public library, not a district adoption cycle. Families could mark more than one source. Library, 66 percent. Bookstore, 62 percent. Websites, 60 percent. Homeschool catalog, 60 percent. Homeschooling organization, 45 percent. Educational publisher, 44 percent. Church, 31 percent. Public school, 25 percent. Private school, 7 percent. Virtual school or curriculum as a source, 3 percent.³⁰ A parent who walks into a library, a used-curriculum sale, and a vendor site in the same month is inside the table. A parent who waits for a district to adopt a program is in a different legal object.

The 2023 First Look did republish a virtual-course table, and it is easy to misread. Among all K–12 students — enrolled plus home — 7 percent took any virtual course. Among those classified as homeschooled, 48 percent took any virtual course and 20 percent took all courses virtually.³¹ Full-time virtual public or private school is a different legal object from parent-directed homeschool. The National Center for Education Statistics now separates them precisely because parents and surveys used to blur them. Pew Research Center, restating the 2023 figures on 20 February 2025, notes an additional 1.8 percent of K–12 students in full-time virtual public or private school whose parents did *not* consider them homeschooled.³² This book will not collapse those rooms. A child in a virtual public charter has a teacher of record, state tests, and, if eligible, an Individualized Education Program. The parent in that room is not the teacher this chapter is installing.

Part-time school is visible and small. In 2016, 18 percent of homeschoolers also attended a school — 15 percent public, 2 percent private (caution), 1 percent college (caution).³³ Dual enrollment is real. It is not the typical path in that table. High-school families using a community-college algebra or calculus course will not show up as a large national percentage and should not be written as the default. They are a release valve when the parent is the ceiling. Section 1.4 will say so. Chapter 12 will say how to put the college transcript on the homeschool record.

The most important 2016 table for this book is the one that asked which mathematics had ever

²⁹Cui and Hanson, NCES 2020-001, Table 3 and related: any tutor or private teacher 23 percent; any local home-school group or co-op 31 percent; any online course 23 percent of homeschoolers, 30 percent of the 9–12 band.

³⁰Cui and Hanson, NCES 2020-001, Table 6, curriculum sources (multiple sources allowed): library 66; bookstore 62; websites 60; homeschool catalog 60; homeschooling organization 45; educational publisher 44; church 31; public school 25; private school 7; virtual school or curriculum 3.

³¹Sempeles and Cui, NCES 2024-113, Table A-8: among all K–12 (enrolled plus home), 7 percent took any virtual course; among those classified homeschool, 48 percent any virtual course, 20 percent all courses virtually.

³²Pew Research Center, “A look at homeschooling in the U.S.,” 20 February 2025, <https://www.pewresearch.org/short-reads/2025/02/20/a-look-at-homeschooling-in-the-us/>. Journalism restating NCES PFI 2023; adds 1.8 percent full-time virtual public or private whose parents did not consider them homeschooled (up from 0.86 percent in 2018–19). Not a primary survey.

³³Cui and Hanson, NCES 2020-001, Table 4: 18 percent of homeschoolers also attended a school (15 percent public; 2 percent private, interpret with caution; 1 percent college, interpret with caution). Dual enrollment visible but small; not the typical path.

been taught to high-school-level homeschoolers during home instruction. Arithmetic, 76 percent. Basic algebra (Algebra I), 79 percent. Geometry, 52 percent. Advanced algebra (Algebra II), 49 percent. Probability, 24 percent. Calculus, 15 percent, a cell the agency flags to interpret with caution.³⁴ Read those numbers as coverage, not as quality, and not as college calculus. Algebra I is common. Algebra II and geometry are not universal. Calculus at home is a minority. The 2012 report, NCES 2016-096, found that 88 percent of high-school-level homeschoolers had been taught basic algebra, 54 percent geometry, about a third advanced algebra, and calculus “not often” — 10 percent, interpret with caution.³⁵ The movement from 88 percent to 79 percent should not be narrated as decline without checking standard errors and questionnaire differences. Both figures say the same operational thing. A book that treats Algebra II as automatic has left the table. A book that treats calculus at home as typical has left it. A book that treats Algebra I as rare has left it the other way.

The same year’s subject-by-grade-band table is why grade labels cannot be prison bars. Arithmetic taught *that year*: 83 percent of K–2, 86 percent of 3–5, 66 percent of 6–8, 31 percent of 9–12. Algebra I that year: suppressed or negligible in K–2, 13 percent of 3–5, 41 percent of 6–8, 33 percent of 9–12. Algebra II that year: 33 percent of 9–12, suppressed below. Geometry that year: 15 percent of 3–5, 19 percent of 6–8, 26 percent of 9–12. Calculus that year: 13 percent of 9–12, caution wherever the agency placed one.³⁶ A non-trivial share of “ninth graders” are still in arithmetic. A non-trivial share of “sixth-to-eighth graders” are in Algebra I. A teaching chapter that says “in sixth grade we do ratios” as if the calendar were the mathematics is using a district sentence in a room that does not run on a district calendar.

Who is in the room, as a labor-force fact, is in the 2023 table. Homeschooling rates were 8.8 percent among students in two-parent families with one parent working full-time and one not in the labor force — the classic “one parent teaching” household — and 1.1 percent among students with two parents both working full-time.³⁷ The 2019 *Condition of Education* indicator, from the previous PFI, found the highest rates among two-parent families with one parent in the labor force (6.6 percent) and among households with three or more children (3.9 percent, versus 2.3 percent with two children and 2.2 percent with one).³⁸ Most *hours* of homeschool mathematics are still parent-supervised in a one-earner or 1.5-earner household. Not every reader is a stay-at-home parent. The

³⁴Cui and Hanson, NCES 2020-001, Table 8 — high-school-level homeschoolers, subjects *ever* taught during home instruction: arithmetic 76; basic algebra (Algebra I) 79; geometry 52; advanced algebra (Algebra II) 49; probability 24; calculus 15 (interpret with caution). Not college calculus and not a completion rate.

³⁵Jeremy Redford, Danielle Battle, and Stacey Bielick, *Homeschooling in the United States: 2012*, NCES 2016-096 (Washington, DC: NCES, 2016), figure 5 and related: 88 percent of high-school-level homeschoolers taught basic algebra; 54 percent geometry; about a third advanced algebra; calculus “not often” (10 percent, interpret with caution). ERIC ED569947, <https://files.eric.ed.gov/fulltext/ED569947.pdf>. Do not narrate 88 → 79 as decline without checking standard errors and questionnaire differences.

³⁶Cui and Hanson, NCES 2020-001, Table 9 — subjects taught *that year* by grade band. Arithmetic: K–2 83; 3–5 86; 6–8 66; 9–12 31. Algebra I: 3–5 13; 6–8 41; 9–12 33. Algebra II: 9–12 33. Geometry: 3–5 15; 6–8 19; 9–12 26. Calculus: 9–12 13. Cells flagged by the agency remain caution cells.

³⁷Sempeles and Cui, NCES 2024-113, Table A-6: two-parent family, one full-time and one not in labor force, 8.8 percent homeschooled; two parents both full-time, 1.1 percent.

³⁸National Center for Education Statistics, “Homeschooled Children and Reasons for Homeschooling,” *Condition of Education*, May 2022 (NHES 2019), <https://nces.ed.gov/programs/coe/indicator/tgk/homeschooled-children>. Home-schooling rate 6.6 percent when two parents, one in the labor force; 3.9 percent in households with three or more children versus 2.3 percent with two and 2.2 percent with one.

1.1 percent is small and not zero. Video and app curricula, evening co-ops, and a partner who teaches the mathematics block after work exist for that household. This book will not write as if every reader has the whole morning. It will write as if most of the hours still have an adult in them who can hear a wrong answer — or who must arrange for someone who can.

Parent expectations in 2016 were more modest than the franchise brochure. Thirty-one percent of homeschool parents expected a bachelor's degree; 30 percent a graduate or professional degree; 13 percent high school only. Enrolled-school parents: 29 percent bachelor's, 43 percent graduate.³⁹ Homeschool parents were less likely, not more, to expect graduate school. Section 1.7 will print the Cardus 2023 adult-attainment rates, which are consistent with more modest bachelor's completion among medium- and long-term homeschoolers and which do not report a mathematics score. The expectation table is not a finding that homeschool mathematics is less ambitious. It is a finding that the parent this book is written for is not, as a population, shopping for a gifted pipeline. Beast Academy and Art of Problem Solving exist for families who are. They are not the modal kitchen.

A *hypothetical* illustration, labelled as such, can put the table in a room without inventing a reported family. A parent of three — ages 8, 11, and 14 — sits the eight-year-old on a mastery program with blocks, the eleven-year-old on a grade-numbered workbook the placement test actually chose, and drives the fourteen-year-old to a Tuesday co-op geometry class.⁴⁰ The empirical pieces that composite combines are in the tables: multi-child households have higher homeschooling rates; 77 percent use a formal curriculum; 31 percent have any co-op instruction; geometry is not universal at the high-school level. The sitting, the driving, and the three named ages are hypothetical. Later chapters may use the same device. They will keep the label.

The adult in that room is the teacher. The book on the table is a tool the adult chose. The legal object facing that adult is not “a math program.” That is the next section.

1.2 The legal object is not a math program

Homeschooling is legal in all fifty states and the District of Columbia. That sentence is load-bearing and easy to over-read. It does not mean there is a federal homeschool statute. There is not. It does not mean the legal object is a mathematics program. It is not. It does not mean the parent must hold a mathematics degree. In the states this research opened, most do not require a teaching credential of any kind, and none of the opened no-notice and notification cards required a mathematics degree.⁴¹ The teaching chapters of this book exist because the law, in most of the country, permits a parent who last passed Algebra I in 1998 to be the teacher of record for Algebra II. That is not a finding that the parent should improvise. It is a finding that the state will not supply the specialist.

³⁹Cui and Hanson, NCES 2020-001, Table 10: homeschool parents — 31 percent expected a bachelor's degree; 30 percent a graduate or professional degree; 13 percent high school only. Enrolled-school parents: 29 percent bachelor's, 43 percent graduate.

⁴⁰Composite *hypothetical*, labelled in the sentence that introduces it. Empirical pieces: NCES 2019 multi-child rate; NCES 2016 formal-curriculum 77 percent and any co-op 31 percent. The sitting, the driving, and the three named ages are not a reported family.

⁴¹HSLDA at-a-glance cards opened for Texas, Illinois, Idaho, California (except option 3), and Massachusetts: teacher qualifications “No.” North Carolina: high-school diploma or equivalent. Pennsylvania option 1: supervisor-education requirement. **Advocacy**, not statutes. None of the opened cards required a mathematics degree.

Regulation ranges from no-notice through notification to local approval. The Home School Legal Defense Association publishes “at a glance” cards. They are advocacy. They are attorney-reviewed for members. They are not statutes. Parents must read their own state’s code. This section uses the cards as a map of *variation*, not as the law of Texas or New York.

No-notice examples opened: Texas, Illinois, Idaho. On those cards: no notification, no assessment, subjects required, no teacher qualification. Texas’s how-to, still advocacy, names mathematics, reading, spelling and grammar, and good citizenship as required subjects, and a written curriculum under the *Leeper* private-school interpretation.⁴² A parent in Texas may legally teach calculus without notifying anyone and without a mathematics credential, provided a bona fide written curriculum includes mathematics. Illinois and Idaho, on the cards, look similar at that grain: subjects yes, notice no, testing no, credential no.

Notification examples: North Carolina, California, Pennsylvania. North Carolina: one-time notice to the Division of Non-Public Education, not to the local district; the person providing academic instruction must have a high-school diploma or equivalent; nine calendar months; an annual nationally standardized test that *must include mathematics* (plus English grammar, reading, and spelling); scores kept available one year.⁴³ The HSLDA card lists state-mandated subjects as no. The test-record requirement is how mathematics enters the compliance list. A family that never taught mathematics would fail the test-record requirement, not a curriculum-naming requirement. California: notification yes; three options; teacher qualifications no except option 3; no assessment on the card.⁴⁴ Pennsylvania: four options; notification yes; teacher qualifications and assessment except option 1. Option 1, on the advocacy how-to: annual affidavit or unsworn declaration to the superintendent by 1 August; portfolio; standardized testing in mathematics and reading in grades 3, 5, and 8; secondary mathematics must include general mathematics, algebra, and geometry, citing 24 P.S. § 13-1327.1.⁴⁵ Option 1 has a supervisor-education requirement. Confirm the statute. Do not treat the card as the statute.

Approval and high-regulation examples: Massachusetts and New York. Massachusetts, on the card: notification yes; assessment yes; subjects yes; teacher qualifications no. Local approval has historically been required. Legislative fights over approval versus notice were in flux on HSLDA’s

⁴²Home School Legal Defense Association, “How to Comply with Texas’s Homeschool Law,” <https://hsllda.org/post/how-to-comply-with-texas-homeschool-law>. **Advocacy.** Required subjects include mathematics; written curriculum per *Leeper* / private-school interpretation. Not a substitute for the Texas Education Code.

⁴³Home School Legal Defense Association, “How to Comply with North Carolina’s Homeschool Law,” <https://hsllda.org/post/how-to-comply-with-north-carolinas-homeschool-law>. **Advocacy.** One-time DNPE notice; instructor high-school diploma or equivalent; nine calendar months; annual nationally standardized test that must measure mathematics (plus grammar, reading, spelling). Confirm against the North Carolina statute.

⁴⁴Home School Legal Defense Association, “California Homeschool Laws (at a glance),” <https://hsllda.org/legal/california>. **Advocacy.** Three options; notification yes; teacher qualifications no except option 3; assessment no.

⁴⁵Home School Legal Defense Association, “Pennsylvania Homeschool Affidavit and Declaration,” <https://hsllda.org/post/pennsylvania-homeschool-affidavit>, and “Pennsylvania Homeschool Laws (at a glance),” <https://hsllda.org/legal/pennsylvania>. **Advocacy** citing 24 P.S. § 13-1327.1: annual affidavit or unsworn declaration to superintendent by 1 August (option 1); portfolio; standardized testing in mathematics and reading in grades 3, 5, and 8; secondary mathematics including general mathematics, algebra, and geometry. Confirm the statute.

news feed in 2026. This book will not freeze a 2026 bill as law.⁴⁶ New York, on the how-to this research opened, is notification plus a plan plus reporting, which functions as high regulation even if the word is “notice.” Annual notice of intent. An Individualized Home Instruction Plan listing syllabi, curriculum materials, textbooks, or a plan of instruction. Hours. Arithmetic in grades 1–6; mathematics credits in grades 7–8 and 9–12. Quarterly reports. Annual assessment — narrative permitted in 1–3; standardized at least every other year in 4–8; standardized every year in 9–12. Named tests include Iowa, California Achievement Test, Stanford, CTBS, MAT (retired), a State Education Department test, PASS. SAT and ACT do *not* qualify. Composite above the 33rd percentile or one year of growth.⁴⁷ A parent in New York must name the mathematics materials, log hours, file quarterly grades, and produce a standardized mathematics-inclusive battery in high school. The same parent in Texas must keep a written curriculum that includes mathematics and need not tell the district. The teaching chapters cannot assume a single paperwork load.

Must a mathematics curriculum be named? Often no. Texas requires *a* written curriculum covering mathematics, not a named title. New York’s plan requires a list of syllabi, curriculum materials, textbooks, or a plan of instruction. Pennsylvania’s affidavit requires an outline of proposed educational objectives by subject area, not a publisher name. North Carolina, on the card, does not mandate subjects but mandates a test that includes mathematics. The legal object is coverage, a record, or a score — not Saxon, not Singapore, not Math-U-See. A later chapter that treats a franchise as compliance has left the statute.

An international contrast keeps the United States kitchen from sounding like the only possible room. Homeschooling is a legal option in the United Kingdom, Canada, Australia, and New Zealand, with provincial and state variation. It is effectively banned in Germany. New South Wales, under the Education Act 1990, requires registered home-schooling programs to be based on New South Wales Education Standards Authority syllabuses, including Mathematics K–10, and an Authorised Person assesses the *program*, not the child.⁴⁸ That is closer to a district scope map than Texas. England’s elective home education is a parental right with local-authority awareness duties but no national mathematics syllabus mandate of the New South Wales type. A 2026 official England registration count was not opened for this research; figures circulating in journalism stay out of this book.⁴⁹ Canada is provincial. Quebec is widely described as more regulated, Alberta and British Columbia as more permissive — confirm on provincial sites before any sentence that names a

⁴⁶Home School Legal Defense Association, “Massachusetts Homeschool Laws (at a glance),” <https://hsllda.org/legal/massachusetts>. **Advocacy.** Notification yes; assessment yes; subjects yes; teacher qualifications no. 2026 legislative fights on approval versus notice were in flux on HSLDA’s news feed. Do not freeze pending legislation as law.

⁴⁷Home School Legal Defense Association, “How to Comply with New York’s Homeschool Law,” <https://hsllda.org/post/how-to-comply-with-new-yorks-homeschool-law>. **Advocacy** reading of 8 NYCRR § 100.10. IHIP, hours, arithmetic/mathematics credits, quarterly reports, named standardized tests, 33rd-percentile-or-growth rule; SAT/ACT do not qualify. Confirm against NYSED regulation before legal advice. This book does not write a 180-day plan from the hours language.

⁴⁸NSW Department of Education, “Home schooling registration at a glance,” <https://www.nsw.gov.au/education-and-training/home-schooling/registration-guidelines/registration-at-a-glance>, and “Curriculum Reform and home schooling,” <https://www.nsw.gov.au/education-and-training/home-schooling/support-material/curriculum-reform-home-schooling>. Australian state government primaries. Registered programs must include Mathematics and be based on NESA Mathematics K–10 syllabuses; Authorised Person assesses the program, not the child.

⁴⁹Official 2026 England elective-home-education registration count: not opened. Bright Heart Education’s “111.7K” figure is journalism, not a statistical agency. **UNVERIFIED** as an official count; left out of the body.

Canadian mathematics requirement. This is a United States book. It does not assume a national curriculum. An Australian New South Wales reader *does* have one. The architecture later chapters use — mathematics first, grade labels as anchors — still works there if each unit can be mapped onto a syllabus outcome. It does not work in a jurisdiction that bans the kitchen.

Federal overlays that are *not* homeschool law still bind the mathematics. The Individuals with Disabilities Education Act, the Family Educational Rights and Privacy Act, the Children’s Online Privacy Protection Act, and the National Collegiate Athletic Association’s eligibility rules are the four that matter for this book’s later chapters. FERPA applies to schools receiving United States Department of Education funds, not to a kitchen. Chapter 3 will say that Grok Bot is not FERPA-ready; this chapter does not sequel that monograph. COPPA’s 2025 amendments and April 2026 compliance date are Chapter 3’s job. NCAA is a private association’s rules, not a statute, and it binds athletic scholarships; Chapter 12 is the export chapter. The overlay that belongs here is IDEA, because parents are sold an “IEP at home” that the regulation does not create.

Under IDEA, children parentally placed in private schools have no individual right to a Free Appropriate Public Education and no individual right to the Individualized Education Program they would have in public school. That is 34 C.F.R. § 300.137(a), current on the eCFR this research day.⁵⁰ Local educational agencies must conduct child find, consult, and spend a proportionate share of IDEA Part B funds on equitable services documented in a *services plan* for those children designated to receive them. Whether homeschoolers are in that bucket depends on whether state law treats homeschools as private elementary or secondary schools. The National Center for Systemic Improvement at WestEd, in a 2025 quick reference, states the fork: if the state recognizes homeschools as private schools, include them in the proportionate-share count; if it does not, the federal private-school provisions may not attach.⁵¹ OSEP’s Questions and Answers 22-01, revised February 2022, is the federal guidance.⁵² Do not promise an Individualized Education Program at home. Do promise: request an evaluation — child find exists even if services do not; know whether your state treats you as a private school; if the parent is the ceiling for a learner with a mathematics disability, buy expertise rather than an AI workaround. Chapter 2 will say the last of those as a teaching rule. This chapter says it as a legal fact.

Pennsylvania adds process that is still not an IEP. Under 24 P.S. § 13-1327(d), as cited on HSLDA’s special-education page — advocacy citing the statute — a student identified under IDEA must have a home program that addresses the student’s needs and is *approved* by a Pennsylvania-certified special education teacher or a licensed clinical or certified school psychologist; written notice of

⁵⁰34 C.F.R. § 300.137(a): parentally placed private-school children with disabilities have no individual right to FAPE or to the IEP they would have in public school. eCFR current as of 27 August 2026 (title last amended 24 July 2026), <https://www.ecfr.gov/current/title-34/subtitle-B/chapter-III/part-300/subpart-B/subject-group-ECFR3556f7ac2fe0a92/section-300.137>. See also §§ 300.130–300.144 (equitable services).

⁵¹National Center for Systemic Improvement (WestEd), *Quick Reference Guide on Parentally-Placed Private School Proportionate Share of IDEA Funds* (2025), https://ncsi.wested.org/wp-content/uploads/2025/01/NCIS_Quick_Reference_Proportionate_Share_FINAL_ADA.pdf. Federally funded technical-assistance center, not binding law. If a state recognizes homeschools as private schools, include those children in the proportionate-share count; if not, the federal private-school provisions may not attach.

⁵²U.S. Department of Education, OSERS/OSEP, *Questions and Answers on Serving Children with Disabilities Placed by Their Parents in Private Schools* (Q&A 22-01), revised February 2022, <https://sites.ed.gov/idea/idea-files/q-and-a-children-with-disabilities-private-schools-parentally-placed/>. Federal guidance: child find, consultation, proportionate share, services plans; no individual FAPE right.

approval is filed with the affidavit.⁵³ That is more process than Texas. It is not FAPE. The Coalition for Responsible Home Education, children's-rights advocacy in the other direction, notes that most states have few provisions ensuring disabled homeschoolers' needs are met; the parent becomes the de facto IEP team.⁵⁴ NCES 2016: 21 percent of homeschoolers had a parent-reported disability; among them, informal teaching style was more common. Special-education notes belong here and as callouts in Chapter 2. They do not belong in a ghetto chapter the anxious parent skips.

The design fact for every later chapter is blunt. In Texas a parent may legally teach calculus without notifying anyone and without a mathematics credential, provided a bona fide written curriculum includes mathematics. In New York the same parent must name the materials, log hours, file quarterly grades, and produce a standardized mathematics-inclusive battery in high school. In Pennsylvania option 1 the parent files an affidavit, keeps a portfolio, tests mathematics in grades 3, 5, and 8, and, if the student is IDEA-identified, obtains an approval letter that is still not an IEP. Most opened states do not require a mathematics degree. Mathematics is often a named subject. It is rarely a named curriculum. HSLDA is how a parent finds the state's code. The code is the authority.

Before teaching anyone, name how many kitchens exist — and refuse a 2026 crown.

1.3 Counts: quote NCES, do not crown 2026

The official federal homeschool rate for 2022–23 is 3.4 percent, about 1,765,000 students. That sentence is the only national headcount this book is allowed to use as NCES. It comes from *Parent and Family Involvement in Education: 2023*, NCES 2024-113, published September 2024, from the National Household Education Surveys Program of 2023, fielded January through August 2023 by the Census Bureau.⁵⁵ Homeschooling, in that survey, means ages 5–17, a K–12 grade equivalent, instruction at home instead of public or private school all or most of the time, excluding those in school more than 25 hours per week and those homeschooled only for temporary illness. Table A-6's footnote is the number to quote: "The total number of homeschooled students in 2023 was approximately 1,765,000." The weighted sample of 19,562 students represents about 53 million K–12 students. Instruction at home — homeschool or full-time virtual — was 5.2 percent. The Institute of Education Sciences press release of 17 September 2024 headlines the broader rise: 5.2 percent versus 3.7 percent in 2018–19.⁵⁶

⁵³Home School Legal Defense Association, "Special Education Provisions for Pennsylvania," <https://hsllda.org/post/special-needs-provisions-for-pennsylvania>. **Advocacy** citing 24 P.S. § 13-1327(d) and § 13-1327.1(e)(1): IDEA-identified student's home program must be approved by a PA-certified special educator or licensed psychologist. More process than Texas; not an IEP with FAPE.

⁵⁴Coalition for Responsible Home Education, "Homeschooling & Disabilities," <https://crhe.org/guides/resources-for-homeschool-parents/disabilities/>. **Advocacy** (children's-rights). Accurate on the FAPE gap; prescriptive IEP-at-home advice is CRHE's stance, not a federal requirement. NCES 2016: 21 percent of homeschoolers had a parent-reported disability.

⁵⁵Sempeles and Cui, NCES 2024-113, First Look. NHES:2023 PFI fielded January–August 2023 by the Census Bureau. Homeschool definition: ages 5–17, K–12 grade equivalent, instruction at home instead of public or private school all or most of the time, excluding those in school more than 25 hours per week and those homeschooled only for temporary illness.

⁵⁶Institute of Education Sciences / NCES, "A Higher Percentage of K–12 Students are Receiving Academic Instruction at Home," press release, 17 September 2024, <https://ies.ed.gov/learn/press-release/higher-percentage-k-12-students-are-receiving-academic-instruction-home>. Agency voice. Same 5.2 percent / 3.4 percent figures; headlines the broader "instruction at home" rise versus 3.7 percent in 2018–19.

The trend inside NHES, not Pulse, is the comparable series. 1998–99: 1.7 percent, about 850,000, telephone era. 2011–12: 3.4 percent adjusted, about 1,773,000, after a mail redesign and a statistical adjustment. 2015–16: about 3.3 percent, 1,690,000. 2018–19: 2.8 percent, about 1,457,000. 2022–23: 3.4 percent, 1,765,000.⁵⁷ Pew Research Center, on 20 February 2025, restates the 3.4 percent figure, notes it is “roughly on par” with 2.8 percent in 2018–19, and adds the 1.8 percent full-time virtual public or private whose parents did not consider them homeschooled, up from 0.86 percent in 2018–19.⁵⁸ Journalism restating NCES is not a second survey. It is a reminder that “instruction at home” and “homeschool” were split for a reason.

The COVID bump is real in the experimental Pulse series and in NCES’s “instruction at home” series. It is not a license to crown a 2026 number. NCES’s *Condition of Education* indicator, May 2022, reports Household Pulse Survey child-level homeschooling at 5.4 percent in 2020–21, and notes that Pulse is experimental, that the cited wave had a 6.6 percent response rate, and that “estimates based on the HPS data cannot be directly compared with those from the NHES.”⁵⁹ Use Pulse as a pandemic-era signal. Do not replace NHES with it. Do not write “homeschooling doubled” as if 2.8 percent and 5.4 percent were the same instrument.

Advocacy has a different number, and this book will print it as advocacy. Brian D. Ray, Braden Hoelzle, and Douglas Pietersma, at the National Home Education Research Institute, estimated 3.408 million K–12 homeschool students in 2024–25 — 6.262 percent of 54,417,758 school-age children from Census QuickFacts dated 1 July 2024. Method, as they describe it: Census Pulse Phase 4.2 Cycle 09 (20 August–16 September 2024) at 6.530 percent homeschooled, plus a 23-state weighted register mean of 4.18 percent, then assumed 15 percent Pulse undercount and 20 percent state-register undercount, then averaged. Range: 3.067 million to 3.749 million. State register examples they publish for 2024–25 include North Carolina 165,243 (9.47 percent of that state’s school-age), Florida 152,871, Pennsylvania 44,692 (2.28 percent), Massachusetts 12,203 (1.21 percent).⁶⁰ Label this advocacy estimation. The 23-state mean is useful as a reminder that notification states see very different registered shares. It is not a national census. Ray’s February 2026 headcount paper is the same shop that publishes the convenience-sample percentiles section 1.7 will refuse to crown. Use NHERI for the estimates they actually compute and for the existence of the claim. Do not use NHERI as NCES.

⁵⁷Trend from NCES *Condition of Education* 2022 (NHES 1999–2019) and NCES 2024-113 (2023). 1998–99: 1.7 percent / 850,000. 2011–12: 3.4 percent adjusted / 1,773,000 (mail redesign). 2015–16: ~3.3 percent / 1,690,000 (NCES 2020-001). 2018–19: 2.8 percent / 1,457,000. 2022–23: 3.4 percent / 1,765,000.

⁵⁸Pew Research Center, “A look at homeschooling in the U.S.,” 20 February 2025. See n. 32. “Roughly on par” is Pew’s phrasing of 3.4 versus 2.8.

⁵⁹NCES, “Homeschooled Children and Reasons for Homeschooling,” *Condition of Education*, May 2022. Household Pulse Survey August 18–30, 2021 wave: 5.4 percent of children reported homeschooled; response rate 6.6 percent. NCES: “estimates based on the HPS data cannot be directly compared with those from the NHES.” Pulse is experimental.

⁶⁰Brian D. Ray, Braden Hoelzle, and Douglas Pietersma, “How Many Homeschool Students are there in the United States During the 2024-2025 School Year?,” National Home Education Research Institute, 27 February 2026, <https://nheri.org/how-many-homeschool-students-are-there-in-the-united-states/>. **Advocacy.** Estimate 3.408 million (6.262 percent of 54,417,758 school-age children from Census QuickFacts 1 July 2024). Method: Pulse Phase 4.2 Cycle 09 (20 Aug–16 Sep 2024) 6.530 percent, plus 23-state weighted register mean 4.18 percent, plus assumed 15 percent Pulse undercount and 20 percent state-register undercount, then averaged. Range 3.067–3.749 million. Not NCES. Whether NC 2024–25’s 165,243 matches DNPE’s own latest PDF is **UNVERIFIED** (NC 2023–24 DNPE report opened in search: 157,642 students / 96,529 schools — different year).

No 2026 national total. NHES:2023 is the latest federal PFI this stream opened. The next NHES cycle had not, as of 30 August 2026, produced a 2025–26 First Look that this research opened. Write “as of the 2022–23 school year, NCES...” and, separately, “advocacy and Pulse estimates for 2024–25 are higher and methodologically distinct.” EdChoice and the Johns Hopkins Homeschool Hub are cited by NHERI as secondary commentators; the Johns Hopkins Pulse page was blocked to this fetch. Do not invent their 2025–26 figures.⁶¹

Who they are, in NHES 2023 Table A-6, is not a gifted enclave and not a single demographic. Homeschooling rates were higher in rural areas (5.2 percent) and towns (5.1 percent) than in cities (2.1 percent) or suburbs (3.1 percent). White non-Hispanic students, 5.1 percent; Hispanic, 1.8 percent; Black non-Hispanic, 1.7 percent; Asian non-Hispanic, 0.7 percent, interpret with caution. Highest parental education: vocational or some college, 4.7 percent homeschooled; bachelor’s, 3.8 percent; graduate, 3.0 percent; less-than-high-school suppressed. Poverty: 2.5 percent below the threshold versus 3.6 percent at or above.⁶² Two-parent, one full-time and one not in the labor force, 8.8 percent, already printed. The book should not pretend homeschool mathematics is only for the rural, only for the White, only for the college-educated, or only for the one-earner household. Those are higher *rates*. They are not the only kitchens.

Reasons, in Table A-7, among the same approximately 1,765,000 students: parents could mark multiple “important” reasons. Concern about the environment of other schools — safety, drugs, negative peer pressure — 83 percent important, 28 percent *most* important. Moral instruction, 75 percent important, 6 percent *most*. Family life together, 72 / 8. Dissatisfaction with academic instruction, 72 / 17. Religious instruction, 53 / 11. Nontraditional approach, 50 / 8. Other special needs, 21 / 7. Physical or mental health problem, 15 / 5. COVID-related reasons did not meet reporting standards as “most important.”⁶³ Academic dissatisfaction is common. It is not the modal *most* important reason. Religious instruction is common as “important” and is 11 percent of “most.” The book should not pretend homeschool mathematics is only for the religious, only for the gifted, or only for the traumatized. A parent who is here because the other school felt unsafe, a parent who is here because the mathematics instruction felt thin, and a parent who is here because the family wanted the hours together are all inside the table. The teaching method this book will lock in Chapter 2 does not change with the reason. The paperwork in section 1.2 might.

Thomas Dee’s Urban Institute analysis of where the kids went during the pandemic enrollment drop is a public-school-exodus paper, not a mathematics-outcome paper. It belongs in a footnote as context for the Pulse bump, not as a 2026 headcount.⁶⁴

⁶¹EdChoice / Johns Hopkins Homeschool Hub figures for 2025–26: not opened; JHU Pulse page blocked to this fetch (Cloudflare). Do not invent. Census Bureau Household Pulse Survey Phase 4.2 Cycle 09 education tables: <https://www.census.gov/data/tables/2024/demo/hhp/cycle09.html>. Federal experimental survey; NCES warns against comparing Pulse to NHES.

⁶²Sempeles and Cui, NCES 2024-113, Table A-6, selected: rural 5.2; town 5.1; suburb 3.1; city 2.1. White non-Hispanic 5.1; Hispanic 1.8; Black non-Hispanic 1.7; Asian non-Hispanic 0.7 (interpret with caution). Highest parental education: vocational/some college 4.7; bachelor’s 3.8; graduate 3.0; less-than-high-school suppressed. Poverty: 2.5 percent below threshold versus 3.6 percent at or above.

⁶³Sempeles and Cui, NCES 2024-113, Table A-7, among ~1,765,000 students; multiple “important” reasons allowed. Environment of other schools 83 percent important / 28 percent *most*; moral instruction 75 / 6; family life 72 / 8; academic dissatisfaction 72 / 17; religious instruction 53 / 11; nontraditional 50 / 8; other special needs 21 / 7; physical or mental health 15 / 5. COVID-related reasons: reporting standards not met as “most important.”

⁶⁴Thomas Dee, “Where the Kids Went: Nonpublic Schooling and Demographic Change During the Pandemic

The design user inside those counts is the parent who stopped at algebra. That sentence is the next section's job. The counts do not tell you whether that parent can teach. They tell you the parent is common, the kitchen is not a district, and anyone selling a 2026 national total is selling a different document than NCES 2024-113.

1.4 Design user: “I stopped at algebra”

The parent this book is built for is not a certified mathematics teacher and is not a fraud for lacking the credential. Most opened states do not require one. The parent this book is built for is the adult who says, in a sentence this research did not invent as a reported quotation, “I stopped at algebra.” That sentence is a design constraint. It is not a marketing persona. It is not a finding that the parent should not teach. It is a finding that the method has to survive an adult whose own working memory still flinches at a fraction, a signed number, or a two-step equation.

Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock published the load-bearing paper in *Psychological Science* in 2015. First- and second-graders of more math-anxious parents learned less mathematics over the school year and were more math-anxious at year's end — *but only if those parents reported frequent homework help*. When help was infrequent, parent anxiety was unrelated to child outcomes. The effect was specific to mathematics. It did not appear in reading.⁶⁵ The authors note that a genetic component is possible; the moderation by homework help points to a social and instructional mechanism. The University of Chicago news write-up of the same team is journalism-adjacent and useful for one operational sentence the paper already implies: do not simply tell anxious parents “get involved”; they need better tools for *how* to help.⁶⁶

This is a school-homework study. It is not a parent-as-sole-teacher randomized trial. No fetched experimental literature on math-anxious parents as full-time instructors was opened for this project. That gap is the highest-priority homeschool-specific hole in the learning-science stream, and this chapter will leave it standing.⁶⁷ A homeschool parent who is the daily instructor cannot “help less often” as a daily strategy the way a school parent can step back from Tuesday's worksheet. The risk is therefore higher, not lower, than in a school household where a math-anxious parent can leave the explaining to someone else. Do not upgrade Maloney et al. into a homeschool experiment. Do not treat the finding as “anxious parents should not teach.” The constructive response the literature actually supports is narrower: the parent learns the day's concept well enough to stay out of the child's working memory; the child generates; the parent does not narrate worry.

Exodus from Public Schools,” Urban Institute, 2023, <https://www.urban.org/research/publication/where-kids-went-nonpublic-schooling-and-demographic-change-during-pandemic>. Think-tank/academic. Public-school enrollment decline and homeschool increase 2019–2021. Not a mathematics-outcome paper. Not a 2026 headcount.

⁶⁵Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, “Intergenerational Effects of Parents' Math Anxiety on Children's Math Achievement and Anxiety,” *Psychological Science* 26, no. 9 (2015): 1480–88, <https://doi.org/10.1177/0956797615592630>. First- and second-graders; frequent-homework-help moderation; math-specific, not reading. School-homework study, not a parent-as-sole-teacher randomized trial.

⁶⁶University of Chicago News, “Parents' math anxiety can undermine children's math achievement,” 2015, <https://news.uchicago.edu/story/parents-math-anxiety-can-undermine-childrens-math-achievement>. University news-office summary of Maloney et al. Journalism-adjacent; the finding is the journal article.

⁶⁷Parent-as-sole-teacher experimental literature: none fetched. Highest-priority homeschool-specific gap (research-stream open question). Maloney is school homework, not full-time instruction. Left standing.

A 2021 follow-up by Fraulein Retanal, Noah Johnston, Sabrina Di Lonardo Burr, Andie Storozuk, Michela DiStefano, and Erin A. Maloney, in *Education Sciences*, found that a controlling-supportive homework-helping style partially mediated the parent-anxiety / child-achievement relation in a survey of parents of 11- to 14-year-olds.⁶⁸ Path analysis on a survey is not a kitchen randomized trial. It names the mechanism to watch. The mechanism is not “parent is bad at mathematics.” It is parent anxiety leaking as controlling help, speed-pressure, or “I’m not a math person, and you take after me.” Replication status of Maloney 2015 is open. Some later work has been mixed on the exact interaction. No registered replication was opened for this project. Flag the 2015 paper as a single large, well-cited field study plus a 2021 mediation follow-up, not as settled kitchen law.⁶⁹

Mark H. Ashcraft and Elizabeth P. Kirk, in the *Journal of Experimental Psychology: General* in 2001, give the adult half of the same working-memory story. High math-anxious college students had smaller working-memory spans especially on computation-based span tasks; under a letter-load plus two-column mental addition, errors rose sharply in the high-anxiety group, more on carry problems. The authors describe a transitory disruption of working memory, likely central-executive, caused by worry competing as a secondary task. Correlation of math anxiety with computation-span: $r = -.40$, Experiments 1 and 3. Verbal span showed little or no anxiety-group difference. High-anxious students also took fewer mathematics courses and earned lower mathematics grades — the avoidance path.⁷⁰ Ashcraft’s college samples are not a K–12 prevalence. This book will not invent one. Ashcraft and Jeremy A. Krause’s later reviews treat the disruption as beginning to matter more as mathematics becomes working-memory intensive, often cited as middle school onward; those reviews were not re-opened for this pass.⁷¹ Sian L. Beilock, Elizabeth A. Gunderson, Gerardo Ramirez, and Susan C. Levine, in 2010, showed that teacher math anxiety predicted first- and second-grade girls’ mathematics achievement over the year. That paper was fetched at secondary level; coefficients remain **UNVERIFIED** here and will not be quoted as if they had been read.⁷² The parent analogue is Maloney, not a laundered teacher-anxiety coefficient.

Chapter 2 will lock the teaching method this constraint requires: prepare so you can hear a wrong answer; then do not take the pencil. Five strands, not a ranking. Conceptual and procedural knowledge iterate. Fractions are the gate. The number line unifies. Worked examples faded for novices; incorrect examples explained; interleaving after a short block; schemas, not keywords; brief re-

⁶⁸Fraulein Retanal, Noah Johnston, Sabrina M. Di Lonardo Burr, Andie Storozuk, Michela DiStefano, and Erin A. Maloney, “Controlling-Supportive Homework Help Partially Explains the Relation between Parents’ Math Anxiety and Children’s Math Achievement,” *Education Sciences* 11, no. 10 (2021): 620, <https://doi.org/10.3390/educsci11100620>. Survey of parents of ages 11–14; path analysis; partial mediator. Not a kitchen randomized trial.

⁶⁹Replication status of Maloney et al. 2015: some later work mixed on the exact interaction; no registered replication opened. Flag as a single large, well-cited field study plus the 2021 mediation follow-up.

⁷⁰Mark H. Ashcraft and Elizabeth P. Kirk, “The Relationships among Working Memory, Math Anxiety, and Performance,” *Journal of Experimental Psychology: General* 130, no. 2 (2001): 224–37. Computation-span $r = -.40$, Experiments 1 and 3. Full 2001 PDF not opened this pass; coefficient from Ashcraft’s later quotation of those experiments and from the learning-science memo. Transitory occupation of the central executive by worry. College samples are not a K–12 prevalence; none invented.

⁷¹Mark H. Ashcraft and Jeremy A. Krause, later reviews (2007 and after), often cited for middle-school onset as mathematics becomes more working-memory intensive. Not re-opened this pass.

⁷²Sian L. Beilock, Elizabeth A. Gunderson, Gerardo Ramirez, and Susan C. Levine, “Female Teachers’ Math Anxiety Affects Girls’ Math Achievement,” *Proceedings of the National Academy of Sciences* (2010). Fetched at secondary level; coefficients **UNVERIFIED** this pass. Not quoted as if read.

trieval of already-known facts; explicit modeling for students already behind. This chapter will not rewrite that lock. It installs the user the lock is built for. A parent who cannot yet hear why $1/2 \div 1/4$ is 2 is not disqualified. That parent is the reason Chapter 5 exists and is the heaviest elementary chapter in the book. A parent who narrates “we’re not math people” while taking the pencil is running the Maloney high-help path. The fix is preparation, not silence, and not a pep talk.

Video and self-teaching curricula exist because of this constraint. Teaching Textbooks, in the publisher’s own words on the homepage this research opened, “does 100% of the teaching” and “100% of the grading,” with audiovisual solutions and a tutor helpline.⁷³ That is a **vendor** claim, not an effectiveness trial, and not a finding that the parent can stop hearing wrong answers. CTCMath is a video-plus-practice system a catalogue reviewer treats as a homeschool option. Khan Academy is free practice. Co-ops, tutors, and dual enrollment are the release valves when the parent is the ceiling for Algebra II through calculus. NCES 2016 already counted them: 23 percent any tutor, 31 percent any co-op, a small share of 9–12 also in college classes. A *hypothetical*, labelled as such: a parent who last passed Algebra I in 1998 hires a weekly tutor for Algebra II, or enrolls the student in College Algebra at the community college, while remaining the homeschool administrator of record.⁷⁴ That is a normal high-school plan, not a failure of the kitchen. Chapter 12 will say how to put the tutor, the co-op teacher, or the college on the transcript. Chapter 3 will say why handing the same student an unconstrained chatbot is Maloney et al. plus a faster pencil. This chapter says only: the design user stopped at algebra; the method has to survive that stop; “help less often” is not available as a daily strategy; “anxious parents should not teach” is not the finding.

Hyde and colleagues, 2006, documented wide variation in the quality of mothers’ scaffolding on fifth-grade mathematics homework. That paper is cited in later work; this chapter uses it only as a named reminder that help is not one behavior.⁷⁵ The parent who sits alongside, asks what the child has tried, and can hear a wrong magnitude is not the parent who takes the pencil and says “this is how I learned it.” Maloney’s frequent-help item does not distinguish those. Retanal’s controlling-supportive style is closer. The kitchen rule is still Chapter 2’s: hear a wrong answer; do not take the pencil.

Before buying Algebra I, the first job is placement — because grade labels lie.

1.5 Grade labels lie; placement is the first job

A “grade 6” book is a publisher’s scope, not a legal grade. Singapore Math Inc., **vendor**, states that Dimensions Math 6 “includes some Pre-algebra topics, but is not a Pre-algebra text in the traditional sense; it is a Grade 6 text,” and that Dimensions 7 and 8 together cover Pre-algebra and Algebra 1 topics with some Geometry. Dimensions 8, on the same vendor FAQ, expects more prior linear-

⁷³Teaching Textbooks homepage, <https://www.teachingtextbooks.com/>, accessed 30 August 2026. **Vendor**. Claims “100% of the teaching” and “100% of the grading”; tutor helpline. Not an effectiveness trial. Price table not displayed on homepage this access — list price **UNVERIFIED** on this primary.

⁷⁴Composite *hypothetical*, labelled in the sentence that introduces it. Empirical pieces: NCES 2016 any tutor 23 percent; small share of 9–12 also in college classes. The 1998 Algebra I date and the weekly tutor are not a reported case.

⁷⁵Janet S. Hyde and colleagues (2006), quality of mothers’ scaffolding on fifth-grade mathematics homework, as cited in later work. Named reminder that help is not one behavior; not re-opened as a primary this pass.

equation work than a traditional Algebra 1.⁷⁶ A parent who buys Dimensions 6 because a child is ten has bought a Grade 6 text with some pre-algebra, not a pre-algebra course and not a legal sixth grade. Editions of Primary Mathematics — US, Standards, 2022, Singapore International — are not interchangeable. The publisher’s placement tests are free PDFs, untimed, about an hour; the publisher instructs parents not to use the test as the sole factor and treats starting at an earlier level as common and acceptable.⁷⁷

Saxon Homeschool, **vendor** Houghton Mifflin Harcourt, uses split names — Math 5/4, 6/5, 7/6, 8/7 — that do not match a calendar grade. The middle-grades placement PDF this research opened uses score bands to place into those books, or into Algebra 1/2 or Algebra 1. The publisher says the test is not infallible and that Saxon is skill-level rather than grade-level; also consider age, prior curriculum, and how quickly the student grasps concepts.⁷⁸ The homeschool sequence HMH publishes runs K, 1, 2, 3, 5/4, 6/5, 7/6, 8/7, Algebra 1, Algebra 2, Advanced Mathematics, Calculus, with 8/7 or Algebra 1/2 as the pre-algebra fork. Geometry is integrated in some classic editions and also exists as a separate course. HMH’s school-side ESSA evidence claims are **vendor** evidence about school implementations, not homeschool randomized trials.⁷⁹

Math-U-See, **vendor** Demme Learning, uses Greek letters uncoupled from calendar grade: Primer, Alpha, Beta, Gamma, Delta, Epsilon, Zeta, then Pre-Algebra through Calculus. Epsilon is a fractions *level*, not “fifth grade.” That is later than a Common Core Grade 3 fractions domain if a family runs one level per calendar year from Primer — or earlier, if Alpha started in kindergarten. The parent is buying a *topic*, not a grade. Placement is an interactive yes/no mastery questionnaire plus a printable pretest; the scoring guide in placement materials uses a mastery threshold in the 80–89 percent range to advance.⁸⁰ A child who cannot yet operate on fractions is not “behind fifth grade.” That child has not finished Epsilon, whatever the birthday.

Beast Academy, **vendor** Art of Problem Solving, is comic-guide plus practice, Levels 1–5, ages about 6–13, “loosely based on the Common Core standards,” “covering grade-level standards in greater depth.” The publisher “generally recommend[s] that students start about a level ‘behind’ their current grade level” because of problem difficulty, and states that levels 4–5 cover many

⁷⁶Singapore Math Inc., School FAQ (Dimensions PK–5 and 6–8), <https://www.singaporemath.com/pages/school-faq>. **Vendor.** Dimensions Math 6 “includes some Pre-algebra topics, but is not a Pre-algebra text in the traditional sense; it is a Grade 6 text.” Dimensions 7 and 8 together cover Pre-algebra and Algebra 1 topics with some Geometry. Dimensions 8 expects more prior linear-equation work than a traditional Algebra 1.

⁷⁷Singapore Math Inc., “Placement Tests,” <https://www.singaporemath.com/pages/placement-tests>. **Vendor.** Untimed PDFs by series/edition; publisher instructs not to use the test as the sole factor; starting at an earlier level common and acceptable. One fetch timed out; URL is the official page.

⁷⁸HMH / Saxon Homeschool, *Middle Grades Math Placement Test*. See n. 24. Publisher: also consider age, prior curriculum, and how quickly the student grasps concepts.

⁷⁹Houghton Mifflin Harcourt, Saxon Math program page, <https://www.hmhco.com/programs/saxon-math>, and *Saxon Homeschool Math Scope and Sequence* (PDF), https://www.hmhco.com/~media/sites/home/education/global/pdf/scope-and-sequence/mathematics/k-12/saxon-math-homeschool/shs_ssintroduction.pdf?la=en. **Vendor.** Sequence: K–3, 5/4, 6/5, 7/6, 8/7, Algebra 1, Algebra 2, Advanced Mathematics, Calculus; 8/7 or Algebra 1/2 as pre-algebra. School-line ESSA “strong/moderate/promising” claims are about school implementations, not homeschool RCTs.

⁸⁰Math-U-See / Demme Learning, curriculum sequence, <https://mathusee.com/products/math-u-see-curriculum/>, and placement tool, <https://curriculumexpress.mathusee.com/>. **Vendor.** Thirteen levels: Primer through Calculus. Epsilon is “Fractions” — a *level*, not a calendar grade. Placement: yes/no mastery items; scoring guide in placement materials uses an ~80–89 percent mastery threshold to advance.

middle-grades standards. A student who completes Level 5, on the vendor page, has the background for AoPS Prealgebra.⁸¹ Relative-difficulty claims beyond that sentence are vendor or secondary. Exact list price on 30 August 2026 was **UNVERIFIED** on the publisher page this access day; the pricing table did not render.⁸² Pedagogically this is the puzzle-hard track, not the default kitchen track. Naming it is not selling it.

A child of ten may place into Singapore 3A, Saxon 6/5, and Math-U-See Gamma simultaneously. That is not a contradiction. It is three different grain-sizes of “grade.” The book organizes by mathematics — number sense, operations, fractions, ratio, expressions, functions, proof, rate of change, data — with grade labels as typical age and sequence anchors. Common Core clusters remain useful as a map (“grade 4 fractions”) even for families who reject Common Core politically. Common Core is not a homeschool mandate. A 2026 state-by-state census of who still uses it is **UNVERIFIED**; this book will not say “most states” without a current count.⁸³

The National Center for Education Statistics already documented the stretch. PFI 2016 Table 9, printed in section 1.1: Algebra I taught that year to 41 percent of the 6–8 homeschool band; arithmetic still taught to 31 percent of 9–12.⁸⁴ Homeschool law rarely forbids a twelve-year-old from taking geometry or a sixteen-year-old from finishing fractions. NCAA and colleges care about the high-school transcript window — ninth-grade start date; Algebra I or higher as core — not about whether Algebra I happened at twelve. Taking Algebra I in eighth grade *and* listing it as a high-school credit is a record-keeping choice with NCAA implications. Read the current Home School Toolkit. Do not invent the pre-ninth rule here. Chapter 12 will export the title. This chapter says only: the calendar is not the mathematics.

Delay is equally normal and, for the math-anxious parent and the student with unfinished arithmetic, often the correct professional judgment. Teaching Textbooks’ gentler pacing is a **vendor** and catalogue reputation, relevant if a family later enters a co-op or public school that uses those labels, not a finding that slower is better. Singapore’s “start earlier in the sequence” is a commercial response to holes. Math-U-See’s Epsilon-as-fractions-year can place formal fraction operations later than Common Core Grades 3–5 if the family runs one level per year from Primer. The National Mathematics Advisory Panel and the Common Core map would call a student who reaches calendar Grade 6 without fraction-as-number work underprepared for ratio and algebra. That is a diagnosis, not a shame.

The Panel’s Finding 15 is the age-lock refusal this section needs and that Chapter 2 will keep. What is developmentally appropriate is “largely contingent on prior opportunities to learn.” Claims that children of a given age cannot learn certain content because they are “too young,” “not in the appro-

⁸¹Art of Problem Solving, Beast Academy books and FAQ. See n. 25. **Vendor.** Level 5 completion as background for AoPS Prealgebra is vendor language. Relative-difficulty claims beyond the “about a level behind” sentence are vendor or secondary.

⁸²Art of Problem Solving, Beast Academy pricing attempt, <https://www.beastacademy.com/pricing>. **Vendor.** Pricing table did not render a public dollar table on 30 August 2026 fetch (page-not-found content in the layout). Third-party 2026 reviews citing ~\$16/month or ~\$100/year: **UNVERIFIED** as list price on the publisher page this access day.

⁸³Common Core State Standards Initiative, *Common Core State Standards for Mathematics*, https://www.corestandards.org/assets/CCSSI_Math%20Standards.pdf. Map, not a homeschool mandate; standards “do not dictate curriculum or teaching methods.” A 2026 state-by-state census of who still uses CCSSM is **UNVERIFIED**. Do not say “most states” without a current count.

⁸⁴Cui and Hanson, NCES 2020-001, Table 9. See n. 26 and n. 36.

prate stage,” or “not ready” have “consistently been shown to be wrong.” Nor are claims justified that children cannot learn particular ideas because their brains are “insufficiently developed,” even if they possess the prerequisite knowledge for learning the ideas.⁸⁵ Finding 30: mathematically gifted students with motivation “appear to be able to learn mathematics much faster... with no harm to their learning, and should be allowed to do so.” Accelerate a child who has the prerequisites. Do not skip fractions because a child is “good at math.” Do not withhold fractions from a nine-year-old who has the prerequisites. Do not accelerate a child who does not. Two-e acceleration protocols (Assouline / Iowa Acceleration Scale) were not opened; effect sizes remain **UNVERIFIED** and will not be invented.⁸⁶

Cardus and Hamlin and Cheng’s duration research, which section 1.7 will use for adult outcomes, has a placement implication already. Homeschooling is often a few years, not a K–12 lockstep. Only 17 percent of ever-homeschooled adults in the Cardus 2023 sample were homeschooled all of K–12; 20 percent were homeschooled for only one year.⁸⁷ A child may enter homeschool in calendar grade 6 missing fraction fluency. The first job is diagnosis, not the assumption of a district year. Placement tests exist because the labels lie.

New York Individualized Home Instruction Plans and some dual-enrollment offices still want a grade. Pennsylvania tests in grades 3, 5, and 8. NCAA wants a ninth-grade start date. The book can be organized by mathematics *and* still tell the parent how to export a grade-labelled transcript. That export is Chapter 12. This chapter’s job is to kill the calendar-grade as a prison before any later chapter uses “Grade 3.”

Once placed, the parent still has to choose a tool — not a franchise.

1.6 Curriculum is a tool, not a franchise

This book is not Saxon. It is not Singapore. It is not Math-U-See, Beast Academy, Art of Problem Solving, Teaching Textbooks, CTCMath, or Khan Academy. Those are options a parent will actually meet. Name them. Compare grain, parent load, and how they place. Do not sell them. No homeschool-curriculum randomized trial comparing them was opened for this project, because none was found.⁸⁸ Robert Kunzman and Milton Gaither, surveying more than two thousand

⁸⁵National Mathematics Advisory Panel, *Foundations for Success: The Final Report of the National Mathematics Advisory Panel* (Washington, DC: U.S. Department of Education, 2008), Finding 15. ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>. What is developmentally appropriate is “largely contingent on prior opportunities to learn.” Age-lock claims (“too young,” “not in the appropriate stage,” “not ready”) have “consistently been shown to be wrong.” Brain-immaturity claims are not justified even when students possess the prerequisite knowledge. Finding 30: gifted students with motivation should be allowed to move faster.

⁸⁶Susan G. Assouline, Nicholas Colangelo, Joyce VanTassel-Baska, and colleagues, *Nation Empowered / Iowa Acceleration Scale literature*. Not opened this pass; effect sizes **UNVERIFIED**.

⁸⁷Albert Cheng and Angela Watson, *Diverse Outcomes for a Diverse Population: Findings About Homeschooled Adults from the Cardus Education Survey* (Cardus, 2024), <https://www.cardus.ca/research/diverse-outcomes-for-a-diverse-population/>. CES fielded 10–22 October 2023; Ipsos KnowledgePanel; 2,350 U.S. adults ages 24–39 who completed at least high school; ~8 percent / $n = 181$ ever homeschooled. Only 17 percent of ever-homeschooled adults were homeschooled all K–12; 20 percent for only one year. Think tank, not a federal statistical agency. Noncausal.

⁸⁸Homeschool-curriculum RCT comparing Saxon, Singapore, Math-U-See, Beast Academy, Teaching Textbooks, CTCMath, or Khan Academy: none opened, because none was found. Kunzman and Gaither 2020 note the thinness of curriculum research relative to demographics and law. School-side Saxon ESSA ratings are vendor/school evidence,

English-language texts in *Other Education* in 2020, note that curriculum research on homeschooling is thin relative to demographics and law.⁸⁹ Cathy Duffy Reviews is a long-running homeschool catalogue with “Top Picks,” publisher-provided review copies, and affiliate links she discloses. Rainbow Resource is a retailer catalogue. Neither is a randomized trial. Choose by fit — parent mathematics confidence, student stance, special needs, budget, college target — not by a phantom homeschool meta-analysis.

Grain first, because “best curriculum” is usually a grain argument in costume. Saxon is incremental and spiral: mixed practice, do not skip levels, an 80 percent mastery rule of thumb to advance, skill-level placement. The parent load is real on the lesson but lighter than a method the parent must first learn; the mixed sets are the point. Singapore’s Dimensions and Primary Mathematics lines are concrete–pictorial–abstract, number bonds, bar models, mastery within a level. The parent teaching load is real: bar models are a method the parent must learn. Math-U-See is mastery with manipulatives — Integer Blocks, “Decimal Street” — video lessons, student-paced, designed for the parent who needs a script and blocks; some materials are lightly Christian, a catalogue observation, not a finding. Beast Academy and Art of Problem Solving are puzzle-hard: comic plus practice, then contest-style texts and classes. Teaching Textbooks is an app that, in **vendor** copy, does the teaching and the grading; catalogue consensus, not a study, describes gentler pacing, sometimes behind typical public-school grade labels. CTCMath, Australian teacher Patrick Murray, is video plus interactive questions, diagnostic tests by topic, a family plan that unlocks all levels; Cathy Duffy quotes a homeschool-discount family price of \$198.50 for twelve months as of the page this research used — **catalogue**, confirm on the vendor site before printing an edition.⁹⁰ Khan Academy is a nonprofit, free mastery-practice map, SAT practice in partnership with College Board, not a complete homeschool program of record unless the parent builds the record.

Parent load is the Maloney constraint in commercial form. A parent who stopped at algebra and buys Singapore without learning bar models has bought a method they cannot hear. A parent who buys Teaching Textbooks because the vendor says it does 100 percent of the teaching has bought a voice; they have not bought a finding that they can stop hearing wrong answers. A parent who buys Math-U-See for the script and the blocks has bought a designed parent load. A parent who buys Beast Academy for a ten-year-old because a neighbor’s ten-year-old uses it has ignored the vendor’s own “about a level behind” sentence. Fit is not a vibe. Fit is whether the adult can hear a wrong answer in that program’s representations, and whether the child will generate the work.

Placement, already named in section 1.5, is part of the comparison because the labels lie in different dialects. Saxon places by score band into split-named books. Singapore places by untimed PDFs and tells you to watch how the student works. Math-U-See places by mastery items and names levels as topics. Beast Academy places by age then unit pretests. Teaching Textbooks’ official

not homeschool trials.

⁸⁹Robert Kunzman and Milton Gaither, “Homeschooling: An Updated Comprehensive Survey of the Research,” *Other Education* 9, no. 1 (2020): 253–336, https://icher.org/files/Kunzman_and_Gaither_An%20Updated_Comprehensive_Survey.pdf. Review of 2,000+ English-language texts. Explicitly treats Ray/NHERI as advocacy-era research. Best single map of the scholarly fight.

⁹⁰Cathy Duffy Reviews, “CTC Math,” <https://cathyduffyreviews.com/homeschool-reviews-core-curricula/math/math-grades-k-6/ctc-math>. **Catalogue / reviewer**, not an RCT. Publisher-provided access; affiliate disclosure. Homeschool pricing quoted: family \$198.50/12 months; individual \$128.50/12 months after homeschool discount. Confirm on ctcmath.com before printing a book edition.

scope-and-sequence PDF was not opened this pass; grade-by-grade topic lists from listicles stay out of this book.⁹¹ RightStart, Math Mammoth, and Life of Fred appear in homeschool roundups; their official scopes were not opened as primaries this pass and remain **UNVERIFIED** as to topic placement. This chapter will not promote those rows to handbook fact.⁹² Khan Academy's live course list as of 30 August 2026 was not opened in full; treat named courses as a free map, not as a researched outcome.

Public-school virtual and charter-at-home are not a homeschool curriculum. NCES 2023: 1.8 percent of students full-time virtual as public or private school students, plus 0.7 percent full-time virtual as homeschoolers.⁹³ Legally these are often public-school enrollments with a teacher of record, state tests, and an Individualized Education Program if eligible. They are not the parent-as-teacher kitchen. Co-ops, umbrella schools, cottage schools, and microschools are legal and social wrappers. NCES 2016: 31 percent any co-op instruction; 61 percent participated in activities with other homeschooled children; 33 percent of families participated in a homeschool group; 16 percent membership in a national homeschooling organization.⁹⁴ Umbrella schools can issue transcripts. They do not magically teach calculus. Dual enrollment is the cleanest external validation of algebra, precalculus, or calculus the kitchen can buy; NCAA treats a college transcript differently, and often more cleanly, than a parent-graded course. Chapter 12 is that export. This chapter names the option so a parent who is the ceiling knows the valve exists.

Time-on-task is the number a blog will invent. NCES 2016 reports *total* hours homeschooled per week in bands — 1–10, 11–24, 25–40, more than 40 — and days per week, not subject minutes.⁹⁵ Most families report a five-day week, with standard errors in the tables. Ethnographies and a 2022 convenience paper on homeschooling routines describe many families with total academic days of about four hours or less, plus uncounted “teachable moments.” An NHERI *Home School Researcher* article, “The Impact of Homeschooling on Math Education,” reports a small-*n* Pearson correlation: less than one hour on mathematics associated with lower computation and total mathematics scores; the difference between 1.0 and 1.5-plus hours was not significant.⁹⁶ Advocacy

⁹¹Teaching Textbooks official scope-and-sequence PDF: not opened this pass. Grade-by-grade topic lists from listicles (HomeschoolLeap, HomeSchoolie, and similar) were not used as primaries and disagreed with each other in search snippets.

⁹²RightStart Mathematics official levels A–H contents; Math Mammoth Light Blue official scope (Maria Miller); Life of Fred official title ladder: **UNVERIFIED** from vendor primaries this pass. Do not promote those rows to handbook fact.

⁹³Sempeles and Cui, NCES 2024-113, and Pew 20 February 2025: 1.8 percent full-time virtual as public or private school students; 0.7 percent full-time virtual as homeschoolers. Do not collapse full-time public virtual into parent-directed homeschool.

⁹⁴Cui and Hanson, NCES 2020-001: 31 percent any co-op instruction; 61 percent participated in activities with other homeschooled children; 33 percent of families participated in a homeschool group; 16 percent membership in a national homeschooling organization.

⁹⁵Cui and Hanson, NCES 2020-001, hours-per-week bands (1–10, 11–24, 25–40, more than 40) and days per week. Not subject minutes. 2012 standard-error table S13 exists; do not invent the percentage-point estimates for those bands from a figure this project did not fully extract.

⁹⁶National Home Education Research Institute / *Home School Researcher*, “The Impact of Homeschooling on Math Education,” <https://nheri.org/home-school-researcher-the-impact-of-homeschooling-on-math-education/>. **Advocacy journal**, small convenience sample. Pearson correlation: less than one hour on mathematics associated with lower computation scores. Not a national time-on-task study. **UNVERIFIED / no national study found** for typical daily homeschool mathematics minutes.

journal, convenience sample, old. It is not a national time-on-task study. **UNVERIFIED / no national study found** for typical daily mathematics minutes in United States homeschools. Do not write “the average homeschooler does 45 minutes of math.” A later chapter that needs a schedule may use a *hypothetical* illustration, labelled hypothetical. A defensible parent-facing range, labelled as practice not data, can note that kitchen mathematics is often a lesson plus a problem set in one sitting, and that video curricula advertise roughly 30–45 minute lessons — **vendor** Teaching Textbooks review literature, not NCES.⁹⁷

Cost, as of 30 August 2026 access, is in the same order of magnitude as a digital mathematics subscription, not a second mortgage — and none of the AI tools is a school-safe product by default. Khan Academy remains free. Khanmigo, **vendor**, \$4 per month or \$44 per year on khanmigo.ai/pricing that day; teachers free; not an autonomous open-web agent; “never gives you the answer” is vendor copy Chapter 3 will not launder as a trial.⁹⁸ Teaching Textbooks list price was not displayed on the homepage this access; third-party 2026 reviews cite about \$43–\$67 per level per year or about \$199 family — **UNVERIFIED** list price on the primary.⁹⁹ CTCMath family, twelve months, homeschool discount: \$198.50 **catalogue** Duffy; individual \$128.50; Rainbow Resource lists a family membership SKU at a pre-discount \$397.¹⁰⁰ Beast Academy Online pricing table did not render; third-party figures about \$16 per month or \$100 per year stay **UNVERIFIED** as list price.¹⁰¹ Singapore, Saxon, and Math-U-See print packages typically sit in the tens to low hundreds of dollars per year in catalogue round-ups, not fetched SKU-by-SKU. AP exam fees are set by College Board plus the host school. CLEP has a published list price on the Calculus page this research opened in the scope stream; confirm at publication. SAT fee waivers exist for eligible homeschooled students. Dual enrollment ranges from free, in some state programs, to ordinary community-college tuition. AI tools at \$44 per year sit below a single digital mathematics subscription. That does not make them a curriculum, FERPA-ready, or appropriate for unsupervised minors. Do not sell Hermes or Grok Bot as homeschool curriculum. Chapter 3 is the AI method. This chapter’s job is the order of magnitude and the refusal.

The operational rule is shorter than the catalog. Pick a tool whose representations you can learn well enough to hear a wrong answer. Place the child by the publisher’s own test, not by birthday. Do not mix Singapore editions. Do not skip Saxon levels because a split name looks like a grade you want to leave. Do not treat Math-U-See Epsilon as fifth grade. Do not start Beast Academy

⁹⁷Video-curriculum lesson-length advertisements (~30–45 minutes): **vendor** Teaching Textbooks review literature and catalogue consensus, not NCES. A parent-facing range labelled as practice not data; not “the average homeschooler does 45 minutes of math.”

⁹⁸Khan Academy, Khanmigo pricing, <https://khanmigo.ai/pricing>, accessed 30 August 2026. **Vendor / nonprofit product**. \$4/month or \$44/year for parents and learners; teachers free. Khan Academy content itself remains free. “Never gives you the answer” is vendor copy. Not FERPA-as-homeschool-product; not an autonomous open-web agent. Chapter 3 will not launder the brochure as a trial.

⁹⁹Teaching Textbooks homepage. See n. 73. Third-party 2026 reviews (Boldly Educated; Think Academy) cite ~\$43–\$67 per level per year or ~\$199 family plan up to five students. **UNVERIFIED** list price on the publisher page this access day.

¹⁰⁰Cathy Duffy Reviews, “CTC Math.” See n. 90. Rainbow Resource Center, “CTCMath Family Membership (12 month subscription),” <https://www.rainbowresource.com/007424.html>. **Catalogue / retailer**. Family membership listed at \$397 pre-discount. Not an effectiveness study.

¹⁰¹Beast Academy pricing page. See n. 82. Third-party ~\$16/month or ~\$100/year: **UNVERIFIED** as 30 August 2026 list price.

on grade-level because the comics look friendly. Do not treat a video that “does 100 percent of the teaching” as a finding that you can stop hearing. Do not treat a catalogue Top Pick as a trial. Do not treat a school-side ESSA rating as a homeschool effect. When the parent is the ceiling, buy a tutor, a co-op seat, or a community-college transcript — a normal plan. When the child has unfinished fractions, go back, whatever the calendar says. Chapter 5 is that gate. This chapter is the refusal to franchise.

The thesis, now that the kitchen is named.

1.7 Thesis, live fight, and the map

A parent who is not a mathematics specialist can take a child from first-grade number sense through high-school calculus if three things hold. The parent understands the concept well enough to hear a wrong answer. The child does the struggle before the tool. Any agent is a supplement hosted and constrained by the parent — not a substitute for the child’s work and not a school-safe product by default. The sources do not prove that thesis as a causal claim about homeschool mathematics. They do not contradict it. They constrain how it can be attempted. Fractions are the gate; Chapter 2 will plant the prediction finding and Chapter 5 will do the teaching. Conceptual and procedural knowledge iterate; Chapter 2 will lock the cycle. Unguarded solvers raise practice scores and cut unaided exams; Chapter 3 will lock the AI lane. Parent mathematics anxiety plus frequent help predicts worse child mathematics in a school-homework study; this chapter already named the design user and refused to upgrade the study into a parent-as-sole-teacher trial. No homeschool-curriculum randomized trial and no 2026 national enrollment total exist to crown a method or a headcount. Promise a path, not a percentile.

The live debate this book will not settle with a convenience sample is the sentence a parent meets first: “homeschoolers score higher in math.” Brian D. Ray’s National Home Education Research Institute has, for decades, published the numbers advocates quote: composites around the 80th percentile, mathematics often a bit lower than reading inside the same sample. The 2009 *Homeschool Progress Report*, hosted by HSLDA, with a 2010 academic write-up, is the specimen the Coalition for Responsible Home Education and the International Center for Home Education Research dissect: participants scored about the 86th percentile overall, **84th in mathematics**, 89th in reading; sample recruited via testing services and networks; disproportionately White, Christian, two-parent, higher-income, higher parental education. Even inside that sample, mathematics lagged reading, and scores associated with parental education, income, expenditure, and structure.¹⁰² Label NHERI advocacy. Use NHERI for the estimates they actually compute and for the existence of the claim. Do not use NHERI percentiles as “what homeschool mathematics does.”

The critique is named and is not a conspiracy. Christopher Lubienksi, Tiffany Puckett, and T. Jameson Brewer, “Does Homeschooling ‘Work’?,” *Peabody Journal of Education*, 2013, and Lu-

¹⁰²Brian D. Ray / National Home Education Research Institute, *Homeschool Progress Report* (2009, HSLDA-hosted) and 2010 academic write-up. **Advocacy.** Specimen: about the 86th percentile overall, 84th in mathematics, 89th in reading; sample recruited via testing services and networks; disproportionately White, Christian, two-parent, higher-income, higher parental education. Coalition for Responsible Home Education, “Research with an Agenda (Ray 2010),” <https://crhe.org/homeschooling-academics-and-demographics-ray-2010/>. **Advocacy critique.** Do not use NHERI percentiles as “what homeschool mathematics does.”

bienski's earlier "A Critical View of Home Education" (2003): advocacy organizations' empirical claims do not support causal inference; selection dominates.¹⁰³ Chelsea McCracken, as discussed on ICHER, "How to Mislead with Data": Ray's sample is not representative of homeschoolers, let alone of all children; only a small percentage of homeschoolers sit the tests in the pools Ray used; volunteering scores into a study selects for success. Her punchline, as ICHER quotes it, is that the result describes "the top-scoring 2–3 percent of homeschooled elementary- and middle-school-age children of wealthy white Christian married couples who volunteer them for standardized tests."¹⁰⁴ Robert Kunzman and Milton Gaither, "Homeschooling: An Updated Comprehensive Survey of the Research," *Other Education* 9, no. 1 (2020): 253–336, reviewing more than two thousand English-language texts, refuse a simple win for homeschooling, note the field's history of advocacy-as-research — explicitly including Ray — and note the recent growth of university-based work. Kunzman, quoted in the *Washington Post* on 11 December 2023 (Moriah Balingit, journalism), said studies suggest homeschooling "does not make a significant difference" in performance and that "one of the best studies found negative results," pointing at Cardus.¹⁰⁵ Rob Reich's 2002 "The Civic Perils of Homeschooling," in *Educational Leadership*, is a civic-isolation argument, not a mathematics paper; it is still part of the legitimacy debate the parent will meet.¹⁰⁶ CRHE is children's-rights advocacy, methodologically aligned with the Ray critiques. Label advocacy in the other direction.

Cardus 2023 is the representative-ish adult survey this book will keep quoting as a rate, not as a mathematics cause. Albert Cheng and Angela Watson, *Diverse Outcomes for a Diverse Population*, Cardus, 2024, on the Cardus Education Survey fielded 10–22 October 2023, Ipsos KnowledgePanel, 2,350 United States adults ages 24–39 who completed at least high school, about 8 percent / $n = 181$ ever homeschooled. Not a statistical agency. A think-tank survey with a reputable panel vendor. They disaggregate short-term (1–2 years), medium (3–7), long-term (8-plus). Only 17 percent of ever-homeschooled adults were homeschooled all of K–12; 20 percent for only one year. Bachelor's or higher: short-term 45 percent versus never-homeschooled 46 percent; medium 29 percent; long-term 27 percent. Demographic controls applied; still **not causal**. Full-time employment and above-median household income: lower for all three homeschool groups than for never-homeschooled. Long-term homeschoolers: highest optimism, gratitude, life satisfaction; lowest depression and anxiety indicators on the CES items. The authors explicitly refuse a causal inter-

¹⁰³Christopher Lubienski, Tiffany Puckett, and T. Jameson Brewer, "Does Homeschooling 'Work'? A Critique of the Empirical Claims and Agenda of Advocacy Organizations," *Peabody Journal of Education* 88, no. 3 (2013), <https://doi.org/10.1080/0161956X.2013.798516>. Peer-reviewed critique. Selection effects; advocacy claims do not support causal inference. Paywalled DOI; bibliographic primary. Lubienski, "A Critical View of Home Education" (2003), earlier piece in the same line.

¹⁰⁴International Center for Home Education Research Reviews, discussing Chelsea McCracken, "HOW TO MISLEAD WITH DATA: A Critique of Brian Ray's Methodology," <https://icher.org/blog/?p=1440>. Scholarly-adjacent blog (Kunzman's ICHER). Not peer-reviewed; methodologically serious. Punchline as ICHER quotes McCracken.

¹⁰⁵Kunzman and Gaither, "Homeschooling: An Updated Comprehensive Survey" (2020). See n. 89. Moriah Balingit, "How a true believer's flawed research helped legitimize home schooling," *Washington Post*, 11 December 2023, <https://www.washingtonpost.com/education/2023/12/11/brian-ray-homeschool-student-outcomes/>. **Journalism**. Quotes Kunzman on Cardus as among the better studies and on the non-significance of homeschooling for achievement. Not a dataset.

¹⁰⁶Rob Reich, "The Civic Perils of Homeschooling," *Educational Leadership* 59, no. 7 (2002), <https://ascd.org/el/articles/the-civic-perils-of-homeschooling>. Academic essay. Civic/democratic critique, not a mathematics-outcome study.

pretation — unmeasured reasons for leaving school, including bullying and safety.¹⁰⁷ Cardus does not report SAT mathematics, calculus completion, or STEM-major rates. It does undermine any sentence of the form “homeschoolers go to college at higher rates.” Medium- and long-term homeschoolers in this representative adult sample completed bachelor’s degrees at *lower* rates. Whether that is a mathematics-preparation problem, a values problem (long-term homeschoolers were less likely to say money is “very important”), a selection problem, or all three is **open**. Do not cite the bachelor’s-rate gaps as the effect of homeschool mathematics, or as proof that homeschooling harms STEM. Daniel Hamlin and Albert Cheng, *Journal of School Choice*, 2022, cited by Cardus: duration of homeschooling is itself a source of heterogeneity. Treating “homeschoolers” as one treatment is a design error.¹⁰⁸

What NCES does *not* show is as load-bearing as what it shows. NCES has never published a nationally representative homeschool NAEP mathematics score. NHES asks subjects taught, not achievement. Any sentence that says “homeschoolers outperform public-school students in mathematics on NAEP” would be false.¹⁰⁹ No named nationally representative source opened for this project gives the share of homeschool graduates who complete a college calculus course or who major in STEM. NCES 2016’s 15 percent is high-school-level ever-taught calculus *during home instruction*, interpret with caution — not college calculus and not a completion rate. That open question is planted for Chapter 12. This chapter will not fill it with Beast Academy anecdotes.

Selection is the honest paragraph. Families who homeschool, who persist, who buy Beast Academy, who sit the SAT, who volunteer scores to NHERI, and who answer Cardus are four different filters. Two-parent one-earner households with a college-educated parent are overrepresented in the NHES homeschool population relative to all children; they are *more* overrepresented in convenience testing samples. Any “homeschool mathematics advantage” that survives a comparison with public-school children of the same parental education, income, and two-parent status has not been shown in a national probability sample this research could open. The teaching chapters promise a path, not a percentile.

College-ready is a documentation problem as much as a teaching problem, and it is Chapter 12’s job. There is no national homeschool diploma. State compliance is not college evidence: Texas legally needs neither portfolio nor test; colleges and the NCAA will still want a transcript. Named university pages this research opened — University of Pittsburgh, Southern Methodist University,

¹⁰⁷Cheng and Watson, *Diverse Outcomes for a Diverse Population* (Cardus, 2024). See n. 87. Bachelor’s or higher: short-term 45 percent versus never-homeschooled 46 percent; medium 29 percent; long-term 27 percent. Demographic controls; authors refuse a causal interpretation. Does not report SAT mathematics, calculus completion, or STEM-major rates. Companion report: Lynn E. Swaner, Albert Cheng, and Jonathan Eckert, *School-Sector Influence on Graduate Outcomes and Flourishing* (Cardus, 2024).

¹⁰⁸Daniel Hamlin and Albert Cheng, “Homeschooling, Perceived Social Isolation, and Life Trajectories: An Analysis of Formerly Homeschooled Adults,” *Journal of School Choice* (2022), <https://doi.org/10.1080/15582159.2022.2028338>. Peer-reviewed. Duration of homeschooling as a heterogeneity dimension; cited by Cardus 2024 for the short/medium/long split. Paywalled DOI.

¹⁰⁹National Center for Education Statistics has never published a nationally representative homeschool NAEP mathematics score. NHES asks subjects taught, not achievement. Any sentence that says homeschoolers outperform public-school students in mathematics on NAEP would be false. STEM-major and college-calculus completion rates for homeschool graduates in a probability sample: no named nationally representative source opened. NCES 2016 15 percent is high-school-level ever-taught calculus during home instruction (interpret with caution), not college calculus. Planted for Chapter 12.

North Central College, Southwest Baptist University — want on the order of three years of high-school mathematics, often through Algebra II, on a transcript a stranger can read. NCAA math core is Algebra I or higher; CLEP is not an NCAA core course; parents cannot order AP exams. Those sentences are planted. They are not dumped. A kitchen that stops at Algebra I because “the state doesn’t require it” has a college problem on every named page. Chapter 12 will export the titles. This chapter says only: the path this book promises has to be documentable. A percentile from a convenience sample is not the document.

The map of the twelve chapters, in short, so the reader can see the kitchen this chapter named before the method lock and the mathematics. Chapter 2, *How Children Learn Mathematics*, locks method: five strands, not a one-way street, the fractions gate as prediction, the number line, the moves that survive, fluency as retrieval not a race, struggle versus rescue without mashing the Panel and Kapur, parent-as-teacher without taking the pencil. Chapter 3, *The Agent Does Not Take the Test*, is one AI chapter, not five, and not a sequel: performance is not learning; Bastani is the mathematics-class replication; ban paste-the-worksheet; the parent holds the key. Chapter 4, grades 1–2: number, operations, place value, the equal sign as a relation. Chapter 5, grades 3–5: the fractions gate, the heaviest elementary chapter. Chapter 6, grades 6–8: ratio, signed numbers, expressions, the function concept. Chapter 7, Algebra I. Chapter 8, Geometry. Chapter 9, Algebra II and precalculus. Chapter 10, calculus — AB, BC, non-AP, CLEP as different objects. Chapter 11, data and chance; statistics is a different college-equivalent, not stealth calculus. Chapter 12, records, transcripts, college-ready. A coda: what an honest book can promise. Open questions named as a list the book will not fill with analogy.

What this chapter has refused, so later chapters do not have to re-argue it. No 2026 national total. No homeschool NAEP mathematics. No “most states require a mathematics degree.” No curriculum A beats B in homeschool. No national mathematics-minutes figure. No Individualized Education Program at home. No collapse of full-time public virtual into parent-directed homeschool. No writing as if every reader is a stay-at-home parent. No selling Hermes or Grok Bot. No treating HSLDA as a statute. No citing Cardus bachelor’s-rate gaps as the effect of homeschool mathematics. No opening with AI. NCES 2016 subject tables are the last opened; NHES:2023 did not republish them.

What this chapter leaves standing, because filling them would be invention. No 2026 national headcount. NHES:2023 microdata on teaching style and mathematics subjects. A homeschool-curriculum randomized trial. Daily mathematics minutes in a probability sample. Parent-as-sole-teacher — Maloney is homework, not full-time instruction. Massachusetts 2026 legislative outcome on approval versus notice. Unschooling mathematics outcomes beyond the 12 percent informal share. STEM-major and college-calculus completion rates for homeschool graduates. Exact 30 August 2026 list prices for Teaching Textbooks and Beast Academy Online. Whether North Carolina’s 2024–25 register count as NHERI published it matches the Division of Non-Public Education’s own latest PDF. AI tools actually used by homeschoolers in 2026 — usage surveys **UNVERIFIED**.

Before any grade band, the parent needs the learning science that will lock method. That is Chapter 2. This chapter named the room. One adult, usually the mother, usually a purchased book, legally allowed in most of the country to teach Algebra II without a mathematics degree, designing for the parent who stopped at algebra, placing by skill rather than by birthday, choosing a tool rather

than a franchise, promising a path rather than a percentile. The kitchen table is not a district. The documents were the proof. The teaching can start.

Chapter 2

How Children Learn Mathematics

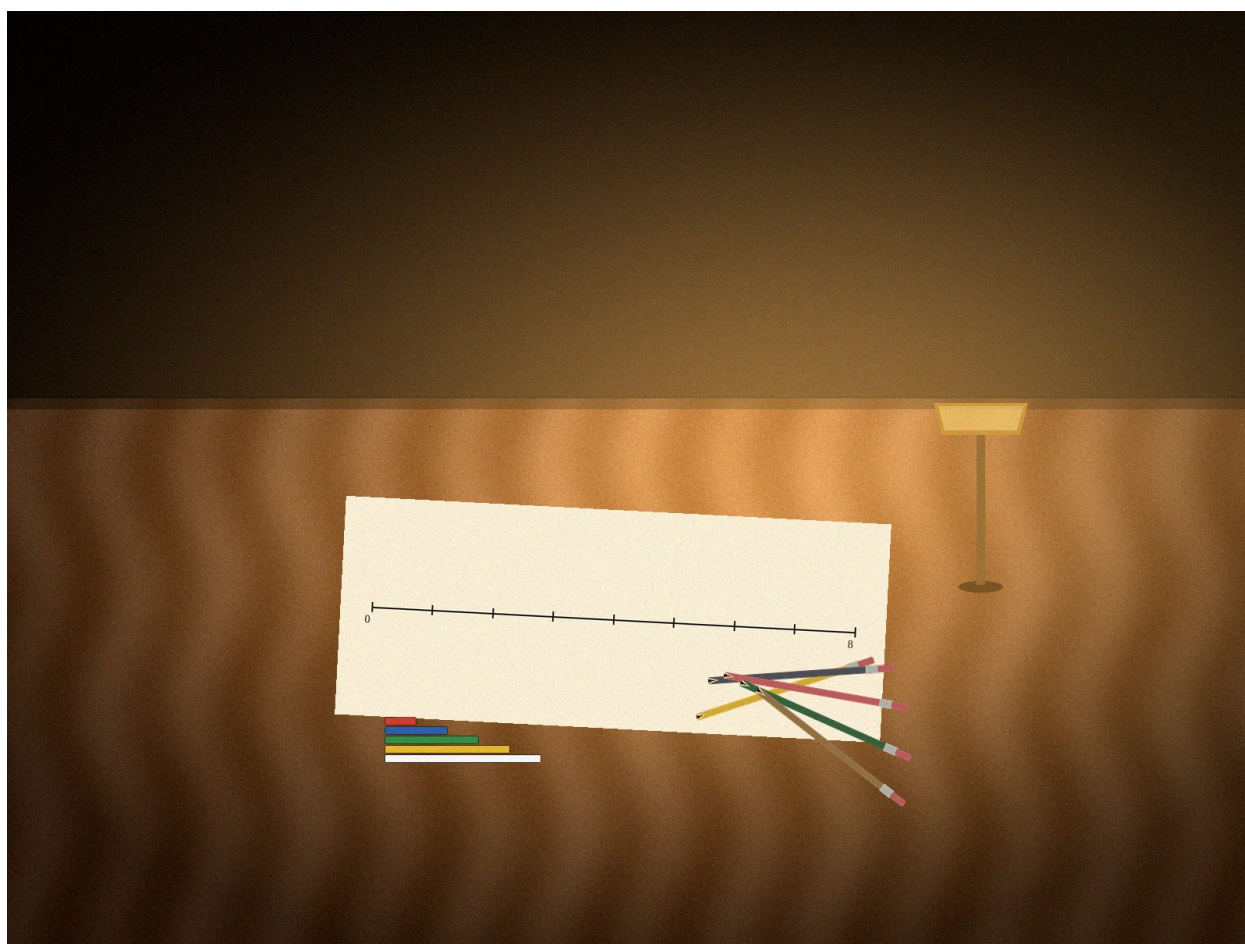


Figure 3: A paper number line, generic wooden counting rods, and five colored pencils on a wooden table.

The oldest quarrel in the psychology of school mathematics is whether children should first understand or first compute. The quarrel is older than any franchise a parent can buy. It is older than Common Core. It is older than the National Mathematics Advisory Panel. Two consensus docu-

ments, written from different committees in different decades, refuse it. They refuse it in language a kitchen teacher can use on Tuesday.

This book's first chapter named the kitchen so this one would not have to. One adult, usually the mother, usually a purchased formal curriculum, legally allowed in most of the country to teach Algebra II without a mathematics degree. The design user stopped at algebra. The legal object is not a math program. Grade labels lie. Curriculum is a tool, not a franchise. Homeschool math-superiority is a fight, not a finding. Those sentences stay in Chapter 1. What this chapter adds is the teaching science that chapter could only plant: how children actually learn mathematics, and which moves survive when the documents are allowed to do the work. Without that lock, Chapters 4 through 11 become folklore with a scope map attached.

Adding It Up, the National Research Council report edited by Jeremy Kilpatrick, Jane Swafford, and Bradford Findell, defines mathematical proficiency as five strands — conceptual understanding, procedural fluency, strategic competence, adaptive reasoning, and productive disposition — and then writes the sentence that later chapters of this book will not let go: the strands “are interwoven and interdependent.”¹¹⁰ The National Mathematics Advisory Panel, seven years later, reached the same conclusion in different words. Finding 10 of *Foundations for Success* says the curriculum “must simultaneously develop conceptual understanding, computational fluency, and problem-solving skills. Debates regarding the relative importance of these aspects of mathematical knowledge are misguided.”¹¹¹ Same refusal. Different letterhead.

This chapter is the method lock for the rest of the book. Every later mathematics chapter — first-grade number bonds, the fractions gate in grades 3–5, ratio and signed numbers in the middle grades, Algebra I, geometry, Algebra II and precalculus, calculus, data and chance — inherits the same rules. Conceptual knowledge and procedural knowledge develop iteratively, not as a one-way street. Fractions, including decimals, percents, and signed fractions, are the elementary content that uniquely predicts later algebra; that is a prediction finding, not a curriculum experiment, and Chapter 5 will do the teaching. The number line is the representation that unifies whole numbers, fractions, decimals, negatives, and later functions. Worked examples help novices and must be faded; incorrect examples, explained, help conceptions. Blocked practice teaches execution; interleaved, spaced practice teaches choosing a strategy and keeping it. Word problems are learned as schemas, not as keyword hunts. Fluency is flexible, accurate, efficient retrieval — not a race and not optional. Timed facts are two objects: brief retrieval of already-known facts in a low-stakes mix, and high-stakes early speed tests used to rank young children. Those are not the same practice. All-encompassing teacher-directed or student-centered prescriptions are not supported. Explicit instruction has consistently positive effects for students already struggling; productive failure is generation-then-instruction on a new concept when priors exist. Those two literatures are not the same comparison. Math anxiety consumes working memory. Parent anxiety, when the parent is the daily helper, is a transmission risk the homeschool kitchen cannot dodge by helping less of-

¹¹⁰National Research Council, *Adding It Up: Helping Children Learn Mathematics*, ed. Jeremy Kilpatrick, Jane Swafford, and Bradford Findell (Washington, DC: National Academy Press, 2001), 116. The five strands “are interwoven and interdependent.” Access date for all URLs in these notes: 30 August 2026.

¹¹¹National Mathematics Advisory Panel, *Foundations for Success: The Final Report of the National Mathematics Advisory Panel* (Washington, DC: U.S. Department of Education, 2008), Finding 10. ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>.

ten. Manipulatives help under guidance and become a crutch if they are not connected to symbols and faded. Kitchen-math transfer is hard. What is “developmentally appropriate” is mostly prior opportunity to learn, not a Piagetian age lock.

The popular story collapses those findings into slogans. Understand first, then fluency will follow. Or: drill first, understanding will arrive. Timed tests cause anxiety. Timed tests are the identity of being good at math. Let them discover. Just tell them. They’ll learn fractions by cooking. Schema instruction raises achievement by a standard deviation and a half. Direct instruction works. Inquiry works. The documents do not say any of those things, not as the slogans say them. This chapter prints the documents. It prints the objects, the sample sizes, the delays, and the nulls. It leaves the open questions standing. It does not invent a kitchen scene to make the science feel warmer. The parent who stopped at algebra is the design user. The job is to hear a wrong answer against five strands, not against a franchise.

A few corrections belong on this page before any later chapter can inherit them. NCTM’s 2014 sentence that “procedural fluency follows and builds on a foundation of conceptual understanding” is a policy stance, not a finding that has been tested against a procedures-then-concepts alternative.¹¹² Rittle-Johnson, Schneider, and Star’s 2015 review is the paper later chapters should cite when a curriculum vendor claims that “research says concepts first.” Siegler, Duncan, Davis-Kean and colleagues’ 2012 coefficients — a one-standard-deviation increase in age-10 fractions knowledge uniquely associated with a 0.15-SD (United Kingdom) / 0.17-SD (United States) increase in high-school algebra after controls — are a prediction, not a randomized trial of a fractions curriculum.¹¹³ Rohrer, Dedrick, and Burgess’s 2014 interleaving result (72 percent versus 38 percent, $d = 1.05$) is a nine-week grade-7 classroom study with a two-week delayed unannounced test; Rohrer, Dedrick, Hartwig, and Cheung’s 2020 result (61 percent versus 38 percent, $d = 0.83$) is a preregistered cluster randomized trial with a one-month delay; Dunlosky, Rawson, Marsh, Nathan, and Willingham’s 2013 “high utility” rating is for practice testing and distributed practice, with interleaving rated moderate.¹¹⁴ Those three objects are not one effect size. Jitendra, Star, Dupuis, and Rodriguez’s 2012 schema-based instruction gains ($g = 1.24$ immediate, $g = 1.27$ at six weeks) are on trained proportion items, with null transfer; Peltier and Vannest’s 2017 meta-analytic $g = 1.57$ is mostly researcher-

¹¹²National Council of Teachers of Mathematics, *Principles to Actions: Ensuring Mathematical Success for All* (Reston, VA: NCTM, 2014), 42: “procedural fluency follows and builds on a foundation of conceptual understanding.” Policy stance, not an experiment tested against a procedures-then-concepts alternative. Flagged as such by Bethany Rittle-Johnson, Michael Schneider, and Jon R. Star, “Not a One-Way Street: Bidirectional Relations Between Procedural and Conceptual Knowledge of Mathematics,” *Educational Psychology Review* 27, no. 4 (2015): 587–97.

¹¹³Robert S. Siegler, Greg J. Duncan, Pamela E. Davis-Kean, Kathryn Duckworth, Amy Claessens, Mimi Engel, Maria Ines Susperreguy, and Meichu Chen, “Early Predictors of High School Mathematics Achievement,” *Psychological Science* 23, no. 7 (2012): 691–97. British Cohort Study $N = 3,677$; U.S. PSID-CDS $N = 599$. Coefficients from the author PDF, <https://sieglertc.columbia.edu/wp-content/uploads/2019/02/Siegler-et-al-PsySci12.pdf>. Prediction, not a curriculum RCT.

¹¹⁴Doug Rohrer, Robert F. Dedrick, and Kaleena Burgess, “The Benefit of Interleaved Mathematics Practice Is Not Limited to Superficially Similar Kinds of Problems,” *Psychonomic Bulletin & Review* 21 (2014): 1323–30, $d = 1.05$, 72% vs. 38%, $n = 140$, 95% CI [0.80, 1.30], ERIC ED548041. Doug Rohrer, Robert F. Dedrick, Marissa K. Hartwig, and Chi-Ngai Cheung, “A Randomized Controlled Trial of Interleaved Mathematics Practice,” *Journal of Educational Psychology* 112, no. 1 (2020): 40–52, 61% vs. 38%, $d = 0.83$, 54 classes. John Dunlosky, Katherine A. Rawson, Elizabeth J. Marsh, Mitchell J. Nathan, and Daniel T. Willingham, “Improving Students’ Learning with Effective Learning Techniques,” *Psychological Science in the Public Interest* 14, no. 1 (2013): 4–58: practice testing and distributed practice rated high-utility; interleaved practice rated moderate-utility.

developed proximal measures, not general achievement.¹¹⁵ IES 2021 Recommendation 6 is brief timed activity on already-known facts; Boaler's 2012 and 2015 argument is about high-stakes early speed tests.¹¹⁶ National Mathematics Advisory Panel Finding 27 is explicit instruction for students already struggling; Kapur 2014 is productive failure as generation-then-instruction.¹¹⁷ Maloney, Ramirez, Gunderson, Levine, and Beilock 2015 is a school-homework study, not a parent-as-sole-teacher randomized trial.¹¹⁸ National Mathematics Advisory Panel Finding 26 is the kitchen-math caution: real-world contexts improve similar real-world items, not computation or equation solving.¹¹⁹ Finding 15 is the age-lock refusal: what is developmentally appropriate is "largely contingent on prior opportunities to learn"; claims that children of a given age cannot learn certain content because they are "too young," "not in the appropriate stage," or "not ready" have "consistently been shown to be wrong."¹²⁰

Those sentences are the lock. The rest of the chapter is the argument that makes them usable at a table where one adult, who may have stopped at algebra, is the teacher.

2.1 Five strands; ranking them is misguided

A parent who buys a program because it advertises "understanding, not facts," or because it advertises "facts first, understanding later," has already accepted a ranking the two consensus documents call misguided. The ranking is the product. The documents are the refusal.

Adding It Up defines mathematical proficiency as five strands, and the definitions are load-bearing,

¹¹⁵Asha K. Jitendra, Jon R. Star, Danielle Dupuis, and Michael C. Rodriguez, "Effectiveness of Schema-Based Instruction for Improving Seventh-Grade Students' Proportional Reasoning: A Randomized Experiment," *Journal of Research on Educational Effectiveness* 5 (2012): immediate $g = 1.24$, six-week retention $g = 1.27$, null transfer, $N = 1,163$, 42 classrooms. Asha K. Jitendra et al., "A Randomized Controlled Trial of the Impact of Schema-Based Instruction on Mathematical Outcomes for Third-Grade Students with Mathematics Difficulties," *Elementary School Journal* 114, no. 2 (2013): 252–76, word-problem $g = 0.46$, district $g = 0.34$, null 8-week retention. Corey Peltier and Kimberly J. Vannest, "A Meta-Analysis of Schema Instruction on the Problem-Solving Performance of Elementary School Students," *Review of Educational Research* (2017): Hedge's $g = 1.57$ immediate problem solving, $g = 1.09$ transfer, 21 studies, 3,408 students — mostly proximal measures; not a general-achievement effect.

¹¹⁶Lynn S. Fuchs et al., *Assisting Students Struggling with Mathematics: Intervention in the Elementary Grades*, WWC 2021006 (Washington, DC: Institute of Education Sciences, 2021), Recommendation 6, strong evidence, 27 studies. Jo Boaler, "Timed Tests and the Development of Math Anxiety," *Education Week / Stanford GSE* (2012), <https://ed.stanford.edu/news/boaler-timed-tests-and-development-math-anxiety>; Boaler, *Fluency Without Fear* (Youcubed working paper, 2015). Different objects: brief timing on already-known facts versus high-stakes early speed tests.

¹¹⁷National Mathematics Advisory Panel, *Foundations for Success*, Finding 27. Manu Kapur, "Productive Failure in Learning Math," *Cognitive Science* 38, no. 5 (2014): 1008–22. Generation-then-instruction versus explicit instruction for students already struggling — not the same comparison.

¹¹⁸Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, "Intergenerational Effects of Parents' Math Anxiety on Children's Math Achievement and Anxiety," *Psychological Science* 26 (2015): 1480–88. School-homework study, not a parent-as-sole-teacher RCT. Moderation by frequent homework help; math-specific, not reading.

¹¹⁹National Mathematics Advisory Panel, *Foundations for Success*, Finding 26: real-world contexts improve similar real-world items; they do not, in the Panel's small high-quality set, improve computation, simple word problems, or equation solving.

¹²⁰National Mathematics Advisory Panel, *Foundations for Success*, Finding 15: what is developmentally appropriate is "largely contingent on prior opportunities to learn"; age-lock claims have "consistently been shown to be wrong." Brain-immaturity claims are not justified even when students possess the prerequisite knowledge.

not decoration. Conceptual understanding is “comprehension of mathematical concepts, operations, and relations.” Procedural fluency is “skill in carrying out procedures flexibly, accurately, efficiently, and appropriately.” Strategic competence is the “ability to formulate, represent, and solve mathematical problems.” Adaptive reasoning is the “capacity for logical thought, reflection, explanation, and justification.” Productive disposition is the “habitual inclination to see mathematics as sensible, useful, and worthwhile, coupled with a belief in diligence and one’s own efficacy.”¹²¹ The committee’s most important sentence is not any one of those five. It is that the strands “are interwoven and interdependent.” Instruction, the committee later wrote, “should not be based on extreme positions that students learn, on one hand, solely by internalizing what a teacher or book says or, on the other hand, solely by inventing mathematics on their own.”¹²² The National Mathematics Advisory Panel quotes the interwoven sentence in a footnote and then restates the point for algebra readiness as Finding 10: conceptual understanding, computational fluency, and problem solving are “mutually supportive”; debates that rank them are “misguided.”¹²³

That is a definition of knowing mathematics that no franchise can split. A child who can recite 8×7 but cannot say why 8×7 equals 7×8 is missing a strand. A child who can draw a beautiful array but cannot retrieve 8×7 when it is needed inside a fraction problem is missing a different strand. A child who can retrieve and can explain the commutative property but cannot formulate a two-step story as an equation is missing strategic competence. A child who can do all of that and still treats a wrong answer as a verdict on talent, not as information, is missing productive disposition. Hear a wrong answer against all five. The first teaching move in this book is that hearing.

Finding 14 of the Panel adds the dispositional piece with experimental backing the committee was willing to name. Changing children’s beliefs from a focus on ability to a focus on effort increases engagement and mathematics outcomes. Public resignation about mathematics, the Panel warned, is “rooted in the erroneous idea that success is largely a matter of inherent talent or ability, not effort.”¹²⁴ That sentence is aimed at the culture, not only at the child. The design user of this book — “I stopped at algebra” — has often inherited the resignation. The constructive response is not a pep talk. It is the parent learning the day’s concept well enough to stay calm, and the child generating the work. Chapter 1 already installed Maloney and colleagues on the transmission risk; this chapter will return to it in section 2.8. The point here is narrower. Productive disposition is a strand of proficiency, not a personality trait the child either has or lacks. Effort beliefs are part of the mathematics, because they change what the child will attempt and what the parent will rescue.

The Common Core State Standards for Mathematics, which this book treats as a map and not as a homeschool mandate, say the same thing in a shorter sentence: “Mathematical understanding and procedural skill are equally important.”¹²⁵ Equally important is not a sequence. It is a refusal to

¹²¹National Research Council, *Adding It Up*, 116, five-strand definitions as quoted. Conceptual understanding, procedural fluency, strategic competence, adaptive reasoning, productive disposition.

¹²²National Research Council, *Adding It Up*, 409, on instruction not based on extreme positions (internalizing only, or inventing only). The five-strand definitions and “interwoven and interdependent” are p. 116; NMAP footnote 2 cites p. 116 for those attributes, not the extreme-positions sentence.

¹²³National Mathematics Advisory Panel, *Foundations for Success*, Finding 10, and the Panel’s quotation of the *Adding It Up* interwoven sentence.

¹²⁴National Mathematics Advisory Panel, *Foundations for Success*, Finding 14, effort beliefs versus the “erroneous idea that success is largely a matter of inherent talent or ability, not effort.”

¹²⁵National Governors Association Center for Best Practices and Council of Chief State School Officers, *Common*

rank. The eight Standards for Mathematical Practice that run through every grade — make sense of problems and persevere; reason abstractly and quantitatively; construct viable arguments and critique the reasoning of others; model with mathematics; use appropriate tools strategically; attend to precision; look for and make use of structure; look for and express regularity in repeated reasoning — rest on NCTM’s process standards and on the *Adding It Up* strands.¹²⁶ They name valued activities. They do not, by themselves, tell a parent how to sequence a Tuesday. NCTM’s *Principles and Standards for School Mathematics* (2000) organizes school mathematics into five content standards and five process standards (problem solving, reasoning and proof, communication, connections, representation) across four grade bands. *Principles to Actions* (2014) added eight mathematics teaching practices and the concepts-before-fluency sentence already flagged as policy.¹²⁷ These are professional-organization standards, not experimental results. A parent can steal the vocabulary. A parent cannot treat a process-standard list as a finding that fluency will arrive after understanding if only the child talks enough.

The operational consequence for the kitchen is blunt. Refuse the franchise that says “we do understanding, they do facts.” Refuse the franchise that says the reverse. Any homeschool program that advertises only one of the five strands is selling a fragment. A video curriculum that “does 100 percent of the teaching” is a vendor claim about who talks, not a finding that the parent can stop hearing wrong answers against the strands. A puzzle-hard program that produces beautiful representations and no retrieval is a fragment. A drill program that produces retrieval and no representation is a fragment. The later grade-band chapters will name Saxon, Singapore, Math-U-See, Beast Academy, Teaching Textbooks, CTCMath, and Khan Academy as tools, by grain and parent load and placement, not as winners of a phantom homeschool meta-analysis. The method lock is prior to the catalog. If the parent cannot say which strand a wrong answer is missing, the catalog cannot help.

Two more refusals belong here so they do not have to be re-argued in every later chapter. First: ranking the strands is misguided, but that does not mean every Tuesday must devote equal minutes to each. Interwoven is not a pie chart. A new two-step equation may spend most of a sitting on a worked example and a completion problem (fluency and conceptual understanding in one object). A mixed quiz at the end of the month may spend most of its minutes on strategic competence — choosing which procedure this problem wants. A child who has just been told “I’m not a math person, and you take after me” needs the dispositional strand repaired before more pages will stick. The parent’s job is to know which strand is thin, not to pretend the five are a schedule.

Second: *Adding It Up* is a 2001 consensus report and the National Mathematics Advisory Panel is a 2008 synthesis. Neither is “the science of 2026 classrooms.” Jere Confrey, Alan Maloney, and Kenny Nguyen published a methodological critique of the Panel the same year it appeared — biased literature selection, inconsistent standards.¹²⁸ The critique is real. The Panel is still the syn-

Core State Standards for Mathematics (Washington, DC: NGA Center / CCSSO, 2010), on understanding and procedural skill equally important. Map, not a homeschool mandate.

¹²⁶*Common Core State Standards for Mathematics*, 6–8, Standards for Mathematical Practice 1–8. Resting on NCTM process standards and the *Adding It Up* strands.

¹²⁷National Council of Teachers of Mathematics, *Principles and Standards for School Mathematics* (Reston, VA: NCTM, 2000), five content and five process standards. NCTM, *Principles to Actions* (2014), eight mathematics teaching practices. Professional-organization standards, not experimental results.

¹²⁸Jere Confrey, Alan P. Maloney, and Kenny H. Nguyen, “Breaching the Conditions for Success for a National

thesis later documents cite for algebra foundations, including the IES practice guides that Chapters 4 through 7 will keep using. This book reports both. It does not treat 2008 as the last word. It does treat 2008 as the still-cited word on the Critical Foundations, on the mutual reinforcement of understanding, fluency, and problem solving, and on the explicit-instruction finding for students already struggling. A parent who discards the Panel because a later blog said “that’s old” is discarding the document the later guides still rest on. A parent who treats the Panel as a religion is ignoring a serious critique. Neither is the job.

If ranking is misguided, order is not a one-way street either. The next section is the iterative model — the finding that kills “concepts first” as if it were an experiment, without installing “procedures first” as the opposite religion.

What the five-strand definition also refuses is the talent folklore that Finding 14 named. A parent who says “we’re not math people” has already ranked productive disposition out of the program. The Panel’s warning was about public resignation, not about a child’s score on a Tuesday. The resignation is useful to a franchise that sells either “just facts” or “just meaning,” because both products need the parent to believe that the other half is someone else’s job. Interwoven means it is not someone else’s job. The parent who stopped at algebra is not disqualified by that stop. The parent is disqualified only by refusing to hear which strand a wrong answer is missing. Chapter 1 already said most hours of homeschool math are still parent-supervised. This chapter says those hours have to be hours in which all five strands can be heard. A purchased book can carry the problems. It cannot hear.

NCTM’s process standards and the Common Core Practices are not a second definition of proficiency. They are activity names that sit on top of the five strands. A child can “communicate” a wrong magnitude and still be missing conceptual understanding. A child can “persevere” on a page of blocked items and still be missing strategic competence — the choosing, not the executing. A parent who treats a Practice poster as the method has swapped a definition for a mood. The method is the strands plus the moves the later sections lock: iteration, the number line, faded examples, interleaving after a short block, schemas not keywords, brief retrieval of known facts, explicit modeling when the child is already behind, generation when the priors are there. The Practices can name the activities. They cannot sequence the Tuesday.

2.2 Conceptual and procedural iterate

Bethany Rittle-Johnson, Michael Schneider, and Jon Star, writing in *Educational Psychology Review* in 2015, put the myth in a sentence this book can keep: “It is a myth that it is a one-way street from conceptual knowledge to procedural knowledge.”¹²⁹ Longitudinal and experimental evidence, they review, shows bidirectional relations. Each type of knowledge predicts gains in the other. Alternative orderings of instruction have “rarely been compared, with limited empirical support for one ordering of instruction over another.”¹³⁰ That is the paper later chapters should cite

Advisory Panel,” *Educational Researcher* (2008). Methodological critique of NMAP 2008; report alongside the Panel.

¹²⁹Rittle-Johnson, Schneider, and Star, “Not a One-Way Street,” 587–97. Full PDF: <https://www.univ-trier.de/fileadmin/fb1/prof/PSY/PAE/Team/Schneider/Rittle-JohnsonEtAl2015.pdf>.

¹³⁰Rittle-Johnson, Schneider, and Star, “Not a One-Way Street,” on alternative orderings rarely compared, with limited empirical support for one ordering over another.

when a curriculum vendor claims that “research says concepts first.” The vendor is usually citing a policy sentence, or citing studies that did not run the comparison the slogan needs.

The iterative model is older than the 2015 review. Rittle-Johnson, Robert Siegler, and Martha Alibali proposed it in 2001 in the *Journal of Educational Psychology*, and they showed the iteration in fifth- and sixth-graders learning decimal fractions. Initial conceptual knowledge predicted gains in procedural knowledge. Gains in procedural knowledge predicted later conceptual knowledge. Correct problem representations mediated the first path. A second experiment manipulated support for correct representation and produced procedural gains.¹³¹ Rittle-Johnson and Alibali had already shown bidirectionality in a brief lesson study in 1999: a procedure lesson improved the concept of mathematical equivalence relative to control, and a concept lesson improved procedures.¹³² The street runs both ways in short lessons and in longer classroom work.

The one classroom experiment that held content constant and varied only order is Rittle-Johnson and Kenneth Koedinger, 2009, two classroom experiments, sixth-grade decimals on an intelligent tutor, $N = 77$ and $N = 26$. An iterative sequence — concept lesson, procedure lesson, concept, procedure — produced more knowledge of arithmetic procedures, including transfer to problems with novel features, than a concepts-first block of the same lessons. Conceptual knowledge was “fairly comparable.” Pretest knowledge of one type predicted gains in the other.¹³³ That is not a license to skip the idea. It is a finding that a long conceptual foundation, with practice withheld until the idea is judged complete, lost to cycling on the outcome the concepts-first slogan claims to protect: procedures that transfer. Schneider, Rittle-Johnson, and Star, in 2011, modeled conceptual knowledge, procedural knowledge, and procedural flexibility as latent variables after a three-day classroom intervention on equation solving. Each type of knowledge predicted gains in the other. The relations were symmetrical.¹³⁴ Flexibility — being able to choose among procedures — is a third outcome, not a luxury. Star’s earlier commentaries had already insisted on that point: a well-connected procedure is what fluency actually is.¹³⁵

The studies often cited for concepts-first do not run the comparison the slogan needs. Cobb and colleagues (1991), Fuson and Briars (1990), Hiebert and Wearne (1996), Kamii and Dominick (1997, 1998): these compared conceptually rich teaching to typical procedural classrooms that gave little or no conceptual instruction. They do not compare concepts-then-procedures to procedures-then-concepts.¹³⁶ A parent who reads “research supports conceptually rich teaching” has read a real

¹³¹Bethany Rittle-Johnson, Robert S. Siegler, and Martha W. Alibali, “Developing Conceptual Understanding and Procedural Skill in Mathematics: An Iterative Process,” *Journal of Educational Psychology* 93, no. 2 (2001): 346–62.

¹³²Bethany Rittle-Johnson and Martha W. Alibali, “Conceptual and Procedural Knowledge of Mathematics: Does One Lead to the Other?,” *Journal of Educational Psychology* 91, no. 1 (1999): 175–89.

¹³³Bethany Rittle-Johnson and Kenneth R. Koedinger, “Iterating between Lessons on Concepts and Procedures Can Improve Mathematics Knowledge,” *British Journal of Educational Psychology* 79 (2009): 483–500. Two classroom experiments, sixth-grade decimals, $N = 77$ and $N = 26$.

¹³⁴Michael Schneider, Bethany Rittle-Johnson, and Jon R. Star, “Relations between Conceptual Knowledge, Procedural Knowledge, and Procedural Flexibility,” *Developmental Psychology* (2011). Cited from the 2015 review; full PDF not opened this pass.

¹³⁵Jon R. Star, “Reconceptualizing Procedural Knowledge,” *Journal for Research in Mathematics Education* 36 (2005): 404–11. Flexibility as part of what fluency is; knowledge type versus quality.

¹³⁶Paul Cobb et al. (1991); Karen C. Fuson and Diane J. Briars (1990); James Hiebert and Diana Wearne (1996); Constance Kamii and Ann Dominick (1997, 1998), as reviewed by Rittle-Johnson, Schneider, and Star, “Not a One-Way Street.” Conceptually rich teaching versus typical procedural classrooms — not concepts-then-procedures versus

finding. A parent who upgrades that finding to “research says withhold procedures until understanding is complete” has added a comparison those papers did not make. Pesek and Kirshner (2000), sometimes cited as showing that early procedures interfere, gave fifth-graders area and perimeter; differences were small and not statistically significant (48 percent versus 42 percent at posttest).¹³⁷ Perry (1991), a brief laboratory study of mathematical equivalence, found that concept-only produced more procedural transfer than concept-plus-procedure or procedure-only. That is a few minutes of instruction, not a curriculum sequence.¹³⁸ The 2015 review is explicit: the ordering question has barely been tested. The one classroom experiment that tested it, holding content constant, favored iteration.

A language trap sits under the slogan. Star (2005, 2007) and Star and Stylianides (2013) argue that “conceptual” and “procedural” must name types of knowledge — principles versus steps — not quality, richly connected versus rote.¹³⁹ Otherwise one cannot even talk about a well-connected procedure. NCTM’s 2014 *Principles to Actions*, at page 42, says “procedural fluency follows and builds on a foundation of conceptual understanding.” The sentence uses “fluency” as an end-state and “conceptual knowledge” as a variable amount. Rittle-Johnson, Schneider, and Star flag that wording as a source of the myth.¹⁴⁰ This book will not treat the NCTM sentence as an experiment. It is a professional-organization policy stance. Policy can be useful. Policy is not a finding tested against a procedures-then-concepts alternative.

For the parent, the operational rule is narrower than either religion. When introducing a new idea, a short conceptual pass — what the symbols mean, a representation, a “why might this work?” — before the first procedure is reasonable, on the thin laboratory evidence (Perry 1991; Matthews and Rittle-Johnson 2009).¹⁴¹ Then cycle: a handful of problems, a return to the idea, a comparison of two methods, more problems. Do not withhold practice for weeks until they understand. Do not run pages of a new algorithm with no representation and no why. Katherine Canobi’s 2009 work on addition and subtraction is the early-grades version of the same point: conceptually sequenced practice supported both conceptual and procedural gains; randomly ordered practice supported only modest procedural gains.¹⁴² Sequence the ideas. Do not withhold the procedures.

The later chapters will apply the cycle to particular objects. Grades 1–2: number bonds and non-canonical equations before, during, and after the first written facts — not a month of “what addition means” with no $8 = 5 + 3$. Grades 3–5: a representation of $3/4$ as a number on a line, then a handful of comparisons, then a return to the line, then unlike-denominator addition as replacing with equivalent fractions that live at the same points — not a quarter of invert-and-multiply with

procedures-then-concepts.

¹³⁷Pesek and Kirshner (2000), fifth-grade area and perimeter, 48% vs. 42% at posttest, small and not statistically significant, as reported by Rittle-Johnson, Schneider, and Star, “Not a One-Way Street.”

¹³⁸Perry (1991), brief laboratory study of mathematical equivalence, as reported by Rittle-Johnson, Schneider, and Star, “Not a One-Way Street.” A few minutes of instruction, not a curriculum sequence.

¹³⁹Star, “Reconceptualizing Procedural Knowledge” (2005); Star (2007); Jon R. Star and Gabriel J. Stylianides (2013), as cited by Rittle-Johnson, Schneider, and Star, “Not a One-Way Street.”

¹⁴⁰NCTM, *Principles to Actions*, 42; Rittle-Johnson, Schneider, and Star, “Not a One-Way Street,” flagging the wording as a source of the one-way-street myth.

¹⁴¹Perry (1991); Matthews and Rittle-Johnson (2009), thin laboratory evidence for a short conceptual pass before the first procedure, as discussed by Rittle-Johnson, Schneider, and Star, “Not a One-Way Street.”

¹⁴²Katherine H. Canobi, “Concept-Procedure Interactions in Children’s Addition and Subtraction,” *Journal of Experimental Child Psychology* 102 (2009): 131–49.

no magnitude, and not a quarter of pies with no written procedure. Algebra I: a short pass on what the equal sign means as a relation, then a faded worked example of a two-step equation, then an incorrect example the child must name, then more equations — not a week of “what is a variable” with no solving, and not twenty pages of solving with no talk about the equal sign. The cycle is the method. The objects change.

Two caveats keep the cycle honest. First, optimal dosage and order of concepts versus procedures for whole topics — not six decimal lessons, not three minutes of equivalence — remains thin. Rittle-Johnson, Schneider, and Star called this out; it is still thin.¹⁴³ This book will not invent a dosage. A parent who cycles in the same week is inside the evidence. A parent who claims a research-based number of concept days before the first procedure is ahead of the literature. Second, flexibility is easy to skip because it does not look like coverage. Comparing two solution methods for the same equation is the move the Rittle-Johnson and Star comparison papers support; those 2007 and 2009 papers were not opened in full for this project, and their exact d values remain **UNVERIFIED** here.¹⁴⁴ The qualitative finding is enough for a kitchen rule: after the child can execute one method, put a second method next to it and ask which is better for this problem, and why. That is flexibility. It is not a luxury unit for the gifted.

If ranking is misguided and order is not a one-way street, the elementary content that uniquely predicts later algebra is why this method chapter cannot skip fractions. The next section plants the gate. Chapter 5 will do the teaching.

The vendor sentence this section is written against is almost always a hybrid. “Research-based, conceptual, mastery, then fluency.” The first adjective is doing the work of the fourth. Rittle-Johnson, Schneider, and Star’s review is what “research-based” would have to survive, and it does not survive as a one-way street. A mastery program that withholds the next procedure until a concept quiz is perfect is running an ordering the one classroom experiment that tested order did not favor. A spiral program that returns to a procedure with no conceptual pass is running the other religion. Iteration is not a brand. It is a weekly cycle the parent can impose on any book: a short why, a handful of items, a return to the why, a comparison of two methods, more items, last month’s type on the same page. The book does not have to advertise the cycle. The parent can write it in the margin.

Flexibility is the strand of procedural knowledge the slogans forget. Schneider, Rittle-Johnson, and Star modeled it as a third latent variable, not as a bonus sticker. A child who can execute the standard algorithm for unlike-denominator addition and cannot choose a benchmark comparison when the problem is “which is larger, $3/8$ or $1/2$?” is not fluent in the *Adding It Up* sense. A child who always cross-multiplies, including when a unit rate would be simpler, and cannot say why $a/b = c/d$ implies $ad = bc$, has a procedure without flexibility. The later chapters will keep asking for the second method. This chapter is why.

¹⁴³Rittle-Johnson, Schneider, and Star, “Not a One-Way Street,” on the thinness of whole-topic dosage and order comparisons. Open question left standing.

¹⁴⁴Bethany Rittle-Johnson and Jon R. Star (2007, 2009), comparison-of-methods papers. Full texts not opened this pass; exact d **UNVERIFIED**. Qualitative finding used: comparison improves flexibility.

2.3 Fractions are the gate (plant for Chapter 5)

The National Mathematics Advisory Panel named three Critical Foundations of Algebra: fluency with whole numbers; fluency with fractions, including decimals, percents, and negative fractions; and particular aspects of geometry and measurement, especially similar triangles, because slope and linear functions “depend logically” on them.¹⁴⁵ Of these, “knowledge of fractions is the most important foundational skill not currently developed among American students.”¹⁴⁶ A nationally representative survey of Algebra I teachers commissioned by the Panel — 743 teachers, 2007 — rated students as having very poor preparation in “rational numbers and operations involving fractions and decimals,” alongside word problems and study habits.¹⁴⁷ That survey is not “teachers today say.” It is the 2007 document the Panel used. This book will not launder it into a 2026 classroom census. It will also not pretend the foundation has since been solved by a new franchise.

Robert Siegler, Greg Duncan, Pamela Davis-Kean, Kathryn Duckworth, Amy Claessens, Mimi Engel, Maria Ines Susperreguy, and Meichu Chen, in *Psychological Science* in 2012, then showed the educational consequence as a prediction. Using the British Cohort Study ($N = 3,677$) and the U.S. Panel Study of Income Dynamics Child Development Supplement ($N = 599$), they found that elementary students’ knowledge of fractions and of whole-number division uniquely predicted algebra knowledge and overall mathematics achievement five or six years later, after controlling for other mathematical knowledge, general intellectual ability or working memory, and family income and education.¹⁴⁸ A one-standard-deviation increase in age-10 fractions knowledge was uniquely associated with a 0.15-SD (United Kingdom) / 0.17-SD (United States) increase in high-school algebra and a 0.16-SD / 0.18-SD increase in overall high-school math.¹⁴⁹ Concurrent correlations of high-school fraction knowledge with overall math were $r = .81$ (United Kingdom) and $r = .87$ (United States).¹⁵⁰ Those numbers are load-bearing for this book. They are also easy to upgrade. This is a *prediction* finding, not a classroom experiment proving that teaching fractions a certain way *causes* algebra success. Do not upgrade it. Grades 3–7 fractions work is not a side topic. Chapter 5 is the heaviest elementary chapter in this book because of that sentence, not because fractions are charming.

The mechanism the prediction sits on is conceptual change, not missing pizza time. Yujing Ni and Yong-Di Zhou, in *Educational Psychologist* in 2005, review whole-number bias: children over-extend whole-number properties to rationals. Unique successors. A finite number of entities in an

¹⁴⁵National Mathematics Advisory Panel, *Foundations for Success*, Critical Foundations of Algebra: whole-number fluency; fraction fluency including decimals, percents, and negative fractions; particular geometry and measurement, especially similar triangles.

¹⁴⁶National Mathematics Advisory Panel, *Foundations for Success*, Executive Summary findings 4 and 12, and Chapter 4: “knowledge of fractions is the most important foundational skill not currently developed among American students.”

¹⁴⁷National Mathematics Advisory Panel, *Foundations for Success*, algebra-teacher survey, 743 teachers, 2007. Weakest incoming preparation: rational numbers and operations involving fractions and decimals. Do not launder into a 2026 classroom census.

¹⁴⁸Siegler et al., “Early Predictors,” after controls for other mathematical knowledge, general intellectual ability or working memory, and family income and education.

¹⁴⁹Siegler et al., “Early Predictors”: 1-SD age-10 fractions $\rightarrow$ 0.15-SD (UK) / 0.17-SD (US) high-school algebra; 0.16-SD / 0.18-SD overall high-school math.

¹⁵⁰Siegler et al., “Early Predictors”: concurrent $r = .81$ (UK) and $r = .87$ (US) of high-school fraction knowledge with overall math.

interval. Multiplication makes bigger. Addition of “pieces” means add the counts you see.¹⁵¹ Ni and Zhou review three origin stories — a domain-specific number mechanism, a discrete-quantity privilege, interference of prior instruction — and conclude the evidence is not yet decisive among them. The instructional upshot they endorse is the one this book will keep: organize fraction teaching so that prior whole-number knowledge is used rather than only fought.¹⁵² “Multiply makes bigger / divide makes smaller” is not cute folklore. It is a whole-number property that fails for multipliers and divisors less than 1. Stella Vosniadou, Xenia Vamvakoussi, and Irini Skopeliti treat it as a central conceptual-change problem.¹⁵³ Common Core 5.NF.5 exists specifically to break it: multiplication as scaling, a fraction greater than 1 enlarges, a fraction less than 1 shrinks.¹⁵⁴ Greer 1994 is the older source sometimes cited for the “multiply makes bigger” data; that paper was not opened for this project, and the data remain **UNVERIFIED** here. Cite Ni and Zhou, Siegler, and 5.NF.5.¹⁵⁵ High-school and community-college students still confuse fraction and whole-number properties.¹⁵⁶ The gate does not close at the end of fifth grade if it was never opened.

Adding numerators and denominators — $1/2 + 1/3 = 2/5$ — is a magnitude failure. The IES fractions practice guide (Siegler, Carpenter, Fennell, Geary, Lewis, Okamoto, Thompson, and Wray, 2010, Recommendation 2) names missing magnitude as “often at the root” of that error and of not seeing fractions as numbers.¹⁵⁷ Pies introduce part-whole. They do not by themselves teach that $3/4$ is a number with a magnitude. The Panel wrote that representing fractions on a number line is “one key mechanism linking conceptual and procedural knowledge.”¹⁵⁸ Siegler, Clarissa Thompson, and Michael Schneider, in *Cognitive Psychology* in 2011, treat numerical development as the progressive extension of the mental number line to new classes of number, and as the learning that many whole-number properties are not properties of number. Fractions are the first large opportunity to learn that. Magnitude estimation of fractions is closely related to correct fraction arithmetic.¹⁵⁹ The next section is the representation. This section is the gate.

¹⁵¹Yujing Ni and Yong-Di Zhou, “Teaching and Learning Fraction and Rational Numbers: The Origins and Implications of Whole Number Bias,” *Educational Psychologist* 40, no. 1 (2005): 27–52.

¹⁵²Ni and Zhou, “Whole Number Bias,” instructional upshot: use prior whole-number knowledge rather than only fight it. Three origin accounts; evidence not decisive among them.

¹⁵³Stella Vosniadou, Xenia Vamvakoussi, and Irini Skopeliti, “The Framework Theory Approach to Conceptual Change,” in *International Handbook of Research on Conceptual Change*, ed. Stella Vosniadou (2008). Cited from Siegler et al. 2012 for persistent whole-number/fraction confusion. Full chapter not opened.

¹⁵⁴*Common Core State Standards for Mathematics*, 5.NF.5, multiplication as scaling. Exists specifically to break “multiply makes bigger.”

¹⁵⁵Greer (1994), “multiply makes bigger” data: paper not opened this pass; **UNVERIFIED**. Cite Ni and Zhou, Siegler, and 5.NF.5 instead.

¹⁵⁶Schneider and Siegler (2010); Vosniadou, Vamvakoussi, and Skopeliti (2008), as cited by Siegler et al., “Early Predictors.”

¹⁵⁷Robert Siegler, Thomas Carpenter, Francis Fennell, David Geary, James Lewis, Yukari Okamoto, Laurie Thompson, and Jonathan Wray, *Developing Effective Fractions Instruction for Kindergarten Through 8th Grade*, NCEE 2010-4039 (Washington, DC: Institute of Education Sciences, 2010), Recommendation 2. Missing magnitude “often at the root” of adding numerators and denominators.

¹⁵⁸National Mathematics Advisory Panel, *Foundations for Success*, on representing fractions on a number line as “one key mechanism linking conceptual and procedural knowledge.”

¹⁵⁹Robert S. Siegler, Clarissa A. Thompson, and Michael Schneider, “An Integrated Theory of Whole Number and Fractions Development,” *Cognitive Psychology* 62, no. 4 (2011): 273–96. Magnitude estimation of fractions closely related to correct fraction arithmetic; see also Steven A. Hecht and Kevin J. Vagi, “Sources of Group and Individual Differences in Emerging Fraction Skills,” *Journal of Educational Psychology* 102 (2010): 843–58.

The IES 2010 guide’s five recommendations are the teaching map Chapter 5 will expand, not a 2026 randomized trial of a homeschool kitchen. Build on informal fraction experiences of young children — sharing, fair shares. Help students recognize that fractions are numbers that expand the number system; use number lines as a central representational tool from the early grades. Help students understand why computational procedures with fractions make sense. Develop understanding of ratios and rates before formal proportions. Address teacher education.¹⁶⁰ Teachers’ own fraction knowledge is often weak. Liping Ma’s 1999 comparison is the one usually cited: U.S. teachers generating few or no explanations of invert-and-multiply versus Japanese and Chinese teachers generating two or three.¹⁶¹ The design user who stopped at algebra is Ma’s U.S. teacher, in a kitchen. That is not an insult. It is why this book’s first thesis condition is that the parent understands the concept well enough to hear a wrong answer. Invert-and-multiply is a procedure a parent can execute and still not hear why $1/2 \div 1/4$ is 2. Chapter 5 will do that hearing. Chapter 6 will do ratio. Chapter 7 will assume the gate was not skipped.

The Panel’s benchmarks for fractions, Table 2, are judgments from curriculum comparison, not experimental placements — the Panel says so.¹⁶² Identify, represent, and compare fractions and decimals on a number line by the end of grade 4. Add and subtract fractions and decimals by the end of grade 5. Multiply and divide by the end of grade 6. All operations with signed fractions by the end of grade 7. Percent, ratio, rate, and proportionality by the end of grade 7. Common Core’s map is close: Grade 3 is the first grade with a Fractions domain (unit fractions, fractions as numbers on a line, denominators 2, 3, 4, 6, 8); Grade 5 is unlike-denominator addition and multiplication as scaling; division of a fraction by a fraction is Grade 6, not Grade 5.¹⁶³ Those are maps. They are not homeschool law. They are not a license to treat a vendor’s “grade 5” book as a legal grade. Math-U-See Epsilon is a fractions *level*, not “fifth grade” — a vendor naming, already installed in Chapter 1.¹⁶⁴ A family that runs one level per calendar year from Primer can sit formal fraction operations later than the map, or earlier if Alpha started in kindergarten. The map and the Panel would call a student who reaches the ratio year without fraction-as-number work underprepared for algebra. Placement is the first job. Skipping the gate because a child is “good at math” is the error Finding 30 does not license. Acceleration is for a child who has the prerequisites. Siegler and colleagues are the reason the prerequisites include fractions.

Fuchs and colleagues’ later fractions-intervention experiments (2013 and after) are the experimental half of the number-line research program. Numerical effect sizes from those papers were not extracted for this project. They remain **UNVERIFIED** here. Do not invent them.¹⁶⁵ The IES 2021 intervention guide’s Recommendation 4 — number lines, strong evidence, 14 studies — is the synthesis this book will use, not a homemade *g*.

¹⁶⁰ Siegler et al., *Developing Effective Fractions Instruction*, Recommendations 1–5.

¹⁶¹ Liping Ma, *Knowing and Teaching Elementary Mathematics* (Mahwah, NJ: Erlbaum, 1999), U.S. versus Chinese teacher knowledge of invert-and-multiply, as cited by Siegler et al., “Early Predictors.” Book not re-opened this pass.

¹⁶² National Mathematics Advisory Panel, *Foundations for Success*, Table 2 fraction benchmarks. Judgments from curriculum comparison, not experimental placements — the Panel says so.

¹⁶³ *Common Core State Standards for Mathematics*, Grade 3 Fractions domain; Grade 5 unlike-denominator addition and multiplication as scaling; fraction $\div$ fraction is Grade 6.

¹⁶⁴ Math-U-See Epsilon as a fractions *level*, not “fifth grade.” Vendor naming; see Chapter 1.

¹⁶⁵ Fuchs et al. fractions-intervention papers (2013 and after): located; numerical effect sizes **not extracted**. **UNVERIFIED** as to coefficients. Do not invent a *g*.

The representation that unifies the gate is the number line. Pies stay. The line is what pies are not.

The plant for Chapter 5 has to stay a plant. This section does not teach unlike-denominator addition. It installs why that teaching is not optional and why pies are not the teaching. A parent who is already anxious about fractions — Ma’s U.S. teacher, in a kitchen — will be tempted to outsource the gate to a video, a solver, or a year of cooking. The video can be a worked example the parent studies first. The solver is Chapter 3’s forbidden object if it takes the child’s problem. The cooking is Finding 26. None of those substitutes for placing $\frac{3}{4}$ on a line and then writing the symbol next to the point. The Siegler coefficients are the reason this book will not let a “good at math” child skip the line. They are also the reason this book will not pretend that a 0.17-SD unique association, after controls, is a treatment effect of a particular fractions curriculum. Prediction is not an RCT. The teaching still has to happen.

Whole-number bias will show up in the parent’s own speech before it shows up on the child’s page. “Multiplying makes it bigger.” “A fourth is smaller than a fifth because four is smaller than five” — or the reverse, “fourths are bigger.” “0.25 is bigger than 0.3 because twenty-five is bigger than three.” Those are representation errors. Rittle-Johnson, Siegler, and Alibali used decimal-fraction items precisely because representation errors of that kind mediate procedure errors. The iterative model and the gate are the same story: fix the representation, and the procedure has something to attach to; run the procedure with a better representation in view, and the concept moves. Chapter 5 will write that cycle on fractions. This chapter only has to make skipping it look as costly as the documents say it is.

2.4 The number line unifies; representations must be connected and faded

Siegler, Thompson, and Schneider’s 2011 integrated theory of numerical development is the cognitive claim under the teaching recommendation. Numerical development is the progressive mapping of more classes of number onto a mental number line. Whole numbers first. Then fractions. Then decimals, which are fractions with denominators 10, 100, 1,000, on the same line. Then negatives. Later, functions — input to output, a graph that is itself a number line in two directions, slope as a rate grounded in similar triangles.¹⁶⁶ The parent who thinks of the number line as a grade-2 decoration for counting on has missed the representation that scales from equal shares through calculus’s function concept. This book will keep returning to it. Chapter 4: number-line diagrams for length. Chapter 5: fractions as numbers, equivalent fractions as the same point, decimals on the same line. Chapter 6: signed numbers as positions and moves; unit rate on a graph through the origin. Chapter 7: the line inside the first linear function. Chapter 10: the limit as arbitrary closeness on that line, not as “plugging in.” The teaching of those later objects is not this chapter’s job. The lock is: one picture, more classes of number.

The IES fractions practice guide, Recommendation 2, is the elementary version: treat fractions as numbers and use number lines as a central tool from the early grades. Part-whole alone does not express magnitude.¹⁶⁷ The 2021 IES intervention guide, Fuchs, Newman-Gonchar, Schumacher, Dougherty, Bucka, Karp, Woodward, Clarke, Jordan, Gersten, Jayanthi, Keating, and Morgan, Rec-

¹⁶⁶ Siegler, Thompson, and Schneider, “Integrated Theory.” Later function and slope uses are the book’s application of the same picture, not a claim that the 2011 paper taught calculus.

¹⁶⁷ Siegler et al., *Developing Effective Fractions Instruction*, Recommendation 2.

ommendation 4, is the same move for struggling elementary students, rated strong evidence on 14 studies.¹⁶⁸ The 2021 guide’s panel advises against cross-multiply-to-compare during intervention because the trick does not build magnitude.¹⁶⁹ Equivalent fractions as the same point on the line — Example 4.5 in that guide — is the move, not a butterfly trick.¹⁷⁰ A parent who teaches cross-multiply as the first comparison has given a procedure with no representation. Iteration, from the last section, would put the line first, a handful of comparisons against 0, 1/2, and 1, then a written method that the child can still point to on the line. The trick can come later, as a compression, if it comes at all.

Representations more generally help under guidance and then become a crutch if they are not connected to symbols and faded. Kira Carbonneau, Scott Marley, and James Selig, in the *Journal of Educational Psychology* in 2013, meta-analyzed 55 studies, $N = 7,237$, kindergarten through college, manipulatives versus abstract symbols only. The overall mean d is approximately 0.37, small-to-moderate, favoring manipulatives.¹⁷¹ That number is not “manipulatives work.” It is versus abstract-only, and it is moderated. Outcome-specific: moderate-to-large on retention ($k = 53$); small on problem solving ($k = 9$), transfer ($k = 13$), and justification ($k = 2$).¹⁷² High-guidance use was associated with higher retention and problem solving; low-guidance was associated with higher transfer. Ages 3–6 showed more difficulty than 7–11 or 12+.¹⁷³ Report the d with the moderators. Do not put “research says use manipulatives” on a catalog page and stop.

The intervention version of the same point is Concrete-Representational-Abstract, also called concrete-semi-concrete-abstract, associated with special-education mathematics — Witzel, Mercer, and Miller 2003 on algebra; Flores 2010 on subtraction — and synthesized as Fuchs and colleagues’ 2021 Recommendation 3, strong evidence, 28 studies.¹⁷⁴ The steps the 2021 guide actually writes are not folklore CRA. Choose representations that match the mathematics: proportional base-ten for place value, length models for fractions. Connect concrete and semi-concrete to the notation in the same visual field, not “blocks today, symbols next month.” Give ample, meaningful practice so representations become thinking tools, not toys. Revisit concrete and semi-concrete after fading; fading is for accuracy, not for a calendar date.¹⁷⁵ Table 3.1 of that guide is the representation-to-topic map a parent can steal. The number line gets its own recommendation because it is the representation that scales from counting to irrationals. Karen Fuson and Diane Briars’s 1990 base-ten-blocks study is the classic place-value version:

¹⁶⁸Fuchs et al., *Assisting Students Struggling* (2021), Recommendation 4, strong evidence, 14 studies.

¹⁶⁹Fuchs et al., *Assisting Students Struggling* (2021), Recommendation 4, panel advice against cross-multiply-to-compare during intervention because it does not build magnitude.

¹⁷⁰Fuchs et al., *Assisting Students Struggling* (2021), Example 4.5, equivalent fractions as the same point on the line.

¹⁷¹Kira J. Carbonneau, Scott C. Marley, and James P. Selig, “A Meta-Analysis of the Efficacy of Teaching Mathematics with Concrete Manipulatives,” *Journal of Educational Psychology* 105, no. 2 (2013): 380–400. Overall mean d approximately 0.37 versus abstract-only; 55 studies, $N = 7,237$. Full PDF not opened; d approximately 0.37 from abstract plus quoting sources. Labelled approximate.

¹⁷²Carbonneau, Marley, and Selig, “Concrete Manipulatives,” outcome-specific pattern from the abstract: larger on retention ($k = 53$) than problem solving ($k = 9$), transfer ($k = 13$), or justification ($k = 2$).

¹⁷³Carbonneau, Marley, and Selig, “Concrete Manipulatives,” guidance and age moderators from the abstract.

¹⁷⁴Bradley S. Witzel, Cecil D. Mercer, and M. David Miller (2003), algebra; Margaret M. Flores (2010), subtraction; Fuchs et al., *Assisting Students Struggling* (2021), Recommendation 3, strong evidence, 28 studies.

¹⁷⁵Fuchs et al., *Assisting Students Struggling* (2021), Recommendation 3 how-to: match the representation to the mathematics; connect concrete and semi-concrete to notation in the same visual field; meaningful practice; fade for accuracy, not a calendar date; revisit after fading.

proportional materials, ten-sticks that are ten ones, connected to the written numeral.¹⁷⁶ 27 is 2 tens and 7 ones, and those tens can be unpacked. $27 + 15$ is not “2 + 1 and 7 + 5” as concatenation. The blocks are a conversation about place value, not a substitute for writing $27 + 15$ or for placing $\frac{3}{4}$ on a line.

Two further cautions keep the blocks honest. First: a manipulative that is too toy-like is hard to see as a symbol. David Uttal, Judy DeLoache, and colleagues’ dual-representation papers are the usual citation. Those papers were not opened in full for this project; exact findings remain **UNVERIFIED** here.¹⁷⁷ The qualitative rule that survives without the pages is the one the 2021 guide already wrote: choose a small, stable set that matches the mathematics; put the blocks next to the numerals; fade when the child is accurate without them; bring them back when a new idea — decimals on the same ten-blocks, fractions on the same line — needs a bridge. Second: virtual versus physical manipulatives is a literature this project did not open. **UNVERIFIED**.¹⁷⁸ Do not claim they are “just as good.” Do not claim they are worse. A screen of movable fraction bars can be the same conversation as tiles if the parent connects them to the written numeral in the same sitting. It can also be a game. The variable is the conversation, not the plastic.

Folklore CRA is a calendar: three weeks of blocks, three weeks of pictures, then symbols forever. The documents do not say that. Folklore the other way is “all children must use manipulatives for weeks” or “manipulatives make it concrete so they will automatically abstract.” Carbonneau’s transfer and justification effects were small. Automatic abstraction is not the finding. The finding is a modest average advantage versus abstract-only, larger on retention, larger under guidance, weaker for the youngest children, and dependent on connecting the object to the symbol. A parent who dumps a bin of bright tiles on the table and leaves the room has not implemented Recommendation 3.

With a representation in hand, the teaching moves that survive are specific — and their effect sizes are not interchangeable. Worked examples, interleaving, and schemas are the next section. They are three objects. They will stay three objects.

The number line also refuses a second folklore: that each new class of number needs a new toy. Fraction circles for fractions, base-ten blocks for decimals, algebra tiles for integers, a graphing app for slope. Some of those tools are well chosen for a first conversation. None of them replaces the line. Decimals are fractions with denominators 10, 100, 1,000, on the same line. Signed numbers are positions and moves on the same line. Slope is a rate that the Panel’s Critical Foundations already tied to similar triangles — a geometric object that lives on the plane the line became. A parent who keeps buying a new manipulative for each chapter and never puts two classes of number on one picture is collecting toys. Carbonneau’s modest d , under guidance, versus abstract-only, is not a shopping list. Fuchs Recommendation 3 is a constraint on the shopping list: few representations, matched to the mathematics, connected to notation, faded, revisited.

¹⁷⁶Karen C. Fuson and Diane J. Briars, “Using a Base-Ten Blocks Learning/Teaching Approach for First- and Second-Grade Place-Value and Multidigit Addition and Subtraction,” *Journal for Research in Mathematics Education* (1990). Cited from Rittle-Johnson, Schneider, and Star 2015. Full PDF not opened.

¹⁷⁷David H. Uttal, Scudder, and Judy S. DeLoache (1997); Uttal, O’Doherty, Newland, Hand, and DeLoache (2009), dual-representation papers. Not opened this pass; exact findings **UNVERIFIED**.

¹⁷⁸Virtual versus physical manipulatives: literature not opened. **UNVERIFIED**.

Place value is the first big compositional idea the line has to live with. 27 is 2 tens and 7 ones. Those tens can be unpacked. The Panel puts composition and decomposition, and the commutative, associative, and distributive properties, inside “fluency with whole numbers,” not as extras. A child who can pack and unpack tens and cannot yet place 27 on a line past 20 is missing the magnitude half of the same idea. A child who can place 27 and cannot unpack it is missing the compositional half. Chapter 4 will do both. The lock is already here: proportional materials next to the numeral, then the line, then the numeral without the materials.

2.5 Teaching moves that survive: examples, interleaving, schemas

Three moves survive the documents with named objects, sample sizes, and delays. They are routinely collapsed in homeschool blogs into “use good practice” or “mix it up” or “make it real.” They are not one move. Their coefficients are not interchangeable. This section prints them separately.

Worked examples, faded, plus incorrect examples explained. Cognitive load theory starts from a narrow working memory and a large long-term memory of schemas. John Sweller’s 1988 paper and the later reviews with Jeroen van Merriënboer and Fred Paas distinguish intrinsic load (the interactivity of the elements being learned) from extraneous load (imposed by poor instructional design).¹⁷⁹ The worked-example effect is the instructional consequence for novices. Sweller and Graham Cooper, in *Cognition and Instruction* in 1985, established it in algebra: studying a fully worked solution produced more learning than asking the novice to solve equivalent problems, because means-ends search consumes working memory that could have been used to extract the schema.¹⁸⁰ That is not “lecture works.” It is a finding about the novice’s working memory on high-element-interactivity material — first two-step equations, first unlike-denominator addition, first formal proof.

The same example that helps a novice becomes redundant for a relative expert and can hurt. That is the expertise-reversal effect (Slava Kalyuga, Paul Ayres, Paul Chandler, and Sweller, 2003).¹⁸¹ The instructional consequence is the guidance-fading effect. Alexander Renkl and Robert Atkinson, with colleagues, developed fading, including backward fading: start with a complete example, move to completion problems with some steps blank, remove the last steps first, then more, until the learner solves conventional problems.¹⁸² Self-explanation is the prompt that makes a worked example more than a script to copy. Michelene Chi, Miriam Bassok, Lewis, Reimann, and Glaser (1989), Renkl (1997), and Rittle-Johnson (2006) are the lineage. Dunlosky and colleagues rate self-explanation as moderate utility: it works, but it is not as robust across conditions as practice

¹⁷⁹John Sweller, “Cognitive Load During Problem Solving: Effects on Learning,” *Cognitive Science* 12 (1988): 257–85. John Sweller, Jeroen J. G. van Merriënboer, and Fred Paas, “Cognitive Architecture and Instructional Design: 20 Years Later,” *Educational Psychology Review* 31 (2019): 261–92.

¹⁸⁰John Sweller and Graham A. Cooper, “The Use of Worked Examples as a Substitute for Problem Solving in Learning Algebra,” *Cognition and Instruction* 2, no. 1 (1985): 59–89.

¹⁸¹Slava Kalyuga, Paul Ayres, Paul Chandler, and John Sweller, “The Expertise Reversal Effect,” *Educational Psychologist* 38 (2003): 23–31.

¹⁸²Alexander Renkl and Robert K. Atkinson, “Structuring the Transition from Example Study to Problem Solving in Cognitive Skill Acquisition: A Cognitive Load Perspective,” *Educational Psychologist* 38 (2003): 15–22. See also Renkl, Atkinson, Maier, and Staley (2002); Atkinson, Derry, Renkl, and Wortham (2000). Backward fading: remove the last steps first.

testing and spacing.¹⁸³

Julie Booth, Karin Lange, Kenneth Koedinger, and Kristie Newton, in *Learning and Instruction* in 2013, then showed what to do with a wrong solution. Two in-vivo Cognitive Tutor experiments: explaining worked examples during guided practice improved conceptual understanding versus guided practice alone. Incorrect examples, alone or mixed with correct ones, were “especially beneficial for fostering conceptual understanding” of two-step linear equations.¹⁸⁴ The paper lists the algebra misconceptions the examples were built to hit: the equal sign as “where the answer goes” (Arthur Baroody and Herbert Ginsburg 1983; Carolyn Kieran 1981); minus signs as “subtract” rather than as modifiers of terms (Vlassis 2004); variables as a single mystery number (Booth 1984; Knuth, Stephens, McNeil, and Alibali 2006; Küchemann 1978).¹⁸⁵ Those misconceptions are already the arithmetic root of algebra. Chapter 4 will write non-canonical equations early so $3 + 4 + 5 = 3 + \underline{\quad}$ is not answered 15 or 12. Chapter 7 will meet the same equal sign inside multi-step equations. The parent’s job in this chapter is the move: include at least one incorrect example per new procedure and ask “what did they think this symbol meant?”

For the parent, the faded sequence on new, high-element-interactivity material is: show one fully worked example, then a second that the child must explain, then a completion problem missing the last step, then a conventional problem. Do not start a brand-new topic with “see if you can figure it out” as the only move if the child is a novice in that topic. That is the situation in which Sweller’s effect is largest. Once the child can execute the procedure, stop showing full examples. Expertise reversal is real. Watching the parent do twenty problems in real time is not studying a worked example. Studying a worked example is the child reading a completed solution and explaining the steps. The generation still belongs to the child.

Interleaving after a short blocked introduction. Four effects are routinely collapsed. They are related and they are not the same. Retrieval, or practice testing: Dunlosky, Katherine Rawson, Elizabeth Marsh, Mitchell Nathan, and Daniel Willingham, in *Psychological Science in the Public Interest* in 2013, give practice testing a high-utility rating, robust across ages, materials, and criterion tasks. The mechanism is not “checking what you know.” Retrieval itself strengthens the memory.¹⁸⁶ Distributed, or spaced, practice: Nicholas Cepeda, Harold Pashler, Edward Vul, John Wixted, and Doug Rohrer’s 2006 *Psychological Bulletin* review is the standard synthesis; Rohrer and Kelli Taylor (2006, 2007) applied spacing to mathematics specifically. Dunlosky and colleagues also rate distributed practice high utility.¹⁸⁷ Spacing typically improves delayed tests

¹⁸³ Michelene T. H. Chi, Miriam Bassok, M. W. Lewis, Peter Reimann, and Robert Glaser (1989); Alexander Renkl (1997); Bethany Rittle-Johnson, “Promoting Transfer: Effects of Self-Explanation and Direct Instruction,” *Child Development* 77 (2006): 1–15. Dunlosky et al., “Effective Learning Techniques,” self-explanation rated moderate utility.

¹⁸⁴ Julie L. Booth, Karin E. Lange, Kenneth R. Koedinger, and Kristie J. Newton, “Using Example Problems to Improve Student Learning in Algebra: Differentiating between Correct and Incorrect Examples,” *Learning and Instruction* 25 (2013): 24–34. ERIC ED543090.

¹⁸⁵ Booth et al., “Using Example Problems,” misconception list: Baroody and Ginsburg (1983); Carolyn Kieran, “Concepts Associated with the Equality Symbol,” *Educational Studies in Mathematics* (1981); Vlassis (2004); Booth (1984); Eric J. Knuth, Ana C. Stephens, Nicole M. McNeil, and Martha W. Alibali, “Does Understanding the Equal Sign Matter? Evidence from Solving Equations,” *Journal for Research in Mathematics Education* 37 (2006): 297–312; Küchemann (1978).

¹⁸⁶ Dunlosky et al., “Effective Learning Techniques,” practice testing rated high utility.

¹⁸⁷ Nicholas J. Cepeda, Harold Pashler, Edward Vul, John T. Wixted, and Doug Rohrer, “Distributed Practice in Verbal Recall Tasks: A Review and Quantitative Synthesis,” *Psychological Bulletin* 132 (2006): 354–80. Doug Rohrer

and can look worse on an immediate test — a feature, not a bug, and the reason a parent who only watches tonight’s homework will underestimate it. Overlearning — continuing practice after one correct performance — has limited additional benefit compared with distributing the same extra trials later (Rohrer and Taylor 2006). Cramming a fact sheet until tonight’s score is 100 percent is not the same as meeting those facts again next week.¹⁸⁸

Interleaving is not “mix in a review problem at the end of the page.” It is arranging practice so that consecutive problems usually require different strategies, forcing the learner to choose a strategy from the problem’s features. Rohrer, Robert Dedrick, and Kaleena Burgess, in *Psychonomic Bulletin & Review* in 2014: grade-7 classroom crossover, $n = 140$, unannounced test two weeks later, interleaved 72 percent versus blocked 38 percent, $d = 1.05$, 95 percent confidence interval $[0.80, 1.30]$. Four superficially dissimilar problem kinds: a four-step linear equation; a proportion word problem; graph $y = mx + b$; slope from two points. Pretest accuracy on these kinds was 0.7 percent for full solutions.¹⁸⁹ Rohrer, Dedrick, Marissa Hartwig, and Chi-Ngai Cheung, in the *Journal of Educational Psychology* in 2020: a preregistered cluster randomized trial, 54 seventh-grade classes, four months of assignments, unannounced test one month later, interleaved 61 percent versus blocked 38 percent, $d = 0.83$. Teachers implemented without training.¹⁹⁰ The authors themselves state the caveat this book will not drop: interleaved assignments also space same-kind items, so d is a joint effect. Most textbooks are almost entirely blocked — Rohrer, Dedrick, and Hartwig, 2020, “The scarcity of interleaved practice in mathematics textbooks.”¹⁹¹

Taylor and Rohrer 2010 is the study that most cleanly isolates interleaving from spacing (both conditions spaced; $d = 1.23$) and that showed blocked-condition errors were mostly wrong-strategy errors.¹⁹² In Rohrer, Dedrick, and Burgess 2014, by contrast, discrimination errors were rare (4.4 percent); the authors propose that interleaving also strengthens the association between a problem kind and its strategy.¹⁹³ Almost all clean classroom randomized trials of interleaving in mathematics are grade 7. Do not export $d = 1.05$ to grade 4 or to calculus. Do not mash $d = 1.05$ from a nine-week classroom study with $d = 0.83$ from a four-month cluster trial with Dunlosky’s “high utility” rating for spacing. Different designs, different delays, interleaving confounded with spacing in the classroom studies. Brunmair and Richter’s later meta-analysis of interleaving is sometimes

and Kelli Taylor, “The Effects of Overlearning and Distributed Practice on the Retention of Mathematics Knowledge,” *Applied Cognitive Psychology* 20 (2006): 1209–24; Rohrer and Taylor, “The Shuffling of Mathematics Practice Problems Boosts Learning,” *Instructional Science* 35 (2007): 481–98. Dunlosky et al., distributed practice rated high utility.

¹⁸⁸Rohrer and Taylor, “Overlearning and Distributed Practice” (2006): limited additional benefit of overlearning versus distributing the same extra trials later.

¹⁸⁹Rohrer, Dedrick, and Burgess, “Benefit of Interleaved Mathematics Practice” (2014). Four kinds: four-step linear; proportion word problem; graph $y = mx + b$; slope from two points. Pretest 0.7% for full solutions.

¹⁹⁰Rohrer, Dedrick, Hartwig, and Cheung, “Randomized Controlled Trial” (2020). Preregistered cluster RCT; teachers implemented without training; one-month delayed unannounced test.

¹⁹¹Doug Rohrer, Robert F. Dedrick, and Marissa K. Hartwig, “The Scarcity of Interleaved Practice in Mathematics Textbooks,” *Educational Psychology Review* 32 (2020): 873–83. Authors’ own caveat: interleaved assignments also space same-kind items, so classroom d is a joint effect.

¹⁹²Kelli Taylor and Doug Rohrer, “The Effects of Interleaving Practice,” *Applied Cognitive Psychology* 24 (2010): 837–48. Both conditions spaced; $d = 1.23$; blocked errors mostly wrong-strategy. Cited from Rohrer, Dedrick, and Burgess 2014; full PDF not opened.

¹⁹³Rohrer, Dedrick, and Burgess (2014): discrimination errors rare (4.4%); authors propose interleaving also strengthens the association between a problem kind and its strategy.

cited as $g = 0.34$ for mathematical tasks. That paper was not opened for this project. The coefficient remains **UNVERIFIED** here. It will not be recycled as if it had been read.¹⁹⁴

For the parent, a weekly shape that respects all four without pretending they are one: day of new skill, a short blocked set, learn to execute. Same week, mixed retrieval of older facts — the testing effect. Later this week and next, the new skill reappears between unlike problems — interleaving plus spacing. Monthly, an unannounced mixed quiz the child does without notes — the Rohrer test. That is not a 180-day plan. It is a shape. The later grade-band chapters will put particular objects into the shape: multiplication and division fact families as soon as both exist; slope-from-two-points and graph- $y = mx + b$ in the same week, not in successive chapters with no return; proportion word problems next to linear equations, which is the 2014 material. A parent whose purchased book is almost entirely blocked — most are — can still interleave by pulling yesterday's and last month's problem types onto the same page. The book does not have to do it. The parent can.

Word problems are schemas, not keyword hunts. Keyword strategies fail because the same word appears in different structures. Fuchs and colleagues, 2021, Table 5.2, list why *all*, *left*, and *total* fail.¹⁹⁵ What does work, repeatedly, is teaching students to identify the underlying structure. Additive: combine, change (increase or decrease), compare. Multiplicative: equal groups, comparison, ratio and rate, percent, proportion. Asha Jitendra's schema-based instruction packages the move: identify the schema, represent it with a diagram, estimate, solve, check.

Jitendra, Star, Danielle Dupuis, and Michael Rodriguez, 2012: seventh-grade randomized trial, 1,163 students in 42 classrooms, schema-based instruction versus regular instruction, proportion problems. Immediate posttest $g = 1.24$. Six-week retention $g = 1.27$. No significant transfer.¹⁹⁶ Jitendra and colleagues, 2013: third-graders with mathematics difficulties, small-group schema-based instruction versus control. Word-problem posttest $g = 0.46$. District achievement $g = 0.34$. No significant retention at 8 weeks.¹⁹⁷ Those two sentences together are the honest effect-size paragraph. Schema-based instruction can produce large gains on the trained problem types and much smaller or null effects on far transfer and delayed retention, depending on sample and measure. Corey Peltier and Kimberly Vannest, 2017, meta-analysis of schema instruction in elementary school: Hedge's $g = 1.57$ on immediate problem solving and $g = 1.09$ on transfer, 21 studies, 3,408 students — a large number that includes many researcher-developed proximal measures.¹⁹⁸ Do not quote $g = 1.57$ as “schema instruction raises math achievement by 1.57 standard deviations.” That is immediate problem-solving on mostly proximal measures. IES 2021 Recommendation 5, strong evidence, 18 studies, and the 2009 RtI guide (Russell Gersten and colleagues) both reject keyword strategies.¹⁹⁹

¹⁹⁴Brunmair and Richter interleaving meta-analysis, sometimes cited as $g = 0.34$ for mathematical tasks. Paper not opened this pass; exact coefficient **UNVERIFIED**. Not recycled.

¹⁹⁵Fuchs et al., *Assisting Students Struggling* (2021), Table 5.2, why keyword strategies (*all*, *left*, *total*) fail.

¹⁹⁶Jitendra, Star, Dupuis, and Rodriguez, “Schema-Based Instruction” (2012), $g = 1.24 / 1.27 /$ null transfer.

¹⁹⁷Jitendra et al., “Schema-Based Instruction” (2013), $g = 0.46 / 0.34 /$ null 8-week retention.

¹⁹⁸Peltier and Vannest, “Meta-Analysis of Schema Instruction” (2017), $g = 1.57 / 1.09$, 21 studies, 3,408 students, mostly proximal.

¹⁹⁹Fuchs et al., *Assisting Students Struggling* (2021), Recommendation 5, strong evidence, 18 studies. Russell Gersten et al., *Assisting Students Struggling with Mathematics: Response to Intervention (RtI) for Elementary and Middle Schools*, NCEE 2009-4060 (Washington, DC: Institute of Education Sciences, 2009), rejecting keyword strategies.

The how-to that transfers to a kitchen table is in the 2021 guide. Introduce one problem type at a time, first as a complete story with all quantities, then as a problem with one unknown. Use role-play or objects to see the action. Vary the cover story so the child is matching structure, not nouns. Never teach “total means add.”²⁰⁰ Chapter 4 will use add-to, take-from, put-together, take-apart, and compare with unknowns in all positions. Chapter 5 will use equal groups, arrays, and comparison. Chapter 6 will use Jitendra’s proportion package, and will not expect free transfer. Teach unit rate and scaling and the proportion equation, and compare them — flexibility again.

Kitchen-math is not the schema work. National Mathematics Advisory Panel Finding 26 is the load-bearing caution, and it belongs in this section because “make it real” is the slogan that tries to replace schemas, retrieval, and the number line with a recipe. Teaching with “real-world” contexts improves performance on similar real-world items. It does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving.²⁰¹ Homeschool culture overclaims kitchen-math. They’ll learn fractions by cooking. Cooking can motivate and can provide a second representation of a fraction the child is already being taught as a number. It does not replace the number-line work, the schema work, or the retrieval. Money is a decimal-and-place-value context, useful, also not a curriculum. Building and making are spatial; they do not automatically produce proof. A child who can double a recipe and cannot place $3/2$ on a line has performed, not learned. That last distinction is Soderstrom and Bjork’s, already a prior-book rule this book will not re-argue at monograph length: current performance is often an unreliable index of learning.²⁰² Barnett and Ceci 2002 is the standard transfer taxonomy, near and far along several dimensions; that paper was not opened here, and the exact dimensions list remains **UNVERIFIED**.²⁰³ The Panel’s Finding 26 is enough. Informal activity is not a substitute for the trained problem types. Transfer from cooking, money, and games to formal items, measured rather than assumed, remains an open question this book will not fill with analogy.²⁰⁴

A parent who has read this far may want a single weekly template. The documents do not sell one, and this book will not invent a 180-day plan to satisfy the want. What they do sell, named and not mashed, is a shape the parent can impose on whatever book is already on the table. New skill: short blocked set plus a faded example. Same week: mixed retrieval of older facts, sometimes timed if they are already known untimed. Later in the week and the next: the new skill between unlike problems. Word problems by type, cover stories varied, no keyword list. At least one incorrect example per new procedure. Monthly: an unannounced mixed page without notes. That shape is Sweller, Renkl, Rohrer, Jitendra, Booth, and Dunlosky on one week, without pretending they are one coefficient. A parent who only implements the shape, and never buys another franchise, has implemented this section.

²⁰⁰Fuchs et al., *Assisting Students Struggling* (2021), Recommendation 5 how-to: one problem type at a time; complete story then one unknown; vary the cover story; never teach “total means add.”

²⁰¹National Mathematics Advisory Panel, *Foundations for Success*, Finding 26.

²⁰²Nicholas C. Soderstrom and Robert A. Bjork, “Learning versus Performance: An Integrative Review,” *Perspectives on Psychological Science* 10, no. 2 (2015): 176–99. Current performance is “often an unreliable index” of learning. Prior-book rule; assumed, not re-argued at monograph length.

²⁰³Susan M. Barnett and Stephen J. Ceci (2002), transfer taxonomy. Paper not opened; exact dimensions list **UNVERIFIED**.

²⁰⁴Transfer from informal activity (cooking, money, games) to formal items, measured rather than assumed: open question. Do not fill with analogy.

Fluency is the strand that homeschool blogs most often turn into a race. The next section separates two objects that the blogs mash.

2.6 Fluency is not a race; timed facts are two objects

Fluency, in the documents that define proficiency, is flexible, accurate, efficient, and appropriate retrieval — not a race and not optional. The National Mathematics Advisory Panel’s Finding 11: computational proficiency depends on automatic, “quick and effortless,” recall of addition and subtraction facts and of multiplication and division facts, and on fluency with standard algorithms, together with understanding of the commutative, associative, and distributive properties.²⁰⁵ *Adding It Up*’s procedural-fluency strand is the same list of adverbs: flexibly, accurately, efficiently, appropriately. A child who only has one brittle procedure is not fluent.²⁰⁶ Arthur Baroody (2003, 2006) distinguishes mastery — correct and reasonably quick — from adaptive expertise, choosing a strategy that fits.²⁰⁷ Retrieval of $6 + 7$ as 13 and knowing it is $6 + 6 + 1$ are both part of fluency. They are not enemies. The iterative model from section 2.2 is why they are not enemies. The fact and the decomposition predict gains in each other.

The IES side of the timed-test argument is an intervention recommendation, not a Mad Minute religion. Gersten, Sybilla Beckmann, Ben Clarke, Anne Foegen, Laurel Marsh, Star, and Bradley Witzel, in the 2009 *RtI* guide, Recommendation 6: about 10 minutes per intervention session on fluent retrieval of basic facts.²⁰⁸ Fuchs and colleagues, 2021, Recommendation 6, strong evidence, 27 studies: “Regularly include timed activities as one way to build students’ fluency.”²⁰⁹ The how-to in that guide — examples in Table 6.1 — is brief, mixed, already-taught material, often with the child graphing their own score, not a high-stakes three-minute 50-item test in grade 1.²¹⁰ Lynn Fuchs, in later interviews, adds the implementation constraint: time a child only on facts they can already produce correctly when untimed; do not hand a lost child a page and a clock.²¹¹ That constraint is the difference between retrieval practice and a ranking ritual.

The Boaler side is a different object. Jo Boaler, in a 2012 *Education Week* / Stanford Graduate School of Education commentary and in the 2015 working paper *Fluency Without Fear*, published through Youcubed, argues that timed tests from age 5 are a “direct cause” of early math anxiety and that fluency in the Common Core has been misread as speed. Flashcards and Mad Minute are “damaging.”²¹² The empirical core she cites is the working-memory disruption under anxiety —

²⁰⁵National Mathematics Advisory Panel, *Foundations for Success*, Finding 11: automatic recall of facts plus fluent standard algorithms plus properties.

²⁰⁶National Research Council, *Adding It Up*, 116, procedural fluency: flexibly, accurately, efficiently, appropriately.

²⁰⁷Arthur J. Baroody (2003, 2006), mastery versus adaptive expertise, as discussed in the learning-science memo. Retrieval of $6 + 7$ as 13 and as $6 + 6 + 1$ are both part of fluency.

²⁰⁸Gersten et al., *RtI* (2009), Recommendation 6: about 10 minutes per intervention session on fluent retrieval of basic facts. School-intervention figure, not a homeschool-minutes study.

²⁰⁹Fuchs et al., *Assisting Students Struggling* (2021), Recommendation 6, strong evidence, 27 studies.

²¹⁰Fuchs et al., *Assisting Students Struggling* (2021), Table 6.1 examples: brief, mixed, already-taught, often with the child graphing their own score.

²¹¹Lynn Fuchs, implementation constraint as reported in later interviews (Hechinger Report “Proof Points”): time only facts the child can already produce correctly untimed. Secondary report of Fuchs’s constraint; not a new experiment.

²¹²Boaler, “Timed Tests” (2012); Boaler, *Fluency Without Fear* (2015 working paper / Youcubed). “Direct cause” of

Ashcraft's program; Sian Beilock's group — plus classroom reports of children describing timed tests as making them nervous, including high-scoring children. *Fluency Without Fear* is a working paper / Youcubed publication, not a journal experiment. Label it as such.²¹³ Almost no one in the research literature argues that children should not come to retrieve single-digit facts. The dispute is whether speed tests used as early high-stakes assessments cause anxiety, and whether timing is necessary or merely bundled in effective interventions.

The bundling caveat is real. Grimes (2024) and a 2025 OSF preprint by Peltier and colleagues, “We’re Talking about Practice,” note that the 27 IES studies generally bundled timing with other intervention components, so timing is not isolated.²¹⁴ The 2025 paper is a preprint. Labelled preprint. Until an isolated timing experiment is opened, the honest parent-facing sentence is this. Retrieval practice of known facts is high-utility (Dunlosky and colleagues 2013). A brief timer can increase the number of successful retrievals per minute for children who already know the facts — Fuchs’s speed-group description. A public, ranked, early-grades speed test is the object Boaler is attacking, and it is not what the IES recommendation describes. Do not claim timed tests “cause” math anxiety as a settled mechanism. Do not claim they are harmless. Report Boaler’s claim, Ashcraft’s working-memory mechanism, and Fuchs’s timed-activity recommendation as a three-part disagreement about what is being timed, in whom, at what stake.

Ashcraft and Elizabeth Kirk, in the *Journal of Experimental Psychology: General* in 2001, measured the working-memory half of that disagreement. High math-anxious college students had smaller working-memory spans especially on computation-based span tasks. Under a letter-load plus two-column mental addition, errors rose sharply in the high-anxiety group, more on carry problems. The authors describe a transitory disruption of working memory, likely central-executive, caused by worry competing as a secondary task. The correlation of math anxiety with computation-span was $r = -.40$ in Experiments 1 and 3. Verbal span showed little or no anxiety-group difference. High-anxious students also took fewer math courses and earned lower math grades — the avoidance path.²¹⁵ Ashcraft’s later reviews treat the effect as beginning to matter more as mathematics becomes working-memory intensive, often cited as middle school onward; that review was not re-opened here.²¹⁶ College samples are not a K–12 base rate. No prevalence of “math anxiety” will be invented in this book.

The maps give the parent dates without giving a timer religion. The Panel: proficient addition and subtraction of whole numbers by the end of grade 3; proficient multiplication and division of whole numbers by the end of grade 5.²¹⁷ Common Core: by the end of grade 2, know from

early math anxiety; fluency misread as speed. Commentary and working paper, not a journal experiment.

²¹³*Fluency Without Fear* labelled working paper / Youcubed publication. Empirical core cited: Ashcraft working-memory disruption; Beilock group; classroom reports.

²¹⁴Grimes (2024); Corey Peltier et al., “We’re Talking about Practice: A Meta-Analysis of Timed Math Practice,” OSF preprint (2025), <https://doi.org/10.35542/osf.io/9bjtg>. Labelled preprint. The 27 IES studies generally bundled timing with other intervention components.

²¹⁵Mark H. Ashcraft and Elizabeth P. Kirk, “The Relationships among Working Memory, Math Anxiety, and Performance,” *Journal of Experimental Psychology: General* 130, no. 2 (2001): 224–37. Computation-span $r = -.40$, Experiments 1 and 3. Full 2001 PDF not opened; coefficient from Ashcraft’s later quotation of those experiments.

²¹⁶Mark H. Ashcraft and Jeremy A. Krause (2007) later reviews, often cited for middle-school onset as mathematics becomes more working-memory intensive. Not re-opened this pass.

²¹⁷National Mathematics Advisory Panel, *Foundations for Success*, Table 2: proficient +/- of whole numbers by end of Grade 3; proficient $\times \div$ of whole numbers by end of Grade 5.

memory all sums of two one-digit numbers (2.OA.2); by the end of grade 3, know from memory all products of two one-digit numbers (3.OA.7).²¹⁸ Grades 1–2 are the strategy years — counting on, making ten, decomposing, the relationship between addition and subtraction — not the “they should already be automatic” years for all facts. Chapter 4 will keep that distinction. Chapter 5 will retrieve multiplication and division as a fact family, interleaved, once both exist. Facts are not a large-language-model job. A model will happily hallucinate 7×8 . Chapter 3 will forbid the model from sitting between the child and the fact; the model may generate the oral script or the shuffled deck, and the parent’s ears do the drill. That sentence is a plant, not an AI chapter.

For the parent, the kitchen rule that survives both objects: retrieve known facts often, sometimes against a clock, after the child can already get them right untimed. Mix the facts. Graph the child’s own score if that helps the child see progress. Do not make the kitchen timer the identity of being good at math. Do not hand a child who still counts all a page of fifty and a clock. Frustration that comes from missing facts is not productive struggle. It is a fluency problem, and the next section will say why that matters for the other kind of struggle.

The two objects stay two objects in every later chapter that could mash them. Chapter 4: retrieve known sums for a few minutes, mixed, once they are known; do not make a fifty-fact test the identity of grade-1 math. Chapter 5: the same rule for products, interleaved with division as soon as both exist. Chapter 6 and after: retrieve the facts inside fraction arithmetic and inside first equations, because a child who still counts on for 8×7 cannot hold a proportion in working memory. The timer, if it is used, is a brief, private, already-known retrieval. The identity of being good at math is five strands, not a kitchen clock.

The isolated effect of the timer remains open. The 2021 recommendation is bundled. The 2025 preprint is the attempt to unbundle. This book will not pretend the unbundling is finished. A parent who wants a research-based reason to retrieve known facts already has Dunlosky’s high-utility rating for practice testing, which does not require a clock. A parent who wants a research-based reason not to rank first-graders with a Mad Minute already has Boaler’s object, Ashcraft’s mechanism, and Fuchs’s own constraint — time only what is already known untimed. Those reasons do not cancel each other. They name two practices.

2.7 Struggle vs rescue: do not mash NMAP 27 with Kapur

The false war is whether mathematics should be teacher-directed or student-centered. The National Mathematics Advisory Panel already refused it. Finding 23: high-quality research does not support exclusive use of either all-encompassing teacher-directed or all-encompassing student-centered prescriptions.²¹⁹ Finding 27: explicit instruction — clear models, extensive practice, think-alouds, extensive feedback — has consistently positive effects for students with mathematical difficulties and for the lowest third of a typical class, on word problems and computation. The same finding adds the clause the slogan-makers drop: this “does not mean that all of a student’s mathematics instruction should be delivered in an explicit fashion.”²²⁰ Later evidence did not resurrect the war. What is

²¹⁸Common Core State Standards for Mathematics, 2.OA.2 and 3.OA.7.

²¹⁹National Mathematics Advisory Panel, *Foundations for Success*, Finding 23.

²²⁰National Mathematics Advisory Panel, *Foundations for Success*, Finding 27, including the clause that this does not mean all instruction should be explicit.

supported is explicit modeling and systematic design for students who are struggling; generation-before-instruction for some new concepts when students have usable priors and will receive the instruction after; comparison of methods; not exclusive fidelity to either slogan.

Manu Kapur's productive failure is the paper inquiry advocates are asked to produce, and it is not Finding 27. The 2014 *Cognitive Science* paper reports two randomized studies of problem-solving before instruction versus instruction before problem-solving. Both sequences produced high procedural knowledge. Productive failure produced significantly greater conceptual understanding and transfer. In the second study, learning from peers' failed attempts outperformed instruction-first but not own-generation. The number of student-generated solutions predicted learning.²²¹ Kapur's productive failure is a designed sequence: an ill-structured problem, generation of suboptimal solutions, then instruction that compares those solutions to the canonical method. It is not "let them be confused all week." It is also not the Panel's struggling-student finding. These two literatures are not the same comparison. Do not use Kapur to cancel Finding 27. A parent can use twenty minutes of "try this with what you already know; we will sort it after" on a new concept the child has priors for, then teach. That is productive failure. A parent who never models for a child who is already behind is not doing productive failure and is contradicting Fuchs and colleagues' 2021 Recommendation 1 and Finding 27.

James Hiebert and Douglas Grouws, in their 2007 handbook chapter, named two features of teaching that support both skill and conceptual understanding: explicit attention to connections among facts, procedures, and ideas, and students' struggle with important mathematics. "Struggle" here is intellectual effort to make sense, not abandonment.²²² Paul Kirschner, Sweller, and Richard Clark, in *Educational Psychologist* in 2006, "Why Minimal Guidance During Instruction Does Not Work," is a review and position paper from cognitive architecture: novices lack the schemas that make unguided problem solving efficient. It is not a new experiment. It is the paper inquiry advocates are asked to answer, and the paper Kapur explicitly contrasts with productive failure.²²³ A parent who treats Kirschner, Sweller, and Clark as "lecture works" has not read the architecture argument. A parent who treats Kapur as a year of unguided discovery has not read the designed sequence.

Jean Stockard, Timothy Wood, Cristy Coughlin, and Caitlin Rasplia Khoury's 2018 *Review of Educational Research* meta-analysis of Direct Instruction curricula — literature 1966–2016, 328 studies, 413 designs, almost 4,000 effects — estimated an effect for the total sample of 0.60, 95 percent confidence interval [0.54, 0.66]. All estimated academic effects were positive and significant; affective outcomes were not significant in metaregressions. Effects were larger with more exposure; little decline at maintenance.²²⁴ This is a meta-analysis of a named family of curricula

²²¹Kapur, "Productive Failure" (2014), two randomized studies; comparable procedural knowledge; greater conceptual understanding and transfer; number of generated solutions predicted outcomes. Study 2: peers' failed attempts outperformed instruction-first but not own-generation.

²²²James Hiebert and Douglas A. Grouws, "The Effects of Classroom Mathematics Teaching on Students' Learning," in *Second Handbook of Research on Mathematics Teaching and Learning*, ed. Frank K. Lester (Charlotte, NC: Information Age / NCTM, 2007). Struggle as intellectual effort to make sense, not abandonment. Full chapter not opened; construct from Kapur 2014 and Rittle-Johnson 2015 citations.

²²³Paul A. Kirschner, John Sweller, and Richard E. Clark, "Why Minimal Guidance During Instruction Does Not Work," *Educational Psychologist* 41, no. 2 (2006): 75–86. Position/review paper, not a new experiment. Contrasted with Kapur 2014 in that paper.

²²⁴Jean Stockard, Timothy W. Wood, Cristy Coughlin, and Caitlin Rasplia Khoury, "The Effectiveness of Direct

(Direct Instruction / DISTAR / later Engelmann programs), not of “any teacher talking.” It is also produced by authors long associated with Direct Instruction. Report the number. Do not launder it into “lecture works.” Project Follow Through, the late-1960s–1970s comparison of more than twenty early-elementary interventions in high-poverty sites, is the ancestor later reviewers describe as showing Direct Instruction (DISTAR, Engelmann and Becker) as the only model with significantly positive impacts across outcome measures. The original Abt / Stebbins evaluation tables were not re-opened for this project. Those tables remain **UNVERIFIED** here.²²⁵ The political afterlife of Follow Through is contested. This book will not recycle blog versions.

The 2021 IES intervention guide’s Recommendation 1 is the current synthesis for students already behind: systematic instruction, strong evidence, 43 studies.²²⁶ Recommendation 2, mathematical language, strong, 16 studies. Recommendation 3, representations, 28. Recommendation 4, number lines, 14. Recommendation 5, word problems, 18. Recommendation 6, timed activities, 27.²²⁷ Those study counts are the guide’s own evidence ratings, not a 2026 census of homeschool kitchens. Evidence ratings in IES guides mean: strong is consistent What Works Clearinghouse–standard studies in diverse samples; moderate is some such evidence with ambiguity; minimal is theory or thin evidence, still included by the panel. A “minimal” rating is not a finding of ineffectiveness.²²⁸ The 2013 IES guide *Teaching Math to Young Children* (Douglas Frye, Baroody, Margaret Burchinal, Sharon Carver, Nancy Jordan, and Judy McDowell), for ages 3–6, recommends teaching number and operations, and geometry, measurement, and data, along a developmental progression; progress-monitoring; talking about the world mathematically; dedicating daily time.²²⁹ Those are not 2026 randomized trials of homeschool kitchens. They are the early-childhood synthesis later chapters will steal from without upgrading.

The parent-scale rule that does not overclaim sits between the two slogans. If the child has the prerequisites and the task is a new idea, ten to twenty minutes of generation — Kapur — then a clear model. If the child is missing prerequisites or is in the lowest third on this topic, model first, then guided practice, then independent — Finding 27; Fuchs Recommendation 1. Frustration that comes from missing facts is not productive; it is a fluency problem, from the last section. Frustration that comes from a well-posed new idea is the Hiebert kind. “Never let them be frustrated” is parenting folklore and contradicts Hiebert, Kapur, and the Panel’s effort-belief finding. “Struggle means I should not help” is the opposite folklore and contradicts the struggling-student literature.

Instruction Curricula: A Meta-Analysis of a Half Century of Research,” *Review of Educational Research* 88, no. 4 (2018): 479–507. Total-sample effect 0.60, 95% CI [0.54, 0.66]; 328 studies. Named DI curriculum family, not “lecture works.” Authors associated with DI.

²²⁵Project Follow Through original Abt / Stebbins evaluation tables: not re-opened. **UNVERIFIED** as to original effect-size tables. Later reviewers (Adams 1996; Bereiter and Kurland 1996; Engelmann, Becker, Carnine, and Gersten 1988; Kennedy 1978) as cited by Stockard et al. 2018. Political afterlife contested (Watkins 1996; Grossen 1996); blog versions not recycled.

²²⁶Fuchs et al., *Assisting Students Struggling* (2021), Recommendation 1, systematic instruction, strong evidence, 43 studies.

²²⁷Fuchs et al., *Assisting Students Struggling* (2021), Recommendations 1–6 study counts: 43 / 16 / 28 / 14 / 18 / 27, taken from the guide.

²²⁸IES evidence-rating definitions as stated in the guides. A “minimal” rating is not a finding of ineffectiveness; Frye et al. 2013, p. 4, as quoted in the REL toolkit.

²²⁹Douglas Frye, Arthur J. Baroody, Margaret Burchinal, Sharon M. Carver, Nancy C. Jordan, and Judy McDowell, *Teaching Math to Young Children*, NCEE 2014-4005 (Washington, DC: Institute of Education Sciences, 2013), ages 3–6. Not a 2026 kitchen RCT.

A short factual note on specific learning disorder belongs here so it is not ghettoed into a later chapter the anxious parent skips. The *Diagnostic and Statistical Manual of Mental Disorders*, fifth edition, folded the old separate “mathematics disorder” into specific learning disorder with specifiers. The mathematics specifier covers persistent difficulty in number sense, memorization of arithmetic facts, accurate or fluent calculation, and/or accurate math reasoning. Criteria, paraphrased, not as a diagnostic manual: persistent difficulty for at least six months despite targeted intervention; skills substantially and quantifiably below age expectations, confirmed by individually administered standardized measures and clinical synthesis; interference with academic or occupational performance or daily life; not better explained by intellectual disability, uncorrected sensory impairment, other mental or neurological disorders, psychosocial adversity, language difference, or lack of adequate instruction. The old IQ-achievement discrepancy requirement of DSM-IV was dropped.²³⁰ DSM-5 estimates specific learning disorders as a class — reading plus writing plus mathematics — at 5–15 percent of school-age children. That is not a dyscalculia base rate.²³¹ Butterworth and others have used stricter, mathematics-specific cuts. One primary-school study applying DSM-5 SLDM criteria (Kinga Morsanyi, Jemma van Bers, Teresa McCormack, and Juhani McGourty) reported 6 percent with persistent SLDM in a sample of 2,421. That is one sample. It is not a universal prevalence.²³² No invented prevalence. A parent who sees persistent, intervention-resistant difficulty with number sense and facts, with otherwise adequate instruction, should seek a comprehensive evaluation — not a blog quiz. Brian Butterworth’s program argues for a core deficit in numerosity coding, distinct from general cognitive ability. David Geary’s mathematical-learning-disability work emphasizes working-memory and retrieval deficits and a distinction between persistent MLD and low achievement. These are research programs, not competing diagnoses a parent can apply at home.²³³ What the intervention guides say to do does not wait for a label. Fuchs 2021 and Gersten 2009 are written for students with or at risk for mathematics disabilities and for the lowest performers. Systematic instruction, language, representations, number line, schema word problems, and brief timed retrieval of known facts are the moves. If the parent is the ceiling for a dyscalculic learner, buy expertise rather than an AI workaround. Chapter 3 will not offer the workaround.

Twice-exceptional learners — gifted plus a specific learning disorder or ADHD — are a real clinical description. This book will not invent a prevalence or a protocol. The Nation Empowered / Iowa Acceleration Scale literature (Susan Assouline, Nicholas Colangelo, Joyce VanTassel-Baska, and colleagues) is the usual gifted-acceleration source; it was not opened here, and its effect sizes remain **UNVERIFIED**.²³⁴ The practical implication that does not overclaim: accelerate in the strand that is ready. A child can be ready for algebraic structure and still need fraction intervention. Do

²³⁰American Psychiatric Association, *Diagnostic and Statistical Manual of Mental Disorders*, 5th ed. (Arlington, VA: American Psychiatric Publishing, 2013), specific learning disorder with impairment in mathematics. Criteria paraphrased from APA public page and Butterworth 2019 preview; not a diagnostic manual reprint.

²³¹DSM-5 estimate of specific learning disorders as a class (reading + writing + mathematics) at 5–15 percent of school-age children. Not a dyscalculia base rate.

²³²Kinga Morsanyi, Jemma H. C. van Bers, Teresa McCormack, and Juhani McGourty, primary-school sample of 2,421 applying DSM-5 SLDM criteria, 6 percent persistent SLDM. One sample, not a universal prevalence. *British Journal of Psychology* lineage.

²³³Brian Butterworth, *The Mathematical Brain / Dyscalculia: From Science to Education* (2019): core deficit in numerosity coding. David C. Geary, MLD program: working-memory and retrieval deficits; persistent MLD versus low achievement. Research programs, not home diagnoses. Geary served on NMAP.

²³⁴Susan G. Assouline, Nicholas Colangelo, Joyce VanTassel-Baska, and colleagues, Nation Empowered / Iowa Acceleration Scale literature. Not opened this pass; effect sizes **UNVERIFIED**.

not use giftedness as a reason to skip the Siegler foundations. Finding 30 is the acceleration warrant, with the motivation-and-prerequisites clause: mathematically gifted students with motivation “appear to be able to learn mathematics much faster... with no harm to their learning, and should be allowed to do so.”²³⁵ Finding 15 is the age-lock refusal, already printed in the opening: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; claims that children of a given age cannot learn certain content because they are “too young,” “not in the appropriate stage,” or “not ready” have “consistently been shown to be wrong.” Nor are claims justified that children cannot learn particular ideas because their brains are “insufficiently developed,” even if they possess the prerequisite knowledge for learning the ideas.²³⁶ Accelerate a child who has the prerequisites. Do not withhold fractions from a nine-year-old who has them. Do not accelerate a child who does not.

English learners in a home where the curriculum is written in a language the child is still learning have a language load on top of the math load. Fuchs 2021 Recommendation 2 is about mathematical language even for monolingual students. Shared, precise vocabulary — say “regroup,” not “borrow”; “greater than,” not “bigger”; “commutative,” not “flip-flop” — is the cheap move. Do not translate away the mathematics into informal English and then expect the child to read a textbook. An English-learner-specific mathematics randomized trial was not opened for this project. Effect sizes remain **UNVERIFIED**.²³⁷

Confrey, Maloney, and Nguyen’s 2008 critique of the Panel is still on the table. Biased literature selection. Inconsistent standards.²³⁸ Report it. The Panel is still the synthesis the 2009, 2010, 2013, and 2021 IES guides rest on for algebra foundations, fractions as numbers, explicit instruction for the struggling, and the refusal of exclusive fidelity. A book that cites only the critique has discarded the document later researchers still use. A book that cites only the Panel has hidden a serious methodological fight. This chapter reports both and then uses the findings later documents still use.

The struggle-versus-rescue fight is also a voice fight in the kitchen. The parent who stopped at algebra often rescues too fast because the child’s stuckness feels like the parent’s old stuckness. Ashcraft and Kirk’s working-memory account is why the rescue feels urgent: worry is already occupying the executive, and the child’s pause looks like danger. Finding 27 is why the rescue, for a child already behind, should be a model and guided practice, not a takeover of the pencil. Kapur is why the rescue, for a new idea with priors, should wait twenty minutes. Hiebert is why some intellectual effort is the point. The same adult has to hold all three without turning any of them into a personality. Preparation is what makes that possible. An unprepared parent has only one move: take the pencil.

The parent is the daily instructor. The pencil is the last thing to take.

²³⁵National Mathematics Advisory Panel, *Foundations for Success*, Finding 30.

²³⁶National Mathematics Advisory Panel, *Foundations for Success*, Finding 15.

²³⁷English-learner-specific mathematics RCT: not opened. Effect sizes **UNVERIFIED**. Fuchs et al. 2021 Recommendation 2 (mathematical language) is the monolingual-inclusive synthesis used instead.

²³⁸Confrey, Maloney, and Nguyen, “Breaching the Conditions for Success” (2008). See n. 128.

2.8 Parent-as-teacher: hear a wrong answer; kitchen-math is not algebra

The thesis of this book has three conditions. The parent understands the concept well enough to hear a wrong answer. The child does the struggle before the tool. Any agent is a supplement hosted and constrained by the parent. The second and third conditions are Chapter 3. This section is the first condition, converted into a kitchen rule, without inventing a parent-as-sole-teacher randomized trial.

Maloney, Gerardo Ramirez, Elizabeth Gunderson, Susan Levine, and Beilock, in *Psychological Science* in 2015, found that first- and second-graders of more math-anxious parents learned less math over the school year and had more math anxiety at year's end — but only if those parents reported frequent homework help. When help was infrequent, parent anxiety was unrelated to child outcomes. The effect was specific to math, not reading.²³⁹ The authors write that a genetic component is possible, but the moderation by homework help points to a social and instructional mechanism. A 2021 follow-up by Fraulein Retanal, Johnston, Di Lonardo Burr, Storozuk, DiStefano, and Maloney, in *Education Sciences*, found that a controlling-supportive homework-helping style partially mediated the parent-anxiety / child-achievement relation in a survey of parents of 11- to 14-year-olds.²⁴⁰ Beilock, Gunderson, Ramirez, and Levine, 2010, showed that teacher math anxiety predicted first- and second-grade girls' math achievement over the year, and gender-ability beliefs; that paper was fetched at secondary level, and its coefficients remain **UNVERIFIED** here.²⁴¹ The parent analogue is the 2015 study.

A homeschool parent who is the daily instructor cannot “help less often” the way a school parent can. The “help less often” clause of Maloney and colleagues is not available as a daily strategy. The risk is therefore higher, not lower, than in a school household where a math-anxious parent can step back from homework. The mechanism to watch is not “parent is bad at math.” It is parent anxiety leaking as controlling help, speed-pressure, or “I’m not a math person, and you take after me.” The constructive response the literature actually supports is: the parent learns the day’s concept well enough to stay out of the child’s working memory — prepare before the lesson; use a worked example you already understand — the child generates; the parent does not narrate worry. Do not treat the finding as “anxious parents should not teach.” Replication status of Maloney 2015 is open. Some later work has been mixed on the exact interaction; no registered replication was opened for this project. Flag it as a single large, well-cited field study plus a 2021 mediation follow-up.²⁴² There is no fetched experimental literature on math-anxious parents as full-time instructors. That is the highest-priority homeschool-specific gap this chapter leaves standing.²⁴³ Video and

²³⁹Maloney, Ramirez, Gunderson, Levine, and Beilock, “Intergenerational Effects” (2015). Frequent-help moderation; math-specific.

²⁴⁰Fraulein Retanal et al., “Controlling-Supportive Homework Help Partially Explains the Relation between Parents’ Math Anxiety and Children’s Math Achievement,” *Education Sciences* 11, no. 10 (2021): 620. Survey of parents of ages 11–14; path analysis; partial mediator.

²⁴¹Sian L. Beilock, Elizabeth A. Gunderson, Gerardo Ramirez, and Susan C. Levine, “Female Teachers’ Math Anxiety Affects Girls’ Math Achievement,” *Proceedings of the National Academy of Sciences* (2010). Fetched at secondary level; coefficients **UNVERIFIED** this pass.

²⁴²Replication status of Maloney et al. 2015: some later work mixed on the exact interaction; no registered replication opened. Single large, well-cited field study plus the 2021 mediation follow-up.

²⁴³Parent-as-sole-teacher experimental literature: none fetched. Highest-priority homeschool-specific gap. Maloney is school homework, not full-time instruction.

self-teaching curricula exist because of this constraint. So do tutors, co-ops, and dual enrollment when the parent is the ceiling. Those are a normal high-school plan, not a failure of the kitchen. Teaching Textbooks' vendor claim that the program "does 100 percent of the teaching" is relevant to Maloney, not a finding that the parent can stop hearing wrong answers.²⁴⁴

Ashcraft and Kirk's $r = -.40$ on computation-span is the adult half of the same working-memory story. Worry occupies the central executive. A parent who is anxious and narrating, while the child is holding a two-step equation in mind, is a second task in the child's working memory. Prepare so you can hear a wrong answer. Then do not take the pencil. The parent who takes over the pencil is the high-help, high-anxiety path.

Generation is the finding that ties several literatures already printed. Retrieval is generation (Dunlosky and colleagues). Self-explanation is generation (Rittle-Johnson 2006; Renkl 1997). Completion problems in fading are generation of the missing steps (Renkl and Atkinson). Productive failure is generation of even wrong solutions, and the number of generated solutions predicted learning (Kapur 2014). Think-alouds in explicit instruction are the child talking, not only the parent (Finding 27; Fuchs Recommendation 1). Maloney and colleagues: the parent who takes over the pencil is the high-help path. The parent-explanation-quality rule is therefore: prepare so that you can hear a wrong answer. Then do not give the next step. Ask the child to show, on a representation or a fading example, what they think the symbol means. If they cannot start, give a worked example and have them explain it. Watching the parent do twenty problems is the novice condition in which Sweller says they should be studying a worked example — but they must study it, not watch a live performance.

The equal sign is already the arithmetic root of algebra, and it is a wrong answer the parent who stopped at algebra can learn to hear this week. Nicole McNeil and Alibali, in *Child Development* in 2005, and McNeil's 2014 *Child Development Perspectives* review, describe three entrenched operational patterns from typical U.S. arithmetic: perform all given operations on all given numbers; the "operations = answer" visual structure, all operations left, blank right; "=" means "the total." Children then fail $3 + 4 + 5 = 3 + \underline{\quad}$ by adding all (answer 15) or adding to the equal sign (answer 12).²⁴⁵ Knuth, Ana Stephens, McNeil, and Alibali, 2006, connect a relational view of "=" to later algebra.²⁴⁶ Baroody and Ginsburg 1983 and Kieran 1981 are the older sources. Children in some other countries, where early instruction does not hammer operations-equals-answer, produce fewer of these errors (Li, Ding, Capraro, and Capraro 2008, as cited by McNeil).²⁴⁷ McNeil, Emily Fyfe, Lori Petersen, April Dunwiddie, and Heather Brletic-Shipley, 2011, and McNeil, Fyfe, and Dunwiddie 2014: practicing $4 = 2 + 2$ formats improved equivalence.²⁴⁸ Chapter 4 will write

²⁴⁴Teaching Textbooks vendor claim that the program "does 100% of the teaching." Vendor copy, labelled vendor. Relevant to the Maloney constraint; not a finding that the parent can stop hearing wrong answers.

²⁴⁵Nicole M. McNeil and Martha W. Alibali, "Why Won't You Change Your Mind? Knowledge of Operational Patterns Hinders Learning and Performance on Equations," *Child Development* 76 (2005): 883–99. Nicole M. McNeil, "A Change-Resistance Account of Children's Difficulties Understanding Mathematical Equivalence," *Child Development Perspectives* (2014).

²⁴⁶Knuth, Stephens, McNeil, and Alibali, "Does Understanding the Equal Sign Matter?" (2006).

²⁴⁷Li, Ding, Capraro, and Capraro (2008), as cited by McNeil, "Change-Resistance Account" (2014). Cross-national comparison of operational-equal-sign errors.

²⁴⁸Nicole M. McNeil, Emily R. Fyfe, Lori A. Petersen, April E. Dunwiddie, and Heather Brletic-Shipley, "Benefits of Practicing $4 = 2 + 2$: Nontraditional Problem Formats Facilitate Children's Understanding of Mathematical

non-canonical equations early: $8 = 3 + 5$, $3 + 5 = 2 + 6$, $6 + 2 = 8$. The parent who only ever writes $3 + 4 = 7$ is training the operator reading. That is a wrong answer waiting for Algebra I. Hear it in grade 1.

Counting is the earlier wrong answer. Rochel Gelman and C. R. Gallistel, in *The Child's Understanding of Number* (1978), described five counting principles. The three how-to-count principles: one-to-one, each item gets exactly one tag; stable order, the tags are always in the same sequence; cardinal, the last tag names the numerosity of the set. The two what-to-count principles: abstraction, any discrete collection can be counted; order-irrelevance, the order of tagging does not change cardinality.²⁴⁹ Gelman and Gallistel originally argued for a principles-first view even in two-year-olds. Later work — for example Sarnecka and Carey 2008, and Bermejo and colleagues 2004, as cited in the Stanford counting-principles brief — shows that producing the last word is not the same as knowing it names the set. A child who recounts when asked “how many?” has not yet got cardinality.²⁵⁰ After a count, ask “how many?” without recounting. Hide the set and ask again. Count sets, not just the recitation. Kindergarten entry knowledge predicts later mathematics for years — Panel Finding 8, and Greg Duncan and colleagues 2007 as cited by Siegler and colleagues 2012.²⁵¹ Mix, Levine, and colleagues’ number-talk program is the usual citation for parent number-word knowledge and caregiver talk. Those papers were fetched at abstract and secondary level; exact coefficients remain **UNVERIFIED** here.²⁵² The homeschool-relevant fact that is well established in the sources that were opened is the kindergarten-entry prediction, not a number-talk *d*.

Dehaene’s approximate number system — *The Number Sense*, and Dehaene, Piazza, Pinel, and Cohen 2003 on the intraparietal sulcus — is the cognitive-neuroscience account of an approximate magnitude system later mapped onto symbols. The book was not re-opened for this project. The instructional claim that survives without the neuroimaging: children need both an approximate sense of magnitude (“about 80, not about 8”) and exact symbolic mapping. Estimation is not a warm-up. It is part of number sense. The Panel’s Critical Foundations include estimating orders of magnitude inside whole-number fluency.²⁵³

Kitchen-math, again, because the homeschool temptation is to treat the last three paragraphs as a license to replace the program with cooking, money, and building. Finding 26 still holds. Real-

Equivalence,” *Child Development* 82 (2011): 1620–33. McNeil, Fyfe, and Dunwiddie (2014).

²⁴⁹Rochel Gelman and C. R. Gallistel, *The Child's Understanding of Number* (Cambridge, MA: Harvard University Press, 1978). Five counting principles from later papers that quote the 1978 text (Sarnecka and Carey 2008; Stanford counting-principles brief). Book itself not re-opened.

²⁵⁰Barbara W. Sarnecka and Susan Carey (2008); Vicente Bermejo and colleagues (2004), as cited in the Stanford counting-principles brief: producing the last word is not cardinality. Starting a count at “three” still produced “seven” as the “how many,” in the Bermejo report as cited.

²⁵¹National Mathematics Advisory Panel, *Foundations for Success*, Finding 8; Greg J. Duncan et al., “School Readiness and Later Achievement,” *Developmental Psychology* 43 (2007): 1428–46, as cited by Siegler et al., “Early Predictors.” Full Duncan PDF not opened.

²⁵²Mix, Levine, and colleagues, early-number and number-talk program (including Mix, Huttenlocher, and Levine 2002; Gunderson and Levine 2011; Levine, Suriyakham, Rowe, Huttenlocher, and Gunderson 2010). Fetched at abstract/secondary level; exact coefficients **UNVERIFIED**.

²⁵³Stanislas Dehaene, *The Number Sense* (1997/2011); Dehaene, Piazza, Pinel, and Cohen (2003), intraparietal sulcus / approximate number system. Book not re-opened. Instructional claim used without neuroimaging: approximate magnitude plus exact symbolic mapping. National Mathematics Advisory Panel, Critical Foundations, whole-number fluency includes estimating orders of magnitude.

world contexts help similar real-world items. They do not, in the Panel’s small high-quality set, raise computation or equation solving. A child who can double a recipe and cannot place $3/2$ on a line has performed, not learned. Unschooling — John Holt’s tradition; 12 percent “mostly or strictly informal” in NCES PFI 2016 — is a real minority practice, not the market majority.²⁵⁴ It is a poor match for NCAA core-course worksheets, New York individualized home instruction plans, and Pennsylvania portfolios unless the parent translates lived math into a course title after the fact. This book’s thesis is closer to structured and mastery. Say so without scorning the 12 percent. Finding 26 is the transfer caution either way. Informal activity can be a second representation. It is not the line, the schema, or the retrieval.

The equal-sign and counting paragraphs are not a tour of grades 1–2. They are the parent-as-teacher test. If the adult can hear 15 and 12 as operational-equal-sign answers, and can hear a recount as a cardinality miss, the first thesis condition is already in use. If the adult cannot hear those, no later chapter will be able to hear a variable used as a single mystery number, or a minus treated only as “subtract,” or a limit treated as “plugging in.” Booth and colleagues built incorrect examples for those later objects. The hearing starts now.

NMAP Finding 15, printed in the opening, is the last refusal this chapter owes the later ones. Grade labels are sequence anchors, not prison bars. Chapter 1 already printed the NCES 2016 stretch: Algebra I in the 6–8 homeschool band at 41 percent that year, arithmetic still in 9–12 at 31 percent. A nine-year-old who has the prerequisites for fraction-as-number work should not be told she is “too young.” A twelve-year-old who does not have those prerequisites should not be accelerated into a course titled Algebra I because a vendor book says grade 8. Finding 30 allows the motivated child with the foundations to move faster. Siegler and colleagues are why the foundations include fractions. The method lock and the age-lock refusal are the same sentence: teach what the prior opportunity has prepared, not what the birthday suggests.

Algebra as generalized arithmetic is the view the later high-school chapters will need and that this method chapter can already name. The letters stand for numbers, and the properties already known for numbers still hold. Booth, Knuth, and Küchemann document the gap: many students treat a variable as a single unknown object, not as a symbol that can take many values. The equal-sign misconception is the arithmetic root of that gap. Function concept is a correspondence, later a set of ordered pairs, later a rule, later a graph. Slope and intercept mixups — taking b as slope because it is the first number the child sees in $y = mx + b$, or reading rise over run off the axes rather than the line — are pervasive in secondary teaching lore. A named prevalence study was not opened here; that prevalence remains **UNVERIFIED**.²⁵⁵ Rohrer, Dedrick, and Burgess treated “graph $y = mx + b$ ” and “slope from two points” as different problem kinds that students confuse when practice is blocked — which is itself evidence that the association from this story to this strategy is weak. Proof and spatial reasoning sit in the same lock without a homeschool experiment behind them. Geometry is not only a vehicle for proof; the Panel treats measurement formulas and similar

²⁵⁴National Center for Education Statistics, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey* (NCES 2020-001), 12 percent of homeschoolers “mostly or strictly informal learning”; 77 percent mostly or strictly formal curriculum. Unschooling as a named minority, not the market majority.

²⁵⁵Slope/intercept mixup prevalence: no named prevalence study opened. **UNVERIFIED** as to a prevalence coefficient. Rohrer, Dedrick, and Burgess (2014) treated the two problem kinds as ones students fail to associate when practice is blocked — used as evidence of a weak story-to-strategy association, not as a prevalence census.

triangles as algebra foundations. Spatial skill's longitudinal prediction of later STEM achievement is usually cited to Wai, Lubinski, and Benbow 2009; that paper was not opened, and its coefficients remain **UNVERIFIED**.²⁵⁶ High-school proof is a new register: definitions, given and prove, the difference between a drawing and a theorem. No homeschool-specific experimental literature on teaching proof was found. The move this chapter can still lock is the one already named: a worked proof with fading, then “what is wrong with this proof,” which is Booth and Sweller applied to arguments, not a geometry-proof randomized trial. Label it as such when Chapter 8 uses it.

The method this chapter locks for Chapters 4 through 11 is now a short list. Iterate conceptual and procedural knowledge; do not wait for full understanding before any practice, and do not drill a procedure with zero talk about why. Hear a wrong answer against five interwoven strands; ranking them is misguided. Treat fractions, including decimals, percents, and signed fractions, as the gate to algebra — a prediction, not a curriculum experiment; Chapter 5 will teach the gate. Use the number line as the representation that scales; connect concrete representations to symbols in the same visual field, then fade. For novices, a fully worked example, then a second the child explains, then a completion problem, then a conventional problem; include an incorrect example and ask what the symbol was taken to mean; stop showing full examples once the child can execute. After a short blocked introduction, mix yesterday's and last month's problem types; retrieve older facts the same week; give an unannounced mixed quiz without notes. Teach problem types, not magic words; do not quote proximal schema *g* values as general achievement. Retrieve known facts often, sometimes against a clock, after they are already right untimed; do not make the timer the identity of being good at math. Model first for a child already behind; use a short generation-then-instruction sequence for a new concept when priors exist; do not mash those comparisons. Prepare so you can hear a wrong answer; then do not take the pencil. Do not claim kitchen-math transfers to algebra. Accelerate a child who has the prerequisites; do not skip fractions because a child is “good at math.”

What this chapter does not lock is a 180-day plan, a franchise winner, a parent-as-sole-teacher experiment, an isolated timer effect, an interleaving *d* for grades 1–5 or for proof and calculus, an optimal dosage of concepts versus procedures for whole topics, a measured transfer from cooking to formal items, or a dyscalculia prevalence. Those remain open. The next lock is the AI lane — one chapter, not five. The model may prepare the adult and the next problem. It may not do the child's problem. That sentence is Chapter 3's. This chapter's sentence, if the family remembers only one, is shorter. Five strands, not a one-way street. Hear the wrong answer. Do not take the pencil.

²⁵⁶Jonathan Wai, David Lubinski, and Camilla P. Benbow (2009), spatial skill and later STEM achievement. Paper not opened; exact coefficients **UNVERIFIED**.

Chapter 3

The Agent Does Not Take the Test



Figure 4: A closed laptop, a paper exam, and a face-down answer key on a wooden table.

Two numbers from a 2025 field experiment already contain the argument this chapter has to keep in view. Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman randomized nearly a thousand Turkish high-school mathematics students into three conditions and then did the thing a kitchen-table parent can do every afternoon: they took the tool away. On

assisted practice, a ChatGPT-like tutor — they call it GPT Base — raised grades 48 percent relative to a control that used notes and a textbook. On the unaided exam that counted toward the course grade, the same students scored 17 percent *worse* than control. A second, guardrailed tutor — GPT Tutor — raised practice grades 127 percent and then, on the unaided exam, was statistically indistinguishable from the students who had never seen the model (point estimate -0.004). GPT Base, asked for the answer, was fully correct only 51 percent of the time. Students copied anyway. They did not perceive that they had learned less.²⁵⁷

That pair — a practice score that looks like mastery, an unaided score that is the opposite — is not a new finding that happened to arrive with large language models. Nicholas C. Soderstrom and Robert A. Bjork, writing in 2015, had already named the confusion this chapter is written against: the goal of instruction is relatively permanent change, which they call *learning*; what we observe during practice is *performance*, “often an unreliable index” of that change. Conditions that boost performance now (massed practice, open notes, a solver at the elbow) can fail to produce learning. Conditions that depress current performance (spacing, testing, generation, delayed feedback) often produce more of it. People “mistakenly interpret their performance during acquisition as a reliable guide to long-term learning.”²⁵⁸ Bastani is the mathematics-class replication of that error, with a chatbot in the seat that used to be occupied by a worked-example booklet or a well-meaning parent who cannot stand to watch a child struggle.

This book is not a sequel to *Autonomous AI and Education*. That earlier book had five chapters of landscape, detectors, companion-harm litigation, and school-product claims to sort. This book gets *one* chapter on the kitchen condition, then an AI lane in every later mathematics chapter that points back here instead of rewriting it. The load-bearing sentence is already in the thesis: a parent who is not a math specialist can take a child from first-grade number sense through high-school calculus if three things hold — the parent understands the concept well enough to hear a wrong answer; the child does the struggle before the tool; and any agent is a supplement hosted and constrained by the parent, not a substitute for the child’s work and not a school-safe product by default. The sources do not prove that thesis as a causal claim about homeschool mathematics. They do not contradict it. They constrain how it can be attempted. Unguarded solvers raise practice scores and cut unaided exams. No year-plus kitchen-table RCT of a parent-constrained tutor exists in the literature opened on 30 August 2026. The popular story that collapses “the AI will tutor” into a path is the thing this chapter has to refuse before any later chapter names a fraction or a derivative.

²⁵⁷Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman, “Generative AI without guardrails can harm learning: Evidence from high school mathematics,” *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122, <https://doi.org/10.1073/pnas.2422633122>; author PDF, accessed 30 August 2026, https://hamsabastani.github.io/education_llm.pdf. Assisted-practice relative gains: GPT Base +48 percent, GPT Tutor +127 percent. Unassisted exam versus control: GPT Base -17 percent; GPT Tutor point estimate -0.004 , not significant. GPT Base fully correct 51 percent of the time on the practice set when asked for the answer. Students used GPT Base as a crutch and did not perceive the harm. These percentages are of control grades, not Cohen’s d .

²⁵⁸Nicholas C. Soderstrom and Robert A. Bjork, “Learning Versus Performance: An Integrative Review,” *Perspectives on Psychological Science* 10, no. 2 (2015), <https://doi.org/10.1177/1745691615569000>; author PDF, accessed 30 August 2026, https://bjorklab.psych.ucla.edu/wp-content/uploads/sites/13/2016/11/soderstorm_ra_learningvsperformance.pdf. The quoted clauses — performance as “often an unreliable index” of learning, and people “mistakenly interpret[ing] their performance during acquisition as a reliable guide to long-term learning” — are from that review. Do not invent a snappier paraphrase as if it were the quote.

What follows is therefore not a product tour and not a second education monograph. Section 3.1 installs Bastani and Soderstrom and Bjork before any logo is named as a helper. Section 3.2 names the landscape so the later chapters cannot collapse an agent, a chatbot, a homework solver, a constrained tutor, and an adaptive practice engine into one word. Section 3.3 prints the ban-versus-allow list the rest of the book will point at. Section 3.4 gives six parent workflows, labelled as practice suggestions, not as a homeschool RCT. Section 3.5 keeps Khanmigo's Tennessee intention-to-treat result from becoming Bloom's two-sigma tutor, and keeps a Harvard physics custom-tutor trial from becoming the kitchen default. Section 3.6 installs the age-and-law constraints — COPPA's 2025 amendments, California A.B. 1159 as a bill still on the Senate floor on 30 August 2026, detectors that fail, companion-harm cases that are allegations — without writing a fifty-state legal manual. Section 3.7 splits what the model is actually good at from what it is not: GSM8K saturation is not a geometry proof; flash facts are not a language-model job; an AP free-response question is unaided work; Wolfram|Alpha is a computer-algebra check, not an explainer. Section 3.8 closes on a session shape and a cheap constrained stack so every later AI lane can reuse a rhythm instead of a pitch. One sentence if the family remembers only one: the model may prepare the adult and the next problem; it may not do the child's problem.

3.1 Performance is not learning; Bastani is the math-class replication

Soderstrom and Bjork's review is the document to put on the table before any vendor page. The paper is an integrative review in *Perspectives on Psychological Science*, not a tutoring trial. Its claim is conceptual and, for a parent, operational. Instruction aims at a change that is relatively permanent and available later, in a new context, without the supports that were present during practice. What a parent sees at 4:15 p.m. — a worksheet filling in, a chatbot producing steps, a child nodding — is performance. Performance is not a fraud. It is also not the thing the parent is trying to buy. The review's useful cruelty is the list of conditions that make performance look good while learning stays thin: massed practice, continuous feedback, open notes, a model at the elbow that removes the need to generate the next step. The useful mercy is the complementary list: spacing, testing, generation, delayed feedback. Those conditions often look worse in the moment. They are the ones that tend to produce the change that is still there on Thursday.²⁵⁹

A homeschool afternoon is built to confuse the two. The child is in the room. The parent can see the page. A tool that completes the page is a tool that ends the afternoon. Soderstrom and Bjork's sentence about people mistaking acquisition performance for a guide to long-term learning is not a metaphor for that afternoon. It is that afternoon. Bastani's students, after using GPT Base, did not think they had learned less. The metacognitive error survived a randomized trial with a graded exam. A parent who treats a finished Saxon lesson as evidence of learning is making the same error with a smaller sample.

The Bastani paper is “Generative AI without guardrails can harm learning: Evidence from high school mathematics,” *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122. The design, from the author-hosted PDF opened on 30 August 2026, is

²⁵⁹Soderstrom and Bjork, “Learning Versus Performance.” Conditions that boost current performance (massed practice, open notes, continuous feedback, a solver at the elbow) can fail to produce learning; spacing, testing, generation, and delayed feedback often look worse in the moment and produce more learning. The review is conceptual, not a tutoring RCT.

a classroom-level randomized controlled trial in a large high school in Turkey, Fall 2023–24, roughly a thousand students in grades 9–11, four ninety-minute sessions covering about 15 percent of a semester’s mathematics. The model is GPT-4-0613. Three arms. GPT Base is a ChatGPT-like tutor prompt plus the problem: the student can ask for the answer and, in practice, does. GPT Tutor is hints only. The hidden prompt contains teacher-written solutions and common-mistake hints and an instruction that the model is in no circumstances to provide the student with the full solution. Control is notes and a textbook. Practice is assisted. The exam is closed-book, paired to the practice items, and counts toward the grade.²⁶⁰

Quote the comparison correctly. Do not mash it with a later paper’s standard deviations.

On assisted practice, relative to control, GPT Base improved grades 48 percent and GPT Tutor improved them 127 percent. On the unassisted exam, GPT Base cut grades 17 percent relative to control (coefficient -0.054 on a 0–1 grade scale, control mean 0.321). GPT Tutor’s exam coefficient was -0.004 , not significant — statistically indistinguishable from control. Those percentages are of control *grades*, not Cohen’s *d*. A later section of this chapter will report Oreopoulos and Low’s Khanmigo results in national percentile ranks and population standard deviations. Those are a different metric on a different intervention. Do not add them.²⁶¹

GPT Base, when asked “What is the answer?”, was fully correct only 51 percent of the time on the practice set. The error breakdown in the paper is 42 percent logical errors and 8 percent arithmetic errors, heterogeneous by problem. The mechanism is not mysterious. GPT Base students sent fewer messages, mostly “what is the answer” or a paste of the question. GPT Tutor students asked for help and attempted answers. The first group used the model as a crutch. They copied solutions. They felt fine. The exam, which is the only phase that resembles a Thursday quiz with the phone in the other room, is where the crutch became a score.²⁶²

Three things this experiment is not, and that this book will not upgrade it into. It is not a homeschool study. The students were in a Turkish high school, in a classroom, with a teacher, for four sessions. It is not GPT-5, and it is not whatever model is current on the day a parent reads this chapter. It is not Khanmigo, which is a constrained product over Khan Academy content, sold with a vendor claim that it never gives the answer. What it *is* is the cleanest field experiment opened on this research day that the kitchen-table pattern — paste the lesson, get the key, feel competent, sit the unaided test — predicts a worse unaided test. The homeschool analogue is not a guess about Turkey. It is a guess about mechanism. The mechanism is access to the solution during acquisition, copying, a metacognitive miss, and a closed test. Photomath, Mathway, Symbolab, and consumer ChatGPT,

²⁶⁰Bastani et al., “Generative AI without guardrails,” author PDF, accessed 30 August 2026. Design: classroom-level RCT, large high school in Turkey, Fall 2023–24, nearly 1,000 students in grades 9–11, four 90-minute sessions covering about 15 percent of a semester’s mathematics, GPT-4-0613. GPT Tutor’s hidden prompt contained teacher-written solutions and common-mistake hints and forbade the full solution. Control used notes and a textbook. Practice assisted; exam closed-book, paired to practice items, counted toward the grade. Pre-registration: AsPredicted 4DL_Q3J, https://aspredicted.org/4DL_Q3J. This is not a homeschool study, not GPT-5, and not Khanmigo.

²⁶¹Bastani et al., author PDF, abstract and Table 1. GPT Base exam coefficient -0.054 on a 0–1 grade (control mean 0.321). Do not mash these grade-relative percents with Oreopoulos and Low’s MAP standard deviations (n. 286).

²⁶²Bastani et al., author PDF. GPT Base correctness when asked “What is the answer?”: 51 percent correct, 42 percent logical errors, 8 percent arithmetic errors, heterogeneous by problem. Mechanism: GPT Base students sent fewer messages, mostly “what is the answer” or a paste of the question; GPT Tutor students asked for help and attempted answers.

Gemini, and Claude are the same mechanism as GPT Base unless the parent constrains them. That sentence is an analogy of mechanism, not a Photomath-named RCT. No Photomath-specific trial of unaided transfer was opened on 30 August 2026. A 2026 Chinese staggered-adoption study reported in secondary journalism was not opened as a PDF on this research day; those reported homework-and-exam figures stay out of this book.²⁶³

GPT Tutor’s hidden trick, which a parent can copy and a child-facing chatbot usually cannot, is labor. The prompt contained the correct solution and the common mistakes. Someone who already knew the mathematics had done the work of anticipating error. In a school trial, that someone is a teacher. In a homeschool, that someone is the parent — or it is no one, and the “tutor” in the child’s tab is GPT Base wearing a friendlier system prompt. A parent-generated hint script, which section 3.4 will describe, is the homeschool analogue. An off-the-shelf ChatGPT tab is GPT Base. The difference is not the logo. The difference is whether the solution is withheld from the child and whether a human who can hear a wrong answer is holding the key.

The exam phase is the other design feature a parent can steal without running a thousand-person trial. Bastani’s students were tested without the model, on items paired to what they had practiced, for a grade. The kitchen analogue is not a midterm. It is five minutes at the end of the session with the phone in another room. Chapter 2 already installed retrieval, spacing, and generation as properties of practice that produce learning rather than a performance that evaporates. This chapter adds the AI-era version of the same demand: if the model was in the room for the struggle, the session is not over until there is an unaided score. That demand is a practice rule grounded in Bastani’s exam phase and in Soderstrom and Bjork’s performance/learning split. It is not a new randomized trial of homeschool exit tickets.

What Bastani licenses, then, is narrower than the internet will sell and more useful than a vibe. It licenses a suspicion of any tool that raises the score on the page the child can see while the tool is open. It licenses a preference for hints that withhold the solution. It licenses an unaided check. It does not license a claim that “AI tutoring works” or that “AI tutoring fails.” GPT Tutor did not beat control on the exam. It also did not harm the exam. GPT Base harmed the exam *after looking like a miracle on practice*. The parent who only watches practice will buy the miracle. The parent who watches the unaided five minutes will not.

The rest of this chapter is the work of not letting that finding get buried under product names. The names matter. They are not the same object. Collapsing them is how a parent ends up running GPT Base and calling it a tutor.

3.2 Product landscape: do not collapse Hermes, Grok, tutors, and adaptive

The error this section exists to prevent is a single word doing five jobs. *Agent*, *chatbot*, *solver*, *tutor*, and *adaptive practice* are not synonyms. A parent who treats them as one stack will run the Bastani

²⁶³No Photomath-named RCT of unaided transfer was opened on 30 August 2026 (UNVERIFIED as a product-specific trial). A 2026 Chinese staggered-adoption study reported in secondary journalism (SCMP / Fortune / WebProNews) was not opened as a PDF this day; those reported figures are not used in this chapter. Knowledge at Wharton, “Without Guardrails, Generative AI Can Harm Education,” 2025, <https://knowledge.wharton.upenn.edu/article/without-guardrails-generative-ai-can-harm-education/>, is university journalism summarizing Bastani, not a substitute for the paper.

crutch inside a tool that can also browse the open web, keep a companion persona, and finish jobs in the parent’s apps. The pages opened on 30 August 2026 do not document a K–12 student-facing autonomous open-web agent sold as a school product. They do document adult agents, teacher-facing educator SKUs, homework solvers, a constrained Socratic tutor over a content library, and adaptive practice engines that sequence items without generating a persona. Name them. Do not collapse them.²⁶⁴

Start with the two objects that share a proper name and are not the same product. **Hermes Agent** is, in the site’s words on hermes-agent.org, “an open-source autonomous AI agent built by Nous Research and released in February 2026.” MIT license. Persistent memory. Automated skill creation. A gateway to Telegram, Discord, Slack, WhatsApp, Signal, and a command line. Self-hosted. Vendor security copy: no telemetry, no tracking; data in `~/.hermes/` on the machine. It can reach a browser, a local terminal, Docker, SSH. It is a parent-machine tool. It is not a child-safety school product. No K–12 student-safety certification was on the pages opened. The honest local and open option for a parent who wants generation and research to stay on a machine they control is also exactly the wrong default to put in a child’s hands, and it is the wrong default twice if the parent turns on the messaging gateway “so the child can ask homework questions on the bus.” That design is a companion plus an open-web agent.²⁶⁵

Hermes 4 is a different object. It is a chat and reasoning model family (70B and 405B in the Portal catalog this day), described in Nous Portal documentation as “strong at math, science, instruction following, schema adherence, roleplay, and long-form writing.” That sentence is vendor copy. The same documentation is explicit that **Hermes 4** is **not recommended for use inside Hermes Agent**, because it is tuned for chat and reasoning, not the rapid tool-calling loop the agent relies on. Portal examples for agent work on this day were Claude Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, and DeepSeek V4 Pro. A parent using Hermes Agent to expand a lesson — pull a public-domain worksheet style, convert it to isomorphic items, write a student sheet and a parent key — should pick an agentic model. A parent who wants a concept explained to the adult can use Hermes 4, or any strong chat model, without opening the tool loop. Hermes 4’s technical-report metrics on MATH and GSM8K were not extracted this day: the arXiv HTML conversion failed on fetch; treat those scores as UNVERIFIED pending the PDF. Even a strong math chat model is still a GPT-Base-class risk if the child can ask it for the answer.²⁶⁶

Three names that share a syllable are the next collapse to refuse. **Grok** is xAI’s conversational assistant. It is a chatbot and reasoner. It is not an education SKU by default. **Grok Bot**, launched

²⁶⁴Role labels used in this section — solver, tutor, agent, adaptive practice — follow the 30 August 2026 product pages in research stream 04, not a standards body. Negative finding on K–12 student-facing autonomous open-web school agents: pages opened that day, not an exhaustive census. See n. 277.

²⁶⁵Nous Research, “Hermes Agent,” <https://hermes-agent.org/>, fetched 30 August 2026. MIT license; released February 2026; persistent memory; Telegram / Discord / Slack / WhatsApp / Signal / CLI; self-hosted; vendor copy “No telemetry, no tracking”; data in `~/.hermes/`. GitHub: <https://github.com/nousresearch/hermes-agent/>. This is a parent-machine tool, not a child-safety school product. No K–12 student-safety certification was on the pages opened.

²⁶⁶Nous Research, “Nous Portal | Hermes Agent,” <https://hermes-agent.nousresearch.com/docs/integrations/nous-portal>, fetched 30 August 2026. Hermes 4 (70B / 405B) described as “strong at math, science, instruction following” — **vendor**. Explicitly not recommended inside Hermes Agent (tuned for chat, not rapid tool-calling). Portal agent-work examples this day: `anthropic/claude-sonnet-4.6`, `openai/gpt-5.5-pro`, `google/gemini-3-pro-preview`, `deepseek/deepseek-v4-pro`. Hermes 4 technical report: arXiv 2508.18255, <https://arxiv.org/abs/2508.18255>; HTML conversion failed on fetch this day; MATH / GSM8K metrics UNVERIFIED pending PDF.

11 August 2026 and expanded to more plans on 26 August 2026, is an always-on agent with its own cloud computer, included with SuperGrok and with Cursor Pro and Teams plans. The introducing post advertises a team of agents that “have their own computer, work inside tools and apps like you do, and keep working 24/7.” Jobs named at launch were sales, website builder, inbox manager, office manager — not tutoring. The FAQ, fetched this day, is load-bearing: bots use a persistent cloud computer; closing the laptop does not stop a background turn; every Bot on the account shares one computer, its files, browser sessions, and logins; the computer is assigned per user, not per Bot; “Do not use separate Bots as a security boundary”; “Grok Bot requires cloud data storage, so Legacy Privacy Mode is not supported.” No education SKU and no FERPA claim were on the pages opened. Cursor Cloud Agents, a related developer surface of isolated cloud VMs for coding, are not a math tutor either. A homeschool parent who uses Grok Bot to generate materials on the parent’s account is using an adult productivity agent. Putting a child’s name, an IEP, or worksheet photographs on that shared cloud computer is a privacy decision the parent owns. It is not a school-compliant deployment.²⁶⁷

El Salvador is a government partnership, not a homeschool SKU and not Grok Bot. xAI’s news post of 11 December 2025 describes a ministry partnership to deploy Grok — the assistant — across more than 5,000 public schools, “delivering personalized learning to over one million students.” Vendor language on that page includes “adaptive, curriculum-aligned tutoring,” with quotes from President Bukele and Elon Musk. Outcome evidence, independent evaluation, and student-data terms were not on the announcement page. Treat the post as an announcement, not an RCT. A U.S. homeschooler cannot opt into that program and should not cite it as evidence that Grok Bot is an approved tutor. Independent evaluation remains UNVERIFIED.²⁶⁸

Khanmigo is the closest widely sold home generative tutor that is *designed* like Bastani’s GPT Tutor. Khan Academy’s learner page, fetched this day, prices it at \$4 per month or \$44 per year. Vendor copy: “Unlike ChatGPT, Khanmigo never gives you the answer.” A parent (18+, U.S. billing address) signs up and can enable access for up to ten children under 18. Classroom student access is only through school or district implementations. The product is a constrained Socratic tutor over Khan content, not an open-web agent. Whether the design holds in the wild is an empirical question, and section 3.5 will not let the brochure answer it. Common Sense Media’s four-star rating is vendor-cited on the page; this book does not launder a rating as an RCT.²⁶⁹

²⁶⁷xAI, “Introducing Grok Bot,” 11 August 2026, <https://x.ai/news/introducing-grok-bot>; “Grok Bot is now included with more plans,” 26 August 2026, <https://x.ai/news/grok-bot-more-plans>; “Grok Bot FAQ,” <https://docs.x.ai/grok-bot/faq>, all fetched 30 August 2026. Load-bearing FAQ sentences: persistent cloud computer; closing the app does not stop a background turn; one computer per user shared across Bots; “Do not use separate Bots as a security boundary”; “Grok Bot requires cloud data storage, so Legacy Privacy Mode is not supported.” No education SKU and no FERPA claim on the pages opened. Cursor Cloud Agents (<https://cursor.com/docs/cloud-agent>) are a related developer surface, formerly Background Agents, not a math tutor. Cursor privacy help: DPA for Teams / Enterprise; individual plans follow the Privacy Policy — not FERPA.

²⁶⁸xAI, “xAI and El Salvador Pioneer the World’s First Nationwide AI Education Program,” 11 December 2025, <https://x.ai/news/el-salvador-partnership>. Vendor: deploy Grok (the assistant, not Grok Bot) across more than 5,000 public schools / over one million students; “adaptive, curriculum-aligned tutoring.” Announcement, not RCT. Independent evaluation UNVERIFIED. CoinDesk, 11 December 2025, is journalism on the same announcement.

²⁶⁹Khan Academy, “Khanmigo for learners,” <https://www.khanmigo.ai/learners>, fetched 30 August 2026. **Vendor** price: \$4/month or \$44/year. **Vendor**: “Unlike ChatGPT, Khanmigo never gives you the answer.” Parent (18+, U.S. billing) enables up to ten children under 18. Classroom student access only through school/district implementations. Common Sense Media four-star rating is vendor-cited on the page, not an RCT.

Then the solvers. **Photomath**'s homepage leads with scan — including word problems — or a smart calculator, after which “A solution will pop up for your exact problem,” plus step-by-step. Tiers displayed this day: Basic at \$0; Monthly at \$9.99; Annually at \$69.99 (USD). Coverage listed from elementary through college. Vendor: “designed for learning.” The pedagogical risk is Bastani’s crutch, not a missing step list. **Mathway**: “Got a math problem? Get the steps.” Type or photo. Vendor: “Every answer comes with a full breakdown.” Subjects from basic math through calculus, statistics, linear algebra, chemistry, physics. Freemium. **Symbolab**: “AI Math Calculator” / “AI Math Solver,” with step-by-step, photo and screenshot, practice problems, quizzes, worksheet generation, and a Chrome “snip and solve.” Solver first, practice second. Vendor language about supporting learning is vendor language. No Photomath-, Mathway-, or Symbolab-named RCT of unaided transfer was opened this day. The transferable evidence is Bastani plus Soderstrom and Bjork.²⁷⁰

Wolfram|Alpha is not an LLM. The homepage, fetched this day, describes “verified, expert-level answers using Wolfram’s breakthrough algorithms and knowledgebase,” built on Wolfram Language. That is the honest check for an antiderivative, a limit, a series test: a computer-algebra system rather than a next-token guess. Step-by-step on Pro is vendor; current Pro pricing was not captured as a live number on the homepage this day and stays UNVERIFIED. Mixing Wolfram and a chat model without a label is how a hallucinated integral survives a parent’s glance.²⁷¹

Adaptive home systems already in use are not autonomous agents. They sequence practice and hints. They do not browse the open web, keep a companion persona, or finish jobs in the parent’s apps. Do not call them “AI tutors” in the Hermes Agent or Grok Bot sense.

ALEKS (Assessment and Learning in Knowledge Spaces), sold by McGraw Hill for K–12, is mastery practice on a knowledge-space map: an Initial Knowledge Check, an ALEKS Pie, periodic Knowledge Checks, free-response emphasis. Vendor claims on the K–12 page include a mastery rate “more than 90 percent of the time” and a 2.7-times proficiency finding for students who reached 75 percent mastery; those are vendor-cited. [aleks.com](https://www.aleks.com) markets Independent Use and a family course library. Homeschool families have used ALEKS for years. It is not new because large language models exist.²⁷²

Khan Academy content and practice remain free. The library is the practice engine; Khanmigo is the chat layer. **IXL Math** lists skills from pre-K through calculus (the math landing page this day listed, for example, 357 first-grade skills, 389 Algebra 1 skills, 310 Geometry skills, 123 Calculus skills). Fun questions, built-in support, awards. Adaptive skill practice and a supplement, not a full

²⁷⁰Photomath, <https://photomath.com/>, fetched 30 August 2026: scan including word problems; “A solution will pop up for your exact problem”; Basic \$0 / Monthly \$9.99 / Annually \$69.99 (USD, this day); **vendor** “designed for learning.” Mathway, <https://www.mathway.com/>: “Got a math problem? Get the steps”; **vendor** “Every answer comes with a full breakdown.” Symbolab, <https://www.symbolab.com/>: “AI Math Calculator” / “AI Math Solver”; solver first, practice add-ons second. No product-specific unaided-transfer RCT opened this day.

²⁷¹Wolfram Research, “Wolfram|Alpha,” <https://www.wolframalpha.com/>, fetched 30 August 2026. Computational engine on Wolfram Language, not an LLM. Pro step-by-step pricing not captured as a live number on the homepage this day (UNVERIFIED current Pro price).

²⁷²McGraw Hill, “ALEKS K–12,” <https://www.mheducation.com/prek-12/program/aleks/MKTSP-GAB02M0.html>, fetched 30 August 2026. Knowledge Space Theory; Initial Knowledge Check; ALEKS Pie. Vendor mastery-rate claims (including “more than 90 percent” and the 2.7-times proficiency finding) are vendor-cited. [aleks.com](https://www.aleks.com) markets Independent Use / family course library. Adaptive mastery practice, not an agent.

curriculum and not an agent. Live homeschool price was not captured on that landing page this day (UNVERIFIED as an invoice number; do not treat third-party blogs as the bill).²⁷³

DreamBox Math sells K–8 family and homeschool plans with a Family Dashboard and a free trial. Devices: browsers and an iPad app; not smartphones, Android tablets, or eReaders. Vendor: “Just one hour each week can drive more than a grade level of math growth”; “only dual-discipline program rated STRONG by Evidence for ESSA.” The grade-level-per-hour claim is vendor. The underlying ESSA study PDF was not opened this day (UNVERIFIED). Intelligent Adaptive Learning is the product’s name for its engine. It is not a generative agent.²⁷⁴

MATHia, from Carnegie Learning, is a grades 6–12 school intelligent-tutoring system: skill-by-skill “AI,” progress meters, LiveLab for teachers, a “Request a Demo” public page. RAND evidence cited on the page is a vendor summary of a Department of Education–funded trial of the *blended Carnegie approach*, not MATHia-alone and not homeschool. Direct family purchase was not documented on the page opened this day (UNVERIFIED). Book language: MATHia is a school ITS; whether a homeschool family can still buy it is unverified on 30 August 2026.²⁷⁵

Educator SKUs are teacher-facing. **ChatGPT for Teachers** (OpenAI, 19 November 2025) is a secure workspace for verified U.S. K–12 educators, free through June 2028, with vendor language about not using chats to train models by default and helping schools meet FERPA requirements. It is for teachers, not a student math-tutor SKU. Homeschool parents are not verified U.S. K–12 educators by default and should not assume the Teachers SKU’s FERPA wrapper. Consumer ChatGPT remains the GPT-Base analogue in Bastani: a student can paste a worksheet and get the key. **Claude for Teachers** (Anthropic, 14 July 2026) is for verified U.S. K–12 educators, free for a year if signed up by 30 June 2027, with a Learning Commons connector, teaching skills, and vendor language that it is “for educators only, consistent with Claude’s 18-and-over policy,” never trains on teacher conversations, and is FERPA-aligned. An update of 28 August 2026: identifiable student records require school or district authorization. Claude is not a student math SKU. Claude Projects — workspaces with files and custom instructions — are the natural parent-side place to keep a curriculum PDF and a hint-script instruction; current Projects UI and limits were not opened as a separate marketing page this day (UNVERIFIED). **Microsoft’s** Math Assistant in OneNote and Math Progress in Teams for Education are school Microsoft 365 surfaces, not a homeschool Copilot

²⁷³Khan Academy content library remains free (Khanmigo page, this day). IXL Learning, “IXL Math,” <https://www.ixl.com/math>, fetched 30 August 2026: skill lists Pre-K through Calculus (357 first-grade, 389 Algebra 1, 310 Geometry, 123 Calculus skills on the landing page this day). Adaptive skill practice, not a full curriculum and not an agent. Live homeschool price not on that URL (UNVERIFIED as an invoice number).

²⁷⁴DreamBox / Discovery Education, “DreamBox Math for Homeschool,” <https://www.dreambox.com/family/homeschool-math>, fetched 30 August 2026. K–8 family plans; browsers and iPad, not smartphones or Android tablets. Vendor: “Just one hour each week can drive more than a grade level of math growth”; ESSA Strong rating is vendor-cited. Underlying ESSA study PDF UNVERIFIED this day. Intelligent Adaptive Learning is the product’s name for its engine, not a generative agent.

²⁷⁵Carnegie Learning, “MATHia,” <https://www.carnegielearning.com/solutions/math/mathia>, fetched 30 August 2026. Grades 6–12 school ITS; “Request a Demo”; LiveLab for teachers. RAND summary on the page is vendor citation of a DoE-funded trial of the blended Carnegie approach, not MATHia-alone and not homeschool. Direct family purchase UNVERIFIED on 30 August 2026. John F. Pane, Beth Ann Griffin, Daniel F. McCaffrey, and Rita Karam, “Effectiveness of Cognitive Tutor Algebra I at Scale,” *Educational Evaluation and Policy Analysis* 36, no. 2 (2014), <https://doi.org/10.3102/0162373713507480>, is the pre-LLM Carnegie Cognitive Tutor evaluation cited inside Oreopoulos and Low — school ITS, not a kitchen.

button that is FERPA by default. Standalone “Microsoft Math Solver” app status (retired versus live) is UNVERIFIED beyond the research page opened this day. Consumer Gemini is a GPT-Base-class assistant. A dedicated Gemini-for-Education feature list, and whether a homeschool parent can buy that SKU without a co-op domain, were not fully extracted this day (UNVERIFIED).²⁷⁶

The negative finding, re-verified on the pages in this stream, is short enough to keep on one line. No page opened on 30 August 2026 documented a district-sold, student-facing, autonomous open-web agent as a school product. ChatGPT for Teachers and Claude for Teachers are teacher-facing. Khanmigo for learners is a constrained tutor over Khan content, parent-gated. Grok Bot and Hermes Agent *are* autonomous open-web agents and are not sold as K–12 student products. SchoolAI Spaces and similar classroom platforms, from the prior education-book landscape, remain teacher-in-the-loop classroom tools, not “give the child an open browser agent.” The negative is the pages in stream 04, not an exhaustive census of every 2026 “AI classmate” SKU. If a later printing needs a stronger negative, those products should be re-fetched. Until then, this book will not invent a school-safe child agent in order to have something to recommend.²⁷⁷

A compact map, so the later chapters can point at a row instead of a paragraph:

Object	Role	Child-safe school product on pages opened 30 Aug 2026?
Hermes Agent	Agent (self-hosted, open web, messaging gateways)	No. Adult/developer tool; no K–12 certification on the pages opened.
Hermes 4	Chat / reasoning model	No. Not recommended inside Hermes Agent; still GPT-Base-class if the child holds it.
Grok	Chat / assistant	No education SKU by default.
Grok Bot	Agent (cloud computer; SuperGrok / Cursor)	No. Cloud storage required; Legacy Privacy Mode unsupported; no FERPA claim.

²⁷⁶OpenAI, “A free version of ChatGPT built for teachers,” 19 November 2025, <https://openai.com/index/chatgpt-for-teachers/>, fetched 30 August 2026: verified U.S. K–12 educators; free through June 2028; vendor FERPA-oriented workspace; not a student math SKU. Anthropic, “Introducing Claude for Teachers,” 14 July 2026, updated 28 August 2026, <https://www.anthropic.com/news/claude-for-teachers>: educators only, 18+, vendor “never trains on teacher conversations,” K–12 DPA / FERPA-aligned; identifiable student records require school/district authorization. Claude Projects current file limits UNVERIFIED (marketing page not opened). Microsoft, “Math Solver,” <https://www.microsoft.com/en-us/research/project/math-solver/>: OneNote Math Assistant and Teams Math Progress are Microsoft 365 Education surfaces; standalone consumer Math Solver app status UNVERIFIED. Gemini for Education feature list and homeschool eligibility UNVERIFIED this day; Google Workspace for Education homeschool setup exists at <https://support.google.com/edu/setup/answer/9473438>.

²⁷⁷Negative finding, re-verified on pages opened 30 August 2026: no district-sold, student-facing, autonomous open-web agent as a school product. Educator SKUs are teacher-facing (nn. 276). Khanmigo is a constrained tutor over Khan content (n. 269). Hermes Agent and Grok Bot are autonomous open-web agents and are not K–12 student products (nn. 265, 267). The negative is this stream, not an exhaustive census of 2026 classroom-agent SKUs.

Object	Role	Child-safe school product on pages opened 30 Aug 2026?
El Salvador Grok-as-tutor	Ministry deployment of the <i>assistant</i>	Announcement, not RCT; not Grok Bot; not a U.S. homeschool SKU.
Khanmigo	Constrained tutor over Khan content	Home SKU is parent-gated; “never gives the answer” is vendor . Not an open-web agent.
Photomath / Mathway / Symbolab	Solver	Solver. Designed-for-learning copy is vendor .
Wolfram Alpha	Computer-algebra / computational engine	Check, not explainer. Not an LLM.
ALEKS / Khan practice / IXL / DreamBox / MATHia	Adaptive practice (MATHia: school ITS)	Not agents. MATHia family purchase UNVERIFIED. DreamBox ESSA underlying study UNVERIFIED.
ChatGPT for Teachers / Claude for Teachers	Teacher-facing educator SKU	FERPA-oriented <i>teacher</i> workspaces. Homeschool parents are not verified educators by default.

Vendor claims in that table stay labelled vendor. A row is not a recommendation. The recommendation, when this chapter has one, is a rule about who holds the key, not a logo.

3.3 Ban vs allow

The kitchen rules follow from Bastani plus Soderstrom and Bjork, not from folklore about logos. A ban is not a moral panic and not a school-district acceptable-use policy. It is a statement about what, during the attempt, turns the session into GPT Base. An allow is not a warrant that the move has been trialed at a kitchen table for a year. The allow list in this section, and the six workflows in the next, are practice suggestions grounded in the documents already named — Bastani’s crutch and hidden-prompt design, Soderstrom and Bjork’s performance/learning split, and, in section 3.5, Oreopoulos and Low’s finding that unprompted student dialogue with a well-designed tutor is rare. They are not a new RCT of homeschool AI. Label them that way every time a later chapter points here.²⁷⁸

Ban, child-facing, during the attempt:

Photograph or paste the assigned worksheet and ask for the key. That is Bastani’s GPT Base prompt, executed with a camera. Photomath’s homepage is built to do it. Mathway and Symbolab sell the

²⁷⁸Ban versus allow follows from Bastani et al. 2025 (nn. 257, 260–262) and Soderstrom and Bjork 2015 (n. 258), plus Oreopoulos and Low’s engagement finding (n. 286). The allow list is a practice suggestion, not a homeschool RCT. No year-plus kitchen GPT-Tutor RCT exists in the literature opened 30 August 2026.

same loop. Consumer ChatGPT, Gemini, and Claude will do it if asked. The child gets a finished page. The parent sees performance. The unaided test is where the 17 percent lives.

“What is the answer?” and “just give me the steps.” Fewer messages, paste the question, copy the solution: that is the mechanism in the paper, not a personality flaw in the child. A solver that prints steps is still a solver. Steps the child did not generate are not generation. Soderstrom and Bjork already had delayed feedback and generation on the side of learning; a step list at the elbow is the other side.

Photo-to-key as a daily driver. The camera is not a pedagogy. It is an interface for the crutch.

Any companion persona — romantic, buddy, coach-with-a-name, “math friend.” Section 3.6 will keep Setzer/Garcia v. Character Technologies and Raine v. OpenAI in their proper legal box: allegations, not verdicts, not a literature on mathematics learning. They are in this chapter because the failure mode is persona plus isolation plus the model as confidant. A math agent with a name, a memory of the child, and a messaging-app gateway is that architecture even if the first week is fractions. The mathematics does not sanitize the design.

Open-web agents in the child’s account: Hermes Agent, Grok Bot, operator-enabled ChatGPT agents. These tools can leave the lesson. They can browse, log in, keep memory, finish jobs. No page opened this day sold them as a K–12 student product. A parent-hosted agent expanding a lesson on the parent’s machine is a different fact pattern from a child with a Telegram gateway.

Using a detector to police the child’s written explanations. Liang, Yuksekgonul, Mao, Wu, and Zou (2023) found an average false-positive rate of 61.22 percent on human TOEFL essays in the arXiv version, 61.3 percent in *Patterns*. A homeschooler who writes a slow, non-native, or simply careful paragraph about why two triangles are similar is exactly the writer those detectors smear. Section 3.6 returns to the number. The ban belongs on this list because the temptation is pedagogical — “I need to know if they wrote it” — and the instrument does not know.²⁷⁹

Allow, parent-hosted, or child only after an attempt and constrained:

The parent asks the model to explain the concept to the *parent*. Not to the child. Not as a lecture the child watches. The adult who has to hear a wrong answer needs the idea, the notation, the two mistakes this age makes, and one tiny problem to solve before sitting down. Bastani’s GPT Tutor worked in part because teachers had already written the solution and the common mistakes into the prompt. The parent is that teacher, or nobody is.

The parent asks for a short lesson outline aligned to the family’s curriculum, then teaches. Claude for Teachers and ChatGPT for Teachers offer unit-planning features; those are vendor educator features, not homeschool RCTs. A parent can still ask for a thirty-minute home sequence — a physical hook, one worked example the parent will write by hand, three student attempts, one exit

²⁷⁹Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou, “GPT detectors are biased against non-native English writers,” arXiv:2304.02819, <https://arxiv.org/abs/2304.02819>; *Patterns* 4, no. 7 (2023), <https://doi.org/10.1016/j.patter.2023.100779>. Seven detectors; 91 TOEFL essays versus 88 U.S. eighth-grade essays; average false-positive rate 61.22 percent (arXiv) / 61.3 percent (*Patterns*) on the TOEFL set; 19.78 percent unanimously flagged; 97.80 percent flagged by at least one detector. A second-round “literary language” prompt dropped detection of ChatGPT essays from as high as 100 percent to 13 percent. Do not use detectors to police a homeschooled’s written mathematics explanations. Full discussion in §3.6.

problem — with answers only in a marked parent key. The outline is a map. It is not a franchise. This book does not replace Saxon, Singapore, Beast Academy, Art of Problem Solving, Math-U-See, Teaching Textbooks, or Khan Academy.

The parent asks for extra isomorphic practice — same structure, new numbers and cover story — and **holds the answer key**. Practice with access to the solution is not unaided learning. Generation plus delayed feedback is on Soderstrom and Bjork’s learning side. Oreopoulos and Low, descriptively, found that practice minutes predicted MAP; chat did not. None of that is a kitchen RCT of isomorphic worksheets. It is why the key stays with the adult.

The parent asks for a hint script: a first hint that is only a question, a second that names the concept without numbers, a third that gives the next algebraic step still short of the final answer, plus one likely mistake and what to say if it appears. The parent reads the hints aloud. That is closer to human tutoring than handing the child Khanmigo and leaving the room.

The parent asks for a hands-on activity — paper, beans, a measuring cup — especially in the elementary years. The model is a materials generator. The mathematics happens in the child’s hands.

The parent pastes the child’s already-wrong work and asks what misconception it is, what question to ask the child (not the correct steps), and what isomorphic item would retest after the talk. The child answers the parent, not the model.

The child uses a constrained tutor — Khanmigo, or a parent-built GPT-Tutor prompt that forbids answers — *after* attempting, with the parent in the room, no paste-the-key. Worked-example study after an attempt, not instead of one.

Do not generate “practice” by photographing tonight’s assigned homework and asking the model to complete it. That is the ban, wearing a worksheet.

The Photomath-specific unaided-transfer RCT remains UNVERIFIED. What transfers is the mechanism. A scan that pops a solution for the exact problem is GPT Base with a camera. Vendor “designed for learning” does not rewrite Bastani. A parent who wants the step list *after* the child has produced a page, as a check against a known-correct key, is running a different session from the parent who lets the camera do the lesson so everyone can get to soccer. The second session is the one with a large-sample causal harm estimate.²⁸⁰

Two columns, then, for the later chapters to point at. Ban during the attempt: paste, “what is the answer,” photo-to-key, companion, open-web agent in the child’s hands, detectors as police. Allow on the parent’s account, or after the attempt: explain to the parent, outline, isomorphic items with the key held back, hint script, hands-on, post-attempt diagnosis, constrained tutor in the room, worked example after. The next section writes the allow list as jobs. The jobs are still not an RCT.

3.4 Parent workflows (practice suggestions, not an RCT)

The thesis again, because every workflow is a test of it. The parent understands the concept well enough to hear a wrong answer. The child does the struggle before the tool. The agent is a sup-

²⁸⁰Photomath-specific unaided-transfer RCT: UNVERIFIED this day (n. 263). Transferable evidence is Bastani’s crutch finding plus the Photomath homepage’s scan-to-solution lead (n. 270).

plement hosted and constrained by the parent. If a workflow quietly moves the struggle onto the model, it is GPT Base and this section will not recommend it. If a workflow assumes a year-plus kitchen trial that does not exist, this section will say so. Each of (a) through (f) is parent-uses-the-agent unless noted. Grounding is named. Where there is no RCT of that exact move, the label is **practice suggestion, not an RCT**.²⁸¹

(a) Explain the concept to the parent first.

Why: a parent who cannot hear a wrong answer cannot use any of the later workflows. GPT Tutor’s hidden prompt contained a teacher-written solution and a bank of common mistakes. That is not a system-prompt trick a child-facing chatbot will perform for free. It is labor by someone who already knows the mathematics. The parent is that someone, or the session has no analogue of GPT Tutor.

Practice suggestion, not an RCT. Prompt pattern, parent account, no child present: “I am teaching [topic] to a [grade] homeschooler using [curriculum]. Explain it to *me* as an adult who last took math in [year]. Give: the idea in one paragraph; the notation I will see; the two mistakes this age makes; one tiny problem I should be able to solve before I sit down with the child; the answer to that problem.” Then the parent solves the tiny problem *without* the model, then checks. The check can be the model, a textbook key, or Wolfram|Alpha if the tiny problem is symbolic. The point of the tiny problem is not a quiz for the adult. It is a test of whether the explanation landed hard enough to hear a wrong answer at the table.

Tools: any capable chat model — ChatGPT, Claude, Gemini, Grok, Hermes 4 as chat, a local vLLM. Prefer not an open-web agent for this job. It is a reasoning job. Hermes Agent’s tool loop is wasted on it and adds a browser the parent did not ask for. Wolfram|Alpha if the parent needs a computer-algebra check on the tiny problem. A *hypothetical* for grain, labelled as such: a parent who last took mathematics in eleventh grade, facing Singapore 6A’s first look at dividing a fraction by a fraction, asks for the idea, the notation, the two mistakes (invert the wrong factor; treat division as “make smaller”), and a one-line problem. The parent solves it on paper. Then they sit down. That sequence is the workflow. It is not a finding that the sequence raises MAP scores.

(b) Generate a short lesson outline.

Practice suggestion, not an RCT. Analogues on the educator SKUs — Claude for Teachers’ “plan a lesson from HQIM,” ChatGPT for Teachers’ unit-planning prompts — are vendor educator features. A homeschool parent is not a verified U.S. K–12 educator by default and should not pretend those FERPA wrappers apply. A parent can still ask: “Thirty-minute home lesson on [topic]. Sequence: hook with a physical object; one worked example I will write by hand; three student attempts of increasing difficulty; one exit problem. Do not print answers in the student-facing section. Put answers in a clearly marked PARENT KEY at the end.”

The outline is a map, not a franchise. This book does not replace named curricula. A model that invents a week of “fun activities” disconnected from the family’s book is not expanding the lesson. It is replacing the book with a second, unvetted one. The parent who already has Saxon,

²⁸¹ Parent workflows (a)–(f) are practice suggestions grounded in Bastani’s hidden-prompt design, Soderstrom and Bjork, and Oreopoulos and Low, not a new RCT of homeschool AI. Composite hypotheticals in this section are labelled *hypothetical* in the sentence that introduces them.

Dimensions, Beast Academy, Math-U-See, Teaching Textbooks, or Art of Problem Solving on the table should name that book in the prompt and ask for a *session* inside it, not a rival scope.

(c) Extra practice with the answer key held by the parent.

Grounded in Bastani (practice with access is not unaided learning), Soderstrom and Bjork (generation and delayed feedback), and Oreopoulos and Low (practice minutes predicted MAP; chat, descriptively, did not). Practice suggestion, not an RCT.²⁸²

Generate five to eight isomorphic items: same structure, new numbers, new cover story. Two files or two pages — student sheet and parent key. The child never sees the key. The parent can grade. The parent can later paste *only the child's work* into the model for diagnosis, which is workflow (f). Do not generate “practice” by taking a photo of tonight’s assigned homework and asking the model to complete it. That is the ban. Isomorphic means *new items of the same type*, not a completed copy of the assigned page.

A model will happily print eight items that look like variety and are actually the same number sentence in a new font. The parent who can hear a wrong answer can also hear a fake distractor. Sample the set. Work one item before the child sees the sheet. If the model’s key is wrong — Bastani’s GPT Base was fully correct only 51 percent of the time on high-school items, with 8 percent arithmetic errors even when the method was right — the sheet is worse than no extra practice. A known-correct key (textbook, parent-worked solution, Wolfram for a symbolic check) still beats a fluent wrong one.

(d) Hint script, not an answer key for the child.

Grounded in Bastani’s GPT Tutor prompt, Figure 1 of the author PDF: withhold the full solution; ask what they have tried; point out a named mistake; escalate information slowly; accept equivalent answers. Practice suggestion, not an RCT.²⁸³

Parent prompt: “Write a hint script for this problem. Level 1: a question only. Level 2: name the concept (for example, ‘parallel lines have equal slope’) without numbers. Level 3: the next algebraic step, still not the final answer. Also list one likely mistake and what to say if I see it. Do not put the final answer anywhere the child might see; put it only under PARENT KEY.” The parent reads the hints aloud. That is closer to human tutoring than handing the child Khanmigo and leaving the room. Oreopoulos and Low, in a mandatory daily intervention block with a teacher present, found that the median student messaged Khanmigo on only a third of practice days and in 17 percent of mistake sessions. Unsupervised dialogue is not the default. A parent holding a script is a different intervention from a chatbot available in a tab.

The script is still only as good as the human who wrote the solution into the request. If the parent has not done workflow (a), the model is guessing at common mistakes. Sometimes the guess is

²⁸²Philip Oreopoulos and Nina Low, “One Click Away: AI Tutoring with Khanmigo in a Two-Year School Experiment,” NBER Working Paper 35620, August 2026, <https://www.nber.org/papers/w35620> and PDF https://www.nber.org/system/files/working_papers/w35620/w35620.pdf, opened 30 August 2026. Descriptively, practice minutes predicted MAP; conversational use was thin (see n. 286). Working paper, not yet journal-peer-reviewed.

²⁸³Bastani et al., author PDF, Figure 1 (GPT Tutor prompt): withhold the full solution; ask what they have tried; point out a named mistake; escalate information slowly; accept equivalent answers. Homeschool analogue is a parent-read hint script, not a child-held chatbot.

useful. Sometimes it is a fluent wrong diagnosis. The parent is the judge of whether the hint names the mathematics in the family's book.

(e) Hands-on activity.

Practice suggestion, not an RCT. Especially grades 1–5: “Give me a ten-minute kitchen-table activity for [fractions / place value / area] using paper, beans, or a measuring cup. No worksheet. What the child should say if they understand. What a wrong move looks like.” The model is a materials generator. The mathematics happens in the child's hands. Chapter 2 already warned that kitchen-math transfer is hard; NMAP Finding 26 is the document, not a vibe. An activity is a representation, not a substitute for the written mathematics the child will have to do later. The allow is generation of the activity. The ban is letting the model narrate the activity while the child watches.

(f) Diagnose a wrong answer the child already produced.

Grounded in Bastani's common-mistake bank; in MATHia and Carnegie vendor claims that their ITS recognizes *why* students err; and in the general intelligent-tutoring literature (Pane and colleagues' 2014 Cognitive Tutor evaluation, cited inside Oreopoulos and Low). No homeschool diagnosis RCT. The Carnegie claim is vendor. Pane is a school ITS trial, not a kitchen.²⁸⁴

Practice suggestion, not an RCT. Photograph the child's work *after* the attempt. Parent-only prompt: “Here is a student's work. Do not praise. Name the most likely misconception in one sentence. Give me one question I should ask the child (not the correct steps). Give me one isomorphic item to retest after we talk.” Then the parent asks the question. The child answers the parent, not the model.

The order is load-bearing. If the photograph happens before the attempt, the session is photo-to-key. If the model addresses the child, the session is a companion. If the parent skips the spoken question and reads the correct steps aloud, the session is a worked example in the wrong place — before generation rather than after. Worked-example study *after* an attempt is on the allow list. Worked-example study *instead* of an attempt is the crutch.

What not to automate sits next to these six jobs, because the jobs are tempting to extend. Oral math-fact drill is the parent's ears, or a dumb spaced-repetition deck, not a language model that will sometimes hallucinate 7×8 . The child's written AP-style free response is unaided work; section 3.7 returns to it. Grading that the parent has not sampled is how a fluent wrong key becomes a transcript. Any “always-on math friend” that messages the child on Telegram via a Hermes gateway is the companion design plus an open-web agent. Those are not workflows. They are the ban, automated.

Child-facing tools, if used at all, have a named evidence grain, and it is small. That is the next section. Khanmigo-the-product is designed like GPT Tutor. Khanmigo-as-used, in the only large school trial opened this day, was mostly unused as a tutor. A custom college-physics tutor is a different object. Neither is a year-plus kitchen RCT. Do not let them become one sentence.

²⁸⁴Carnegie / MATHia vendor claim that the ITS recognizes why students err: vendor (n. 275). Pane et al. 2014 (n. 275) is school Cognitive Tutor, cited inside Oreopoulos and Low. No homeschool diagnosis RCT.

3.5 Khanmigo ITT is not Bloom's 2-sigma; Kestin is college physics

Three objects get sold as one sentence: “AI tutors work.” The documents do not say that. They say three different things about three different systems, in three different populations, on three different outcomes. Oreopoulos and Low measured assignment to Khan Academy plus Khanmigo in Tennessee middle schools, on MAP mathematics, over two years. Kestin, Miller, Klales, Milbourne, and Ponti measured a custom, course-specific physics tutor against an in-class active-learning lesson for Harvard undergraduates, on an immediate post-test. Bastani, already installed, measured GPT Base and GPT Tutor against notes and a textbook for Turkish high-school mathematics students, on assisted practice and then an unaided exam. Collapse those three and a parent will buy a \$4 chatbot, walk away, and expect a Harvard effect size. Keep them apart and the kitchen advice is almost boring: extra structured practice can move a score a little; a purpose-built tutor can beat a lecture for motivated older students on a short test; an unguarded chatbot can raise the homework grade and cut the exam. None of those sentences is Bloom's two-sigma tutor. None is a year-plus homeschool RCT. No such RCT exists in the literature opened on 30 August 2026.²⁸⁵

Philip Oreopoulos and Nina Low, “One Click Away: AI Tutoring with Khanmigo in a Two-Year School Experiment,” NBER Working Paper 35620, August 2026, is a two-year cluster randomized trial in eighteen Tennessee middle schools. Assignment was to Khan Academy plus Khanmigo. Quote the abstract accurately, then quote Table 5, and do not invent a standalone 0.06 as the intention-to-treat effect.

The abstract: assignment raises math achievement by 1.3 national percentile ranks per term, or about 0.06 to 0.08 standard deviations over a school year; the implied effect of a full year of active participation reaches 0.14 standard deviations. “These gains resemble those from Khan Academy practice without AI assistance.” Ninety-six percent of students tried Khanmigo at least once, but the median student messaged it on only a third of the days they practiced, and in only 17 percent of the exercise sessions in which they made a mistake. Of messages, 14.5 percent contained a mathematical question or a step of reasoning.²⁸⁶

Table 5, pre-registered, pooled ITT, baseline-adjusted: 1.264 national percentile ranks per term (standard error 0.597); 0.040 population standard deviations per term; 0.050 control-group standard deviations per term. Year 2: 0.084 population SD per school year (SE 0.041). Stacked years: 0.062 per school year. Cost in the district deployment was about \$15 per student per year. Message mix, from the paper: 39.4 percent bare answers to Khanmigo's questions, 24.2 percent button clicks, 12.6 percent low-effort or “what's the answer,” 9.3 percent off-task.

²⁸⁵No year-plus kitchen-table RCT of a parent-constrained GPT-Tutor workflow was opened on 30 August 2026. Teacher-replacement slogan is folklore: no opened RCT shows an LLM replacing a parent or teacher for K–12 mathematics.

²⁸⁶Oreopoulos and Low, NBER w35620, abstract, accessed 30 August 2026. Two-year cluster RCT, 18 Tennessee middle schools. Quote the abstract: assignment raises math achievement by 1.3 national percentile ranks per term, or about 0.06 to 0.08 standard deviations over a school year; implied effect of a full year of active participation reaches 0.14 standard deviations. “These gains resemble those from Khan Academy practice without AI assistance.” 96 percent tried Khanmigo at least once; median student messaged it on only a third of practice days and in 17 percent of mistake sessions; 14.5 percent of messages contained a mathematical question or a step of reasoning. Do not invent standalone 0.06 as the ITT. Pre-registration: AEA RCT Registry AEARCTR-0013519, <https://www.socialscienceregistry.org/trials/13519>.

Do not quote an invented 0.06 SD as the ITT. The paper’s own abstract uses 0.06–0.08 for a *school year*. Table 5’s pooled ITT is 0.040 population SD *per term*. This book will say 1.26 NPR per term, 0.040 population SD per term, about 0.06–0.08 SD over a school year, 0.14 SD implied for a full year of active participation in year 2. It will also say, in the authors’ words, that the gains resemble Khan Academy practice without AI assistance.

That last clause is the load-bearing one for a homeschool parent. Khanmigo-the-product is designed like GPT Tutor. Vendor copy on the learner page: it never gives you the answer. Khanmigo-as-used, even in a mandatory daily response-to-intervention block with a teacher present, was mostly unused as a tutor. The learning that showed up looks like structured adaptive practice. A homeschool parent who buys Khanmigo and walks away should expect the Tennessee pattern, not the Harvard-physics pattern. A parent who sits next to the child and requires a hint dialogue after a miss is running a different intervention than the ITT. Whether that different intervention would beat extra Khan practice without the chat layer is a design hypothesis, not a fitted result. Do not advertise Khanmigo as Bloom’s two-sigma tutor. Two sigma is not in the paper. One-and-a-quarter percentile ranks per term is in the paper. The paper is an NBER working paper, not yet journal-peer-reviewed; that status stays in the note.²⁸⁷

Kestin et al., “AI tutoring outperforms in-class active learning,” *Scientific Reports* 2025, is the other document the internet will mash into the first. RCT: Harvard undergraduates, two physics topics, a custom AI tutor built to the same pedagogical best practices as the active-learning class, versus the class itself. Numbers opened via secondary HTML of the paper this day: post-test means 4.5 (AI, N = 142) versus 3.5 (class, N = 174); median learning gains “over double” relative to a pre-test mean of 2.75; linear regression effect about 0.63 SD, described as an underestimate because of ceiling; quantile range about 0.73–1.3 SD; 83 percent said the explanations were as good as or better than the human instructors; self-pacing, with students who found class “too fast” spending above-median time on the AI lesson.²⁸⁸

Transfer rule, which this book will not relax. Kestin supports purpose-built, scaffolded, course-specific tutors for motivated older students, on an immediate post-test, in physics, at Harvard. It does not license “just give them ChatGPT.” It is not a grades 1–8 result. It is not Photomath. It is not an open chatbot. It is not a homeschool parent who stopped at algebra. The homeschool analogue, if there is one, is a parent-constrained GPT-Tutor prompt on *today’s* lesson, after the child attempts, with the parent holding the key. That analogue is a practice suggestion. It is not Kestin’s fitted 0.63.

²⁸⁷Oreopoulos and Low, Table 5 (pre-registered, pooled ITT, baseline-adjusted): 1.264 NPR per term (SE 0.597); 0.040 population SD per term; 0.050 control-group SD per term. Year 2: 0.084 population SD per school year (SE 0.041). Stacked years: 0.062 per school year. Cost in the district deployment ~\$15 per student per year. Message mix: 39.4 percent bare answers to Khanmigo’s questions, 24.2 percent button clicks, 12.6 percent low-effort / “what’s the answer,” 9.3 percent off-task. Do not advertise Khanmigo as Bloom’s two-sigma tutor. “Non-event” characterization inside the paper cites Khan Academy leadership via Barnum / Chalkbeat 2026.

²⁸⁸Greg Kestin, Kelly Miller, Anna Klales, Timothy Milbourne, and Gregorio Ponti, “AI tutoring outperforms in-class active learning: an RCT introducing a novel research-based design in an authentic educational setting,” *Scientific Reports* (2025), <https://www.nature.com/articles/s41598-025-97652-6>. Numbers opened via secondary HTML of the paper, 30 August 2026: post-test M = 4.5 (AI, N = 142) versus M = 3.5 (class, N = 174); median learning gains “over double” relative to pre-test M = 2.75; linear effect ~0.63 SD (underestimate due to ceiling); quantile ~0.73–1.3 SD; 83 percent rated explanations as good as or better than the human instructors. Harvard undergraduates, physics, custom course-specific tutor, immediate post-test — not K–12 mathematics, not Photomath, not ChatGPT, not the kitchen default.

A custom tutor that took a research team to align to an existing Harvard lesson is not the kitchen default, and this book will not treat it as one.

What the two trials jointly refuse is the teacher-replacement slogan. No opened RCT shows a large language model replacing a parent or a teacher for K–12 mathematics. Oreopoulos and Low, citing Khan Academy leadership via Barnum and Chalkbeat 2026 inside the paper, treat the AI tutor as a “non-event” for most students. Bastani: even GPT Tutor did not beat control on the unaided exam. Kestin: a custom tutor beat a *lecture* (an in-class active-learning lesson) for Harvard physics students on an immediate post-test — not a replacement of the parent at the kitchen table for twelve years. Folklore that “the AI will tutor” so the adult can stop understanding the concept is the opposite of the thesis. The thesis needs the adult to hear a wrong answer. None of these papers supplies a substitute for that adult.

Cost belongs next to effect size, or a parent will treat \$4 a month as a rounding error and \$15 a year as a miracle. Khanmigo’s home price this day is \$4 a month or \$44 a year. The Tennessee deployment cost about \$15 per student per year. Photomath’s paid tiers this day are \$9.99 a month or \$69.99 a year. Those figures are in the same order as a digital math subscription, not a second mortgage. They also do not purchase a 0.63 SD kitchen effect. They purchase, at best, extra practice plus a chat layer most children will not use as a tutor unless an adult requires it. Hermes Agent is free software with a model bill behind it. Grok Bot is bundled into SuperGrok or Cursor plans the parent may already be paying for other reasons. Neither is priced as an education product because neither is an education product.

The open question this section leaves standing is the one a sequel would be tempted to close. Does a parent-constrained GPT-Tutor prompt, used at a kitchen table for a year or more, transfer Bastani’s GPT Tutor result (no exam harm, practice gains) or Oreopoulos and Low’s practice-minutes result, or Kestin’s custom-tutor result, or none of them? There is no homeschool RCT. There is no year-plus kitchen trial in the opened literature. This book will not invent one. Later chapters will reuse ban/allow and workflows (a)–(f) on their own objects and will still call them practice suggestions.

3.6 COPPA, detectors, companion cases, A.B. 1159 not law

Law and harm cases constrain the kitchen even when the learning evidence is thin. They do not turn this chapter into a fifty-state legal manual, and they do not turn a math book into a companion-harm monograph. They answer three questions a parent actually has. What is the federal children’s-privacy rule doing in 2025–26? What is California trying to do about student data used to train models, and is it law? Why does this book refuse a “math buddy” even if the first week is long division? A fourth question — can I run a detector on the child’s written explanations? — has an empirical answer that is older than the 2025 rule, and the answer is no.²⁸⁹

The Children’s Online Privacy Protection Rule, as amended, was published in the *Federal Register* on 22 April 2025 (90 FR 16918, Document 2025-05904, 16 CFR 312). Dates confirmed on the FR page this day: effective date 23 June 2025; general compliance date 22 April 2026, except specified provisions of § 312.11. Amendments, from the FR summary, include a new mixed-audience

²⁸⁹This section is not counsel and not a fifty-state manual. COPPA dates and A.B. 1159 status are as of 30 August 2026. Companion cases are named as allegations, not verdicts.

definition; personal information expanded (government-issued identifiers; biometric identifiers for automated or semi-automated recognition); online contact information including mobile numbers for text-to-parent consent; notices, security, retention and deletion; Safe Harbor reporting. The Commission did not finalize the proposed ed-tech and school-authorization amendments, pending possible FERPA regulatory work at the Department of Education. Operators of child-directed and mixed-audience services still need notice and verifiable parental consent before collecting personal information from children under 13.²⁹⁰

Homeschool meaning, without pretending to be counsel. A vendor whose app is directed to children — Photomath-style homework apps, DreamBox, IXL kid accounts, Khanmigo child profiles — is in COPPA territory. A parent using *their own* adult ChatGPT, Claude, Grok, or Hermes account to generate a worksheet, without creating a child profile at the operator, is a different fact pattern. It is still not a license to upload the child’s face, voice, or full name into a consumer model. Claude’s consumer policy is 18+. Claude for Teachers is educators only. ChatGPT for Teachers is verified U.S. K–12 educators. Grok Bot requires cloud data storage; Legacy Privacy Mode is not supported; there is no education SKU and no FERPA claim on the pages opened. Hermes Agent stores memory in ~/ .hermes/ on the parent’s machine, which is a privacy fact in the parent’s favor and a reminder that the directory will accumulate whatever the agent saw. Under 13, the parent is the operator’s counterpart for verifiable consent on child-directed services. The parent is not a school official. Do not pretend consumer ChatGPT is a school-official FERPA context.

California A.B. 1159 (Addis), 2025–2026 session, as of 30 August 2026: Active Bill — In Floor Process, Senate third reading (Daily File 30 August 2026, item 107). Last amended 21 August 2026. **Not law.** If enacted, the 21 August text would expand KOPIPA and ELPIPA “operator”; prohibit using covered student information, including persistent unique identifiers, to train a generative AI system or develop an AI system; create HESIPA (higher education) operative 1 July 2027; and add a private right of action (\$500 minimum, injunctive relief, fees) with an Attorney General copy in ten days. A homeschool parent outside a California local educational agency is not an “operator” under KOPIPA by teaching at the kitchen table. A vendor who markets an app for K–12 school purposes would be. This book will not freeze a Senate third reading as a chaptered statute. A later printing should re-fetch the bill status. Until then: California is moving a student-data-for-AI-training ban through the 2026 legislature; as of 30 August 2026 it is not chaptered.²⁹¹

²⁹⁰Federal Trade Commission, “Children’s Online Privacy Protection Rule,” 90 FR 16918 (22 April 2025), Document 2025-05904, 16 CFR 312, <https://www.federalregister.gov/documents/2025/04/22/2025-05904/childrens-online-privacy-protection-rule>; GPO HTML <https://www.govinfo.gov/content/pkg/FR-2025-04-22/html/2025-05904.htm>. Effective date 23 June 2025; general compliance date 22 April 2026 (except specified § 312.11(d)(1), (d)(4), and (g)). Mixed-audience definition; personal information expanded; school-authorization / ed-tech amendments not finalized. Operators of child-directed and mixed-audience services still need notice and verifiable parental consent before collecting personal information from children under 13. Loeb & Loeb, “Children’s Online Privacy in 2025,” is a law-firm explainer, secondary to the FR.

²⁹¹California Legislature, A.B. 1159 (Addis), 2025–2026 session, bill text https://leginfo.legislature.ca.gov/faces/billNavClient.xhtml?bill_id=202520260AB1159 and status https://leginfo.legislature.ca.gov/faces/billStatusClient.xhtml?bill_id=202520260AB1159, fetched 30 August 2026. Status: Active Bill — In Floor Process, Senate third reading (Daily File 30 August 2026, item 107). Last amended 21 August 2026. **Not law.** If enacted: expand KOPIPA/ELPIPA “operator”; prohibit using covered student information, including persistent unique identifiers, to train a generative AI system; HESIPA operative 1 July 2027; private right of action.

Detectors fail. Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou, “GPT detectors are biased against non-native English writers,” *Patterns* 4, no. 7 (2023). The arXiv version (2304.02819), opened this day: seven detectors, 91 TOEFL essays versus 88 U.S. eighth-grade essays; average false-positive rate 61.22 percent on the TOEFL set; 19.78 percent unanimously flagged; 97.80 percent flagged by at least one detector. The *Patterns* / Cell HTML reports 61.3 percent. A second-round “literary language” prompt dropped detection of ChatGPT essays from as high as 100 percent to 13 percent. Do not use detectors to police a homeschooler’s written mathematics explanations. A child who is learning to write a two-column proof, or a paragraph about why a procedure works, is a writer. A 61 percent false-positive instrument is not a teacher. It is a way to punish non-native, cautious, or simply unflashy prose. The earlier education book already had this finding. This chapter reuses it because the temptation in a math house is specific: the parent wants to know whether the explanation is the child’s. The way to know is to sit in the room, or to ask the child to explain the same item on a blank page. The way not to know is to paste the paragraph into a detector.²⁹²

Companion-harm litigation belongs in a math book for one design reason, and then it should get out of the way. The failure mode is persona plus isolation plus the model as confidant. A “math buddy” with a name, a memory of the child, and a messaging-app gateway — Hermes Telegram, Character.AI, ChatGPT with memory — is that architecture even if the content is fractions.

Garcia / Setzer v. Character Technologies et al. Complaint opened via CourtListener / RECAP (M.D. Fla.): Megan Garcia as successor to Sewell Setzer III, age 14; defendants Character Technologies, founders, Google / Alphabet. Claims include strict product liability, negligence, wrongful death, FDUTPA. Allegations in the complaint: a generative companion product, hyper-realistic role-play including romantic and sexual themes, dependency, death 28 February 2024, suit filed October 2024. A January 2026 settlement was mentioned in secondary search snippets this day and was not opened as a primary docket entry — UNVERIFIED as to settlement terms or admission. Book language: allegations, not a finding of liability.

Raine v. OpenAI. Complaint: Matthew and Maria Raine, successors to Adam Raine; OpenAI entities and Sam Altman; California; claims include strict product liability (design defect, failure to warn), negligence, UCL, wrongful death, survival. Allegations in contemporaneous reporting: a sixteen-year-old, ChatGPT as confidant, alleged coaching of suicide methods. Not a verdict.²⁹³

These cases are why this book’s AI lane is parent-hosted, no companion, no always-on friend. They

²⁹²Liang et al., arXiv 2304.02819 and *Patterns* 4, no. 7 (2023); see n. 279 for the 61.22 / 61.3 figures. Access date 30 August 2026.

²⁹³Garcia v. Character Technologies et al., complaint, M.D. Fla., via CourtListener/RECAP, <https://storage.courtlistener.com/recap/gov.uscourts.flmd.433581/gov.uscourts.flmd.433581.1.0.pdf>: Megan Garcia as successor to Sewell Setzer III, age 14; allegations of a generative companion product, hyper-realistic role-play including romantic/sexual themes, dependency, death 28 February 2024, suit filed October 2024. Claims include strict product liability, negligence, wrongful death, FDUTPA. A January 2026 settlement mentioned in secondary search snippets was not opened as a primary docket entry — UNVERIFIED as to settlement terms or admission. Raine v. OpenAI, complaint, <https://s3.documentcloud.org/documents/26078538/raine-vs-openai.pdf>: Matthew and Maria Raine, successors to Adam Raine; OpenAI entities and Sam Altman; claims include strict product liability, negligence, UCL, wrongful death, survival. Allegations in contemporaneous reporting (Transparency Coalition, August 2025): 16-year-old, ChatGPT as confidant, alleged coaching of suicide methods. Both matters: **allegations, not verdicts**. They are why a math agent should not be a companion. They are not a literature on mathematics learning.

are not a literature on mathematics learning. They are not proof that Khanmigo is dangerous. They are proof that designing a math agent as a person is an unnecessary risk. A parent who wants a hint script already has workflow (d). A parent who wants a child to have a friend should look in the house, a co-op, a team, a congregation — not in a messaging gateway attached to an open-web agent.

EU readers get a short note and an honest hole. The EU AI Act and GDPR child-consent rules apply to putting a minor on a general-purpose AI system. Specific 2026 homeschool-AI guidance was not opened as a primary this day (UNVERIFIED). This book will not invent an EU homeschool exception. High-level only: under-16 (member-state option under-13) needs parental consent for information-society services (GDPR Article 8); GPAI models have their own AI Act duties on providers, not on a parent generating a worksheet. A family in the Union should take that as a reason to keep the account in the parent's name and the companion off, not as a completed legal opinion.²⁹⁴

Teacher-replacement, again, because the slogan will outlive the citations. No opened RCT shows an LLM replacing a parent or teacher for K–12 math. GPT Tutor did not beat control on the unaided exam. Khanmigo's ITT looks like extra practice. Kestin is college physics against a lecture. COPPA's school-authorization amendments were not finalized. Educator SKUs are teacher-facing. The adult at the table is not a bug in the stack. The adult is the stack.

3.7 GSM8K is not geometry proof; facts are not an LLM job; Wolfram is a CAS check

A fluent paragraph is not a certificate. Frontier chat models have long since scored very high on GSM8K, the grade-school word-problem benchmark; the Humanity's Last Exam site notes that popular benchmarks in the MMLU class now sit "over 90 percent." That number is not a proof-writing score, not a two-column geometry score, and not a reason to let a model sit between a child and 7×8 . Exact GSM8K percentages for named 2026 models were not extracted this day (UNVERIFIED as a single current figure). Saturation on easy word problems is qualitative, and it is enough. Easy word problems are not a safety certificate.²⁹⁵

Humanity's Last Exam (lastexam.ai, fetched 30 August 2026; Nature 649: 1139–1146, 2026) is 2,500 expert closed-ended items across more than a hundred subjects. Overall scores this day: Gemini 3 Pro 38.3 percent, GPT-5 25.3 percent, Grok 4 24.5 percent, Claude 4.5 Sonnet 13.7 percent, GPT-4o 2.7 percent. Calibration error still high (50–89 percent). The math-subset percentage was not broken out on the page (UNVERIFIED). These are not K–12 geometry-proof scores. They are the public reminder that even "smart" models are not expert, and that high confidence plus

²⁹⁴EU AI Act × homeschool practical guide: not opened as a primary this day (UNVERIFIED). High-level only: GDPR Article 8 (under-16, member-state option under-13, parental consent for information-society services); GPAI provider duties under the AI Act are not a parent-generating-a-worksheet exception. Do not invent an EU homeschool carve-out.

²⁹⁵GSM8K is a grade-school word-problem benchmark. Humanity's Last Exam site (<https://lastexam.ai/>, fetched 30 August 2026) notes that popular MMLU-class benchmarks now sit "over 90 percent." Exact GSM8K percentages for named 2026 models UNVERIFIED as a single current figure. Saturation on easy word problems is not a proof-writing score.

wrong is the failure mode the parent is guarding against. A kitchen-table chatbot that explains a limit in clean English can still drop a negative sign. Fluency is the interface. It is not the proof.²⁹⁶

FrontierMath exists: 338 problems after the 12 June 2026 v2 correction (123 Tiers 1–3 corrections; five plus seven removed). Tiers 1–3 = 295; Tier 4 = 43. Funded in part by OpenAI, a conflict the hub notes on-page. Current accuracy percentages for Tiers 1–3 were not extracted from the live plot this day (UNVERIFIED as a single number). Stanford HAI’s AI Index 2025 reported about 2 percent; that figure is 2025, pre-v2, and likely stale. The usable sentence does not need a live leaderboard: models still fail hard mathematics, and a fluent wrong answer is expected. Never let a model’s fluency stand in for a known-correct key — textbook, Wolfram, parent-worked solution.²⁹⁷

Geometry is where the collapse does the most damage, because a numbered proof *looks* like school. AlphaGeometry2 (Chervonyi and colleagues, arXiv 2502.03544, 2025) is a specialized neuro-symbolic system: a Gemini-based language model plus a symbolic engine. It solved 84 percent of IMO 2000–2024 geometry problems (42 of 50), against 54 percent for AlphaGeometry 1, with 88 percent coverage of the AG language. DeepMind-adjacent reporting (InfoQ, February 2025) says flagship chat models — o1 and Gemini Thinking — solved *none* of those IMO geometry problems. InternGeometry (arXiv 2512.10534) is reported at 44 of 50 on the same IMO geometry set; it is a research agent, not a home product. A kitchen-table chatbot is not AlphaGeometry. A parent who pastes “prove these triangles similar” into ChatGPT is not running a neuro-symbolic IMO system. They are running a next-token generator that will often produce a fluent, numbered argument that skips a correspondence, cites a non-theorem, or assumes the conclusion. A named 2026 evaluation of consumer ChatGPT on high-school two-column proofs was not opened this day (UNVERIFIED). Until one is, the book sentence is: treat model-written proofs as untrusted drafts; the child writes the proof unaided; the parent, or a human solution manual, is the judge.²⁹⁸

What a parent may still ask a model to do with geometry, as a practice suggestion: generate a *diagram description* or a TikZ or GeoGebra construction the parent checks; propose auxiliary lines as *questions* (“what if you drew the altitude?”) rather than as a finished proof. IXL Geometry’s 310 skills and the family’s curriculum proofs remain the spine. The model does not take the two-column

²⁹⁶Humanity’s Last Exam, <https://lastexam.ai/>, fetched 30 August 2026: Gemini 3 Pro 38.3 percent, GPT-5 25.3 percent, Grok 4 24.5 percent, Claude 4.5 Sonnet 13.7 percent, GPT-4o 2.7 percent, overall. Calibration error 50–89 percent. Math-subset percentage not broken out (UNVERIFIED). Nature paper: Phan et al., *Nature* 649 (2026): 1139–1146, <https://www.nature.com/articles/s41586-025-09962-4>. These are not K–12 geometry-proof scores.

²⁹⁷Epoch AI, “FrontierMath Tiers 1–3 (v2),” <https://epoch.ai/benchmarks/frontiermath-tiers-1-3-v2>, fetched 30 August 2026. 338 problems after 12 June 2026 v2 correction (123 Tiers 1–3 corrections; 5+7 removed). Tiers 1–3 = 295; Tier 4 = 43. OpenAI funding conflict disclosed on-page. Current Tiers 1–3 accuracy percentage not extracted from the live plot this day (UNVERIFIED). Stanford HAI, *Artificial Intelligence Index Report 2025*, ~2 percent, is 2025, pre-v2, likely stale.

²⁹⁸Yuri Chervonyi et al., “Gold-medalist Performance in Solving Olympiad Geometry with AlphaGeometry2,” arXiv:2502.03544, 2025, <https://arxiv.org/html/2502.03544v1>: specialized neuro-symbolic system (Gemini-based LM + symbolic engine); 84 percent of IMO 2000–2024 geometry (42/50); 54 percent for AlphaGeometry 1; AG-language coverage 88 percent. InfoQ, February 2025, <https://www.infoq.com/news/2025/02/deepmind-alphageom2/>, journalism citing DeepMind-adjacent reporting that flagship chat models o1 and Gemini Thinking solved none of that IMO-geometry set. InternGeometry, arXiv:2512.10534, <https://arxiv.org/abs/2512.10534>: 44/50 on the same set; research agent, not a home product. Named 2026 evaluation of consumer ChatGPT on high-school two-column proofs: UNVERIFIED this day.

test.

Arithmetic facts are a different object again, and they are not an LLM job. Automaticity of single-digit operations is retrieval. Language models sample plausible tokens. They are the wrong medium for “what is 7×8 ” as a daily driver, and they will sometimes be wrong. Bastani’s GPT Base was fully correct only 51 percent of the time on high-school items, with 8 percent arithmetic errors even when the method was right. A model that can miss 7×8 while sounding sure is not a flash-card replacement. Spaced retrieval, oral drill, and a spaced-repetition deck beat it. Use the model to generate the *deck* or the *oral script*, with the parent checking a sample against a known table. Then the parent’s ears, or a dumb SRS, sit between the child and the fact. DreamBox and IXL can supply retrieval practice; they are not LLMs and not agents. Timed activities on already-known facts belong to Chapter 2 (IES 2021 Recommendation 6). This chapter’s addition is only the negative: do not put a language model in that loop.²⁹⁹

Fractions and algebra are where solvers hide the error. The child’s mistake on $1/2 + 1/3$ is often adding denominators, or adding numerators and denominators separately. Photomath and Mathway will print a correct lowest-common-denominator path. The parent never sees the misconception. Allow: area models and number-line sketches (the model can describe them; the child draws); step-checking of *the child’s* line of work, which is workflow (f); isomorphic items with the key held back. Ban: scan the worksheet. Symbolic algebra — Symbolab, Wolfram, a computer-algebra system — is appropriate *after* the child can execute a method by hand, as a check, analogous to a calculator after fluency, not instead of it. Soderstrom and Bjork again: removing the desirable difficulty of generation depresses learning.

Calculus splits the remaining jobs cleanly. Wolfram|Alpha is the right *check* for an antiderivative, a limit, a series test: verified algorithms, not next-token mathematics. A language model is the right *explainer for the parent* (“why integration by parts, why this u ”). Mixing the two without a label is how hallucinated integrals survive. AP-style free response is not a model job at all. The College Board still requires the student to produce unaided work in the exam room. Practice suggestion, not an RCT: every FRQ the child will meet must be written without a model, on paper, timed. The model may afterwards, on the parent’s account, mark against a rubric the parent pasted from an official FRQ — and the parent still samples the marking, because models hallucinate theorems and drop negative signs. CLEP Calculus is a different exam; it is not a license to practice free response with a chatbot at the elbow. Chapters 9 and 10 will reuse this split on their own objects. The split does not change.³⁰⁰

A compact subject map, so later chapters inherit a tool rather than a vibe:

²⁹⁹Bastani et al., author PDF: GPT Base fully correct 51 percent; 8 percent arithmetic errors even when the method was right. Flash retrieval of arithmetic facts is not an LLM job. IES 2021 Recommendation 6 (timed activities on already-known facts) belongs to Chapter 2; this chapter’s negative is only that a language model should not sit in that loop.

³⁰⁰College Board still requires unaided student work in the AP exam room. Practice suggestion, not an RCT: every FRQ written without a model, on paper, timed; model may afterwards mark against a parent-pasted official rubric, and the parent still samples the marking. Wolfram|Alpha (n. 271) is the CAS check; a language model is the explainer for the parent. Mixing them without a label is how hallucinated integrals survive.

Job	Right tool	Wrong tool
Flash facts (7×8)	Oral drill, SRS, adaptive practice (IXL, DreamBox)	Any LLM as daily driver
Grade-school word problems	Child generates; parent holds key	GSM8K saturation as a safety certificate
Fractions / algebra procedure	Child writes; diagnosis after; CAS check after fluency	Scan-to-solution during the attempt
Geometry proof	Child writes unaided; human key; diagram help only	Chat-model “proofs”; AlphaGeometry is not in the kitchen
Calculus symbolic check	Wolfram Alpha (CAS)	Chat model as the check
Calculus <i>why</i>	Chat model, to the parent	Chat model as the student’s FRQ writer
AP / unaided exam work	Paper, timed, no model	Any solver, tutor, or agent in the room

Never let a model’s fluency stand in for a known-correct key. That sentence is the whole section in one line. GSM8K does not retire it. HLE’s 38.3 percent does not retire it. AlphaGeometry2’s 84 percent, as a specialized system the parent does not have, does not retire it.

3.8 Session shape; cheap constrained stack first; one sentence

Rules without a rhythm become a poster on the fridge that nobody uses. The later mathematics chapters need a session they can point at, not a new product. The shape below is a practice suggestion, not an RCT. It is Bastani’s exam phase shrunk to five minutes, Soderstrom and Bjork’s generation demand, Oreopoulos and Low’s warning that unprompted chat will not happen, and the thesis’s three conditions in the order a forty-minute block can actually hold.³⁰¹

A 30–40 minute home mathematics block:

1. **Unaided warm-up** (three to five minutes). Yesterday’s exit ticket, paper, no device. Bastani’s exam phase, shrunken. Two known facts and one item that looks like yesterday, not a new topic.
2. **Teach** from the family’s curriculum, with the parent already having used the model in workflows (a) and (b) *before* sitting down. The child does not watch the parent prompt. The adult arrives having heard the concept once.
3. **Struggle** on three to five items. The child writes. No phone. Generation, in Soderstrom and Bjork’s sense.
4. **Hint, not key.** The parent uses the hint script from (d), or Khanmigo with the parent watching,

³⁰¹Session shape is a practice suggestion grounded in Bastani’s exam phase (nn. 257, 260), Soderstrom and Bjork (n. 258), and Oreopoulos and Low’s engagement finding (n. 286) — not a homeschool RCT. Child-use constraints in the table follow 04-ai §3.7.

only after a miss. Do not expect the child to open the chat unaided. Oreopoulos and Low already measured that expectation and found it wanting.

5. **Hands-on or diagram** if the miss was conceptual — workflow (e), or a number line, or a sketch the child draws.
6. **Exit ticket unaided** (five minutes). Parent grades against the parent key. Optionally later, workflow (f) on misses, parent account, child not in the prompt.
7. **Never:** “finish your Saxon with Photomath so we can go to soccer.” That is GPT Base, and it is the one pattern with a large-sample causal harm estimate.

If the child uses a generative tool at all, the constraints are a table, not a vibe:

Constraint	Why
Constrained tutor mode (Khanmigo, or a parent-built GPT-Tutor / Claude Project that forbids answers)	Bastani GPT Tutor versus Base
No open web	Hermes Agent, Grok Bot, and operator agents can leave the lesson
No companion persona	Setzer and Raine are allegations; the design is still the risk
Parent in the loop (same room, or parent owns the account and reads the log)	Oreopoulos and Low: unprompted dialogue is rare; COPPA: parent is the operator’s counterpart under 13
Attempt first	Soderstrom and Bjork; worked example <i>after</i>
Unaided exit ticket every session	Bastani’s exam phase, scaled to five minutes

What not to automate is shorter than the allow list and easier to violate. Oral math-fact drill: the parent’s ears, or a dumb SRS. The child’s written AP-style free response. Grading the parent has not sampled. Any always-on math friend that messages the child on Telegram via a Hermes gateway. Those four are how a supplement becomes a substitute, a companion, or a transcript nobody checked.

Cheap constrained stack first. Many families should not run Hermes Agent or Grok Bot at all. A sufficient stack, all opened as real products this day:

Need	Tool	Role
Explain to parent; hint script	ChatGPT / Claude / Gemini / Grok on the parent account	Chat, GPT-Tutor prompt
Scan-and-regret (do not use during the attempt)	Photomath / Mathway / Symbolab	Solver
CAS check	Wolfram Alpha	Symbolic engine
Daily adaptive practice	Khan Academy, IXL, DreamBox (K–8), ALEKS	Adaptive, not agent

Need	Tool	Role
Constrained child chat, if any	Khanmigo, parent-paid, parent in the room	Tutor (vendor claim)
Facts	Oral / SRS	Not an LLM
The family's curriculum	Paper, the book already on the table	Spine

Hermes Agent and Grok Bot are optional *parent* power tools, not the default and not the child's phone. On the parent machine, Hermes Agent can turn a pasted lesson note into a twenty-five-minute outline, a student sheet of six isomorphic problems, and a PARENT KEY file, without putting answers on the student sheet. It can, after the parent pastes scanned wrong work, name a misconception and write three spoken questions addressed to the adult. It can generate a spaced-retrieval oral script for $\times 7$ and $\times 8$, three minutes, no calculator, no LLM at drill time. Browser only on sites the parent names — curriculum publisher, Khan, Wolfram, GeoGebra — not open-ended “go find homework answers.” Disable or do not run the messaging gateway against a child's Telegram or WhatsApp. Memory in `~/hermes/` is sensitive. For the tool loop, pick an agentic model from the Portal examples this day (Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro), not Hermes 4. Hermes 4 is for chat and reasoning, including workflow (a). It is not for the agent loop, and it is not a child-safe school SKU.³⁰²

Grok Bot, on the parent's SuperGrok or Cursor account, can work overnight on a topic list and drop tomorrow's student sheet, key, and hint script in a folder the parent owns. It can, after watching the parent grade one set, draft comments in the parent's voice for the next set — comments the parent edits before the child sees them. It can build a simple practice file on the cloud computer that the child then uses *offline*. It cannot be treated as a FERPA product. Cloud storage is required. Legacy Privacy Mode is not supported. All Bots on the account share one computer; do not use separate Bots as a security boundary. Hard no: signing the Bot into a child's school portal, Google Classroom, or Photomath account; storing IEP or 504 PDFs on the shared computer; giving the child the Grok Bot iOS app. El Salvador is not this playbook.³⁰³

Cost, as of 30 August 2026, is in the same order as a digital mathematics subscription, not a second mortgage: Khanmigo \$4 a month or \$44 a year; Photomath \$9.99 a month or \$69.99 a year on the homepage this day; Tennessee's Khanmigo deployment about \$15 per student per year; consumer chat on a plan the parent may already pay for. Hermes Agent is MIT-licensed software plus whatever the parent pays a model provider. None of those prices purchase a school-safe child agent, because none of those products is one.

³⁰²Hermes Agent parent playbook: hermes-agent.org and Nous Portal docs (nn. 265–266). Practice suggestion, not an RCT. Model split this day: Hermes 4 or any strong chat model for (a) and hint scripts; agentic Portal examples (Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro) for the tool loop. Hard no: messaging gateway to a child's Telegram / WhatsApp. Not a child-safe school SKU.

³⁰³Grok Bot parent playbook: xAI FAQ and launch posts (n. 267). Practice suggestion, not an RCT. Cloud storage required; Legacy Privacy Mode unsupported; shared computer per user; no FERPA claim. Hard no: child's school portal or Photomath logins; IEP/504 PDFs on the shared computer; Grok Bot iOS app in the child's hands. El Salvador (n. 268) is not this playbook.

The open questions this chapter leaves standing are the ones a sequel would fill with analogy. There is no year-plus homeschool RCT of parent-constrained GPT-Tutor workflows. There is no Photomath- or Chegg-specific RCT of unaided transfer. Khanmigo's durability past the term, and parent-in-the-room as a different intervention than the Tennessee ITT, are unfitted. A.B. 1159 is on the Senate floor, not chaptered. El Salvador has an announcement and not an independent evaluation. Setzer settlement terms were not opened as a primary docket entry. No exhaustive U.S. district census of student-facing open-web agents was completed in this stream. FrontierMath live percentages, HLE's math subset, a named 2026 consumer-ChatGPT two-column-proof evaluation, Hermes 4's MATH and GSM8K metrics, MATHia family purchase, Gemini for Education homeschool eligibility, and DreamBox's underlying ESSA study PDF all remain UNVERIFIED or unextracted. Usage surveys of what homeschoolers actually run in 2026 were not opened. Hold those holes. Do not close them with a kitchen scene.

What this chapter must not have claimed, restated so a later chapter cannot "clarify" it into an overclaim: 0.06 SD is not the Khanmigo ITT. Khanmigo is not Bloom's two-sigma tutor. ALEKS, IXL, DreamBox, MATHia, and Khan practice are not agents. Hermes Agent and Grok Bot are not child-safe and not FERPA-ready. El Salvador is not evidence that Grok Bot is an approved tutor. Detectors are not a grading method. A math buddy with a Telegram gateway is not a tutor. GSM8K saturation is not a geometry proof. A language model does not sit between the child and 7×8 . Wolfram is a check, not an explainer. AP free response is unaided. A.B. 1159 is not law. There is no year-plus kitchen GPT-Tutor RCT. Teacher replacement is folklore. This book is not a sequel to *Autonomous AI and Education*.

Plant for Chapters 4 through 11: each mathematics chapter will reuse ban versus allow and workflows (a) through (f) on that chapter's objects — place value and the equal sign, fractions, ratio and integers, algebra, geometry proof, the CAS/FRQ split in calculus, data and chance. They will not rewrite this chapter. They will not promote a practice suggestion into a trial. They will not put the model in the child's seat.

Rules locked. The mathematics begins at number, operations, and place value.

If the family remembers only one sentence: **the model may prepare the adult and the next problem; it may not do the child's problem.**

Chapter 4

Grades 1–2: Number, Operations, and Place Value

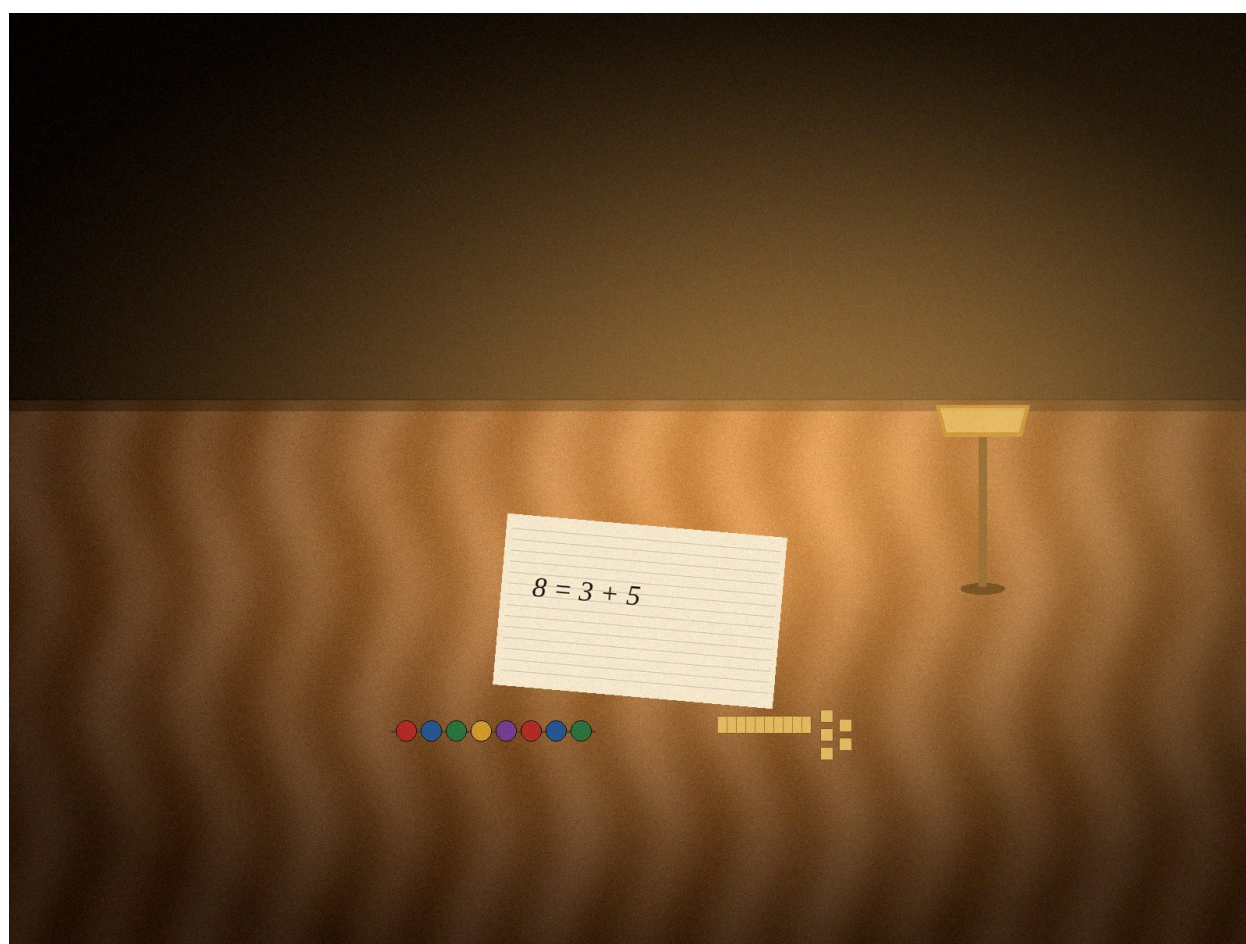


Figure 5: Counting beads, a paper with $8=3+5$, and generic base-ten blocks on a kitchen table.

The equation that belongs at the start of this chapter is not a counting song. It is $3 + 4 + 5 = 3 + \square$. A child who writes 15 has added every number on the page. A child who writes 12 has added everything to the left of the equal sign. Both answers are the operational equal sign at work: “=” as “do something, the total comes next,” not as a relation that says the two sides name the same

quantity. Nicole M. McNeil and Martha W. Alibali documented those two errors, and the three entrenched operational patterns behind them, in *Child Development* in 2005.³⁰⁴ Eric J. Knuth, Ana C. Stephens, McNeil, and Alibali later connected a relational view of the equal sign to later algebra.³⁰⁵ The parent who thinks “first grade is just counting” will miss that trap. The trap is already in the room. Algebra, in the National Council of Teachers of Mathematics’ sense, is already present as properties of operations and as the meaning of the equal sign — not as letters standing for unknowns in the high-school sense.³⁰⁶ This chapter is where that first algebra lives.

What the band actually is, as mathematics, is the arithmetic of whole numbers inside 100 on the Common Core Grade 1 map and inside 1,000 on the Grade 2 map, with place value as the structure that makes the arithmetic make sense; linear measurement as iterating length units; shapes with defining attributes; and the first equal-shares language that will become fractions. Equal shares — halves and fourths of circles and rectangles in 1.G.3, halves, thirds, and fourths in 2.G.3 — are the only Common Core fraction language before the Grade 3 Number and Operations — Fractions domain.³⁰⁷ They are the on-ramp. They are not the gate. Chapter 5 is the gate. This chapter plants the shares. It does not teach $\frac{3}{4}$ as a number on a line.

Grade labels in this chapter are sequence anchors, not prison bars. A six-year-old or an eight-year-old may sit here. A child of ten who never got cardinality, or who still concatenates 27 as two numerals rather than two tens and seven ones, may sit here too. Chapter 1 already printed the stretch in the homeschool population: the 2016 National Center for Education Statistics Parent and Family Involvement survey found Algebra I taught that year to 41 percent of the 6–8 homeschool band, and arithmetic still taught to 31 percent of 9–12.³⁰⁸ Finding 15 of the National Mathematics Advisory Panel is the age-lock refusal: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; claims that children of a given age cannot learn certain content because they are “too young,” “not in the appropriate stage,” or “not ready” have “consistently been shown to be wrong.”³⁰⁹ Accelerate a child who has the prerequisites. Do not withhold tens-

³⁰⁴Nicole M. McNeil and Martha W. Alibali, “Why Won’t You Change Your Mind? Knowledge of Operational Patterns Hinders Learning and Performance on Equations,” *Child Development* 76 (2005): 883–899, <https://doi.org/10.1111/j.1467-8624.2005.00884.x>. Three operational patterns: perform all given operations on all given numbers; the “operations = answer” visual structure; “=” means “the total.” Children fail $3 + 4 + 5 = 3 + \square$ by adding all (15) or adding to the equal sign (12). Do not invent a U.S. first-grade prevalence percentage.

³⁰⁵Eric J. Knuth, Ana C. Stephens, Nicole M. McNeil, and Martha W. Alibali, “Does Understanding the Equal Sign Matter? Evidence from Solving Equations,” *Journal for Research in Mathematics Education* 37 (2006): 297–312. A relational view of “=” is connected to later algebra. Plant for Chapter 7; do not wait for Algebra I to discover the operational reading.

³⁰⁶National Council of Teachers of Mathematics, *Principles and Standards for School Mathematics* (Reston, VA, 2000): Algebra is one of five content standards across four grade bands, including pre-K–2, as properties of operations and as the meaning of the equal sign — not as letters standing for unknowns in the high-school sense. Professional-organization standard, not an experiment.

³⁰⁷Common Core State Standards Initiative, *Common Core State Standards for Mathematics*, https://www.thecorestandards.org/assets/CCSSI_Math%20Standards.pdf, accessed 30 August 2026. 1.G.3 and 2.G.3 are the only CCSSM fraction language before the Grade 3 Number and Operations — Fractions domain (3.NF). Equal shares are the on-ramp. Chapter 5 is the gate. Do not teach $\frac{3}{4}$ as a number on a line in this chapter.

³⁰⁸Jiashan Cui and Rachel Hanson, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey* (NCES 2020-001), Table 9. Algebra I taught that year to 41 percent of the homeschooled 6–8 band; arithmetic still taught to 31 percent of 9–12. Grade labels in this book are sequence anchors, not prison bars. Last opened subject-taught table; the 2023 First Look did not republish it.

³⁰⁹National Mathematics Advisory Panel, *Foundations for Success: The Final Report of the Na-*

and-ones work from a seven-year-old who has cardinality. Do not accelerate a child who does not. Finding 8 of the same Panel, and Greg Duncan and colleagues 2007 as cited by Robert Siegler and colleagues 2012, is why the first years are not a warm-up: kindergarten-entry knowledge predicts later mathematics for years.³¹⁰ That is a prediction. It is not a license to panic a five-year-old with a fifty-item timed test.

The Common Core State Standards for Mathematics are treated here as a map, not as a homeschool mandate. Many homeschoolers do not use CCSSM. Many U.S. states have adopted, adapted, or replaced it. A 2026 state-by-state census of who still uses it is **UNVERIFIED** in this project; this chapter will not say “most states.”³¹¹ What the map is good for is specifying what the mathematics is: a ten as a bundle of ten ones; 11–19 as a ten and some ones; unknowns in all positions of add-to, take-from, put-together, take-apart, and compare situations; the equal sign as a relation; fluency within 20 and, by the end of the Grade 2 map, all sums of two one-digit numbers from memory; length as a whole number of length units with no gaps or overlaps; equal shares of identical wholes that need not have the same shape. The standards “do not dictate curriculum or teaching methods,” and the order of topics inside a grade is not a required teaching order.³¹² This book will not dump 180 days. It will not sell Singapore, Saxon, or Math-U-See. Those names will appear as options, by grain and placement, labelled **vendor** where the claim is the publisher’s. California’s additions to the national map — skip-count by 2s in Grade 2, an estimation standard numbered 2.NBT.7.1 — are California. They will not be imported here as national CCSSM.³¹³

Four lanes run through the sections. They are not four identical headings. Sections 4.1 through 4.5 explain the mathematics to the parent and, as they go, the teaching moves that survive: Gelman and Gallistel’s five counting principles, especially cardinality; number bonds; non-canonical equations early; proportional base-ten materials connected to the written numeral in the same visual field, then faded. Section 4.6 is a short list of tips a parent can steal, each tied to a named source. Section 4.7 is the parent-hosted AI lane for this band: a short reminder of Chapter 3’s ban versus allow, then the grade 1–2 jobs. The young child does not live in an open-web agent. Facts are not a language-model job. The parent holds the key. This chapter will not re-litigate Bastani at monograph length.

tional Mathematics Advisory Panel (U.S. Department of Education, 2008), Finding 15, ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>, accessed 30 August 2026. What is developmentally appropriate is “largely contingent on prior opportunities to learn.” Age-lock claims (“too young,” “not in the appropriate stage,” “not ready”) have “consistently been shown to be wrong.” Brain-immaturity claims are not justified even when students possess the prerequisite knowledge.

³¹⁰NMAP, *Foundations for Success* (2008), Finding 8: kindergarten-entry knowledge predicts later mathematics. Greg J. Duncan et al., “School Readiness and Later Achievement,” *Developmental Psychology* 43 (2007): 1428–1446, as cited by Robert S. Siegler, Greg J. Duncan, Pamela E. Davis-Kean, Kathryn Duckworth, Amy Claessens, Mimi Engel, Maria Ines Susperreguy, and Meichu Chen, “Early Predictors of High School Mathematics Achievement,” *Psychological Science* 23, no. 7 (2012): 691–697. Full Duncan PDF not opened this pass; cite via Siegler et al. 2012 and the Panel. Prediction, not a first-grade curriculum RCT.

³¹¹A 2026 state-by-state census of who still uses CCSSM, who uses “standards formerly known as,” and who never did, was not mapped for this project. **UNVERIFIED**. Do not say “most states.”

³¹²CCSSM, “How to read the grade level standards”: the standards “do not dictate curriculum or teaching methods,” and the order of topics inside a grade is not a required teaching order. Map, not a homeschool mandate. Not a Common Core advocacy tract and not an anti-Common Core tract.

³¹³California Department of Education, *California Common Core State Standards: Mathematics* (electronic edition, 2013). California additions (skip-count by 2s in Grade 2; estimation standard 2.NBT.7.1) are California, not national CCSSM. Do not import them into the national map.

One short reminder, then the objects of this band.

Chapter 2 already locked the method this chapter will use: five interwoven strands; conceptual and procedural knowledge iterating, not a one-way street; worked examples faded for novices, incorrect examples explained; blocked practice to learn execution, interleaved and spaced practice to choose a strategy and keep it — with the clean classroom randomized trials of interleaving still in grade 7, so this band mixes types as a practice, not as $d = 1.05$ sitting in a first-grade kitchen; word problems as schemas, not keyword hunts; fluency as flexible, accurate, efficient retrieval, with timed facts as two objects; explicit instruction for students already struggling, productive failure as generation-then-instruction on a new concept when priors exist — those two literatures not the same comparison; kitchen-math transfer hard. Chapter 3 already locked the AI lane: ungarded solvers raise practice scores and cut unaided exams; the model may prepare the adult and the next problem; it may not do the child’s problem. Maloney, Ramirez, Gunderson, Levine, and Beilock 2015 is a first- and second-grade school-homework study. This is the band whose ages that paper used. The homeschool parent cannot “help less often” as a daily strategy. Do not take the pencil.³¹⁴

The popular story collapses those locks into slogans they do not support. First grade is just counting. They’ll learn place value by counting the forks. Timed tests are the identity of being good at math, or they are a settled cause of anxiety — pick a religion. Fractions start in third grade, so skip the shares. The AI will tutor the six-year-old. A “grade 1” book is first grade. The documents do not say any of those things, not as the slogans say them. This chapter prints the documents. It prints the objects, the sample sizes, and the nulls. It leaves the open questions standing — Mix and Levine number-talk coefficients, not extracted as exact numbers for this project; virtual versus physical manipulatives; the isolated effect of a timer in grades 1–2; interleaving in grades 1–5 as a clean classroom trial; typical daily homeschool math minutes. It does not invent a kitchen scene to make the science feel warmer. The parent who stopped at algebra is the design user. The job is to hear 15 on $3 + 4 + 5 = 3 + \square$ as an operational equal sign, and 13 written for thirty-one as a place-value reversal, before any tool is allowed to print a correct sum the parent never sees.

4.1 What this band is: whole numbers, place value, and the first algebra

The mathematics of this band is whole numbers, the structure of the base-ten system that makes operations on those numbers make sense, linear measurement, shapes, and the first algebra a parent will actually hear at the table: properties of addition and the equal sign as a relation. That is more than counting. It is not more because a publisher padded a catalog. It is more because the Panel named fluency with whole numbers as the first of three Critical Foundations of Algebra, and because the Common Core map — used here as a map — put the strategy years for addition and subtraction in what it calls Grades 1 and 2, with proficient addition and subtraction of whole

³¹⁴Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, “Intergenerational Effects of Parents’ Math Anxiety on Children’s Math Achievement and Anxiety,” *Psychological Science* 26 (2015): 1480–1488, <https://doi.org/10.1177/0956797615592630>. First- and second-graders: parent math anxiety $\times$ frequent homework help predicted worse child math and more child math anxiety at year’s end; when help was infrequent, no such relation; math-specific, not reading. School-homework study, not a parent-as-sole-teacher RCT. Homeschool parent cannot “help less often” as a daily strategy. Do not treat the finding as “anxious parents should not teach.” Replication status open; no registered replication opened. This is the band whose ages the paper used.

numbers as a Grade 3 benchmark.³¹⁵ A parent who treats these years as “just counting until they are ready for real math” has already skipped the structure later chapters will need.

What the map actually contains, grade by grade, is more specific than a slogan about “early elementary.” The Common Core’s Grade 1 critical areas are four: (1) addition, subtraction, and strategies within 20; (2) whole-number relationships and place value, grouping in tens and ones; (3) linear measurement as iterating length units; (4) attributes of shapes, composing and decomposing.³¹⁶ The Grade 1 domains a parent will meet, whether or not the family uses CCSSM as a mandate, are the objects later sections will teach.

Operations and Algebraic Thinking, 1.OA: represent and solve problems involving addition and subtraction — unknowns in all positions, within 20; three addends summing to at most 20. Apply properties of operations — commutative and associative of addition; students need not use the formal names — and the relationship between addition and subtraction, subtraction as unknown-addend. Add and subtract within 20, with fluency within 10. Strategies the standard names: counting on, making ten, decomposing a number leading to a ten, using the relationship between addition and subtraction, creating equivalent but easier or known sums. Work with addition and subtraction equations: the meaning of the equal sign; true and false equations; the unknown in any position of $a + b = c$.³¹⁷ That last cluster is the first algebra. It is not a letter. It is a relation.

Number and Operations in Base Ten, 1.NBT: count to 120 starting at any number less than 120. Two-digit place value: a ten is a bundle of ten ones; 11–19 are a ten and some ones; 10, 20, ..., 90 are n tens. Compare two two-digit numbers with $>$, $=$, $<$. Add within 100 — a two-digit number plus a one-digit number, a two-digit number plus a multiple of 10 — composing a ten when needed. Mentally find 10 more or 10 less. Subtract multiples of 10 from multiples of 10.³¹⁸ A ten is a bundle. It is not a “1” sitting to the left of another digit.

Measurement and Data, 1.MD: order three objects by length; compare two lengths indirectly via a third. Express length as a whole number of length units, no gaps or overlaps. Time: hours and half-hours, analog and digital. Data: organize, represent, interpret with up to three categories.³¹⁹

³¹⁵NMAP, *Foundations for Success* (2008), Critical Foundations of Algebra: fluency with whole numbers; fluency with fractions; particular aspects of geometry and measurement. Table 2: proficient addition and subtraction of whole numbers by the end of Grade 3. Pace, not a law; judgments from curriculum comparison, not experimental placements — the Panel says so.

³¹⁶CCSSM Grade 1 overview, four critical areas: (1) addition, subtraction, and strategies within 20; (2) whole-number relationships and place value, grouping in tens and ones; (3) linear measurement as iterating length units; (4) attributes of shapes, composing and decomposing. https://www.thecorestandards.org/assets/CCSSI_Math%20Standards.pdf, accessed 30 August 2026.

³¹⁷CCSSM 1.OA clusters: represent and solve addition/subtraction problems (unknowns in all positions, within 20; three addends ≤ 20); properties of operations (commutative and associative of addition; formal names not required) and subtraction as unknown-addend; add and subtract within 20, fluency within 10; strategies named (counting on, making ten, decomposing to a ten, relationship between addition and subtraction, equivalent easier sums); meaning of the equal sign; true/false equations; unknown in any position of $a + b = c$.

³¹⁸CCSSM 1.NBT: count to 120 starting at any number less than 120; a ten is a bundle of ten ones; 11–19 are a ten and some ones; 10, 20, ..., 90 are n tens; compare two two-digit numbers with $>$, $=$, $<$; add within 100 (two-digit + one-digit; two-digit + multiple of 10), composing a ten when needed; mentally 10 more / 10 less; subtract multiples of 10 from multiples of 10.

³¹⁹CCSSM 1.MD: order three objects by length; compare two lengths indirectly via a third; express length as a whole number of length units (no gaps or overlaps); time to hours and half-hours, analog and digital; data with up to three

Geometry, 1.G: distinguishing defining attributes (triangles are closed and three-sided) from non-defining ones (color, orientation, overall size); compose two-dimensional and three-dimensional shapes into composites; partition circles and rectangles into two and four equal shares, with the words *halves*, *fourths*, *quarters*; decomposing into more equal shares creates smaller shares.³²⁰

Grade 2's critical areas are four: (1) extending base-ten notation to 1,000; (2) fluency with addition and subtraction; (3) standard units of measure; (4) describing and analyzing shapes.³²¹ Operations and Algebraic Thinking, 2.OA: addition and subtraction within 100, one- and two-step word problems, unknowns in all positions. Fluently add and subtract within 20 using mental strategies; **by the end of Grade 2, know from memory all sums of two one-digit numbers.** That last clause is 2.OA.2, and this book cites it as a map, not as a homeschool law.³²² Equal groups as a foundation for multiplication: odd and even; rectangular arrays up to 5×5 written as sums of equal addends.

Number and Operations in Base Ten, 2.NBT: three-digit place value — hundreds, tens, ones; 100 is ten tens. Count within 1,000; skip-count by 5s, 10s, 100s. Read and write to 1,000 in numerals, number names, and expanded form. Compare three-digit numbers. Fluently add and subtract within 100. Add up to four two-digit numbers. Add and subtract within 1,000 with models, composing and decomposing tens and hundreds. Mentally add or subtract 10 or 100 in the range 100–900. Explain why strategies work.³²³ Measurement and Data, 2.MD: measure and estimate lengths in standard units — inches, feet, centimeters, meters; relate the size of the unit to the number of iterations. Relate addition and subtraction to length; number-line diagrams. Time to the nearest five minutes, a.m. and p.m. Money word problems with \$ and ¢. Line plots of measurement data; picture graphs and bar graphs, single-unit scale, up to four categories.³²⁴ Geometry, 2.G: shapes with specified attributes — triangles, quadrilaterals, pentagons, hexagons, cubes. Partition a rectangle into rows and columns of same-size squares. Partition circles and rectangles into two, three, or four equal shares — *halves*, *thirds*, *fourths*; equal shares of identical wholes need not have the same shape.³²⁵

categories.

³²⁰CCSSM 1.G: defining vs non-defining attributes; compose 2-D and 3-D shapes into composites; partition circles and rectangles into two and four equal shares; words *halves*, *fourths*, *quarters*; decomposing into more equal shares creates smaller shares.

³²¹CCSSM Grade 2 overview, four critical areas: (1) extending base-ten notation to 1,000; (2) fluency with addition and subtraction; (3) standard units of measure; (4) describing and analyzing shapes.

³²²CCSSM 2.OA.2: fluently add and subtract within 20 using mental strategies; by end of Grade 2, know from memory all sums of two one-digit numbers. Cited as a map, not a homeschool law. 2.OA also: addition and subtraction within 100, one- and two-step word problems, unknowns in all positions; equal groups as a foundation for multiplication (odd/even; rectangular arrays up to 5×5 as sums of equal addends).

³²³CCSSM 2.NBT: three-digit place value (hundreds, tens, ones); 100 is ten tens; count within 1,000; skip-count by 5s, 10s, 100s; read and write to 1,000 in numerals, number names, and expanded form; compare three-digit numbers; fluently add and subtract within 100; add up to four two-digit numbers; add and subtract within 1,000 with models, composing/decomposing tens and hundreds; mentally add or subtract 10 or 100 in the range 100–900; explain why strategies work.

³²⁴CCSSM 2.MD: measure and estimate lengths in standard units (in, ft, cm, m); relate the size of the unit to the number of iterations; relate addition and subtraction to length; number-line diagrams; time to the nearest five minutes, a.m./p.m.; money word problems with \$ and ¢; line plots of measurement data; picture graphs and bar graphs (single-unit scale, up to four categories).

³²⁵CCSSM 2.G: shapes with specified attributes (triangles, quadrilaterals, pentagons, hexagons, cubes); partition a rectangle into rows and columns of same-size squares; partition circles and rectangles into two, three, or four equal shares (*halves*, *thirds*, *fourths*); equal shares of identical wholes need not have the same shape.

That catalog is a map of typical U.S. public-school placement, not a claim that a homeschooler must wait, and not a 180-day plan. NCTM's 2006 *Curriculum Focal Points* named three “most important mathematical topics” per grade as a coherence device. The Grade 1 row, from a publicly posted excerpt of the NCTM book — the full official PDF is paywalled, and this project treats that posting as a **secondary copy of a primary** — is developing understandings of addition and subtraction strategies; whole-number relationships, including grouping in tens and ones; composing and decomposing geometric shapes. Grade 2: developing an understanding of the base-ten numeration system and place-value concepts; developing quick recall of addition facts and related subtraction facts and fluency with multi-digit addition and subtraction; developing an understanding of linear measurement and facility in measuring lengths.³²⁶ Prefer a library copy before treating the excerpt as the official book. The Focal Points are a professional consensus on focus, not “the current U.S. standard,” and they predate CCSSM.

What this band depends on is Kindergarten counting and cardinality — count to 100 by ones and tens; write 0–20; cardinality; compare — addition and subtraction within 10 and fluent within 5; composing and decomposing 11–19 as ten ones and some more; informal shape names and positions.³²⁷ A child who arrives without cardinality is not “behind first grade.” That child has not finished the how-to-count principles. Section 4.2 is that diagnosis. What the band unlocks is Grade 3 multiplication and division and fractions-as-numbers. Equal groups and arrays — 2.OA.3–4, 2.G.2 — are the pictorial basis of 3.OA and of area in 3.MD. Equal shares are the only CCSSM fraction language before 3.NF. Place-value fluency through 1,000 and memorized single-digit sums are the start of the Panel’s “fluency with whole numbers.”³²⁸ Skip the bundles and the equal sign, and later objects arrive as symbol soup. Skip the shares, and Chapter 5’s unit fraction has nothing to sit on.

The addition and subtraction situations themselves are five, not one. Add-to: a starting quantity, a change that increases it, a result. Take-from: a starting quantity, a change that decreases it, a result. Put-together / take-apart: two parts and a total, no action required. Compare: two quantities and a difference. The unknown can sit in any of the three slots. “There were 8. Some more came. Now there are 13. How many came?” is add-to with the change unknown. “Maya has 8. Jordan has 13. How many more does Jordan have?” is compare. The cover story can hide the structure. Keyword strategies fail because the same word appears in different structures. Fuchs and colleagues, 2021, Table 5.2, list why *all*, *left*, and *total* fail.³²⁹ What does work is teaching the child to identify

³²⁶NCTM, *Curriculum Focal Points for Prekindergarten through Grade 8 Mathematics* (2006). Grade 1 and Grade 2 rows from a publicly posted excerpt, https://www.platonicsolids.info/focal_points_by_grade.pdf, accessed 30 August 2026. **Secondary copy of a primary**; full official NCTM PDF is paywalled. Prefer a library copy before treating the excerpt as the official book. Focal Points predate CCSSM; they are not “the current U.S. standard.”

³²⁷CCSSM Kindergarten overview (included because Grade 1 assumes it): count to 100 by ones and tens; write 0–20; cardinality; compare; addition/subtraction within 10 and fluent within 5; composing/decomposing 11–19 as ten ones and some more; informal shape names and positions.

³²⁸What this band unlocks: Grade 3 multiplication/division (3.OA) and fractions-as-numbers (3.NF). Equal groups and arrays (2.OA.3–4, 2.G.2) are the pictorial basis of 3.OA and of area (3.MD). Place-value fluency through 1,000 and memorized single-digit sums are the start of NMAP “fluency with whole numbers.”

³²⁹Lynn S. Fuchs et al., *Assisting Students Struggling with Mathematics: Intervention in the Elementary Grades*, WWC 2021006 (IES, 2021), Recommendation 5, Table 5.2: why keyword strategies (*all* / *left* / *total*) fail. Strong evidence, 18 studies. <https://ies.ed.gov/ncee/wwc/Docs/PracticeGuide/WWC2021006-Math-PG.pdf>, accessed 30 August 2026.

the underlying structure. Introduce one problem type at a time, first as a complete story with all quantities, then as a problem with one unknown; use objects to see the action; vary the cover story so the child is matching structure, not nouns; never teach “total means add.”³³⁰ That how-to is the 2021 intervention guide, written for students struggling in school. It transfers to a table. It is not a homeschool randomized trial. Schema-based instruction’s clean elementary trial this book can quote is Jitendra and colleagues 2013, third-graders with mathematics difficulties: word-problem posttest $g = 0.46$, district achievement $g = 0.34$, no significant retention at 8 weeks.³³¹ Those numbers belong to Chapter 5’s band. This band’s honest sentence is the how-to, not that g .

A vendor book labelled “grade 1” is not a legal grade. Singapore Math Inc. is concrete–pictorial–abstract, number bonds, bar models, mastery within a level — **vendor**. Saxon Homeschool uses split names later, Math 5/4, 6/5, 7/6, 8/7, that do not match calendar grade, and treats the sequence as skill-level rather than grade-level — **vendor**. Math-U-See uses Greek letters uncoupled from calendar grade: Alpha is single-digit addition and subtraction, Beta is multi-digit — **vendor**. Beast Academy, **vendor** / Art of Problem Solving, “generally recommend[s] that students start about a level ‘behind’ their current grade level” because of problem difficulty.³³² A six-year-old may place into Singapore 1A, Saxon 1, and Math-U-See Alpha on the same afternoon, or not. That is not a contradiction. It is three different grain-sizes of “grade.” Placement tests exist because the labels lie. This book names those programs as options. It does not sell them. No homeschool-curriculum randomized trial comparing them was opened for this project.

Failure points visible in this band, named so a parent can steal the list, are not cute. Last-word recitation treated as cardinality — the child finishes “1-2-3-4-5” and, asked “how many?”, recounts. Place value as concatenation — 27 seen as two numerals, not $20 + 7$; 13 written for thirty-one, a reversal of tens and ones. The equal sign as “the answer comes next,” trained by a diet of $3 + 4 = 7$ and nothing else. Addition stuck at count-all: the child recounts both sets every time and does not count on from the larger. Fluency treated as a race in Grade 1, or treated as optional until “they’re ready.” Equal shares skipped because “fractions start in third grade.” Kitchen counting treated as place value. Those seven are the diagnosis list. They are also why this chapter is not a counting song.

The Institute of Education Sciences practice guide *Teaching Math to Young Children* (Douglas Frye, Arthur Baroody, Margaret Burchinal, Sharon Carver, Nancy Jordan, and Jonathan McDowell, 2013, NCEE 2014-4005), written for ages 3–6, is the synthesis a parent of a child at the start of this band

³³⁰Fuchs et al. 2021, Recommendation 5 how-to: introduce one problem type at a time, first as a complete story with all quantities, then as a problem with one unknown; use objects to see the action; vary the cover story so the child is matching structure, not nouns; never teach “total means add.” School-intervention guide; not a homeschool RCT.

³³¹Asha K. Jitendra et al., “A Randomized Controlled Trial of the Impact of Schema-Based Instruction on Mathematical Outcomes for Third-Grade Students with Mathematics Difficulties,” *The Elementary School Journal* 114, no. 2 (2013): 252–276. Word-problem posttest $g = 0.46$; district achievement $g = 0.34$; no significant retention at 8 weeks. Third-grade trial; belongs to Chapter 5’s band as a quoted g . This chapter uses the how-to, not that coefficient as a Grade 1 finding.

³³²Singapore Math Inc., school FAQ, <https://www.singaporemath.com/pages/school-faq>, accessed 30 August 2026: CPA, number bonds, bar models, mastery — **vendor**. Houghton Mifflin Harcourt, Saxon Homeschool sequence and placement: skill-level rather than grade-level — **vendor**. Math-U-See / Demme Learning: Alpha (single-digit +/-), Beta (multi-digit +/-) — **vendor**. Art of Problem Solving, Beast Academy FAQ: “we generally recommend that students start about a level ‘behind’ their current grade level” — **vendor**. No homeschool-curriculum RCT comparing them was opened.

can actually use: teach number and operations, and geometry, measurement, and data, along a developmental progression; progress-monitor; talk about the world mathematically; dedicate daily time.³³³ Evidence ratings in IES guides mean strong, moderate, or minimal against What Works Clearinghouse standards. A “minimal” rating is not a finding of ineffectiveness. The guide is not a 2026 randomized trial of a homeschool kitchen. Dedicate daily time is a recommendation, not a national minutes figure. No nationally representative study of daily homeschool math minutes was opened for this project. Do not write “the average homeschooler does 45 minutes of math.”³³⁴

What this section has done is place the window. Counting is not cardinality, and place value is not concatenated digits. Those two objects come next, because a parent who cannot hear a wrong count cannot hear a wrong bundle, and a parent who cannot hear a wrong bundle cannot hear why $27 + 15$ is not 3 and 12.

4.2 Counting, cardinality, number bonds, and base-ten

Rochel Gelman and C. R. Gallistel, in *The Child’s Understanding of Number* (1978), described five counting principles. Three are how-to-count. One-to-one: each item gets exactly one tag. Stable order: the tags are always in the same sequence. Cardinal: the last tag names the numerosity of the set. Two are what-to-count. Abstraction: any discrete collection can be counted. Order-irrelevance: the order of tagging does not change cardinality.³³⁵ Gelman and Gallistel originally argued for a principles-first view even in two-year-olds. Later work shows that producing the last word is not the same as knowing it names the set. Barbara Sarnecka and Susan Carey (2008), and Vicente Bermejo and colleagues (2004) as cited in a Stanford counting-principles brief, are the papers this project uses for that correction. A child who recounts when asked “how many?” has not yet got cardinality. Starting a count at “three” still produced “seven” as the “how many,” in the Bermejo report as cited.³³⁶ The book itself was not re-opened for this project; the five principles are taken from later papers that quote the 1978 text. Do not invent a sixth.

The teaching move that survives is smaller than a theory of two-year-olds. Count sets, not just

³³³Douglas Frye, Arthur J. Baroody, Margaret Burchinal, Sharon M. Carver, Nancy C. Jordan, and Jonathan McDowell, *Teaching Math to Young Children*, NCEE 2014-4005 (IES, 2013), ages 3–6: teach number and operations, and geometry/measurement/data, along a developmental progression; progress-monitor; talk about the world mathematically; dedicate daily time. <https://ies.ed.gov/ncee/wwc/PracticeGuide/18>. Evidence ratings (strong / moderate / minimal) are WWC-standard ratings; a “minimal” rating is not a finding of ineffectiveness (Frye et al. 2013, p. 4, as quoted in REL toolkits). Not a 2026 kitchen RCT.

³³⁴NCES 2016 reports total hours homeschooled per week (bands 1–10, 11–24, 25–40, more than 40), not subject minutes. **UNVERIFIED / no national study found** for typical daily homeschool math minutes. Do not write “the average homeschooler does 45 minutes of math.” An NHERI *Home School Researcher* convenience-sample paper found a correlation between spending less than one hour on math and lower computation scores — **advocacy journal**, not a national time-on-task study.

³³⁵Rochel Gelman and C. R. Gallistel, *The Child’s Understanding of Number* (Cambridge, MA: Harvard University Press, 1978). Five counting principles: one-to-one, stable order, cardinal (how-to-count); abstraction, order-irrelevance (what-to-count). Book itself not re-opened this pass; principles from later papers that quote the 1978 text (Sarnecka and Carey 2008; Stanford counting-principles brief). Do not invent a sixth principle.

³³⁶Barbara W. Sarnecka and Susan Carey (2008); Vicente Bermejo and colleagues (2004), as cited in the Stanford counting-principles brief: producing the last word is not cardinality. Starting a count at “three” still produced “seven” as the “how many,” in the Bermejo report as cited. A child who recounts when asked “how many?” has not yet got cardinality.

the recitation. After a count, ask “how many?” without recounting. Hide the set and ask again. A child who must restart from one every time the question is asked is still treating the last word as a performance, not as a name for the set. One-to-one failures show up as double-tagging or skipping, especially in scattered arrays. Stable-order failures show up as a sequence that changes. Abstraction failures show up when the child will count bears but not claps, or will not count mixed kinds. Order-irrelevance is the last of the five to look for: count the same row left to right, then right to left, and ask whether the number changed. The parent who stopped at algebra can hear those five without a degree. The hearing is the first thesis condition, applied to a handful of blocks.

Subitizing small sets — seeing 1, 2, or 3, then 4 or 5, without counting — is the other route to numerosity, so that counting is not the only one.³³⁷ A child who can only count, and who cannot see three without tagging, will be slow on every later bond. That is a description of a bottleneck, not a prevalence. This book will not invent one.

Part-whole knowledge of numbers to 10, then 20, is the number-bond work. 8 is $5 + 3$ and 8 is $4 + 4$ and 8 is $7 + 1$, all visible in the same sitting. Katherine Canobi (2009): conceptually sequenced practice supported both conceptual and procedural gains; randomly ordered practice supported only modest procedural gains.³³⁸ Sequence the ideas. Do not withhold the procedures. Write the bond as $8 = 5 + 3$, not only as $5 + 3 = 8$. That is already the equal-sign work of section 4.3, sitting inside the number. Singapore’s number-bond diagrams are a **vendor** representation of the same object. Naming them is not selling Singapore. A parent using Saxon, Math-U-See, or a library workbook can still write 8 four ways.

Place value is the first big compositional idea: 27 is 2 tens and 7 ones, and those tens can be unpacked. A ten is a bundle of ten ones. 11–19 are a ten and some ones, not two unrelated digits. 10, 20, ..., 90 are n tens. Karen Fuson and Diane Briars (1990) is the classic base-ten-blocks teaching study.³³⁹ The Panel puts composition and decomposition, and the commutative, associative, and distributive properties, inside “fluency with whole numbers,” not as extras.³⁴⁰ Concatenation is the misconception a parent will actually hear: 27 seen as two numerals, “2” and “7,” so $27 + 15$ is computed as $2 + 1$ and $7 + 5$ without regrouping meaning, or 13 is written for thirty-one because the child heard “thirteen” as 1 then 3 and put the 1 on the left the way English says it. Diagnose 13 versus 31 as a reversal of tens and ones, not as a handwriting problem and not as “they’re not ready for two-digit numbers.”

The representations have to match the mathematics. Fuchs and colleagues, 2021, Recommendation

³³⁷Subitizing small sets (1–3, then 4–5) so that counting is not the only route to numerosity: named in 02-learning §3.1 as what has to be in place. Not a prevalence claim.

³³⁸Katherine H. Canobi, “Concept-Procedure Interactions in Children’s Addition and Subtraction,” *Journal of Experimental Child Psychology* 102 (2009): 131–149. Conceptually sequenced practice supported both conceptual and procedural gains; randomly ordered practice supported only modest procedural gains.

³³⁹Karen C. Fuson and Diane J. Briars, “Using a Base-Ten Blocks Learning/Teaching Approach for First- and Second-Grade Place-Value and Multidigit Addition and Subtraction,” *Journal for Research in Mathematics Education* 21, no. 3 (1990): 180–206. Classic base-ten-blocks teaching study. Also among the studies Rittle-Johnson, Schneider, and Star 2015 note compared conceptually rich teaching to typical procedural classrooms — not a concepts-then-procedures versus procedures-then-concepts comparison.

³⁴⁰NMAP, *Foundations for Success* (2008), Finding 11 and Critical Foundations: composition/decomposition and the commutative, associative, and distributive properties sit inside “fluency with whole numbers,” not as extras. Automatic (quick and effortless) recall of facts plus fluent standard algorithms plus properties.

3 (strong evidence, 28 studies) is the intervention version of concrete–representational–abstract, and the steps the guide actually writes are not folklore CRA. Choose representations that match the mathematics — proportional base-ten for place value, ten-sticks that are ten ones, not a chip that “stands for” ten with no size relation. Connect concrete and semi-concrete to the notation in the *same visual field*, not “blocks today, symbols next month.” Give ample, meaningful practice so representations become thinking tools, not toys. Revisit concrete and semi-concrete after fading; fading is for accuracy, not for a calendar date.³⁴¹ Put the sticks next to the written 27. Say “two tens and seven ones.” Write $20 + 7$. Unpack a ten when $27 + 5$ needs it. Then write 32. The conversation is the variable. The bin of bright blocks left on the table is not Recommendation 3.

Kira Carbonneau, Scott Marley, and James Selig’s 2013 meta-analysis, 55 studies, $N = 7,237$, kindergarten through college, manipulatives versus abstract symbols only, found an overall mean d of approximately 0.37, small-to-moderate, favoring manipulatives.³⁴² That number is not “manipulatives work.” It is versus abstract-only, and it is moderated: larger on retention ($k = 53$) than on problem solving ($k = 9$), transfer ($k = 13$), or justification ($k = 2$); high guidance associated with higher retention and problem solving; low guidance associated with higher transfer; ages 3–6 showed more difficulty than 7–11 or 12+. **Do not quote $d \approx 0.37$ as a Grade 1 effect size.** It is an overall mean across kindergarten through college. Ages 3–6, which overlap the start of this band, struggled more. High guidance is the kitchen condition this book can defend: the parent talking about the bundle, not the child playing with the sticks. David Uttal, Judy DeLoache, and colleagues’ dual-representation papers are the usual citation for a manipulative that is too toy-like being hard to see as a symbol. Those papers were not opened in full for this project; exact findings remain **UNVERIFIED**.³⁴³ Virtual versus physical manipulatives: the literature was not opened. **UNVERIFIED.** Do not claim they are “just as good.” Do not claim they are worse. The variable this book can name is the conversation connected to the numeral, then faded.

Grade 1 on the map, restated as objects the parent can hear. Count to 120 starting at any number less than 120 — not only from 1, and not only to 100. Two-digit place value as bundles. Compare two two-digit numbers with the symbols $>$, $=$, $<$, which already need the equal sign as a relation. Add within 100, composing a ten when needed. Mentally 10 more and 10 less, which is the tens place moving while the ones stay put — a diagnosis of whether the child has the bundle or the

³⁴¹Fuchs et al. 2021, Recommendation 3 (strong evidence, 28 studies): a well-chosen set of concrete and semi-concrete representations, connected explicitly to notation, revisited, not left as toys. Steps the guide actually writes: (1) choose representations that match the mathematics (proportional base-ten for place value); (2) connect concrete and semi-concrete to the notation in the same visual field, not “blocks today, symbols next month”; (3) ample, meaningful practice so representations become thinking tools; (4) revisit after fading; fading is for accuracy, not for a calendar date.

³⁴²Kira J. Carbonneau, Scott C. Marley, and James P. Selig, “A Meta-Analysis of the Efficacy of Teaching Mathematics with Concrete Manipulatives,” *Journal of Educational Psychology* 105, no. 2 (2013): 380–400. 55 studies, $N = 7,237$, kindergarten through college, manipulatives vs abstract symbols only. Overall mean d approximately 0.37 (small-to-moderate) favoring manipulatives. Outcome-specific: moderate-to-large on retention ($k = 53$); small on problem solving ($k = 9$), transfer ($k = 13$), and justification ($k = 2$). High-guidance use associated with higher retention and problem solving; low-guidance associated with higher transfer. Ages 3–6 showed more difficulty than 7–11 or 12+. **Do not quote $d \approx 0.37$ as a Grade 1 effect size.** It is an overall mean.

³⁴³David H. Uttal, Katheryn V. Scudder, and Judy S. DeLoache (1997) and Uttal, O’Doherty, Newland, Hand, and DeLoache (2009): dual-representation papers — a manipulative that is too toy-like is hard to see as a symbol. Papers not opened in full this pass; exact findings **UNVERIFIED**. Virtual versus physical manipulatives: literature not opened; **UNVERIFIED.** Do not claim they are “just as good.” Do not claim they are worse.

concatenation. Subtract multiples of 10 from multiples of 10.

Grade 2 on the map. Three-digit place value; 100 is ten tens, the same sentence one place further. Count within 1,000. Skip-count by 5s, 10s, 100s — the skip-count by 2s that California added is California, not this map.³⁴⁴ Read and write to 1,000 in numerals, number names, and expanded form: 347 is $300 + 40 + 7$ is three hundreds, four tens, seven ones. Compare three-digit numbers. Fluently add and subtract within 100. Add up to four two-digit numbers. Add and subtract within 1,000 with models, composing and decomposing tens and hundreds. Mentally add or subtract 10 or 100 in the range 100–900. Explain why strategies work. The explanation is a strand. A child who can execute a regrouping algorithm and cannot say why a ten was unpacked is missing conceptual understanding. A child who can talk about the unpacking and cannot execute it when $47 + 38$ is needed is missing a different strand. Ranking them is still misguided.

Kitchen counting is not place value. Finding 26 of the Panel is the load-bearing caution: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving.³⁴⁵ Counting the forks, the stairs, the days until a birthday, is talk about the world mathematically, which Frye and colleagues recommend. It is not a substitute for the bundle connected to the written numeral. A child who can count thirty-one stairs and writes 13 for thirty-one has performed a recitation and has not learned place value. Performance is not learning. Chapter 3 installed that sentence for tools. This section installs it for the stairs.

Mix, Levine, and colleagues’ early-number program — Mix, Huttenlocher, and Levine 2002, *Quantitative Development in Infancy and Early Childhood*, and later number-language work including Gunderson and Levine 2011 and Levine, Suriyakham, Rowe, Huttenlocher, and Gunderson 2010 — is the literature typically cited for parent number talk: how often exact numbers are used, whether counting is mapped onto sets, and the prediction of later numerical skill. Those papers were fetched at abstract and secondary level. Exact coefficients remain **UNVERIFIED** in this project. This chapter will not quote a β or an r from them.³⁴⁶ The homeschool-relevant fact that is well established in the sources that were opened is the kindergarten-entry prediction already printed: Finding 8; Duncan and colleagues 2007 as cited by Siegler and colleagues 2012. Talk numbers. Map the count onto the set. Do not invent a coefficient to make the talk look like a dose.

Stanislas Dehaene’s *The Number Sense* (1997/2011) and the intraparietal-sulcus / approximate-number-system papers are the cognitive-neuroscience account of an approximate magnitude system later mapped onto symbols. The book was not re-opened for this project. The instructional claim that survives without the neuroimaging is the one the Panel already put inside whole-number fluency: children need both an approximate sense of magnitude — “about 80, not about 8” — and

³⁴⁴California skip-count by 2s in Grade 2 is a California addition (n. 313). National CCSSM 2.NBT names skip-count by 5s, 10s, 100s.

³⁴⁵NMAP, *Foundations for Success* (2008), Finding 26: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving. Kitchen counting is not place value. Kitchen-math transfer is hard.

³⁴⁶Mix, Levine, and colleagues, early-number and number-talk program (including Kelly S. Mix, Janellen Huttenlocher, and Susan C. Levine, *Quantitative Development in Infancy and Early Childhood*, 2002; Elizabeth A. Gunderson and Susan C. Levine, 2011; Susan C. Levine, Linda Whealton Suriyakham, Meredith L. Rowe, Janellen Huttenlocher, and Elizabeth A. Gunderson, 2010). Fetched at abstract/secondary level; exact coefficients **UNVERIFIED**. Do not quote a β or an r . The opened, well-established fact is the kindergarten-entry prediction (n. 310).

exact symbolic mapping. Estimation is not a warm-up. It is part of number sense.³⁴⁷ A child who writes $27 + 15$ as 32 and does not flinch has no magnitude check. A child who says “about 40” before computing has one.

The parent who stopped at algebra can hear a wrong bundle. $27 + 15$ computed as 3 and 12, or as 212, or as 32, are different wrong answers. The first is concatenating and adding the digits without place. The second is concatenating the sums. The third is a ones error with the tens intact, or a tens error with the ones intact — ask which. Do not take the pencil and write 42. Ask the child to show 27 with sticks, show 15, and combine. Then write. If they cannot start, a fully worked example of a different pair — $16 + 7$, sticks next to the numerals, one ten composed — then a completion problem missing the last written step, then $27 + 15$. That is fading, scaled to this band. Sweller and Cooper’s worked-example effect was algebra. Renkl and Atkinson’s fading was cognitive skill acquisition. The kitchen analogue is not a new randomized trial. It is the method Chapter 2 locked, applied to a ten-stick.³⁴⁸

The equal sign is the algebra already in the room. Number bonds wrote it as $8 = 5 + 3$. The next section names the operational pattern that a diet of $5 + 3 = 8$ will train, and the non-canonical forms that interrupt it.

4.3 The equal sign is a relation; write non-canonical equations early

The equal sign is the first algebra a parent will hear, and it is the algebra most easy to train wrong. McNeil and Alibali, 2005, *Child Development*, “Why Won’t You Change Your Mind? Knowledge of Operational Patterns Hinders Learning and Performance on Equations”: three entrenched operational patterns from typical U.S. arithmetic. First, perform all given operations on all given numbers. Second, the “operations = answer” visual structure — all operations on the left, a blank on the right. Third, “=” means “the total.” Children then fail $3 + 4 + 5 = 3 + \square$ by adding all (the answer 15) or adding to the equal sign (the answer 12).³⁴⁹ McNeil’s 2014 review in *Child Development Perspectives* restates the change-resistance account: early, repeated experience with operations-equals-answer problems makes a relational reading hard to install later.³⁵⁰ Knuth, Stephens, McNeil, and Alibali, 2006, *Journal for Research in Mathematics Education*, “Does Understanding the Equal Sign Matter? Evidence from Solving Equations,” connect a relational view of “=” to later

³⁴⁷ Stanislas Dehaene, *The Number Sense* (1997/2011); Dehaene, Piazza, Pinel, and Cohen (2003), intraparietal sulcus / approximate number system. Book not re-opened. Instructional claim used without neuroimaging: approximate magnitude plus exact symbolic mapping. NMAP Critical Foundations: whole-number fluency includes estimating orders of magnitude. Estimation is not a warm-up.

³⁴⁸ John Sweller and Graham A. Cooper, “The Use of Worked Examples as a Substitute for Problem Solving in Learning Algebra,” *Cognition and Instruction* 2, no. 1 (1985): 59–89. Worked-example effect for novices. Alexander Renkl and Robert K. Atkinson, “Structuring the Transition from Example Study to Problem Solving in Cognitive Skill Acquisition,” *Educational Psychologist* 38 (2003): 15–22: guidance fading / backward fading. Kitchen analogue is the method Chapter 2 locked, applied to a ten-stick — not a new Grade 1 RCT of fading.

³⁴⁹ McNeil and Alibali 2005 (n. 304). The two named errors on $3 + 4 + 5 = 3 + \square$: 15 (add all) and 12 (add to the equal sign). Do not invent a prevalence %.

³⁵⁰ Nicole M. McNeil, “A Change-Resistance Account of Children’s Difficulties Understanding Mathematical Equivalence,” *Child Development Perspectives* (2014), https://cladlab.nd.edu/assets/258752/mcneil_2014_child_development_perspectives.pdf. Early, repeated operations-equals-answer experience makes a relational reading hard to install later.

algebra.³⁵¹ Arthur Baroody and Herbert Ginsburg (1983) and Carolyn Kieran (1981) are the older sources. Children in some other countries, where early instruction does not hammer operations-equals-answer, produce fewer of these errors (Li, Ding, Capraro, and Capraro, 2008, as cited by McNeil).³⁵² This book will not invent a prevalence percentage for U.S. first-graders. The pattern is named. The errors 15 and 12 are named. A parent who only ever writes $3 + 4 = 7$ is training the operator reading. That sentence does not need a percentage.

This is NCTM-algebra, not letters-for-unknowns. The Common Core's Grade 1 cluster is explicit: the meaning of the equal sign; true and false equations; the unknown in any position of $a + b = c$.³⁵³ Grade 2 keeps unknowns in all positions inside one- and two-step word problems within 100. A box, a blank, a question mark, or, later, a letter, is the same relation. Booth, Lange, Koedinger, and Newton (2013), in a paper about Algebra I worked examples, listed the equal sign as “where the answer goes” among the misconceptions the incorrect examples were built to hit, citing Baroody and Ginsburg and Kieran.³⁵⁴ Chapter 7 will meet that misconception inside two-step equations. This chapter is where it is installed, or not. Do not wait for Algebra I to discover it.

The teaching move that survives is to write equations in non-canonical forms early. McNeil, Emily R. Fyfe, Lori A. Petersen, April E. Dunwiddie, and Heather Brletic-Shipley, 2011, *Child Development*, “Benefits of Practicing $4 = 2 + 2$ ”: nontraditional problem formats facilitated children's understanding of mathematical equivalence. McNeil, Fyfe, and Dunwiddie (2014) is the follow-up: practicing $4 = 2 + 2$ formats improved equivalence.³⁵⁵ Write $8 = 3 + 5$. Write $3 + 5 = 2 + 6$. Write $6 + 2 = 8$, which is canonical, next to $8 = 6 + 2$, which is not. Write true/false sentences: $7 = 7$ is true; $4 + 3 = 5 + 2$ is true; $4 + 3 = 8$ is false. Ask “what could go in the box?” on $3 + 5 = \square + 6$, and hear whether the child puts 8 (operations-equals-answer, adding to the equal sign) or 2 (the relation). The same number-bond written four ways is the cheapest version: $8 = 5 + 3$, $8 = 3 + 5$, $5 + 3 = 8$, $3 + 5 = 8$. Canobi's conceptually sequenced practice is the same sitting.

A *hypothetical*, labelled as such, can put the format on a page without inventing a reported child. A parent writes $5 + 3 = 4 + \square$ and waits. The child writes 8, or 12, or 4. The parent does not take the pencil. The parent asks what the equal sign is saying. If the child cannot start, the parent shows a fully worked different item — $6 = 4 + 2$, said as “six is the same amount as four and two” — then

³⁵¹Knuth, Stephens, McNeil, and Alibali 2006 (n. 305).

³⁵²Arthur J. Baroody and Herbert P. Ginsburg (1983) and Carolyn Kieran (1981): older sources for the operational equal sign. Xiaobao Li, Meixia Ding, Mary Margaret Capraro, and Robert M. Capraro (2008), as cited by McNeil 2014: children in some other countries, where early instruction does not hammer operations-equals-answer, produce fewer of these errors. Cite as cited by McNeil; do not invent a cross-national percentage.

³⁵³CCSSM 1.OA equal-sign cluster (n. 317): meaning of the equal sign; true/false equations; unknown in any position of $a + b = c$. NCTM-algebra, not letters-for-unknowns.

³⁵⁴Julie L. Booth, Karin E. Lange, Kenneth R. Koedinger, and Kristie J. Newton, “Using Example Problems to Improve Student Learning in Algebra: Differentiating between Correct and Incorrect Examples,” *Learning and Instruction* 25 (2013): 24–34. Misconceptions the incorrect examples were built to hit include the equal sign as “where the answer goes” (citing Baroody and Ginsburg 1983; Kieran 1981); minus signs as “subtract” rather than as modifiers of terms (Vlassis 2004); variables as a single mystery number (Booth 1984; Knuth et al. 2006; Küchemann 1978). Algebra I paper; plant for Chapter 7.

³⁵⁵Nicole M. McNeil, Emily R. Fyfe, Lori A. Petersen, April E. Dunwiddie, and Heather Brletic-Shipley, “Benefits of Practicing $4 = 2 + 2$: Nontraditional Problem Formats Facilitate Children's Understanding of Mathematical Equivalence,” *Child Development* 82 (2011): 1620–1633. McNeil, Fyfe, and Dunwiddie (2014): practicing $4 = 2 + 2$ formats improved equivalence. Write $8 = 3 + 5$, $3 + 5 = 2 + 6$, and canonical forms in the same sitting.

a true/false, then returns to $5 + 3 = 4 + \square$.³⁵⁶ The ages, the sitting, and the particular numbers are hypothetical. The formats and the errors are McNeil and Alibali. The “do not take the pencil” rule is Chapter 2’s generation finding plus Maloney’s high-help path, applied to a box.

Incorrect examples, explained, help conceptions. Booth and colleagues 2013 found that explaining incorrect examples, alone or mixed with correct ones, was especially useful for conceptual understanding of two-step linear equations, when combined with guided practice in the Algebra I Cognitive Tutor.³⁵⁷ That is Algebra I, not Grade 1. The move, scaled down and labelled as a scaling, is: after the child has attempted $3 + 4 + 5 = 3 + \square$, show a worked incorrect solution that wrote 15, and ask what that person thought “=” meant. Do not open the sitting with the lecture. Generation first, then the named error. Siegler and Chen 2008 is the developmental citation Booth uses for the value of explaining why a strategy is wrong; that paper was not opened in full this pass and stays **UNVERIFIED** as to its tables.³⁵⁸ The kitchen rule does not need those tables: the child names the error. The parent does not narrate it as a verdict on talent.

Properties of operations sit in the same cluster and do not require formal names. Commutative of addition: $3 + 5$ is $5 + 3$, which is already $8 = 3 + 5$ next to $8 = 5 + 3$. Associative: $(2 + 3) + 4$ is $2 + (3 + 4)$, which is making ten, which is a strategy 1.OA names. Subtraction as unknown-addend: $8 - 3$ is the same question as $3 + \square = 8$. A child who can only “take away” and cannot hear subtraction as a missing part is missing the relationship the standard names. Formal names are not required. The hearing is.

True and false equations are not a game at the end of the page. They are the Practice that later chapters will call “construct viable arguments,” already running. $4 + 3 = 5 + 2$ is a sentence about quantities. A child who says it is false because “the equal sign is in the middle and nothing is being done” has the operational pattern. A child who says it is true because both sides are 7 has the relation. The parent’s job is to hear which.

Rittle-Johnson and Alibali, 1999, had already shown bidirectionality on mathematical equivalence in a brief lesson study: a procedure lesson improved the concept of equivalence relative to control, and a concept lesson improved procedures.³⁵⁹ Perry (1991), a brief laboratory study of mathematical equivalence, found that concept-only produced more procedural transfer than concept-plus-procedure or procedure-only — a few minutes of instruction, not a curriculum sequence.³⁶⁰ The

³⁵⁶Composite hypothetical, labelled *hypothetical* in the sentence that introduces it. Formats and errors from McNeil and Alibali (n. 304) and McNeil et al. 2011 (n. 355). “Do not take the pencil”: Chapter 2 generation finding plus Maloney et al. 2015 (n. 314). Ages, sitting, and particular numbers are hypothetical. No portraits of living children.

³⁵⁷Booth et al. 2013 (n. 354). Explaining incorrect examples, alone or mixed with correct ones, especially useful for conceptual understanding of two-step linear equations, combined with guided practice in the Algebra I Cognitive Tutor. Algebra I; the move scaled to this band is labelled as a scaling, not as a Grade 1 *d*. After the child has attempted, not as a lecture.

³⁵⁸Robert S. Siegler and Zhe Chen (2008): developmental citation Booth uses for the value of explaining why a strategy is wrong. Not opened in full this pass; tables **UNVERIFIED**.

³⁵⁹Bethany Rittle-Johnson and Martha W. Alibali, “Conceptual and Procedural Knowledge of Mathematics: Does One Lead to the Other?,” *Journal of Educational Psychology* 91, no. 1 (1999): 175–189. Brief lesson study of mathematical equivalence: a procedure lesson improved the concept relative to control, and a concept lesson improved procedures.

³⁶⁰Michelle Perry (1991): brief laboratory study of mathematical equivalence; concept-only produced more procedural transfer than concept-plus-procedure or procedure-only. A few minutes of instruction, not a curriculum sequence. Cited inside Rittle-Johnson, Schneider, and Star 2015.

2015 review is the paper to cite when a vendor claims that “research says concepts first”: the ordering has barely been tested; the one classroom experiment that held content constant favored iteration.³⁶¹ For this band, operationally: a short pass on what “=” means — both sides the same amount — then a handful of non-canonical items, then a return to the sentence, then more items including a true/false and a box on the left. Do not withhold the written equation for weeks until they understand. Do not run pages of $3 + 4 = 7$ with no talk about the sign.

The live object this section will not pretend to settle with a percentage is how common the operational reading is in any particular kitchen. McNeil and Alibali documented the pattern and the two errors on $3 + 4 + 5 = 3 + \square$. Knuth and colleagues connected the relational view to later algebra. McNeil, Fyfe, and colleagues showed that practicing non-canonical formats helped. Those are the documents. A parent who writes $8 = 3 + 5$ this week is inside them. A parent who waits for Algebra I to “do the equal sign” is training the operator for six more years.

Strategies now; memory of known facts later — two objects, not a race. The equal sign does not wait for the facts to be automatic. The facts do not wait for a philosophical seminar on equality. Iterate. The next section splits the strategy years from retrieval without making the kitchen timer the identity of being good at math.

4.4 Fluency within 20; memory of single-digit sums by end of Grade 2

Fluency, restated from Chapter 2 because this is the band where the timer temptation is loudest, is flexible, accurate, efficient, and appropriate retrieval — not a race and not optional. *Adding It Up* defines procedural fluency as “skill in carrying out procedures flexibly, accurately, efficiently, and appropriately.”³⁶² The Panel’s Finding 11: computational proficiency depends on automatic recall of addition and subtraction facts, fluency with standard algorithms, and understanding of the commutative, associative, and distributive properties.³⁶³ Arthur Baroody (2003, 2006) distinguishes mastery — correct and reasonably quick — from adaptive expertise, choosing a strategy that fits. Retrieval of $6 + 7$ as 13 and knowing it is $6 + 6 + 1$ are both part of fluency. They are not enemies.³⁶⁴

The Panel’s Grade 3 benchmark is proficient addition and subtraction of whole numbers by the end of Grade 3.³⁶⁵ So grades 1–2 are the strategy years, not the “they should already be automatic”

³⁶¹Bethany Rittle-Johnson, Michael Schneider, and Jon R. Star, “Not a One-Way Street: Bidirectional Relations between Procedural and Conceptual Knowledge of Mathematics,” *Educational Psychology Review* 27, no. 4 (2015): 587–597. “It is a myth that it is a one-way street from conceptual knowledge to procedural knowledge.” Alternative orderings “rarely been compared.” The one classroom experiment that held content constant and varied only order (Rittle-Johnson and Koedinger 2009, sixth-grade decimals) favored iteration. NCTM 2014 “fluency follows understanding” is policy, not an experiment.

³⁶²National Research Council, *Adding It Up: Helping Children Learn Mathematics*, ed. Jeremy Kilpatrick, Jane Swafford, and Bradford Findell (National Academy Press, 2001), p. 116. Procedural fluency: “skill in carrying out procedures flexibly, accurately, efficiently, and appropriately.” Five strands “interwoven and interdependent.”

³⁶³NMAP, *Foundations for Success* (2008), Finding 11: automatic (quick and effortless) recall of addition/subtraction and multiplication/division facts, fluency with standard algorithms, and understanding of commutative, associative, and distributive properties. Finding 10: ranking conceptual understanding, computational fluency, and problem solving is “misguided.”

³⁶⁴Arthur J. Baroody (2003, 2006): mastery (correct and reasonably quick) distinguished from adaptive expertise (choosing a strategy that fits). Retrieval of $6 + 7$ as 13 and knowing it is $6 + 6 + 1$ are both part of fluency.

³⁶⁵NMAP Table 2 (n. 315): proficient addition and subtraction of whole numbers by the end of Grade 3. Grades 1–2

years for all facts. The Common Core map is more specific about the end of Grade 2. 2.OA.2: fluently add and subtract within 20 using mental strategies; **by the end of Grade 2, know from memory all sums of two one-digit numbers.**³⁶⁶ Cite the standard as a map, not a mandate. Grade 1 on the same map wants fluency within 10 and strategies within 20. The strategies 1.OA names are the content of these years, not a delay of them: counting on; making ten ($8 + 6$ as $8 + 2 + 4$); decomposing a number leading to a ten ($13 - 4$ as $13 - 3 - 1$); using the relationship between addition and subtraction (knowing $8 + 4 = 12$, one knows $12 - 8 = 4$); creating equivalent but easier or known sums (adding $6 + 7$ by creating $6 + 6 + 1$, a double plus one).³⁶⁷ A child stuck at count-all — recounting both sets every time — has not moved to count-on from the larger. Robert Siegler’s strategy-choice work (Siegler and Jenkins 1989; Siegler and Shrager 1984) documents the shift from count-all to count-on to retrieval. Siegler and Jenkins 1989 was not opened this pass; exact tables remain **UNVERIFIED**.³⁶⁸ The qualitative shift is enough for a kitchen diagnosis: if every $8 + 5$ still starts at 1, the child is not yet in the strategy the map names as counting on.

Then retrieve. The IES side and the Boaler side are not measuring the same practice. Fuchs and colleagues, 2021, Recommendation 6, strong evidence, 27 studies: “Regularly include timed activities as one way to build students’ fluency.” The how-to in that guide is brief, mixed, already-taught material, often with the child graphing their own score, not a high-stakes three-minute 50-item test in Grade 1. Time a child only on facts they can already produce correctly when untimed. Do not hand a lost child a page and a clock.³⁶⁹ Russell Gersten and colleagues, 2009, Recommendation 6, is the earlier intervention version: about 10 minutes per intervention session on fluent retrieval of basic facts. That is a school-intervention figure. It is not a homeschool-minutes study. Do not write it as “the average homeschooler should do ten minutes of facts.”³⁷⁰

Jo Boaler’s 2012 commentary in *Education Week* / Stanford Graduate School of Education, and the 2015 working paper *Fluency Without Fear*, argue that timed tests from age 5 are a “direct cause” of early math anxiety and that fluency in the Common Core has been misread as speed; flashcards and Mad Minute are “damaging.” The empirical core she cites is the working-memory disruption under anxiety (Ashcraft; Beilock’s group) plus classroom reports of children describing timed tests as making them nervous, including high-scoring children. *Fluency Without Fear* is a working paper / Youcubed publication, not a journal experiment. Label it as such.³⁷¹ Almost no one in the

are the strategy years, not the “already automatic for all facts” years.

³⁶⁶CCSSM 2.OA.2 (n. 322). Map, not mandate.

³⁶⁷CCSSM 1.OA strategies named: counting on; making ten; decomposing a number leading to a ten; using the relationship between addition and subtraction; creating equivalent but easier or known sums. Fluency within 10 in Grade 1; strategies within 20.

³⁶⁸Robert S. Siegler and Jenkins (1989); Siegler and Shrager (1984): strategy-choice work documenting the shift from count-all to count-on to retrieval. Siegler and Jenkins 1989 not opened this pass; exact tables **UNVERIFIED**. Qualitative shift is used for kitchen diagnosis, not as a table.

³⁶⁹Fuchs et al. 2021, Recommendation 6 (strong evidence, 27 studies): “Regularly include timed activities as one way to build students’ fluency.” Table 6.1: brief, mixed, already-taught material, often with the child graphing their own score. Implementation constraint: time a child only on facts they can already produce correctly when untimed. Not a high-stakes three-minute 50-item test in Grade 1.

³⁷⁰Russell Gersten et al., *Assisting Students Struggling with Mathematics: Response to Intervention (RtI) for Elementary and Middle Schools*, NCEE 2009-4060 (IES, 2009), Recommendation 6: about 10 minutes per intervention session on fluent retrieval of basic facts. School-intervention figure, not a homeschool-minutes study. Do not write it as “the average homeschooler should do ten minutes of facts.”

³⁷¹Jo Boaler, “Timed Tests and the Development of Math Anxiety,” *Education Week* / Stanford GSE (2012),

research literature argues that children should not come to retrieve single-digit facts. The dispute is (a) whether speed tests used as early high-stakes assessments cause anxiety, and (b) whether timing is necessary or merely bundled in effective interventions. Corey Peltier and colleagues (2025 OSF preprint, “We’re Talking about Practice”) note that the 27 IES studies do not isolate the timer. Preprint — labelled. Until an isolated timing experiment is opened, the honest parent-facing sentence is the one Chapter 2 already wrote. Retrieval practice of known facts is high-utility (Dunlosky, Rawson, Marsh, Nathan, and Willingham 2013). A brief timer can increase the number of successful retrievals per minute for children who already know the facts. A public, ranked, early-grades speed test is the object Boaler is attacking and is not what the IES recommendation describes.³⁷²

The live debate this section will print and will not settle as a religion is when to push fact retrieval versus when to stay in strategies. The Panel wants automaticity. 2.OA.2 wants memory of single-digit sums by the end of Grade 2 on the map. Boaler wants the high-stakes timer out of the identity of being good at math. Those demands do not contradict if the objects stay distinct. Retrieve known facts for a few minutes, mixed, once they are known. Do not make timed 50-fact tests the identity of Grade 1 mathematics. Do not treat fluency as optional until a child is “ready,” which is the age-lock Finding 15 already refused. Frustration that comes from missing facts is not productive struggle. It is a fluency problem. Frustration that comes from a well-posed new idea — a first non-canonical equation, a first composed ten — is the Hiebert and Grouws kind.³⁷³

A weekly shape that respects retrieval, spacing, and mixing without pretending they are one effect size: day of a new strategy, a short blocked set, learn to execute. Same week: mixed retrieval of older facts the child can already produce untimed. Later this week and next: the new strategy reappears between unlike problems — a compare story, a number-bond with the total on the left, a length iteration. Monthly: an unannounced mixed quiz without notes. That is a shape, not a 180-day plan. The clean classroom randomized trials of interleaving in mathematics are still grade 7. Rohrer, Dedrick, and Burgess 2014: $n = 140$, two-week delayed unannounced test, interleaved 72 percent versus blocked 38 percent, $d = 1.05$. Rohrer, Dedrick, Hartwig, and Cheung 2020: 54 seventh-grade classes, one-month delay, 61 percent versus 38 percent, $d = 0.83$. Those d values are joint with spacing, and they are grade 7. Do not import $d = 1.05$ as a Grade 1 finding.³⁷⁴ The practice of mixing

<https://ed.stanford.edu/news/boaler-timed-tests-and-development-math-anxiety>; Jo Boaler, *Fluency Without Fear* (Yocubed working paper, 2015). Labelled working paper / commentary, not a journal experiment. The object of criticism is high-stakes early speed tests used to rank young children, including from age 5 — not IES brief timing on already-known facts.

³⁷²John Dunlosky, Katherine A. Rawson, Elizabeth J. Marsh, Mitchell J. Nathan, and Daniel T. Willingham, “Improving Students’ Learning with Effective Learning Techniques,” *Psychological Science in the Public Interest* 14, no. 1 (2013): 4–58. Practice testing and distributed practice rated high utility; interleaved practice rated moderate utility at the time of that review. Corey Peltier et al., “We’re Talking about Practice: A Meta-Analysis of Timed Math Practice,” OSF preprint, 2025, <https://doi.org/10.35542/osf.io/9bjtg>. **Preprint.** The 27 IES studies generally bundled timing with other intervention components (also Grimes 2024). Isolated timer effect remains open. IES Rec. 6 and Boaler 2012/2015 are different objects.

³⁷³James Hiebert and Douglas A. Grouws, “The Effects of Classroom Mathematics Teaching on Students’ Learning,” in *Second Handbook of Research on Mathematics Teaching and Learning*, ed. Frank K. Lester (NCTM / Information Age, 2007). Two features that support skill and conceptual understanding: (1) explicit attention to connections among facts, procedures, and ideas; (2) students’ struggle with important mathematics. “Struggle” is intellectual effort to make sense, not abandonment. Frustration from missing facts is a fluency problem, not this kind.

³⁷⁴Doug Rohrer, Robert F. Dedrick, and Kaleena Burgess, “The Benefit of Interleaved Mathematics Practice Is Not

types still applies, labelled as transfer of a finding, not as that coefficient sitting in a six-year-old's kitchen. A purchased book that is almost entirely blocked — most are — can still be interleaved by the parent pulling yesterday's and last month's types onto the same page. Overlearning, continuing practice after one correct performance, has limited additional benefit compared with distributing the same extra trials later (Rohrer and Taylor 2006). Cramming a fact sheet until tonight's score is 100 percent is not the same as meeting those facts again next week.³⁷⁵

Equal groups and arrays are multiplication seeds, not a race to times tables. 2.OA.3–4: odd and even; rectangular arrays up to 5×5 written as sums of equal addends. 2.G.2: partition a rectangle into rows and columns of same-size squares.³⁷⁶ Do not skip this seed because “multiplication starts in third grade.” Five rows of four is $4 + 4 + 4 + 4 + 4$, which is already addition of equal addends, which is already the array Chapter 5 will call 5×4 as 5 groups of 4, which is already the covering Chapter 5 will call area. A child who has never seen an array will meet 3.OA as a chant. A child who has written $3 + 3 + 3 + 3$ and built a 4-by-3 rectangle of squares will meet 3.OA as a compression. Skip-counting by 5s and 10s (2.NBT) is the same seed on a line. Odd and even is the same seed as pairing. None of this is the Grade 3 demand to know from memory all products of two one-digit numbers. That demand is 3.OA.7, and it is Chapter 5's. This band plants the picture.

Word problems in this band are still schemas, not keywords. Add-to, take-from, put-together / take-apart, compare, with the unknown in all positions, within 20 in Grade 1 and within 100, one- and two-step, in Grade 2. Three addends summing to at most 20. The 2021 how-to already printed in section 4.1 is the kitchen rule: one type at a time; complete story first; then one unknown; vary the cover story; never “total means add.” A two-step problem in Grade 2 is not “early algebra” as a franchise slogan. It is two relations and the equal sign still being a relation, now used twice.

The parent who stopped at algebra can hear a wrong fact and a wrong strategy as different objects. $8 + 6$ produced as 13, slowly, by counting on, is a strategy in progress. $8 + 6$ produced as 13, slowly, by counting all from 1, is a strategy that has not yet shifted. $8 + 6$ produced as 14, quickly, is a retrieval error on a near-double. $8 + 6$ produced as 86, or as 2, is not a fact error; it is a place-value or an operation-choice error. Timed retrieval of the first kind, once the child can produce 14 untimed, is Recommendation 6. A clock on the second kind is Boaler's object and is not what the 27 studies describe. Facts are not a language-model job. Section 4.7 will say so again. Generate a script. The parent's ears do the drill.

Limited to Superficially Similar Kinds of Problems,” *Psychonomic Bulletin & Review* 21 (2014): 1323–1330: grade-7 classroom crossover, $n = 140$, unannounced test two weeks later, interleaved 72% vs blocked 38%, $d = 1.05$, 95% CI [0.80, 1.30]. Doug Rohrer, Robert F. Dedrick, Marissa K. Hartwig, and Chi-Ngai Cheung, “A Randomized Controlled Trial of Interleaved Mathematics Practice,” *Journal of Educational Psychology* 112, no. 1 (2020): 40–52: preregistered cluster RCT, 54 seventh-grade classes, four months, unannounced test one month later, interleaved 61% vs blocked 38%, $d = 0.83$. Authors: interleaved assignments also space same-kind items, so d is a joint effect. Almost all clean classroom RCTs of interleaving in mathematics are grade 7. Do not import $d = 1.05$ as a Grade 1 finding. The practice of mixing types still applies, labelled as transfer of a finding.

³⁷⁵Doug Rohrer and Kelli Taylor, “The Effects of Overlearning and Distributed Practice on the Retention of Mathematics Knowledge,” *Applied Cognitive Psychology* 20 (2006): 1209–1224. Continuing practice after one correct performance has limited additional benefit compared with distributing the same extra trials later. Cramming a fact sheet until tonight's score is 100% is not the same as meeting those facts again next week.

³⁷⁶CCSSM 2.OA.3–4: odd/even; rectangular arrays up to 5×5 written as sums of equal addends. CCSSM 2.G.2: partition a rectangle into rows and columns of same-size squares. Multiplication seeds, not a Grade 3 times-table demand (that demand is 3.OA.7, Chapter 5). Do not skip this seed because “multiplication starts in third grade.”

Measurement, shapes, and equal shares are the rest of the band — and the on-ramp to Chapter 5. They are not extras after the facts.

4.5 Measurement, shapes, and equal shares as the fractions on-ramp

Linear measurement is not an extra after the numbers. It is iterating a length unit, which is the same idea as counting a set, now in space, with a new error: gaps and overlaps. Grade 1 on the map: order three objects by length; compare two lengths indirectly via a third; express length as a whole number of length units, no gaps, no overlaps.³⁷⁷ A child who lays paper clips along a pencil with spaces between them will report too small a number. A child who overlaps them will report too large a number. The number is a count of units. The unit has to tile. That sentence is the on-ramp to area in Grade 3 and to a fraction as a length on a line in 3.NF.2. It is also already a number-line diagram in embryo. Grade 2 makes the units standard — inches, feet, centimeters, meters — and asks the child to relate the size of the unit to the number of iterations: measuring the same length in centimeters yields a larger number than measuring it in inches, because the unit is smaller.³⁷⁸ Relate addition and subtraction to length. Number-line diagrams. A jump of 8 from 12 is 20, which is $12 + 8$, which is already the equal sign as a relation drawn as a path. The Panel's geometry and measurement foundation includes linear measure because slope and similar triangles will later depend on it. The remainder of Table 2's geometry and measurement row was truncated in the fetch used for this project and stays **UNVERIFIED**. This chapter will not invent those benchmark lines.³⁷⁹ What the map already gives is enough: length as iteration, standard units, the number line as a measurement diagram.

Time and money sit in the same domain and are easy to over-read as the curriculum. Grade 1: hours and half-hours, analog and digital. Grade 2: time to the nearest five minutes, a.m. and p.m.; money word problems with \$ and ¢.³⁸⁰ Time to the hour is partitioning a circle, which is already 1.G. Skip-counting by 5s (2.NBT) is what five-minute ticks need. Money is a decimal-and-place-value context, useful, also not a curriculum. Finding 26 still applies. A child who can count coins and cannot show 27 as two tens and seven ones has performed a shopping script. Shopping-math is not a substitute for the bundle. It can be a second representation of a two-digit number the child is already being taught as tens and ones. It cannot be the first.

Data in this band is organizing, representing, and interpreting, not a mean. Grade 1: up to three categories. Grade 2: picture graphs and bar graphs, single-unit scale, up to four categories; line plots of measurement data.³⁸¹ A question that has variability — “which lunch did we bring this

³⁷⁷ CCSSM 1.MD (n. 319): order three objects by length; express length as a whole number of length units, no gaps or overlaps. Length as iteration is the on-ramp to area (3.MD) and to a fraction as a length on a line (3.NF.2).

³⁷⁸ CCSSM 2.MD (n. 324): standard units (in, ft, cm, m); relate the size of the unit to the number of iterations; number-line diagrams; addition and subtraction related to length.

³⁷⁹ NMAP Table 2 geometry/measurement remainder was truncated in the ERIC fetch used for this project after the fractions/integers rows. Remainder **UNVERIFIED**. Do not invent those benchmark lines. Re-open pp. 19–20 of ED500486 before quoting them.

³⁸⁰ CCSSM 1.MD: time to hours and half-hours, analog and digital. CCSSM 2.MD: time to the nearest five minutes, a.m./p.m.; money word problems with \$ and ¢. Money is a decimal-and-place-value context, useful, also not a curriculum (Finding 26, n. 345).

³⁸¹ CCSSM 1.MD: data with up to three categories. CCSSM 2.MD: picture graphs and bar graphs, single-unit scale, up to four categories; line plots of measurement data. Not a mean. Chapter 11 is the through-line.

week?” — is the spirit of later statistical thinking. GAISE II page-level claims were not opened for this project and stay **UNVERIFIED**. This section will not quote a GAISE page. It will say: a category count the child made is better than a worksheet of already-tallied bars with no question.³⁸² Chapter 11 is the through-line. This band is three categories and a bar.

Geometry is defining attributes, composition, and equal shares — not a cute craft. Grade 1: distinguishing defining attributes from non-defining ones. A triangle is closed and three-sided; color, orientation, and overall size are not what make it a triangle. A child who rejects an inverted triangle as “not a triangle” is using a prototype, not a definition. Compose two-dimensional and three-dimensional shapes into composites. Grade 2: shapes with specified attributes — triangles, quadrilaterals, pentagons, hexagons, cubes. Partition a rectangle into rows and columns of same-size squares, the array seed already named.³⁸³ NCTM Focal Points Grade 1, from the third-party posting, includes composing and decomposing geometric shapes as one of three focal points. Prefer a library copy before treating the posting as the official book.³⁸⁴ Spatial play, building, and making are spatial. They do not automatically produce the defining-attribute distinction, and they do not automatically produce proof. Finding 26 again. Wai, Lubinski, and Benbow (2009) is the usual citation for spatial skill predicting later STEM achievement; that paper was not opened, and exact coefficients remain **UNVERIFIED**. This chapter will not invent a *r*.³⁸⁵

Equal shares are the only prior fraction work this book will let this band claim, and they are load-bearing because Chapter 5 sits on them. 1.G.3: partition circles and rectangles into two and four equal shares; words *halves*, *fourths*, *quarters*; decomposing into more equal shares creates smaller shares. 2.G.3: partition circles and rectangles into two, three, or four equal shares — *halves*, *thirds*, *fourths*; equal shares of identical wholes need not have the same shape.³⁸⁶ Do not skip equal-shares work because “fractions start in third grade.” The Grade 3 Fractions domain, 3.NF, limited to denominators 2, 3, 4, 6, and 8, defines a unit fraction $1/b$ as one part when a whole is partitioned into b equal parts, and a/b as a copies of $1/b$, both as numbers on a line.³⁸⁷ Without equal shares, “equal parts” is a phrase with no picture. With only pizza, “equal parts” is a picture with no magnitude. Chapter 5 will put the share on a line. This chapter plants the share.

The teaching move that survives, and that the IES 2010 fractions guide names as Recommendation

³⁸²American Statistical Association, *Guidelines for Assessment and Instruction in Statistics Education* (Pre-K–12 GAISE I 2005, rev. 2007; Pre-K–12 GAISE II 2020; 2016 is the college GAISE report). Page-level claims **UNVERIFIED** this project. Spirit without those pages: a question that has variability, not a worksheet of already-tallied bars with no question.

³⁸³CCSSM 1.G and 2.G (nn. 320, 325). Defining vs non-defining attributes; compose 2-D and 3-D shapes; Grade 2 shapes with specified attributes; array partition of a rectangle.

³⁸⁴NCTM Focal Points Grade 1 (n. 326): composing and decomposing geometric shapes as one of three focal points. Third-party posting of NCTM 2006 text; prefer a library copy.

³⁸⁵Jonathan Wai, David Lubinski, and Camilla P. Benbow (2009), spatial skill and later STEM achievement. Paper not opened; exact coefficients **UNVERIFIED**. Do not invent an *r*. Building and making are spatial; they do not automatically produce the defining-attribute distinction or proof (Finding 26).

³⁸⁶CCSSM 1.G.3 and 2.G.3 (n. 307). The only CCSSM fraction language before 3.NF. Do not skip equal-shares work because “fractions start in third grade.”

³⁸⁷CCSSM 3.NF (limited to denominators 2, 3, 4, 6, 8): a unit fraction $1/b$ is one part when a whole is partitioned into b equal parts; a/b is a copies of $1/b$; fractions as numbers on the number line. Plant for Chapter 5. Do not teach the gate here.

1, is to build on informal fraction experiences of young children — sharing, fair shares.³⁸⁸ Two children, one sandwich, halves. Four children, one rectangle of paper, fourths. The words. The observation that four fourths are smaller pieces than two halves of the same whole. Grade 2's sentence that equal shares of identical wholes need not have the same shape — a half as a cut through the diagonal of a rectangle, a half as a cut through the midline — is the beginning of the later idea that the same quantity can look different. It is not yet equivalence of $1/2$ and $2/4$ as the same point on a line. Do not teach that gate here. Do not skip the shares that make the gate walkable.

Cooking is not fractions. Finding 26 is the caution, and Chapter 5 will say it again because that is where the folklore is loudest. Cooking can motivate and can provide a second representation of a share the child is already being taught. Halving a recipe does not place $1/2$ on a line and does not teach that $1/2 + 1/2 = 1$ as two copies of a unit fraction. A child who can halve a recipe and cannot say that two of four equal shares are a half of the same whole has performed, not learned.³⁸⁹ Unschooling cooking-and-games versus the 77 percent formal-curriculum majority in NCES 2016 is a real minority practice, 12 percent mostly or strictly informal, not the market majority.³⁹⁰ It is a poor match for a New York Individualized Home Instruction Plan and a Pennsylvania portfolio unless the parent translates lived mathematics into a course title after the fact. NMAP Finding 26 is the transfer caution either way. This book's thesis is closer to structured or mastery work. That is a design choice, not a scorn. The twelve percent are not a failure of the kitchen. They are a poor match for the equal-sign formats and the bundles this chapter has argued the documents support.

Frye and colleagues 2013, already named: number and operations plus geometry, measurement, and data along a developmental progression; talk about the world mathematically; dedicate daily time. Talk about the world mathematically is the forks and the stairs and the sandwich, as talk, as a second representation. It is not the bundle, the non-canonical equation, or the equal share as a taught partition. Dedicate daily time is a recommendation for ages 3–6 in a school-facing guide. It is not a 2026 kitchen RCT and not a minutes figure.

What this section has done is keep measure and shape from being treated as extras, and plant the only fraction language the map allows before 3.NF. Tips that survive this band come next, then the AI lane — child writes, parent holds the key.

³⁸⁸Robert Siegler, Thomas Carpenter, Francis Fennell, David Geary, James Lewis, Yukari Okamoto, Laura Thompson, and Jonathan Wray, *Developing Effective Fractions Instruction for Kindergarten Through 8th Grade*, NCEE 2010-4039 (IES, 2010), Recommendation 1: build on informal fraction experiences of young children (sharing, fair shares). Recommendation 2 (number lines as a central tool) is Chapter 5's job.

³⁸⁹NMAP Finding 26 (n. 345). Cooking can motivate and can provide a second representation of a share the child is already being taught. It does not replace equal-shares teaching, and it does not place $1/2$ on a line. Nicholas C. Soderstrom and Robert A. Bjork, "Learning Versus Performance: An Integrative Review," *Perspectives on Psychological Science* 10, no. 2 (2015): current performance is "often an unreliable index" of learning. A child who can halve a recipe and cannot say that two of four equal shares are a half of the same whole has performed, not learned. Full discussion of Soderstrom and Bjork in Chapter 3; this chapter uses the distinction.

³⁹⁰Cui and Hanson, NCES 2020-001: 77 percent of homeschooled students taught with a "mostly or strictly formal curriculum"; 12 percent "mostly or strictly informal learning." Unschooling is a named minority, not the market majority. This book's thesis is closer to structured or mastery work — a design choice, not a scorn.

4.6 Tips that survive evidence in grades 1–2

A short list a parent can steal, each tied to a named source. This is not a 180-day plan. It is not a franchise. It is the moves of sections 4.2 through 4.5, compressed so they can be read on a Tuesday.

After a count, ask “how many?” without recounting. If the child restarts from one, cardinality is not yet the last word as a name for the set. Hide the set and ask again. Count sets, not just the recitation. Gelman and Gallistel; Sarnecka and Carey; Bermejo as cited.³⁹¹

Write $8 = 5 + 3$ and $8 = 4 + 4$ in the same sitting. Number bonds make part-whole visible; Canobi 2009 is why the sequence of ideas matters. Write the total on the left some of the time. That is already the equal sign.³⁹²

Write at least some equations with the unknown on the left or with operations on both sides. $8 = 3 + 5$. $3 + 5 = 2 + 6$. $3 + 4 + 5 = 3 + \square$. True and false sentences. McNeil and Alibali 2005; McNeil, Fyfe, Petersen, Dunwiddie, and Brletic-Shipley 2011; McNeil 2014. A diet of $3 + 4 = 7$ trains the operator. Knuth and colleagues 2006 is why the relation is not a luxury unit.³⁹³

Connect sticks-and-beans — proportional ten-sticks that are ten ones, not a chip that “stands for” ten — to the written numeral in the same visual field, then fade. Fuchs and colleagues 2021, Recommendation 3; Fuson and Briars 1990. Fading is for accuracy, not for a calendar date. Bring the sticks back when a new idea needs a bridge. Carbonneau, Marley, and Selig $d \approx 0.37$ is overall, kindergarten through college, versus abstract-only, moderated by guidance, age, and outcome. It is not a Grade 1 d . High guidance. Then notation. Then fade. Blocks are a conversation about place value, not a substitute for writing $27 + 15$.³⁹⁴

Do not take the pencil. Retrieval is generation. Self-explanation is generation. Completion problems in fading are generation of the missing steps. Think-alouds in explicit instruction are the child talking, not only the parent. Maloney and colleagues 2015: the parent who takes over the pencil is the high-help, high-anxiety path, in a first- and second-grade sample. The homeschool parent cannot “help less often” as a daily strategy. Prepare so you can hear a wrong answer. Then do not give the next step.³⁹⁵

Retrieve known facts mixed, briefly, after they are known untimed. Fuchs and colleagues 2021, Recommendation 6. Gersten and colleagues 2009, about 10 minutes per intervention session, is a school-intervention figure, not a homeschool-minutes study. Do not make the timer the identity of being good at math. Do not hand a child who still count-alls a page of fifty and a clock. IES

³⁹¹Gelman and Gallistel 1978 (n. 335); Sarnecka and Carey 2008 and Bermejo et al. 2004 as cited (n. 336). After a count, ask “how many?” without recounting; hide the set and ask again.

³⁹²Canobi 2009 (n. 338). Number bonds: $8 = 5 + 3$ and $8 = 4 + 4$ in the same sitting; total on the left some of the time.

³⁹³McNeil and Alibali 2005 (n. 304); McNeil et al. 2011 (n. 355); McNeil 2014 (n. 350); Knuth et al. 2006 (n. 305). Non-canonical equations early. A diet of $3 + 4 = 7$ trains the operator.

³⁹⁴Fuchs et al. 2021 Recommendation 3 (n. 341); Fuson and Briars 1990 (n. 339); Carbonneau, Marley, and Selig 2013 (n. 342). Connect proportional ten-sticks to the written numeral in the same field, then fade. $d \approx 0.37$ is overall, not a Grade 1 d .

³⁹⁵Dunlosky et al. 2013 (retrieval as generation); Renkl and Atkinson fading (n. 348); NMAP Finding 27 think-alouds; Maloney et al. 2015 (n. 314). Do not take the pencil. Fraulein Retanal et al., *Education Sciences* 11, no. 10 (2021): 620: controlling-supportive homework-helping style partially mediated the parent-anxiety / child-achievement relation in a survey of parents of 11- to 14-year-olds. Path analysis on a survey, not a kitchen RCT.

and Boaler are different objects. Print both. Do not pick a religion.³⁹⁶ Isolated effect of the timer remains open. The 2025 Peltier preprint is the attempt to unbundle; labelled preprint.³⁹⁷

Arrays and equal groups as multiplication seeds, not a times-table race. 2.OA.3–4, 2.G.2. Five rows of four written as $4 + 4 + 4 + 4 + 4$. Do not skip this seed because “multiplication starts in third grade.” Odd and even as pairing. Skip-count by 5s, 10s, 100s as jumps on a line.³⁹⁸

Equal shares of circles and rectangles before Grade 3 pies-as-fractions. 1.G.3, 2.G.3. Halves, thirds, fourths. More shares, smaller pieces. Equal shares of identical wholes need not have the same shape. IES 2010 Recommendation 1: build on informal sharing. Do not teach $3/4$ as a number on a line here. Do not skip the shares. Chapter 5 is the gate.³⁹⁹

Do not claim kitchen counting “is” place value. Do not claim cooking “is” fractions. Finding 26: real-world contexts help similar real-world items; they do not, in the Panel’s small high-quality set, raise computation, simple word problems, or equation solving. Talk about the world mathematically (Frye and colleagues 2013). Then write 27. Then partition the rectangle. The talk is a second representation, not the structure.⁴⁰⁰ Unschooling cooking-and-games versus the 77 percent formal-curriculum majority is a live cultural debate this book will not settle with a trial it does not have. Transfer from informal activity to formal items, measured rather than assumed, is an open question. Do not fill it with analogy.⁴⁰¹

When to push fact retrieval versus when to stay in strategies is a live debate. The Panel’s automaticity and 2.OA.2’s memory of single-digit sums by the end of Grade 2 on the map sit on one side. Boaler’s object — high-stakes early speed tests — sits on the other. The objects are different. Retrieve known facts. Do not rank six-year-olds with a Mad Minute. Do not treat fluency as optional.⁴⁰²

Hear a wrong answer against five strands. A child who can recite $8 + 7$ but cannot say why $8 + 7$ equals $7 + 8$ is missing conceptual understanding. A child who can draw the number bond and cannot retrieve $8 + 7$ when it is needed inside $18 + 7$ is missing a different strand. A child who can retrieve and cannot formulate a compare story as an equation is missing strategic competence.

³⁹⁶Fuchs et al. 2021 Recommendation 6 (n. 369); Gersten et al. 2009 Recommendation 6 (n. 370); Boaler 2012/2015 (n. 371). Retrieve known facts mixed, briefly, after they are known untimed. Do not make the timer the identity of being good at math.

³⁹⁷Isolated timer effect: Peltier et al. 2025 OSF preprint (n. 372). Labelled preprint. Until an isolated timing experiment is opened, print both objects.

³⁹⁸CCSSM 2.OA.3–4 and 2.G.2 (n. 376). Arrays and equal groups as multiplication seeds. Skip-count by 5s, 10s, 100s (2.NBT). Odd and even as pairing.

³⁹⁹CCSSM 1.G.3 / 2.G.3 (n. 307); IES 2010 Recommendation 1 (n. 388). Equal shares before Grade 3 pies-as-fractions. Do not teach $3/4$ as a number on a line here. Chapter 5 is the gate.

⁴⁰⁰NMAP Finding 26 (n. 345); Frye et al. 2013 (n. 333). Talk about the world mathematically. Then write 27. Then partition the rectangle. Kitchen counting is not place value. Cooking is not fractions.

⁴⁰¹Transfer from informal activity (cooking, money, games) to formal items, measured rather than assumed: open question this book leaves standing (02-learning open questions; brief open question 20). Do not fill with analogy. Barnett and Ceci 2002 is the standard transfer taxonomy; paper not opened; exact dimensions **UNVERIFIED**. NCES 2016 77 percent formal / 12 percent informal (n. 390).

⁴⁰²Live debate: NMAP automaticity (Finding 11; Table 2 Grade 3 +/- benchmark; CCSSM 2.OA.2) versus Boaler’s object (high-stakes early speed tests). Print both. Do not pick a religion. Retrieve known facts. Do not rank six-year-olds with a Mad Minute. Do not treat fluency as optional.

A child who treats a wrong 15 on $3 + 4 + 5 = 3 + \square$ as a verdict on talent is missing productive disposition. Ranking the strands is misguided. Hearing which is thin is the job.⁴⁰³

Explicit instruction for a child already struggling with this content — clear models, extensive practice, think-alouds, extensive feedback — has consistently positive effects (Finding 27; Fuchs and colleagues 2021, Recommendation 1). Productive failure is generation-then-instruction on a new concept when priors exist (Kapur 2014). Those two literatures are not the same comparison. Do not mash them. If the child is missing cardinality, model counting a set, then guided practice, then independent. If the child has cardinality and the task is a first composed ten, ten to twenty minutes of generation with what they already know, then a clear model, is Kapur scaled and labelled as a scaling, not as a Grade 1 RCT.⁴⁰⁴

Accelerate a child who has the prerequisites. Do not withhold tens-and-ones from a seven-year-old who has them. Do not accelerate a child who does not. Finding 15. Finding 30 licenses faster movement for mathematically gifted students with motivation; it does not license skipping the equal sign because a child is “good at math.”⁴⁰⁵ Two-e acceleration protocols were not opened; effect sizes remain **UNVERIFIED**.⁴⁰⁶

A purchased book is a tool. Singapore’s number bonds, Math-U-See’s integer blocks, Saxon’s mixed practice, Beast Academy’s puzzle-hard pictures, Teaching Textbooks’ video, Khan Academy’s free map: name them by grain and parent load. Do not sell them. No homeschool-curriculum randomized trial was opened. Teaching Textbooks’ “does 100% of the teaching” is a **vendor** claim, relevant to Maloney, not a finding that the parent can stop hearing 15 on $3 + 4 + 5 = 3 + \square$.⁴⁰⁷

Do not write a 180-day plan. The shape — short blocked introduction, mixed retrieval of known facts, non-canonical equations in the same week, sticks next to numerals then faded, equal shares before the gate, unaided exit ticket — is a shape. The parent imposes it on whatever book is on the

⁴⁰³NRC, *Adding It Up* (2001), p. 116 (n. 362); NMAP Finding 10 (n. 363). Hear a wrong answer against five strands. Ranking them is misguided.

⁴⁰⁴NMAP Finding 27: explicit instruction (clear models, extensive practice, think-alouds, extensive feedback) has consistently positive effects for students with mathematical difficulties and for the lowest third of a typical class — and this “does not mean that all of a student’s mathematics instruction should be delivered in an explicit fashion.” Fuchs et al. 2021 Recommendation 1 (strong, 43 studies): systematic instruction. Manu Kapur, “Productive Failure in Learning Math,” *Cognitive Science* 38, no. 5 (2014): 1008–1022: problem-solving first, then instruction, produced comparable procedural knowledge to instruction-first but greater conceptual understanding and transfer. These two literatures are not the same comparison. Do not mash them. Finding 23: high-quality research does not support exclusive use of either all-encompassing teacher-directed or student-centered prescriptions. Jere Confrey, Alan P. Maloney, and Kenny H. Nguyen, “Breaching the Conditions for Success for a National Advisory Panel,” *Educational Researcher* (2008): methodological critique of the Panel. Report both.

⁴⁰⁵NMAP Finding 15 (n. 309); Finding 30: mathematically gifted students with motivation “appear to be able to learn mathematics much faster... with no harm to their learning, and should be allowed to do so.” Accelerate a child who has the prerequisites. Do not skip the equal sign because a child is “good at math.”

⁴⁰⁶Susan G. Assouline, Nicholas Colangelo, Joyce VanTassel-Baska, and colleagues, *Nation Empowered / Iowa Acceleration Scale* literature. Not opened this pass; effect sizes **UNVERIFIED**. 2e (gifted + SLD or ADHD) is a real clinical description; this book will not invent a prevalence or a protocol.

⁴⁰⁷Teaching Textbooks vendor claim that the program “does 100% of the teaching” and “100% of the grading”: **vendor** copy, labelled vendor. Relevant to the Maloney constraint; not a finding that the parent can stop hearing 15 on $3 + 4 + 5 = 3 + \square$. Homeschool-curriculum RCT: none opened (n. 332). Cathy Duffy Reviews and Rainbow Resource are **catalogue / reviewer**, not independent effectiveness trials.

table.

The model may prepare the adult and the next story. It may not sit between the child and $7 + 8$. That is the last section.

4.7 Parent-hosted AI for grades 1–2

Chapter 3 already locked the kitchen condition. Unguarded solvers raise practice scores and cut unaided exams. Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman, *PNAS* 2025: GPT Base, a ChatGPT-like tutor, raised assisted practice grades 48 percent relative to control, then cut unaided exam grades 17 percent. GPT Tutor, hints only, with the solution in a hidden prompt, raised practice 127 percent and was statistically indistinguishable from control on the unaided exam (point estimate -0.004). GPT Base, asked for the answer, was fully correct only 51 percent of the time on the practice set; 8 percent of the errors were arithmetic even when the method was right.⁴⁰⁸ This chapter will not re-argue that trial. It is not a homeschool study, not a six-year-old sample, not GPT-5, and not Khanmigo. It is the cleanest field experiment opened on 30 August 2026 that the pattern — paste the page, get the key, feel competent, sit the unaided test — predicts a worse unaided test. Nicholas C. Soderstrom and Robert A. Bjork, 2015: current performance is “often an unreliable index” of learning.⁴⁰⁹ The six-year-old with a solver on a fact sheet is the same pattern with smaller numbers. One short reminder, then the objects of this band.

The young child does not live in an open-web agent. That sentence is age, law, and product, not a vibe. The Children’s Online Privacy Protection Rule, as amended, published 22 April 2025, 90 FR 16918, effective 23 June 2025, general compliance **22 April 2026**: operators of child-directed and mixed-audience services still need notice and verifiable parental consent before collecting personal information from children under 13.⁴¹⁰ A first-grader is under 13. Photomath-style homework apps, DreamBox, IXL kid accounts, and Khanmigo child profiles are in COPPA territory. A parent using their own adult ChatGPT, Claude, Grok, or Hermes account to generate a worksheet, without creating a child profile at the operator, is a different fact pattern. It is still not a license to upload

⁴⁰⁸Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman, “Generative AI without Guardrails Can Harm Learning: Evidence from High School Mathematics,” *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122, <https://doi.org/10.1073/pnas.2422633122>; author PDF https://hamsabastani.github.io/education_llm.pdf, accessed 30 August 2026. GPT Base: assisted practice +48% relative to control; unaided exam -17% . GPT Tutor: practice $+127\%$; exam point estimate -0.004 , not significant. GPT Base fully correct 51% of the time when asked for the answer; 8% arithmetic errors even when the method was right. Classroom-level RCT, large high school in Turkey, Fall 2023–24, $\sim 1,000$ students in grades 9–11, four 90-minute sessions, GPT-4-0613. Not a homeschool study, not a six-year-old sample, not GPT-5, not Khanmigo. Percentages are of control grades, not Cohen’s d . Full discussion: Chapter 3. This chapter is a pointer, not a sequel.

⁴⁰⁹Nicholas C. Soderstrom and Robert A. Bjork, “Learning Versus Performance: An Integrative Review,” *Perspectives on Psychological Science* 10, no. 2 (2015), <https://doi.org/10.1177/1745691615569000>. Current performance is “often an unreliable index” of learning. People “mistakenly interpret their performance during acquisition as a reliable guide to long-term learning.” Bastani is the mathematics-class replication. Full discussion: Chapter 3.

⁴¹⁰Federal Trade Commission, “Children’s Online Privacy Protection Rule,” 90 FR 16918 (22 April 2025), Document 2025-05904, 16 CFR 312, <https://www.federalregister.gov/documents/2025/04/22/2025-05904/childrens-online-privacy-protection-rule>. Effective date 23 June 2025; general compliance date 22 April 2026 (except specified § 312.11(d)(1), (d)(4), and (g)). Operators of child-directed and mixed-audience services still need notice and verifiable parental consent before collecting personal information from children under 13. Full discussion: Chapter 3 §3.6.

the child’s face, voice, or full name into a consumer model.⁴¹¹ Claude’s consumer policy is 18+; Claude for Teachers is verified U.S. K–12 educators only. Homeschool parents are not verified U.S. K–12 educators by default and should not assume a FERPA wrapper. Hermes Agent is a self-hosted adult agent with an open web and messaging gateways; no K–12 student-safety certification was on the pages opened. Grok Bot requires cloud data storage; Legacy Privacy Mode is not supported; no education SKU and no FERPA claim on the pages opened.⁴¹² Do not put either in a six-year-old’s hands. Do not run a Telegram or WhatsApp gateway “so they can ask homework questions.” Companion-harm cases — *Garcia v. Character Technologies* (Setzer) and *Raine v. OpenAI* — are **allegations, not verdicts**. They are why a math agent should not be a companion. They are not a literature on mathematics learning.⁴¹³ A “math buddy” with a name, a memory of the child, and a messaging-app gateway is that architecture even if the first week is number bonds.

What a parent should ban versus allow follows from Bastani and Soderstrom and Bjork, not from folklore. The allow list is a practice suggestion, not a homeschool randomized trial. No year-plus kitchen-table RCT of a parent-constrained tutor exists in the literature opened on 30 August 2026.⁴¹⁴

Allow, parent-hosted, adult account, child not in the loop until the page is on paper.

Parent-first explanation. Ask the model to explain place value, or the equal sign as a relation, *to the parent*. “Explain to an adult who last passed Algebra I why 27 is two tens and seven ones, and why a child might write 13 for thirty-one. Do not write a script for the child. Name the misconception.” Bastani’s GPT Tutor worked in part because teachers had already written the solution and common mistakes into the hidden prompt. The parent is that teacher. Prefer a chat or reasoning model, not an open-web agent, for this job. Hermes 4, or any strong chat model, for the explanation; Hermes 4 is not recommended inside Hermes Agent for the tool loop.⁴¹⁵

⁴¹¹A parent using an adult ChatGPT / Claude / Grok / Hermes account to generate a worksheet, without creating a child profile at the operator, is a different fact pattern from a child-directed app. Still not a license to upload the child’s face, voice, or full name into a consumer model. Claude consumer policy is 18+; Claude for Teachers is verified U.S. K–12 educators only. Homeschool parents are not verified U.S. K–12 educators by default.

⁴¹²Hermes Agent, <https://hermes-agent.org/>, fetched 30 August 2026: open-source MIT autonomous agent, Nous Research, released February 2026; self-hosted; messaging gateways; data in `~/hermes/`; no K–12 student-safety certification on the pages opened. Adult/developer tool, not a child-safety school product. xAI, “Grok Bot FAQ,” <https://docs.x.ai/grok-bot/faq>, fetched 30 August 2026: persistent cloud computer; “Grok Bot requires cloud data storage, so Legacy Privacy Mode is not supported”; no education SKU and no FERPA claim on the pages opened. Full distinctions: Chapter 3 §3.2. Do not put either in a six-year-old’s hands. Do not run a Telegram/WhatsApp gateway for homework questions.

⁴¹³*Garcia v. Character Technologies et al.*, complaint, M.D. Fla., via CourtListener/RECAP: allegations, not a verdict. *Raine v. OpenAI*, complaint: allegations, not a verdict. Setzer settlement terms **UNVERIFIED** as a primary docket entry. Why a math agent should not be a companion. Not a literature on mathematics learning. Full discussion: Chapter 3 §3.6.

⁴¹⁴Parent workflows in this section are practice suggestions grounded in Bastani, Soderstrom and Bjork, and Oreopoulos and Low, not a new RCT of homeschool AI. No year-plus kitchen-table RCT of a parent-constrained GPT-Tutor workflow was opened on 30 August 2026 (04-ai open questions). Label them that way. Chapter 3 §3.3–3.4 is the lock; this section is the grade 1–2 pointer.

⁴¹⁵Chapter 3 workflow (a): explain the concept to the *parent* first. Practice suggestion, not an RCT. Bastani’s GPT Tutor worked in part because teachers had already written the solution and common mistakes into the hidden prompt. The parent is that teacher. Nous Portal docs, fetched 30 August 2026: Hermes 4 is “strong at math, science, instruction following” — **vendor** — and **not recommended for use inside Hermes Agent** (tuned for chat, not rapid tool-calling).

Isomorphic stories with the key held back. “Write six add-to, take-from, and compare stories within 20, unknown in different positions. Student sheet: stories only. Parent key: structure name, equation, answer, in a separate block. Do not put answers on the student sheet. Vary the cover story. Do not use ‘total means add.’” Do not generate “practice” by completing tonight’s assigned homework. That is the ban. Chapter 3 workflow (c).⁴¹⁶

A ten-minute beans-and-sticks activity. “Describe a hands-on activity using ten-sticks and ones for composing a ten, $27 + 5$, with the written numeral in the same visual field. Materials list. What the adult says. What the child does. What would count as the child generating. Do not address the child.” The model describes. The child handles the sticks. Carbonneau $d \approx 0.37$ remains overall, not a Grade 1 d . Virtual versus physical **UNVERIFIED**. The variable is the conversation.⁴¹⁷

Diagnose after the attempt. Paste — type, do not photograph a face — the child’s written $3 + 4 + 5 = 3 + \square$ with the answer 15, or 12, or the reversal 13 for thirty-one. “Name the misconception. Write three spoken questions for me. Do not address the child. Do not give the correct answer as the first line.” For 15, the sentence is McNeil and Alibali’s operational pattern, perform all operations on all numbers, not “they cannot add.” For 12, adding to the equal sign. For 13 versus 31, tens and ones reversed, concatenation, not handwriting. The child answers the parent, not the model. Chapter 3 workflow (f).⁴¹⁸

Generate an oral fact script or a shuffled fact sheet. Parent checks a sample against a known table — a printed addition chart, not the model’s arithmetic. Facts are not a language-model job. Bastani: 8 percent arithmetic errors even when the method was right. A language model will happily hallucinate $7 + 8$. Generate the script. Then the parent’s ears do the drill, or a dumb spaced-repetition deck the parent loaded from a known-correct table does. IES 2021 Recommendation 6 belongs to the parent’s ears, not to a chatbot conducting oral drill.⁴¹⁹

Ban. Photomath on the fact sheet. Mathway, Symbolab, consumer ChatGPT, Gemini, and Claude in the child’s hands, asked “what is the answer.” Paste-the-worksheet. Photo-to-key. The model conducting oral drill. A companion “math friend.” An open-web agent — Hermes Agent with a gateway, Grok Bot on an iPhone — in a child’s hands. Scan-the-page will print a correct 42 for $27 + 15$ and the parent never sees whether the child concatenated, reversed, or composed a ten.⁴²⁰

Prefer a chat/reasoning model for parent explanation.

⁴¹⁶Chapter 3 workflow (c): extra isomorphic practice with the answer key held by the parent. Do not generate “practice” by completing tonight’s assigned homework. That is the ban. Add-to / take-from / compare schemas, not keywords (n. 329).

⁴¹⁷Chapter 3 workflow (e): hands-on activity as materials generation. The model describes; the child handles the sticks. Carbonneau $d \approx 0.37$ is overall, not a Grade 1 d (n. 342). Virtual vs physical **UNVERIFIED** (n. 343). Fuchs et al. 2021 Recommendation 3 (n. 341): same visual field.

⁴¹⁸Chapter 3 workflow (f): diagnose a wrong answer the child already produced. Practice suggestion, not an RCT. For 15 on $3 + 4 + 5 = 3 + \square$: McNeil and Alibali operational pattern, perform all operations on all numbers (n. 304), not “they cannot add.” For 12: adding to the equal sign. For 13 versus 31: tens and ones reversed, concatenation, not handwriting. The child answers the parent, not the model.

⁴¹⁹Bastani et al. 2025: GPT Base fully correct 51%; 8% arithmetic errors even when the method was right. Flash retrieval of arithmetic facts is not an LLM job. Chapter 3 §3.7 / 04-ai load-bearing finding 12: use the model to generate the deck or the oral script, not to sit between the child and the fact. IES 2021 Recommendation 6 belongs to the parent’s ears, or a dumb SRS loaded from a known-correct table, not to a language model conducting oral drill.

⁴²⁰Ban during the attempt: paste-the-worksheet, photo-to-key, “what is the answer,” companion persona, open-web agent in the child’s hands, the model conducting oral drill. Follows from Bastani et al. 2025 (n. 408) and Soderstrom

Photomath’s homepage, fetched 30 August 2026, leads with scan and “A solution will pop up for your exact problem.” Designed-for-learning copy is **vendor**. No Photomath-specific unaided-transfer RCT was opened. The transferable evidence is Bastani’s crutch plus that homepage.⁴²¹

Adaptive practice engines already in use — DreamBox, IXL, Khan Academy practice, ALEKS — sequence items. They are not agents in the Hermes Agent or Grok Bot sense and not language models sitting between the child and $7 + 8$. They can supply retrieval practice. DreamBox’s “one hour each week can drive more than a grade level of math growth” is **vendor**; the underlying ESSA study PDF was not opened and stays **UNVERIFIED**. IXL live homeschool price was not captured as an invoice number. MATHia family purchase **UNVERIFIED**. Khanmigo is a constrained Socratic tutor over Khan content, parent-gated, **vendor** “never gives you the answer,” \$4 per month or \$44 per year on the page opened 30 August 2026. Oreopoulos and Low, NBER w35620, August 2026: assignment to Khan Academy plus Khanmigo in 18 Tennessee middle schools raised MAP math by 0.040 population standard deviations per term; gains resemble Khan without AI; conversational use was thin. That trial is middle school, not Grade 1. A homeschool parent who buys Khanmigo for a six-year-old and walks away should expect the Tennessee pattern — extra practice, thin chat — not a personal tutor, and should still not put an open-web agent in that child’s hands.⁴²²

Session shape, scaled to this band, from Chapter 3. Practice suggestion, not an RCT. Unaided warm-up: two known facts, one non-canonical equation. Teach. Struggle. Hint not key. Unaided exit ticket — five minutes, phone in another room, two items isomorphic to what was taught. If the model was in the room for the adult’s preparation, the session is not over until there is an unaided score the child produced.⁴²³ “Finish your facts with Photomath so we can go to soccer” is GPT Base, the pattern with a large-sample causal harm estimate, applied to a six-year-old.

A hypothetical, labelled as such: a parent of a six-year-old, adult account, asks a chat model for

and Bjork 2015 (n. 409). Photomath, Mathway, Symbolab, and consumer ChatGPT/Gemini/Claude are solvers or general assistants unless the parent constrains them.

⁴²¹Photomath, <https://photomath.com/>, fetched 30 August 2026: scan including word problems; “A solution will pop up for your exact problem”; Basic \$0 / Monthly \$9.99 / Annually \$69.99 (USD, this day); **vendor** “designed for learning.” Mathway, <https://www.mathway.com/>: “Got a math problem? Get the steps.” Symbolab, <https://www.symbolab.com/>: solver first. No product-specific unaided-transfer RCT opened this day. Transferable evidence is Bastani’s crutch plus solver homepages that lead with scan-to-solution.

⁴²²Adaptive practice engines (DreamBox, IXL, Khan Academy practice, ALEKS) sequence items; they are not agents in the Hermes Agent / Grok Bot sense. DreamBox grade-level-per-hour claim is **vendor**; underlying ESSA study PDF **UNVERIFIED**. IXL live homeschool price **UNVERIFIED** as an invoice number. MATHia family purchase **UNVERIFIED**. Khan Academy, “Khanmigo for learners,” <https://www.khanmigo.ai/learners>, fetched 30 August 2026: **vendor** “Unlike ChatGPT, Khanmigo never gives you the answer”; **vendor** price \$4/month or \$44/year; parent-gated home SKU, not an open-web agent. Philip Oreopoulos and Nina Low, “One Click Away: AI Tutoring with Khanmigo in a Two-Year School Experiment,” NBER Working Paper 35620, August 2026, <https://www.nber.org/papers/w35620>: two-year cluster RCT, 18 Tennessee middle schools; 1.264 NPR per term (SE 0.597); 0.040 population SD per term; about 0.06 to 0.08 SD over a school year; “These gains resemble those from Khan Academy practice without AI assistance.” Conversational use thin (median student messaged on a third of practice days; 17% of mistake sessions; 14.5% of messages mathematical). Do not invent standalone 0.06 as the ITT. Middle school, not Grade 1. Working paper, not yet journal-peer-reviewed. Full discussion: Chapter 3 §3.5.

⁴²³Session shape: Chapter 3 §3.8 / 04-ai §9.4, scaled to this band. Practice suggestion, not an RCT. Grounded in Bastani’s exam phase (n. 408), Soderstrom and Bjork (n. 409), and Oreopoulos and Low’s engagement finding (n. 422). Unaided warm-up (two known facts, one non-canonical equation), teach, struggle, hint not key, unaided exit ticket. “Finish your facts with Photomath so we can go to soccer” is GPT Base.

six compare-stories within 20, receives a student sheet and a parent key, sits the child with beans for one of the stories, does not paste the child's worksheet into a solver, diagnoses a written 15 on $3 + 4 + 5 = 3 + \square$ after the attempt, and ends with two unaided items.⁴²⁴ The sitting, the age, and the particular stories are hypothetical. The ban, the allow, the McNeil pattern, and the COPPA fact pattern are not.

Detectors fail. Liang, Yuksekgonul, Mao, Wu, and Zou, 2023: average false-positive rate 61.22 percent (arXiv) / 61.3 percent (*Patterns*) on human TOEFL essays. Do not use detectors to police a homeschooler's written number-bond explanations.⁴²⁵ California A.B. 1159, as of 30 August 2026, is an active bill on Senate third reading, **not law**. A homeschool parent outside a California LEA is not an "operator" under KOIPA by teaching at the kitchen table.⁴²⁶

The cheap constrained stack for this band is smaller than a product tour. Parent account, chat model, for explanations and for student-sheet-plus-key. Sticks and beans in the same visual field as the numeral. A printed fact table the parent trusts. An adaptive practice engine if the family already has one, for retrieval, not as a tutor-replacement. No child profile on a companion app. No open-web agent. No solver during the attempt. The child writes. The parent holds the key.

Equal shares were the on-ramp. Chapter 5 is the gate. A student who reaches the calendar year a publisher calls Grade 3 without equal-shares language, without memorized single-digit sums, without tens as bundles, and without the equal sign as a relation is underprepared for arrays, for 3.NF, and for every later object that assumes those four. This chapter's job was to make those four hearable. The next chapter's job is a fraction as a number on a line, not a pizza.

⁴²⁴Composite hypothetical, labelled *hypothetical* in the sentence that introduces it. Ban/allow, McNeil pattern, and COPPA fact pattern are from named sources (nn. 304, 408, 410, 414–419). The sitting, the age, and the particular stories are hypothetical. No portraits of living children.

⁴²⁵Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou, "GPT Detectors Are Biased against Non-Native English Writers," arXiv:2304.02819; *Patterns* 4, no. 7 (2023), <https://doi.org/10.1016/j.patter.2023.100779>. Average false-positive rate 61.22% (arXiv) / 61.3% (*Patterns*) on human TOEFL essays. Do not use detectors to police a homeschooler's written number-bond explanations. Full discussion: Chapter 3 §3.6.

⁴²⁶California Legislature, A.B. 1159 (Addis), 2025–2026 session, bill text https://leginfo.ca.gov/faces/billNavClient.xhtml?bill_id=202520260AB1159 and status https://leginfo.ca.gov/faces/billStatusClient.xhtml?bill_id=202520260AB1159, fetched 30 August 2026. Status: Active Bill — In Floor Process, Senate third reading (Daily File 30 August 2026, item 107). Last amended 21 August 2026. **Not law**. A homeschool parent outside a California LEA is not an "operator" under KOIPA by teaching at the kitchen table. Full discussion: Chapter 3 §3.6.

Chapter 5

Grades 3–5: Multiplicative Reasoning and Fractions

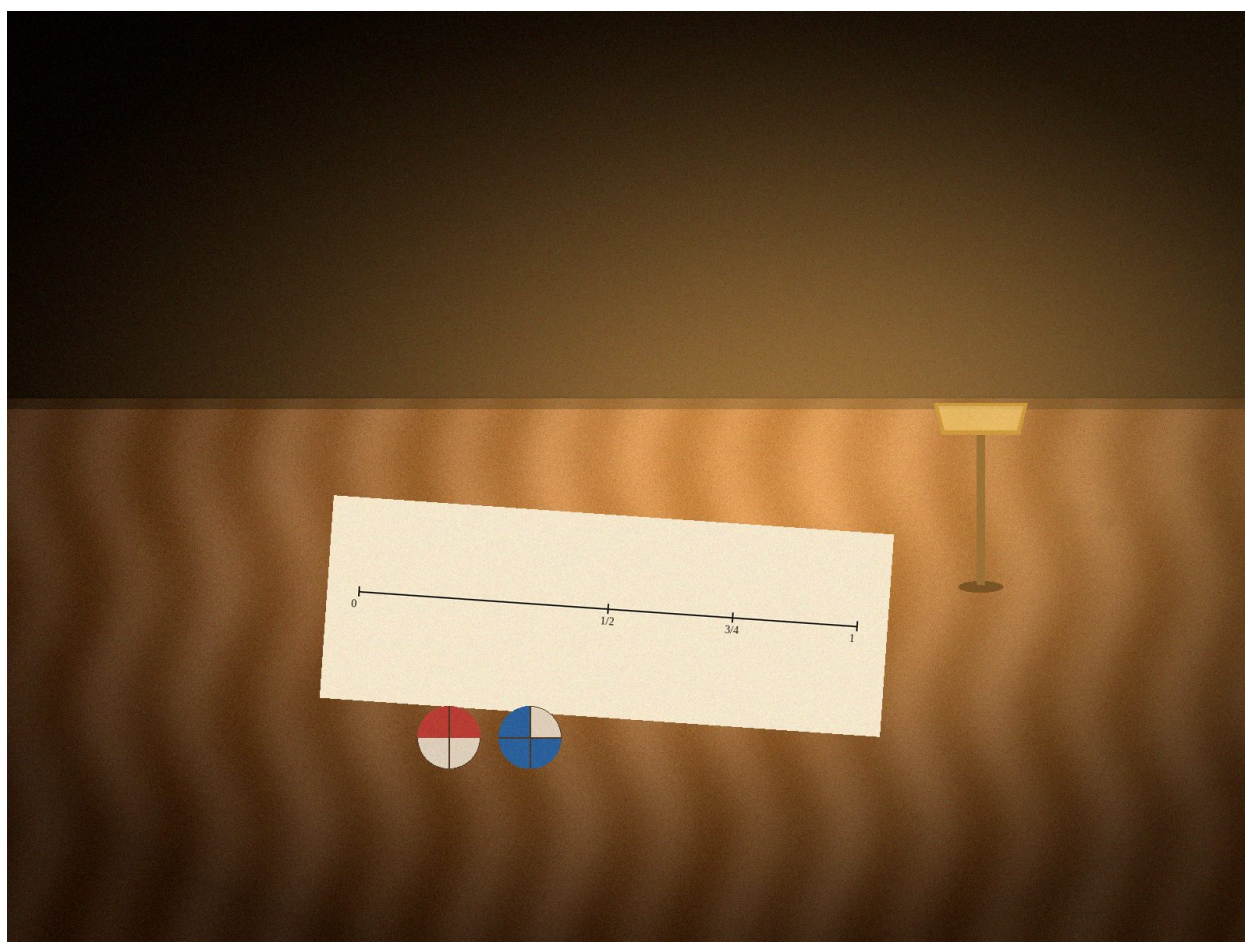


Figure 6: A paper number line marked $\frac{1}{2}$ and $\frac{3}{4}$, and generic fraction circles on a wooden table.

Two documents already contain the argument this chapter has to keep in view. The first is a sentence from the National Mathematics Advisory Panel’s 2008 final report, *Foundations for Success*: “knowledge of fractions is the most important foundational skill not currently developed among

American students.”⁴²⁷ The Panel had named three Critical Foundations of Algebra — fluency with whole numbers; fluency with fractions, including decimals, percents, and negative fractions; and particular aspects of geometry and measurement — and then said, of those three, which one U.S. students most often lack.⁴²⁸ A nationally representative survey of Algebra I teachers the Panel commissioned in 2007, 743 teachers, rated incoming preparation as poorest in “rational numbers and operations involving fractions and decimals,” alongside word problems and study habits.⁴²⁹ That survey is not “teachers today say.” It is the 2007 document the Panel used. This chapter will not launder it into a 2026 classroom census. It will also not pretend the foundation has since been solved by a new franchise.

The second document is not a paper. It is a line of work a parent will see, this year or next, on a page the child has just finished: $1/2 + 1/3 = 2/5$. The error looks like a forgotten rule. It is a magnitude failure. The child has added the counts that are visible — two numerators, two denominators — as if the symbols were whole numbers sitting side by side. Yujing Ni and Yong-Di Zhou named that over-extension whole-number bias.⁴³⁰ The IES fractions practice guide, Recommendation 2, named missing magnitude as “often at the root” of adding numerators and denominators and of not seeing fractions as numbers.⁴³¹ Pies introduce part-whole. They do not by themselves teach that $3/4$ is a number with a magnitude, a location on a line, a quantity that can be compared with $2/3$ without a butterfly. The Panel wrote that representing fractions on a number line is “one key mechanism linking conceptual and procedural knowledge.”⁴³² Chapter 2 planted that sentence. This chapter does the teaching.

This is the band the Panel treats as the make-or-break window for the Critical Foundations: the four operations on whole numbers, taken toward fluency with the standard algorithms; fractions as numbers on the number line, then operations on fractions; decimals as fractions with denominators 10 and 100, then as place value; area in what the Common Core map calls Grade 3, angle measure

⁴²⁷National Mathematics Advisory Panel, *Foundations for Success: The Final Report of the National Mathematics Advisory Panel* (U.S. Department of Education, 2008), Executive Summary findings 4 and 12, and Chapter 4, ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>, accessed 30 August 2026. The quoted clause — “knowledge of fractions is the most important foundational skill not currently developed among American students” — is the Panel’s. Prefer the Panel’s sentence over a journalism paraphrase.

⁴²⁸NMAP, *Foundations for Success* (2008), Critical Foundations of Algebra: fluency with whole numbers; fluency with fractions, including decimals, percents, and negative fractions; particular aspects of geometry and measurement. Of these three, fractions are named as the most important foundational skill not currently developed.

⁴²⁹NMAP, *Foundations for Success* (2008), nationally representative survey of Algebra I teachers, 743 teachers, 2007: poorest incoming preparation in “rational numbers and operations involving fractions and decimals,” alongside word problems and study habits. These are 2008 data, not a 2026 classroom census. Do not write “teachers today say.”

⁴³⁰Yujing Ni and Yong-Di Zhou, “Teaching and Learning Fraction and Rational Numbers: The Origins and Implications of Whole Number Bias,” *Educational Psychologist* 40, no. 1 (2005): 27–52, https://doi.org/10.1207/s15326985ep4001_3. Whole-number bias: unique successors; a finite number of entities in an interval; multiplication makes bigger; addition of “pieces” means add the counts you see. $1/2 + 1/3 = 2/5$ is the counts-you-see error.

⁴³¹Robert Siegler, Thomas Carpenter, Francis Fennell, David Geary, James Lewis, Yukari Okamoto, Laura Thompson, and Jonathan Wray, *Developing Effective Fractions Instruction for Kindergarten Through 8th Grade*, NCEE 2010-4039 (IES / What Works Clearinghouse, 2010), Recommendation 2, https://ies.ed.gov/ncee/wwc/docs/practiceguide/fractions_pg_093010.pdf, accessed 30 August 2026. Missing magnitude is “often at the root” of adding numerators and denominators and of not seeing fractions as numbers.

⁴³²NMAP, *Foundations for Success* (2008): representing fractions on a number line is “one key mechanism linking conceptual and procedural knowledge.”

in Grade 4, volume in Grade 5; the first quadrant of the coordinate plane in Grade 5.⁴³³ Robert Siegler, Greg Duncan, Pamela Davis-Kean, Kathryn Duckworth, Amy Claessens, Mimi Engel, Maria Ines Susperreguy, and Meichu Chen, in *Psychological Science* in 2012, is why the chapter is the heaviest elementary chapter in the book. Using the British Cohort Study ($N = 3,677$) and the U.S. Panel Study of Income Dynamics Child Development Supplement ($N = 599$), they found that elementary students' knowledge of fractions and of whole-number division uniquely predicted algebra knowledge and overall mathematics achievement five or six years later, after controlling for other mathematical knowledge, general intellectual ability or working memory, and family income and education.⁴³⁴ A one-standard-deviation increase in age-10 fractions knowledge was uniquely associated with a 0.15-SD (United Kingdom) / 0.17-SD (United States) increase in high-school algebra, and a 0.16-SD / 0.18-SD increase in overall high-school math.⁴³⁵ Concurrent correlations of high-school fraction knowledge with overall math were $r = .81$ (United Kingdom) and $r = .87$ (United States).⁴³⁶ Those numbers are load-bearing. They are also easy to upgrade. This is a *prediction* finding, not a classroom experiment proving that teaching fractions a certain way *causes* algebra success. Do not upgrade it. Grades 3–5 fractions work is not a side topic, and it is not charming. It is the gate.

Grade labels in this chapter are sequence anchors, not prison bars. A child may sit this band at calendar grade 2 or at calendar grade 6. Chapter 1 already printed the stretch in the homeschool population: the 2016 NCES Parent and Family Involvement survey found Algebra I taught that year to 41 percent of the 6–8 homeschool band, and arithmetic still taught to 31 percent of 9–12.⁴³⁷ Finding 15 of the Panel is the age-lock refusal: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; claims that children of a given age cannot learn certain content because they are “too young,” “not in the appropriate stage,” or “not ready” have “consistently been shown to be wrong.”⁴³⁸ Accelerate a child who has the prerequisites. Do not

⁴³³Common Core State Standards Initiative, *Common Core State Standards for Mathematics*, https://www.thecorestandards.org/assets/CCSSI_Math%20Standards.pdf, accessed 30 August 2026. Grade 3–5 overviews: four operations on whole numbers; fractions as numbers then operations; decimals as tenths/hundredths then place value; area (Grade 3), angle measure (Grade 4), volume (Grade 5); first quadrant (Grade 5). CCSSM is a map, not a homeschool mandate.

⁴³⁴Robert S. Siegler, Greg J. Duncan, Pamela E. Davis-Kean, Kathryn Duckworth, Amy Claessens, Mimi Engel, Maria Ines Susperreguy, and Meichu Chen, “Early Predictors of High School Mathematics Achievement,” *Psychological Science* 23, no. 7 (2012): 691–697, <https://doi.org/10.1177/0956797612440101>; author PDF <https://sieglertc.columbia.edu/wp-content/uploads/2019/02/Siegler-et-al-PsySci12.pdf>, accessed 30 August 2026. British Cohort Study $N = 3,677$; U.S. PSID-CDS $N = 599$. Elementary fractions and whole-number division uniquely predicted algebra and overall mathematics five or six years later after other mathematical knowledge, general intellectual ability or working memory, and family income and education. This is a *prediction* finding, not a fractions-curriculum RCT.

⁴³⁵Siegler et al., “Early Predictors” (2012). A 1-SD increase in age-10 fractions knowledge uniquely associated with a 0.15-SD (UK) / 0.17-SD (US) increase in high-school algebra and a 0.16-SD / 0.18-SD increase in overall high-school math, after the controls in n. 434.

⁴³⁶Siegler et al., “Early Predictors” (2012). Concurrent correlations of high-school fraction knowledge with overall math: $r = .81$ (UK) and $r = .87$ (US).

⁴³⁷National Center for Education Statistics, Parent and Family Involvement in Education survey (NHES:2016 / PFI 2016), as used in this book’s Chapter 1: Algebra I taught that year to 41 percent of the 6–8 homeschool band; arithmetic still taught to 31 percent of 9–12. Grade labels are sequence anchors, not prison bars. The 2023 First Look (NCES 2024-113) did not republish those teaching-style or subject tables.

⁴³⁸NMAP, *Foundations for Success* (2008), Finding 15: what is developmentally appropriate is “largely contingent

withhold fractions from a nine-year-old who has them. Do not accelerate a child who does not. Skipping the gate because a child is “good at math” is the error Finding 30 does not license.⁴³⁹ Siegler and colleagues are the reason the prerequisites include fractions.

The Common Core State Standards for Mathematics are treated here as a map, not as a homeschool mandate. Many homeschoolers do not use CCSSM. Many U.S. states have adopted, adapted, or replaced it. A 2026 state-by-state census of who still uses it is **UNVERIFIED** in this project; this chapter will not say “most states.”⁴⁴⁰ What the map is good for is specifying, grade by grade, what the mathematics is: 5×7 as 5 groups of 7; a unit fraction $1/b$ as one of b equal parts, a/b as a copies of $1/b$, both as numbers on a line; equivalent fractions as the same point; unlike-denominator addition as replacing with equivalent like-denominator fractions; multiplication as scaling — the sentence that kills “multiply makes bigger”; division of a fraction by a fraction *not* a Grade 5 requirement, but a Grade 6 one; decimals as tenths and hundredths on the same line; area as covering with unit squares. The map does not dictate teaching methods. The order of topics inside a grade is not a required teaching order.⁴⁴¹ This book will not dump 180 days. It will not sell Singapore, Saxon, or Math-U-See. Those names will appear as options, by grain and placement, labelled **vendor** where the claim is the publisher’s.

Four lanes run through the sections. They are not four identical headings. Sections 5.2 through 5.5 explain the mathematics to the parent — multiplicative reasoning; fractions as numbers; operations and whole-number bias; decimals, area, volume, the first quadrant. Section 5.6 is teaching guidance: concrete representations connected to symbols in the same visual field, then faded; the number line as the central fraction tool; interleaving after a short blocked introduction, labelled as transfer of a grade-7 finding, not as a grade-4 *d*. Section 5.7 is tips that survive evidence, including the cooking caution. Section 5.8 is the parent-hosted AI lane for this band: a short reminder of Chapter 3’s ban versus allow, then the grade 3–5 jobs. The child does not live in an open-web agent. Facts are not a language-model job. The parent holds the key.

Chapter 2 already locked the method this chapter will use: five interwoven strands; conceptual and procedural knowledge iterating, not a one-way street; the number line as the representation that scales; worked examples faded for novices, incorrect examples explained; blocked practice to learn execution, interleaved and spaced practice to choose a strategy and keep it; word problems as schemas, not keyword hunts; fluency as flexible, accurate, efficient retrieval, with timed facts as two objects; explicit instruction for students already struggling, productive failure as generation-then-instruction on a new concept when priors exist — those two literatures not the same compar-

on prior opportunities to learn”; age-lock claims (“too young,” “not in the appropriate stage,” “not ready”) have “consistently been shown to be wrong.” Brain-immaturity claims are not justified even when students possess the prerequisite knowledge.

⁴³⁹NMAP, *Foundations for Success* (2008), Finding 30: mathematically gifted students with motivation “appear to be able to learn mathematics much faster... with no harm to their learning, and should be allowed to do so.” Acceleration requires prerequisites; Siegler et al. 2012 is why those include fractions. Do not skip the gate because a child is “good at math.”

⁴⁴⁰State-by-state status of CCSSM in 2026 (who still uses it, who uses “standards formerly known as,” who never did): **UNVERIFIED** in this project (01-scope open question 8). Do not say “most states” without a current count. CCSSM is treated as a widely used map, not a mandate.

⁴⁴¹CCSSM, “How to read the grade level standards,” https://www.thecorestandards.org/assets/CCSSI_Math%20Standards.pdf. The standards “do not dictate curriculum or teaching methods”; the order of topics inside a grade is not a required teaching order.

ison; kitchen-math transfer hard. Chapter 3 already locked the AI lane: unguarded solvers raise practice scores and cut unaided exams; the model may prepare the adult and the next problem; it may not do the child's problem. This chapter will not re-litigate Bastani at monograph length. One short reminder, then the objects of this band.

The popular story collapses those locks into slogans they do not support. They'll learn fractions by cooking. Multiply always makes bigger, until a cute exception. Invert and multiply, and do not ask why. Cross-multiply to compare. Pizza is fractions. The times tables are a chant, and then we are done. The AI will tutor. A "grade 5" book is fifth grade. The documents do not say any of those things, not as the slogans say them. This chapter prints the documents. It prints the objects, the sample sizes, the delays, and the nulls. It leaves the open questions standing — Fuchs and colleagues' 2013 and 2014 fractions-intervention coefficients, not extracted for this project; the remainder of the Panel's Table 2 geometry and measurement row, truncated in the fetch; transfer from cooking and money to formal items, measured rather than assumed; interleaving in grades 3–5 as a clean classroom trial; an isolated timer effect. It does not invent a kitchen scene to make the science feel warmer. The parent who stopped at algebra is the design user. The job is to hear $2/5$ as a magnitude failure, and $5 \times 1/2$ as a shrink, before any tool is allowed to print a correct least-common-denominator path the parent never sees.

5.1 What this band is, and why it is the heaviest elementary chapter

The mathematics of this band is four operations on whole numbers, taken toward fluency; fractions as numbers, then operations on fractions; decimals as a notation for tenths and hundredths, then as place value; and measurement that is also a model of multiplication — area, then angle, then volume — plus the first coordinate plane. That is a lot of mathematics. It is not a lot because a publisher padded a catalog. It is a lot because the Panel named this window as the place the Critical Foundations are either built or not, and because Siegler and colleagues then showed that what is built here still predicts algebra years later, after the controls a skeptic would demand.⁴⁴²

What the map actually contains, grade by grade, is more specific than a slogan about "upper elementary." The Common Core's Grade 3 critical areas are four: multiplication and division and strategies within 100; fractions, especially unit fractions; rectangular arrays and area; two-dimensional shapes.⁴⁴³ The Grade 3 domains a parent will meet, whether or not the family uses CCSSM as a mandate, are the objects later sections will teach.

Operations and Algebraic Thinking, 3.OA: interpret products — 5×7 as 5 groups of 7 — and quotients, both partitive (how many in each group) and measurement (how many groups). Word problems within 100: equal groups, arrays, measurement quantities. The unknown in any position of $a \times b = c$. Properties — commutative, associative, distributive — with formal names not required. Division as unknown-factor. Fluently multiply and divide within 100; by the end of Grade 3, know from memory all products of two one-digit numbers. That last clause is 3.OA.7, and this book cites

⁴⁴²NMAP Critical Foundations window plus Siegler et al. 2012 (nn. 427–428, 434–435). The prediction is after controls; it is not a treatment effect of a named fractions curriculum.

⁴⁴³CCSSM Grade 3 overview, four critical areas: (1) multiplication and division and strategies within 100; (2) fractions, especially unit fractions; (3) rectangular arrays and area; (4) two-dimensional shapes.

it as a map, not as a homeschool law.⁴⁴⁴ Two-step word problems with a letter for the unknown. Arithmetic patterns explained with properties.

Number and Operations in Base Ten, 3.NBT, is the whole-number remainder of Grade 2, not a new universe: round to 10 or 100; fluently add and subtract within 1,000; multiply one-digit numbers by multiples of 10 from 10 to 90.

Number and Operations — Fractions, 3.NF, is the first Fractions domain in the K–8 map. It is limited to denominators 2, 3, 4, 6, and 8. A unit fraction $1/b$ is one part when a whole is partitioned into b equal parts. a/b is a copies of $1/b$. Fractions as numbers on the number line. Equivalence in special cases; simple equivalent fractions; whole numbers as fractions; compare fractions with the same numerator or the same denominator, same whole.⁴⁴⁵ Equal shares in 1.G.3 and 2.G.3 — halves, thirds, fourths of circles and rectangles — are the only CCSSM fraction language before this domain. They are the on-ramp. They are not the gate. Chapter 4 was the on-ramp. This chapter is the gate.

Measurement and Data, 3.MD: time to the nearest minute and elapsed time; liquid volume and mass in one-step problems, same units; scaled picture and bar graphs; line plots with halves and fourths of an inch; area — the unit square, tiling, area of a rectangle as the product of side lengths, the distributive property via area, area as additive; perimeter of polygons, including same perimeter with different area and the reverse. Geometry, 3.G: shared attributes define larger categories (quadrilaterals); partition shapes into equal-area parts expressed as unit fractions.

Grade 4's critical areas are three: multi-digit multiplication and division with multi-digit dividends; fraction equivalence, addition and subtraction with like denominators, multiplication of a fraction by a whole number; classifying geometric figures by properties.⁴⁴⁶ 4.OA adds multiplication as comparison — “times as many,” distinguished from additive comparison — multistep whole-number problems, interpreting remainders, factor pairs from 1 to 100, prime and composite, number and shape patterns. 4.NBT, whole numbers to 1,000,000: a digit in one place represents ten times what it represents to the right; read, write, compare, round; fluently add and subtract multi-digit whole numbers using the standard algorithm; multiply up to four digits by one digit, and two two-digit numbers, with equations, arrays, area models; quotients and remainders, up to four-digit dividends and one-digit divisors, with the same representations. 4.NF, denominators 2, 3, 4, 5, 6, 8, 10, 12, 100: why $a/b = (n \times a)/(n \times b)$; compare fractions with unlike denominators by common denominators or numerators or a benchmark such as $1/2$, same whole; build fractions from unit fractions; add and subtract like denominators including mixed numbers; multiply a fraction by a whole number; decimal notation for tenths and hundredths; compare decimals to hundredths. Unlike-denominator addition in general is *not* a Grade 4 requirement. 4.MD: relative sizes of units; convert larger to smaller; area and perimeter formulas for rectangles; line plots in $1/2$, $1/4$, $1/8$ units;

⁴⁴⁴CCSSM 3.OA.7: fluently multiply and divide within 100; by end of Grade 3, know from memory all products of two one-digit numbers. Cited as a map, not a homeschool law.

⁴⁴⁵CCSSM 3.NF (denominators 2, 3, 4, 6, 8): unit fraction $1/b$; a/b as a copies of $1/b$; fractions as numbers on the number line; equivalence in special cases; whole numbers as fractions; compare same numerator or same denominator, same whole.

⁴⁴⁶CCSSM Grade 4 overview, three critical areas: (1) multi-digit multiplication and division with multi-digit dividends; (2) fraction equivalence, addition/subtraction with like denominators, multiplication of a fraction by a whole number; (3) classifying geometric figures by properties.

angles — the one-degree angle as $1/360$ of a circle, measure with a protractor, angle measure as additive. 4.G: points, lines, line segments, rays, right, acute, and obtuse angles, perpendicular and parallel lines; classify two-dimensional figures; right triangles; line symmetry.

Grade 5's critical areas are three: fluency with addition and subtraction of fractions, multiplication of fractions, and division of unit fractions by whole numbers and of whole numbers by unit fractions; two-digit divisors, decimals in the place-value system, operations with decimals to hundredths, fluency with whole-number and decimal operations; volume.⁴⁴⁷ 5.OA: parentheses, brackets, braces; write and interpret numerical expressions without evaluating; two numerical patterns, ordered pairs, graph in the first quadrant. 5.NBT: a digit represents 10 times the place to its right and $1/10$ the place to its left; patterns of zeros and decimal-point placement with powers of 10; whole-number exponents for powers of 10; read, write, compare decimals to thousandths; round decimals; fluently multiply multi-digit whole numbers using the standard algorithm; quotients with up to four-digit dividends and two-digit divisors (strategies and area models; the standard algorithm for division is a Grade 6 fluency); add, subtract, multiply, divide decimals to hundredths, with explanation. 5.NF: add and subtract fractions with unlike denominators, including mixed numbers, by replacing with equivalent like-denominator fractions; word problems; benchmark estimates — reject $2/5 + 1/2 = 3/7$ because $3/7$ is less than $1/2$; a fraction as division, $a/b = a \div b$; multiply fraction $\times$ whole and fraction $\times$ fraction; area of a rectangle with fractional side lengths; multiplication as scaling — a fraction greater than 1 enlarges, a fraction less than 1 shrinks; division limited to unit-fraction $\div$ whole and whole $\div$ unit-fraction. Division of a fraction by a fraction is *not* a Grade 5 requirement. It is Grade 6.⁴⁴⁸ 5.MD: convert among units within a system; line plots in $1/2$, $1/4$, $1/8$ and fraction operations on the data; volume — unit cube, packing, $V = l \times w \times h$ and $V = B \times h$ for right rectangular prisms, volume as additive. 5.G: coordinate plane, origin, ordered pairs, graph in the first quadrant; hierarchy of two-dimensional figures — squares are rectangles, so squares have four right angles.

That catalog is a map of typical U.S. public-school placement, not a claim that a homeschooler must wait, and not a 180-day plan. NCTM's 2006 *Curriculum Focal Points* named three “most important mathematical topics” per grade as a coherence device. The Grade 3–5 rows, from a publicly posted excerpt of the NCTM book — the full official PDF is paywalled, and this project treats that posting as a **secondary copy of a primary** — put meanings and equivalence of fractions in Grade 3, decimals in Grade 4, addition and subtraction of fractions in Grade 5, with multiplication and division of fractions held for Grade 6.⁴⁴⁹ Prefer a library copy before treating the excerpt as the official book. The Focal Points are a professional consensus on focus, not “the current U.S. standard,” and they predate CCSSM.

⁴⁴⁷CCSSM Grade 5 overview, three critical areas: (1) fluency with addition/subtraction of fractions, multiplication of fractions, division of unit fractions by wholes and wholes by unit fractions; (2) two-digit divisors, decimals in the place-value system, operations with decimals to hundredths; (3) volume.

⁴⁴⁸CCSSM 5.NF: unlike-denominator addition/subtraction; fraction as division; fraction $\times$ fraction; multiplication as scaling (5.NF.5); division limited to unit-fraction $\div$ whole and whole $\div$ unit-fraction. Division of a fraction by a fraction is Grade 6 (6.NS), not Grade 5.

⁴⁴⁹National Council of Teachers of Mathematics, *Curriculum Focal Points for Prekindergarten through Grade 8 Mathematics* (2006). Grade lists used here from a publicly posted excerpt, https://www.platonicsolids.info/focal_points_by_grade.pdf, accessed 30 August 2026 — **secondary copy of a primary**; full NCTM book paywalled. Grade 3 meanings/equivalence; Grade 4 decimals; Grade 5 $+/-$ fractions; Grade 6 $\times \div$ fractions. Prefer a library copy before treating the excerpt as the official book.

The Panel’s Table 2 benchmarks give a pace, not a law, and they are judgments from curriculum comparison, not experimental placements — the Panel says so.⁴⁵⁰ By the end of Grade 3: proficient addition and subtraction of whole numbers. By the end of Grade 5: proficient multiplication and division of whole numbers; proficient comparing of fractions, decimals, and common percents, and addition and subtraction of fractions and decimals. Identify, represent, and compare fractions and decimals on a number line by the end of Grade 4. Multiply and divide fractions and decimals by the end of Grade 6. The geometry and measurement remainder of Table 2 was truncated in the fetch used for this project. That remainder stays **UNVERIFIED**. This chapter will not invent those benchmark lines.⁴⁵¹

What this band depends on is Chapter 4’s work: place value through 1,000, memorized single-digit sums, equal-shares language, arrays as multiplication seeds. What it unlocks is Grade 6 ratio and rate, division of fractions by fractions, and the rational-number system including negatives; and, years later, high-school algebra’s rational expressions and linear equations over the rationals. Area models for the distributive property become polynomial multiplication. The first quadrant becomes four quadrants, then a graph of a function. Skip the gate and those later objects arrive as symbol soup.

The prediction that makes the chapter heavy, restated with the controls visible, is Siegler and colleagues 2012. Age-10 fractions knowledge uniquely predicted high-school algebra after other arithmetic, IQ or working memory, and family background were in the model. The United Kingdom coefficient was 0.15 SD of algebra per 1 SD of fractions; the United States coefficient was 0.17 SD.⁴⁵² Concurrent high-school correlations of fraction knowledge with overall math were $r = .81$ and $r = .87$. Those are not small associations. They are also not a treatment effect. Fuchs and colleagues’ later fractions-intervention experiments, 2013 and after, are the experimental half of that research program. Numerical effect sizes from those papers were not extracted for this project. They remain **UNVERIFIED** here. Do not invent a g .⁴⁵³ The IES 2021 intervention guide’s Recommendation 4 — number lines, strong evidence, 14 studies — is the synthesis this chapter will use, not a homemade coefficient.

A vendor book labelled “grade 5” is not a legal grade. Singapore Math Inc. states that Dimensions Math 6 “includes some Pre-algebra topics, but is not a Pre-algebra text in the traditional sense; it is a Grade 6 text” — **vendor**. Saxon’s homeschool sequence uses split names, Math 5/4, 6/5, 7/6, 8/7, that do not match calendar grade. Math-U-See uses Greek letters uncoupled from calendar

⁴⁵⁰NMAP, *Foundations for Success* (2008), Table 2. Benchmarks are judgments from curriculum comparison, not experimental placements — the Panel says so. Partial rows used here: proficient $+/-$ of whole numbers by end of Grade 3; identify/represent/compare fractions and decimals by end of Grade 4; proficient $\times\div$ of whole numbers and $+/-$ of fractions and decimals by end of Grade 5; $\times\div$ of fractions and decimals by end of Grade 6.

⁴⁵¹NMAP Table 2 geometry/measurement remainder: truncated in the ERIC fetch used for this project (01-scope open question 3 / brief open question 11). **UNVERIFIED** until pp. 19–20 of ED500486 are re-opened. Do not invent those benchmark lines.

⁴⁵²Siegler et al., “Early Predictors” (2012); see nn. 434–435. UK 0.15 SD / US 0.17 SD of high-school algebra per 1 SD of age-10 fractions after controls. Do not upgrade from prediction to a fractions-curriculum RCT.

⁴⁵³Lynn S. Fuchs et al., fractions-intervention experiments (2013 and after), including “Improving At-Risk Learners’ Understanding of Fractions,” *Journal of Educational Psychology* (2013), author PDF located at <https://sieglertc.columbia.edu/wp-content/uploads/2019/08/5023-Fuchs13.pdf>. Numerical effect sizes were **not extracted** for this project (01-scope open question 10; brief open question 17). **UNVERIFIED** as coefficients here. Do not invent a g . Use Fuchs et al. 2021 Recommendation 4 (n. 482) as the synthesis.

grade: Epsilon is a fractions *level*, not “fifth grade” — **vendor**. If a family runs one Math-U-See level per calendar year from Primer, formal fraction operations can sit later than the CCSSM Grade 3–5 window — or earlier, if Alpha started in kindergarten.⁴⁵⁴ Beast Academy, **vendor** / Art of Problem Solving, “generally recommend[s] that students start about a level ‘behind’ their current grade level” because of problem difficulty, and states that levels 4–5 cover many middle-grades standards. A child of ten may place into Singapore 3A, Saxon 6/5, and Math-U-See Gamma simultaneously. That is not a contradiction. It is three different grain-sizes of “grade.” Placement tests exist because the labels lie. This book names those programs as options. It does not sell them. No homeschool-curriculum randomized trial comparing them was opened for this project.

Failure points visible in this band, named in the scope memo and inherited here, are five, and they are not cute. Multiplication facts not known from memory by the end of Grade 3 — 3.OA.7 is explicit; the Panel requires automatic recall to free working memory. Fractions treated only as pizza slices, never as numbers on a line — 3.NF.2 is the number-line standard the Panel later calls a key mechanism. Unlike-denominator addition delayed past Grade 5. “Multiplication always makes bigger” — 5.NF.5 exists specifically to break this. Standard algorithms used without place-value explanation, or, conversely, only strategies with no path to a general algorithm.⁴⁵⁵ Those five are the diagnosis list a parent can steal before buying the next book. They are also why this chapter is long. Algebra I will be symbol soup if this chapter thins.

The live debates this chapter will not pretend to settle belong on the page so they are not settled by a franchise. When to introduce the standard algorithms versus extended strategy work: the Panel wants both automaticity and properties; CCSSM places fluency with the standard algorithm for multi-digit addition and subtraction in Grade 4, for multi-digit multiplication in Grade 5, and for the division algorithm in Grade 6.⁴⁵⁶ Print both demands. Do not pick a franchise’s religion. Whether Fuchs 2013 and 2014 coefficients should appear: papers located, numbers not copied; open those PDFs before quoting a *g*. Concepts-first versus iterative for fractions specifically: Rittle-Johnson, Schneider, and Star 2015 still applies — it is a myth that it is a one-way street — and Rittle-Johnson and Koedinger 2009 was a decimal study, not a fractions-curriculum randomized trial.⁴⁵⁷ Cross-multiply-to-compare as a parent “trick” versus the IES 2021 intervention panel’s

⁴⁵⁴Math-U-See / Demme Learning, curriculum sequence, <https://mathusee.com/products/math-u-see-curriculum/>, accessed 30 August 2026. Epsilon is a fractions *level*, **vendor**, not “fifth grade.” Singapore Math Inc., School FAQ, <https://www.singaporemath.com/pages/school-faq>; Dimensions Math 6 “includes some Pre-algebra topics, but is not a Pre-algebra text in the traditional sense; it is a Grade 6 text” — **vendor**. Houghton Mifflin Harcourt, Saxon Home-school Math Scope and Sequence: 5/4, 6/5, 7/6, 8/7 are skill names, not calendar grades. Art of Problem Solving, Beast Academy FAQ, <https://beastacademy.com/faq/books>: “we generally recommend that students start about a level ‘behind’ their current grade level” — **vendor**. No homeschool-curriculum RCT comparing these was opened.

⁴⁵⁵Failure points from 01-scope Grades 3–5: (a) multiplication facts not from memory by end of Grade 3 (3.OA.7; NMAP automatic recall); (b) fractions treated only as pizza, never as numbers on a line (3.NF.2; NMAP key-mechanism sentence); (c) unlike-denominator addition delayed past Grade 5; (d) “multiplication always makes bigger” (5.NF.5); (e) standard algorithms without place-value explanation, or only strategies with no path to a general algorithm.

⁴⁵⁶NMAP Finding 11 (automaticity plus properties plus standard algorithms) and CCSSM fluencies: Grade 4 standard algorithm for multi-digit $\div$ (4.NBT); Grade 5 for multi-digit $\times$ (5.NBT); Grade 6 for the division algorithm (6.NS). Live debate: when the algorithm arrives versus extended strategy work. Print both demands; do not pick a franchise’s religion.

⁴⁵⁷Bethany Rittle-Johnson, Michael Schneider, and Jon R. Star, “Not a One-Way Street: Bidirectional Relations between Procedural and Conceptual Knowledge of Mathematics,” *Educational Psychology Review* 27, no. 4 (2015): 587–597. “It is a myth that it is a one-way street from conceptual knowledge to procedural knowledge.” Bethany Rittle-

magnitude objection: for intervention, magnitude wins on the page this book can defend.⁴⁵⁸

What this section has done is place the window. Multiplicative reasoning comes next, because arrays are also the area model and the on-ramp to fraction multiplication. The gate is not the times table. The gate is a fraction as a number. The times table is what frees working memory so the gate can be walked through.

5.2 Multiplicative reasoning: groups, arrays, area, and facts from memory

Multiplication, for a parent who stopped at algebra, is often a chant. 5×7 is “thirty-five,” retrieved, or not, and the conversation ends. The map does not end there. 3.OA asks the parent to interpret a product: 5×7 as 5 groups of 7. Not 5 and 7 somehow fused. Not a rhyme. Five groups, seven in each group, thirty-five in all.⁴⁵⁹ That sentence is the first teaching move of this band, and it is a sentence the parent has to be able to hear wrong. A child who can recite 5×7 and cannot show five groups of seven, or a 5-by-7 array of unit squares, or five jumps of seven on a number line, is missing a strand. A child who can draw the array and cannot retrieve 5×7 when it is needed inside $3/5$ of 35 has missing a different strand. Chapter 2’s five strands are already in the room. Ranking them is still misguided.

Division is the inverse, and it is two situations, not one. Partitive division asks how many in each group: 35 cookies, 5 friends, how many each. Measurement division asks how many groups: 35 cookies, 7 to a bag, how many bags. Both are unknown-factor problems. $5 \times \square = 35$ and $\square \times 7 = 35$. The unknown in any position of $a \times b = c$ is the algebraic seed, the same seed Chapter 4 planted with addition and subtraction unknowns in all positions. A letter for the unknown in a two-step word problem, 3.OA.8, is not “early algebra” as a franchise slogan. It is the equal sign still being a relation, now inside a product.⁴⁶⁰

Word problems in this band are equal groups, arrays, and measurement quantities, then, in Grade 4, multiplicative comparison: “times as many,” distinguished from additive comparison. “Maria has 3 times as many as Sam” is not “Maria has 3 more than Sam.” The cover story can hide the structure. Keyword strategies fail because the same word appears in different structures. Fuchs and colleagues, 2021, Table 5.2, list why *all*, *left*, and *total* fail.⁴⁶¹ What does work, repeatedly, is teaching the child to identify the underlying structure. Additive: combine, change, compare. Multiplicative: equal groups, comparison, and, late in the band, the first ratio language. Asha

Johnson and Kenneth R. Koedinger, “Iterating between Lessons on Concepts and Procedures Can Improve Mathematics Knowledge,” *British Journal of Educational Psychology* 79 (2009): 483–500: sixth-grade *decimals* on an intelligent tutor, $N = 77$ and $N = 26$; iterative sequence beat a concepts-first block on procedures including transfer. Do not pretend 2009 was a fractions-curriculum RCT.

⁴⁵⁸Fuchs et al., *Assisting Students Struggling with Mathematics: Intervention in the Elementary Grades*, WWC 2021006 (IES, 2021), Recommendation 4 obstacles: the panel advises against cross-multiply-to-compare during intervention because it does not build magnitude. <https://ies.ed.gov/ncee/wwc/Docs/PracticeGuide/WWC2021006-Math-PG.pdf>, accessed 30 August 2026.

⁴⁵⁹CCSSM 3.OA: interpret products (5×7 as 5 groups of 7) and quotients (partitive and measurement division).

⁴⁶⁰CCSSM 3.OA.8: two-step word problems with a letter for the unknown. Equal-sign-as-relation is the McNeil and Alibali / Knuth et al. lineage already locked in Chapter 2; it is now inside a product.

⁴⁶¹Fuchs et al. 2021, Recommendation 5, Table 5.2: why keyword strategies (*all* / *left* / *total*) fail. Strong evidence, 18 studies. Russell Gersten et al., *Assisting Students Struggling with Mathematics: Response to Intervention (RtI) for Elementary and Middle Schools*, NCEE 2009-4060 (IES, 2009), also rejects keyword strategies.

Jitendra's schema-based instruction packages the move: identify the schema, represent it with a diagram, estimate, solve, check.

Jitendra and colleagues, 2013, is the third-grade randomized trial this section can actually quote. Third-graders with mathematics difficulties, small-group schema-based instruction versus control. Word-problem posttest $g = 0.46$. District achievement $g = 0.34$. No significant retention at 8 weeks.⁴⁶² Those three numbers together are the honest paragraph. Schema-based instruction can move trained word-problem items. The district test moved less. Eight weeks later the retention was not significant. Do not quote the 0.46 as a general-achievement miracle. Jitendra, Jon Star, Danielle Dupuis, and Michael Rodriguez, 2012, is a seventh-grade proportion trial — $g = 1.24$ immediate, $g = 1.27$ at six weeks, null transfer — and it belongs to Chapter 6, not here, except as a warning not to mash the two g values.⁴⁶³ Corey Peltier and Kimberly Vannest's 2017 meta-analytic $g = 1.57$ on immediate elementary problem solving is mostly researcher-developed proximal measures. Do not quote it as “schema instruction raises math achievement by 1.57 standard deviations.”⁴⁶⁴ The kitchen how-to that survives is in the 2021 IES guide: introduce one problem type at a time, first as a complete story with all quantities, then as a problem with one unknown; use objects to see the action; vary the cover story so the child is matching structure, not nouns; never teach “total means add.”⁴⁶⁵

The representations of multiplication have to be connected, not sequenced as a calendar. Equal groups — five piles of seven. Arrays — five rows of seven unit squares. Skip jumps on a number line — five jumps of seven. Area of a rectangle — a 5-by-7 rectangle tiled with unit squares, the product as covering. The written fact — $5 \times 7 = 35$. Concrete-Representational-Abstract, as Fuchs and colleagues actually wrote it in 2021 Recommendation 3, is not folklore CRA. Choose representations that match the mathematics. Connect concrete and semi-concrete to the notation in the *same visual field*, not “blocks today, symbols next month.” Give ample, meaningful practice so representations become thinking tools, not toys. Revisit after fading; fading is for accuracy, not for a calendar date.⁴⁶⁶ Kira Carbonneau, Scott Marley, and James Selig's 2013 meta-analysis, 55 studies,

⁴⁶²Asha K. Jitendra et al., “A Randomized Controlled Trial of the Impact of Schema-Based Instruction on Mathematical Outcomes for Third-Grade Students with Mathematics Difficulties,” *The Elementary School Journal* 114, no. 2 (2013): 252–276, <https://doi.org/10.1086/673199>. Word-problem posttest $g = 0.46$; district achievement $g = 0.34$; no significant retention at 8 weeks. Trained items, not a general-achievement miracle.

⁴⁶³Asha K. Jitendra, Jon R. Star, Danielle Dupuis, and Michael C. Rodriguez, “Effectiveness of Schema-Based Instruction for Improving Seventh-Grade Students' Proportional Reasoning: A Randomized Experiment,” *Journal of Research on Educational Effectiveness* 5 (2012), <https://doi.org/10.1080/19345747.2012.725804>. Immediate $g = 1.24$; six-week retention $g = 1.27$; no significant transfer. Do not mash with Jitendra 2013 (n. 462). Belongs to Chapter 6 except as a warning.

⁴⁶⁴Corey Peltier and Kimberly J. Vannest, “A Meta-Analysis of Schema Instruction on the Problem-Solving Performance of Elementary School Students,” *Review of Educational Research* (2017), <https://doi.org/10.3102/0034654317720163>. Hedge's $g = 1.57$ immediate problem solving / $g = 1.09$ transfer; 21 studies, 3,408 students; many researcher-developed proximal measures. Do not quote $g = 1.57$ as general achievement.

⁴⁶⁵Fuchs et al. 2021, Recommendation 5 how-to: one problem type at a time; complete story then one unknown; vary cover stories; never teach “total means add.”

⁴⁶⁶Fuchs et al. 2021, Recommendation 3 (strong evidence, 28 studies): choose representations that match the mathematics; connect concrete and semi-concrete to notation in the same visual field; ample meaningful practice; revisit after fading. Not folklore CRA (three weeks of blocks, then pictures, then symbols forever). Table 3.1 is the representation-to-topic map.

$N = 7,237$, kindergarten through college, manipulatives versus abstract symbols only, found an overall mean d of approximately 0.37, small-to-moderate, favoring manipulatives.⁴⁶⁷ That number is not “manipulatives work.” It is versus abstract-only, and it is moderated: larger on retention ($k = 53$) than on problem solving, transfer, or justification; high guidance associated with higher retention and problem solving; ages 3–6 more difficulty than 7–11 or 12+. A bin of bright tiles left on the table is not Recommendation 3. The tiles next to the written 5×7 , in the same sitting, with the parent asking “show me the five groups,” is.

Area, 3.MD, is not an extra after the times tables. It is the array, made measurement. The unit square is the unit. Tiling a rectangle with unit squares, no gaps, no overlaps, is the covering. The area of a rectangle as the product of side lengths is then a compression of that covering: 5 rows of 7 unit squares. The formula should be said as “rows of unit squares,” not as “multiply the numbers you see.” The distributive property via area — a 5-by-7 rectangle split as 5-by-5 plus 5-by-2 — is the same property the child will later use to multiply 14×23 , and later still to multiply polynomials. Area as additive: two rectangles sharing a side, total area the sum. Perimeter of polygons sits in the same cluster and is a different quantity. Same perimeter, different area, and the reverse, are the problems that break the “multiply the numbers you see” habit for both measures at once.⁴⁶⁸ Tile, then array, then formula. A 3-by-4 rectangle that is “12” with no unit squares is a child who has the chant and not the covering. That failure point is named in the learning memo as common teaching lore; a single named prevalence study was not opened, and stays **UNVERIFIED** as to prevalence. The IES fractions guide and *Adding It Up* both treat area models as something that must be taught, not assumed.⁴⁶⁹

Properties are not extras, and formal names are not required. Commutative: 5×7 is 7×5 , five groups of seven is a 5-by-7 array rotated into a 7-by-5 array, same covering. Associative: $(2 \times 3) \times 4$ is $2 \times (3 \times 4)$. Distributive: 6×7 as $6 \times (5 + 2)$, which is the area split. The Panel’s Finding 11 puts those properties inside computational proficiency, together with automatic recall of facts and fluency with standard algorithms.⁴⁷⁰ A child who can retrieve 8×7 but cannot say why 8×7 equals 7×8 is missing a strand. A child who can draw the rotation and cannot retrieve 8×7 inside a fraction problem is missing a different one. Hear both.

Then retrieve. 3.OA.7: fluently multiply and divide within 100; by the end of Grade 3, know from

⁴⁶⁷Kira J. Carbonneau, Scott C. Marley, and James P. Selig, “A Meta-Analysis of the Efficacy of Teaching Mathematics with Concrete Manipulatives,” *Journal of Educational Psychology* 105, no. 2 (2013): 380–400, <https://doi.org/10.1037/a0031084>. 55 studies, $N = 7,237$, K through college, manipulatives vs abstract-only. Overall mean $d \approx 0.37$ (small-to-moderate). Outcome-specific: moderate-to-large on retention ($k = 53$); small on problem solving ($k = 9$), transfer ($k = 13$), and justification ($k = 2$). High guidance associated with higher retention and problem solving; low guidance with higher transfer; ages 3–6 more difficulty than 7–11 or 12+. Do not quote $d \approx 0.37$ as “manipulatives work” without the moderators.

⁴⁶⁸CCSSM 3.MD: unit square; tiling; area of a rectangle as product of side lengths; distributive property via area; area as additive; perimeter of polygons; same perimeter / different area and vice versa.

⁴⁶⁹Area-as-chant-without-unit: **UNVERIFIED** as to a single named prevalence study (02-learning §3.2). IES 2010 fractions guide and National Research Council, *Adding It Up: Helping Children Learn Mathematics*, ed. Jeremy Kilpatrick, Jane Swafford, and Bradford Findell (National Academy Press, 2001), both treat area models as something that must be taught, not assumed.

⁴⁷⁰NMAP, *Foundations for Success* (2008), Finding 11: computational proficiency depends on automatic (“quick and effortless”) recall of addition/subtraction and multiplication/division facts, fluency with standard algorithms, and understanding of commutative, associative, and distributive properties.

memory all products of two one-digit numbers. Cite the standard as a map, not a mandate. The Panel's Finding 11 is the same demand in different letterhead: automatic, "quick and effortless," recall of multiplication and related division facts, to free working memory.⁴⁷¹ NMAP's Grade 5 benchmark is proficient multiplication and division of whole numbers by the end of Grade 5 — multi-digit, not only the facts.⁴⁷² Grades 3 is the facts-from-memory year on the map; Grades 4 and 5 are the multi-digit years. 4.NBT: multiply up to four digits by one digit, and two two-digit numbers, with equations, arrays, area models. 5.NBT: fluently multiply multi-digit whole numbers using the standard algorithm; quotients with up to four-digit dividends and two-digit divisors, strategies and area models, with the standard algorithm for division held as a Grade 6 fluency.⁴⁷³

The live debate is when the standard algorithm arrives versus how long to stay in extended strategies. Print both demands. The Panel wants automaticity *and* properties. CCSSM places the standard algorithm for multi-digit addition and subtraction as Grade 4 fluency, for multi-digit multiplication as Grade 5 fluency, and for division as Grade 6 fluency. A franchise that withholds a general algorithm until a child has invented one, with no path to a general method, is one failure point. A franchise that runs the standard algorithm as a chant with no place-value explanation is the other. Neither is the method Chapter 2 locked. Iterate: a short conceptual pass on what the partial products mean — the area model of 14×23 as $(10 + 4) \times (20 + 3)$, four rectangles — then a handful of problems, then a return to the model, then more problems. Do not withhold practice for weeks until they understand. Do not run pages of a new algorithm with no representation and no why.⁴⁷⁴

Fluency, restated from Chapter 2 because this is the band where the timer temptation is loudest, is flexible, accurate, efficient, and appropriate retrieval — not a race and not optional. Timed facts are two objects. IES 2021 Recommendation 6, strong evidence, 27 studies: regularly include timed activities as one way to build fluency, on already-taught material, brief, mixed, often with the child graphing their own score. Time a child only on facts they can already produce correctly when untimed.⁴⁷⁵ Jo Boaler's 2012 commentary and 2015 working paper *Fluency Without Fear* attack a different object: high-stakes early speed tests used to rank young children. The working paper is a working paper, not a journal experiment.⁴⁷⁶ The 27 IES studies bundled timing with other intervention components; a 2025 OSF preprint by Peltier and colleagues is the attempt to unbundle,

⁴⁷¹NMAP Finding 11 (n. 470) and CCSSM 3.OA.7 (n. 444). Automatic recall frees working memory for later fraction and multi-digit work.

⁴⁷²NMAP Table 2: proficient multiplication and division of whole numbers by end of Grade 5 — multi-digit, not only the facts.

⁴⁷³CCSSM 4.NBT (multi-digit $\times$ and $\div$ with representations); 5.NBT (standard algorithm for multi-digit $\times$; quotients with two-digit divisors via strategies/area models); 6.NS (standard algorithm for division as fluency).

⁴⁷⁴Iterative model: Rittle-Johnson, Schneider, and Star 2015 (n. 457); Rittle-Johnson and Koedinger 2009 (decimals). Operational rule from Chapter 2: short conceptual pass, then cycle; do not withhold practice for weeks; do not run a new algorithm with no representation and no why.

⁴⁷⁵Fuchs et al. 2021, Recommendation 6 (strong evidence, 27 studies): "Regularly include timed activities as one way to build students' fluency." Table 6.1: brief, mixed, already-taught material; often the child graphs their own score. Implementation constraint (Fuchs, later interviews): time only facts the child can already produce correctly untimed. Gersten et al. 2009 Recommendation 6: about 10 minutes per intervention session on fluent retrieval.

⁴⁷⁶Jo Boaler, "Timed Tests and the Development of Math Anxiety," *Education Week* / Stanford GSE (2012); Jo Boaler, *Fluency Without Fear* (Youcubed working paper, 2015). Labelled working paper / commentary, not a journal experiment. The object of criticism is high-stakes early speed tests used to rank young children — not IES brief timing on already-known facts.

labelled preprint.⁴⁷⁷ The honest kitchen sentence is the one Chapter 2 already wrote. Retrieve known facts often, sometimes against a clock, after the child can already get them right untimed. Mix the facts. Do not make the kitchen timer the identity of being good at math. Do not hand a child who still skip-counts a page of fifty and a clock. Frustration that comes from missing facts is not productive struggle. It is a fluency problem.

Interleave multiplication and division as soon as both exist. Fact families — 5×7 , 7×5 , $35 \div 5$, $35 \div 7$ — force a choice of operation from the problem's features. That is the *practice* of mixing types. The clean classroom randomized trials of interleaving in mathematics are still grade 7. Rohrer, Dedrick, and Burgess 2014: $n = 140$, two-week delayed unannounced test, interleaved 72 percent versus blocked 38 percent, $d = 1.05$. Rohrer, Dedrick, Hartwig, and Cheung 2020: 54 seventh-grade classes, one-month delay, 61 percent versus 38 percent, $d = 0.83$. Those d values are joint with spacing, and they are grade 7. Do not import $d = 1.05$ as a grade-4 finding.⁴⁷⁸ The practice of mixing types still applies, labelled as transfer of a finding, not as that coefficient sitting in a fourth-grade kitchen. Day of a new fact family: a short blocked set, learn to execute. Same week: mixed retrieval of older facts. Later this week and next: the new family reappears between unlike problems — a perimeter item, a unit-fraction comparison, a two-digit product. Monthly: an unannounced mixed quiz without notes. That is a shape, not a 180-day plan. A purchased book that is almost entirely blocked — most are — can still be interleaved by the parent pulling yesterday's and last month's types onto the same page.

3.NBT is the whole-number remainder that multiplicative reasoning sits on: round to 10 or 100; fluently add and subtract within 1,000; multiply one-digit numbers by multiples of 10. Place value is still the structure. 40×3 is four tens, three times, twelve tens. A child who treats 40×3 as " 4×3 with a zero stuck on" may get 120 and still not have the tens. Grade 4's "a digit in one place represents ten times what it represents to the right" is the same structure, one place further. Grade 5's "and $1/10$ the place to its left" is the decimal half of the same sentence, waiting in section 5.5.

Multi-digit multiplication and division, Grades 4 and 5, are the area model grown up. 23×14 is $(20 + 3) \times (10 + 4)$, four partial products, four rectangles. The standard algorithm is a compression of those partial products, not a replacement for them. Division with a remainder, Grade 4, is a measurement or partitive situation that does not come out even; interpreting the remainder is a schema problem, not a "R means leftover" chant. Two-digit divisors, Grade 5, are the last whole-number operation of the band. The standard division algorithm as fluency is Grade 6 on the map. Do not pretend a Grade 5 student who can find $1,248 \div 24$ with an area model has failed the year

⁴⁷⁷The 27 IES studies generally bundled timing with other intervention components (Grimes 2024; Corey Peltier et al., "We're Talking about Practice: A Meta-Analysis of Timed Math Practice," OSF preprint, 2025, <https://doi.org/10.35542/osf.io/9bjtg>). **Preprint.** Isolated timer effect remains open.

⁴⁷⁸Doug Rohrer, Robert F. Dedrick, and Kaleena Burgess, "The Benefit of Interleaved Mathematics Practice Is Not Limited to Superficially Similar Kinds of Problems," *Psychonomic Bulletin & Review* 21 (2014): 1323–1330: grade-7 classroom crossover, $n = 140$, unannounced test two weeks later, interleaved 72% vs blocked 38%, $d = 1.05$, 95% CI [0.80, 1.30]. Doug Rohrer, Robert F. Dedrick, Marissa K. Hartwig, and Chi-NGai Cheung, "A Randomized Controlled Trial of Interleaved Mathematics Practice," *Journal of Educational Psychology* 112, no. 1 (2020): 40–52: preregistered cluster RCT, 54 seventh-grade classes, four months, unannounced test one month later, interleaved 61% vs blocked 38%, $d = 0.83$. Authors: interleaved assignments also space same-kind items, so d is a joint effect. Almost all clean classroom RCTs of interleaving in mathematics are grade 7. Do not import $d = 1.05$ as a grade-4 finding. The *practice* of mixing types still applies, labelled as transfer of a finding.

because the compact algorithm has not yet been automatic. Do not pretend an algorithm with no model is fluency as *Adding It Up* defined it: flexible, accurate, efficient, *appropriate*.

What this section has not done is write a times-table program. Oral retrieval, a shuffled deck, covering the answers, a brief timer on already-known facts: those are Chapter 2's fluency objects, applied here. A language model will happily hallucinate 7×8 . Chapter 3 forbade the model from sitting between the child and the fact; the model may generate the oral script or the shuffled sheet, and the parent's ears do the drill. Section 5.8 will repeat that ban for this band. The gate is not the times table. The times table is what makes the gate walkable. The gate is a fraction as a number, on a line.

5.3 Fractions as numbers on a line

A unit fraction $1/b$ is one part when a whole is partitioned into b equal parts. a/b is a copies of $1/b$. Those two sentences are 3.NF.1, and they are the mathematics a parent who stopped at algebra has to be able to hear before any procedure is run.⁴⁷⁹ $1/4$ is one of four equal parts of a specified whole. $3/4$ is three copies of that $1/4$. $3/4$ is not “three and four.” It is not a pizza slice that happens to be drawn. It is a number. The next standard, 3.NF.2, puts that number on a line. Partition the interval from 0 to 1 into b equal parts. Each part is $1/b$. Iterate from 0. Three iterations of $1/4$ land at $3/4$. That point is the number $3/4$. The same point is the number you will later call 0.75, and $6/8$, and $75/100$. Equivalence is the same point, not a butterfly.⁴⁸⁰

Pies introduce part-whole. They do not by themselves teach that $3/4$ is a number with a magnitude. That is the load-bearing distinction of this section, and it is not a preference about manipulatives. The IES fractions practice guide — Siegler, Thomas Carpenter, Francis Fennell, David Geary, James Lewis, Yukari Okamoto, Laura Thompson, and Jonathan Wray, 2010, NCEE 2010-4039 — Recommendation 2: help students recognize that fractions are numbers that expand the number system; use number lines as a central representational tool from the early grades. Part-whole alone does not express magnitude. Missing magnitude is “often at the root” of adding numerators and denominators and of not seeing fractions as numbers.⁴⁸¹ The 2021 IES intervention guide, Fuchs and colleagues, Recommendation 4, is the same move for struggling elementary students, rated strong evidence on 14 studies.⁴⁸² The Panel wrote that representing fractions on a number line is “one key mechanism linking conceptual and procedural knowledge.”⁴⁸³ Three documents, three letterheads, one representation.

Siegler, Clarissa Thompson, and Michael Schneider, in *Cognitive Psychology* in 2011, supply the cognitive claim under the teaching recommendation. Numerical development is the progressive

⁴⁷⁹CCSSM 3.NF.1: a unit fraction $1/b$ is one part when a whole is partitioned into b equal parts; a/b is a copies of $1/b$.

⁴⁸⁰CCSSM 3.NF.2: fractions as numbers on the number line. Partition 0 to 1 into b equal parts; each part is $1/b$; a copies land at a/b . Equivalence as the same point is the Grade 4 extension (4.NF.1) already planted here.

⁴⁸¹Siegler et al., *Developing Effective Fractions Instruction* (2010), Recommendation 2 (n. 431). Treat fractions as numbers; use number lines as a central tool. Part-whole alone does not express magnitude.

⁴⁸²Lynn S. Fuchs et al., *Assisting Students Struggling with Mathematics: Intervention in the Elementary Grades*, WWC 2021006 (IES, 2021), Recommendation 4: number lines for struggling elementary students, strong evidence, 14 studies.

⁴⁸³NMAP, *Foundations for Success* (2008), number-line mechanism sentence (n. 432).

extension of the mental number line to new classes of number, and the learning that many whole-number properties are not properties of number. Fractions are the first large opportunity to learn that.⁴⁸⁴ Whole numbers have unique successors. Between 3 and 4 there is no whole number. Between 3 and 4 there are infinitely many fractions. Multiplication of whole numbers, by a whole number greater than 1, makes bigger. Multiplication by $1/2$ makes smaller. Addition of “pieces” that a child can see — two numerators, two denominators — is not addition of magnitudes. Magnitude estimation of fractions is closely related to correct fraction arithmetic.⁴⁸⁵ A child who cannot place $3/4$ on a line between 0 and 1, nearer to 1 than to 0, nearer to $1/2$ than to 0, is the child who will write $1/2 + 1/3 = 2/5$, because $2/5$ is what you get when you add the counts you see, and there is no magnitude in the room to reject it.

Equal shares in Grade 1 and Grade 2 were the on-ramp, and they are the only CCSSM fraction language before 3.NF. 1.G.3: partition circles and rectangles into two and four equal shares; the words *halves*, *fourths*, *quarters*; decomposing into more equal shares creates smaller shares. 2.G.3: two, three, or four equal shares — *halves*, *thirds*, *fourths*; equal shares of identical wholes need not have the same shape.⁴⁸⁶ IES 2010 Recommendation 1 is to build on those informal experiences of sharing and fair shares.⁴⁸⁷ A child who has never partitioned a rectangle into four equal parts, and said “fourths,” and noticed that eight equal parts of the same whole are smaller, arrives at 3.NF without the informal residue the guide wants used. Do not skip equal-shares work because “fractions start in third grade.” Chapter 4 was that work. This chapter assumes it, or diagnoses its absence.

Grade 3 NF is limited to denominators 2, 3, 4, 6, and 8. That is a curriculum-comparison judgment about which unit fractions a typical third-grader has the partitioning skill to mean, not a law of nature and not a homeschool mandate. Unit fractions first. Then non-unit fractions as copies. Then the line. Then equivalence in special cases — $1/2 = 2/4 = 3/6$, because they are the same point — and whole numbers as fractions: $3 = 3/1 = 6/2$, same point as the whole number 3 on the same line. Compare fractions with the same numerator or the same denominator, same whole: $1/4$ compared with $1/8$ is a same-numerator comparison that the more-parts-are-smaller observation from equal shares already prepares; $3/8$ compared with $5/8$ is a same-denominator comparison that “more copies of the same unit fraction” already prepares.⁴⁸⁸ Same whole is not a throwaway. $3/4$ of a small bar and $3/4$ of a large bar are not the same magnitude. The map is explicit. The parent who compares across different wholes has left the standard.

⁴⁸⁴Robert S. Siegler, Clarissa A. Thompson, and Michael Schneider, “An Integrated Theory of Whole Number and Fractions Development,” *Cognitive Psychology* 62, no. 4 (2011): 273–296, <https://doi.org/10.1016/j.cogpsych.2011.03.001>. Numerical development as progressive mapping of more classes of number onto a mental number line, and as learning that many whole-number properties are not properties of number. Fractions are the first large opportunity.

⁴⁸⁵Siegler, Thompson, and Schneider 2011; Steven A. Hecht and Kevin J. Vagi, “Sources of Group and Individual Differences in Emerging Fraction Skills,” *Journal of Educational Psychology* 102 (2010): 843–858. Magnitude estimation of fractions closely related to correct fraction arithmetic.

⁴⁸⁶CCSSM 1.G.3 and 2.G.3: equal shares (halves, thirds, fourths). The only CCSSM fraction language before the Grade 3 NF domain. 2.G.3: equal shares of identical wholes need not have the same shape.

⁴⁸⁷Siegler et al. 2010, Recommendation 1: build on informal fraction experiences of young children (sharing, fair shares).

⁴⁸⁸CCSSM 3.NF.3: equivalence in special cases; whole numbers as fractions; compare same numerator or same denominator, same whole. Denominators limited to 2, 3, 4, 6, 8.

Grade 4 extends the denominators — 2, 3, 4, 5, 6, 8, 10, 12, 100 — and asks why $a/b = (n \times a)/(n \times b)$. Multiplying numerator and denominator by the same n is partitioning each unit fraction into n smaller unit fractions and taking n times as many of them. The point on the line does not move. That is equivalence as the same point, not as a butterfly, not as “criss-cross,” not as a trick whose only warrant is that it gets the worksheet right. Compare fractions with unlike denominators by finding a common denominator or a common numerator, or by a benchmark such as $1/2$, still the same whole.⁴⁸⁹ Benchmarks 0, $1/2$, and 1 before any cross-multiply-to-compare. The 2021 guide’s panel explicitly advises against cross-multiply-to-compare during intervention because the trick does not build magnitude.⁴⁹⁰ That is a live debate a parent will meet in a forum: the trick is fast; the line is slow. For a child already behind, magnitude wins on the page this book can defend. For a child who can already place both fractions on a line, a later compression into a comparison algorithm is iteration, not a religion. The trick as the *first* comparison is a procedure with no representation. Chapter 2’s iterative model would put the line first, a handful of comparisons against 0, $1/2$, and 1, then a written method the child can still point to on the line.

Equivalent fractions as the same point is Example 4.5 in the 2021 guide.⁴⁹¹ A parent can steal it without buying an intervention kit. Draw 0 to 1. Mark $1/2$. Partition each half into two: $2/4$ lands on $1/2$. Partition each half into three: $3/6$ lands on $1/2$. The child sees the point not moving. Then write $1/2 = 2/4 = 3/6$ as names for that point. The butterfly — cross-multiply to test equivalence — can wait until the point is believed. A child who “cross-multiplies” $2/3$ and $3/4$, gets 8 and 9, and concludes $2/3 < 3/4$, may be right this time and still not know that $2/3$ is past $1/2$ and $3/4$ is farther past $1/2$. Right answers are performance. The line is the magnitude.

The teacher-knowledge problem is the design user’s problem. IES 2010 Recommendation 5 addresses teacher education: teachers’ own fraction knowledge is often weak. Liping Ma’s 1999 *Knowing and Teaching Elementary Mathematics* is the comparison usually cited — U.S. teachers generating few or no explanations of invert-and-multiply versus Japanese and Chinese teachers generating two or three.⁴⁹² Siegler and colleagues 2012 discuss that comparison. Invert-and-multiply is not even a Grade 5 requirement; fraction $\div$ fraction is Grade 6. The point arrives early anyway. A parent who can execute invert-and-multiply and cannot hear why $1/2 \div 1/4$ is 2 — how many one-fourths fit in one-half; two of them — is Ma’s U.S. teacher, in a kitchen. That is not an insult. It is why this book’s first thesis condition is that the parent understands the concept well enough to hear a wrong answer. Section 5.4 will do operations. Section 5.8 will let a model explain invert-and-multiply to the *parent*, not to the child, and then demand that the parent solve a tiny problem without the model before sitting down. The line is what the parent has to be able to point at when the child’s invert-and-multiply has inverted the wrong factor.

⁴⁸⁹CCSSM 4.NF: why $a/b = (n \times a)/(n \times b)$; compare unlike denominators by common denominators/numerators or a benchmark such as $1/2$, same whole. Denominators 2, 3, 4, 5, 6, 8, 10, 12, 100. Unlike-denominator addition in general is not a Grade 4 requirement.

⁴⁹⁰Fuchs et al. 2021, Recommendation 4 obstacles (n. 458): against cross-multiply-to-compare during intervention because the trick does not build magnitude. Benchmarks 0, $1/2$, 1 first.

⁴⁹¹Fuchs et al. 2021, Example 4.5: equivalent fractions as the same point on the line, not as a butterfly trick.

⁴⁹²Liping Ma, *Knowing and Teaching Elementary Mathematics* (Erlbaum, 1999). The comparison usually cited: U.S. teachers generating few or no explanations of invert-and-multiply versus Japanese and Chinese teachers generating two or three. Discussed in Siegler et al. 2012 and in IES 2010 Recommendation 5 (teacher knowledge). The design user who stopped at algebra is Ma’s U.S. teacher, in a kitchen.

Length models match fraction magnitude in a way area models and set models do not, by themselves. Fuchs 2021 Recommendation 3: choose representations that match the mathematics; length models for fractions.⁴⁹³ A paper strip folded into four equal parts is a length. Placing that strip against a number line, in the same visual field, with $3/4$ written at the third fold, is the connection. Fraction tiles that show $1/2$ as a piece twice as long as $1/4$ are a length model if they are used as lengths. They are a toy if they are used as a puzzle whose only win-state is filling a box. Carbonneau's $d \approx 0.37$ was versus abstract-only, larger on retention, larger under guidance. Automatic abstraction is not the finding. The finding is a modest average advantage that depends on connecting the object to the symbol.⁴⁹⁴ Fade when the child can mark $3/4$ on a blank 0-to-1 line without the tiles. Bring the tiles back when mixed numbers, or unlike-denominator addition, needs a bridge. Folklore CRA is three weeks of tiles, three weeks of pictures, then symbols forever. The documents do not say that.

Whole numbers as fractions, 3.NF.3, is the sentence that later makes $2 = 2/1 = 8/4$ available inside unlike-denominator work, and that later still makes a rational number a point on the same line as an integer. A child who thinks “fractions are the numbers less than one” has not got $5/4$, or 2, as a fraction. Improper fractions and mixed numbers are Grade 4 building-from-unit-fractions work: $5/4$ is five copies of $1/4$, which is one whole and one $1/4$, which is $1 \frac{1}{4}$, which is a point to the right of 1 on the same line.⁴⁹⁵ The pizza that only ever shows a slice inside a single pie trains the “less than one” habit. The line does not.

Compare with the same numerator is the comparison whole-number bias most aggressively sabotages. $1/4$ and $1/5$: the child compares 4 and 5 and says fourths are bigger, or says $1/5$ is bigger because 5 is bigger. Both errors treat the denominator as a whole-number count rather than as the size of the unit fraction. More equal parts of the same whole makes each part smaller. That sentence is already in 1.G.3. It has to be said again, on the line: $1/5$ is to the left of $1/4$ between 0 and 1. Benchmarks catch the error before a procedure does. $1/4$ is less than $1/2$; $1/5$ is also less than $1/2$, and closer to 0. A child who can say that, pointing, is not yet fluent at comparison. They have a magnitude to hear a wrong answer against.

The 2010 guide's remaining recommendations belong in this chapter even though they overflow this section. Recommendation 3: help students understand why computational procedures with fractions make sense — section 5.4. Recommendation 4: develop understanding of ratios and rates before formal proportions — Chapter 6, planted here as the thing the gate unlocks. Recommendation 5: teacher knowledge, already named.⁴⁹⁶ The five recommendations are a teaching map, not a 2026 randomized trial of a homeschool kitchen. Fuchs 2013 and 2014 are the experimental half; coefficients not copied; **UNVERIFIED** as numbers in this book.⁴⁹⁷

⁴⁹³Fuchs et al. 2021, Recommendation 3: choose representations that match the mathematics; length models for fractions. Number line gets its own Recommendation 4 because it scales from counting to irrationals.

⁴⁹⁴Carbonneau, Marley, and Selig 2013 (n. 467). Modest average advantage versus abstract-only, larger on retention, larger under guidance; transfer and justification effects small. Automatic abstraction is not the finding.

⁴⁹⁵CCSSM 4.NF: build fractions from unit fractions; mixed numbers; improper fractions as copies of a unit fraction past 1 on the same line.

⁴⁹⁶Siegler et al. 2010, five recommendations: (1) informal fraction experiences; (2) fractions as numbers / number lines; (3) procedures that make sense; (4) ratios/rates before formal proportion; (5) teacher knowledge. Recommendation 4 is planted for Chapter 6.

⁴⁹⁷Fuchs et al. 2013/2014 numerical effect sizes: **UNVERIFIED** as coefficients in this book (n. 453). Do not invent

A parent who teaches only pies will get part-whole language and will not get magnitude. A parent who teaches only the line, with no sharing and no equal parts of a length or an area, will get a picture that is not connected to the informal experiences Recommendation 1 wants used. Part-whole first is fine. It is not sufficient. The line is what pies are not. Equivalent fractions are the same point. Benchmarks 0, $1/2$, 1 before the trick. The design user who stopped at algebra is the teacher whose own invert-and-multiply was a chant. The line is how that adult hears a wrong answer. Operations are where whole-number bias goes to work.

5.4 Operations on fractions; “multiply makes bigger” is whole-number bias

“Multiply makes bigger; divide makes smaller” is not cute folklore. It is a whole-number property that fails for multipliers and divisors less than 1, and it is a central conceptual-change problem in the whole-number-bias literature. Ni and Zhou, in *Educational Psychologist* in 2005, review the over-extension of whole-number properties to rationals: unique successors; a finite number of entities in an interval; multiplication makes bigger; addition of “pieces” means add the counts you see.⁴⁹⁸ They review three origin stories — a domain-specific number mechanism, a discrete-quantity privilege, interference of prior instruction — and conclude the evidence is not yet decisive among them. The instructional upshot they endorse is the one this chapter keeps: organize fraction teaching so that prior whole-number knowledge is used rather than only fought.⁴⁹⁹ Stella Vosniadou, Xenia Vamvakoussi, and Irini Skopeliti treat “multiply makes bigger” as a central conceptual-change problem.⁵⁰⁰ Common Core 5.NF.5 exists specifically to break it: multiplication as scaling. A fraction greater than 1 enlarges. A fraction less than 1 shrinks.⁵⁰¹ Greer 1994 is the older source sometimes cited for the “multiply makes bigger” data; that paper was not opened for this project, and the data remain **UNVERIFIED** here. Cite Ni and Zhou, Siegler, and 5.NF.5.⁵⁰² High-school and community-college students still confuse fraction and whole-number properties.⁵⁰³ The gate does not close at the end of fifth grade if it was never opened.

Scaling is the sentence the parent has to be able to hear. $4 \times 1/2$ is four copies of one-half, which is 2, which is smaller than 4. $1/2 \times 4$ is half of four, which is 2, which is smaller than 4. The commutative property still holds; the whole-number expectation that a product exceeds each factor does not. $3/2 \times 4$ is three-halves of four, which is 6, which is larger than 4, because $3/2$ is greater than 1. The test is not “did they get 2.” The test is “did they flinch when the product was smaller

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⁴⁹⁸Ni and Zhou 2005 (n. 430). Whole-number bias properties listed in the text.

⁴⁹⁹Ni and Zhou 2005: three origin stories (domain-specific number mechanism; discrete-quantity privilege; interference of prior instruction) — evidence not yet decisive among them. Instructional upshot they endorse: organize fraction teaching so prior whole-number knowledge is used rather than only fought.

⁵⁰⁰Stella Vosniadou, Xenia Vamvakoussi, and Irini Skopeliti, “The Framework Theory Approach to Conceptual Change,” in *International Handbook of Research on Conceptual Change*, ed. Stella Vosniadou (2008). “Multiply makes bigger / divide makes smaller” as a central conceptual-change problem.

⁵⁰¹CCSSM 5.NF.5: multiplication as scaling. A fraction > 1 enlarges; a fraction < 1 shrinks. The standard exists specifically to break “multiply always makes bigger.”

⁵⁰²Brian Greer 1994 is the older source sometimes cited for “multiply makes bigger” data. Paper not opened for this project. Data **UNVERIFIED**. Cite Ni and Zhou 2005, Siegler et al., and CCSSM 5.NF.5 instead.

⁵⁰³Michael Schneider and Robert S. Siegler, 2010; Vosniadou, Vamvakoussi, and Skopeliti 2008. High-school and community-college students still confuse fraction and whole-number properties. The gate does not close at the end of fifth grade if it was never opened.

than 4, and can they say why.” 5.NF.5 is that flinch, written as a standard. A parent who treats the shrink as a cute exception — “sometimes multiply makes smaller, isn’t that wild” — has left the whole-number bias in place and laminated a footnote onto it.

Adding numerators and denominators — $1/2 + 1/3 = 2/5$ — is the diagnosis object of this chapter. It is the counts-you-see error Ni and Zhou named. It is the magnitude failure the 2010 guide named. $2/5$ is less than $1/2$. $1/2 + 1/3$ cannot be less than $1/2$. A child who can hear that, against a benchmark, has a reason to reject $2/5$ before any least-common-denominator procedure is run. Grade 5.NF.2 is explicit about that estimate: reject $2/5 + 1/2 = 3/7$ because $3/7$ is less than $1/2$.⁵⁰⁴ The procedure without the estimate is a chant. The estimate without a line to point at is a slogan. Iterate: mark $1/2$ and $1/3$ on a 0-to-1 line; see that their join is past $1/2$ and short of 1; then replace with equivalent like-denominator fractions; then add the copies of the common unit fraction.

Unlike-denominator addition, Grade 5, is replacing with equivalent like-denominator fractions, including mixed numbers.⁵⁰⁵ That sentence is the mathematics. It is not “find the LCD” as a first move. Equivalence as the same point, from section 5.3, is the prerequisite. $1/2$ is $3/6$. $1/3$ is $2/6$. Three copies of $1/6$ plus two copies of $1/6$ is five copies of $1/6$, which is $5/6$, which is the point you already estimated past $1/2$. Like-denominator addition, Grade 4 — including mixed numbers — is the same move with the unit fraction already common: $3/8 + 2/8$ is five copies of $1/8$.⁵⁰⁶ Build fractions from unit fractions first. Add the copies. Unlike-denominator work is the same addition, after a replacement that does not move the points. A parent who teaches “LCD, then add numerators, keep the denominator” as a chant, with no replacement-as-same-point, will get worksheets right and will not hear $2/5$ as a magnitude failure. A parent who never reaches unlike-denominator addition by the end of the Grade 5 window, on the map, has delayed the operation the Panel placed as a Grade 5 benchmark.⁵⁰⁷

Multiplication of a fraction by a whole number is Grade 4: $3 \times 1/4$ is three copies of $1/4$, which is $3/4$, which the child already knows how to say. Multiplication of a fraction by a fraction is Grade 5: $1/2 \times 1/3$ is a half of a third, which is $1/6$, which is also a third of a half. The area model, already in 5.2 as covering, is the representation: a unit square, one side halved, the other thirded, the overlap $1/6$ of the unit square. 5.NF.4 includes the area of a rectangle with fractional side lengths, so the Grade 3 covering is now a covering with fractional sides.⁵⁰⁸ Iterate: a representation, a handful of products, a return to “does this enlarge or shrink?”, more products. Rittle-Johnson and Koedinger 2009, the classroom experiment that held content constant and varied only order, was a *decimal* study. An iterative sequence beat a concepts-first block on procedures that transfer. Conceptual knowledge was fairly comparable.⁵⁰⁹ The iteration finding applies. Do not pretend it

⁵⁰⁴ CCSSM 5.NF.2: word problems; benchmark estimates — e.g., reject $2/5 + 1/2 = 3/7$ because $3/7 < 1/2$.

⁵⁰⁵ CCSSM 5.NF.1: add/subtract fractions with unlike denominators, including mixed numbers, by replacing with equivalent like-denominator fractions.

⁵⁰⁶ CCSSM 4.NF: add/subtract like denominators including mixed numbers; build fractions from unit fractions. $3/8 + 2/8$ is five copies of $1/8$.

⁵⁰⁷ NMAP Table 2: add/subtract fractions and decimals by end of Grade 5. Delaying unlike-denominator addition past that window is a named failure point (n. 455).

⁵⁰⁸ CCSSM 5.NF.4: multiply fraction $\times$ whole and fraction $\times$ fraction; area of a rectangle with fractional side lengths. The Grade 3 covering (3.MD) is now a covering with fractional sides.

⁵⁰⁹ Rittle-Johnson and Koedinger 2009 (n. 457). Sixth-grade decimals; iterative sequence produced more knowledge of arithmetic procedures, including transfer, than a concepts-first block; conceptual knowledge “fairly comparable.”

was a fractions-curriculum randomized trial. Rittle-Johnson, Schneider, and Star 2015 still applies: it is a myth that it is a one-way street from conceptual knowledge to procedural knowledge.⁵¹⁰ Do not wait for full understanding of scaling before any product is written. Do not drill invert-and-multiply — or even fraction $\times$ fraction as “multiply numerators, multiply denominators” — with zero talk about why.

Division in this band is limited, and the limit is load-bearing. Grade 5 has unit-fraction $\div$ whole and whole $\div$ unit-fraction. $1/4 \div 3$ is one-fourth partitioned into three equal parts, each $1/12$. $3 \div 1/4$ is how many one-fourths fit in 3, which is 12. Division of a fraction by a fraction is *not* a Grade 5 requirement. It is Grade 6.⁵¹¹ The Panel’s Table 2: add and subtract fractions and decimals by the end of Grade 5; multiply and divide by the end of Grade 6.⁵¹² A homeschool program that teaches invert-and-multiply in what it calls Grade 5 is not breaking a law. It is ahead of the map. A program that never reaches fraction $\div$ fraction by the ratio year is behind the Panel’s benchmark. Invert-and-multiply is a compression the parent must be able to hear wrong. $1/2 \div 1/4$: how many one-fourths in one-half? Two. Invert $1/4$, get $4/1$, multiply $1/2 \times 4/1 = 2$. The 2 matches the measurement-division meaning. A child who inverts the $1/2$ instead, and gets $1/4 \times 2/1 = 1/2$, has executed a chant on the wrong factor. A parent who cannot hear that $1/2$ is not the number of one-fourths in one-half has not got the meaning, and Ma’s comparison is the document that says U.S. teachers often cannot generate that explanation.⁵¹³ IES 2010 Recommendation 3: help students understand why computational procedures with fractions make sense.⁵¹⁴ The compression is allowed. The compression as the only meaning is not.

Common-denominator addition is the other compression a parent has to hear. Finding a common denominator is finding a common unit fraction. $1/2 + 1/3$ is not “the LCD is 6” as a first sentence. It is “halves and thirds can both be named as sixths, because $1/2$ and $3/6$ are the same point, and $1/3$ and $2/6$ are the same point.” Then five sixths. A child who finds an LCD, adds, and cannot place the sum on a line has performed. A child who places $1/2$ and $1/3$, estimates past $1/2$, then finds sixths, has a magnitude to check the $5/6$ against. Iteration, not a one-way street.

Mixed numbers are not a separate species. $1\ 3/4$ is $1 + 3/4$, which is $4/4 + 3/4$, which is $7/4$, which is seven copies of $1/4$, which is a point between 1 and 2. Adding mixed numbers with like denominators is Grade 4; with unlike denominators, Grade 5. The failure modes are predictable. Adding the whole numbers and adding the fractions separately without regrouping when the fraction sum exceeds 1. Treating $1\ 3/4$ as $13/4$ by concatenating, or as $1/34$. The line catches concatenation: $13/4$ is $3\ 1/4$, a different point from $1\ 3/4$. The same whole-number bias that concatenates 27 as “2

Iteration finding applies; it was not a fractions-curriculum RCT.

⁵¹⁰Rittle-Johnson, Schneider, and Star 2015 (n. 457). Quote: “It is a myth that it is a one-way street from conceptual knowledge to procedural knowledge.”

⁵¹¹CCSSM 5.NF.7: division limited to unit-fraction $\div$ whole and whole $\div$ unit-fraction. Fraction $\div$ fraction is Grade 6 (6.NS). Do not treat invert-and-multiply as a Grade 5 requirement.

⁵¹²NMAP Table 2: add/subtract fractions and decimals by end of Grade 5; multiply/divide by end of Grade 6; all operations with signed fractions by end of Grade 7.

⁵¹³Ma, *Knowing and Teaching Elementary Mathematics* (1999) (n. 492). Invert-and-multiply as a compression the parent must be able to hear wrong: $1/2 \div 1/4$ is how many one-fourths in one-half (2), which matches invert-and-multiply when the right factor is inverted.

⁵¹⁴Siegler et al. 2010, Recommendation 3: help students understand why computational procedures with fractions make sense.

and 7” rather than $20 + 7$ is at work. Use it. Do not only fight it.⁵¹⁵

Word problems on operations are still schemas, not keywords. “Altogether” does not mean add. “Of” does not always mean multiply. “Shared” does not always mean divide. Equal groups, comparison, and, late, the first “times as much” fraction language. Estimate first against a benchmark. Then represent. Then compute. Then check the estimate. NMAP Finding 26 still holds: real-world contexts improve similar real-world items; they do not, in the small high-quality set the Panel admitted, improve computation, simple word problems, or equation solving.⁵¹⁶ A recipe that doubles $\frac{3}{4}$ cup is a second representation of $2 \times \frac{3}{4}$. It is not the line, and it is not the schema. Section 5.7 will say so at tip length.

Decimals belong in the next section as the same numbers in a different notation, but one bias cousin belongs here because it is whole-number bias in decimal clothes. Rittle-Johnson, Siegler, and Alibali, 2001, used decimal-fraction items because representation errors of this kind mediate procedure errors: $0.25 > 0.3$ because $25 > 3$, or $0.25 > 0.250$ because it “looks longer.”⁵¹⁷ More digits, bigger, is unique-successors-and-counts-you-see applied to the digits after the point. Tenths and hundredths on the same line as $\frac{1}{10}$ and $\frac{1}{100}$ catch it. Section 5.5 will put them there.

The equal sign is still live. McNeil and Alibali’s operational patterns — perform all given operations on all given numbers; operations-equals-answer visual structure; “=” means “the total” — now sit inside $a/b = c/\square$ and inside the first equations with a letter.⁵¹⁸ $\frac{1}{2} = \square/8$ is not “do something.” It is a relation: the same point, a missing numerator. A child who adds 1, 2, and 8, or who writes 4 because “double the 2, so double the 1,” without being able to say “same point, $\frac{4}{8}$ is $\frac{1}{2}$,” is running a procedure that happens to work for doubling and will fail for $\frac{1}{2} = \square/6$. Non-canonical forms still help: $\frac{3}{4} = \frac{3}{4}$, $\frac{6}{8} = \frac{3}{4}$, $\frac{3}{4} = \frac{1}{2} + \frac{1}{4}$. Chapter 4 wrote $8 = 3 + 5$. This band writes $1 = \frac{2}{3} + \frac{1}{3}$.

What the parent has to hear, compressed. $\frac{1}{2} + \frac{1}{3}$ is not $\frac{2}{5}$, because $\frac{2}{5}$ is less than $\frac{1}{2}$, and because $\frac{1}{2}$ and $\frac{1}{3}$ are points whose join is past $\frac{1}{2}$. $3 \times \frac{1}{4}$ is three copies of $\frac{1}{4}$, not a mysterious shrink or enlargement. $\frac{1}{2} \times \frac{1}{3}$ is a shrink, because $\frac{1}{2}$ is less than 1, and 5.NF.5 is why. $3 \div \frac{1}{4}$ is 12, which is bigger, because measurement division by a unit fraction less than 1 asks how many

⁵¹⁵Ni and Zhou 2005 (n. 430): use prior whole-number knowledge rather than only fight it. Concatenation of mixed numbers ($1 \frac{3}{4}$ read as $\frac{13}{4}$ or $\frac{1}{34}$) is the place-value concatenation error in fraction clothes.

⁵¹⁶NMAP, *Foundations for Success* (2008), Finding 26: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving. Kitchen-math transfer is hard.

⁵¹⁷Bethany Rittle-Johnson, Robert S. Siegler, and Martha W. Alibali, “Developing Conceptual Understanding and Procedural Skill in Mathematics: An Iterative Process,” *Journal of Educational Psychology* 93, no. 2 (2001): 346–362, <https://doi.org/10.1037/0022-0663.93.2.346>. Fifth- and sixth-graders learning decimal fractions; representation errors ($0.25 > 0.3$ because $25 > 3$; “more digits, bigger”) mediate procedure errors. Iteration: initial conceptual knowledge predicted procedural gains; procedural gains predicted later conceptual knowledge.

⁵¹⁸Nicole M. McNeil and Martha W. Alibali, “Why Won’t You Change Your Mind? Knowledge of Operational Patterns Hinders Learning and Performance on Equations,” *Child Development* 76 (2005): 883–899; Nicole M. McNeil, “A Change-Resistance Account of Children’s Difficulties Understanding Mathematical Equivalence,” *Child Development Perspectives* (2014). Three operational patterns: perform all given operations on all given numbers; “operations = answer” visual structure; “=” means “the total.” Eric J. Knuth, Ana C. Stephens, Nicole M. McNeil, and Martha W. Alibali, “Does Understanding the Equal Sign Matter? Evidence from Solving Equations,” *Journal for Research in Mathematics Education* 37 (2006): 297–312: relational view of “=” connected to later algebra. Now live inside $a/b = c/\square$.

small pieces fit, and “divide makes smaller” is the bias. Invert-and-multiply, when it arrives in Grade 6, is a compression of that measurement meaning. A parent who can run the compression and cannot hear the meaning is not yet the teacher of this band. IES Recommendation 3 is that hearing. Ma 1999 is why it cannot be assumed.

Decimals, area, volume, and the first quadrant sit on the same line and the same array. They are not extras after fractions. They are the rest of the Critical Foundations window.

5.5 Decimals, area, volume, and the first quadrant

Decimals in this band are fractions with denominators 10 and 100, written in place-value notation, then compared and operated on as place value. They are not a new species of number. 4.NF introduces decimal notation for tenths and hundredths and comparison of decimals to hundredths. 5.NBT puts decimals into the place-value system through thousandths: a digit represents 10 times the place to its right and $1/10$ the place to its left; read, write, compare, round; add, subtract, multiply, divide decimals to hundredths, with explanation.⁵¹⁹ The same line that held $3/4$ now holds 0.75, and $75/100$, and $7/10 + 5/100$. If the line is missing, the decimal is a whole-number lookalike, and whole-number bias has a new costume.

“More digits, bigger” is that costume. Rittle-Johnson, Siegler, and Alibali, 2001, used decimal-fraction items precisely because representation errors of this kind mediate procedure errors. $0.25 > 0.3$ because $25 > 3$. $0.25 > 0.250$ because it “looks longer.”⁵²⁰ Both errors treat the digits after the point as a whole number, or treat length of the numeral as magnitude. The line catches both. 0.3 is $3/10$, three jumps of $1/10$ from 0, past $1/4$, short of $1/2$. 0.25 is $25/100$, which is $1/4$, which is the same point as 0.250 if the child has hundredths and thousandths on the same line. Trailing zeros do not move the point. That sentence is equivalence, restated for decimals. Iterate: a short pass on 0.3 and $3/10$ as the same point, a handful of comparisons against 0, 0.5, and 1, a return to the line, more comparisons. The 2009 Rittle-Johnson and Koedinger classroom experiments were sixth-grade decimals on an intelligent tutor. Iteration beat a concepts-first block on procedures that transferred.⁵²¹ This is the band those experiments were about, one year earlier on the map. Do not withhold decimal operations until the idea is judged complete. Do not run pages of decimal addition with no tenths-as-fractions and no line.

Place value to the right of the point is the Grade 5 NBT sentence the parent has to hear. A digit represents 10 times the place to its right and $1/10$ the place to its left. 0.4 is four tenths. 0.40 is four tenths and zero hundredths, same point. 0.04 is four hundredths, a different point, one-tenth as large as 0.4. Patterns of zeros and decimal-point placement with powers of 10 — $0.4 \times 10 = 4$, $0.4 \times 100 = 40$, $4 \div 10 = 0.4$ — are the same structure as “ 40×3 is twelve tens,” one direction further. Whole-number exponents for powers of 10 appear in 5.NBT as the notation for that structure, not

⁵¹⁹CCSSM 4.NF: decimal notation for tenths and hundredths; compare decimals to hundredths. CCSSM 5.NBT: a digit represents 10 times the place to its right and $1/10$ the place to its left; decimals through thousandths; operations with decimals to hundredths, with explanation.

⁵²⁰Rittle-Johnson, Siegler, and Alibali 2001 (n. 517). Decimal “more digits, bigger” is the whole-number-bias cousin.

⁵²¹Rittle-Johnson and Koedinger 2009 (n. 457, n. 509). Sixth-grade decimals on an intelligent tutor; iteration beat concepts-first on procedures that transferred. This band is those experiments one year earlier on the map.

as a high-school topic smuggled down.⁵²² A child who “moves the point” as a chant, and cannot say that multiplying by 10 is exchanging tenths for ones, has a procedure. Fluency as *Adding It Up* defined it includes knowing when the chant is appropriate. The chant with no place-value explanation is one failure point of this band. Only strategies, with no path to a general way of multiplying 1.4×0.3 , is the other.

Operations with decimals to hundredths, Grade 5, are the four operations, with explanation. Addition and subtraction of 0.25 and 0.3 are addition and subtraction of 25/100 and 30/100, or of 1/4 and 3/10 after a common unit is found. Multiplication of 0.2×0.3 is $2/10 \times 3/10 = 6/100 = 0.06$, a shrink, because both factors are less than 1, which is 5.NF.5 in decimal clothes. Division of $1.2 \div 0.3$ is how many three-tenths in twelve-tenths, which is 4. The solver will print a correct alignment of decimal points. The parent who never sees the child’s $0.25 + 0.3 = 0.28$ — added the visible tails, whole-number bias again — has let a tool hide the misconception. Section 5.8 will ban that hide.

Money is a decimal-and-place-value context, useful, also not a curriculum. \$0.25 is a quarter, which is 1/4 of a dollar, which is 0.25, which is a point. Counting coins can motivate hundredths. It does not, by itself, teach that 0.25 is less than 0.3, or that $0.25 + 0.3$ is 0.55 rather than 0.28. NMAP Finding 26 is the caution, already locked: real-world contexts improve similar real-world items; they do not, in the Panel’s small high-quality set, improve computation or equation solving.⁵²³ A child who can make change and cannot place 0.3 on a line has performed, not learned. Soderstrom and Bjork’s performance/learning split is the same sentence Chapter 3 used for a solver at the elbow. It applies to a cash register too.

Area was already in 5.2 as the array made measurement. It returns here because Grade 4 writes the formulas for rectangles and Grade 5 extends the covering to fractional side lengths. Tile, then array, then formula, with the formula said as “rows of unit squares.” Perimeter versus area is a classic confusion: a child multiplies when the problem asked for the distance around, or adds when the problem asked for covering. Pesek and Kirshner 2000, sometimes cited as showing that early procedures interfere, gave fifth-graders area and perimeter; differences between conditions were small and not statistically significant (48 percent versus 42 percent at posttest).⁵²⁴ That paper does not settle a sequence. It does warn a parent not to cite it as “research says withhold the formulas.” Same perimeter, different area, and the reverse, from 3.MD, are the problems that force a choice of quantity. Interleave them with each other after a short blocked introduction of each, labelled as transfer of Rohrer’s practice, not as $d = 1.05$ in grade 3.⁵²⁵

Angle measure is Grade 4 MD, and it is a fraction of a turn. The one-degree angle is 1/360 of a circle. A right angle is 90 one-degree angles, 90/360 of a turn, 1/4 of a turn. Measure with a

⁵²²CCSSM 5.NBT: patterns of zeros and decimal-point placement with powers of 10; whole-number exponents for powers of 10.

⁵²³NMAP Finding 26 (n. 516). Money is a decimal-and-place-value context, useful, also not a curriculum.

⁵²⁴Pesek and Kirshner (2000), as reviewed in Rittle-Johnson, Schneider, and Star 2015: fifth-graders, area and perimeter; differences small and not statistically significant (48% vs 42% at posttest). Sometimes cited as showing that early procedures interfere; the 2015 review flags the differences as small and non-significant. Do not cite it as “research says withhold the formulas.”

⁵²⁵Rohrer et al. 2014 and 2020 (n. 478): grade-7 classroom RCTs. Interleave area and perimeter after a short blocked introduction of each, labelled as transfer of the *practice*, not as $d = 1.05$ in grade 3.

protractor. Angle measure is additive: two adjacent angles joining a line sum to 180 one-degree angles.⁵²⁶ The fraction language is not ornamental. A child who has $1/b$ as a unit fraction can hear $1/360$ as a unit angle. A child who has only “the protractor says 90” has a reading, not a magnitude. 4.G puts the angle vocabulary — right, acute, obtuse — next to parallel and perpendicular lines, classification of two-dimensional figures, right triangles, and line symmetry. Shared attributes define larger categories, already in 3.G: a square is a rectangle, so a square has four right angles. Grade 5 G continues the hierarchy. The geometry of this band is classification by properties, not proof. Proof readiness, Chapter 2 already said, is CCSSM Practice 3 built from the equal sign through informal geometric arguments, not two-column theater.

Volume is Grade 5 MD, and it is the array grown one dimension. The unit cube is the unit. Packing a right rectangular prism with unit cubes, no gaps, no overlaps, is the filling. $V = l \times w \times h$ is the product of three edge lengths, rows of rows of unit cubes. $V = B \times h$ is the same product, with B the area of the base — the Grade 3 covering — times the number of layers. Volume as additive: two prisms joined, total volume the sum.⁵²⁷ Tile (pack), then array (layers of unit cubes), then formula, with the formula said as “layers of unit cubes,” not as “multiply the three numbers you see.” A child who multiplies $3 \times 4 \times 5$ and writes “60” with no unit cubes has the chant. Fractional edge lengths wait for Grade 6; this band’s prisms are whole-number edges on the map. The unlock is later Cavalieri and, much later, the integral as accumulation. The parent does not need those names this year. The parent needs to hear “60” without a unit as a missing covering.

The first quadrant of the coordinate plane is Grade 5 G and 5.OA together. Origin, ordered pairs, graph in the first quadrant. Two numerical patterns generate ordered pairs; the pairs sit as points.⁵²⁸ This is not yet a function, and it is not yet four quadrants. Grade 6 will put negatives on the same plane; Grade 8 will name a function as each input exactly one output. What this band owes those later objects is the habit of a pair as a location, and a location as a pair. A child who treats (3, 5) as “3 and 5” in either order has concatenation again. The first coordinate is distance from the vertical axis; the second, from the horizontal. Same visual field: the grid, the pair written, the point marked. Interleave “plot this pair” with “name the pair of this point” as soon as both exist, so the child must choose a direction of translation.

These topics unlock Grade 6 ratio — a unit rate on a graph through the origin is an ordered-pair pattern the first quadrant already made possible — and Grade 8 functions. They are not extras after fractions. A program that spends the year on fraction worksheets and never packs a prism, never marks a first-quadrant pair, never writes 0.3 as $3/10$ on the same line as $3/10$, has taught a fragment of the window. The Panel’s geometry and measurement remainder of Table 2 stays **UNVERIFIED**; this section will not invent a Grade 4 or Grade 5 geometry benchmark the fetch truncated.⁵²⁹ The CCSSM clusters above are the map this book can defend.

⁵²⁶CCSSM 4.MD: one-degree angle as $1/360$ of a circle; measure with a protractor; angle measure as additive. 4.G: right/acute/obtuse; perpendicular and parallel; classify 2-D figures; right triangles; line symmetry.

⁵²⁷CCSSM 5.MD: volume — unit cube; packing; $V = l \times w \times h$ and $V = B \times h$ for right rectangular prisms; volume as additive. Fractional edge lengths wait for Grade 6 (6.G).

⁵²⁸CCSSM 5.G and 5.OA: coordinate plane, origin, ordered pairs; graph in the first quadrant; two numerical patterns generating ordered pairs.

⁵²⁹NMAP Table 2 geometry/measurement remainder **UNVERIFIED** (n. 451). Do not invent Grade 4 or Grade 5 geometry benchmarks the fetch truncated.

Line plots in $1/2$, $1/4$, and $1/8$ units, Grades 3 through 5, are the data strand sitting on the same fractions. Grade 5 asks for fraction operations on the data in the plot.⁵³⁰ That is not a statistics course. It is a reason to treat a measured length as a number on a line, collected, then added or averaged as a fraction. GAISE page-level claims were not opened for this project and stay **UNVERIFIED**. The spirit a parent can steal without those pages: a question that has variability — how long are these ten bean sprouts, to the nearest quarter inch — not a worksheet of mean, median, and mode with no data story. Chapter 6 and Chapter 11 will do more. This band plants the line plot as a number line with tallies.

What this section has done is keep decimals on the same line as fractions, and keep area and volume as measurement that also models multiplication. The first quadrant is a pair of number lines. Perimeter is not area. Angle is a fraction of a turn. Money is a context, not a curriculum. The Panel's truncated geometry row is not filled by analogy. Teaching guidance for the band comes next, now that the mathematics is on the table: connect concrete to notation in the same visual field, then fade; the line as the central fraction tool; interleave after a short block, labelled as transfer; magnitude before tricks.

5.6 Teaching guidance: CRA, interleave, magnitude before tricks

Chapter 2 locked the moves. This section puts this band's objects into them. It is not a 180-day plan. It is not a franchise. It is the conversion of named recommendations into a kitchen sequence a parent who stopped at algebra can run without pretending a homeschool randomized trial exists.

Connect concrete to notation in the same visual field, then fade. Fuchs and colleagues, 2021, Recommendation 3, strong evidence, 28 studies: choose representations that match the mathematics — proportional base-ten for place value, length models for fractions, unit squares for area, unit cubes for volume. Connect concrete and semi-concrete to the written symbols in the same sitting, same field of view, not “tiles this month, numerals next month.” Practice until the representation is a thinking tool. Fade when the child is accurate without it. Revisit when a new idea needs a bridge.⁵³¹ Carbonneau, Marley, and Selig's overall $d \approx 0.37$ versus abstract-only is the meta-analytic background, small-to-moderate, larger on retention than on transfer or justification, larger under guidance.⁵³² Report it with the moderators. A parent who dumps fraction tiles on the table and leaves the room has not implemented Recommendation 3. A parent who folds a paper strip into fourths, lays it on a drawn 0-to-1 line, writes $3/4$ at the third fold, and asks the child to say what the 3 and the 4 mean, has. Virtual versus physical manipulatives is a literature this project did not open. **UNVERIFIED**. Do not claim they are just as good. Do not claim they are worse. The variable is the conversation, not the plastic.⁵³³

⁵³⁰CCSSM 3.MD, 4.MD, 5.MD: line plots in $1/2$, $1/4$, $1/8$ units; Grade 5 fraction operations on the data. GAISE II page-level claims **UNVERIFIED** this project (02-learning). Spirit without those pages: a question that has variability, not a worksheet of mean/median/mode with no data story.

⁵³¹Fuchs et al. 2021, Recommendation 3 (n. 466). Connect in the same visual field; fade for accuracy, not for a calendar date; revisit when a new idea needs a bridge.

⁵³²Carbonneau, Marley, and Selig 2013 (n. 467). Report $d \approx 0.37$ with the moderators.

⁵³³Virtual versus physical manipulatives: literature not opened. **UNVERIFIED** (02-learning open questions). Do not claim they are “just as good.” Do not claim they are worse. The variable is the conversation, not the plastic. David Uttal, Judy DeLoache, and colleagues' dual-representation papers are the usual citation for toy-like manipulatives being hard to see as symbols; those papers were not opened in full; exact findings **UNVERIFIED**.

The number line is the central fraction tool. IES 2010 Recommendation 2; Fuchs 2021 Recommendation 4, strong evidence, 14 studies; the Panel’s “key mechanism” sentence.⁵³⁴ Part-whole first is fine. It is not sufficient. Benchmarks 0, $\frac{1}{2}$, 1 before cross-multiply-to-compare. Equivalent fractions as the same point, Example 4.5 of the 2021 guide, not a butterfly. The 2021 panel’s objection to cross-multiply-to-compare during intervention is a magnitude objection: the trick does not build the thing the error is missing.⁵³⁵ Live debate, named: a parent who already has a child placing both fractions, estimating against $\frac{1}{2}$, and then wants a faster comparison algorithm is iterating, not defying the guide. A parent who teaches the trick as the first comparison, to a child who cannot mark $\frac{3}{5}$ on a blank line, is the case the panel wrote against. For intervention, magnitude wins on the page this book can defend.

Iterate concepts and procedures. Do not wait for full understanding before any practice. Do not drill invert-and-multiply — or unlike-denominator addition, or decimal-point-moving — with zero talk about why. Rittle-Johnson, Schneider, and Star 2015: it is a myth that it is a one-way street. The one classroom experiment that held content constant and varied only order, Rittle-Johnson and Koedinger 2009, was decimals, and favored cycling.⁵³⁶ Operational rule for this band, inherited from Chapter 2: when introducing a new idea, a short conceptual pass — what the symbols mean, a representation, a “why might this work?” — then a handful of problems, a return to the idea, a comparison of two methods, more problems. Optimal dosage for whole fraction topics, not six decimal lessons, remains thin. This book will not invent a number of concept days.⁵³⁷ A parent who cycles in the same week is inside the evidence. A parent who claims a research-based quarter of pies with no written procedure, or a research-based quarter of invert-and-multiply with no magnitude, is ahead of the literature, in opposite directions.

Worked-example fading for the first unlike-denominator lesson. New, high-element-interactivity material is the situation in which Sweller and Cooper’s 1985 worked-example effect is largest: studying a fully worked solution produces more learning for a novice than unguided problem solving, because means-ends search consumes working memory that could have been used to extract the schema.⁵³⁸ Show one fully worked unlike-denominator addition, on the line and in symbols, in the same visual field. Then a second that the child must explain. Then a completion problem missing the last step — the sum of the copies of the common unit fraction left blank. Then a conventional problem. Include an incorrect example of $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$ and ask what they thought the symbols meant. Julie Booth, Karin Lange, Kenneth Koedinger, and Kristie Newton, 2013, found

⁵³⁴IES 2010 Recommendation 2 (n. 431); Fuchs et al. 2021 Recommendation 4, strong, 14 studies (n. 482); NMAP number-line mechanism sentence (n. 432). Three documents, one representation.

⁵³⁵Fuchs et al. 2021 Recommendation 4 obstacles (n. 458, n. 490). Live debate: trick as first comparison versus trick as later compression after magnitude is in place. For intervention, magnitude wins on the page this book can defend.

⁵³⁶Rittle-Johnson, Schneider, and Star 2015 (n. 457); Rittle-Johnson and Koedinger 2009 (n. 457). Cycle in the same week.

⁵³⁷Rittle-Johnson, Schneider, and Star 2015 called out that optimal dosage and order of concepts versus procedures for whole topics (not six decimal lessons; not three minutes of equivalence) remains thin. Still thin. This book will not invent a number of concept days.

⁵³⁸John Sweller and Graham A. Cooper, “The Use of Worked Examples as a Substitute for Problem Solving in Learning Algebra,” *Cognition and Instruction* 2, no. 1 (1985): 59–89. Worked-example effect for novices: studying a fully worked solution produced more learning than unguided problem solving, because means-ends search consumes working memory. High-element-interactivity material (first unlike-denominator addition) is the situation in which the effect is largest.

that explaining incorrect examples, alone or mixed with correct ones, was especially beneficial for conceptual understanding of two-step linear equations.⁵³⁹ That paper is Algebra I. The move, scaled to this band and labelled as transfer of a finding, not as a grade-4 d , is: name the magnitude error, not “they forgot LCD.” Once the child can execute the replacement-and-add, stop showing full examples. Expertise reversal is real.⁵⁴⁰ Watching the parent do twenty problems in real time is not studying a worked example. The child has to study it, and then generate.

Schema-based word problems for equal groups, compare, and, late in the band, early ratio language — not keywords. Jitendra 2013, third grade, mathematics difficulties: $g = 0.46$ on word problems, $g = 0.34$ on a district test, null retention at 8 weeks.⁵⁴¹ Introduce one problem type at a time. Complete story first, then one unknown. Vary the cover story. Never teach “total means add.” Multiplicative comparison, Grade 4 — “times as many” — is a new schema, not a bigger add-to. Distinguish it from “3 more than” in the same week after a short blocked introduction of each, so the child must choose.

Interleave multiplication and division as soon as both exist; interleave like-denominator addition with comparison against $1/2$; interleave area and perimeter; interleave “plot the pair” with “name the pair.” After a short blocked introduction of a new skill, mix yesterday’s and last month’s types. The clean classroom RCTs of interleaving in mathematics are still grade 7. Rohrer 2014: 72 percent versus 38 percent, $d = 1.05$, two-week delay, $n = 140$. Rohrer 2020: 61 percent versus 38 percent, $d = 0.83$, 54 classes, one-month delay. Authors note that interleaved assignments also space same-kind items, so d is a joint effect. Most textbooks are almost entirely blocked.⁵⁴² Do not quote $d = 1.05$ as a grade-3 finding. The *practice* of mixing types still applies, labelled as transfer of a finding. Brunmair and Richter’s later meta-analysis of interleaving is a different and broader literature; that paper was not opened; the exact coefficient remains **UNVERIFIED** here. This chapter will not quote a g .⁵⁴³ A weekly shape that respects retrieval, spacing, and interleaving without mashing them: day of new skill, short blocked set. Same week, mixed retrieval of older facts. Later this week and next, the new skill between unlike problems. Monthly, an unannounced mixed quiz without notes.

⁵³⁹Julie L. Booth, Karin E. Lange, Kenneth R. Koedinger, and Kristie J. Newton, “Using Example Problems to Improve Student Learning in Algebra: Differentiating between Correct and Incorrect Examples,” *Learning and Instruction* 25 (2013): 24–34. Two in-vivo Cognitive Tutor experiments: explaining incorrect examples, alone or mixed with correct ones, especially beneficial for conceptual understanding of two-step linear equations. Algebra I paper; the move scaled to this band is labelled as transfer of a finding, not as a grade-4 d . Name the magnitude error, not “they forgot LCD.”

⁵⁴⁰Slava Kalyuga, Paul Ayres, Paul Chandler, and John Sweller, “The Expertise Reversal Effect,” *Educational Psychologist* 38 (2003): 23–31. The worked example that helps a novice becomes redundant for a relative expert and can hurt. Stop showing full examples once the child can execute. Alexander Renkl and Robert K. Atkinson, “Structuring the Transition from Example Study to Problem Solving,” *Educational Psychologist* 38 (2003): 15–22: guidance fading / backward fading.

⁵⁴¹Jitendra et al. 2013 (n. 462). $g = 0.46$ word problems; $g = 0.34$ district; null 8-week retention.

⁵⁴²Rohrer et al. 2014 and 2020 (n. 478). Doug Rohrer, Robert F. Dedrick, and Marissa K. Hartwig, “The Scarcity of Interleaved Practice in Mathematics Textbooks,” *Educational Psychology Review* 32 (2020): 873–883: most textbooks almost entirely blocked. d is a joint interleaving-plus-spacing effect. Grade 7. Do not quote $d = 1.05$ as a grade-3 finding.

⁵⁴³Brunmair and Richter interleaving meta-analysis: paper not opened for this project. Exact coefficient **UNVERIFIED**. Do not quote a g for mathematical tasks. Do not recycle a cited g as if the paper had been read.

Brief timed retrieval of *already-known* multiplication facts. Fuchs 2021 Recommendation 6, strong evidence, 27 studies. Gersten and colleagues 2009: about 10 minutes per intervention session on fluent retrieval.⁵⁴⁴ Time only facts the child can already produce correctly untimed. Mix them. Graph the child's own score if that helps the child see progress. Do not make a public, ranked, 50-item Mad Minute the identity of being good at math. That is Boaler's object, and it is not what the IES recommendation describes. The 27 studies bundled timing; the isolated timer effect remains open; the 2025 Peltier preprint is labelled preprint.⁵⁴⁵ Facts are not a language-model job. Section 5.8 will say so again.

Explicit instruction for students already behind. NMAP Finding 27: clear models, extensive practice, think-alouds, extensive feedback, consistently positive effects for students with mathematical difficulties and for the lowest third of a typical class, on word problems and computation — and this “does not mean that all of a student's mathematics instruction should be delivered in an explicit fashion.”⁵⁴⁶ Fuchs 2021 Recommendation 1, systematic instruction, strong evidence, 43 studies.⁵⁴⁷ Do not mash with Kapur 2014. Productive failure is generation-then-instruction on a new concept when priors exist; both sequences produced high procedural knowledge, and productive failure produced greater conceptual understanding and transfer in those two randomized studies.⁵⁴⁸ A parent can use ten to twenty minutes of “try this with what you already know; we will sort it after” on a new idea — first unlike-denominator addition, for a child who already has equivalence as the same point — then teach. A parent who never models for a child who cannot yet place $1/2$ on a line is not doing productive failure and is contradicting Finding 27. Frustration from missing facts is a fluency problem. Frustration from a well-posed new idea is Hiebert and Grouws's intellectual effort.⁵⁴⁹ “Never let them be frustrated” is folklore. “Struggle means I should not help” is the opposite folklore.

⁵⁴⁴Fuchs et al. 2021 Recommendation 6, strong, 27 studies (n. 475). Gersten et al. 2009 Recommendation 6: about 10 minutes per intervention session on fluent retrieval of basic facts.

⁵⁴⁵Bundling caveat and Peltier et al. 2025 OSF preprint (n. 477). Isolated timer effect remains open. Boaler's object (n. 476) is not what the IES recommendation describes.

⁵⁴⁶NMAP, *Foundations for Success* (2008), Finding 27: explicit instruction (clear models, extensive practice, think-alouds, extensive feedback) has consistently positive effects for students with mathematical difficulties and for the lowest third of a typical class, on word problems and computation — and this “does not mean that all of a student's mathematics instruction should be delivered in an explicit fashion.” Finding 23: high-quality research does not support exclusive use of either all-encompassing teacher-directed or student-centered prescriptions. Jere Confrey, Alan P. Maloney, and Kenny H. Nguyen, “Breaching the Conditions for Success for a National Advisory Panel,” *Educational Researcher* (2008), is a methodological critique of the Panel (biased literature selection, inconsistent standards). Report both. The Panel is still the synthesis later IES guides cite.

⁵⁴⁷Fuchs et al. 2021, Recommendation 1: systematic instruction, strong evidence, 43 studies. Written for students with or at risk for mathematics disabilities and for the lowest performers.

⁵⁴⁸Manu Kapur, “Productive Failure in Learning Math,” *Cognitive Science* 38, no. 5 (2014): 1008–1022, <https://doi.org/10.1111/cogs.12107>. Two randomized studies: problem-solving before instruction vs instruction before problem-solving. Both sequences produced high procedural knowledge; productive failure produced significantly greater conceptual understanding and transfer. Designed sequence (ill-structured problem, generation of suboptimal solutions, then instruction that compares them), not “let them be confused all week,” and not NMAP Finding 27. Do not mash.

⁵⁴⁹James Hiebert and Douglas A. Grouws, “The Effects of Classroom Mathematics Teaching on Students' Learning,” in *Second Handbook of Research on Mathematics Teaching and Learning*, ed. Frank K. Lester (NCTM / Information Age, 2007). Two features that support skill and conceptual understanding: explicit attention to connections among facts, procedures, and ideas; students' struggle with important mathematics. “Struggle” is intellectual effort to make sense, not abandonment.

Mathematical language, Fuchs Recommendation 2, strong evidence, 16 studies, even for monolingual students: say “greater than,” not only “bigger”; “unit fraction,” not “the little number”; “equivalent,” not “the same” if “the same” is ambiguous between same point and same picture; “regroup,” not “borrow.” Shared, precise vocabulary is the cheap move.⁵⁵⁰ An English-learner-specific mathematics RCT was not opened; effect sizes **UNVERIFIED**. Do not translate the mathematics away into informal English and then expect the child to read a textbook that says *numerator*.

Cross-multiply-to-compare as a parent trick versus the intervention panel’s magnitude objection, restated as a decision rule. If the child cannot place both fractions on a line, do not teach the trick. If the child can place both, can estimate against 0, $\frac{1}{2}$, and 1, and is ready for a compression, the trick can arrive as a compression, with the line still in the field. The butterfly as the *identity* of equivalence is the failure. Flexibility — being able to choose among a benchmark, a common denominator, and, later, a cross-product — is a third outcome, not a luxury. Schneider, Rittle-Johnson, and Star 2011 modeled conceptual knowledge, procedural knowledge, and procedural flexibility as latent variables; each predicted gains in the other.⁵⁵¹ After the child can execute one comparison method, put a second next to it and ask which is better for this pair, and why. Rittle-Johnson and Star’s 2007 and 2009 comparison-of-methods papers were not opened in full; exact *d* values **UNVERIFIED**. The qualitative finding is enough for a kitchen rule.⁵⁵²

A session shape that does not invent a trial: unaided warm-up, three to five minutes — two known facts, one fraction on a line, one like-denominator sum. Teach from the family’s book, parent already prepared. Struggle on three to five items, child writes, no phone. Hint, not key, on a miss. Hands-on or a line if the miss was conceptual. Exit ticket unaided, five minutes, parent grades against a parent-held key. That shape is a practice suggestion grounded in Chapter 2’s retrieval and generation findings and in Chapter 3’s unaided-exam demand, scaled. It is not a homeschool RCT.⁵⁵³ Chapter 3’s full session shape, including the AI-era version, is section 5.8. This section’s job was the mathematics-teaching moves. Tips that survive evidence, including the cooking caution, come next.

5.7 Tips that survive evidence — and cooking is not the line

The tips in this section are stealable sentences, each sourced. They are not a program. They are not a 180-day plan. They are the corrections this chapter owes a parent who will otherwise repeat the popular story.

⁵⁵⁰Fuchs et al. 2021, Recommendation 2: mathematical language, strong evidence, 16 studies. Table 2.1 examples: say “regroup,” not “borrow”; “greater than,” not only “bigger.” EL-specific mathematics RCT not opened; effect sizes **UNVERIFIED**.

⁵⁵¹Michael Schneider, Bethany Rittle-Johnson, and Jon R. Star, “Relations between Conceptual Knowledge, Procedural Knowledge, and Procedural Flexibility,” *Developmental Psychology* (2011), <https://doi.org/10.1037/a0024997>. After a three-day classroom intervention on equation solving, each type of knowledge predicted gains in the other; relations symmetrical. Flexibility is a third outcome, not a luxury.

⁵⁵²Bethany Rittle-Johnson and Jon R. Star, comparison-of-methods papers (2007, 2009): comparison improves flexibility. Papers not opened in full for this project; exact *d* **UNVERIFIED**. Qualitative kitchen rule: after the child can execute one method, put a second next to it and ask which is better for this problem, and why.

⁵⁵³Session shape as a practice suggestion grounded in Chapter 2 (retrieval, generation, fading) and Chapter 3 (Bastani exam phase, unaided exit ticket), scaled to this band. Not a homeschool RCT. No year-plus kitchen GPT-Tutor RCT was opened on 30 August 2026.

Do not treat “multiply makes bigger” as cute folklore. It is Ni and Zhou whole-number bias — unique successors, finite entities in an interval, multiplication makes bigger, add the counts you see — and 5.NF.5 exists to break it. A fraction greater than 1 enlarges. A fraction less than 1 shrinks. The flinch when $4 \times 1/2$ is 2 is the conception. The cute exception leaves the bias in place.⁵⁵⁴ Vosniadou, Vamvakoussi, and Skopeliti treat the property as a conceptual-change problem, not a forgotten footnote. Greer 1994 data were not opened; do not cite them.⁵⁵⁵

Equivalent fractions are the same point, not a butterfly trick. $1/2$, $2/4$, and $3/6$ land at the same location on a 0-to-1 line. IES 2010 Recommendation 2; Fuchs 2021 Example 4.5; the Panel’s number-line mechanism sentence.⁵⁵⁶ The butterfly can arrive later as a compression, if it arrives, after the point is believed. Cross-multiply-to-compare as the first comparison, during intervention, is the move the 2021 panel advises against because it does not build magnitude.⁵⁵⁷

Tile, then array, then formula. Area of a rectangle is rows of unit squares. Volume of a right rectangular prism is layers of unit cubes. Perimeter is the distance around, a different quantity. Same perimeter, different area, and the reverse, force the choice. A 3-by-4 rectangle that is “12” with no unit is the chant without the covering.⁵⁵⁸

Cooking can be a second representation of a fraction already being taught as a number. It does not replace the line. NMAP Finding 26: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving.⁵⁵⁹ Homeschool culture overclaims kitchen-math. They’ll learn fractions by cooking. A child who can double a recipe and cannot place $3/2$ on a line has performed, not learned. That last distinction is Soderstrom and Bjork’s: current performance is “often an unreliable index” of learning.⁵⁶⁰ Transfer from cooking, money, and games to formal items, measured rather than assumed, remains an open question this book will not fill with analogy.⁵⁶¹ Money is a decimal-and-place-value context, useful, also not a curriculum. Building and making are spatial; they do not automatically produce proof, and they do not automatically produce $3/4$ as a point.

Do not skip 3.NF number-line work. Pies introduce part-whole. They do not by themselves teach

⁵⁵⁴Ni and Zhou 2005 (n. 430); CCSSM 5.NF.5 (n. 501). Do not treat “multiply makes bigger” as cute folklore.

⁵⁵⁵Vosniadou, Vamvakoussi, and Skopeliti 2008 (n. 500). Greer 1994 data **UNVERIFIED** (n. 502). Do not cite Greer 1994.

⁵⁵⁶IES 2010 Recommendation 2 (n. 431); Fuchs et al. 2021 Example 4.5 (n. 491); NMAP number-line mechanism sentence (n. 432). Equivalent fractions as the same point.

⁵⁵⁷Fuchs et al. 2021 Recommendation 4 obstacles (n. 458). Cross-multiply-to-compare as first comparison during intervention is the move the panel advises against.

⁵⁵⁸CCSSM 3.MD and 5.MD (nn. 468, 527). Tile, then array, then formula, said as “rows of unit squares” / “layers of unit cubes.”

⁵⁵⁹NMAP Finding 26 (n. 516). Cooking can motivate and can provide a second representation of a fraction already being taught as a number. It does not replace the number-line work, the schema work, or the retrieval.

⁵⁶⁰Nicholas C. Soderstrom and Robert A. Bjork, “Learning Versus Performance: An Integrative Review,” *Perspectives on Psychological Science* 10, no. 2 (2015), <https://doi.org/10.1177/1745691615569000>. Current performance is “often an unreliable index” of learning. A child who can double a recipe and cannot place $3/2$ on a line has performed, not learned. Full discussion in Chapter 3; this chapter uses the distinction, it does not re-argue the review.

⁵⁶¹Transfer from informal activity (cooking, money, games) to formal items, measured rather than assumed: open question this book leaves standing (02-learning open questions; brief open question 20). Do not fill with analogy. Barnett and Ceci 2002 is the standard transfer taxonomy; paper not opened; exact dimensions **UNVERIFIED**.

that $3/4$ is a number with a magnitude. Equal shares in 1.G and 2.G are the only prior fraction language on the map. They are the on-ramp. They are not the gate.⁵⁶²

Do not delay unlike-denominator addition past Grade 5 on the map. The Panel's Table 2: add and subtract fractions and decimals by the end of Grade 5; multiply and divide by the end of Grade 6. Division of a fraction by a fraction is Grade 6, not Grade 5.⁵⁶³ Ahead of the map is a placement decision. Behind the Panel's benchmark, into the ratio year without unlike-denominator addition, is underprepared.

Do not treat a "grade 5" vendor book as a legal grade. Math-U-See Epsilon is a fractions *level*, **vendor**, not "fifth grade." Saxon 5/4 and 6/5 are skill names. Beast Academy's vendor advice is to start about a level behind. Dimensions 6 is a Grade 6 text with some pre-algebra topics, not a pre-algebra text, **vendor**. A child of ten may sit this chapter's mathematics at calendar grade 2 or 6. Grade labels are anchors. NCES 2016 already showed the stretch.⁵⁶⁴ Placement tests exist because the labels lie. No homeschool-curriculum RCT was opened to crown a winner. Name the options. Do not sell.

Do not let a solver hide the $1/2 + 1/3 = 2/5$ error. Photomath, Mathway, and Symbolab will print a correct least-common-denominator path. The parent never sees the counts-you-see misconception. Chapter 3 banned paste-the-worksheet and photo-to-key during the attempt. The mechanism is Bastani's GPT Base: assisted practice looks like mastery; the unaided exam is where the crutch becomes a score. No Photomath-named RCT of unaided transfer was opened; the transferable evidence is the mechanism plus the homepages that lead with scan-to-solution.⁵⁶⁵ Diagnose the child's already-wrong work. Name "adding the counts you see," not "they forgot LCD."

Diagnose fractions, the equal sign, and — when Chapter 6 arrives — signed numbers, before buying Algebra I. The Panel's 2007 algebra-teacher survey, 743 teachers, rated incoming preparation poorest in rational numbers and operations with fractions and decimals.⁵⁶⁶ That is 2008 data, still the best nationally representative teacher-voice primary on incoming Algebra I gaps, not a 2026 replication. Do not write "teachers today say." The equal sign is already live in this band, inside $a/b = c/\square$. McNeil and Alibali's operational patterns do not wait for Algebra I.⁵⁶⁷ Signed numbers

⁵⁶²CCSSM 3.NF.2 (n. 480); 1.G.3 / 2.G.3 (n. 486); IES 2010 Recommendation 2 (n. 431). Do not skip number-line work. Pies introduce part-whole; they do not by themselves teach that $3/4$ is a number with a magnitude.

⁵⁶³NMAP Table 2 (n. 450, n. 512); CCSSM 5.NF.7 vs 6.NS (n. 511). Do not delay unlike-denominator addition past Grade 5 on the map. Fraction $\div$ fraction is Grade 6.

⁵⁶⁴Vendor grain (n. 454); NCES PFI 2016 stretch (n. 437). A "grade 5" book is a publisher's scope, not a legal grade. Math-U-See Epsilon is a fractions *level*. Do not treat it as "fifth grade."

⁵⁶⁵Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman, "Generative AI without Guardrails Can Harm Learning: Evidence from High School Mathematics," *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122, <https://doi.org/10.1073/pnas.2422633122>; author PDF https://hamsabastani.github.io/education_llm.pdf, accessed 30 August 2026. GPT Base: assisted practice +48% relative to control; unaided exam -17%. GPT Tutor: practice +127%; exam point estimate -0.004, not significant. GPT Base fully correct 51% of the time when asked for the answer. Photomath-named RCT of unaided transfer: **UNVERIFIED**. Transferable evidence is Bastani's crutch plus solver homepages that lead with scan-to-solution (n. 587). Full product landscape and ban/allow list: Chapter 3. This chapter is a pointer, not a sequel.

⁵⁶⁶NMAP 2007 algebra-teacher survey, 743 teachers (n. 429). Poorest incoming preparation in rational numbers and operations with fractions and decimals. 2008 data; not a 2026 replication. Diagnose fractions (and, in Chapter 6, signed numbers) and the equal sign before buying Algebra I.

⁵⁶⁷McNeil and Alibali 2005; Knuth et al. 2006 (n. 518). Operational equal sign now inside $a/b = c/\square$. Signed numbers

wait for Chapter 6. Do not skip them later because an Algebra I book is on the shelf.

Concepts-first versus iterative, for fractions specifically: Rittle-Johnson 2015 still applies. Rittle-Johnson and Koedinger 2009 was decimals. Cycle in the same week. Do not withhold practice until they understand. Do not drill a new procedure with no representation and no why.⁵⁶⁸ NCTM 2014's "procedural fluency follows and builds on a foundation of conceptual understanding" is a policy stance, not a finding tested against a procedures-then-concepts alternative.⁵⁶⁹

Retrieve known multiplication facts often, sometimes against a clock, after they are already right untimed. 3.OA.7 is the map: from memory, all products of two one-digit numbers, by the end of Grade 3. The Panel wants automatic recall to free working memory. Mix multiplication with division as soon as both exist. Do not make the timer the identity of being good at math.⁵⁷⁰ Do not let a language model sit between the child and 7×8 . Generate a deck or an oral script; the parent's ears do the drill.

Interleave after a short blocked introduction. Mix types. Do not quote Rohrer's $d = 1.05$ as a grade-4 finding. The practice transfers; the coefficient stays in grade 7.⁵⁷¹

Model first for a child already behind on this topic (Finding 27; Fuchs Recommendation 1). Use a short generation-then-instruction sequence for a new idea when priors exist (Kapur). Do not mash those comparisons.⁵⁷² Accelerate a child who has the prerequisites. Do not skip fractions because a child is "good at math." Finding 15 is the age-lock refusal. Finding 30 is the acceleration warrant with the motivation-and-prerequisites clause. Siegler and colleagues are why the prerequisites include fractions.⁵⁷³

Prepare so you can hear a wrong answer; then do not take the pencil. Maloney, Ramirez, Gunderson, Levine, and Beilock 2015 is a school-homework study: parent math anxiety $\times$ frequent help predicted worse child math, math-specific. A homeschool parent cannot "help less often" as a daily strategy. The finding is not "anxious parents should not teach." The constructive response is preparation, generation, and not narrating fear. Replication status is open. There is no fetched experimental literature on math-anxious parents as full-time instructors. That gap stays standing.⁵⁷⁴

wait for Chapter 6.

⁵⁶⁸Rittle-Johnson, Schneider, and Star 2015; Rittle-Johnson and Koedinger 2009 (n. 457). Cycle in the same week.

⁵⁶⁹National Council of Teachers of Mathematics, *Principles to Actions: Ensuring Mathematical Success for All* (Reston, VA, 2014), p. 42: "procedural fluency follows and builds on a foundation of conceptual understanding." Policy stance, not a finding tested against a procedures-then-concepts alternative. Rittle-Johnson, Schneider, and Star 2015 flag the wording as a source of the one-way-street myth.

⁵⁷⁰CCSSM 3.OA.7 (n. 444); NMAP Finding 11 (n. 470); Fuchs et al. 2021 Recommendation 6 (n. 475); Boaler 2012/2015 (n. 476). Retrieve known facts often, sometimes against a clock, after they are already right untimed. Do not make the timer the identity of being good at math. Facts are not an LLM job (n. 585).

⁵⁷¹Rohrer et al. 2014 $d = 1.05$ is grade 7, two-week delay, $n = 140$ (n. 478). The practice of mixing types transfers; the coefficient stays in grade 7.

⁵⁷²NMAP Finding 27 (n. 546); Fuchs et al. 2021 Recommendation 1 (n. 547); Kapur 2014 (n. 548). Do not mash explicit instruction for students already struggling with productive failure as generation-then-instruction on a new concept when priors exist.

⁵⁷³NMAP Findings 15 and 30 (nn. 438–439); Siegler et al. 2012 (nn. 434–435). Accelerate a child who has the prerequisites. Do not skip fractions because a child is "good at math."

⁵⁷⁴Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, "Intergenerational Effects of Parents' Math Anxiety on Children's Math Achievement and Anxiety," *Psychological Science* 26 (2015): 1480–1488, <https://doi.org/10.1177/0956797615592630>. School-homework study: parent math anxiety

Do not write a 180-day plan from this chapter. The map specifies objects, not Tuesdays. The family's book specifies Tuesdays. This book specifies which wrong answers to hear, which representations to connect, which slogans to refuse, and which open questions not to fill with analogy.

The solver will print a correct LCD path. That is the problem. The next section is the AI lane for this band: parent-first explanation, isomorphic practice with the parent holding the key, diagnosis after the attempt, a number-line activity the child draws. Ban paste-the-worksheet. The child does not live in an open-web agent. Facts are not an LLM job. Practice suggestions, not an RCT. Chapter 3 already ran the trial that matters. This section will not sequel it.

5.8 Parent-hosted AI for grades 3–5

Chapter 3 already ran the argument this section is not going to sequel. Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman randomized nearly a thousand Turkish high-school mathematics students. A ChatGPT-like tutor, GPT Base, raised assisted-practice grades 48 percent relative to control and then cut unaided exam grades 17 percent. A guardrailed tutor, GPT Tutor — hints, no full solution, teacher-written answers and common mistakes in the hidden prompt — raised practice 127 percent and was statistically indistinguishable from control on the unaided exam (point estimate -0.004). GPT Base, asked for the answer, was fully correct only 51 percent of the time. Students copied anyway. They did not perceive that they had learned less.⁵⁷⁵ Nicholas Soderstrom and Robert Bjork had already named the confusion: performance during acquisition is “often an unreliable index” of learning.⁵⁷⁶ That pair is the kitchen condition. This section applies it to fractions, facts, and the first quadrant. It does not rewrite the product landscape, the COPPA dates, or the companion-harm cases. Those stay in Chapter 3. One reminder, then the grade 3–5 jobs.

The jobs are **practice suggestions, not an RCT**. No year-plus kitchen-table trial of a parent-constrained tutor exists in the literature opened on 30 August 2026.⁵⁷⁷ Grounding is named: Bastani's crutch and hidden-prompt design; Soderstrom and Bjork's performance/learning split; Oreopoulos and Low's finding that unprompted student dialogue with a well-designed tutor is rare; Chapter 3's ban-versus-allow list. Composite hypotheticals, if used, are labelled *hypothetical* in

× frequent help predicted worse child math and more child math anxiety; when help was infrequent, no such relation; math-specific, not reading. Not a parent-as-sole-teacher RCT. Homeschool parent cannot “help less often” as a daily strategy. Do not treat the finding as “anxious parents should not teach.” Replication status open; no registered replication opened. Fraulein Retanal et al., *Education Sciences* 11, no. 10 (2021): 620: controlling-supportive homework-helping style partially mediated the relation in a survey of parents of 11- to 14-year-olds. Highest-priority homeschool-specific gap: no fetched experimental literature on math-anxious parents as full-time instructors.

⁵⁷⁵Bastani et al. 2025 (n. 565). Short reminder, not a second monograph. Percentages are of control grades, not Cohen's d . Design: classroom-level RCT, large high school in Turkey, Fall 2023–24, ~1,000 students in grades 9–11, four 90-minute sessions, GPT-4-0613. Not a homeschool study, not GPT-5, not Khanmigo. Mechanism: GPT Base as a crutch (copying solutions; fewer messages, mostly “what is the answer”).

⁵⁷⁶Soderstrom and Bjork 2015 (n. 560). Performance during acquisition is “often an unreliable index” of learning. People “mistakenly interpret their performance during acquisition as a reliable guide to long-term learning.” Bastani's student-perception finding is the AI-era replication of that metacognitive error.

⁵⁷⁷Parent workflows in this section are practice suggestions grounded in Bastani, Soderstrom and Bjork, and Oreopoulos and Low, not a new RCT of homeschool AI. No year-plus kitchen-table RCT of a parent-constrained GPT-Tutor workflow was opened on 30 August 2026 (04-ai open questions). Label them that way. Chapter 3 §3.3–3.4 is the lock; this section is the grade 3–5 pointer.

the sentence that introduces them.

Allow, parent-hosted, or child only after an attempt and constrained.

Parent-first explanation of a fraction procedure. Workflow (a) from Chapter 3, aimed at this band's objects. Parent account, no child present: "I am teaching unlike-denominator addition to a home-schooler using [the family's book]. Explain it to *me* as an adult who last took math in [year]. Give: the idea in one paragraph; the notation I will see; the two mistakes this age makes; one tiny problem I should be able to solve before I sit down with the child; the answer to that problem." Then the parent solves the tiny problem *without* the model, then checks.⁵⁷⁸ The tiny problem for this band is why $1/2 + 1/3$ is not $2/5$. The idea is magnitude: $2/5$ is less than $1/2$; the join of $1/2$ and $1/3$ is past $1/2$. The notation is equivalent fractions as the same point, then copies of a common unit fraction. The two mistakes are adding the counts you see, and finding an LCD with no point to check against. A *hypothetical*, labelled as such: a parent who last passed algebra in 1998, facing a Grade 5 lesson on $2/5 + 1/2$, asks for that explanation, solves $1/4 + 1/3$ on paper, checks, and only then sits down. That sequence is the workflow. It is not a finding that the sequence raises a score. Prefer a chat model for this job, not an open-web agent. Hermes 4 as chat, or any capable reasoner. Hermes Agent's tool loop is wasted on it. Wolfram|Alpha is not yet the right check for $1/2 + 1/3$; the right check is a line the parent marks, or a known-correct key.

Isomorphic practice with the parent-held key. Workflow (c). Generate five to eight unlike-denominator items, same structure, new numbers, new cover story. Two pages: student sheet, parent key. The child never sees the key. Do not generate "practice" by photographing tonight's assigned page and asking the model to complete it. That is the ban, wearing a worksheet.⁵⁷⁹ Isomorphic means new items of the same type. $1/2 + 1/5$ after $1/2 + 1/3$ is a cousin, not a key to the assigned problem. Grounding: practice with access to the solution is not unaided learning (Bastani); generation plus delayed feedback sits on Soderstrom and Bjork's learning side; Oreopoulos and Low, descriptively, found that practice minutes predicted MAP and chat did not.⁵⁸⁰ None of that is a kitchen RCT of isomorphic fraction sheets. It is why the key stays with the adult.

Hint script that names the magnitude error. Workflow (d). Bastani's GPT Tutor prompt withheld the full solution, asked what they had tried, pointed out a named mistake, escalated slowly.⁵⁸¹ Parent prompt: "Write a hint script for $1/2 + 1/3$. Level 1: a question only. Level 2: name the concept — fractions are numbers with magnitude; $2/5$ is less than $1/2$ — without giving the sum. Level 3: the next step, still not the final answer: replace with sixths. Also list one likely mistake

⁵⁷⁸Chapter 3 workflow (a): explain the concept to the *parent* first. Practice suggestion, not an RCT. Bastani's GPT Tutor worked in part because teachers had already written the solution and common mistakes into the hidden prompt. The parent is that teacher. Prefer a chat/reasoning model, not an open-web agent, for this job.

⁵⁷⁹Chapter 3 workflow (c): extra isomorphic practice with the answer key held by the parent. Do not generate "practice" by completing tonight's assigned homework. That is the ban.

⁵⁸⁰Bastani et al. 2025 (practice with access $\neq$ unaided learning); Soderstrom and Bjork 2015 (generation and delayed feedback); Philip Oreopoulos and Nina Low, "One Click Away: AI Tutoring with Khanmigo in a Two-Year School Experiment," NBER Working Paper 35620, August 2026, <https://www.nber.org/papers/w35620>. Descriptively, practice minutes predicted MAP; chat did not. Working paper, not yet journal-peer-reviewed.

⁵⁸¹Bastani et al., author PDF, Figure 1 (GPT Tutor prompt): withhold the full solution; ask what they have tried; point out a named mistake; escalate information slowly; accept equivalent answers. Homeschool analogue is a parent-read hint script, not a child-held chatbot. Chapter 3 workflow (d).

— adding the counts you see — and what to say if I see $2/5$. Do not put the final answer anywhere the child might see; put it only under PARENT KEY.” The parent reads the hints aloud. That is closer to human tutoring than handing the child Khanmigo and leaving the room. Oreopoulos and Low: in a two-year cluster randomized trial in 18 Tennessee middle schools, assignment to Khan Academy plus Khanmigo raised MAP math by 1.264 national percentile ranks per term (SE 0.597), 0.040 population SD per term, about 0.06 to 0.08 SD over a school year; gains resembled Khan practice without AI; the median student messaged Khanmigo on only a third of practice days and in 17 percent of mistake sessions; 14.5 percent of messages contained a mathematical question or a step of reasoning.⁵⁸² Do not invent standalone 0.06 as the intention-to-treat. Do not advertise Khanmigo as Bloom’s two-sigma tutor. A parent who sits next to the child and requires a hint dialogue after a miss is running a different intervention than that ITT.

Diagnose a wrong student paper after the attempt. Workflow (f). Photograph the child’s work *after* the page is produced. Parent-only prompt: “Here is a student’s work. Do not praise. Name the most likely misconception in one sentence. Give me one question I should ask the child (not the correct steps). Give me one isomorphic item to retest after we talk.” Then the parent asks the question. The child answers the parent, not the model.⁵⁸³ For $1/2 + 1/3 = 2/5$, the sentence is “adding the counts you see,” not “they forgot LCD.” For $0.25 > 0.3$, the sentence is “more digits, bigger.” For $4 \times 1/2 = 8$, the sentence is “multiply makes bigger,” whole-number bias, 5.NF.5. For $3/4$ compared with $3/8$ by “fourths are bigger because 4 is bigger,” the sentence is same-numerator magnitude, more parts smaller. The model is a misconception namer. The conversation is the parent’s.

Generate a number-line activity. Workflow (e). “Give me a ten-minute kitchen-table activity for placing $3/4$, $1/2$, and 1 on a 0-to-1 line, using a paper strip or a drawn line. No worksheet. What the child should say if they understand. What a wrong mark looks like.” The model describes. The child draws. Area models the same way: the model can describe a unit square halved on one side and thirded on the other; the child draws the $1/6$ overlap. A screen of movable fraction bars can be the same conversation if the parent connects them to the written numeral in the same sitting. Virtual versus physical remains **UNVERIFIED**. The variable is the conversation.⁵⁸⁴

Generate a fact-family oral script. Do not let the model sit between the child and 7×8 . Flash retrieval of arithmetic facts is not an LLM job. Bastani: GPT Base fully correct 51 percent of the time, with 8 percent arithmetic errors even when the method was right.⁵⁸⁵ A language model

⁵⁸²Oreopoulos and Low, NBER w35620, abstract and Table 5, accessed 30 August 2026. Two-year cluster RCT, 18 Tennessee middle schools. Assignment raises math achievement by 1.264 NPR per term (SE 0.597); 0.040 population SD per term; about 0.06 to 0.08 SD over a school year; implied effect of a full year of active participation 0.14 SD. “These gains resemble those from Khan Academy practice without AI assistance.” 96% tried Khanmigo at least once; median student messaged it on only a third of practice days and in 17% of mistake sessions; 14.5% of messages contained a mathematical question or a step of reasoning. Do not invent standalone 0.06 as the ITT. Do not advertise Khanmigo as Bloom’s two-sigma tutor. Pre-registration: AEA RCT Registry AEARCTR-0013519.

⁵⁸³Chapter 3 workflow (f): diagnose a wrong answer the child already produced. Practice suggestion, not an RCT. Name the misconception; ask a question; retest with an isomorphic item. The child answers the parent, not the model. For $1/2 + 1/3 = 2/5$, the sentence is “adding the counts you see” (Ni and Zhou 2005), not “they forgot LCD.”

⁵⁸⁴Chapter 3 workflow (e): hands-on activity as materials generation. The model describes; the child draws. Virtual vs physical manipulatives **UNVERIFIED** (n. 533). Carbonneau $d \approx 0.37$ is versus abstract-only, under guidance (n. 467).

⁵⁸⁵Bastani et al. 2025: GPT Base fully correct 51%; 8% arithmetic errors even when the method was right. Flash retrieval of arithmetic facts is not an LLM job. Chapter 3 §3.7 / 04-ai load-bearing finding 12: use the model to generate

will happily hallucinate a product. Use the model to generate a three-minute oral script for the $\times 7$ and $\times 8$ families, mixed with the related quotients, no calculator, no LLM at drill time. The parent checks a sample against a known table. DreamBox and IXL can supply retrieval practice; they are adaptive engines, not agents, and not a substitute for the parent's ears.⁵⁸⁶

Ban, child-facing, during the attempt.

Paste-the-worksheet. Photograph the assigned Saxon or Singapore or Math-U-See page and ask for the key. Photomath's homepage leads with scan, including word problems, after which "A solution will pop up for your exact problem." Mathway: "Got a math problem? Get the steps." Symbolab sells the same loop. Vendor "designed for learning" does not rewrite Bastani.⁵⁸⁷ The solver will print a correct LCD path for $1/2 + 1/3$. The parent never sees $2/5$. That hide is the pedagogical failure this band cannot afford, because $2/5$ is the magnitude misconception the rest of the chapter exists to make hearable. "What is the answer?" and "just give me the steps" are the same ban. Steps the child did not generate are not generation.

Companion persona. A "math buddy" with a name, a memory of the child, and a messaging-app gateway is the architecture Chapter 3 refused. Setzer/Garcia v. Character Technologies and Raine v. OpenAI are allegations, not verdicts, and not a literature on mathematics learning. They are why a fraction agent should not be a person.⁵⁸⁸

Open-web agents in the child's hands. Hermes Agent and Grok Bot can browse, keep memory, finish jobs. No page opened on 30 August 2026 sold them as a K–12 student product. A parent-hosted agent expanding a lesson on the parent's machine is a different fact pattern. Putting a child's worksheet photographs on Grok Bot's shared cloud computer is a privacy decision the parent owns; Legacy Privacy Mode is not supported; there is no FERPA claim on the pages opened.⁵⁸⁹ El Salvador's ministry deployment of Grok the assistant is not this playbook and not an RCT.

Detectors as police of the child's written "why $1/2 + 1/3$ is $5/6$." Liang and colleagues 2023: average false-positive rate 61.22 percent (arXiv) / 61.3 percent (*Patterns*) on human TOEFL essays. A

the deck or the oral script, not to sit between the child and the fact. IES 2021 Recommendation 6 belongs to the parent's ears, or a dumb SRS, not to a language model.

⁵⁸⁶ Adaptive practice engines (DreamBox, IXL, Khan Academy practice, ALEKS) sequence items; they are not agents in the Hermes Agent / Grok Bot sense. DreamBox grade-level-per-hour claim is **vendor**; underlying ESSA study PDF **UNVERIFIED**. IXL live homeschool price **UNVERIFIED** as an invoice number. MATHia family purchase **UNVERIFIED**. Chapter 3 §3.2.

⁵⁸⁷ Photomath, <https://photomath.com/>, fetched 30 August 2026: scan including word problems; "A solution will pop up for your exact problem"; Basic \$0 / Monthly \$9.99 / Annually \$69.99 (USD, this day); **vendor** "designed for learning." Mathway, <https://www.mathway.com/>: "Got a math problem? Get the steps." Symbolab, <https://www.symbolab.com/>: solver first. No product-specific unaided-transfer RCT opened this day. Ban during the attempt: paste-the-worksheet, photo-to-key, "what is the answer."

⁵⁸⁸ Garcia v. Character Technologies et al., complaint, M.D. Fla., via CourtListener/RECAP: allegations, not a verdict. Raine v. OpenAI, complaint: allegations, not a verdict. Setzer settlement terms **UNVERIFIED**. Why a math agent should not be a companion. Not a literature on mathematics learning. Full discussion: Chapter 3 §3.6.

⁵⁸⁹ xAI, "Grok Bot FAQ," <https://docs.x.ai/grok-bot/faq>, fetched 30 August 2026: persistent cloud computer; "Grok Bot requires cloud data storage, so Legacy Privacy Mode is not supported"; one computer per user shared across Bots; no education SKU and no FERPA claim on the pages opened. Hermes Agent, <https://hermes-agent.org/>: self-hosted adult agent, open web, messaging gateways; no K–12 student-safety certification on the pages opened. El Salvador (xAI news, 11 December 2025) is a ministry deployment of Grok the assistant, announcement not RCT, not Grok Bot, not a U.S. homeschool SKU. Full distinctions: Chapter 3 §3.2.

careful, slow, or non-native paragraph about unit fractions is exactly the writer those detectors smear.⁵⁹⁰

Symbolic algebra solvers are not yet the right check in this band. The right check is a parent-held key or a number line the child marks. Wolfram|Alpha becomes the honest check later, for an antiderivative, not for $3/4 \times 2/5$. Mixing a chat model's fluency with a known-correct key, without a label, is how a wrong sum survives a parent's glance.

If the child uses a generative tool at all. Constrained tutor mode — Khanmigo, or a parent-built GPT-Tutor prompt that forbids answers — after an attempt, parent in the room, no paste-the-key. No open web. No companion. Unaided exit ticket every session. Khanmigo's "never gives the answer" is **vendor**. Bastani had to put the solution in the hidden prompt and still built a separate product to stop answer-giving.⁵⁹¹ Under 13, COPPA applies to operators of child-directed or mixed-audience services; a parent using an adult tool on a parent account to make a worksheet is a different fact pattern from creating a child account on a companion app. Do not pretend consumer ChatGPT is a school-official FERPA context. Chapter 3 has the dates: COPPA amendments, Federal Register 22 April 2025, effective 23 June 2025, general compliance 22 April 2026.⁵⁹²

Session shape, practice suggestion, not an RCT, scaled to this band from Chapter 3's kitchen block.

1. Unaided warm-up, three to five minutes, paper, no device: two known facts, one fraction on a line, one like-denominator sum — Bastani's exam phase, shrunken.
2. Teach from the family's book, the parent having used workflow (a) or (b) *before* sitting down.
3. Struggle on three to five items, child writes. No phone. Generation.
4. Hint, not key: parent uses the hint script, or Khanmigo with the parent watching, only after a miss.
5. Hands-on or a line if the miss was conceptual — workflow (e).
6. Exit ticket unaided, five minutes. Parent grades against the parent key. Later, workflow (f) on misses.
7. Never: "finish your fractions with Photomath so we can go to soccer." That is GPT Base, and it is the one pattern with a large-sample causal harm estimate.⁵⁹³

⁵⁹⁰Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou, "GPT Detectors Are Biased against Non-Native English Writers," arXiv:2304.02819; *Patterns* 4, no. 7 (2023), <https://doi.org/10.1016/j.patter.2023.100779>. Average false-positive rate 61.22% (arXiv) / 61.3% (*Patterns*) on human TOEFL essays. Do not use detectors to police a homeschooled's written mathematics explanations.

⁵⁹¹Khan Academy, "Khanmigo for learners," <https://www.khanmigo.ai/learners>, fetched 30 August 2026. **Vendor:** "Unlike ChatGPT, Khanmigo never gives you the answer." **Vendor** price \$4/month or \$44/year. Parent-gated home SKU, not an open-web agent. Whether the design holds in the wild is Oreopoulos and Low (n. 582), not the brochure. Bastani's GPT Tutor required a hidden prompt containing the solution and common mistakes.

⁵⁹²Federal Trade Commission, "Children's Online Privacy Protection Rule," 90 FR 16918 (22 April 2025), Document 2025-05904, 16 CFR 312. Effective date 23 June 2025; general compliance date 22 April 2026. Full discussion: Chapter 3 §3.6. A parent using an adult tool on a parent account to generate a worksheet is a different fact pattern from creating a child account on a companion app. Homeschool parents are not verified U.S. K–12 educators by default and should not assume ChatGPT for Teachers or Claude for Teachers FERPA wrappers.

⁵⁹³Session shape: Chapter 3 §3.8 / 04-ai §9.4, scaled to this band. Practice suggestion, not an RCT. Grounded in Bastani's exam phase (n. 575), Soderstrom and Bjork (n. 576), and Oreopoulos and Low's engagement finding (n. 582). "Finish your fractions with Photomath so we can go to soccer" is GPT Base, the pattern with a large-sample causal harm estimate.

Hermes Agent on the parent machine, if the parent already runs it: lesson outlines, isomorphic sheets, parent keys, oral scripts, misconception names. Disable or do not run a messaging gateway against a child's Telegram. Skills and memory live in `~/ .hermes/`; treat that directory as sensitive. For the tool loop — convert a parent-typed topic list into a student sheet plus key — an agentic model, not Hermes 4. For parent explanation, Hermes 4 as chat is fine. Hard no: the child asking Hermes homework questions on the bus.⁵⁹⁴ Grok Bot on the parent account, if the parent already pays SuperGrok or Cursor Pro: overnight generation of tomorrow's sheet plus key plus hint script, dropped in a folder the parent owns. Cloud storage required. Hard no: the child's Photomath login, an IEP PDF, the Grok Bot iOS app in the child's hands.⁵⁹⁵ Many families should not run either agent. A chat model on the parent account, a known-correct key, a line the child draws, and oral facts are a sufficient stack.

If the family only remembers one sentence from this section, Chapter 3 already wrote it: the model may prepare the adult and the next problem; it may not do the child's problem. The grade 3–5 version is narrower. The model may explain to the parent why $1/2 + 1/3$ is not $2/5$. It may not print the LCD path onto the child's page. Facts stay with the parent's ears. The child marks the line.

A student who reaches the ratio year without fraction-as-number work is underprepared for Grade 6. That is Chapter 6's first job. This chapter's job was the gate: multiplicative reasoning connected to area; facts from memory as a map, not a mandate; fractions as numbers on a line; operations that break whole-number bias rather than laminate it; decimals on the same line; volume as layers of unit cubes; the first quadrant as a pair of lines; CRA connected and faded; interleaving after a short block, labelled as transfer; cooking as a second representation, not the line; a parent-hosted model that does not take the test. Siegler and colleagues 2012 remains a prediction. Fuchs 2013 coefficients remain unextracted. The Panel's geometry row remains truncated. Those stay standing. The parent who can hear $2/5$ as a magnitude failure, and $5 \times 1/2$ as a shrink, has walked the gate. Algebra, later, will have something to attach to.

⁵⁹⁴Hermes Agent parent playbook: Chapter 3 §3.8 / 04-ai §9.1. Practice suggestion, not an RCT. Model split this day (Nous Portal): Hermes 4 or any strong chat model for parent explanation and hint scripts; agentic examples (Claude Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro) for the tool loop. Hermes 4 is **not recommended inside Hermes Agent** (tuned for chat, not rapid tool-calling). Hard no: messaging gateway to a child's Telegram/WhatsApp. Not a child-safe school SKU. Memory in `~/ .hermes/` is sensitive.

⁵⁹⁵Grok Bot parent playbook: Chapter 3 §3.8 / 04-ai §9.2. Practice suggestion, not an RCT. Included with SuperGrok and Cursor Pro/Teams as of the 26 August 2026 expansion post. Cloud storage required; Legacy Privacy Mode unsupported; shared computer per user; no FERPA claim. Hard no: child's school portal or Photomath logins; IEP/504 PDFs on the shared computer; Grok Bot iOS app in the child's hands. El Salvador is not this playbook (n. 589).

Chapter 6

Grades 6–8: Ratio, Signed Numbers, Expressions, Early Functions

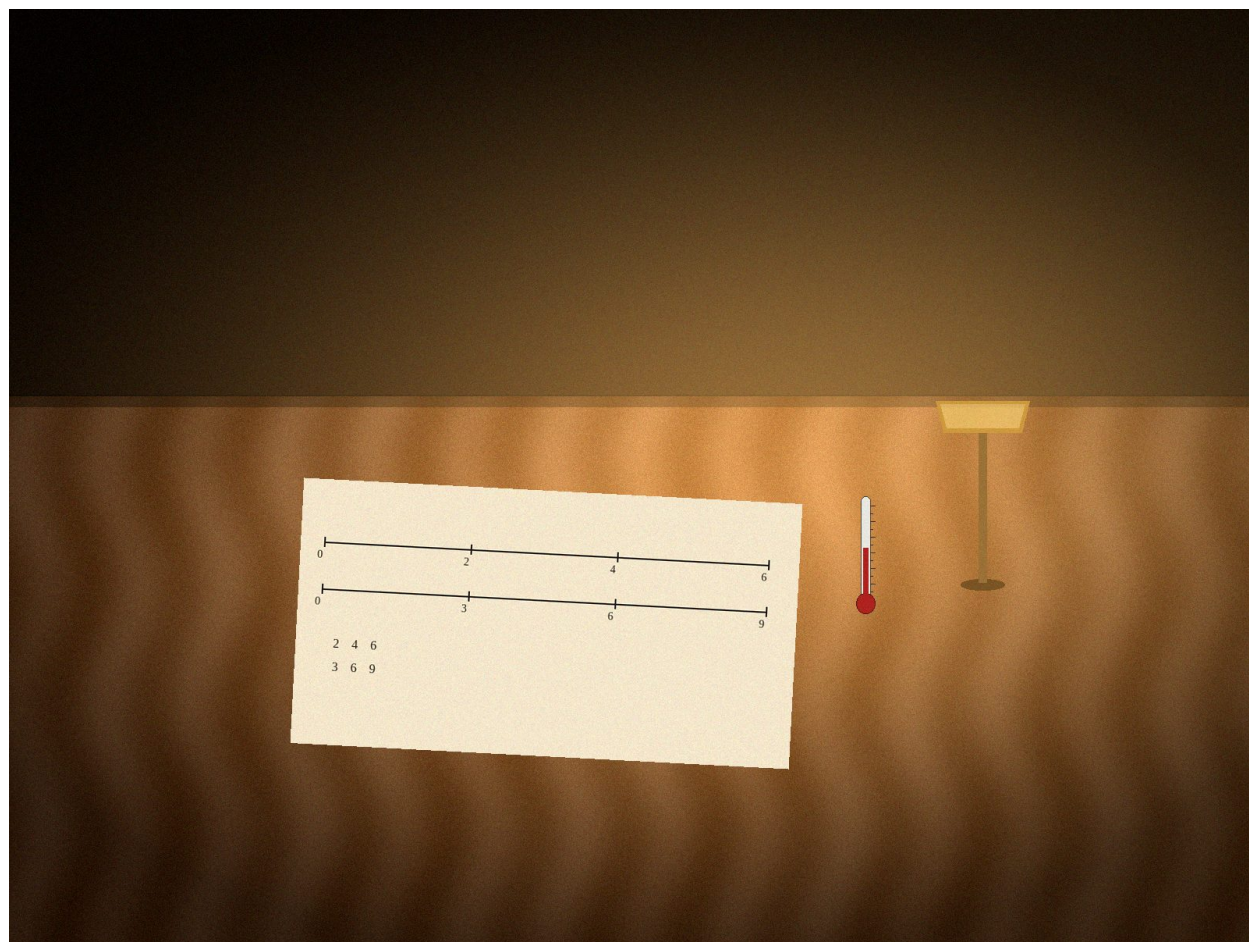


Figure 7: A double number line, a thermometer, and a ratio table drawn on paper.

Two publisher sentences, labelled **vendor**, already contain the argument this chapter has to keep in view. Singapore Math Inc. tells parents that Dimensions Math 6 “includes some Pre-algebra topics,

but is not a Pre-algebra text in the traditional sense; it is a Grade 6 text.”⁵⁹⁶ Beast Academy, on the Art of Problem Solving pages opened this research day, already puts integers and expressions in Level 5A, fractions in 5B, ratios and rates in 5C, and percents, square roots, and exponents in 5D, and “generally recommend[s] that students start about a level ‘behind’ their current grade level” because of problem difficulty.⁵⁹⁷ Those two sentences are not a ranking. They are a warning. A parent buying a “grade 6” book is buying a publisher’s own scope, not a legal grade and not a finished pre-algebra. Dimensions 6 is not pre-algebra. Dimensions 7 and 8 together, the same vendor says, cover pre-algebra and Algebra I topics with some geometry.⁵⁹⁸ Beast Academy Level 5 is not “fifth grade.” Saxon’s homeschool sequence uses split names — Math 5/4, 6/5, 7/6, 8/7 — that do not match calendar grade.⁵⁹⁹ Math-U-See’s Pre-Algebra level is when negatives and solving for the unknown are the focus — a *topic*, not a birthday.⁶⁰⁰ A child of eleven may sit in all four of those rooms on the same afternoon. That is not a contradiction. It is four different grain-sizes of “grade.”

The National Center for Education Statistics already knew the sequence was stretched. In the 2016 Parent and Family Involvement tables, Algebra I was taught *that year* to 41 percent of the home-schooled 6–8 band; arithmetic was still taught to 31 percent of the 9–12 band.⁶⁰¹ Grade labels in this chapter are sequence anchors, not prison bars. A child may sit this band at calendar grade 5 or at calendar grade 9. Finding 15 of the National Mathematics Advisory Panel is still the age-lock refusal: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; claims that children of a given age cannot learn certain content because they are “too young,” “not in the appropriate stage,” or “not ready” have “consistently been shown to be wrong.”⁶⁰² Acceler-

⁵⁹⁶Singapore Math Inc., School FAQ, <https://www.singaporemath.com/pages/school-faq>, accessed 30 August 2026. Dimensions Math 6 “includes some Pre-algebra topics, but is not a Pre-algebra text in the traditional sense; it is a Grade 6 text.” **Vendor.**

⁵⁹⁷Art of Problem Solving, Beast Academy books, <https://beastacademy.com/books>, and Beast Academy FAQ, <https://beastacademy.com/faq/books>, accessed 30 August 2026. Level 5A: 3-D Solids / Integers / Expressions & Equations; 5B: Statistics / Factors & Multiples / Fractions; 5C: Sequences / Ratios & Rates / Decimals; 5D: Percents / Square Roots / Exponents. Vendor: “we generally recommend that students start about a level ‘behind’ their current grade level.” **Vendor.**

⁵⁹⁸Singapore Math Inc., School FAQ (n. 596). Dimensions 7 and 8 together cover Pre-algebra and Algebra 1 topics with some Geometry. **Vendor.**

⁵⁹⁹Houghton Mifflin Harcourt, Saxon Homeschool Math Scope and Sequence, https://www.hmhco.com/~media/sites/home/education/global/pdf/scope-and-sequence/mathematics/k-12/saxon-math-homeschool/shs_ssintroduction.pdf?la=en, accessed 30 August 2026. Sequence: Math K–3, 5/4, 6/5, 7/6, 8/7, Algebra 1, Algebra 2, Advanced Mathematics, Calculus. Split names are skill-level, not calendar grade. **Vendor.**

⁶⁰⁰Math-U-See / Demme Learning, curriculum sequence, <https://mathusee.com/products/math-u-see-curriculum/>, accessed 30 August 2026. Pre-Algebra level: negatives, order of operations, solving for the unknown. **Vendor.** A *topic*, not a birthday.

⁶⁰¹Jiashan Cui and Rachel Hanson, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey*, NCES 2020-001 (Washington, DC: NCES, 2019), Table 9 (subject taught *that year* by grade band). Algebra I taught *that year* to 41 percent of the homeschooled 6–8 band; arithmetic still taught to 31 percent of 9–12. Last opened subject-by-band table; NHES:2023 did not republish it. <https://nces.ed.gov/pubs2020/2020001.pdf>

⁶⁰²National Mathematics Advisory Panel, *Foundations for Success: The Final Report of the National Mathematics Advisory Panel* (U.S. Department of Education, 2008), Finding 15; ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>, accessed 30 August 2026. What is developmentally appropriate is “largely contingent on prior opportunities to learn.” Age-lock claims (“too young,” “not in the appropriate stage,” “not ready”) have “consistently been shown to be wrong.” Brain-immaturity claims are not justified even when

ate a child who has the prerequisites. Do not withhold ratio work from an eleven-year-old who has fractions as numbers. Do not accelerate a child who does not. Skipping signed-fraction operations because an Algebra I book is already on the shelf is the error Finding 30 does not license.⁶⁰³ Chapter 5 was the fractions gate. This chapter is the last chance the Panel named to close the Critical Foundations before “authentic algebra.”⁶⁰⁴ Diagnose first. Then this band. Do not start Algebra I here as if the course were finished.

What the band actually is, as mathematics, is the move from arithmetic to algebra. Ratios and proportional relationships. The rational-number system, including negatives, on the same number line the child has already used for wholes, fractions, and decimals. Expressions and equations, and, in what the Common Core map calls Grade 8, the function concept. Informal bivariate data. Euclidean geometry of congruence, similarity, and the Pythagorean theorem.⁶⁰⁵ That is a lot of mathematics. It is not a lot because a publisher padded a catalog. It is a lot because proportional reasoning is algebra readiness, because signed fractions are still the Panel’s Critical Foundation, and because Grade 8 opens Functions — 8.F is the first Functions domain in K–8 — and calculus’s $f'(x)$ later has nothing to attach to if that concept never forms.⁶⁰⁶ Chapter 7 will be Algebra I. Chapter 10 will be calculus. This chapter plants both. It does not write them.

The Common Core State Standards for Mathematics are treated here as a map, not as a homeschool mandate. Many homeschoolers do not use CCSSM. Many U.S. states have adopted, adapted, or replaced it. A 2026 state-by-state census of who still uses it is **UNVERIFIED** in this project; this chapter will not say “most states.”⁶⁰⁷ What the map is good for is specifying what the mathematics is: a ratio as a relationship, not a pair of counts; a unit rate a/b associated with $a:b$; a constant of proportionality k in $y = kx$, graph through the origin; percent as a rate per 100; division of a fraction by a fraction — the operation the map withheld from Grade 5; integers and signed rationals as numbers with opposites and absolute value, in four quadrants; an expression as a number that depends on x , not as “do it”; solving as answering which values make the equation true; a function as assigning to each input exactly one output; slope as a unit rate, grounded in similar triangles; congruence via rigid motions; similarity via dilations; the Pythagorean theorem; scatter plots and informal linear models. The standards “do not dictate curriculum or teaching methods,” and the order of topics inside a grade is not a required teaching order.⁶⁰⁸ This book will not dump 180 days. It will not sell Singapore, Saxon, Math-U-See, or Beast Academy. Those names will appear

students possess the prerequisite knowledge.

⁶⁰³NMAP, *Foundations for Success* (2008), Finding 30: mathematically gifted students with motivation “appear to be able to learn mathematics much faster... with no harm to their learning, and should be allowed to do so.” Not a license to skip the Critical Foundations because a child is “good at math.”

⁶⁰⁴NMAP, *Foundations for Success* (2008), Chapter 4 and the Critical Foundations of Algebra: fluency with whole numbers; fluency with fractions including decimals, percents, and negative fractions; particular aspects of geometry and measurement. This band is the last chance the Panel named to close those foundations before authentic algebra.

⁶⁰⁵Common Core State Standards Initiative, *Common Core State Standards for Mathematics*, https://www.thecorestandards.org/assets/CCSSI_Math%20Standards.pdf, accessed 30 August 2026, Grades 6–8 overviews. Map, not a homeschool mandate. Grade 6–8 critical areas and domains as listed in the body.

⁶⁰⁶CCSSM 8.F: first Functions domain in K–8. A function assigns to each input exactly one output. Plant for Chapter 10: without this concept, calculus’s $f'(x)$ has nothing to attach to. 01-scope load-bearing 13.

⁶⁰⁷2026 state-by-state census of who still uses CCSSM, who uses “standards formerly known as,” and who never did: **UNVERIFIED** (01-scope open question 8). Do not say “most states.”

⁶⁰⁸CCSSM, “How to read the grade level standards”: the standards “do not dictate curriculum or teaching methods”; order of topics inside a grade is not a required teaching order. Same PDF as n. 605.

as options, by grain and placement, labelled **vendor** where the claim is the publisher's.

Four lanes run through the sections. They are not four identical headings. Sections 6.1 through 6.5 explain the mathematics to the parent — what the band is and why the publisher label lies; ratio, rate, percent, and proportion as a relationship; signed numbers on the same line, with rules as compressions of moves; expressions, equations, and the function concept; congruence, similarity, the Pythagorean theorem, and informal bivariate data, planted for Chapter 11. Section 6.6 is teaching guidance and the tips that survive: this is the band where Doug Rohrer, Robert Dedrick, and Kaleena Burgess's 2014 grade-7 classroom crossover lives — interleaved 72 percent versus blocked 38 percent, $d = 1.05$, on an unannounced test two weeks later — and where Asha Jitendra, Jon Star, Danielle Dupuis, and Michael Rodriguez's 2012 seventh-grade proportion trial lives, $g = 1.24$ immediate, $g = 1.27$ at six weeks, null transfer.⁶⁰⁹ Those two numbers are not the same object. Do not mash them. Section 6.7 is the parent-hosted AI lane for this band: a short reminder of Chapter 3's ban versus allow, then the grade 6–8 jobs. Parent-first ratio tables and signed-number number-line diagnosis. Isomorphic practice, parent-held key. Ban solvers for the child's homework. Facts and integer rules are not a language-model job to memorize by chat. The child does not live in an open-web agent. The parent holds the key.

Chapter 2 already locked the method this chapter will use: five interwoven strands; conceptual and procedural knowledge iterating, not a one-way street; the number line as the representation that scales from counting to irrationals; worked examples faded for novices, incorrect examples explained; blocked practice to learn execution, interleaved and spaced practice to choose a strategy and keep it — and the clean classroom randomized trials of interleaving are *here*, in grade 7, not in grade 4 and not in calculus; word problems as schemas, not keyword hunts; fluency as flexible, accurate, efficient retrieval; explicit instruction for students already struggling, productive failure as generation-then-instruction on a new concept when priors exist — those two literatures not the same comparison; kitchen-math transfer hard. Chapter 3 already locked the AI lane: unguarded solvers raise practice scores and cut unaided exams; the model may prepare the adult and the next problem; it may not do the child's problem. This chapter will not re-litigate Bastani at monograph length. One short reminder, then the objects of this band. Chapter 5 already did the fractions gate. This chapter will not dump it. A child who cannot place $3/4$ and $-3/4$ on the same line, who still adds numerators and denominators, who treats 2:3 as “two and three,” is not ready for Algebra I regardless of calendar grade. Diagnose that. Then teach this band.

The popular story collapses those locks into slogans they do not support. Cross-multiply and you are done. A negative times a negative is a positive, because the teacher said so. Functions are $y = mx + b$. A “grade 6” book is sixth grade, and Dimensions 6 is pre-algebra. Algebra I in calendar grade 8 is the same object as the Common Core's Grade 8 expressions, equations, and functions. Singapore or England is “two years ahead.” They'll learn percent by shopping. The AI will tutor

⁶⁰⁹Doug Rohrer, Robert F. Dedrick, and Kaleena Burgess, “The Benefit of Interleaved Mathematics Practice Is Not Limited to Superficially Similar Kinds of Problems,” *Psychonomic Bulletin and Review* 21 (2014): 1323–1330, <https://files.eric.ed.gov/fulltext/ED548041.pdf>, accessed 30 August 2026. Grade-7 classroom crossover, $n = 140$, unannounced test two weeks later, interleaved 72% vs blocked 38%, $d = 1.05$, 95% CI [0.80, 1.30]. Asha K. Jitendra, Jon R. Star, Danielle Dupuis, and Michael C. Rodriguez, “Effectiveness of Schema-Based Instruction for Improving Seventh-Grade Students' Proportional Reasoning: A Randomized Experiment,” *Journal of Research on Educational Effectiveness* 5 (2012), <https://doi.org/10.1080/19345747.2012.725804>. Immediate $g = 1.24$; six-week retention $g = 1.27$; no significant transfer; 1,163 students, 42 classrooms. Do not mash the d with the g . Full objects in nn. 652, 697.

the pre-algebra worksheet. The documents do not say any of those things, not as the slogans say them. This chapter prints the documents. It prints the objects, the sample sizes, the delays, and the nulls. It leaves the open questions standing — slope/intercept mixup prevalence, not a named study opened; GAISE II page-level claims, not opened; interleaving without the spacing confound; compacted 7–8 versus delay as a live pathway debate, not a kitchen randomized trial; Appendix A’s integrated Mathematics I–III topic-by-topic, only partly extracted. It does not invent a kitchen scene to make the science feel warmer. The parent who stopped at algebra is the design user. The job is to hear 2:3 treated as a pair of counts, and “a negative times a negative is a positive” recited with no move on a line, before any tool is allowed to print a solved proportion the parent never sees.

A compacted path through Grades 7–8 exists so that calculus is reachable in senior year. CCSSM Appendix A, published by Achieve in partnership with the Common Core writing team, is explicit: compacted versions of the traditional and integrated pathways complete Grade 7, Grade 8, and the first high-school course in Grades 7–8 so that a student can reach Calculus or another college-level course by senior year. “The pathways and courses are models, not mandates.”⁶¹⁰ The Panel recommended that all *prepared* students have access to an authentic algebra course by Grade 8; “authentic” means the Major Topics of School Algebra, not a course that is algebra in name only.⁶¹¹ A non-accelerated student who starts Algebra I in Grade 9 finishes Algebra II in Grade 11 and has senior year for precalculus, statistics, or a first calculus course — not both precalculus and BC unless the fourth course is overloaded.⁶¹² That sentence is a pathway fact. It is not a mandate to compact. Compact only if the gates are open: fractions including signed fractions, integer operations, the equal sign as a relation. Siegler, Duncan, Davis-Kean, Duckworth, Claessens, Engel, Susperreguy, and Chen 2012 is still the prediction that makes the gate load-bearing: elementary fraction and division knowledge uniquely predicted high-school algebra five or six years later, after the controls a skeptic would demand.⁶¹³ Do not skip the gate because calculus is the destination. Chapter 10 will say that again. This chapter is where the destination becomes possible, or not.

⁶¹⁰Achieve / CCSS writing team, *Appendix A: Designing High School Mathematics Courses Based on the Common Core State Standards*, Oregon ODE copy, <https://www.oregon.gov/ode/educator-resources/standards/mathematics/Documents/math-appendix-a-model-course-pathways.pdf>, accessed 30 August 2026. Compacted traditional and integrated pathways complete Grade 7 + Grade 8 + the first high-school course in Grades 7–8 so a student can reach Calculus or another college-level course by senior year. “The pathways and courses are models, not mandates.”

⁶¹¹NMAP, *Foundations for Success* (2008): all *prepared* students should have access to an authentic algebra course by Grade 8. “Authentic” means the Major Topics of School Algebra (Table 1), not a course that is algebra in name only.

⁶¹²Appendix A (n. 610): the regular three-year pathway plus a fourth course reaches precalculus as a junior and calculus as a senior *only if* high-school content started in Grade 8. A non-accelerated student who starts Algebra I in Grade 9 has senior year for precalculus, statistics, or a first calculus course — not both precalculus and BC unless the fourth course is overloaded. Plant for Chapter 10. Not a mandate to compact.

⁶¹³Robert S. Siegler, Greg J. Duncan, Pamela E. Davis-Kean, Kathryn Duckworth, Amy Claessens, Mimi Engel, Maria Ines Susperreguy, and Meichu Chen, “Early Predictors of High School Mathematics Achievement,” *Psychological Science* 23, no. 7 (2012): 691–697, <https://doi.org/10.1177/0956797612440101>; author PDF <https://sieglertc.columbia.edu/wp-content/uploads/2019/02/Siegler-et-al-PsySci12.pdf>, accessed 30 August 2026. British Cohort Study $N = 3,677$; U.S. PSID-CDS $N = 599$. A 1-SD increase in age-10 fractions knowledge uniquely associated with 0.15-SD (UK) / 0.17-SD (US) increase in high-school algebra after other arithmetic, IQ or working memory, and family background. Prediction, not a curriculum RCT. Full discussion in Chapter 5; this chapter uses the prediction as the reason not to skip the gate in order to compact.

6.1 What this band is, and why a “grade 6” book is a publisher’s scope

The mathematics of this band is the move from arithmetic to algebra, named more specifically than a slogan about “middle school.” The Common Core’s Grade 6 critical areas, from the official PDF, are four: connecting ratio and rate to whole-number multiplication and division and using ratio and rate to solve problems; completing understanding of division of fractions and extending number to the system of rational numbers, which includes negative numbers; writing, interpreting, and using expressions and equations; developing understanding of statistical thinking. Students also extend area, surface area, and volume — right rectangular prisms with fractional edge lengths; polygons in the coordinate plane.⁶¹⁴ That is the first year of the band on the map. It is not pre-algebra as a high-school course. It is the year ratio language arrives, fraction-by-fraction division arrives, and negatives arrive as numbers.

Grade 6 domains and clusters, as the map writes them, are the objects later sections will teach.

Ratios and Proportional Relationships, 6.RP: ratio language; unit rate a/b associated with $a:b$; ratio and rate reasoning with tables, tape diagrams, double number lines, and equations; unit pricing, constant speed; percent as a rate per 100; convert measurement units.⁶¹⁵

The Number System, 6.NS: divide fractions by fractions — the operation CCSSM withheld from Grade 5. Fluently divide multi-digit numbers using the standard algorithm; fluently add, subtract, multiply, and divide multi-digit decimals using the standard algorithm for each. Greatest common factor and least common multiple; the distributive property with common factors. Rational numbers: positive and negative integers and other rationals; opposites; absolute value; the four quadrants; order and compare; reflections across axes.⁶¹⁶

Expressions and Equations, 6.EE: write and evaluate numerical expressions with whole-number exponents; write, read, and evaluate expressions with variables; identify parts of an expression; evaluate at specific values; generate equivalent expressions; identify when two expressions are equivalent. Reason about and solve one-variable equations and inequalities; understand solving as a process of answering “which values make the equation true.” Represent and analyze quantitative relationships between dependent and independent variables — tables, graphs, equations such as $y = 3x$.⁶¹⁷

Geometry, 6.G: area of triangles, special quadrilaterals, and polygons by composing and decom-

⁶¹⁴CCSSM Grade 6 overview / critical areas (four), official PDF (n. 605). Connecting ratio and rate; completing division of fractions and extending to rationals including negatives; expressions and equations; statistical thinking. Also area, surface area, volume (right rectangular prisms with fractional edge lengths; polygons in the coordinate plane).

⁶¹⁵CCSSM 6.RP: ratio language; unit rate a/b associated with $a:b$; tables, tape diagrams, double number lines, equations; unit pricing, constant speed; percent as rate per 100; convert measurement units.

⁶¹⁶CCSSM 6.NS: fraction $\div$ fraction (withheld from Grade 5); fluent multi-digit division by the standard algorithm; fluent four operations on multi-digit decimals by the standard algorithm for each; GCF/LCM; distributive property with common factors; rational numbers including negatives; opposites; absolute value; four quadrants; order and compare; reflections across axes.

⁶¹⁷CCSSM 6.EE: numerical expressions with whole-number exponents; expressions with variables; parts of an expression; evaluate at specific values; equivalent expressions; one-variable equations and inequalities; solving as answering “which values make the equation true”; dependent and independent variables (tables, graphs, equations such as $y = 3x$).

posing; volume of right rectangular prisms with fractional edge lengths; draw polygons in the coordinate plane; surface area via nets.⁶¹⁸

Statistics and Probability, 6.SP: statistical variability; statistical questions. Distributions: center (median, mean), spread (interquartile range, mean absolute deviation), shape (clusters, peaks, gaps, symmetry). Summarize numerical data sets in context.⁶¹⁹ That last domain is not a worksheet of mean, median, and mode with no data story. It is the on-ramp to Chapter 11. A skipped K–8 data strand is AP Statistics Unit 1 retrofit. Plant that. Do not dump Chapter 11 here.

Grade 7's critical areas are four: proportional relationships; operations with rational numbers and working with expressions and linear equations; scale drawings, informal geometric constructions, area, surface area, and volume; drawing inferences about populations based on samples.⁶²⁰ 7.RP: analyze proportional relationships; unit rate including ratios of fractions; constant of proportionality k in $y = kx$; graphs of proportional relationships through the origin; percent problems — tax, tip, markup, markdown, simple interest, percent increase and decrease; scale drawings. 7.NS: add, subtract, multiply, and divide rational numbers, including negatives; interpret sums in contexts such as temperature and debt; $p + q$ as distance $|q|$ from p in the positive or negative direction; conversion of a rational number to a decimal, terminating or repeating. 7.EE: generate equivalent expressions — expand, factor; solve multi-step real-life problems with positive and negative rationals; solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$; inequalities of the same forms. 7.G: draw, construct, describe geometrical figures; scale drawings; two-dimensional figures from slicing three-dimensional figures; angle relationships from intersecting lines; area and circumference of a circle; area, volume, and surface area of objects composed of triangles, quadrilaterals, polygons, cubes, and right prisms. 7.SP: random sampling to draw inferences about a population; informal comparative inferences about two populations; chance processes and probability models — sample spaces, compound events, simulations.⁶²¹

Grade 8's critical areas are three: expressions and equations, including modeling an association in bivariate data with a linear equation, and solving linear equations and systems; the concept of a function; two- and three-dimensional space using distance, angle, similarity, congruence, and the Pythagorean theorem.⁶²² 8.NS: know that there are numbers that are not rational; approximate irrationals by rationals — $\sqrt{2}$ is the usual example. 8.EE: integer exponents and scientific notation; square and cube roots; connections among proportional relationships, lines, and linear equations; slope as unit rate; similar triangles to explain why the slope is the same between any two distinct

⁶¹⁸CCSSM 6.G: area of triangles, special quadrilaterals, polygons by composing/decomposing; volume of right rectangular prisms with fractional edge lengths; polygons in the coordinate plane; surface area via nets.

⁶¹⁹CCSSM 6.SP: statistical variability; statistical questions; distributions — center (median, mean), spread (IQR, MAD), shape (clusters, peaks, gaps, symmetry); summarize numerical data sets in context. Plant for Chapter 11. Not a worksheet of mean/median/mode with no data story.

⁶²⁰Common Core State Standards Initiative, Grade 7 Introduction, <https://www.corestandards.org/Math/Content/7/introduction/>, accessed 30 August 2026. Four critical areas: proportional relationships; operations with rationals and linear equations; scale drawings / informal constructions / area-volume; inferences from samples.

⁶²¹CCSSM Grade 7 domains 7.RP, 7.NS, 7.EE, 7.G, 7.SP as clustered in the body. Official PDF (n. 605) and Grade 7 introduction (n. 620).

⁶²²Common Core State Standards Initiative, Grade 8 Introduction, <https://www.corestandards.org/Math/Content/8/introduction/>, accessed 30 August 2026. Three critical areas: expressions and equations including linear models of bivariate data and systems; the concept of a function; two- and three-dimensional space using distance, angle, similarity, congruence, and the Pythagorean theorem.

points on a non-vertical line; $y = mx + b$. Analyze and solve linear equations in one variable, including those with infinitely many solutions or no solution, and pairs of simultaneous linear equations — graphically, algebraically, and by inspection: $x + y = 5$ and $x - y = 1$. 8.F, the first Functions domain in K–8: a function assigns to each input exactly one output; compare functions represented in different ways; linear versus nonlinear; interpret $y = mx + b$ as defining a linear function, m as rate of change, b as initial value; sketch a graph from a qualitative description. 8.G: congruence via rotations, reflections, translations; similarity via dilations; informal arguments for triangle angle-sum and exterior-angle theorems, and for angle relationships with parallel lines cut by a transversal; Pythagorean theorem and its converse; distance between points in the coordinate plane; volume of cylinders, cones, and spheres. 8.SP: bivariate data — scatter plots; informal linear models; clustering, outliers, positive and negative association; intercept, slope, and frequency tables of bivariate categorical data.⁶²³

That catalog is a map of typical U.S. public-school placement, not a claim that a homeschooler must wait, and not a 180-day plan. It is not Algebra I.⁶²⁴ A full Algebra I course — linear, exponential, and quadratic functions — is a *high-school* course on the traditional pathway unless the compacted pathway started in Grade 7. “Grade 8 algebra” in U.S. speech often means either (a) CCSSM Grade 8 EE/F or (b) a full Algebra I taken in Grade 8. Those are not the same course.⁶²⁵ Do not treat a calendar-grade-8 Algebra I as identical to the map’s Grade 8 expressions, equations, and functions. Chapter 7 is the course. This chapter is the door diagnostic.

NCTM’s 2006 *Curriculum Focal Points* named three “most important mathematical topics” per grade as a coherence device. The Grade 6–8 rows, from a publicly posted excerpt of the NCTM book — the full official PDF is paywalled, and this project treats that posting as a **secondary copy of a primary** — put fluency with multiplication and division of fractions and decimals, connecting ratio and rate to multiplication and division, and writing, interpreting, and using mathematical expressions and equations in Grade 6; proportionality including similarity, formulas for surface area and volume, and operations on all rational numbers and solving linear equations in Grade 7; linear functions and systems of linear equations, analyzing two- and three-dimensional space using distance and angle including the Pythagorean theorem, and analyzing and summarizing data sets in Grade 8.⁶²⁶ NCTM notes that programs offering algebra in Grade 8 should still not omit Grade 7–8

⁶²³CCSSM Grade 8 domains 8.NS, 8.EE, 8.F, 8.G, 8.SP as clustered in the body. 8.F is the first Functions domain in K–8.

⁶²⁴CCSSM Grades 6–8 catalog (nn. 614–623) is a map of typical U.S. public-school placement, not a 180-day plan and not a full Algebra I. Linear, exponential, and quadratic functions are the high-school Algebra I object unless the compacted pathway started in Grade 7 (n. 610).

⁶²⁵01-scope international note and high-school Algebra I band: “Grade 8 algebra” is ambiguous across systems and even inside U.S. speech. CCSSM Grade 8 EE/F is not a full Algebra I (linear, exponential, and quadratic functions). Appendix A compacted Algebra I in Grade 8 is the other object. Do not treat them as identical.

⁶²⁶National Council of Teachers of Mathematics, *Curriculum Focal Points for Prekindergarten through Grade 8 Mathematics* (2006). Grade lists used here from a publicly posted excerpt, https://www.platonicsolids.info/focal_points_by_grade.pdf, accessed 30 August 2026 — **secondary copy of a primary**; full NCTM book paywalled. Grade 6: fluency with $\times \div$ of fractions and decimals; connecting ratio and rate to $\times \div$; writing, interpreting, and using expressions and equations. Grade 7: proportionality including similarity; surface area/volume formulas; operations on all rationals and solving linear equations. Grade 8: linear functions and systems; two- and three-dimensional space including the Pythagorean theorem; analyzing and summarizing data sets. Prefer a library copy before treating the excerpt as the official book.

focal-point content; it can be moved into Grade 6–7 or into high school, but not dropped.⁶²⁷ Prefer a library copy before treating the excerpt as the official book. The Focal Points are a professional consensus on focus, not “the current U.S. standard,” and they predate CCSSM. The warning is still the one a kitchen that “starts Algebra I early” has to hear: similarity, signed rationals, and the Pythagorean theorem are not optional extras the Algebra I book will magically supply.

The Panel’s Table 2 benchmarks give a pace, not a law, and they are judgments from curriculum comparison, not experimental placements — the Panel says so.⁶²⁸ By the end of Grade 6: proficient multiplication and division of fractions and decimals; proficient operations with positive and negative integers. By the end of Grade 7: proficient operations with positive and negative fractions; problems involving percent, ratio, rate, and proportionality.⁶²⁹ The geometry and measurement remainder of Table 2 was truncated in the fetch used for this project. That remainder stays **UNVERIFIED**. This chapter will not invent those benchmark lines.⁶³⁰ Signed fractions are inside the Critical Foundation the Panel named as the one U.S. students most often lack: “knowledge of fractions is the most important foundational skill not currently developed among American students,” and the Foundation includes decimals, percents, and *negative* fractions.⁶³¹ Chapter 5 taught unsigned fractions as numbers. This chapter has to put the negatives on the same line and close operations on signed fractions before Algebra I. A nationally representative survey of Algebra I teachers the Panel commissioned in 2007, 743 teachers, rated incoming preparation as poorest in “rational numbers and operations involving fractions and decimals,” alongside word problems and study habits.⁶³² That survey is not “teachers today say.” It is the 2007 document the Panel used. Still the best nationally representative teacher-voice primary on *incoming* Algebra I gaps this project opened. Not a 2026 replication. Do not start Algebra I while those operations are still fragile.

What this band depends on is Chapter 5’s work: fractions as numbers on the line, unlike-denominator addition, multiplication of fractions, unit-fraction division, multiplication fluency, place value through decimals. What it unlocks is Algebra I — linear and exponential functions, quadratic equations — and Geometry — rigid motions, similarity, proof. Grade 8 functions are the conceptual prerequisite for F-IF in high school. Slope-from-similar-triangles is the geometric

⁶²⁷NCTM Focal Points (n. 626): programs offering algebra in Grade 8 should still not omit Grade 7–8 focal-point content; it can be moved into Grade 6–7 or into high school, but not dropped.

⁶²⁸NMAP, *Foundations for Success* (2008), Table 2. Benchmarks are judgments from curriculum comparison, not experimental placements — the Panel says so.

⁶²⁹NMAP Table 2: by end of Grade 6, proficient $\times \div$ of fractions and decimals and operations with positive and negative integers; by end of Grade 7, proficient operations with positive and negative fractions and problems involving percent, ratio, rate, and proportionality.

⁶³⁰NMAP Table 2 geometry/measurement remainder: truncated in the ERIC fetch used for this project (01-scope open question 3). **UNVERIFIED** until pp. 19–20 of ED500486 are re-opened. Do not invent those benchmark lines.

⁶³¹NMAP, *Foundations for Success* (2008), Executive Summary findings 4 and 12, and Chapter 4: “knowledge of fractions is the most important foundational skill not currently developed among American students.” The Critical Foundation includes decimals, percents, and negative fractions.

⁶³²NMAP, *Foundations for Success* (2008), nationally representative survey of Algebra I teachers commissioned by the Panel, 743 teachers, 2007. Poorest incoming preparation in “rational numbers and operations involving fractions and decimals,” alongside word problems and study habits. 2008 data; not a 2026 replication. Do not write “teachers today say.” Still the best nationally representative teacher-voice primary on *incoming* Algebra I gaps this project opened.

prerequisite the Panel names for linear functions.⁶³³ Skip the gate and those later objects arrive as symbol soup. Skip signed fractions because the Algebra I book is on the shelf and the soup is the same soup the 743 teachers described.

A vendor book labelled “grade 6” is not a legal grade. Restate the grain, because this is the chapter where the parent is most likely to buy “pre-algebra” by accident.

Singapore Math Inc., **vendor**: Dimensions Math 6 “includes some Pre-algebra topics, but is not a Pre-algebra text in the traditional sense; it is a Grade 6 text.” Dimensions 7 assumes students are new to integers and simple equations but goes further than a traditional pre-algebra. Dimensions 7 and 8 together cover pre-algebra and Algebra 1 with some geometry. Dimensions 8 expects more prior linear-equation work than a traditional Algebra 1.⁶³⁴ A U.S. parent using Dimensions 7–8 is buying something closer to Singapore’s earlier algebra, not to CCSSM Grade 7. Primary Mathematics editions are not interchangeable with Dimensions; the publisher says so. Placement tests are recommended — **vendor**.

Saxon Homeschool, **vendor** / Houghton Mifflin Harcourt: the sequence uses split names, Math 5/4, 6/5, 7/6, 8/7, then Algebra 1, Algebra 2, Advanced Mathematics, Calculus. Alternate: 8/7 or Algebra 1/2 as pre-algebra. Geometry is integrated in Algebra 1, Algebra 2, and Advanced Math in classic editions; a separate Geometry course also exists. Compacted students may reach Algebra 1 in Grade 8. HMH placement tests set the starting book by score and treat Saxon as skill-level rather than grade-level.⁶³⁵ Do not transcript “Saxon 8/7” without a parenthetical a stranger can map onto pre-algebra or Algebra I. Chapter 12 will say that as an export rule. This chapter says it as a placement rule.

Math-U-See, **vendor** / Demme Learning: Greek letters uncoupled from calendar grade. Epsilon is a fractions *level*. Zeta is decimals and percents. Pre-Algebra is when negatives, order of operations, and solving for the unknown are the focus.⁶³⁶ If a family ran one level per calendar year from Primer, negatives can sit later than the CCSSM Grade 6 window — or earlier, if Alpha started in kindergarten. The parent is buying a *topic*, not a grade.

Beast Academy / Art of Problem Solving, **vendor**: Levels 1–5, ages about 6–13, then AoPS Pre-algebra. Level 5A: 3-D solids, integers, expressions and equations. 5B: statistics, factors and multiples, fractions. 5C: sequences, ratios and rates, decimals. 5D: percents, square roots, exponents. Vendor: a student who completes Level 5 has the background for AoPS Prealgebra. Vendor: “we

⁶³³NMAP, *Foundations for Success* (2008), Critical Foundations, geometry and measurement: similar triangles, because slope and linear functions “depend logically” on them. CCSSM 8.EE: similar triangles to explain why slope is the same between any two distinct points on a non-vertical line.

⁶³⁴Singapore Math Inc., School FAQ (n. 596). Dimensions 7 assumes students are new to integers and simple equations but goes further than a traditional pre-algebra; Dimensions 7+8 together = pre-algebra + Algebra 1 + some geometry; Dimensions 8 expects more prior linear-equation work than a traditional Algebra 1. **Vendor**. Editions (Primary Mathematics US / Standards / 2022 / SIE vs Dimensions) are not interchangeable. Placement tests recommended — **vendor**.

⁶³⁵HMH Saxon Homeschool scope (n. 599). Alternate: 8/7 or Algebra 1/2 as pre-algebra. Geometry integrated in Alg 1, Alg 2, and Advanced Math in classic editions; a separate Geometry course also exists. Compacted students may reach Algebra 1 in Grade 8. HMH placement tests set the starting book by score; Saxon treated as skill-level rather than grade-level. **Vendor**. Transcript as a course a stranger can map (Chapter 12).

⁶³⁶Math-U-See / Demme Learning (n. 600). Epsilon = Fractions *level*; Zeta = decimals and percents; Pre-Algebra = negatives, order of operations, solving for the unknown. **Vendor**. Calendar placement depends on when Alpha started.

generally recommend that students start about a level ‘behind’ their current grade level.” Levels 4–5 cover many middle-grades standards.⁶³⁷ Beast Academy is commonly described in secondary comparisons as sitting above same-age peers; that relative-difficulty claim is **vendor/secondary**, not a standards finding. Do not upgrade it.

Teaching Textbooks, RightStart, Math Mammoth, and Life of Fred topic placements that were not opened on a vendor primary remain **UNVERIFIED**. Do not promote those rows to handbook fact.⁶³⁸ No homeschool-curriculum randomized trial comparing any of these programs was opened for this project. Choose by fit — parent math confidence, student stance, special needs, budget, college target — not by a phantom homeschool meta-analysis. Cathy Duffy Reviews and Rainbow Resource are **catalogue / reviewer**, not independent effectiveness trials.

An international note, short, so a parent using a non-U.S. sequence can translate “grade 8 algebra” without inventing a two-year gap. In Singapore, TIMSS 2015 Encyclopedia and the Ministry of Education G2/G3 syllabuses: by the end of Grade 8 (Secondary 2), students are to have algebraic expressions and formulas, algebraic manipulation of linear and quadratic, and functions and graphs, linear and quadratic.⁶³⁹ G3 is the most demanding core route. **Translation:** a Singapore Secondary 2 G3 student is typically into *quadratic functions*, which U.S. CCSSM places in Algebra I / Grade 9, or Grade 8 compacted Algebra I. A U.S. parent using Dimensions 7–8 is buying something closer to this earlier algebra, not to CCSSM Grade 7. In England, Key Stage 3 is Years 7–9 (U.S. Grades 6–8). KS3 algebra already includes algebraic notation, collecting like terms, expanding binomials, linear equations in one variable, coordinates in four quadrants, graphs of linear *and quadratic* functions, $y = mx + c$, approximate solutions of simultaneous linear equations from graphs, arithmetic and geometric sequences. KS3 geometry includes Pythagoras and trigonometric ratios in similar right triangles.⁶⁴⁰ **Translation:** “Year 8 algebra” in England already expects graphs of simple quadratics, which U.S. CCSSM saves for Algebra I. Do not say that therefore Singapore or England is “two years ahead” as a single number. Different systems package quadratics, proof, and

⁶³⁷Beast Academy books and FAQ (n. 597). Vendor: a student who completes Level 5 has the background for AoPS Prealgebra; levels 4–5 cover many middle-grades standards. Relative-difficulty claim that Beast Academy sits above same-age peers is **vendor/secondary**, not a standards finding. Do not upgrade it.

⁶³⁸Teaching Textbooks, RightStart, Math Mammoth Light Blue, and Life of Fred official scopes: not opened on vendor primaries this project (01-scope open question 5). **UNVERIFIED**. Do not promote those rows to handbook fact. Cathy Duffy Reviews and Rainbow Resource are **catalogue / reviewer**, not independent effectiveness trials. No homeschool-curriculum RCT comparing named programs was opened.

⁶³⁹Ministry of Education, Singapore, *G2 and G3 Mathematics Syllabuses* (implementation from 2020 Secondary One), <https://www.moe.gov.sg/-/media/files/secondary/fsbb/syllabus/2020-g2-and-g3-mathematics-syllabuses.pdf>, accessed 30 August 2026; IEA TIMSS 2015 Encyclopedia, Singapore mathematics curriculum, <https://timss2015.org/encyclopedia/countries/singapore/the-mathematics-curriculum-in-primary-and-lower-secondary-grades/>, accessed 30 August 2026. By the end of Grade 8 (Secondary 2): algebraic expressions and formulas, algebraic manipulation (linear and quadratic), functions and graphs (linear and quadratic). G3 is the most demanding core route. **Translation:** Secondary 2 G3 is typically into quadratic functions, which U.S. CCSSM places in Algebra I / Grade 9 (or Grade 8 compacted Algebra I).

⁶⁴⁰UK Department for Education, National curriculum in England: mathematics programmes of study, <https://www.gov.uk/government/publications/national-curriculum-in-england-mathematics-programmes-of-study/national-curriculum-in-england-mathematics-programmes-of-study>, accessed 30 August 2026. Key Stage 3 (Years 7–9 / U.S. Grades 6–8) algebra includes algebraic notation, collecting like terms, expanding binomials, linear equations in one variable, four quadrants, graphs of linear *and quadratic* functions, $y = mx + c$, approximate solutions of simultaneous linear equations from graphs, arithmetic and geometric sequences. KS3 geometry includes Pythagoras and trigonometric ratios in similar right triangles.

statistics differently. Say instead: if your non-U.S. book is doing quadratic graphs in what it calls Grade 8, you are not behind a U.S. Algebra I student — you are on a different train, often earlier on the algebra car.⁶⁴¹

The live pathway debate this section will not pretend to settle belongs on the page so it is not settled by a franchise. Compacted 7–8, so that Algebra I is finished in Grade 8 and calculus is reachable senior year, versus delay until fraction and integer fluency is real. Appendix A and the Panel’s Grade 8 authentic-algebra recommendation are the compacting documents. Siegler and colleagues 2012 and the Panel’s 2007 teacher survey are the delay documents. Diagnose. Do not calendar. NMAP Finding 15 still refuses the age lock; Finding 30 still allows prepared, motivated students to move faster, with no harm to their learning.⁶⁴² Neither finding licenses a skip of signed fractions because a child is “good at math.” Failure points visible in this band, named in the scope memo and inherited here, are four, and they are not cute. Proportional reasoning treated as a cross-multiply trick rather than a constant of proportionality visible in a table and a graph through the origin. Negatives as “rules” without a number line or a context. Functions introduced as “ $y = mx + b$ ” only, so the general “each input one output” idea never forms. Algebra I started while fraction and integer operations are still fragile — the teacher survey’s top preparation gap.⁶⁴³ Those four are the diagnosis list a parent can steal before buying the next book. They are also why this chapter is this long. Algebra I will be symbol soup if this chapter thins.

What this section has done is place the window and kill the calendar grade. Ratio comes next, because it is the proportional-reasoning gate, and because a ratio treated as a pair of counts is the misconception this band is built to break.

6.2 Ratio, rate, percent, and proportion — a relationship, not a pair of counts

A ratio is a relationship. That sentence is the first teaching move of this band, and it is a sentence the parent who stopped at algebra has to be able to hear wrong. 2:3 is not “two and three.” It is not a pair of counts sitting side by side, the way $2 + 3$ is two and three combined. It is a relationship between two quantities that stays the same when both are scaled: 2:3, 4:6, 6:9, 20:30. The child who treats 2:3 as “two and three,” so that $2:3 = 4:6$ is a mystery, has the misconception Jitendra, Star, and colleagues built a seventh-grade randomized trial to hit.⁶⁴⁴ The IES fractions practice guide, Recommendation 4, had already named the instructional order: develop understanding of

⁶⁴¹01-scope international note: do not say Singapore or England is “two years ahead” as a single number. Different systems package quadratics, proof, and statistics differently. If a non-U.S. book is doing quadratic graphs in what it calls Grade 8, the student is on a different train, often earlier on the algebra car — not “behind” a U.S. Algebra I student.

⁶⁴²NMAP Findings 15 and 30 (nn. 602–603). Diagnose; do not calendar. Compacted 7–8 versus delay until fraction/integer fluency is real is a live pathway debate, not a kitchen RCT.

⁶⁴³Failure points from 01-scope Grades 6–8: (a) proportional reasoning as a cross-multiply trick rather than a constant of proportionality visible in a table and a graph through the origin; (b) negatives as “rules” without number-line or context; (c) functions introduced as “ $y = mx + b$ ” only; (d) Algebra I started while fraction/integer operations are still fragile — the teacher survey’s top preparation gap (n. 632).

⁶⁴⁴Jitendra, Star, Dupuis, and Rodriguez 2012 (n. 609). Ratio as a pair of counts, not a relationship, is the hard middle-school problem the schema-based proportion trial exists to hit. Jitendra, Star et al. 2009 and 2011 are earlier papers in the same program; 2012 is the seventh-grade RCT this chapter quotes.

ratios and rates *before* formal proportions.⁶⁴⁵ Cross-multiply is a consequence, not a belief. The child who always cross-multiplies, including when a unit rate or a scaling-up would be simpler, and who cannot say why $a/b = c/d$ implies $ad = bc$, has the trick and not the relationship. Failure point (a) from the last section is this one. It is the load-bearing failure of the band.

What the map actually contains is more specific than “do some ratios.” 6.RP asks for ratio language, a unit rate a/b associated with a ratio $a:b$ with $b \neq 0$, and ratio and rate reasoning with tables, tape diagrams, double number lines, and equations. Unit pricing. Constant speed. Percent as a rate per 100. Convert measurement units.⁶⁴⁶ 7.RP then asks the child to *analyze* proportional relationships: unit rate including ratios of fractions; the constant of proportionality k in $y = kx$; graphs of proportional relationships through the origin; percent problems — tax, tip, markup, markdown, simple interest, percent increase and decrease; scale drawings.⁶⁴⁷ Percent is not a fourth operation. It is a rate per 100. 3:4 is 75 per 100 when the whole is 100. A child who can compute 25 percent of 80 by a chant and cannot say that 25 per 100 is the same relationship as 1 per 4 has the chant and not the rate. The Panel’s Grade 7 benchmark is proficient problems involving percent, ratio, rate, and proportionality — a judgment from curriculum comparison, not an experimental placement.⁶⁴⁸

A unit rate is the ratio with a denominator of one. If 12 miles in 3 hours, the unit rate is 4 miles per 1 hour. That 4 is also the k in $y = kx$ when x is hours and y is miles, and it is the slope of the line through the origin that graphs the relationship, and it is the number the child should see repeating in a ratio table: 1 hour, 4 miles; 2 hours, 8 miles; 3 hours, 12 miles; $\frac{1}{2}$ hour, 2 miles. Tables, tape diagrams, double number lines, and equations are not four franchises. They are four representations of the same relationship, and they belong in the same sitting the way Chapter 5 put arrays, equal groups, skip jumps, and the written fact in the same sitting. A table that is only “multiply both columns by 2” without ever naming k is a procedure. A graph through the origin with no table is a picture. An equation $y = 4x$ with no graph is a chant. The parent who can hear a wrong answer hears all three.

Fraction divided by fraction lives here. The map withheld it from Grade 5. 5.NF.7 limited division to unit-fraction $\div$ whole and whole $\div$ unit-fraction. 6.NS is where $1/2 \div 1/4$ becomes how many one-fourths are in one-half, which is 2, and where invert-and-multiply is the compression of that measurement-division meaning, not a magic flip.⁶⁴⁹ Liping Ma’s comparison, already cited in Chapter 5, is the parent-knowledge problem: U.S. teachers generating few or no explanations of invert-and-multiply versus Japanese and Chinese teachers generating two or three.⁶⁵⁰ The design

⁶⁴⁵Robert Siegler, Thomas Carpenter, Francis Fennell, David Geary, James Lewis, Yukari Okamoto, Laurie Thompson, and Jonathan Wray, *Developing Effective Fractions Instruction for Kindergarten Through 8th Grade*, NCEE 2010-4039 (IES, 2010), Recommendation 4: develop understanding of ratios and rates before formal proportions. https://ies.ed.gov/ncee/wwc/docs/practiceguide/fractions_pg_093010.pdf, accessed 30 August 2026.

⁶⁴⁶CCSSM 6.RP (n. 615).

⁶⁴⁷CCSSM 7.RP: analyze proportional relationships; unit rate including ratios of fractions; constant of proportionality k in $y = kx$; graphs through the origin; percent problems (tax, tip, markup, markdown, simple interest, percent increase/decrease); scale drawings.

⁶⁴⁸NMAP Table 2 Grade 7 benchmark (n. 629). Judgment from curriculum comparison, not an experimental placement.

⁶⁴⁹CCSSM 5.NF.7 vs 6.NS: division limited to unit-fraction $\div$ whole and whole $\div$ unit-fraction in Grade 5; fraction $\div$ fraction is Grade 6. Invert-and-multiply as a compression of measurement division (how many $1/4$ s in $1/2$), not a magic flip. Chapter 5 planted this; this chapter does the operation.

⁶⁵⁰Liping Ma, *Knowing and Teaching Elementary Mathematics* (Erlbaum, 1999). The comparison usually cited: U.S.

user who stopped at algebra is Ma's U.S. teacher, in a kitchen. The parent has to be able to hear "flip the second fraction" as a slogan and ask, on a number line or with a measurement story, how many $1/4$ s fit in $1/2$. Then the rule is a compression. Then practice. Iterate. Do not withhold the procedure for weeks. Do not run a page of invert-and-multiply with no measurement meaning and no line.

Word problems in this band still reject keywords. Fuchs and colleagues, 2021, Table 5.2, listed why *all*, *left*, and *total* fail in the additive cases; the multiplicative cases are worse, because "of" and "per" and "times" appear in structures that are not the same.⁶⁵¹ What does work, repeatedly, is teaching the child to identify the underlying structure. Additive: combine, change, compare. Multiplicative: equal groups, comparison, ratio, rate, percent, proportion. Jitendra's schema-based instruction packages the move: identify the schema, represent it with a diagram, estimate, solve, check. Teach unit rate, and scaling, and the proportion equation, and *compare* them. Flexibility — being able to choose among procedures — is a third outcome, not a luxury. Chapter 2 already said that, citing Schneider, Rittle-Johnson, and Star 2011. This band is where the parent will watch a child reach for cross-multiply on a problem that is a unit rate in one step.

Jitendra, Star, Dupuis, and Rodriguez 2012 is the trial this section can actually quote, and it is a seventh-grade trial, which is why it lives here and not in Chapter 5. Seventh-grade randomized experiment, 1,163 students in 42 classrooms, schema-based instruction versus regular instruction, proportion problems. Immediate posttest $g = 1.24$. Six-week retention $g = 1.27$. No significant transfer.⁶⁵² Those three numbers together are the honest paragraph. Schema-based instruction can produce large gains on the trained problem types. Six weeks later the trained items were still there. Transfer to untrained items was not significant. Do not quote the 1.24 as a general-achievement miracle. Do not mash it with Rohrer's $d = 1.05$, which is a different design, a different comparison — interleaved versus blocked practice of four problem kinds — and a different outcome. Jitendra et al. 2013 was a third-grade trial with students who had mathematics difficulties: word-problem posttest $g = 0.46$, district achievement $g = 0.34$, no significant retention at 8 weeks. Corey Peltier and Kimberly Vannest's 2017 meta-analytic $g = 1.57$ on immediate elementary problem solving, $g = 1.09$ on transfer, 21 studies, 3,408 students, is a large number that includes many researcher-developed proximal measures. Do not treat it as a general-achievement effect.⁶⁵³ The kitchen how-

teachers generating few or no explanations of invert-and-multiply versus Japanese and Chinese teachers generating two or three. Discussed in Siegler et al. 2012 and in IES 2010 Recommendation 5 (teacher knowledge). The design user who stopped at algebra is Ma's U.S. teacher, in a kitchen.

⁶⁵¹Lynn S. Fuchs et al., *Assisting Students Struggling with Mathematics: Intervention in the Elementary Grades*, WWC 2021006 (IES, 2021), Recommendation 5, Table 5.2: why keyword strategies fail. Strong evidence, 18 studies. <https://ies.ed.gov/ncee/wwc/Docs/PracticeGuide/WWC2021006-Math-PG.pdf>, accessed 30 August 2026. Multiplicative keywords (*of*, *per*, *times*) fail for the same structural reason.

⁶⁵²Jitendra, Star, Dupuis, and Rodriguez 2012 (n. 609). Immediate posttest $g = 1.24$; six-week retention $g = 1.27$; no significant transfer. Trained proportion items, not a general-achievement miracle. Do not mash with Rohrer $d = 1.05$ (n. 609, n. 697).

⁶⁵³Asha K. Jitendra et al., "A Randomized Controlled Trial of the Impact of Schema-Based Instruction on Mathematical Outcomes for Third-Grade Students with Mathematics Difficulties," *The Elementary School Journal* 114, no. 2 (2013): 252–276, <https://doi.org/10.1086/673199>. Word-problem posttest $g = 0.46$; district achievement $g = 0.34$; no significant retention at 8 weeks. Corey Peltier and Kimberly J. Vannest, "A Meta-Analysis of Schema Instruction on the Problem-Solving Performance of Elementary School Students," *Review of Educational Research* (2017), <https://doi.org/10.3102/0034654317720163>. Hedge's $g = 1.57$ immediate problem solving / $g = 1.09$ transfer; 21 studies, 3,408 students; many researcher-developed proximal measures. Do not quote $g = 1.57$ as general achievement.

to that survives is the one the 2021 IES guide already wrote and Chapter 5 already stole: introduce one problem type at a time, first as a complete story with all quantities, then as a problem with one unknown; vary the cover story so the child is matching structure, not nouns; never teach a magic word.

Do not teach cross-multiply as a belief. The implication $a/b = c/d \Rightarrow ad = bc$ is true, when b and d are not zero, because both ratios equal the same k , or because multiplying both sides by bd is a legal move on an equation. A child who can execute the cross and cannot say either of those sentences has a trick. The IES 2021 intervention panel advised against cross-multiply-to-compare fractions during intervention because the trick does not build magnitude.⁶⁵⁴ Comparing $2/3$ and $3/5$ by placing them on a line, or by finding a common denominator, or by a benchmark such as $1/2$, is still the magnitude work. Cross-multiply as a proportion-solving move, *after* the relationship is visible in a table and a graph, is a compression. Before that, it is a slogan. Star's flexibility work, already locked in Chapter 2, is the reason to teach unit rate *and* scaling *and* the proportion equation, and to ask which one fits. A 2:3 recipe scaled to 8:12 is a scaling. A "how many miles in 1 hour" from 12 miles in 3 hours is a unit rate. A missing value in $2/3 = \square/12$ can be either, or a proportion equation. The child who has only one brittle procedure is not fluent. Fluency, restated from Chapter 2 because this is the band where the trick temptation is loudest, is flexible, accurate, efficient, and appropriate.

Rohrer, Dedrick, and Burgess 2014 treated "graph $y = mx + b$ " and "slope from two points" as different problem kinds that students fail to associate when practice is blocked.⁶⁵⁵ That finding belongs fully in section 6.6, where the trial lives. It is planted here because proportional reasoning is how slope arrives. Slope as unit rate, 8.EE, is the Grade 8 compression of 7.RP's k . Similar triangles explain why the slope is the same between any two distinct points on a non-vertical line — the Panel's geometry foundation, the load-bearing skill for linear functions.⁶⁵⁶ A child who can graph $y = 2x + 1$ from a chant and cannot find the slope between (1, 3) and (3, 7), or who can find that slope and cannot see it as the k of a proportional relationship on a translated line, has two tricks and not one relationship. Interleaving those kinds is the teaching move. Naming them as kinds is the parent's job first.

Kitchen-math does not replace this work. The Panel's Finding 26 is still the load-bearing caution: teaching with "real-world" contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving.⁶⁵⁷ Shopping-math and recipe-math can motivate and can provide a second representation of a percent or a ratio the child is already being taught as a relationship. They do not replace the table, the double number line, the schema work, or the retrieval. A child who can compute a 20 percent tip and cannot write $20/100 = 1/5$ as the same rate has performed,

⁶⁵⁴Fuchs et al. 2021, Recommendation 4 obstacles: the panel advises against cross-multiply-to-compare during intervention because it does not build magnitude.

⁶⁵⁵Rohrer, Dedrick, and Burgess 2014 (n. 609, n. 697). Four problem kinds included "graph $y = mx + b$ " and "slope from two points" as kinds students fail to associate when practice is blocked. Plant for §6.6.

⁶⁵⁶NMAP similar-triangles prerequisite (n. 633); CCSSM 8.EE slope as unit rate. 7.RP's k is the Grade 7 object; slope is the Grade 8 compression, free to have a b .

⁶⁵⁷NMAP, *Foundations for Success* (2008), Finding 26: teaching with "real-world" contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving. Kitchen-math transfer is hard.

not learned. Nicholas C. Soderstrom and Robert A. Bjork, 2015: current performance is “often an unreliable index” of learning.⁶⁵⁸ Chapter 3 used that distinction for AI. This section uses it for a receipt.

What this section has done is name the relationship. Signed numbers are the other Critical Foundation this band must close. They live on the same line. The rules are compressions of moves, or they are slogans, and slogans will not survive Algebra I.

6.3 Signed numbers on the same line; rules as compressions of moves

Integers and signed rationals are numbers. That sentence is as load-bearing as “a ratio is a relationship,” and it is easier to miss, because the parent who stopped at algebra often remembers signed numbers as a list of slogans. A negative times a negative is a positive. Two minuses make a plus. Keep-change-change. Those slogans are compressions of moves on a line, or they are folklore, and folklore will not survive the first two-step equation that has a negative coefficient. Chapter 5 already installed the number line as the representation that unifies whole numbers, fractions, and decimals. Siegler, Thompson, and Schneider 2011 treat numerical development as progressively mapping more classes of number onto a mental number line, and as learning that many whole-number properties are not properties of number.⁶⁵⁹ This band maps the negatives onto the same line. Pies do not do that work. Integer chips can, under guidance, as a second representation. The line is the one that scales to irrationals in Grade 8 and to a graph of a function the year after that. The IES 2021 intervention guide’s Recommendation 4 — number lines, strong evidence, 14 studies — was written for struggling elementary students. The representation does not retire when the child turns eleven.⁶⁶⁰

What the map actually contains is more specific than “do some negatives.” 6.NS introduces rational numbers: positive and negative integers and other rationals; opposites; absolute value; the four quadrants; order and compare; reflections across axes.⁶⁶¹ A number and its opposite are the same distance from zero on opposite sides. Absolute value is that distance. -3 is less than -1 , which is the first sentence a child who treats “more digits” or “the minus makes it small” as a slogan will get wrong. Place -3 , -1 , 0 , $1/2$, and 2 on the same line. If the child cannot, the rules have nothing to attach to. Grade 5’s first quadrant becomes four quadrants here: $(-2, 3)$ is two left and three up from the origin, a reflection of $(2, 3)$ across the y -axis. That is not a graphing trick. It is the number line, crossed with itself.

⁶⁵⁸Nicholas C. Soderstrom and Robert A. Bjork, “Learning Versus Performance: An Integrative Review,” *Perspectives on Psychological Science* 10, no. 2 (2015), <https://doi.org/10.1177/1745691615569000>. Current performance is “often an unreliable index” of learning. Full discussion in Chapter 3; this chapter uses the distinction for a receipt and for a solver.

⁶⁵⁹Robert S. Siegler, Clarissa A. Thompson, and Michael Schneider, “An Integrated Theory of Whole Number and Fractions Development,” *Cognitive Psychology* 62, no. 4 (2011): 273–296, <https://doi.org/10.1016/j.cogpsych.2011.03.001>. Numerical development as progressive mapping of more classes of number onto a mental number line, and as learning that many whole-number properties are not properties of number. This band maps negatives onto the same line.

⁶⁶⁰Fuchs et al. 2021, Recommendation 4: number lines for struggling elementary students, strong evidence, 14 studies. IES 2010 Recommendation 2: fractions as numbers; number lines as a central tool. The representation scales from counting to irrationals; it does not retire in this band.

⁶⁶¹CCSSM 6.NS rational-number cluster (n. 616).

7.NS then asks for all four operations on rational numbers, including negatives. Interpret sums in contexts — temperature, debt. $p + q$ as the number located a distance $|q|$ from p , in the positive or negative direction depending on the sign of q . Convert a rational number to a decimal, terminating or repeating.⁶⁶² The Panel’s Table 2: proficient operations with positive and negative integers by the end of Grade 6; proficient operations with positive and negative fractions by the end of Grade 7.⁶⁶³ Signed fractions are inside the Critical Foundation. Do not skip them because the Algebra I book is on the shelf. The 743-teacher survey’s top content gap was rational numbers and operations with fractions and decimals. That is this work, unfinished, arriving in Algebra I as symbol soup.⁶⁶⁴

Operations as moves, then rules as compressions. Addition of a positive is a move to the right; addition of a negative is a move to the left. $p + q$ is a distance $|q|$ from p . Subtraction is adding the opposite: $p - q = p + (-q)$. That is not a slogan. It is the definition that makes $3 - (-2)$ a move of 2 to the right from 3, landing on 5, and that makes $3 - 5$ a move of 5 to the left from 3, landing on -2 . A child who recites “two minuses make a plus” and cannot show $3 - (-2)$ on a line has the slogan. A child who can show the move and cannot yet retrieve $3 - (-2) = 5$ as a fact is in the iterative window Chapter 2 already described: a short conceptual pass, then a handful of problems, then a return to the line, then more problems. Do not withhold practice for weeks until they understand. Do not run pages of keep-change-change with no line and no why.

Multiplication and division are the slogans that fail most loudly. “A negative times a negative is a positive” is true, and it is the sentence a parent who stopped at algebra is most likely to remember without being able to hear as a move. Two justifications the parent must be able to give, neither of them a kitchen scene, both of them mathematics.

The pattern justification. $3 \times 2 = 6$. $2 \times 2 = 4$. $1 \times 2 = 2$. $0 \times 2 = 0$. $(-1) \times 2 = -2$. $(-2) \times 2 = -4$. The products decrease by 2 each time the first factor decreases by 1. Continue the pattern: $(-3) \times 2 = -6$, which is also “a negative times a positive is a negative.” Now hold the first factor at -3 and decrease the second: $(-3) \times 2 = -6$. $(-3) \times 1 = -3$. $(-3) \times 0 = 0$. $(-3) \times (-1) = 3$. The products increase by 3 each time the second factor decreases by 1, because we have been subtracting 3 and now we are subtracting a negative, which is a move the other way. A negative times a negative is a positive because the pattern that already held for whole numbers continues to hold. That is generalized arithmetic, which is what algebra is. Booth, Knuth, and Küchemann’s variable-as-mystery-number work, waiting in the next section, is the same idea in letters.⁶⁶⁵

The number-line / directed-magnitude justification. Multiplication by a positive stretches or shrinks and keeps direction. Multiplication by a negative stretches or shrinks and *reverses* direction. $(-3) \times (-2)$ is: take the directed magnitude -2 , stretch it by 3, reverse it, land at $+6$. Two reversals return the original direction. This is the scaling sentence from 5.NF.5, now with a sign. “Multiply makes bigger” already failed for multipliers less than 1. It fails again for negatives: $(-3) \times 2$ is

⁶⁶²CCSSM 7.NS: four operations on rationals including negatives; interpret sums in contexts; $p + q$ as distance $|q|$ from p in the positive or negative direction; conversion of a rational to a decimal (terminating or repeating).

⁶⁶³NMAP Table 2 (n. 629): integers by end of Grade 6; signed fractions by end of Grade 7.

⁶⁶⁴NMAP 2007 algebra-teacher survey (n. 632). Do not skip signed-fraction operations because the Algebra I book is on the shelf.

⁶⁶⁵Pattern justification as generalized arithmetic: NMAP Major Topics of School Algebra treat symbols and expressions as the first topic; algebra as generalized arithmetic is the view that letters stand for numbers and the properties already known still hold (02-learning §1.6). Booth / Knuth / Küchemann variable-as-mystery-number cluster: n. 675.

-6 , which is smaller, not bigger, than -3 if “bigger” means greater on the line. Whole-number bias, named by Ni and Zhou 2005, still operates in this band. Unique successors fail for rationals — there is no “next” fraction after $1/2$. A finite number of entities in an interval fails — infinitely many rationals between 0 and 1. Multiplication makes bigger fails for $|m| < 1$ and for negatives. Addition of “pieces” as adding the counts you see fails for signed fractions the same way it failed for $1/2 + 1/3 = 2/5$.⁶⁶⁶ Instructional upshot Ni and Zhou endorse: organize teaching so prior whole-number knowledge is *used* rather than only fought. The pattern justification above is that use. The line is the fight the bias loses.

Division follows from multiplication as unknown-factor, the same inverse Chapter 5 already used. $(-6) \div (-2) = 3$ because $(-2) \times 3 = -6$. $(-6) \div 2 = -3$ because $2 \times (-3) = -6$. The slogans compress those inverses. A child who has only the slogans and cannot write the unknown-factor sentence will lose the sign in the first rational-coefficient equation of Algebra I. Julie L. Booth, Kenneth R. Koedinger, and colleagues, 2013, list minus signs treated as “subtract” rather than as modifiers of terms — Vlassis 2004 is the source they cite — among the algebra misconceptions incorrect examples were built to hit.⁶⁶⁷ That misconception is already available in this band. $3 + -2$ is not “three plus minus two, so subtract, I guess.” It is $3 + (-2)$, a sum of a number and a signed number. Parentheses around a negative, early and often, are a notation habit that later saves the minus-as-subtract error. Hear it now.

Contexts help *similar* contexts. Temperature below zero, debt, elevation below sea level, a football loss of yards, are the usual ones, and they are useful for addition and subtraction as moves. The Panel’s Finding 26 still applies: real-world contexts improve performance on similar real-world items; they do not, in the Panel’s small high-quality set, raise computation or equation solving.⁶⁶⁸ Debt will not teach $(-3) \times (-2)$. The line and the pattern will. Use the context as a second representation of a sum the child is already being taught as a move. Do not replace the line with a story.

Worked examples, faded, for the first pass through a new signed-number procedure. Chapter 2 already locked this for high-element-interactivity material. Show one fully worked example on the line, then a second the child must explain, then a completion problem missing the last step, then

⁶⁶⁶Yujing Ni and Yong-Di Zhou, “Teaching and Learning Fraction and Rational Numbers: The Origins and Implications of Whole Number Bias,” *Educational Psychologist* 40, no. 1 (2005): 27–52, https://doi.org/10.1207/s15326985ep4001_3. Whole-number properties that fail for rationals: unique successors; a finite number of entities in an interval; multiplication makes bigger; addition of “pieces” as adding the counts you see. Instructional upshot they endorse: organize teaching so prior whole-number knowledge is used rather than only fought. Stella Vosniadou, Xenia Vamvakoussi, and Irini Skopeliti, “The Framework Theory Approach to Conceptual Change,” in *International Handbook of Research on Conceptual Change*, ed. Stella Vosniadou (2008): “multiply makes bigger / divide makes smaller” as a central conceptual-change problem. CCSSM 5.NF.5 already broke “multiply always makes bigger” for $|m| < 1$; negatives break it again.

⁶⁶⁷Julie L. Booth, Karin E. Lange, Kenneth R. Koedinger, and Kristie J. Newton, “Using Example Problems to Improve Student Learning in Algebra: Differentiating between Correct and Incorrect Examples,” *Learning and Instruction* 25 (2013): 24–34, <https://files.eric.ed.gov/fulltext/ED543090.pdf>, accessed 30 August 2026. Explaining incorrect examples (alone or with correct ones) especially useful for conceptual understanding of two-step linear equations, combined with guided practice in the Algebra I Cognitive Tutor. Misconceptions the examples were built to hit: equal sign as “where the answer goes” (Baroody and Ginsburg 1983; Kieran 1981); minus signs as “subtract” rather than as modifiers of terms (Joëlle Vlassis, 2004); variables as a single mystery number (Booth 1984; Knuth et al. 2006; Dietmar Küchemann 1978).

⁶⁶⁸NMAP Finding 26 (n. 657). Temperature, debt, and elevation help similar contexts; they do not teach $(-3) \times (-2)$.

a conventional problem.⁶⁶⁹ Include at least one incorrect example and ask what they thought the symbol meant. “They subtracted 5 from 3 and wrote 2, and then stuck a minus on it” is a diagnosis. “They forgot the rule” is not. Once the child can execute, stop showing full examples. Expertise reversal is real. Then retrieve. Mixed signed-number items, including signed fractions, on a short quiz without notes. The Panel wants operations with signed fractions by the end of Grade 7. That is not a race. It is a gate.

Manipulatives — integer chips, two-color counters — help under guidance and then become a crutch if they are not connected to symbols and faded. Carbonneau, Marley, and Selig 2013: overall d approximately 0.37 versus abstract-only; larger on retention than on transfer or justification; high guidance associated with higher retention and problem solving.⁶⁷⁰ That number is not “chips work.” Put the chips next to the written $-3 + 5$ and the same sum on a line, in the same visual field. Fade when the child is accurate without them. Bring them back when signed-fraction addition needs a bridge. Fuchs and colleagues 2021, Recommendation 3: choose representations that match the mathematics; connect concrete and semi-concrete to the notation in the same visual field, not “chips today, symbols next month”; fade for accuracy, not for a calendar date.⁶⁷¹

Do not start Algebra I while fraction and integer operations are still fragile. Restate that, because this is the sentence the calendar will try to overwrite. NCES 2016: 41 percent of the 6–8 home-school band were being taught Algebra I *that year*.⁶⁷² Some of those students had the gates. Some did not. The Panel’s 2007 survey is why the diagnostic is content, not a birthday. Can the child add, subtract, multiply, and divide signed fractions? Place $-3/4$ and $2/3$ on the same line? Treat $3 + -2$ as a sum of a number and a signed number, not as a slogan? Treat the equal sign as a relation, still, now inside $-2x + 5 = 11$? If not, the Algebra I book is early. Diagnose. Then this band. Chapter 7 will say it again as a door diagnostic. This chapter is where the door is built.

A *hypothetical*, labelled as such, can put the diagnostic in a room without inventing a reported family. A parent whose twelve-year-old placed into a book titled Algebra I on a publisher test, and who then watches the child write $1/2 + (-1/3) = 1/6$, or $1/2 + (-1/3) = -1/5$, or $1/2 + (-1/3) = 1/5$, is watching a magnitude-and-sign failure, not a reason to “push through the algebra.”⁶⁷³ The

⁶⁶⁹John Sweller and Graham A. Cooper, “The Use of Worked Examples as a Substitute for Problem Solving in Learning Algebra,” *Cognition and Instruction* 2, no. 1 (1985): 59–89. Worked-example effect for novices. Alexander Renkl and Robert K. Atkinson, “Structuring the Transition from Example Study to Problem Solving in Cognitive Skill Acquisition: A Cognitive Load Perspective,” *Educational Psychologist* 38 (2003): 15–22; guidance fading / backward fading. Slava Kalyuga, Paul Ayres, Paul Chandler, and John Sweller, “The Expertise Reversal Effect,” *Educational Psychologist* 38 (2003): 23–31: the same worked example that helps a novice can hurt a relative expert. Full discussion in Chapter 2; this chapter applies the sequence to signed-number procedures and two-step equations.

⁶⁷⁰Kira J. Carbonneau, Scott C. Marley, and James P. Selig, “A Meta-Analysis of the Efficacy of Teaching Mathematics with Concrete Manipulatives,” *Journal of Educational Psychology* 105, no. 2 (2013): 380–400, <https://doi.org/10.1037/a0031084>. 55 studies, $N = 7,237$, K through college, manipulatives vs abstract-only. Overall mean $d \approx 0.37$. Larger on retention than on transfer or justification; high guidance associated with higher retention and problem solving. Do not quote $d \approx 0.37$ as “chips work” without the moderators.

⁶⁷¹Fuchs et al. 2021, Recommendation 3 (strong evidence, 28 studies): choose representations that match the mathematics; connect concrete and semi-concrete to notation in the same visual field; fade for accuracy, not for a calendar date.

⁶⁷²NCES 2020-001, Table 9 (n. 601). 41 percent of the 6–8 homeschooled band taught Algebra I *that year*. Coverage, not quality, and not a finding that those students had the gates.

⁶⁷³Hypothetical labelled in the sentence that introduces it. Empirical pieces: publisher placement tests exist because labels lie (Chapter 1); NMAP 2007 survey (n. 632); Ni and Zhou 2005 (n. 666). The sitting, the age, and the three

empirical pieces that composite combines are in the documents: publisher placement tests exist because labels lie; the 2007 survey named rational numbers as the top incoming gap; Ni and Zhou named adding the counts you see. The sitting, the age, and the three wrong sums are hypothetical. The teaching move is not. Put $-1/3$ and $1/2$ on the line. Ask which side of zero the sum lives on, and about how far. Then compute. Then retrieve. Then, and only then, return to the Algebra I book, or do not.

What this section has done is put the negatives on the line Chapter 5 already owned. Expressions, equations, and 8.F are next: algebra as generalized arithmetic, and a function as more than a line.

6.4 Expressions, equations, and the function concept (8.F)

An expression is a number that depends on x . It is not an instruction to “do it.” $3x + 2$, when x is 4, is 14. When x is -1 , it is -1 . The expression is the number; the x is a symbol that can take many values. That sentence is algebra as generalized arithmetic: the letters stand for numbers, and the properties already known for numbers still hold.⁶⁷⁴ It is also the sentence many students never form. Booth, Lange, Koedinger, and Newton 2013, building on Booth 1984, Knuth, Stephens, McNeil, and Alibali 2006, and Küchemann 1978, document the gap: many students treat a variable as a single mystery number, not as a symbol that can take many values.⁶⁷⁵ $3x + 2$ treated as “do it” is the operational equal sign, still alive, now wearing a letter. Nicole M. McNeil and Martha W. Alibali’s three entrenched operational patterns — perform all given operations on all given numbers; the “operations = answer” visual structure; “=” means “the total” — do not retire in Grade 6. Knuth and colleagues connected a relational view of the equal sign to later algebra. The equal sign is still a relation, now inside $a/b = c/\square$ and inside $-2x + 5 = 11$.⁶⁷⁶ Hear $3x + 2$ as a number. Hear “=” as “names the same quantity.” Those two hearings are the door to this section.

What the map actually contains, grade by grade, is more specific than “early algebra.” 6.EE: write and evaluate numerical expressions with whole-number exponents; write, read, and evaluate expressions with variables; identify parts of an expression — terms, factors, coefficients — so that $3(2 + x)$ is three times a sum, not a blob; evaluate at specific values; generate equivalent expressions; identify when two expressions are equivalent. Reason about and solve one-variable equations and inequalities; understand solving as a process of answering “which values make the equation true.” Represent and analyze quantitative relationships between dependent and independent variables —

wrong sums are hypothetical.

⁶⁷⁴Algebra as generalized arithmetic: NMAP Table 1, Symbols and Expressions; 02-learning §1.6. Letters stand for numbers; properties already known still hold.

⁶⁷⁵Booth et al. 2013 (n. 667), citing Booth 1984; Eric J. Knuth, Ana C. Stephens, Nicole M. McNeil, and Martha W. Alibali, “Does Understanding the Equal Sign Matter? Evidence from Solving Equations,” *Journal for Research in Mathematics Education* 37 (2006): 297–312; Küchemann 1978. Variable as a single mystery number, not as a symbol that can take many values.

⁶⁷⁶Nicole M. McNeil and Martha W. Alibali, “Why Won’t You Change Your Mind? Knowledge of Operational Patterns Hinders Learning and Performance on Equations,” *Child Development* 76 (2005): 883–899; Nicole M. McNeil, “A Change-Resistance Account of Children’s Difficulties Understanding Mathematical Equivalence,” *Child Development Perspectives* (2014). Three operational patterns: perform all given operations on all given numbers; “operations = answer” visual structure; “=” means “the total.” Knuth et al. 2006 (n. 675); relational view of “=” connected to later algebra. Now live inside $a/b = c/\square$ and $-2x + 5 = 11$. Full discussion in Chapter 4; this chapter does not dump it.

tables, graphs, equations such as $y = 3x$.⁶⁷⁷ Solving is not “move this to that side.” Solving is asking which numbers, substituted for the letter, make the two sides name the same quantity. A child who can mechanically isolate x and cannot substitute back to check has a procedure and not the relation. Write $3x + 2 = 14$, try $x = 4$, both sides 14, true. Try $x = 3$, $11 = 14$, false. Then the inverse steps are a compression of that search, not a dance.

7.EE: generate equivalent expressions — expand, factor; solve multi-step real-life problems with positive and negative rationals; solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$; inequalities of the same forms.⁶⁷⁸ Those two forms are the first two-step linear equations. Booth and colleagues 2013 used exactly those, in the Algebra I Cognitive Tutor, as the objects on which explaining incorrect examples helped conceptual understanding.⁶⁷⁹ The misconceptions the examples were built to hit are already this band’s: the equal sign as “where the answer goes” (Baroody and Ginsburg 1983; Kieran 1981); minus signs as “subtract” rather than as modifiers of terms (Vlassis 2004); variables as a single mystery number. Worked-example fading for the first two-step equation; incorrect examples explained. Combined with guided practice in that paper, not as a kitchen-table Cognitive Tutor — MATHia family purchase remains **UNVERIFIED** as of 30 August 2026, and school RCTs are not homeschool results.⁶⁸⁰ The move that transfers is the parent showing one fully worked $px + q = r$ on paper, then a second the child must explain, then a completion problem missing the last step, then a conventional problem, then an incorrect example: “they added q instead of adding the opposite of q . What did they think the plus sign meant?”

8.EE is the linear-equation year on the map: integer exponents and scientific notation; square and cube roots; connections among proportional relationships, lines, and linear equations; slope as unit rate; similar triangles to explain why the slope is the same between any two distinct points on a non-vertical line; $y = mx + b$. Analyze and solve linear equations in one variable, including those with infinitely many solutions or no solution, and pairs of simultaneous linear equations — graphically, algebraically, and by inspection.⁶⁸¹ Infinitely many solutions and no solution are the first time the child meets an equation as a claim that can be always true or never true, not only sometimes true. $2x + 3 = 2x + 5$ is never true. $2x + 3 = 2(x + 1) + 1$ is always true. Substituting still decides. The operational equal sign, which wanted a number on the right, has nowhere to go. That is a feature. Systems of two linear equations — $x + y = 5$ and $x - y = 1$ — are three representations again: a graph of two lines and their intersection; an algebraic elimination or substitution; an inspection of structure. They are not Algebra I’s full systems cluster. They are the on-ramp. Chapter 7 will formalize. This chapter plants.

⁶⁷⁷CCSSM 6.EE (n. 617).

⁶⁷⁸CCSSM 7.EE: equivalent expressions (expand, factor); multi-step problems with signed rationals; $px + q = r$ and $p(x + q) = r$; inequalities of the same forms.

⁶⁷⁹Booth et al. 2013 (n. 667). Two-step linear equations in the Algebra I Cognitive Tutor; incorrect examples especially beneficial for conceptual understanding.

⁶⁸⁰MATHia family purchase: **UNVERIFIED** as of 30 August 2026 (04-ai; Chapter 3). Cognitive Tutor / MATHia school RCTs are not homeschool results. Do not treat Booth 2013 as a kitchen-table Cognitive Tutor trial. The transferable move is fading plus incorrect-example explanation.

⁶⁸¹CCSSM 8.EE: integer exponents and scientific notation; square and cube roots; proportional relationships, lines, and linear equations; slope as unit rate; similar triangles and constant slope; $y = mx + b$; linear equations in one variable including infinitely many solutions or no solution; pairs of simultaneous linear equations (graphically, algebraically, by inspection).

8.F is the first Functions domain in K–8, and it is the sentence this section exists to protect. A function assigns to each input exactly one output.⁶⁸² That is the concept. It is not “ $y = mx + b$.” Linear versus nonlinear. Compare functions represented in different ways — table, graph, equation, verbal. Interpret $y = mx + b$ as defining a linear function, m as rate of change, b as initial value. Sketch a graph from a qualitative description. Do not introduce functions as “ $y = mx + b$ ” only, so the general “each input one output” idea never forms. Failure point (c) from section 6.1 is this one. A vertical-line test, later, is a compression of “each input exactly one output.” A machine, a table, a set of ordered pairs, a graph, a rule, are representations of the same correspondence. Early functions: input-output tables and graphs and “the rule,” same three representations in one sitting. If the table says $0 \rightarrow 1, 1 \rightarrow 3, 2 \rightarrow 5, 3 \rightarrow 7$, the rule is “double and add 1,” the graph is a line with slope 2 and intercept 1, and the equation is $y = 2x + 1$. Ask all three in the same sitting. Then ask one that is not linear: $0 \rightarrow 0, 1 \rightarrow 1, 2 \rightarrow 4, 3 \rightarrow 9$. The rule is “square.” The graph is not a line. The function concept survived. $y = mx + b$ did not swallow it.

Slope as unit rate, grounded in similar triangles, is the Panel’s geometry foundation for linear functions. Similar triangles explain why the slope is the same between any two distinct points on a non-vertical line.⁶⁸³ Rise over run is a unit rate: vertical change per one unit of horizontal change. It is 7.RP’s k , now free to have a b . A proportional relationship is the special case $b = 0$, graph through the origin. $y = mx + b$ is that relationship translated. A child who treats b as the slope because it is the first number they see, or who reads rise/run off the axes rather than the line, has a mixup this book cannot quantify. Slope/intercept mixup prevalence is **UNVERIFIED** as to a named prevalence study; the learning memo did not open a single primary that quantified it.⁶⁸⁴ Rohrer, Dedrick, and Burgess 2014 treated “graph $y = mx + b$ ” and “slope from two points” as different problem kinds that students fail to associate when practice is blocked — which is itself evidence that the association “this story to this strategy” is weak.⁶⁸⁵ Do not start the function concept as “the graph looks like.” A graph that “looks like a line” is not yet a function. A scatter that “looks like a line” is an association, 8.SP, not a function until someone specifies the rule. Plant Chapter 11. Do not dump it.

Without this function concept, calculus’s $f'(x)$ has nothing to attach to. That is a plant for Chapter 10, and it is not a scare. Calculus is a new operation on functions the student already knows. A student with a shaky function concept will experience calculus as symbol soup. A student who can move among formula, graph, table, and verbal, in this band, has something for a derivative to operate on.⁶⁸⁶ Do not skip 8.F because an Algebra I book is on the shelf. Algebra I’s F-IF — function notation, domain and range, key features, average rate of change — assumes the correspondence already exists. Grade 8 linear equations, systems, functions, and slope *unlock* Algebra I. They are

⁶⁸²CCSSM 8.F: a function assigns to each input exactly one output; compare functions represented in different ways; linear vs nonlinear; interpret $y = mx + b$ as defining a linear function, m as rate of change, b as initial value; sketch a graph from a qualitative description. Do not introduce functions as “ $y = mx + b$ ” only.

⁶⁸³NMAP similar-triangles prerequisite (n. 633); CCSSM 8.EE.

⁶⁸⁴Slope/intercept mixup prevalence: **UNVERIFIED** as to a named prevalence study (02-learning §1.6). Teaching lore; do not invent a percentage.

⁶⁸⁵Rohrer, Dedrick, and Burgess 2014 (n. 609, n. 697). Evidence that the association between those two problem kinds is weak when practice is blocked — not a prevalence study of the mixup.

⁶⁸⁶01-scope calculus-door gaps, labelled as inference from the maps, not a survey: weak function concept (cannot move among formula, graph, table, verbal). Plant for Chapter 10. Do not skip 8.F because an Algebra I book is on the shelf.

not identical to a full Algebra I taken in Grade 8. “Grade 8 algebra” is ambiguous: CCSSM Grade 8 EE/F versus a full Algebra I. Those are not the same course. Restate that, because the NCES 41 percent will tempt the parent to treat them as the same. A compacted pathway that finishes Algebra I in Grade 8, Appendix A’s model, includes the 8.F concept *and* linear, exponential, and quadratic functions. A year that only graphs $y = mx + b$ and calls it Algebra I has not done Appendix A’s Algebra I and has not done 8.F either, if the correspondence never formed.

Integer exponents and scientific notation, 8.EE, are the other structure work of the year. 10^3 is $10 \times 10 \times 10$. 10^0 is 1, as the pattern of dividing by 10 demands. 10^{-1} is $1/10$, the same pattern, now on the other side of zero in the exponent, which is the signed-number line applied to a count of factors. Square roots and cube roots: $\sqrt{2}$ is not rational; approximate it by rationals. 8.NS is the first time the number line is admitted to have points that are not ratios of integers.⁶⁸⁷ That is not a curiosity. It is why the Pythagorean theorem, next section, produces a length the child cannot write as a fraction, and why calculus later needs completeness the child will not be asked to name. For now: $\sqrt{2}$ lives on the line, between 1.4 and 1.5, closer to 1.41. Place it. Do not only press a calculator key.

Dependent and independent variables, 6.EE, are the language 8.F will need. y depends on x . Input, output. The rule. A child who can fill a table from $y = 3x$ and cannot say which column is the input has the chant. A child who can say “each input one output” and cannot produce a table has the slogan. Three representations, one sitting. Then a qualitative sketch: “the water in the tub rises steadily, then the tap is off, then it drains faster than it filled.” Sketch. That is 8.F.5. It is also the first modeling the child will do that is not a word problem with one unknown. NMAP Finding 26 still applies: context helps similar contexts. The tub sketch helps the next sketch. It does not replace the table.

What this section has done is name the expression, the equation, and the function, and refuse to let $y = mx + b$ swallow the last of those. Geometry of this band is the other half of authentic-algebra readiness. Similarity is how slope was justified. Do not drop it because a family “starts Algebra I early.”

6.5 Congruence, similarity, the Pythagorean theorem; informal bivariate data

Geometry in this band is not an extra, and data is not a Friday worksheet. NCTM’s Focal Points warning, already quoted, is the sentence a kitchen that “starts Algebra I early” has to hear: if algebra is offered in Grade 8, do not omit Grade 7–8 focal-point content; it can be moved, but not dropped.⁶⁸⁸ Similarity is the load-bearing skill for slope — the last section already used it — and for later right-triangle trigonometry. Informal bivariate data in Grade 8 is the on-ramp to Algebra I’s S.ID and to Chapter 11. Skip either because the Algebra I book is on the shelf and the later courses spend a semester retrofitting.

What the map actually contains, year by year. Grade 6 geometry is still measurement-adjacent: area of triangles, special quadrilaterals, and polygons by composing and decomposing; volume

⁶⁸⁷CCSSM 8.NS: numbers that are not rational; approximate irrationals by rationals (e.g., $\sqrt{2}$). CCSSM 8.EE integer exponents and scientific notation; square and cube roots.

⁶⁸⁸NCTM Focal Points warning (n. 627).

of right rectangular prisms with fractional edge lengths, which is Chapter 5’s unit cube now with fractional sides; polygons in the coordinate plane; surface area via nets.⁶⁸⁹ Grade 6 statistics is distributions: a statistical question is one whose answer varies; center (median, mean), spread (interquartile range, mean absolute deviation), shape (clusters, peaks, gaps, symmetry); summarize numerical data sets in context.⁶⁹⁰ Mean, median, and mode as a triad with no data story is not this domain. A question that has variability — how long did these sunflower seedlings grow in a week? — is. GAISE, the American Statistical Association’s Guidelines for Assessment and Instruction in Statistics Education, frames school statistics as formulate questions, collect, analyze, interpret, with variability as the central idea. GAISE II was not opened in full for this project; page-level claims stay **UNVERIFIED**. Treat the spirit, not a quoted recommendation list.⁶⁹¹ Chapter 11 will gather the through-line. This section plants the Grade 6–8 spine.

Grade 7 geometry: draw, construct, describe geometrical figures; scale drawings; two-dimensional figures from slicing three-dimensional figures; angle relationships from intersecting lines; area and circumference of a circle; area, volume, and surface area of objects composed of triangles, quadrilaterals, polygons, cubes, and right prisms.⁶⁹² Scale drawings are 7.RP in geometric clothes: a scale factor is a k . A child who can compute a 1:20 scale and cannot say that every length in the drawing is $1/20$ of the corresponding length in the object has a chant and not a dilation-in-waiting. Circumference and area of a circle — $C = 2\pi r$, $A = \pi r^2$ — are the first formulas the child will meet that are not a rectangle. π is a number on the line, not a button. Informal arguments now; Cavalieri and informal limits wait for high-school Geometry. Grade 7 statistics: random sampling to draw inferences about a population; informal comparative inferences about two populations; chance processes and probability models — sample spaces, compound events, simulations.⁶⁹³ A sample is not a population. A voluntary-response sample is a biased sample. Those two sentences are Unit 1 of AP Statistics, years early, and they are 7.SP. Do not skip them because “we’ll do AP Stats later.”

Grade 8 geometry is the transformation year on the map: congruence via rotations, reflections, translations; similarity via dilations; informal arguments for triangle angle-sum and exterior-angle theorems, and for angle relationships with parallel lines cut by a transversal; Pythagorean theorem and its converse; distance between points in the coordinate plane; volume of cylinders, cones, and spheres.⁶⁹⁴ Congruence means there is a rigid motion — or a composition of rigid motions —

⁶⁸⁹CCSSM 6.G (n. 618). Fractional-edge volume is Chapter 5’s unit cube extended.

⁶⁹⁰CCSSM 6.SP (n. 619).

⁶⁹¹American Statistical Association, *Guidelines for Assessment and Instruction in Statistics Education*. Pre-K–12 GAISE I 2005 (rev. 2007); Pre-K–12 GAISE II 2020. The 2016 date belongs to the *college* GAISE report revision, not the school document. Cited inside the AP Statistics CED (Fall 2026) as the alignment for statistical practices. ASA primary not separately opened this pass; page-level claims **UNVERIFIED**. Treat as spirit — formulate questions, collect, analyze, interpret; variability as the central idea — not as quoted recommendations. Plant for Chapter 11.

⁶⁹²CCSSM 7.G: draw, construct, describe; scale drawings; 2-D from slicing 3-D; angle relationships from intersecting lines; area and circumference of a circle; area, volume, and surface area of composed objects.

⁶⁹³CCSSM 7.SP: random sampling to draw inferences about a population; informal comparative inferences about two populations; chance processes and probability models (sample spaces, compound events, simulations).

⁶⁹⁴CCSSM 8.G: congruence via rotations, reflections, translations; similarity via dilations; informal arguments for triangle angle-sum and exterior-angle theorems, and for angle relationships with parallel lines cut by a transversal; Pythagorean theorem and converse; distance in the coordinate plane; volume of cylinders, cones, and spheres. Plant for Chapter 8 (rigid motions, proof) and Chapter 10 (informal limits made precise).

that takes one figure onto the other. That is an explicit difference from many older high-school geometry courses that began with axioms and two-column proof of triangle congruence. Appendix A's high-school Geometry will formalize from this definition. Chapter 8 will say so. This chapter is the informal argument. A drawing is not a theorem. Two triangles that look the same on the page are not yet congruent. The child has to name the motion, or the correspondence of vertices, or both.

Similarity via dilations is the proportional-reasoning gate, used. A dilation about a point with scale factor k produces a figure whose corresponding lengths are k times the original and whose corresponding angles are equal. AA similarity, later, is a compression. For now: corresponding angles equal, corresponding sides proportional. Scale drawings were the Grade 7 preview. Slope from similar triangles is the Grade 8 use. Right-triangle trigonometry, in high-school Geometry, is the next use. Do not skip similarity because trigonometry “is Algebra II.” It is not. Chapter 8 will need this. Chapter 9's unit circle will need Chapter 8's right-triangle ratios. The chain starts here.

The Pythagorean theorem is a statement about right triangles: if a triangle is right-angled, the square on the hypotenuse equals the sum of the squares on the legs. Its converse: if that equation holds, the triangle is right-angled. Distance between two points in the coordinate plane is the theorem, applied to a horizontal and a vertical leg. $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ is not a formula to memorize as a blob. It is a right triangle the child can draw on the grid, with a hypotenuse the child has already been told may be irrational. Place $\sqrt{5}$ on the line, between 2 and 3. Volume of cylinders, cones, and spheres extends Grade 5's $V = B \times h$ and Grade 7's prisms. The informal arguments — dissection, stacking, a cone as one-third of a cylinder with the same base and height — are the on-ramp to the informal-limit arguments high-school Geometry will make, and that calculus will make precise. Plant Chapter 8. Plant Chapter 10. Do not write them.

Grade 8 statistics is bivariate: scatter plots; informal linear models; clustering, outliers, positive and negative association; intercept, slope, and frequency tables of bivariate categorical data.⁶⁹⁵ An association is not a function until someone specifies a rule, and it is not causation ever, by itself. Algebra I's S.ID will add residuals and correlation-versus-causation as named clusters. AP Statistics Unit 5 will make regression a course. This is the scatter, the “looks like a line,” the informal $y = mx + b$ fitted by eye. Statistical thinking remains a question that has variability, not a worksheet of mean, median, and mode. NCES 2016: probability ever-taught to 24 percent of high-school-level homeschoolers.⁶⁹⁶ That is not a statistics-course completion rate. It is a reminder that the data strand is easy to drop. A skipped K–8 data strand is AP Statistics Unit 1 retrofit. Chapter 11 exists because statistics is a different college-equivalent from calculus, and because CCSSM already placed it from Grade 6. Do not stuff it into this paragraph and call the strand done. Do not skip it because this chapter is “the algebra chapter.”

Kitchen building and making are spatial. They do not automatically produce congruence, similarity, or a proof. Finding 26 is still the caution: real-world contexts help similar real-world items. A well-built birdhouse is not a theorem. Use the building as a second representation of a net or a scale factor the child is already being taught. Do not replace 8.G with a project.

⁶⁹⁵CCSSM 8.SP: bivariate data — scatter plots; informal linear models; clustering, outliers, positive/negative association; intercept, slope, and frequency tables of bivariate categorical data. An association is not a function until someone specifies a rule, and is not causation by itself. Plant for Algebra I S.ID and Chapter 11.

⁶⁹⁶NCES 2020-001: probability ever-taught to 24 percent of high-school-level homeschoolers. Not a statistics-course completion rate. A reminder that the data strand is easy to drop.

What this section has done is keep geometry and data from being dropped. Teaching moves that actually live in this band — including Rohrer, whose classroom trials are grade 7 — come next. This is their home chapter. The d stays here. It is not exported to grade 4, and it is not exported to calculus.

6.6 Teaching guidance and tips: Rohrer lives here; do not start Algebra I on fragile fractions

The clean classroom randomized trials of interleaved mathematics practice are grade 7. That is why this section exists, and why Chapters 4 and 5 were not allowed to import a d . Doug Rohrer, Robert F. Dedrick, and Kaleena Burgess, 2014, *Psychonomic Bulletin and Review*: grade-7 classroom crossover, $n = 140$, unannounced test two weeks later, interleaved 72 percent versus blocked 38 percent, $d = 1.05$, 95 percent CI [0.80, 1.30]. Four superficially dissimilar problem kinds: a four-step linear equation; a proportion word problem; graph $y = mx + b$; slope from two points. Pretest accuracy on these kinds was 0.7 percent for full solutions.⁶⁹⁷ Those are the objects of this band. A four-step linear equation is 7.EE. A proportion word problem is 7.RP. Graph $y = mx + b$ and slope from two points are 8.EE / 8.F, and they are the two kinds the authors treated as ones students fail to associate when practice is blocked. Discrimination errors on the delayed test were rare, 4.4 percent; the authors propose that interleaving also strengthens the association between a problem kind and its strategy.⁶⁹⁸ Taylor and Rohrer 2010 is the study that most cleanly isolates interleaving from spacing — both conditions spaced; $d = 1.23$ — and that showed blocked-condition errors were mostly wrong-strategy errors.⁶⁹⁹ Do not mash 2010's $d = 1.23$ with 2014's $d = 1.05$. Different designs, different isolation.

Rohrer, Dedrick, Marissa K. Hartwig, and Chi-Ngai Cheung, 2020, *Journal of Educational Psychology*: preregistered cluster randomized trial, 54 seventh-grade classes, four months of assignments, unannounced test one month later, interleaved 61 percent versus blocked 38 percent, $d = 0.83$. Teachers implemented without training.⁷⁰⁰ The authors themselves state the caveat this book will keep stating: interleaved assignments also space same-kind items, so the d is a joint effect of interleaving and spacing. Most textbooks are almost entirely blocked — Rohrer, Dedrick, and Hartwig 2020, *Educational Psychology Review*, “The scarcity of interleaved practice in mathemat-

⁶⁹⁷Rohrer, Dedrick, and Burgess 2014 (n. 609). Grade 7, $n = 140$, counterbalanced crossover, four superficially dissimilar kinds (four-step linear equation; proportion word problem; graph $y = mx + b$; slope from two points). Unannounced test two weeks later: interleaved 72% vs blocked 38%, $d = 1.05$, 95% CI [0.80, 1.30]. Pretest accuracy 0.7% for full solutions. This is the home chapter for that d . Do not export it to grade 4 or to calculus.

⁶⁹⁸Rohrer, Dedrick, and Burgess 2014: discrimination errors rare (4.4%); authors propose that interleaving also strengthens the association between a problem kind and its strategy.

⁶⁹⁹Kelli Taylor and Doug Rohrer, “The Effects of Interleaving Practice,” *Applied Cognitive Psychology* 24 (2010): 837–848. Both conditions spaced; $d = 1.23$; blocked-condition errors mostly wrong-strategy errors. The study that most cleanly isolates interleaving from spacing. Do not mash $d = 1.23$ with 2014's $d = 1.05$.

⁷⁰⁰Doug Rohrer, Robert F. Dedrick, Marissa K. Hartwig, and Chi-Ngai Cheung, “A Randomized Controlled Trial of Interleaved Mathematics Practice,” *Journal of Educational Psychology* 112, no. 1 (2020): 40–52, <https://doi.org/10.1037/edu0000367>. Preregistered cluster RCT, 54 seventh-grade classes, four months of assignments, unannounced test one month later, interleaved 61% vs blocked 38%, $d = 0.83$. Teachers implemented without training. Authors note that interleaved assignments also space same-kind items, so d is a joint effect. Do not mash with 2014's $d = 1.05$.

ics textbooks.”⁷⁰¹ Do not mash $d = 1.05$ (2014, two-week delay, $n = 140$, crossover) with $d = 0.83$ (2020, one-month delay, 54 classes, cluster RCT) with Dunlosky, Rawson, Marsh, Nathan, and Willingham 2013’s “high utility” rating for distributed practice. Different designs, different delays, interleaving confounded with spacing in the classroom studies. Brunmair and Richter’s later meta-analysis of interleaving, sometimes cited as a g for mathematical tasks, was not opened in full for this project. Exact coefficient **UNVERIFIED**. Do not recycle it.⁷⁰² Almost all clean classroom RCTs of interleaving in mathematics are grade 7. Do not export $d = 1.05$ to grade 4. Do not export it to calculus. Use it here.

What the parent does with that, operationally, is not a 180-day plan. Day of a new skill: a short blocked set, learn to execute. Same week: mixed retrieval of older facts and last month’s types. Later this week and next: the new skill reappears between unlike problems — a signed-fraction sum, a unit-rate, a “is this a function?”, a scale-drawing length. Monthly: an unannounced mixed quiz the child does without notes, which is the Rohrer test. A purchased book that is almost entirely blocked — most are — can still be interleaved by the parent pulling yesterday’s integers onto the same page as last month’s unit rates. Interleaving homework is a parent job. Section 6.7 will say the model can shuffle kinds; the parent holds the key. The child does the mixed page without a solver.

Schema-based proportion instruction, restated from 6.2 because the g has to sit next to the d without being added to it. Jitendra, Star, Dupuis, and Rodriguez 2012: $g = 1.24$ immediate, $g = 1.27$ at six weeks, null transfer, seventh grade, 1,163 students, 42 classrooms, trained proportion items.⁷⁰³ Teach the schema, represent it, estimate, solve, check. Teach unit rate *and* scaling *and* the proportion equation, and compare them. Do not expect free transfer. Do not quote Peltier and Vannest $g = 1.57$ as “schema instruction raises math achievement by 1.57 standard deviations.” That is immediate problem solving on mostly proximal measures.⁷⁰⁴ Word problems in this band still reject keywords. Never teach “per means divide” as a magic word. “Per” lives in unit rate *and* in “percent” *and* in “miles per hour squared” later. The structure, not the word.

Signed numbers on a line before slogans. Three representations of a function in one sitting. Worked-example fading for the first two-step linear equation; incorrect examples, Booth and colleagues 2013, hit the equal sign, minus as subtract, variable as a single mystery number — already Algebra I’s misconceptions, arriving early.⁷⁰⁵ For a child already behind on this topic, explicit, systematic instruction with models, practice, think-alouds, and feedback: NMAP Finding 27, and Fuchs et al. 2021 Recommendation 1. That “does not mean that all of a student’s mathematics instruction

⁷⁰¹Doug Rohrer, Robert F. Dedrick, and Marissa K. Hartwig, “The Scarcity of Interleaved Practice in Mathematics Textbooks,” *Educational Psychology Review* 32 (2020): 873–883. Most textbooks are almost entirely blocked. A purchased book can still be interleaved by the parent pulling unlike types onto the same page.

⁷⁰²Brunmair and Richter interleaving meta-analysis, sometimes cited as $g = 0.34$ for mathematical tasks: full paper not opened for this project (02-learning). Exact coefficient **UNVERIFIED**. Do not recycle the g . John Dunlosky, Katherine A. Rawson, Elizabeth J. Marsh, Mitchell J. Nathan, and Daniel T. Willingham, “Improving Students’ Learning with Effective Learning Techniques,” *Psychological Science in the Public Interest* 14, no. 1 (2013): 4–58, rate practice testing and distributed practice high utility and interleaved practice moderate utility (less classroom generality at the time of that review). A utility rating is not $d = 1.05$.

⁷⁰³Jitendra, Star, Dupuis, and Rodriguez 2012 (n. 609, n. 652).

⁷⁰⁴Peltier and Vannest 2017 (n. 653). Do not quote $g = 1.57$ as general achievement.

⁷⁰⁵Booth et al. 2013 (n. 667). Already Algebra I’s misconceptions, arriving in this band.

should be delivered in an explicit fashion.”⁷⁰⁶ Productive failure, Kapur 2014, is a different comparison: generation-then-instruction on a new concept when students have usable priors. Do not mash Finding 27 with Kapur. A child missing signed-fraction operations needs a model first. A child who has the operations and is meeting “which values make this identity always true?” can spend ten to twenty minutes generating, then receive the instruction. Frustration from missing facts is not productive. It is a fluency problem.

Adaptive practice engines — IXL, Khan Academy practice, ALEKS, DreamBox — sequence items. They are not agents in the Hermes Agent / Grok Bot sense, and they are not a substitute for the parent hearing the wrong constant of proportionality.⁷⁰⁷ DreamBox grade-level-per-hour claims are **vendor**; underlying ESSA study PDF **UNVERIFIED**. IXL live homeschool price **UNVERIFIED** as an invoice number. MATHia family purchase **UNVERIFIED**. Khan Academy content is free; Khanmigo is a paid constrained tutor, **vendor** “never gives you the answer,” **vendor** \$4/month or \$44/year as of 30 August 2026. Oreopoulos and Low, NBER w35620, August 2026: assignment to Khan Academy plus Khanmigo in 18 Tennessee middle schools raised MAP math by 0.040 population standard deviations per term; about 0.06 to 0.08 over a school year; authors: “These gains resemble those from Khan Academy practice without AI assistance.”⁷⁰⁸ Do not invent standalone 0.06 as the intention-to-treat. Do not advertise Khanmigo as Bloom’s two-sigma tutor. If Khanmigo is in the house, expect the Tennessee pattern — median student messaged it on only a third of practice days and in 17 percent of mistake sessions — unless the parent requires the hint dialogue. Full product landscape: Chapter 3. This section names the engines so they are not mistaken for a parent who can hear k .

Tips a parent can steal, each tied to a named source, not to folklore.

Do not teach cross-multiply as a belief. The implication is a consequence of a constant k or of a legal move on an equation. Magnitude first; compression after. IES 2010 Recommendation 4; IES

⁷⁰⁶NMAP, *Foundations for Success* (2008), Finding 27: explicit instruction (clear models, extensive practice, think-alouds, extensive feedback) has consistently positive effects for students with mathematical difficulties and for the lowest third of a typical class, on word problems and computation — and this “does not mean that all of a student’s mathematics instruction should be delivered in an explicit fashion.” Fuchs et al. 2021, Recommendation 1 (strong evidence, 43 studies): systematic instruction. Manu Kapur, “Productive Failure in Learning Math,” *Cognitive Science* 38, no. 5 (2014): 1008–1022: generation-then-instruction on a new concept; comparable procedural knowledge, greater conceptual understanding and transfer. Different comparisons. Do not mash.

⁷⁰⁷Adaptive practice engines (DreamBox, IXL, Khan Academy practice, ALEKS, MATHia) sequence items; they are not agents in the Hermes Agent / Grok Bot sense. Chapter 3 §3.2. DreamBox grade-level-per-hour claim is **vendor**; underlying ESSA study PDF **UNVERIFIED**. IXL live homeschool price **UNVERIFIED** as an invoice number. MATHia family purchase **UNVERIFIED**.

⁷⁰⁸Philip Oreopoulos and Nina Low, “One Click Away: AI Tutoring with Khanmigo in a Two-Year School Experiment,” NBER Working Paper 35620, August 2026, <https://www.nber.org/papers/w35620>. Two-year cluster RCT, 18 Tennessee middle schools. Assignment raises math achievement by 1.264 NPR per term (SE 0.597); 0.040 population SD per term; about 0.06 to 0.08 SD over a school year; implied effect of a full year of active participation 0.14 SD. “These gains resemble those from Khan Academy practice without AI assistance.” 96% tried Khanmigo at least once; median student messaged it on only a third of practice days and in 17% of mistake sessions; 14.5% of messages contained a mathematical question or a step of reasoning. Do not invent standalone 0.06 as the ITT. Do not advertise Khanmigo as Bloom’s two-sigma tutor. Working paper, not yet journal-peer-reviewed. Pre-registration: AEA RCT Registry AEARCTR-0013519. Khanmigo **vendor** price \$4/month or \$44/year, <https://www.khanmigo.ai/learners>, accessed 30 August 2026; **vendor** “Unlike ChatGPT, Khanmigo never gives you the answer.”

2021 Recommendation 4 obstacles; Jitendra 2012.⁷⁰⁹

Do not start Algebra I while fraction and integer operations are still fragile. Diagnose signed fractions, integer operations, and the equal sign before buying the book. The 2007 survey of 743 Algebra I teachers is the teacher-voice primary. Siegler and colleagues 2012 is the prediction. NMAP Table 2 is the pace, not a law.⁷¹⁰

Do not introduce functions as “ $y = mx + b$ ” only. 8.F is each input exactly one output. Three representations in one sitting. Linear versus nonlinear. Without this concept, $f'(x)$ has nothing to attach to.⁷¹¹

Do not treat Dimensions 6 as pre-algebra. **Vendor:** it is a Grade 6 text with some pre-algebra topics. Dimensions 7 and 8 together are the pre-algebra-plus-Algebra-I package. Saxon 5/4–8/7 are skill names. Math-U-See Pre-Algebra is a negatives-and-unknowns *level*. Beast Academy starts about a level behind. Placement tests exist because the labels lie.⁷¹²

Do not treat calendar-grade-8 Algebra I as identical to CCSSM Grade 8 EE/F. Those are not the same course. Compacted 7–8, Appendix A, is a model, not a mandate. Calculus senior year on a CCSSM-aligned path requires that high-school content started in Grade 8. Compact only if the gates are open.⁷¹³

Do not skip signed-fraction operations because the Algebra I book is on the shelf. Negative fractions are inside the Critical Foundation. The Panel’s Grade 7 benchmark is operations with signed fractions and percent, ratio, rate, proportionality.⁷¹⁴

Do not skip Grade 7–8 geometry and data because a family “starts Algebra I early.” NCTM Focal Points: do not omit the content. Similarity is slope’s justification and trigonometry’s gate. 8.SP is Chapter 11’s on-ramp.⁷¹⁵

Do not mash Jitendra’s $g = 1.24$ with Rohrer’s $d = 1.05$. Different comparisons, different outcomes, both grade 7, both this chapter. Do not quote Peltier and Vannest $g = 1.57$ as general achievement.⁷¹⁶

Do not import Rohrer’s d into grade 4 or into calculus. The practice of mixing types transfers, labelled as transfer of a finding. The coefficient stays in grade 7.⁷¹⁷

Do not replace schema work with shopping-math. Finding 26: context helps similar contexts. A child who can compute a tip and cannot write a unit rate has performed, not learned.⁷¹⁸

⁷⁰⁹IES 2010 Recommendation 4 (n. 645); Fuchs et al. 2021 Recommendation 4 obstacles (n. 654); Jitendra et al. 2012 (n. 609).

⁷¹⁰NMAP 2007 survey (n. 632); Siegler et al. 2012 (n. 613); NMAP Table 2 (n. 629). Diagnose signed fractions, integer operations, and the equal sign before buying Algebra I. Plant for Chapter 7.

⁷¹¹CCSSM 8.F (n. 682). Plant for Chapter 10.

⁷¹²Vendor grain: nn. 596–600, 634–637. A “grade 6” book is a publisher’s scope, not a legal grade.

⁷¹³Appendix A compacted pathways (nn. 610, 612). Models, not mandates. Compact only if the gates are open.

⁷¹⁴NMAP Table 2 Grade 7 (n. 629); Critical Foundation includes negative fractions (n. 631).

⁷¹⁵NCTM Focal Points warning (n. 627); CCSSM 8.G / 8.SP; plant for Chapters 8 and 11.

⁷¹⁶Jitendra 2012 $g = 1.24 / 1.27 /$ null transfer (n. 609); Rohrer 2014 $d = 1.05$ (n. 697); Peltier and Vannest $g = 1.57$ (n. 653). Different comparisons, different outcomes. Do not mash. Do not quote 1.57 as general achievement.

⁷¹⁷Almost all clean classroom RCTs of interleaving in mathematics are grade 7 (02-learning). The practice of mixing types transfers, labelled as transfer of a finding. The coefficient stays in grade 7.

⁷¹⁸NMAP Finding 26 (n. 657); Soderstrom and Bjork 2015 (n. 658).

Do not make the kitchen timer the identity of being good at this band’s mathematics, and do not abandon retrieval. Fluency is still flexible, accurate, efficient, appropriate. Brief timed retrieval of *already-known* integer facts and fraction operations, mixed, after they are right untimed, is IES 2021 Recommendation 6. Boaler’s object is a different object: high-stakes early speed tests. Facts and integer rules are not a language-model job.⁷¹⁹

Do not take the pencil. Maloney, Ramirez, Gunderson, Levine, and Beilock 2015 remains a school-homework study, not a parent-as-sole-teacher randomized trial: parent math anxiety plus frequent help predicted worse child math, math-specific; when help was infrequent, no such relation. A homeschool parent cannot “help less often” as a daily strategy. Prepare so you can hear a wrong *k*. Then the child generates.⁷²⁰ Retanal and colleagues 2021: a controlling-supportive homework-helping style partially mediated the relation in a survey of parents of 11- to 14-year-olds — this band’s ages. Replication status of Maloney 2015 is open; no registered replication opened. Flag as a single large, well-cited field study plus a 2021 mediation follow-up. Highest-priority homeschool-specific gap: no fetched experimental literature on math-anxious parents as full-time instructors.

When the parent is the ceiling, tutor, co-op, and a later dual-enrollment course are release valves, not a failure of the kitchen. NCES 2016: 23 percent any tutor; 31 percent any local homeschool group or co-op.⁷²¹ Video curricula exist because of the Maloney constraint. Teaching Textbooks’ “does 100% of the teaching” is a **vendor** claim — relevant to a parent who is anxious, not a finding that the parent can stop hearing wrong answers. Keep the parent as supervisor who can still hear a wrong inverse later. Do not outsource hearing.

What this section has done is put Rohrer in his home and refuse to start Algebra I on a fragile gate. The AI lane, next, reuses Chapter 3 on this band’s objects. The model may prepare the adult and the next problem. It may not do the child’s problem.

6.7 Parent-hosted AI for grades 6–8

One short reminder, then the objects of this band. Hamsa Bastani and colleagues, 2025: a ChatGPT-like tutor raised assisted practice grades 48 percent relative to control and then cut the unaided

⁷¹⁹Fuchs et al. 2021, Recommendation 6 (strong evidence, 27 studies): regularly include timed activities as one way to build fluency, on already-taught material, brief, mixed. Jo Boaler, “Timed Tests and the Development of Math Anxiety,” *Education Week / Stanford GSE* (2012); *Fluency Without Fear* (2015 working paper / Youcubed) — labelled working paper. Different objects. Facts and integer rules are not an LLM job (n. 735). Full fluency discussion in Chapter 2 and Chapter 5.

⁷²⁰Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, “Intergenerational Effects of Parents’ Math Anxiety on Children’s Math Achievement and Anxiety,” *Psychological Science* 26 (2015): 1480–1488, <https://doi.org/10.1177/0956797615592630>. School-homework study: parent math anxiety × frequent help predicted worse child math and more child math anxiety; when help was infrequent, no such relation; math-specific, not reading. Not a parent-as-sole-teacher RCT. Homeschool parent cannot “help less often” as a daily strategy. Do not treat the finding as “anxious parents should not teach.” Replication status open; no registered replication opened. Fraulein Retanal et al., *Education Sciences* 11, no. 10 (2021): 620: controlling-supportive homework-helping style partially mediated the relation in a survey of parents of 11- to 14-year-olds — this band’s ages. Highest-priority homeschool-specific gap: no fetched experimental literature on math-anxious parents as full-time instructors.

⁷²¹NCES 2020-001: any tutor 23 percent; any local homeschool group or co-op 31 percent. Release valves, not a failure of the kitchen. Teaching Textbooks **vendor** “does 100% of the teaching” is a vendor claim, relevant to Maloney, not a finding that the parent can stop hearing wrong answers.

exam 17 percent. A guardrailed tutor that withheld the full solution raised practice 127 percent and was statistically indistinguishable from control on the unaided exam. GPT Base, asked for the answer, was fully correct 51 percent of the time. Students copied anyway.⁷²² Soderstrom and Bjork 2015: current performance is “often an unreliable index” of learning.⁷²³ Full design, product landscape, ban-versus-allow list, six parent workflows, COPPA dates, detectors, companion-harm cases as allegations: Chapter 3. This section is a pointer, not a sequel. The workflows below are practice suggestions grounded in Bastani, Soderstrom and Bjork, and Oreopoulos and Low, not a new randomized trial of homeschool AI. No year-plus kitchen-table RCT of a parent-constrained GPT-Tutor workflow was opened on 30 August 2026.⁷²⁴

Allow, parent-hosted, child doing the work, parent holding the key.

Parent-first explanation of why a negative times a negative is a positive, with a number-line or pattern justification the parent must be able to hear. Chapter 3 workflow (a): explain the concept to the *parent* first. Bastani’s GPT Tutor worked in part because teachers had already written the solution and common mistakes into the hidden prompt. The parent is that teacher. Prefer a chat or reasoning model, not an open-web agent, for this job.⁷²⁵ Ask the model to give the pattern justification and the directed-magnitude justification from section 6.3, and then to name the wrong slogan the child is likely to recite. The parent reads that before the lesson. The child does not. The same workflow for why a ratio is a relationship, not a pair of counts: ask the model to build a ratio table for 2:3, a double number line, a graph through the origin, and the equation $y = (2/3)x$, and to say what a child who treats 2:3 as “two and three” will write for 4:□. The parent checks the table. Models hallucinate arithmetic. GPT Base had 8 percent arithmetic errors even when the method was right.⁷²⁶ A ratio table is a facts-adjacent object. The parent holds the key.

Isomorphic proportion items, parent-held key. Chapter 3 workflow (c): extra practice with the answer key held by the parent. Do not generate “practice” by completing tonight’s assigned homework. That is the ban.⁷²⁷ Ask the model for five proportion problems that share a structure with tonight’s type and differ in cover story — a speed, a recipe, a scale drawing, a percent increase, a

⁷²²Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman, “Generative AI without Guardrails Can Harm Learning: Evidence from High School Mathematics,” *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122, <https://doi.org/10.1073/pnas.2422633122>; author PDF https://hamsabastani.github.io/education_llm.pdf, accessed 30 August 2026. GPT Base: assisted practice +48% relative to control; unaided exam −17%. GPT Tutor: practice +127%; exam point estimate −0.004, not significant. GPT Base fully correct 51% of the time when asked for the answer; 8% arithmetic errors even when the method was right. Design: classroom-level RCT, large high school in Turkey, Fall 2023–24, ~1,000 students in grades 9–11, four 90-minute sessions, GPT-4-0613. Not a homeschool study, not GPT-5, not Khanmigo. Percentages are of control grades, not Cohen’s *d*. Full discussion: Chapter 3. This chapter is a pointer, not a sequel.

⁷²³Soderstrom and Bjork 2015 (n. 658).

⁷²⁴Parent workflows in this section are practice suggestions grounded in Bastani, Soderstrom and Bjork, and Oreopoulos and Low, not a new RCT of homeschool AI. No year-plus kitchen-table RCT of a parent-constrained GPT-Tutor workflow was opened on 30 August 2026 (04-ai open questions). Chapter 3 §3.3–3.4 is the lock; this section is the grade 6–8 pointer.

⁷²⁵Chapter 3 workflow (a): explain the concept to the *parent* first. Practice suggestion, not an RCT. Prefer a chat/reasoning model, not an open-web agent, for this job.

⁷²⁶Bastani et al. 2025 (n. 722): 8% arithmetic errors even when the method was right. A ratio table is a facts-adjacent object. The parent checks.

⁷²⁷Chapter 3 workflow (c): extra isomorphic practice with the answer key held by the parent. Do not generate “practice” by completing tonight’s assigned homework. That is the ban.

unit price — and for a worked key the child never sees. The child does two of them unaided. The parent diagnoses. Jitendra 2012’s null transfer is why the cover stories have to vary and why the parent should not expect a new schema to appear for free.⁷²⁸

A hint script that names “ratio as a pair of counts.” Chapter 3 workflow (d): Bastani’s GPT Tutor analogue. Withhold the full solution; ask what they have tried; point out a named mistake; escalate information slowly; accept equivalent answers.⁷²⁹ The homeschool analogue is a parent-read script, not a child-held chatbot. “You treated 2:3 as two and three. What stays the same when both parts of a ratio are scaled?” Then wait. Then a unit-rate prompt. Then, if needed, a table with one row filled. Not the missing number.

Diagnose a cross-multiply-on-everything paper. Chapter 3 workflow (f): diagnose a wrong answer the child already produced. Name the misconception; ask a question; retest with an isomorphic item. The child answers the parent, not the model.⁷³⁰ For a page on which every item was cross-multiplied, including a unit-rate that was one division, the sentence is “cross-multiply as a belief, not a consequence,” not “they forgot to simplify.” For $1/2 + (-1/3) = -1/5$, the sentence is “adding the counts you see, with a sign stuck on,” Ni and Zhou 2005, not “they forgot LCD.” For a graph of $y = 2x + 1$ on which the child plotted slope 1 and intercept 2, the sentence is “slope/intercept mixup,” labelled as teaching lore, prevalence **UNVERIFIED**, and the retest is “slope from (0, 1) and (1, 3).” Generate extra linear versus nonlinear graph sketches the parent checks. The model can describe a graph from a qualitative story; the parent verifies; the child sketches unaided.

Interleaving homework is a parent job. Mix yesterday’s integers with last month’s unit rates. The model can shuffle kinds — “give me a mixed page: two signed-fraction operations, two unit rates, one ‘is this a function?’, one slope from two points, one four-step linear equation, answer key separately.” The parent holds the key. The child does the page without a solver. That mixed page is the Rohrer assignment, in a kitchen, labelled as a practice suggestion, not as $d = 1.05$ sitting on the table.⁷³¹ Rohrer’s teachers implemented without training in 2020. The parent can implement without a product.

Hands-on as materials generation, Chapter 3 workflow (e): the model describes a double number line or a four-quadrant grid; the child draws. Carbonneau $d \approx 0.37$ is versus abstract-only, under guidance. Virtual versus physical manipulatives remain **UNVERIFIED**.⁷³²

Ban. Paste-the-prealgebra-worksheet. Photo-to-key. “What is the answer.” Open-web agent in the child’s hands. Companion persona. Photomath, Mathway, and Symbolab lead with scan-to-solution; no product-specific unaided-transfer RCT was opened this research day; the transferable

⁷²⁸Jitendra et al. 2012 null transfer (n. 609). Vary cover stories; do not expect a new schema to appear for free.

⁷²⁹Bastani et al., author PDF, Figure 1 (GPT Tutor prompt): withhold the full solution; ask what they have tried; point out a named mistake; escalate information slowly; accept equivalent answers. Homeschool analogue is a parent-read hint script, not a child-held chatbot. Chapter 3 workflow (d).

⁷³⁰Chapter 3 workflow (f): diagnose a wrong answer the child already produced. Practice suggestion, not an RCT. Name the misconception; ask a question; retest with an isomorphic item. The child answers the parent, not the model.

⁷³¹Rohrer et al. 2014 / 2020 (nn. 697, 700). Mixed page as a practice suggestion, not as $d = 1.05$ sitting on the table. 2020: teachers implemented without training.

⁷³²Chapter 3 workflow (e): hands-on activity as materials generation. The model describes; the child draws. Virtual vs physical manipulatives **UNVERIFIED**. Carbonneau $d \approx 0.37$ is versus abstract-only, under guidance (n. 670).

evidence is Bastani’s crutch plus the homepages.⁷³³ Hermes Agent and Grok Bot can browse the open web; they are adult and developer tools, not a child-safe school SKU. No documented K–12 student-facing autonomous open-web agent as a school product on the pages opened 30 August 2026. Hard no: messaging gateway to a child’s Telegram or WhatsApp; Grok Bot iOS app in the child’s hands; IEP or 504 PDFs on a shared Grok Bot computer; the child’s school portal or Photomath logins in an agent. El Salvador’s Grok-as-tutor is a ministry deployment of Grok *the assistant*, announcement not RCT, not Grok Bot, not a U.S. homeschool SKU.⁷³⁴ Adaptive is not an agent. IXL and Khan sequence items; they do not replace the parent hearing a wrong *k*.

Facts and integer rules are not an LLM job to memorize by chat. Flash retrieval of arithmetic facts, and of “a negative times a negative is a positive” as a retrieved compression, belongs to the parent’s ears, a dumb spaced-repetition deck, or a short mixed oral. Use the model to generate the deck or the oral script, not to sit between the child and the fact. Chapter 3 §3.7 / 04-ai load-bearing finding 12. GPT Base was fully correct 51 percent of the time, with 8 percent arithmetic errors even when the method was right. A language model that will happily hallucinate 7×8 is the wrong object to put between a child and an integer rule.⁷³⁵

Detectors fail. Liang, Yuksekgonul, Mao, Wu, and Zou 2023: average false-positive rate 61.22 percent (arXiv) / 61.3 percent (*Patterns*) on human TOEFL essays. Do not use detectors to police a homeschooler’s written explanations of a proportion or a signed-number move.⁷³⁶ Companion-harm cases — *Garcia v. Character Technologies*, *Raine v. OpenAI* — are allegations, not verdicts. They are why a math agent should not be a companion. Not a literature on mathematics learning. Full discussion: Chapter 3 §3.6. COPPA amended Rule published 22 April 2025, general compliance 22 April 2026. A parent using an adult tool on a parent account to generate a worksheet is a different fact pattern from creating a child account on a companion app. Homeschool parents are not verified U.S. K–12 educators by default and should not assume ChatGPT for Teachers or

⁷³³Photomath, <https://photomath.com/>, fetched 30 August 2026: scan including word problems; “A solution will pop up for your exact problem”; Basic \$0 / Monthly \$9.99 / Annually \$69.99 (USD, this day); **vendor** “designed for learning.” Mathway, <https://www.mathway.com/>: “Got a math problem? Get the steps.” Symbolab, <https://www.symbolab.com/>: solver first. No product-specific unaided-transfer RCT opened this day. Ban during the attempt: paste-the-worksheet, photo-to-key, “what is the answer.” Transferable evidence is Bastani’s crutch (n. 722) plus the homepages.

⁷³⁴xAI, “Grok Bot FAQ,” <https://docs.x.ai/grok-bot/faq>, fetched 30 August 2026: persistent cloud computer; “Grok Bot requires cloud data storage, so Legacy Privacy Mode is not supported”; one computer per user shared across Bots; no education SKU and no FERPA claim on the pages opened. Hermes Agent, <https://hermes-agent.org/>: self-hosted adult agent, open web, messaging gateways; no K–12 student-safety certification on the pages opened. El Salvador (xAI news, 11 December 2025) is a ministry deployment of Grok *the assistant*, announcement not RCT, not Grok Bot, not a U.S. homeschool SKU. No documented K–12 student-facing autonomous open-web agent as a school product on the pages opened 30 August 2026. Full distinctions: Chapter 3 §3.2. Hard no: messaging gateway to a child’s Telegram/WhatsApp; Grok Bot iOS app in the child’s hands; IEP/504 PDFs on the shared computer.

⁷³⁵Bastani et al. 2025: GPT Base fully correct 51%; 8% arithmetic errors even when the method was right. Flash retrieval of arithmetic facts, and of integer-rule compressions, is not an LLM job. Chapter 3 §3.7 / 04-ai load-bearing finding 12: use the model to generate the deck or the oral script, not to sit between the child and the fact. IES 2021 Recommendation 6 belongs to the parent’s ears, or a dumb SRS, not to a language model.

⁷³⁶Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou, “GPT Detectors Are Biased against Non-Native English Writers,” arXiv:2304.02819; *Patterns* 4, no. 7 (2023), <https://doi.org/10.1016/j.patter.2023.100779>. Average false-positive rate 61.22% (arXiv) / 61.3% (*Patterns*) on human TOEFL essays. Do not use detectors to police a homeschooler’s written mathematics explanations.

Claude for Teachers FERPA wrappers.⁷³⁷ California A.B. 1159 is not law as of 30 August 2026 (Senate third reading). Do not freeze it as a statute.

Session shape, Chapter 3 §3.8, scaled to this band. Practice suggestion, not an RCT. Unaided warm-up: one signed-fraction operation, one unit-rate, one “is this a function?” Teach. Struggle. Hint not key. Unaided exit ticket: a mixed three-item check without the model, which is Bastani’s exam phase in five minutes. “Finish your pre-algebra with Photomath so we can go to soccer” is GPT Base, the pattern with a large-sample causal harm estimate.⁷³⁸ If Khanmigo is in the house, expect the Tennessee pattern unless the parent requires the hint dialogue. 96 percent tried it at least once; the median student messaged it on only a third of practice days and in 17 percent of mistake sessions; 14.5 percent of messages contained a mathematical question or a step of reasoning.⁷³⁹ Practice minutes predicted MAP; chat did not, descriptively. Working paper, not yet journal-peer-reviewed.

A *hypothetical*, labelled as such, can put the ban in a room. A parent whose thirteen-year-old photographs tonight’s 7.RP worksheet into Photomath, copies the green steps, and scores 100 percent on the page, then cannot find the unit rate on tomorrow’s unaided three-item check, is running GPT Base.⁷⁴⁰ The empirical pieces are Bastani’s crutch, Photomath’s scan-to-solution homepage, and Soderstrom and Bjork’s performance/learning split. The age, the domain code, and the 100 percent are hypothetical. The move is not: take the phone, put the line and the table on paper, and run the unaided check.

Hermes Agent, if used, lives on the parent machine. Model split this day (Nous Portal): Hermes 4 or any strong chat model for parent explanation and hint scripts; agentic examples — Claude Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro — for the tool loop. Hermes 4 is **not recommended inside Hermes Agent** (tuned for chat, not rapid tool-calling). Hermes 4 ≠ Hermes Agent. Not a child-safe school SKU. Memory in `~/hermes/` is sensitive.⁷⁴¹ Grok Bot, if used, is included with SuperGrok and Cursor Pro/Teams as of the 26 August 2026 expansion post. Cloud storage required; Legacy Privacy Mode unsupported; shared computer per user; no education SKU and no FERPA claim on the pages opened. Grok ≠ Grok Bot ≠ El Salvador Grok-as-tutor. Hard no:

⁷³⁷*Garcia v. Character Technologies et al.*, complaint, M.D. Fla., via CourtListener/RECAP: allegations, not a verdict. *Raine v. OpenAI*, complaint: allegations, not a verdict. Why a math agent should not be a companion. Not a literature on mathematics learning. Full discussion: Chapter 3 §3.6. Federal Trade Commission, “Children’s Online Privacy Protection Rule,” 90 FR 16918 (22 April 2025), Document 2025-05904, 16 CFR 312. Effective date 23 June 2025; general compliance date 22 April 2026. Homeschool parents are not verified U.S. K–12 educators by default. California A.B. 1159 is not law as of 30 August 2026 (Senate third reading).

⁷³⁸Session shape: Chapter 3 §3.8 / 04-ai §9.4, scaled to this band. Practice suggestion, not an RCT. Grounded in Bastani’s exam phase (n. 722), Soderstrom and Bjork (n. 658), and Oreopoulos and Low’s engagement finding (n. 708). “Finish your pre-algebra with Photomath so we can go to soccer” is GPT Base.

⁷³⁹Oreopoulos and Low, NBER w35620 (n. 708). Tennessee pattern: 96% tried; median messaged on a third of practice days and in 17% of mistake sessions; 14.5% of messages contained a mathematical question or a step of reasoning. Descriptively, practice minutes predicted MAP; chat did not.

⁷⁴⁰Hypothetical labelled in the sentence that introduces it. Empirical pieces: Bastani’s crutch (n. 722); Photomath homepage (n. 733); Soderstrom and Bjork (n. 658). The age, the domain code, and the 100 percent are hypothetical.

⁷⁴¹Hermes Agent parent playbook: Chapter 3 §3.8 / 04-ai §9.1. Practice suggestion, not an RCT. Model split this day (Nous Portal): Hermes 4 or any strong chat model for parent explanation and hint scripts; agentic examples (Claude Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro) for the tool loop. Hermes 4 is **not recommended inside Hermes Agent** (tuned for chat, not rapid tool-calling). Hermes 4 ≠ Hermes Agent. Hard no: messaging gateway to a child’s Telegram/WhatsApp. Not a child-safe school SKU. Memory in `~/hermes/` is sensitive.

the child's hands.⁷⁴²

Diagnose fractions, including signed, integer operations, and the equal sign, before buying Algebra I. That sentence is the last teaching move of this chapter and the first of Chapter 7. The Panel's 2007 survey, Siegler and colleagues 2012, 8.F, and Rohrer's four problem kinds are the diagnostic objects. A child who can operate on signed rationals, see a ratio as a relationship, treat "=" as a relation, and assign to each input exactly one output is ready for the course Appendix A calls a more ambitious Algebra I than has generally been offered.⁷⁴³ A child who cannot is not ready, regardless of calendar grade, regardless of a publisher placement test, regardless of a chatbot that can print the next chapter's worked examples while the child watches. The agent does not take the test. The child does. Chapter 7 is the course. This chapter was the gate, closed or not.

⁷⁴²Grok Bot parent playbook: Chapter 3 §3.8 / 04-ai §9.2. Practice suggestion, not an RCT. Included with Super-Grok and Cursor Pro/Teams as of the 26 August 2026 expansion post. Cloud storage required; Legacy Privacy Mode unsupported; shared computer per user; no FERPA claim. Grok ≠ Grok Bot ≠ El Salvador Grok-as-tutor. Hard no: the child's hands. El Salvador is not this playbook (n. 734).

⁷⁴³Appendix A Algebra I: "The fundamental purpose of this course is to formalize and extend the mathematics that students learned in the middle grades. Because it is built on the middle grades standards, this is a more ambitious version of Algebra I than has generally been offered." Oregon ODE copy (n. 610). Door diagnostic: operate on signed rationals; see a ratio as a relationship; treat "=" as a relation; assign to each input exactly one output. Plant for Chapter 7. The agent does not take the test.

Chapter 7

Algebra I

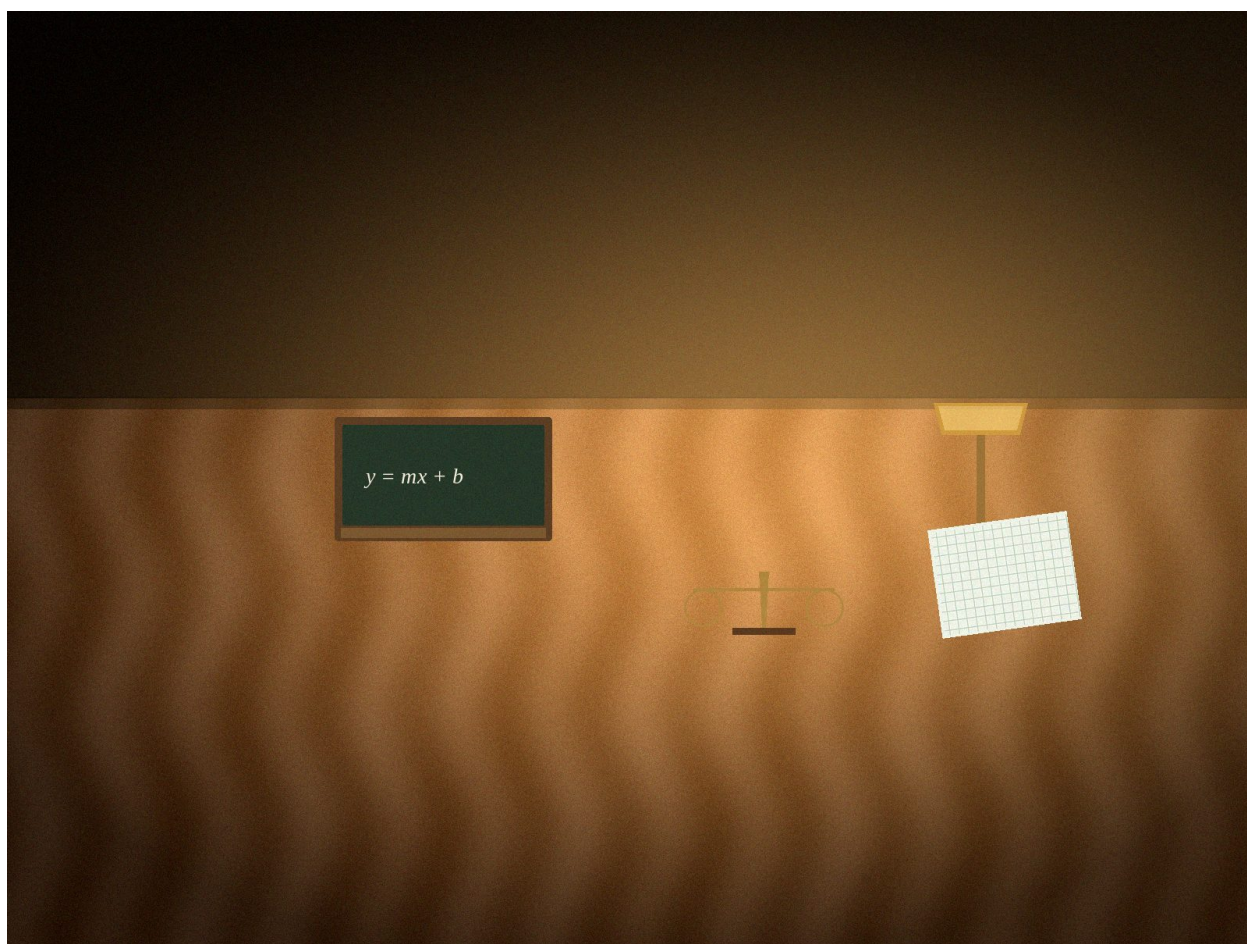


Figure 8: A chalkboard with $y=mx+b$, a balance scale, and a sheet of graph paper.

Two sentences from a model pathway, not a kitchen scene, already contain the course this chapter has to name. Achieve, in partnership with the Common Core writing team, published Appendix A as four model high-school pathways, not as a mandate. Of Algebra I the document says: “The fundamental purpose of this course is to formalize and extend the mathematics that students learned

in the middle grades. Because it is built on the middle grades standards, this is a more ambitious version of Algebra I than has generally been offered.”⁷⁴⁴ That is the object. It is not a birthday. It is not a publisher’s grade. It is not “the year after pre-algebra” as a franchise slogan. It is a more ambitious first high-school algebra course than has generally been offered, built on the middle-grades work Chapter 6 just finished, and it is a model, not a law. The pathways and courses are models, not mandates. The typical U.S. traditional organization is Algebra I, then Geometry, then Algebra II. An international-style integrated sequence of Mathematics I–II–III also exists. Compacted versions of each complete Grade 7, Grade 8, and the first high-school course in Grades 7–8 so that a student can reach Calculus or another college-level course by senior year.⁷⁴⁵ This chapter is the traditional first course. Integrated Mathematics I is the alternative packaging of much of this content plus geometry clusters. Name it. Do not write three integrated chapters. Appendix A’s integrated pathway was only partly extracted for this project; topic-by-topic Mathematics I remains **UNVERIFIED** pending a re-opening of Appendix A pages 49–90.⁷⁴⁶

When the course typically appears is a calendar fact, not a developmental law. Grade 9 in the non-compacted traditional pathway. Grade 8 in the compacted pathway, after a compacted Grade 7 that absorbs some Grade 8 content. Some students take it in Grade 7.⁷⁴⁷ The National Mathematics Advisory Panel recommended that all *prepared* students have access to an authentic algebra course by Grade 8; “authentic” means the Major Topics of School Algebra, not a course that is algebra in name only.⁷⁴⁸ Finding 15 of the Panel is still the age-lock refusal: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; claims that children of a given age cannot learn certain content because they are “too young,” “not in the appropriate stage,” or “not ready” have “consistently been shown to be wrong.”⁷⁴⁹ Accelerate a child who has the prerequisites. Do not withhold authentic algebra from a thirteen-year-old who has them. Do not start the course to keep up with public-school Grade 9 if the gates are closed.

⁷⁴⁴Achieve / Common Core writing team, *Appendix A: Designing High School Mathematics Courses Based on the Common Core State Standards*, Oregon Department of Education copy, <https://www.oregon.gov/ode/educator-resources/standards/mathematics/Documents/math-appendix-a-model-course-pathways.pdf>, accessed 30 August 2026. Algebra I: “The fundamental purpose of this course is to formalize and extend the mathematics that students learned in the middle grades. Because it is built on the middle grades standards, this is a more ambitious version of Algebra I than has generally been offered.” Pathways and courses are models, not mandates.

⁷⁴⁵Appendix A (n. 744). Four model pathways: Traditional (Algebra I, Geometry, Algebra II), Integrated (Mathematics I, II, III), and compacted versions of each that complete Grade 7 + Grade 8 + the first high-school course in Grades 7–8 so a student can reach Calculus or another college-level course by senior year.

⁷⁴⁶Integrated Mathematics I–III topic-by-topic from Appendix A was only partly extracted (01-scope open question 7). Topic-by-topic Mathematics I remains **UNVERIFIED** pending a re-opening of Appendix A pp. 49–90. Name the pathway; do not write three integrated chapters.

⁷⁴⁷Appendix A (n. 744); 01-scope, Algebra I band. Grade 9 in the non-compacted traditional pathway; Grade 8 in the compacted pathway; some students take it in Grade 7. Calendar placement is typical U.S. public-school placement, not a developmental law.

⁷⁴⁸National Mathematics Advisory Panel, *Foundations for Success: The Final Report of the National Mathematics Advisory Panel* (U.S. Department of Education, 2008), ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>, accessed 30 August 2026. Recommendation that all *prepared* students have access to an authentic algebra course by Grade 8; “authentic” means the Major Topics of School Algebra (Table 1), not a course that is algebra in name only.

⁷⁴⁹NMAP, *Foundations for Success* (2008), Finding 15: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; age-lock claims (“too young,” “not in the appropriate stage,” “not ready”) have “consistently been shown to be wrong.” Brain-immaturity claims are not justified even when students possess the prerequisite knowledge.

The National Center for Education Statistics already knew the sequence was stretched. In the 2016 Parent and Family Involvement tables, Algebra I was taught *that year* to 41 percent of the home-schooled 6–8 band and to 33 percent of the 9–12 band; arithmetic was still taught to 31 percent of the 9–12 band. Of high-school-level homeschoolers, 79 percent had ever been taught basic algebra (Algebra I).⁷⁵⁰ Grade labels in this book are sequence anchors, not prison bars. A child may sit this course at calendar grade 7 or at calendar grade 10. The 2023 First Look did not republish the subject tables. Whether the 2023 microdata still show the same shares is an open question this chapter will not fill. Algebra I at home is common. Algebra II and geometry are not universal. Calculus at home is a minority. A book that treats Algebra I as rare has left the table. A book that treats a calendar-grade-8 Algebra I as identical to the Common Core’s Grade 8 expressions, equations, and functions has left it the other way. Those are not the same course. Chapter 6 said so. This chapter is the course.

The parent arriving underprepared should diagnose before buying an Algebra I book. The diagnostic is content, not a birthday. Can the student operate on rationals, including signed fractions? See structure in an expression? Treat the equal sign as a relation, now inside a multi-step equation? A student who cannot is not ready, regardless of calendar grade.⁷⁵¹ The National Mathematics Advisory Panel named three Critical Foundations of Algebra: fluency with whole numbers; fluency with fractions, including decimals, percents, and negative fractions; and particular aspects of geometry and measurement, especially similar triangles, because slope and linear functions “depend logically” on them. Of these, “knowledge of fractions is the most important foundational skill not currently developed among American students.”⁷⁵² A nationally representative survey of Algebra I teachers the Panel commissioned in 2007 — 743 teachers — rated incoming preparation as poorest in “rational numbers and operations involving fractions and decimals,” alongside word problems and study habits.⁷⁵³ That survey is not “teachers today say.” It is the 2007 document the Panel used. Still the best nationally representative teacher-voice primary on *incoming* Algebra I gaps this project opened. Not a 2026 replication. Chapter 5 was the fractions gate. Chapter 6 closed signed numbers and the equal sign inside two-step equations and planted 8.F. This chapter will not rewrite them. It will use them as a door.

⁷⁵⁰Jiashan Cui and Rachel Hanson, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey*, NCES 2020-001 (Washington, DC: NCES, 2019), Table 9: Algebra I taught *that year* to 41 percent of the 6–8 homeschool band and 33 percent of 9–12; arithmetic still taught to 31 percent of 9–12. Table 8: high-school-level homeschoolers, subjects *ever* taught — basic algebra (Algebra I) 79 percent. <https://nces.ed.gov/pubs2020/2020001.pdf>. The 2023 First Look (NCES 2024-113) did not republish those subject tables.

⁷⁵¹01-scope, Algebra I “Depends on” and “Common gaps.” A student who cannot operate on rationals, cannot see structure in an expression, or treats the equal sign as “compute now” is not ready, regardless of calendar grade. Door diagnostic: fractions (including signed), integer operations, equal sign as relation. Chapters 5 and 6 do the teaching; this chapter uses the door.

⁷⁵²NMAP, *Foundations for Success* (2008), Executive Summary findings 4 and 12, and Chapter 4. Three Critical Foundations: fluency with whole numbers; fluency with fractions (including decimals, percents, and negative fractions); particular aspects of geometry and measurement. Quoted clause: “knowledge of fractions is the most important foundational skill not currently developed among American students.”

⁷⁵³NMAP, *Foundations for Success* (2008), nationally representative survey of Algebra I teachers, 743 teachers, 2007: poorest incoming preparation in “rational numbers and operations involving fractions and decimals,” alongside word problems and study habits. These are 2008 data, not a 2026 classroom census. Do not write “teachers today say.” Still the best nationally representative teacher-voice primary on *incoming* Algebra I gaps this project opened.

Four lanes run through the sections. They are not four identical headings. Sections 7.1 through 7.4 explain the mathematics to the parent — what the course is and when it appears, with the door diagnostic; algebra as generalized arithmetic, variables, the equal sign inside multi-step equations; linear, exponential, and quadratic families, systems, and modeling; descriptive statistics as the on-ramp to Chapter 11, not as AP Statistics. Section 7.5 is teaching guidance: fading worked examples, incorrect examples explained (Booth), comparison of methods, readiness. Section 7.6 is the tips that survive: gates, course titles on the transcript, contexts. Kitchen-math is not a linear function. Section 7.7 is the parent-hosted AI lane: a parent-first explanation of a transformation; isomorphic items, parent-held key; diagnose a wrong student paper; ban Photomath and solver-for-the-child on tonight's set. Symbolic tools are not the explainer. Point to Chapter 3. AP-style work later is unaided; even here, the exit ticket is unaided.

Chapter 2 already locked the method this chapter will use: five interwoven strands; conceptual and procedural knowledge iterating; worked examples faded for novices, incorrect examples explained; blocked practice to learn execution, interleaved practice to choose a strategy; word problems as schemas, not keywords; explicit instruction for students already struggling, productive failure as generation-then-instruction on a new concept when priors exist — those two literatures not the same comparison; kitchen-math transfer hard. Chapter 3 already locked the AI lane: unguarded solvers raise practice scores and cut unaided exams; the model may prepare the adult and the next problem; it may not do the child's problem. This chapter will not re-litigate Bastani at monograph length. One short reminder, then the objects of Algebra I. Chapter 4 already named the operational equal sign so this chapter would not have to discover it. Chapter 6 already planted 8.F. Do not mash the National Mathematics Advisory Panel's Finding 27 with Kapur 2014. Do not sell Saxon or Art of Problem Solving. Do not dump Chapter 9's polynomials and complex numbers here, and do not dump Chapter 12's NCAA worksheet. Plant both. No homeschool Algebra I randomized trial was opened for this project. Do not invent one.

The popular story collapses those locks into slogans they do not support. Algebra I is ninth grade. Letters stand for mystery numbers you solve for once. The equal sign means “compute now.” Functions are $y = mx + b$. Kitchen shopping is a linear function. Descriptive statistics can wait for the AP year. The video curriculum does 100 percent of the teaching, so the parent can stop hearing a wrong inverse. Photomath will tutor tonight's set. The documents do not say any of those things, not as the slogans say them. This chapter prints the documents. It prints the objects, the sample sizes, and the nulls. It leaves the open questions standing — Rittle-Johnson and Star 2007 and 2009 exact d **UNVERIFIED**; slope/intercept mixup prevalence **UNVERIFIED**; MATHia family purchase **UNVERIFIED**; Integrated Mathematics I topic-by-topic **UNVERIFIED**; no homeschool-curriculum randomized trial; authentic Grade 8 Algebra I versus waiting as a live pathway debate, not a kitchen experiment. It does not invent a kitchen scene to make the science feel warmer. The parent who stopped at algebra is still the design user. The job is to hear $(x + 2)(x + 3)$ written as $x^2 + 5x + 6$, and “=” treated as “the answer goes here,” before any tool is allowed to print a completed square the parent never sees.

The National Mathematics Advisory Panel's Major Topics of School Algebra are the catalog of what “algebra through Algebra II” contains, independent of course packaging: symbols and expressions; linear equations; quadratic equations; functions — linear, quadratic, polynomial, simple nonlinear, exponential, logarithmic, trigonometric, and fitting models to data; algebra of polynomi-

als, including complex numbers and the fundamental theorem of algebra; combinatorics and finite probability.⁷⁵⁴ This chapter is the first half. Linear, exponential, and quadratic functions live here. Polynomials beyond quadratic, complex numbers, logarithms, and trigonometry wait for Chapter 9. Combinatorics and finite probability are named so they are not forgotten; they are not this year's dump. A homeschool Algebra I that never reaches a function concept has not done the ambitious course Appendix A described. A homeschool Algebra I that pretends to have finished school algebra because the book's last chapter was the quadratic formula has not covered the catalog. The split is the honest one.

7.1 What the course is, when it appears, and the door diagnostic

Algebra I, as Appendix A writes it, has five critical areas. First: relationships between quantities and reasoning with equations — linear and simple exponential. Second: linear and exponential relationships, function notation, domain and range, systems. Third: descriptive statistics — shape, center, spread; two-way tables; linear models, residuals, correlation versus causation. Fourth: expressions and equations, including quadratics — structure, factoring, completing the square, quadratic formula with real solutions. Fifth: quadratic functions and modeling, compared with linear and exponential; absolute value, step, and piecewise-defined functions.⁷⁵⁵ Complex solutions of quadratics are recognized but not developed until Algebra II. That last clause is a boundary this chapter will keep. Completing the square for a real vertex is Algebra I. The complex plane is Chapter 9.

The content catalog, with Appendix A's limits, is the parent checklist, not a 180-day plan. N.Q: units and quantities. N.RN.1–3: rational exponents; sums and products of rationals and irrationals. A.SSE: structure of linear, exponential, and quadratic expressions; factor, complete the square, exponent properties. A.APR.1: arithmetic on linear and quadratic polynomials. A.CED: create linear, quadratic, exponential equations; constraints. A.REI: explain solving as reasoning; linear inequalities; quadratics with real solutions; linear systems and linear-quadratic systems; graphs of equations as solution sets. F.IF: function concept and notation; interpret key features; average rate of change; graph linear, quadratic, exponential, absolute value, step, piecewise. F.BF: build linear, exponential, quadratic functions; arithmetic and geometric sequences; vertical translations. F.LE: distinguish linear versus exponential growth; construct linear and exponential functions; parameters in context. S.ID.1–3, 5–9: univariate plots; two-way tables; scatter plots; linear fit; correlation versus causation.⁷⁵⁶ That is a year. It is not a lot because a publisher padded a catalog. It is a lot

⁷⁵⁴NMAP, *Foundations for Success* (2008), Table 1, p. 16 of the ERIC PDF. Major Topics of School Algebra: Symbols and Expressions; Linear Equations; Quadratic Equations; Functions (linear, quadratic, polynomial, simple nonlinear, exponential, logarithmic, trigonometric, and fitting models to data); Algebra of Polynomials (including complex numbers and the fundamental theorem of algebra); Combinatorics and Finite Probability. This chapter is the first half; polynomials beyond quadratic, complex numbers, logarithms, and trigonometry wait for Chapter 9.

⁷⁵⁵Appendix A (n. 744), Algebra I five critical areas: (1) relationships between quantities and reasoning with equations (linear and simple exponential); (2) linear and exponential relationships, function notation, domain and range, systems; (3) descriptive statistics; (4) expressions and equations including quadratics (structure, factoring, completing the square, quadratic formula with real solutions); (5) quadratic functions and modeling compared with linear and exponential; absolute value, step, and piecewise-defined functions. Complex solutions of quadratics recognized, not developed until Algebra II.

⁷⁵⁶Appendix A (n. 744) and 01-scope Algebra I content catalog with Appendix A limits: N.Q; N.RN.1–3; A.SSE; A.APR.1; A.CED; A.REI; F.IF; F.BF; F.LE; S.ID.1–3, 5–9. Not a 180-day plan. CCSSM is a map, not a homeschool mandate.

because the course formalizes the middle grades and then adds exponential and quadratic families the Common Core's Grade 8 did not require as a full course.

What the course depends on is the work Chapters 5 and 6 already named. The Panel's Critical Foundations: whole-number fluency, fraction fluency including signed fractions, decimals, and percents, similar triangles and linear measure for slope. The Common Core's Grade 8: linear equations, systems, functions, slope.⁷⁵⁷ A student who cannot operate on rationals, cannot see structure in an expression, or treats the equal sign as "compute now" is not ready, regardless of calendar grade. Restate that, because the calendar will try to overwrite it. The 743-teacher survey's top content gap was rational numbers. Siegler, Duncan, Davis-Kean, Duckworth, Claessens, Engel, Susperreguy, and Chen, 2012, is still the prediction that makes the gate load-bearing: elementary fraction and division knowledge uniquely predicted high-school algebra five or six years later, after the controls a skeptic would demand — a 1-SD increase in age-10 fractions knowledge uniquely associated with a 0.15-SD (United Kingdom) / 0.17-SD (United States) increase in high-school algebra.⁷⁵⁸ That is a *prediction* finding, not a classroom experiment proving that teaching fractions a certain way causes algebra success. Do not upgrade it. Do not skip the gate because an Algebra I book is on the shelf. Do not dump Chapter 5 here.

What the course unlocks is Geometry — coordinate proof, completing the square for circles and parabolas — and Algebra II — polynomial, rational, radical, logarithmic, trigonometric functions; complex numbers; inference. Without Algebra I's function concept, calculus's $f'(x)$ has nothing to attach to.⁷⁵⁹ That sentence is a plant for Chapters 8, 9, and 10. It is not a scare. Calculus is a new operation on functions the student already knows. A student with a shaky function concept will experience calculus as symbol soup. A student who can move among formula, graph, table, and verbal, in this course, has something for a derivative to operate on. Do not skip F-IF because the quadratic formula is the chapter the parent remembers as "algebra."

The door diagnostic, stealable, is three questions, not a publisher placement test. The placement test exists because labels lie; it is not a substitute for the three questions.

First: fractions, including signed. Can the child add, subtract, multiply, and divide signed fractions? Place $-3/4$ and $2/3$ on the same number line? Treat $1/2 + (-1/3)$ as a magnitude-and-sign problem, not as adding the counts you see? Chapter 5 taught unsigned fractions as numbers. Chapter 6 put the negatives on the same line and closed operations on signed fractions. The Panel's Table 2: proficient operations with positive and negative integers by the end of Grade 6; proficient

⁷⁵⁷01-scope, Algebra I "Depends on": NMAP Critical Foundations; CCSSM Grade 8 linear equations, systems, functions, slope. Slope-from-similar-triangles is the geometric prerequisite the Panel names for linear functions.

⁷⁵⁸Robert S. Siegler, Greg J. Duncan, Pamela E. Davis-Kean, Kathryn Duckworth, Amy Claessens, Mimi Engel, Maria Ines Susperreguy, and Meichu Chen, "Early Predictors of High School Mathematics Achievement," *Psychological Science* 23, no. 7 (2012): 691–697, <https://doi.org/10.1177/0956797612440101>; author PDF <https://sieglertc.columbia.edu/wp-content/uploads/2019/02/Siegler-et-al-PsySci12.pdf>, accessed 30 August 2026. British Cohort Study $N = 3,677$; U.S. PSID-CDS $N = 599$. A 1-SD increase in age-10 fractions knowledge uniquely associated with a 0.15-SD (UK) / 0.17-SD (US) increase in high-school algebra after other mathematical knowledge, general intellectual ability or working memory, and family income and education. *Prediction* finding, not a fractions-curriculum RCT. Do not upgrade it.

⁷⁵⁹01-scope, Algebra I "Unlocks": Geometry (coordinate proof, completing the square for circles/parabolas) and Algebra II. Without Algebra I's function concept, calculus's $f'(x)$ has nothing to attach to. Plant for Chapters 8–10, not a dump of those chapters.

operations with positive and negative fractions by the end of Grade 7 — judgments from curriculum comparison, not experimental placements.⁷⁶⁰ If those operations are still fragile, the Algebra I book is early. Diagnose. Then Chapter 6's band. Then this course.

Second: signed numbers more generally, already inside the first question, named separately because the slogans fail loudly. Can the child treat $3 + -2$ as a sum of a number and a signed number, not as “keep-change-change”? Write $(-3) \times (-2) = 6$ as a pattern that continues whole-number arithmetic, or as a directed-magnitude reversal, not as a chant? Parentheses around a negative, early and often, are a notation habit that later saves the minus-as-subtract error Julie L. Booth, Kenneth R. Koedinger, and colleagues listed among the algebra misconceptions incorrect examples were built to hit — Vlassis 2004 is the source they cite.⁷⁶¹ Hear it at the door.

Third: the equal sign as a relation. Can the child treat “=” as “names the same quantity,” now inside $-2x + 5 = 11$, and not as “compute now, the answer goes on the right”? Nicole M. McNeil and Martha W. Alibali's three entrenched operational patterns — perform all given operations on all given numbers; the “operations = answer” visual structure; “=” means “the total” — do not retire in Grade 8. Eric J. Knuth, Martha W. Alibali, Nicole M. McNeil, Ana Stephens, and colleagues connected a relational view of the equal sign to later algebra.⁷⁶² Chapter 4 installed the operational pattern in grades 1–2. Chapter 2 locked the method. This chapter will not rewrite them. It will use them. A child who can mechanically isolate x and cannot substitute back to check has a procedure and not the relation. Write $-2x + 5 = 11$, try $x = 3$, both sides 11, true. Try $x = 2$, $1 = 11$, false. Then the inverse steps are a compression of that search, not a dance. If the child cannot do the search, the book is early.

A *hypothetical*, labelled as such, can put the diagnostic in a room without inventing a reported family. A parent whose thirteen-year-old placed into a book titled Algebra I on a publisher test, and who then watches the child write $1/2 + (-1/3) = 1/5$, or $3 + -2$ as “three plus minus two, so subtract, I guess,” or $3 + 4 + 5 = 3 + \square$ as 15, is watching a gate failure, not a reason to “push through the algebra.”⁷⁶³ The empirical pieces that composite combines are in the documents: publisher place-

⁷⁶⁰NMAP, *Foundations for Success* (2008), Table 2: proficient operations with positive and negative integers by end of Grade 6; proficient operations with positive and negative fractions by end of Grade 7. Judgments from curriculum comparison, not experimental placements (the Panel says so). Geometry/measurement remainder of Table 2 was truncated in the fetch used for this project — remainder **UNVERIFIED**.

⁷⁶¹Julie L. Booth, Karin E. Lange, Kenneth R. Koedinger, and Kristie J. Newton, “Using Example Problems to Improve Student Learning in Algebra: Differentiating between Correct and Incorrect Examples,” *Learning and Instruction* 25 (2013): 24–34. Misconception list includes minus signs treated as “subtract” rather than as modifiers of terms; Joëlle Vlassis, “Making Sense of Signs of Subtraction and the Equal Sign in Algebra,” as cited by Booth et al. (Vlassis 2004). Combined with guided practice in the Algebra I Cognitive Tutor in that paper — school RCT, not a homeschool result.

⁷⁶²Nicole M. McNeil and Martha W. Alibali, “Why Won't You Change Your Mind? Knowledge of Operational Patterns Hinders Learning of Arithmetic Equations,” *Child Development* 76, no. 4 (2005): 883–899. Three entrenched operational patterns: perform all given operations on all given numbers; “operations = answer” visual structure; “=” means “the total.” Eric J. Knuth, Ana C. Stephens, Nicole M. McNeil, and Martha W. Alibali, “Does Understanding the Equal Sign Matter? Evidence from Solving Equations,” *Journal for Research in Mathematics Education* 37, no. 4 (2006): 297–312, connect a relational view of the equal sign to later algebra. Full teaching of the operational pattern: Chapters 2 and 4. This chapter uses the door, and does not rewrite those chapters.

⁷⁶³Composite *hypothetical*, labelled in the sentence that introduces it. Empirical pieces: publisher placement tests exist because labels lie (Chapter 1; 01-scope load-bearing 14); NMAP 2007 teacher survey (n. 753); McNeil and Alibali operational patterns (n. 762); Yujing Ni and Yong-Di Zhou, “Teaching and Learning Fraction and Rational Numbers:

ment tests exist because labels lie; the 2007 survey named rational numbers as the top incoming gap; McNeil and Alibali named the operational equal sign; Ni and Zhou named adding the counts you see. The sitting, the age, and the three wrong marks are hypothetical. The teaching move is not. Put the fractions on the line. Ask which values make the equation true. Then, and only then, return to the Algebra I book, or do not.

When the course appears, restated against the NCES tables so the calendar cannot pretend to be the mathematics. In the non-compacted traditional pathway, Grade 9. In the compacted pathway, Grade 8. Some students take it in Grade 7. NCES 2016: 41 percent of the 6–8 homeschool band were being taught Algebra I *that year*; 33 percent of the 9–12 band were; 79 percent of high-school-level homeschoolers had ever been taught it. Arithmetic was still being taught that year to 31 percent of 9–12.⁷⁶⁴ Some of those 41 percent had the gates. Some did not. The Panel’s 2007 survey is why the diagnostic is content. Finding 30 still allows prepared, motivated students to move faster, with no harm to their learning.⁷⁶⁵ Neither finding licenses a skip of signed fractions because a child is “good at math.” Authentic Grade 8 Algebra I — the Panel’s recommendation, Appendix A’s compacted model — versus waiting until fraction and integer fluency is real, is the live debate this section will not pretend to settle. Diagnose. Do not calendar.

Integrated Mathematics I is the other train. Appendix A publishes it as a model pathway that distributes the same high-school clusters across Mathematics I, II, and III rather than concentrating algebra in two years and geometry in one. A family on that train will not have a year titled Algebra I. They will still need the function concept, the linear and exponential and quadratic families, the systems, and the descriptive statistics this chapter names. They will also need geometry clusters this chapter will not write. Name the pathway. Transcript the course as Integrated Mathematics I, or Mathematics I, with a parenthetical a stranger can map onto Algebra I plus geometry — a plant for Chapter 12, not a dump of the NCAA worksheet.⁷⁶⁶ Do not write three integrated chapters. Topic-by-topic Mathematics I from Appendix A pages 49–90 remains **UNVERIFIED**. If the family’s book is integrated, map tonight’s lesson onto the objects in sections 7.2 through 7.4 and onto Chapter 8’s geometry objects. The mathematics does not change with the cover.

The National Collegiate Athletic Association’s math core courses are Algebra I or higher. Transcript the course as “Algebra I,” not “Saxon 8/7” without a parenthetical.⁷⁶⁷ That sentence is a plant for Chapter 12. This chapter says it as a naming rule. A stranger reading a transcript — a college

The Origins and Implications of Whole Number Bias,” *Educational Psychologist* 40, no. 1 (2005): 27–52. The sitting, the age, and the three wrong marks are not a reported family.

⁷⁶⁴Cui and Hanson, NCES 2020-001, Tables 8–9 (n. 750). Restated so the calendar cannot pretend to be the mathematics.

⁷⁶⁵NMAP, *Foundations for Success* (2008), Finding 30: mathematically gifted students with motivation “appear to be able to learn mathematics much faster... with no harm to their learning, and should be allowed to do so.” Acceleration requires prerequisites. Do not skip signed fractions because a child is “good at math.”

⁷⁶⁶Appendix A integrated pathway (nn. 744–746). Transcript plant for Chapter 12: course title a stranger can map onto Algebra I plus geometry. Not a dump of the NCAA worksheet.

⁷⁶⁷NCAA Eligibility Center, Home School Toolkit 2025–26: math core courses are Algebra I or higher. Division I: 16 core courses including 3 years math (Algebra I or higher); minimum 2.3 core GPA. Division II: 16 core including 2 years math (Algebra I or higher); minimum 2.2 core GPA. Homeschool packet: transcript, Administrator and Accordance Statement, one Core-Course Worksheet per core course. Audited, CLEP, or credit-by-exam courses are not NCAA-approved core courses. NCAA is a private association’s eligibility rules, not a statute. Full export rules: Chapter 12.

admissions officer, an NCAA eligibility staffer, a dual-enrollment registrar — needs a course title that maps onto the traditional pathway or the integrated one. “Fun with numbers” does not. “Saxon Algebra 1” with a parenthetical “Algebra I” does. Vendor names are tools. They are not transcript objects. Do not sell Saxon. Do not sell Art of Problem Solving. Name them as options, labelled **vendor**, by grain and placement, the way Chapters 1 and 6 already did.

Singapore Math Inc., **vendor**, tells parents that Dimensions 7 and 8 together cover pre-algebra and Algebra 1 topics with some geometry, and that Dimensions 8 expects more prior linear-equation work than a traditional Algebra 1. A U.S. parent using Dimensions 7–8 is buying something closer to Singapore’s earlier algebra, not to the Common Core’s Grade 7.⁷⁶⁸ Beast Academy / Art of Problem Solving, **vendor**, puts a student who completes Level 5 into AoPS Prealgebra, then Algebra. Teaching Textbooks, **vendor**, “does 100% of the teaching” and “100% of the grading” — a vendor claim, relevant to the parent who stopped at algebra, not a finding that the parent can stop hearing a wrong inverse.⁷⁶⁹ Math-U-See’s Algebra 1, **vendor** / Demme Learning, is a named level after Pre-Algebra. No homeschool-curriculum randomized trial comparing any of these was opened for this project. Choose by fit — parent math confidence, student stance, special needs, budget, college target — not by a phantom homeschool meta-analysis. Cathy Duffy Reviews and Rainbow Resource remain **catalogue** / **reviewer**, not independent effectiveness trials.

An international note, short, inherited from Chapter 6 so a parent using a non-U.S. sequence can translate “Algebra I” without inventing a two-year gap. A Singapore Secondary 2 (U.S. Grade 8) G3 student is typically into quadratic functions, which U.S. Common Core places in Algebra I / Grade 9, or Grade 8 compacted Algebra I. England’s Key Stage 3 already expects graphs of simple quadratics. Do not say that therefore Singapore or England is “two years ahead” as a single number. Different systems package quadratics, proof, and statistics differently. If the non-U.S. book is doing quadratic graphs in what it calls Grade 8, the family is not behind a U.S. Algebra I student — they are on a different train, often earlier on the algebra car.⁷⁷⁰

Authentic versus name-only is the Panel’s distinction, and it is the one a kitchen that titles a year “Algebra I” has to survive. A course that graphs $y = mx + b$, solves two-step equations with whole-number coefficients, and never reaches a function concept, an exponential table, or a quadratic with real solutions, is algebra in name. Appendix A’s ambitious version is not that course. Neither is a video year the parent never hears. The 2007 survey’s poorest incoming preparation was rational numbers; a name-only Algebra I that starts anyway produces the symbol soup those teachers described. Authenticity is content, not a cover. The Major Topics catalog is the independent

⁷⁶⁸Singapore Math Inc., School FAQ, <https://www.singaporemath.com/pages/school-faq>, accessed 30 August 2026. **Vendor**. Dimensions 7 and 8 together cover pre-algebra and Algebra 1 with some geometry; Dimensions 8 expects more prior linear-equation work than a traditional Algebra 1. Dimensions 6 is a Grade 6 text with some pre-algebra, not a pre-algebra text.

⁷⁶⁹Teaching Textbooks homepage, fetched 30 August 2026. **Vendor** claim: the program “does 100% of the teaching” and “100% of the grading.” Relevant to the Maloney constraint (n. 812); not a finding that the parent can stop hearing a wrong inverse. Art of Problem Solving, Beast Academy books and FAQ, <https://beastacademy.com/books>, accessed 30 August 2026. **Vendor**. A student who completes Level 5 has the background for AoPS Prealgebra. Math-U-See / Demme Learning, curriculum sequence, <https://mathusee.com/products/math-u-see-curriculum/>, accessed 30 August 2026. **Vendor**. Algebra 1 is a named level after Pre-Algebra.

⁷⁷⁰01-scope, International note. Singapore MOE G2/G3 syllabuses and TIMSS 2015 Encyclopedia: Secondary 2 (U.S. Grade 8) G3 typically into quadratic functions. UK Department for Education, national curriculum KS3 algebra: graphs of linear and quadratic functions. Do not say Singapore or England is “two years ahead” as a single number.

check. This year's half of the catalog is symbols and expressions through quadratic, linear equations and systems, linear and exponential and quadratic functions, and the descriptive-statistics on-ramp. Polynomials beyond quadratic, complex numbers, and the rest of the function list wait. A parent who can hear $(x + 2)(x + 3)$ written as $x^2 + 5$, who can hear "=" as compute-now, and who will not start while signed fractions are still fragile, is already doing the job the thesis named: understand the concept well enough to hear a wrong answer. The child still has to do the struggle before the tool. Section 7.7 will say how an agent is hosted. This section says what the year is.

What this section has done is place the course, refuse the calendar, and install the door. The mathematics, explained to the parent who stopped at algebra, comes next: letters stand for numbers, and the properties already known still hold.

7.2 Algebra as generalized arithmetic; variables; the equal sign inside multi-step equations

Algebra is generalized arithmetic. That sentence is the first teaching move of this course, and it is a sentence the parent who stopped at algebra has to be able to hear wrong. The letters stand for numbers, and the properties already known for numbers still hold.⁷⁷¹ $3(x + 2)$ is three times a sum, the distributive property the child already used on $3(10 + 2) = 36$, now with a letter. $a + b = b + a$ is the commutative property of addition, already known for $3 + 5 = 5 + 3$, now written so it is visible as a property of numbers, not as a trick for a particular pair. The child who treats $3(x + 2)$ as a blob, or who writes $3(x + 2) = 3x + 2$ because "the 3 only hits the first thing," has not generalized the arithmetic. The child who can expand and cannot say that the 3 hits both terms because multiplication distributes over addition has a procedure and not the generalization. Iterate. A short conceptual pass: the properties still hold. Then a handful of expansions. Then a return to the property. Then more. Do not withhold practice for weeks. Do not run pages of FOIL with no property and no why.

A variable is not a single mystery number. That is the gap Booth, Knuth, and Dietmar Küchemann documented, and it is the gap this course exists to close. Many students treat a variable as a single unknown object, not as a symbol that can take many values.⁷⁷² $3x + 2$, when x is 4, is 14. When x is -1 , it is -1 . The expression is the number; the x is a symbol that can take many values. $3x + 2$ treated as "do it" is the operational equal sign, still alive, now wearing a letter. Chapter 6 already named the expression as a number that depends on x . This chapter will not dump that section. It will use it. Function notation later this year — $f(x) = 3x + 2$ — is the same idea with a name for the correspondence. Domain is the set of inputs the correspondence will accept. Range is the set of outputs it produces. Those two words are not decorations. They are how the parent hears a wrong answer on "can x be -2 in $f(x) = 1/x$?" No. Division by zero is not a number. The domain excludes it. A child who can compute $f(3)$ and cannot say what f is has a chant.

The equal sign is a relation, now inside multi-step equations. McNeil and Alibali's three opera-

⁷⁷¹02-learning §1.6. Algebra as generalized arithmetic: letters stand for numbers and the properties already known for numbers still hold. Booth et al. 2013 and Knuth et al. 2006 document the gap (nn. 761–762).

⁷⁷²Booth et al. 2013 (n. 761), building on Lesley Booth 1984; Knuth et al. 2006 (n. 762); Dietmar Küchemann 1978. Many students treat a variable as a single mystery number, not as a symbol that can take many values. Küchemann 1978 was cited inside Booth et al.; this project did not re-open the 1978 primary.

tional patterns, already locked in Chapters 2 and 4, do not retire because the book's cover says Algebra I.⁷⁷³ Perform all given operations on all given numbers: $3 + 4 + 5 = 3 + \square$ becomes 15. The “operations = answer” visual structure: all operations left, blank right, so $11 = -2x + 5$ looks backwards. “=” means “the total.” Knuth, Stephens, McNeil, and Alibali connected a relational view of the equal sign to later algebra success. The equal sign is still a relation, now inside $-2x + 5 = 11$ and inside $(x + 2)(x + 3) = x^2 + 5x + 6$. Solving is not “move this to that side.” Solving is asking which numbers, substituted for the letter, make the two sides name the same quantity. Infinitely many solutions and no solution, planted in Chapter 6's 8.EE, are the first time the child meets an equation as a claim that can be always true or never true, not only sometimes true. $2x + 3 = 2x + 5$ is never true. $2x + 3 = 2(x + 1) + 1$ is always true. Substituting still decides. The operational equal sign, which wanted a number on the right, has nowhere to go. That is a feature.

Minus signs as modifiers of terms, not only as “subtract.” Joëlle Vlassis, 2004, as cited by Booth, Lange, Koedinger, and Newton 2013, is the source for the misconception the incorrect examples were built to hit.⁷⁷⁴ $3 + -2$ is not “three plus minus two, so subtract, I guess.” It is $3 + (-2)$, a sum of a number and a signed number. $-2x$ is a term whose coefficient is -2 , not a subtraction waiting to happen. $5 - (-2x)$ is $5 + 2x$, because subtracting a negative is adding the opposite, the move Chapter 6 already put on the line. A child who recites “two minuses make a plus” and cannot show the term is missing the modifier. Parentheses around a negative, early and often. Hear $3x - 2 = 10$ as $3x + (-2) = 10$ when the child starts “moving the 2” with the wrong sign. The incorrect example is the teaching move. Section 7.5 will lock it. This section names the object.

Structure in expressions, A.SSE, is seeing $(x + 2)(x + 3)$ as a product, not as “FOIL magic.” FOIL — first, outer, inner, last — is a mnemonic for a special case of the distributive property applied twice. It fails the moment the product is $(x + 2)(x + 3 + x)$, or $(x + 2)(x + 3)(x + 4)$, or $(2x - 1)(x^2 + x + 1)$. The parent who stopped at algebra often remembers FOIL and does not remember why $(x + 2)(x + 3) = x^2 + 5x + 6$. Two justifications the parent must be able to give, neither of them a kitchen scene, both of them mathematics.

The area-model justification. A rectangle with sides $x + 2$ and $x + 3$ has area $(x + 2)(x + 3)$. Split each side: an x -by- x square, an x -by-3 rectangle, a 2-by- x rectangle, a 2-by-3 rectangle. Areas: x^2 , $3x$, $2x$, 6. Sum: $x^2 + 5x + 6$. That is Chapter 4's array and Chapter 5's area as covering, now with a letter and a variable side. Put the rectangle next to the written product in the same visual field. Fade when the child is accurate without it.

The distributive-property justification. $(x + 2)(x + 3) = (x + 2)x + (x + 2)3 = x^2 + 2x + 3x + 6 = x^2 + 5x + 6$. The $x + 2$ hits both terms of $x + 3$, because multiplication distributes over addition. A child who writes $(x + 2)(x + 3) = x^2 + 5$ has added the insides and forgotten the product of the constants, or has added 2 and 3 and written them as a constant with no x term. That error is the one section 7.7 will ask a model to help the parent hear. It is not “they forgot FOIL.” It is “they did not distribute, or they treated the binomials as two numbers to be added.” Name the misconception. Then an isomorphic item. Then the child generates.

⁷⁷³McNeil and Alibali 2005; Knuth et al. 2006 (n. 762). Pointer to Chapters 2 and 4. Do not rewrite those chapters here.

⁷⁷⁴Vlassis 2004 as cited by Booth et al. 2013 (n. 761). Minus as a unary operator / modifier of terms, not only “subtract.”

Seeing structure also means seeing $x^2 + 5x + 6$ as $(x + 2)(x + 3)$, the reverse move, factoring, which is the same product read the other way. Completing the square is a different rewriting: $x^2 + 6x + 5 = (x + 3)^2 - 4$, because $x^2 + 6x$ wants a 9 to be a square, and 5 is 9 minus 4. The quadratic formula is the completing-the-square move, done once on $ax^2 + bx + c = 0$ and then used as a compression. Three rewritings of the same quadratic, not three franchises. Compare them. Rittle-Johnson and Star’s comparison-of-methods papers support the flexibility move; those 2007 and 2009 papers were not opened in full for this project, and their exact d values remain **UNVERIFIED**.⁷⁷⁵ The qualitative finding is enough for a kitchen rule: after the child can execute one method, put a second method next to it and ask which is better for this problem, and why. Factor $x^2 + 5x + 6$. Complete the square on $x^2 + 6x + 5$. Use the quadratic formula on $x^2 + x + 1 = 0$ and notice the discriminant is negative — real solutions do not exist; complex solutions are recognized, not developed. That last notice is the boundary. Do not dump Chapter 9.

Function notation, domain, and range, already named, now as objects the parent can hear wrong. $f(x)$ is not f times x . It is the output of f at input x . $f(x + 1)$ is the output at an input one more than x , a horizontal move, not “add 1 to the formula and hope.” $f(x) + 1$ is a vertical move. Those two are the first transformations this course will actually teach. A parent-first explanation of a transformation is section 7.7’s first allow. The mathematics lives here. $y = f(x) + k$ shifts the graph of f up by k if k is positive, down if negative. $y = f(x - h)$ shifts the graph of f right by h if h is positive. The minus in $x - h$ is the one the child will get wrong, because “minus means left.” It does not. $f(x - 2)$ takes the same output when the input is two larger, which is a shift right. Hear that. A short conceptual pass, a graph, a table, then a handful of items. Iterate. Absolute value, step, and piecewise-defined functions are Appendix A’s fifth critical area, not extras. $|x|$ is the distance from zero, already Chapter 6’s absolute value, now as a function whose graph is a V. A step function is constant on intervals and jumps. A piecewise function is different rules on different parts of the domain. Three representations in one sitting: table, graph, formula. Do not introduce functions as “ $y = mx + b$ ” only, so the general “each input exactly one output” idea never forms. Chapter 6’s 8.F was the correspondence. This course’s F-IF is the notation, the features, the average rate of change. Average rate of change of f from a to b is $[f(b) - f(a)] / (b - a)$. That is slope between two points on the graph of f , whether or not f is linear. For a linear function it is constant. For a quadratic it is not. That difference is how the three families will be compared in the next section.

A student who cannot operate on rationals, cannot see structure in an expression, or treats the equal sign as “compute now” is not ready, regardless of calendar grade. Restate the door, because this section is where the parent who bought the book anyway will watch the soup. Signed-fraction coefficients in $-(2/3)x + 1/4 = 5/6$ are not a new algebra topic. They are Chapter 6’s operations, unfinished, arriving as symbol soup. The 743 teachers already said so in 2007. Diagnose. Then the families.

What this section has done is name algebra as generalized arithmetic, a variable as a symbol that can take many values, and the equal sign as a relation inside multi-step equations. Linear, exponential, quadratic — three families, not one chant — come next.

⁷⁷⁵Bethany Rittle-Johnson and Jon R. Star, comparison-of-methods papers, 2007 and 2009: comparison improves flexibility. Papers not opened in full for this project; exact d **UNVERIFIED**. Qualitative finding is enough for the kitchen rule: after the child can execute one method, put a second next to it. Flagged in 02-learning §3.4 and in the book brief’s Chapter 7 do-not-overclaim.

7.3 Linear, exponential, quadratic, systems — and modeling

Three function families, compared, are the spine of this course. Linear. Exponential. Quadratic. Appendix A's first, second, and fifth critical areas are those families, plus systems, plus the absolute-value and piecewise extras already named.⁷⁷⁶ A child who can graph $y = 2x + 1$ from a chant and cannot say how that growth differs from $y = 2^x$, or from $y = x^2$, has one family and not a course. Distinguishing the families is F.LE and F.IF together: a linear function has a constant rate of change; an exponential function has a constant multiplicative rate; a quadratic has a constant second difference, a vertex, and a graph that is a parabola. Three representations in one sitting, inherited from Chapter 6's 8.F and now required of all three families: table, graph, formula, and a verbal description. If the table says $0 \rightarrow 3, 1 \rightarrow 5, 2 \rightarrow 7, 3 \rightarrow 9$, the first differences are constant 2, the rule is “start at 3, add 2 each time,” the graph is a line with slope 2 and intercept 3, and the equation is $y = 2x + 3$. If the table says $0 \rightarrow 3, 1 \rightarrow 6, 2 \rightarrow 12, 3 \rightarrow 24$, the ratios are constant 2, the rule is “start at 3, multiply by 2 each time,” the graph is not a line, and the equation is $y = 3 \cdot 2^x$. If the table says $0 \rightarrow 0, 1 \rightarrow 1, 2 \rightarrow 4, 3 \rightarrow 9$, the first differences are 1, 3, 5 and the second differences are constant 2, the rule is “square,” the graph is a parabola, and the equation is $y = x^2$. Ask all three in the same sitting once each family exists. Then mix them. Interleave problem kinds that share a surface and differ in strategy. Rohrer, Dedrick, and Burgess 2014 treated “graph $y = mx + b$ ” and “slope from two points” as different problem kinds that students fail to associate when practice is blocked — grade 7, $n = 140$, interleaved 72 percent versus blocked 38 percent, $d = 1.05$, two-week delay.⁷⁷⁷ That d stays in Chapter 6. Do not re-quote it as if Rohrer ran Algebra I. The practice of mixing types transfers, labelled as transfer of a finding. The coefficient does not.

Linear functions first, because they are the family Chapter 6 already planted and this course has to formalize. $y = mx + b$. m is the rate of change, slope as unit rate, grounded in similar triangles — the Panel's geometry foundation.⁷⁷⁸ b is the initial value, the intercept, the output when the input is zero. A proportional relationship is the special case $b = 0$, graph through the origin, Chapter 6's $y = kx$. $y = mx + b$ is that relationship translated. Slope/intercept mixup — taking b as slope because it is the first number the child sees, or reading rise/run off the axes rather than the line — is pervasive in secondary teaching lore. This project did not open a single primary that quantified prevalence. Flag as **UNVERIFIED** as to a named prevalence study.⁷⁷⁹ Rohrer's materials are themselves evidence that the association “this story to this strategy” is weak when practice is blocked. Do not start the function concept as “the graph looks like.” A graph that “looks like a line” is not yet a function.

⁷⁷⁶Appendix A Algebra I critical areas 1, 2, and 5 (n. 755). Linear vs exponential vs quadratic; systems; absolute value, step, piecewise.

⁷⁷⁷Doug Rohrer, Robert F. Dedrick, and Kaleena Burgess, “The Benefit of Interleaved Mathematics Practice Is Not Limited to Superficially Similar Kinds of Problems,” *Psychonomic Bulletin & Review* 21 (2014): 1323–1330. Grade-7 classroom crossover, $n = 140$, interleaved 72 percent vs blocked 38 percent, $d = 1.05$, 95 percent CI [0.80, 1.30], unannounced test two weeks later. Four kinds included “graph $y = mx + b$ ” and “slope from two points.” The d stays in Chapter 6. Do not re-quote it as if Rohrer ran Algebra I. Transfer of the *practice* of mixing types, labelled as transfer of a finding.

⁷⁷⁸NMAP, *Foundations for Success* (2008), Critical Foundations, geometry/measurement: similar triangles, because slope and linear functions “depend logically” on them. Chapter 6 planted slope as unit rate; this course formalizes $y = mx + b$.

⁷⁷⁹Slope/intercept mixup prevalence: **UNVERIFIED** as to a named prevalence study. 02-learning §1.6: no single primary quantifying prevalence was opened. Rohrer et al. 2014 materials are evidence that the association is weak when practice is blocked.

A scatter that “looks like a line” is an association, S.ID, not a function until someone specifies the rule. Plant Chapter 11. Do not dump it.

Systems of linear equations are three representations again: a graph of two lines and their intersection; an algebraic elimination or substitution; an inspection of structure. Chapter 6 planted $x + y = 5$ and $x - y = 1$. This course formalizes. Two lines may meet at one point (one solution), be parallel (no solution), or be the same line (infinitely many). Substituting still decides. Linear inequalities, in one variable and in two, extend the same relation: the solution is a set, a ray on a line or a half-plane, not a single number. Linear-quadratic systems — a line and a parabola — are Appendix A’s reach, not Algebra II’s. Graph them. Algebra them. Real intersection points, zero or one or two. Complex intersections wait.

Exponential functions are the family the parent who stopped at algebra is most likely to remember as “the one with the exponent,” without being able to hear as a constant multiplicative rate. $y = a \cdot b^x$, $a \neq 0$, $b > 0$, $b \neq 1$. Arithmetic sequences have a constant difference; geometric sequences have a constant ratio. F.BF plants both as discrete versions of linear and exponential. Simple interest is linear in time if the interest is a fixed amount; compound interest is exponential if the interest is a fixed rate on the current amount. Distinguishing those two is F.LE, and it is a modeling move, not a worksheet of “write the formula.” A population that doubles every so many years, a radioactive quantity that halves, a bank balance at a stated rate: similar contexts. The Panel’s Finding 26 still applies: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving.⁷⁸⁰ Compound interest will not teach 2^x . The table of ratios, the graph that is not a line, and the comparison with $2x$ will. Use the context as a second representation of a growth the child is already being taught as a family. Do not replace the family with a story.

Quadratic functions are the family this course adds that Grade 8 on the non-compacted map did not require as a full course. $y = ax^2 + bx + c$, $a \neq 0$. A parabola. A vertex. An axis of symmetry. Real zeros, zero or one or two, visible as x -intercepts. Completing the square puts the vertex in sight: $y = a(x - h)^2 + k$, vertex (h, k) . Factoring puts the zeros in sight: $y = a(x - r)(x - s)$, zeros r and s . The quadratic formula puts the zeros in sight when factoring is unfriendly: $x = [-b \pm \sqrt{b^2 - 4ac}] / (2a)$. The discriminant $b^2 - 4ac$ tells how many real zeros: positive, two; zero, one; negative, none real. Complex solutions are recognized — “the square root of a negative is not real; we will have a name for it later” — and not developed.⁷⁸¹ That is the honest boundary. A parent who pushes the quadratic formula past a negative discriminant into “just write i ” has started Chapter 9. A parent who pretends the discriminant is always positive has lied about quadratics.

Three methods, compared, not ranked. Factor when the numbers are friendly. Complete the square when the vertex is the question, or when the formula is being derived. Use the formula when the numbers are not friendly. Flexibility is a third outcome, not a luxury. Schneider, Rittle-Johnson,

⁷⁸⁰NMAP, *Foundations for Success* (2008), Finding 26: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving. Kitchen-math is not a linear function. Do not claim kitchen-math transfers to algebra.

⁷⁸¹Appendix A (n. 755): complex solutions of quadratics recognized, not developed until Algebra II. Boundary this chapter keeps. Chapter 9 develops complex numbers.

and Star modeled it as a latent variable after a three-day classroom intervention on equation solving; each type of knowledge predicted gains in the other.⁷⁸² After the child can execute one method, put a second next to it. “Which is better for $x^2 + 5x + 6 = 0$? For $x^2 + 6x - 1 = 0$? For $2x^2 + 3x - 7 = 0$?” That is the comparison move. Exact d from Rittle-Johnson and Star 2007 and 2009 remains **UNVERIFIED**. The kitchen rule does not need the d .

Modeling, in this course, is specifying a family, fitting it to a situation or to data, and checking against the situation. F.LE and S.ID share the job. A verbal description — “the water in the tub rises steadily, then the tap is off, then it drains faster than it filled” — is a qualitative sketch, 8.F.5 inherited, now asked of piecewise and of the three families. A table of years and populations may be closer to exponential than to linear; residuals from a linear fit that fan or curve are a reason to try a different family. Residuals belong fully in section 7.4. They are planted here because modeling is not “write an equation from a word problem and stop.” NMAP Finding 26’s caution still applies: context helps similar contexts. Kitchen-math is not a linear function. A recipe that doubles is a scaling, a proportional relationship, $y = kx$, if both quantities scale; it is not $y = mx + b$ with a nonzero intercept unless something is being added that does not scale. Shopping-math can be a percent, a unit rate, or a linear cost with a fixed fee. Name the family. Do not call every kitchen story “algebra.” Soderstrom and Bjork 2015: current performance is “often an unreliable index” of learning.⁷⁸³ A child who can compute a 20 percent tip and cannot write $y = 0.20x$ as a linear function through the origin, or who writes $y = 0.20x + 20$ because “there is always a 20,” has performed, not modeled.

Interleave kinds that share a surface. A page that is only “solve the quadratic by factoring” teaches execution. A page that mixes “is this linear, exponential, or quadratic?,” “solve by factoring,” “solve by formula,” “fit a line to these four points,” and “does this system have one solution, none, or infinitely many?” teaches choosing. Day of a new skill: a short blocked set, learn to execute. Same week: mixed retrieval of older facts and last month’s types. Later this week and next: the new skill reappears between unlike problems. Monthly: an unannounced mixed quiz the child does without notes. A purchased book that is almost entirely blocked — most are; Rohrer, Dedrick, and Hartwig 2020 documented the scarcity of interleaved practice in mathematics textbooks — can still be interleaved by the parent pulling yesterday’s linear system onto the same page as last month’s exponential table.⁷⁸⁴ Interleaving homework is a parent job. Section 7.7 will say the model can shuffle kinds; the parent holds the key. The child does the mixed page without a solver.

Absolute value, step, and piecewise-defined functions, restated as graphs the parent can hear wrong.

⁷⁸²Michael Schneider, Bethany Rittle-Johnson, and Jon R. Star, “Relations among Conceptual Knowledge, Procedural Knowledge, and Procedural Flexibility in Two Samples Differing in Prior Knowledge,” *Developmental Psychology* 47, no. 6 (2011): 1525–1538. Conceptual, procedural, and flexibility as latent variables after a three-day classroom intervention on equation solving; each predicted gains in the other; relations symmetrical. Flexibility is a third outcome, not a luxury. Cited in Chapter 2; used here, not rewritten.

⁷⁸³Nicholas C. Soderstrom and Robert A. Bjork, “Learning Versus Performance: An Integrative Review,” *Perspectives on Psychological Science* 10, no. 2 (2015), <https://doi.org/10.1177/1745691615569000>. Current performance is “often an unreliable index” of learning. Full discussion: Chapter 3.

⁷⁸⁴Doug Rohrer, Robert F. Dedrick, and Marissa K. Hartwig, “The Scarcity of Interleaved Practice in Mathematics Textbooks,” *Educational Psychology Review* (2020). Most textbooks are almost entirely blocked. Rohrer, Dedrick, Hartwig, and Chi-Ngai Cheung, 2020, *Journal of Educational Psychology*: preregistered cluster RCT, 54 seventh-grade classes, interleaved 61 percent vs blocked 38 percent, $d = 0.83$, one-month delay. Do not mash $d = 1.05$ with $d = 0.83$. Both stay in Chapter 6 as coefficients; the practice of mixing types is used here.

$y = |x - 2|$ is a V with vertex at (2, 0), because absolute value is distance. $y = |x| + 2$ is a V shifted up. $y = |x + 2|$ is a V shifted left, the same minus-means-left error as $f(x + 2)$. A step function used for postage or for a tax bracket is constant on an interval and then jumps; the graph has open and closed circles the child must attend to, which is domain and range in a picture. Piecewise: one rule for $x < 0$, another for $x \geq 0$, is how a “rises then drains” tub becomes an equation. Do not treat these as extras after the quadratic formula. Appendix A put them in the fifth critical area. A course that never graphs a V has not done the ambitious version.

Arithmetic and geometric sequences are the discrete versions of the first two families, and they belong in the same sitting as the continuous-looking formulas. An arithmetic sequence 3, 5, 7, 9, ... has first term 3 and common difference 2; the n th term is $3 + (n - 1) \cdot 2$, which is linear in n . A geometric sequence 3, 6, 12, 24, ... has first term 3 and common ratio 2; the n th term is $3 \cdot 2^{(n - 1)}$, which is exponential in n . F.BF asks the child to build functions from those sequences. A child who can write the explicit formula and cannot say which family the sequence belongs to has a chant. A child who can say “geometric, so exponential” and cannot produce the formula has the slogan. Three representations again: the list, the formula, the graph of the discrete points. Do not treat sequences as a separate franchise chapter. They are the families, counted.

Average rate of change is how the three families are told apart when a table is all the child has. From $x = 0$ to $x = 2$ on $y = 2x + 1$, the average rate is 2, and it is 2 on every interval, because the function is linear. On $y = 2^x$, from 0 to 2 the average rate is $(4 - 1)/2 = 1.5$; from 2 to 4 it is $(16 - 4)/2 = 6$. The rate is not constant. On $y = x^2$, from 0 to 2 it is 2; from 2 to 4 it is 6. Also not constant; the second differences of a quadratic table are. A child who computes one average rate and declares the function linear has not compared families. Interleave “compute the average rate on this interval” with “which family has a constant average rate?” Rohrer’s lesson, now on Algebra I objects, labelled as transfer of a finding.

What is *not* yet Algebra II, named so this chapter does not dump Chapter 9. Polynomials beyond quadratic. Rational expressions as arithmetic of fractions in letters, beyond A.APR.1’s linear and quadratic polynomials. Complex numbers and the fundamental theorem of algebra. Logarithms as solutions of exponentials. Trigonometric functions on the unit circle. The remainder theorem. Inverse functions as a general machine. Those objects wait. A homeschool Algebra I that omits the function concept has not done this course. A homeschool Algebra I that pretends to have finished school algebra because the quadratic formula appeared has not covered the Panel’s Major Topics. The first half is this chapter. The rest is Chapter 9.

What this section has done is put three families on the table and refuse to let $y = mx + b$ swallow them. Descriptive statistics are in this course. They are not “Chapter 11’s problem.”

7.4 Descriptive statistics in Algebra I — the on-ramp to Chapter 11

Algebra I already includes S.ID.1–3 and 5–9. Shape, center, spread. Two-way tables. Scatter plots. Linear fit. Residuals. Correlation versus causation.⁷⁸⁵ Chapter 11 is the through-line and the AP course. These clusters are the on-ramp. Do not skip them because “stats is Chapter 11.” A skipped

⁷⁸⁵Appendix A Algebra I; CCSSM S.ID.1–3, 5–9. Do not skip because “stats is Chapter 11.” The Algebra I clusters are the on-ramp; Chapter 11 is the through-line and the AP course.

K–8 data strand will show up here as “we never asked a question that has variability.” Chapter 6 planted Grade 6 distributions, Grade 7 sampling, and Grade 8 bivariate association. This section uses them. It does not write AP Statistics.

What the clusters actually contain is more specific than “do some averages.” S.ID.1–3: represent data with plots on the real number line — dot plots, histograms, box plots; compare center (median, mean) and spread (interquartile range, standard deviation) of two or more data sets, using shape, outliers, and the context; interpret differences in shape, center, and spread in the context of the data, accounting for possible effects of extreme data points.⁷⁸⁶ Mean, median, and mode as a triad with no data story is not this cluster. A question that has variability — how long did these sunflower seedlings grow in a week? — is. GAISE, the American Statistical Association’s Guidelines for Assessment and Instruction in Statistics Education, frames school statistics as formulate questions, collect, analyze, interpret, with variability as the central idea. GAISE II was not opened in full for this project; page-level claims stay **UNVERIFIED**. Treat the spirit, not a quoted recommendation list.⁷⁸⁷ Chapter 11 will gather the through-line.

S.ID.5: two-way frequency tables for two categorical variables; relative frequencies; possible association. S.ID.6: represent two quantitative variables on a scatter plot; describe how the variables are related; fit a function to the data, informally; informally assess the fit by plotting and analyzing residuals; fit a linear function for a scatter plot that suggests a linear association. S.ID.7: interpret the slope and the intercept of a linear model in the context of the data. S.ID.8: compute (using technology) and interpret the correlation coefficient of a linear fit. S.ID.9: distinguish between correlation and causation.⁷⁸⁸ Those last four are the modeling half of section 7.3, now with data that did not come from a clean function. A residual is the difference between an observed y and the y the line predicts. A residual plot that looks random is consistent with a linear model being reasonable. A residual plot that curves is a reason to doubt the linear family. Correlation r is a number between -1 and 1 that measures the strength and direction of a linear association. $r = 0.9$ does not mean X causes Y . Random assignment versus random sampling is the design distinction Chapter 11 and AP Statistics Unit 1 will make precise. This course already says the slogan: correlation is not causation. Hear a child say “the correlation is 0.9 so X causes Y ” and the sentence is the slogan, not “they computed r wrong.”

AP Statistics’ official prerequisite as of the Fall 2026 Course and Exam Description is first-year algebra. College Board: “any secondary school student who has successfully completed a first-year algebra course.” The former second-year-algebra prerequisite is gone in this CED.⁷⁸⁹ This course is that prerequisite. AP Statistics is “equivalent to a one-semester, introductory, **non calculus-based**

⁷⁸⁶ CCSSM S.ID.1–3, as listed in 01-scope Algebra I catalog and the Statistics band. Shape, center (median, mean), spread (IQR, standard deviation), outliers, context.

⁷⁸⁷ American Statistical Association, GAISE / GAISE II, cited inside the AP Statistics CED as alignment for statistical practices. GAISE II was not opened in full for this project; page-level claims **UNVERIFIED**. Treat the spirit (formulate, collect, analyze, interpret; variability as the central idea), not a quoted recommendation list.

⁷⁸⁸ CCSSM S.ID.5–9, as listed in 01-scope. Two-way tables; scatter plots; linear fit; residuals; interpret slope and intercept in context; correlation coefficient (using technology); correlation versus causation.

⁷⁸⁹ College Board, *AP Statistics Course and Exam Description*, Effective Fall 2026, <https://apcentral.collegeboard.org/media/pdf/ap-statistics-course-and-exam-description.pdf>, accessed 30 August 2026. Prerequisite: “any secondary school student who has successfully completed a first-year algebra course.” Former second-year-algebra prerequisite gone in this CED. This course is that prerequisite.

college course in statistics.” The revised framework encourages taking both AP Statistics and AP Calculus when the student has the background. It is not published as a substitute for calculus in STEM calculus sequences.⁷⁹⁰ Do not treat AP Statistics as stealth Algebra II. Do not treat this section as stealth AP Statistics. Do not treat this section as stealth calculus. The five AP Statistics units, effective Fall 2026 — exploring one-variable data and collecting data; probability, random variables, and distributions; inference for categorical data, proportions; inference for quantitative data, means; regression analysis — wait for Chapter 11. Inference, p -values, experimental design at AP grain, the normal model as S.ID.4 (an Algebra II cluster), are not this year’s dump. Shape, center, spread, two-way tables, a linear fit, residuals, and correlation-versus-causation are.

A skipped K–8 data strand is Unit 1 retrofit later, and it is a hole in this course now. Grade 6 asked for a statistical question, a distribution, a median and a mean, an interquartile range and a mean absolute deviation, a shape. Grade 7 asked for random sampling and informal comparative inferences and probability models. Grade 8 asked for scatter plots and informal linear models. If those years were “we’ll do stats later,” this year’s S.ID is the later, and it will feel like a new subject. It is not a new subject. It is the on-ramp. Teach it. A *hypothetical*, labelled as such: a parent who skipped every data lesson in the middle-grades book because “the algebra chapters are the real math,” and who then meets a scatter plot in the Algebra I book with no vocabulary for association, is meeting a hole the map already named.⁷⁹¹ The empirical pieces are the cluster list and the K–8 spine. The skipping and the sitting are hypothetical. The move is not: ask a question that has variability, plot, name the shape, fit a line only if the scatter suggests one, and say out loud that the line is a model, not a cause.

Kitchen surveys are real data. They do not by themselves teach S.ID. Finding 26 analogue: similar-context help, not transfer to a residual plot the child has not been taught to read. A family height-versus-age scatter can be a second representation of a linear association the child is already being taught as a fit. It does not replace the residual. It does not replace “correlation is not causation.” Use it. Do not crown it.

What this section has done is keep S.ID from being dropped. Teaching moves for a symbol-heavy first pass come next. This is their home for Algebra I’s first two-step equation, first completing of the square, first system.

7.5 Teaching guidance: fading, incorrect examples, comparison, readiness

Worked examples help novices; they must be faded; incorrect examples, explained, help conceptions. That lock is Chapter 2’s. This section converts it into Algebra I’s first symbol-heavy procedures.

John Sweller and Graham Cooper, 1985, established the worked-example effect in algebra:

⁷⁹⁰College Board, AP Statistics CED, Effective Fall 2026 (n. 789), and *AP Statistics Revised Course Framework*, <https://apcentral.collegeboard.org/media/pdf/ap-statistics-revised-course-framework.pdf>. Equivalent to a one-semester, introductory, **non calculus-based** college course. Framework encourages taking both AP Statistics and AP Calculus when the student has the background. Not published as a substitute for calculus in STEM calculus sequences. Do not treat AP Statistics as stealth Algebra II; do not treat this section as stealth AP Statistics.

⁷⁹¹Composite *hypothetical*, labelled in the sentence that introduces it. Empirical pieces: Algebra I S.ID cluster list (n. 785); K–8 data spine in 01-scope (6.SP, 7.SP, 8.SP). The skipping and the sitting are not a reported family.

studying worked solutions outperformed unguided problem solving for novices, because means-ends search consumes working memory that could have been used to extract the schema.⁷⁹² The expertise-reversal effect, Kalyuga, Ayres, Chandler, and Sweller 2003, is why the same full example that helps a novice becomes redundant for a relative expert and can hurt.⁷⁹³ The instructional consequence is guidance fading, Renkl and Atkinson and colleagues: start with a complete example, move to completion problems with some steps blank, fade from the end of the solution first — backward fading — and end with conventional problems.⁷⁹⁴ For the parent, operationally, on a first two-step equation, a first completing of the square, a first system: show one fully worked example on paper, then a second the child must explain, then a completion problem missing the last step, then a conventional problem. Do not start a brand-new topic with “see if you can figure it out” as the only move if the child is a novice in that topic. That is the situation in which Sweller’s effect is largest. Once the child can execute, stop showing full examples. Expertise reversal is real.

Incorrect examples, explained, are the Algebra I-specific finding. Julie L. Booth, Karin E. Lange, Kenneth R. Koedinger, and Kristie J. Newton, 2013, *Learning and Instruction*, two in-vivo Cognitive Tutor experiments: explaining worked examples during guided practice improved conceptual understanding versus guided practice alone. Incorrect examples, alone or mixed with correct ones, were “especially beneficial for fostering conceptual understanding.” Combined with guided practice in the Algebra I Cognitive Tutor in that paper.⁷⁹⁵ Do not treat Cognitive Tutor or MATHia school randomized trials as homeschool results. MATHia family purchase remains **UNVERIFIED** as of 30 August 2026.⁷⁹⁶ The move that transfers is the parent showing one fully worked $px + q = r$ on paper, then a second the child must explain, then a completion problem, then a conventional problem, then an incorrect example: “they added q instead of adding the opposite of q . What did they think the plus sign meant?” The paper lists the misconceptions the examples were built to hit: the equal sign as “where the answer goes” (Baroody and Ginsburg 1983; Kieran 1981); minus signs as “subtract” rather than as modifiers of terms (Vlassis 2004); variables as a single mystery number (Booth 1984; Knuth et al. 2006; Küchemann 1978).⁷⁹⁷ Include at least one incorrect example per new procedure and ask “what did they think this symbol meant?” “They forgot the rule” is not a

⁷⁹²John Sweller and Graham A. Cooper, “The Use of Worked Examples as a Substitute for Problem Solving in Learning Algebra,” *Cognition and Instruction* 2, no. 1 (1985): 59–89. Worked-example effect for novices; cognitive-load account. Cited in Chapter 2 §2.3 / 02-learning §1.3; converted here to Algebra I’s first symbol-heavy procedures.

⁷⁹³Slava Kalyuga, Paul Ayres, John Chandler, and John Sweller, “The Expertise Reversal Effect,” *Educational Psychologist* 38, no. 1 (2003): 23–31. The same worked example that helps a novice becomes redundant for a relative expert and can hurt. Once the child can execute, stop showing full examples.

⁷⁹⁴Alexander Renkl and Robert K. Atkinson, guidance fading / backward fading (Renkl and Atkinson 2003; Renkl, Atkinson, Maier, and Staley 2002; Atkinson, Derry, Renkl, and Wortham 2000), as synthesized in 02-learning §1.3. Start complete, then completion problems, fade from the end, end with conventional problems.

⁷⁹⁵Booth, Lange, Koedinger, and Newton, “Using Example Problems” (2013) (n. 761). Explaining worked examples during guided practice improved conceptual understanding versus guided practice alone. Incorrect examples, alone or mixed with correct ones, “especially beneficial for fostering conceptual understanding.” Two in-vivo Cognitive Tutor experiments, Algebra I, two-step linear equations.

⁷⁹⁶MATHia family purchase **UNVERIFIED** as of 30 August 2026 (04-ai; book brief Chapter 7). Do not treat Cognitive Tutor / MATHia school RCTs as homeschool results.

⁷⁹⁷Booth et al. 2013 misconception list (n. 761): equal sign as “where the answer goes” (Arthur J. Baroody and Herbert P. Ginsburg 1983; Carolyn Kieran 1981); minus as “subtract” (Vlassis 2004); variables as a single mystery number (Booth 1984; Knuth et al. 2006; Küchemann 1978). “They forgot the rule” is not a diagnosis.

diagnosis. “They treated the equal sign as the place the answer goes, so they added everything they saw” is.

Self-explanation is the prompt that makes a worked example more than a script to copy. Chi, Bassok, Lewis, Reimann, and Glaser 1989; Renkl 1997; Rittle-Johnson 2006. Dunlosky, Rawson, Marsh, Nathan, and Willingham 2013 rate self-explanation as moderate utility: it works, but it is not as robust across conditions as practice testing and spacing.⁷⁹⁸ Ask “why did they add the opposite of q here?” not “what is the next step?” The child generates the why. The parent does not narrate it.

Compare two solution methods for the same equation. Rittle-Johnson and Star 2007 and 2009: comparison improves flexibility. Exact d **UNVERIFIED** this pass; the papers were not opened in full.⁷⁹⁹ After the child can execute one method, put a second next to it and ask which is better for this problem, and why. Substitution versus elimination on a system. Factoring versus completing the square versus the formula on a quadratic. Unit rate versus scaling versus the proportion equation on a leftover middle-grades item that still appears. Flexibility is a third outcome, not a luxury. Do not wait for a “gifted unit.”

Interleave kinds that share a surface, restated from section 7.3 because the parent will buy a blocked book. Day of a new skill: a short blocked set. Same week: mixed retrieval. Later this week and next: the new skill between unlike problems. Monthly: an unannounced mixed quiz without notes. Almost all clean classroom randomized trials of interleaving in mathematics are grade 7. Do not export $d = 1.05$ to this course as if it were an Algebra I coefficient. Use the practice. Leave the coefficient in Chapter 6.

Schema instruction for the word problems Algebra I teachers said students could not do. The 2007 survey named word problems next to rational numbers and study habits.⁸⁰⁰ Jitendra, Star, Dupuis, and Rodriguez 2012 is the method, not a promise of $g = 1.24$ on Algebra I items. That g was seventh-grade proportion, 1,163 students, 42 classrooms, trained items, null transfer.⁸⁰¹ Teach the schema, represent it, estimate, solve, check. Algebra I word problems are still structures: a linear comparison, a mixture, a motion, a work, a system of two constraints, an exponential growth, a projectile that is a quadratic in time. Introduce one type at a time, first as a complete story with all quantities, then as a problem with one unknown; vary the cover story so the child is matching structure, not nouns; never teach a magic word. “Per” lives in unit rate and in percent and in “miles per hour squared” later. The structure, not the word. Do not quote Peltier and Vannest $g = 1.57$ as “schema instruction raises math achievement by 1.57 standard deviations.” That is

⁷⁹⁸Michelene T. H. Chi, Miriam Bassok, Matthew W. Lewis, Peter Reimann, and Robert Glaser, 1989; Alexander Renkl, 1997; Bethany Rittle-Johnson, 2006 — self-explanation citations as in 02-learning §1.3. John Dunlosky, Katherine A. Rawson, Elizabeth J. Marsh, Mitchell J. Nathan, and Daniel T. Willingham, “Improving Students’ Learning with Effective Learning Techniques,” *Psychological Science in the Public Interest* 14, no. 1 (2013): 4–58: self-explanation rated moderate utility; practice testing and distributed practice rated high. Do not mash those ratings with Rohrer’s d .

⁷⁹⁹Rittle-Johnson and Star 2007, 2009 (n. 775). Exact d **UNVERIFIED**.

⁸⁰⁰NMAP 2007 Algebra I teacher survey (n. 753): word problems named next to rational numbers and study habits.

⁸⁰¹Asha K. Jitendra, Jon R. Star, Danielle N. Dupuis, and Michael C. Rodriguez, seventh-grade schema-based instruction RCT, 2012: 1,163 students, 42 classrooms, proportion problems, immediate posttest $g = 1.24$, six-week retention $g = 1.27$, no significant transfer. The method, not a promise of $g = 1.24$ on Algebra I items. Do not mash with Rohrer $d = 1.05$.

immediate elementary problem solving on mostly proximal measures.⁸⁰² Do not mash Jitendra's g with Rohrer's d . Different comparisons, different outcomes, both already used in Chapter 6.

Explicit instruction for students already behind. Finding 27 of the Panel: explicit instruction — clear models, extensive practice, think-alouds, extensive feedback — has consistently positive effects for students with mathematical difficulties and for the lowest third of a typical class, on word problems and computation — and this “does not mean that all of a student's mathematics instruction should be delivered in an explicit fashion.”⁸⁰³ Fuchs and colleagues 2021, Recommendation 1, strong evidence, 43 studies: systematic instruction.⁸⁰⁴ A child missing signed-fraction operations, or still treating the equal sign as “compute now,” needs a model first. Productive failure, Kapur 2014, is a different comparison: generation-then-instruction on a new concept when students have usable priors; two randomized studies produced comparable procedural knowledge to instruction-first, greater conceptual understanding and transfer; number of student-generated solutions predicted outcomes.⁸⁰⁵ Do not mash Finding 27 with Kapur. A child who has the operations and is meeting “which values make this identity always true?” can spend ten to twenty minutes generating, then receive the instruction. Frustration from missing facts is not productive. It is a fluency problem. Finding 23 already refused the exclusive war: high-quality research does not support exclusive use of either all-encompassing teacher-directed or student-centered prescriptions.⁸⁰⁶ Jere Confrey, Alan P. Maloney, and Kenny H. Nguyen, 2008, published a methodological critique of the Panel — biased literature selection, inconsistent standards. The critique is real. The Panel is still the synthesis later documents cite. Report both.⁸⁰⁷

Readiness is the door diagnostic, restated as a teaching rule. Do not start the course to keep up with public-school Grade 9 if the gates are closed. Diagnose fractions including signed, integer operations, and the equal sign before buying the book. Chapter 5 and Chapter 6 are the repair, not a pep talk and not a skip. Finding 15 still refuses the age lock. Finding 30 still allows prepared, motivated students to move faster.⁸⁰⁸ Compacted 7–8 so that Algebra I is finished in Grade 8,

⁸⁰²Corey Peltier and Kimberly J. Vannest, 2017 meta-analysis of schema instruction in elementary school: Hedges's $g = 1.57$ immediate problem solving, $g = 1.09$ transfer, 21 studies, 3,408 students — mostly researcher-developed proximal measures. Do not quote as “schema instruction raises math achievement by 1.57 SD.”

⁸⁰³NMAP, *Foundations for Success* (2008), Finding 27: explicit instruction (clear models, extensive practice, think-alouds, extensive feedback) has consistently positive effects for students with mathematical difficulties and for the lowest third of a typical class — and this “does not mean that all of a student's mathematics instruction should be delivered in an explicit fashion.”

⁸⁰⁴Lynn S. Fuchs et al., *Assisting Students Struggling with Mathematics: Intervention in the Elementary Grades*, WWC 2021006 (IES, 2021), Recommendation 1, strong evidence, 43 studies: systematic instruction. K–6 intervention guide; used here as the struggling-student finding, not as an Algebra I RCT.

⁸⁰⁵Manu Kapur, “Productive Failure in Learning Math,” *Cognitive Science* 38, no. 5 (2014): 1008–1022. Problem-solving first, then instruction, produced comparable procedural knowledge to instruction-first but greater conceptual understanding and transfer; number of student-generated solutions predicted outcomes. Different comparison from NMAP Finding 27. Do not mash them.

⁸⁰⁶NMAP, *Foundations for Success* (2008), Finding 23: high-quality research does not support exclusive use of either all-encompassing teacher-directed or student-centered prescriptions.

⁸⁰⁷Jere Confrey, Alan P. Maloney, and Kenny H. Nguyen, “Breaching the Conditions for Success for a National Advisory Panel,” *Educational Researcher* (2008): methodological critique of NMAP (biased literature selection, inconsistent standards). The critique is real; the Panel is still the synthesis later documents cite. Report both. NMAP is the still-cited algebra-foundations synthesis, not “the science of 2026 classrooms.”

⁸⁰⁸NMAP Findings 15 and 30 (nn. 749, 765). Compacted 7–8 versus waiting: a pathway choice, not a kitchen RCT. Live debate this chapter leaves standing.

Appendix A's model, is a pathway choice, not a kitchen randomized trial. Waiting until the gates are open is the other choice. Diagnose. Do not calendar.

When the parent is the ceiling, the release valves are tutor, co-op, dual enrollment. NCES 2016: 23 percent any instruction by a tutor or private teacher; 31 percent any from a local homeschool group or co-op; a small share of 9–12 also in college classes.⁸⁰⁹ That is a normal high-school plan, not a failure of the kitchen. A *hypothetical*, labelled as such: a parent who last passed Algebra I in 1998 hires a weekly tutor for this course, or enrolls the student in College Algebra at the community college the following year, while remaining the homeschool administrator of record.⁸¹⁰ The empirical pieces are the 23 and 31 percent and the dual-enrollment cell. The year 1998 and the sitting are hypothetical. Chapter 12 will say how to put the tutor, the co-op teacher, or the college on the transcript. This chapter says: do not outsource hearing. Video and self-teaching curricula exist because of the Maloney constraint. Teaching Textbooks, **vendor**, “does 100% of the teaching” and “100% of the grading.”⁸¹¹ That is a vendor claim, not an effectiveness trial, and not a finding that the parent can stop hearing a wrong inverse. Whether homeschool Algebra I taught from a video curriculum produces the parent who can hear a wrong answer — no homeschool randomized trial. Keep the parent as supervisor who can still hear a wrong inverse later. Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, 2015: first- and second-graders of more math-anxious parents learned less mathematics over the school year and were more math-anxious at year's end — but only if those parents reported frequent homework help. When help was infrequent, parent anxiety was unrelated to child outcomes. Math-specific, not reading.⁸¹² That is a school-homework study, not a parent-as-sole-teacher randomized trial. A homeschool parent who is the daily instructor cannot “help less often” as a daily strategy. The constructive response the literature actually supports is narrower: the parent learns the day's concept well enough to stay out of the child's working memory; the child generates; the parent does not narrate worry. Do not treat the finding as “anxious parents should not teach.” Replication status of Maloney 2015 is open; no registered replication was opened for this project. Flag as a single large, well-cited field study plus a 2021 mediation follow-up by Retanal and colleagues, on parents of 11- to 14-year-olds, this band's ages.⁸¹³ Prepare so you can hear $(x + 2)(x + 3)$ written

⁸⁰⁹Cui and Hanson, NCES 2020-001: any instruction by a tutor or private teacher, 23 percent; any from a local homeschool group or co-op, 31 percent; any online course 23 percent of homeschoolers, 30 percent of the 9–12 band. Dual enrollment visible but small (Table 4: 1 percent college, interpret with caution). Release valves, not a failure of the kitchen.

⁸¹⁰Composite *hypothetical*, labelled in the sentence that introduces it. Empirical pieces: NCES 2016 tutor 23 percent / co-op 31 percent (n. 809); dual-enrollment cell. The year 1998 and the sitting are not a reported family. Chapter 12 for how to put the tutor, co-op teacher, or college on the transcript.

⁸¹¹Teaching Textbooks **vendor** claim (n. 769). Not an effectiveness trial.

⁸¹²Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, “Intergenerational Effects of Parents' Math Anxiety on Children's Math Achievement and Anxiety,” *Psychological Science* 26, no. 9 (2015): 1480–1488. First- and second-graders of more math-anxious parents learned less math over the year and were more math-anxious at year's end — **but only if those parents reported frequent homework help**. Math-specific, not reading. School-homework study, not a parent-as-sole-teacher RCT. Do not treat as “anxious parents should not teach.” Highest-priority homeschool-specific gap: no fetched experimental literature on math-anxious parents as full-time instructors.

⁸¹³Fraulein Retanal, Noah Johnston, Sabrina Di Lonardo Burr, Andie Storozuk, Michela DiStefano, and Erin A. Maloney, *Education Sciences* (2021): controlling-supportive homework-helping style partially mediated the parent-anxiety / child-achievement relation in a survey of parents of 11- to 14-year-olds. Path analysis on a survey, not a kitchen RCT. Replication status of Maloney 2015 is open; no registered replication opened. Flag as a single large,

as $x^2 + 5$. Then do not take the pencil.

Adaptive practice engines — IXL, Khan Academy practice, ALEKS, DreamBox, Cognitive Tutor / MATHia — sequence items. They are not agents in the Hermes Agent / Grok Bot sense, and they are not a substitute for the parent hearing a wrong expansion.⁸¹⁴ DreamBox grade-level-per-hour claims are **vendor**; underlying ESSA study PDF **UNVERIFIED**. IXL live homeschool price **UNVERIFIED** as an invoice number. MATHia family purchase **UNVERIFIED**. Khan Academy content is free; Khanmigo is a paid constrained tutor, **vendor** “never gives you the answer,” **vendor** \$4/month or \$44/year as of 30 August 2026. Oreopoulos and Low, NBER w35620, August 2026: assignment to Khan Academy plus Khanmigo in 18 Tennessee middle schools raised MAP math by 0.040 population standard deviations per term; about 0.06 to 0.08 over a school year; authors: “These gains resemble those from Khan Academy practice without AI assistance.”⁸¹⁵ Do not invent standalone 0.06 as the intention-to-treat. Do not advertise Khanmigo as Bloom’s two-sigma tutor. If Khanmigo is in the house, expect the Tennessee pattern unless the parent requires the hint dialogue. Full product landscape: Chapter 3. This section names the engines so they are not mistaken for a parent who can hear a wrong inverse.

What this section has done is convert Chapter 2 into Algebra I’s first symbol-heavy sitting. Tips, including the transcript title and the kitchen-math refusal, come next.

7.6 Tips that survive: gates, titles, contexts

Tips a parent can steal, each tied to a named source, not to folklore.

Do not start the course to keep up with public-school Grade 9 if the National Mathematics Advisory Panel’s gates are closed. Diagnose fractions, including signed, integer operations, and the equal sign before buying the book. The 2007 survey of 743 Algebra I teachers is the teacher-voice primary. Siegler and colleagues 2012 is the prediction. Table 2 is the pace, not a law.⁸¹⁶ Chapter 5 and Chapter 6 are the repair. This chapter is the course.

Do not treat a calendar-grade-8 Algebra I as identical to the Common Core’s Grade 8 expressions, equations, and functions. Those are not the same course. Compacted 7–8, Appendix A, is a model, not a mandate. Calculus senior year on a Common Core-aligned path requires that high-school

well-cited field study plus a 2021 mediation follow-up.

⁸¹⁴Adaptive practice engines (DreamBox, IXL, Khan Academy practice, ALEKS, Cognitive Tutor / MATHia) sequence items; they are not agents in the Hermes Agent / Grok Bot sense. DreamBox grade-level-per-hour claim is **vendor**; underlying ESSA study PDF **UNVERIFIED**. IXL live homeschool price **UNVERIFIED** as an invoice number. MATHia family purchase **UNVERIFIED** (n. 796). Full product landscape: Chapter 3.

⁸¹⁵Philip Oreopoulos and Nina Low, “One Click Away: AI Tutoring with Khanmigo in a Two-Year School Experiment,” NBER Working Paper 35620, August 2026, <https://www.nber.org/papers/w35620>. Two-year cluster RCT, 18 Tennessee middle schools; 0.040 population SD per term; about 0.06 to 0.08 SD over a school year; “These gains resemble those from Khan Academy practice without AI assistance.” Conversational use thin (median student messaged on a third of practice days; 17 percent of mistake sessions; 14.5 percent of messages mathematical). Do not invent standalone 0.06 as the ITT. Working paper, not yet journal-peer-reviewed. Khanmigo **vendor** “never gives you the answer”; **vendor** \$4/month or \$44/year as of 30 August 2026, <https://www.khanmigo.ai/learners>. Full discussion: Chapter 3 §3.5.

⁸¹⁶Door diagnostic restated as a stealable tip. Sources: nn. 751–753, 758, 760.

content started in Grade 8. Compact only if the gates are open.⁸¹⁷ Authentic Grade 8 Algebra I versus waiting is a live pathway debate, not a kitchen randomized trial.

Transcript the course as “Algebra I,” not “Saxon 8/7” without a parenthetical. The National Collegiate Athletic Association’s math core courses are Algebra I or higher. Division I: 16 core courses including 3 years of math (Algebra I or higher); Division II: 16 core including 2 years of math (Algebra I or higher). Homeschool packet: transcript, Administrator and Accordance Statement, one Core-Course Worksheet per core course.⁸¹⁸ Named university pages this project opened — Pittsburgh, three years of mathematics; SMU, suggested Algebra I, Geometry, Algebra II; North Central, three years including algebra and geometry, one course beyond geometry preferred — want a transcript with course titles, grades, and credits.⁸¹⁹ Plant for Chapter 12. Do not dump the NCAA worksheet here. Integrated-path families transcript Integrated Mathematics I or Mathematics I, with a parenthetical a stranger can map. Saxon classic editions, **vendor**, integrate some geometry inside Algebra 1; a separate Geometry course also exists. The transcript still needs a course a stranger can map onto Algebra I, and later onto Geometry. Do not sell the franchise.

Do not skip descriptive statistics because “stats is Chapter 11.” S.ID.1–3 and 5–9 are in this course. They are the on-ramp. AP Statistics’ Fall 2026 prerequisite is first-year algebra. This course is that prerequisite. Do not treat AP Statistics as stealth Algebra II. Do not treat this section as stealth AP Statistics.⁸²⁰

Do not replace symbol sense with kitchen-math. Finding 26: real-world contexts improve performance on similar real-world items; they do not, in the Panel’s small high-quality set, raise computation, simple word problems, or equation solving.⁸²¹ Kitchen-math is not a linear function. A doubled recipe is a scaling if both quantities scale. A shopping total with a coupon may be linear with a negative intercept. Name the family. Shopping will not teach completing the square. The area model and the distributive property will. Use the context as a second representation. Do not crown it. Soderstrom and Bjork: current performance is often an unreliable index of learning. A child who can compute a tip and cannot write a linear function through the origin has performed, not learned.

Schema instruction for word problems; never a keyword. The 2007 survey named word problems. Jitendra is the method, not a promise of $g = 1.24$ on this year’s items. Null transfer in that trial is why cover stories have to vary and why the parent should not expect a new schema to appear for free.⁸²²

⁸¹⁷Appendix A compacted pathway (n. 745); 01-scope international note and “Grade 8 algebra” ambiguity. CCSSM Grade 8 EE/F versus a full Algebra I taken in Grade 8 are not the same course.

⁸¹⁸NCAA Home School Toolkit 2025–26 (n. 767). Transcript as “Algebra I,” not “Saxon 8/7” without a parenthetical. Plant for Chapter 12.

⁸¹⁹University of Pittsburgh: minimum 3 years of mathematics; Home School Supplemental Form. SMU: suggested three mathematics (Algebra I, Geometry, Algebra II). North Central College: recommended 3 years mathematics including algebra and geometry, one course beyond geometry preferred. Pattern, not a rule: three years through Algebra II is the floor most of these pages name. Full treatment: Chapter 12. Do not dump those worksheets here.

⁸²⁰Algebra I S.ID (n. 785); AP Statistics Fall 2026 prerequisite (n. 789). Do not skip descriptive statistics because “stats is Chapter 11.”

⁸²¹NMAP Finding 26 (n. 780). Kitchen-math is not a linear function. Shopping will not teach completing the square.

⁸²²Jitendra et al. 2012 (n. 801): null transfer is why cover stories have to vary. IES 2021 Recommendation 5 / Fuchs et al. 2021 Table 5.2: keyword strategies fail. Schema vs keywords locked in Chapter 2; applied here.

Worked-example fading for the first pass through a new symbol-heavy procedure; incorrect examples explained; then the child generates. Booth and colleagues 2013, combined with guided practice in a school Cognitive Tutor, not as a kitchen-table Cognitive Tutor. Compare two methods. Interleave kinds that share a surface. Do not mash Finding 27 with Kapur.⁸²³

Do not claim a named Algebra I curriculum wins in homeschool. No homeschool-curriculum randomized trial was opened. HMH's school-side Saxon ESSA ratings are **vendor** evidence about school implementations, not homeschool results. Teaching Textbooks' "100% of the teaching" is **vendor**. Cathy Duffy and Rainbow Resource are **catalogue / reviewer**. Choose by fit.⁸²⁴

Do not treat the 2007 teacher survey as 2026. Write "the Panel's 2007 nationally representative survey of Algebra I teachers found." Do not write "teachers today say."⁸²⁵

Do not treat Cognitive Tutor or MATHia school randomized trials as homeschool results. MATHia family purchase **UNVERIFIED**.⁸²⁶

Do not introduce functions as " $y = mx + b$ " only. 8.F was each input exactly one output. F-IF is notation, features, average rate of change, three families. Without the function concept, $f(x)$ has nothing to attach to.⁸²⁷

Do not dump polynomials beyond quadratic, complex numbers, logarithms, or trigonometry into this year and call it Algebra I. Those wait for Chapter 9. Recognize complex solutions of quadratics; do not develop them.⁸²⁸

Do not make the kitchen timer the identity of being good at this course's mathematics, and do not abandon retrieval. Fluency is still flexible, accurate, efficient, appropriate. Brief timed retrieval of already-known fraction operations, signed-number facts, and linear-equation moves, mixed, after they are right untimed, is IES 2021 Recommendation 6 applied here. Boaler's object is a different object: high-stakes early speed tests.⁸²⁹ Facts are not a language-model job.

Do not take the pencil. Maloney 2015 remains a school-homework study. Prepare so you can hear a wrong expansion. Then the child generates.⁸³⁰ When the parent is the ceiling, tutor, co-op, and dual enrollment are release valves, not a failure of the kitchen. Keep the parent as supervisor who

⁸²³Booth et al. 2013 (n. 795); NMAP Finding 27 vs Kapur 2014 (nn. 803, 805). Do not mash.

⁸²⁴No homeschool-curriculum RCT was opened (03-homeschool load-bearing 4; Kunzman and Gaither 2020: curriculum research on homeschooling is thin). HMH school-side Saxon ESSA ratings are **vendor** evidence about school implementations. Teaching Textbooks "100% of the teaching" is **vendor** (n. 769). Cathy Duffy Reviews and Rainbow Resource are **catalogue / reviewer**, not independent effectiveness trials. Choose by fit.

⁸²⁵NMAP 2007 survey (n. 753). Write "the Panel's 2007 nationally representative survey of Algebra I teachers found." Do not write "teachers today say."

⁸²⁶MATHia family purchase **UNVERIFIED** (n. 796). Cognitive Tutor school RCTs are not homeschool results.

⁸²⁷CCSSM 8.F (Chapter 6); F-IF in this course (n. 756). Without the function concept, $f(x)$ has nothing to attach to (n. 759).

⁸²⁸NMAP Table 1 (n. 754); Appendix A Algebra II cluster homes in 01-scope. Polynomials beyond quadratic, complex numbers, logarithms, trigonometry: Chapter 9. Recognize complex solutions; do not develop them.

⁸²⁹Fuchs et al. 2021 Recommendation 6, strong evidence, 27 studies: brief timed activities on *already-known* facts as one way to build fluency. Jo Boaler, 2012 / 2015 "Fluency Without Fear": high-stakes early speed tests — a different object. Live debate locked in Chapter 2. Facts are not a language-model job (n. 845).

⁸³⁰Maloney et al. 2015 (n. 812). Prepare so you can hear a wrong expansion; then the child generates. Do not take the pencil.

can still hear a wrong inverse later. Do not outsource hearing.

What this section has done is put stealable sentences on the page. Bastani’s kitchen pattern has a name in this course: paste the Algebra worksheet. The AI lane, next, reuses Chapter 3 on this course’s objects. The model may prepare the adult and the next problem. It may not do the child’s problem.

7.7 Parent-hosted AI for Algebra I

One short reminder, then the objects of this course. Hamsa Bastani and colleagues, 2025: a ChatGPT-like tutor raised assisted practice grades 48 percent relative to control and then cut the unaided exam 17 percent. A guardrailed tutor that withheld the full solution raised practice 127 percent and was statistically indistinguishable from control on the unaided exam. GPT Base, asked for the answer, was fully correct 51 percent of the time, with 8 percent arithmetic errors even when the method was right. Students copied anyway.⁸³¹ Soderstrom and Bjork 2015: current performance is “often an unreliable index” of learning.⁸³² Full design, product landscape, ban-versus-allow list, six parent workflows, COPPA dates, detectors, companion-harm cases as allegations: Chapter 3. This section is a pointer, not a sequel. The workflows below are practice suggestions grounded in Bastani, Soderstrom and Bjork, and Oreopoulos and Low, not a new randomized trial of homeschool AI. No year-plus kitchen-table RCT of a parent-constrained GPT-Tutor workflow was opened on 30 August 2026. No homeschool Algebra I RCT of any kind was opened. Do not invent one.⁸³³

Allow, parent-hosted, child doing the work, parent holding the key.

Parent-first explanation of a transformation, and of why $(x + 2)(x + 3)$ is not $x^2 + 5$. Chapter 3 workflow (a): explain the concept to the *parent* first. Bastani’s GPT Tutor worked in part because teachers had already written the solution and common mistakes into the hidden prompt. The parent is that teacher. Prefer a chat or reasoning model, not an open-web agent, for this job.⁸³⁴ Ask the model to give the area-model justification and the distributive-property justification from section 7.2, and then to name the wrong expansion the child is likely to write. Ask it, separately, why $y =$

⁸³¹Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman, “Generative AI without Guardrails Can Harm Learning: Evidence from High School Mathematics,” *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122, <https://doi.org/10.1073/pnas.2422633122>; author PDF https://hamsabastani.github.io/education_llm.pdf, accessed 30 August 2026. GPT Base: assisted practice +48% relative to control; unaided exam −17%. GPT Tutor: practice +127%; exam point estimate −0.004, not significant. GPT Base fully correct 51% of the time when asked for the answer; 8% arithmetic errors even when the method was right. Classroom-level RCT, large high school in Turkey, Fall 2023–24, ~1,000 students in grades 9–11, four 90-minute sessions, GPT-4-0613. Not a homeschool study, not GPT-5, not Khanmigo. Percentages are of control grades, not Cohen’s d . Full discussion: Chapter 3.

⁸³²Soderstrom and Bjork 2015 (n. 783).

⁸³³Parent workflows in this section are practice suggestions grounded in Bastani, Soderstrom and Bjork, and Oreopoulos and Low, not a new RCT of homeschool AI. No year-plus kitchen-table RCT of a parent-constrained GPT-Tutor workflow was opened on 30 August 2026 (04-ai open questions). No homeschool Algebra I RCT of any kind was opened. Do not invent one.

⁸³⁴Chapter 3 workflow (a): explain the concept to the *parent* first. Practice suggestion, not an RCT. Bastani’s GPT Tutor worked in part because teachers had already written the solution and common mistakes into the hidden prompt. Prefer a chat/reasoning model, not an open-web agent, for this job. Nous Portal docs, fetched 30 August 2026: Hermes 4 is “strong at math, science, instruction following” — **vendor** — and **not recommended for use inside Hermes Agent**.

$f(x - 2)$ shifts right, not left, with a table and a graph the parent can check. Ask it why completing the square turns $x^2 + 6x + 5$ into $(x + 3)^2 - 4$, with the missing 9 visible. The parent reads that before the lesson. The child does not. Models hallucinate arithmetic. GPT Base had 8 percent arithmetic errors even when the method was right. An expansion is a facts-adjacent object. The parent holds the key. Symbolic algebra — Symbolab, Wolfram|Alpha, a computer-algebra system — is not the explainer. Wolfram is the honest *check* after the child can execute a method by hand, analogous to a calculator after fluency, not instead of it.⁸³⁵ A language model is the explainer for the parent. Mixing the two without a label is how a fluent wrong expansion survives. Point to Chapter 3. Do not sequel it.

Isomorphic items, parent-held key. Chapter 3 workflow (c): extra practice with the answer key held by the parent. Do not generate “practice” by completing tonight’s assigned homework. That is the ban.⁸³⁶ Ask the model for five items that share a structure with tonight’s type and differ in cover story or in the numbers — five completing-the-square items with even, then odd, linear coefficients; five systems, one with no solution, one with infinitely many; five “which family is this?” tables — and for a worked key the child never sees. The child does two of them unaided. The parent diagnoses. Jitendra 2012’s null transfer is why the surface has to vary and why the parent should not expect a new schema to appear for free.

A hint script that withholds the full solution. Chapter 3 workflow (d): Bastani’s GPT Tutor analogue. Withhold the full solution; ask what they have tried; point out a named mistake; escalate information slowly; accept equivalent answers.⁸³⁷ The homeschool analogue is a parent-read script, not a child-held chatbot. “You wrote $(x + 2)(x + 3) = x^2 + 5$. What does the 2 have to multiply besides the x ?” Then wait. Then the area rectangle with one cell blank. Then, if needed, the first line of the distributive expansion. Not the finished polynomial. Khanmigo or a parent-built GPT-Tutor prompt only after an attempt, parent in the room. Khanmigo, **vendor**, “never gives you the answer.” That is a design goal, not a guarantee. Bastani had to put the solution in the *hidden* prompt and still built a separate product to stop answer-giving.⁸³⁸ Expect the Tennessee pattern — median student messaged Khanmigo on only a third of practice days and in 17 percent of mistake sessions; 14.5 percent of messages contained a mathematical question or a step of reasoning — unless the parent requires the hint dialogue.⁸³⁹

Diagnose the child’s line of work. Chapter 3 workflow (f): diagnose a wrong answer the child

⁸³⁵Wolfram|Alpha, <https://www.wolframalpha.com/>, fetched 30 August 2026: “Compute verified, expert-level answers using Wolfram’s breakthrough algorithms and knowledgebase.” Honest *check* for CAS work after the child can execute by hand, analogous to a calculator after fluency. Step-by-step on Pro is **vendor**; current Pro price **UNVERIFIED** this access day. A language model is the explainer for the parent. Mixing the two without a label is how a fluent wrong expansion survives. 04-ai §4.2 CAS-after-fluency. Symbolic tools are not the explainer.

⁸³⁶Chapter 3 workflow (c): extra isomorphic practice with the answer key held by the parent. Do not generate “practice” by completing tonight’s assigned homework. That is the ban. Follows from Bastani et al. 2025 (n. 831).

⁸³⁷Chapter 3 workflow (d): Bastani GPT Tutor analogue. Withhold the full solution; ask what they have tried; point out a named mistake; escalate information slowly; accept equivalent answers. Homeschool analogue is a parent-read script, not a child-held chatbot.

⁸³⁸Khanmigo **vendor** “never gives you the answer” is a design goal, not a guarantee (04-ai load-bearing finding 7). Bastani had to put the solution in the *hidden* prompt and still built a separate product to stop answer-giving.

⁸³⁹Oreopoulos and Low, NBER w35620 (n. 815). 96 percent tried Khanmigo at least once; median student messaged it on only a third of practice days and in 17 percent of mistake sessions; 14.5 percent of messages contained a mathematical question or a step of reasoning.

already produced. Name the misconception; ask a question; retest with an isomorphic item. The child answers the parent, not the model.⁸⁴⁰ For $(x + 2)(x + 3) = x^2 + 5$, the sentence is “they added the insides and did not distribute,” not “they forgot FOIL.” For $3 + -2$ treated as a slogan, the sentence is minus-as-subtract, Vlassis 2004 as cited by Booth. For $3 + 4 + 5 = 3 + \square$ as 15, the sentence is McNeil and Alibali’s perform-all-operations pattern, still alive, now inside an Algebra I warm-up. For a graph of $y = 2x + 1$ on which the child plotted slope 1 and intercept 2, the sentence is “slope/intercept mixup,” labelled as teaching lore, prevalence **UNVERIFIED**, and the retest is “slope from (0, 1) and (1, 3).” For “the correlation is 0.9 so X causes Y ,” the sentence is S.ID.9, not “they computed r wrong.” GPT Base was fully correct only 51 percent of the time. Do not let fluency of a wrong expansion survive because the model said it smoothly.

Hands-on as materials generation, Chapter 3 workflow (e): the model describes an area rectangle for $(x + 2)(x + 3)$, or a four-quadrant grid for a system, or a qualitative sketch prompt for a piecewise graph; the child draws. Carbonneau, Marley, and Selig 2013: overall d approximately 0.37 versus abstract-only, under guidance. Virtual versus physical manipulatives remain **UNVERIFIED**.⁸⁴¹ Put the rectangle next to the written product in the same visual field. Fade.

Interleaving homework is a parent job. Mix yesterday’s linear system with last month’s exponential table and a quadratic to factor. The model can shuffle kinds — “give me a mixed page: two two-step linear equations with signed-fraction coefficients, one system, one ‘which family?’, one complete-the-square, one correlation-versus-causation item, answer key separately.” The parent holds the key. The child does the page without a solver. That mixed page is the Rohrer assignment, in a kitchen, labelled as a practice suggestion, not as $d = 1.05$ sitting on the table.⁸⁴²

Ban. Paste-the-Algebra-worksheet. Photo-to-key. “What is the answer.” Open-web agent in the child’s hands. Companion persona. Photomath, Mathway, and Symbolab lead with scan-to-solution; no product-specific unaided-transfer RCT was opened this research day; the transferable evidence is Bastani’s crutch plus the homepages.⁸⁴³ Photomath, fetched 30 August 2026: scan including word problems; “A solution will pop up for your exact problem.” Mathway: “Got a math problem? Get the steps.” Symbolab: solver first. Ban Photomath and solver-for-the-child on tonight’s set. Ban paste-the-Saxon-Algebra-worksheet — the outline’s name for the same crutch,

⁸⁴⁰Chapter 3 workflow (f): diagnose a wrong answer the child already produced. Practice suggestion, not an RCT. The child answers the parent, not the model. Named misconceptions: distributive failure on $(x + 2)(x + 3)$; Vlassis minus-as-subtract (n. 774); McNeil and Alibali operational equal sign (n. 762); slope/intercept mixup, prevalence **UNVERIFIED** (n. 779); S.ID.9 correlation vs causation (n. 788). GPT Base fully correct 51 percent (n. 831).

⁸⁴¹Chapter 3 workflow (e): hands-on as materials generation. The model describes; the child draws. Kira J. Carbonneau, Scott C. Marley, and James P. Selig, 2013: 55 studies, overall mean d approximately 0.37 vs abstract-only. Virtual vs physical manipulatives **UNVERIFIED**. Fuchs et al. 2021 Recommendation 3: same visual field; fade for accuracy.

⁸⁴²Mixed page as Rohrer assignment in a kitchen: practice suggestion, not $d = 1.05$ sitting on the table (n. 777). Rohrer et al. 2020 teachers implemented without training. The parent can implement without a product.

⁸⁴³Photomath, <https://photomath.com/>, fetched 30 August 2026: scan including word problems; “A solution will pop up for your exact problem”; Basic \$0 / Monthly \$9.99 / Annually \$69.99 (USD, this day); **vendor** “designed for learning.” Mathway, <https://www.mathway.com/>: “Got a math problem? Get the steps.” Symbolab, <https://www.symbolab.com/>: solver first. No product-specific unaided-transfer RCT opened this day. Transferable evidence is Bastani’s crutch plus solver homepages that lead with scan-to-solution. Ban Photomath / solver-for-the-child on tonight’s set. Ban paste-the-Algebra-worksheet (outline’s “paste-the-Saxon-Algebra-worksheet” names the crutch, not a franchise).

which this book will not convert into a franchise endorsement. The ban is the solver on tonight's assigned items, whatever the cover says. Symbolic tools are not the explainer, restated: Wolfram or a CAS after the child can execute by hand, as a check. Not during the attempt. Not as the lesson.

Hermes Agent and Grok Bot can browse the open web; they are adult and developer tools, not a child-safe school SKU. No documented K–12 student-facing autonomous open-web agent as a school product on the pages opened 30 August 2026. Hard no: messaging gateway to a child's Telegram or WhatsApp; Grok Bot iOS app in the child's hands; IEP or 504 PDFs on a shared Grok Bot computer; the child's school portal or Photomath logins in an agent. El Salvador's Grok-as-tutor is a ministry deployment of Grok *the assistant*, announcement not RCT, not Grok Bot, not a U.S. homeschool SKU.⁸⁴⁴ Adaptive is not an agent. IXL and Khan sequence items; they do not replace the parent hearing a wrong expansion.

Facts and fluent algebra moves are not an LLM job to memorize by chat. Flash retrieval of signed-number facts, of “a negative times a negative is a positive” as a retrieved compression, of linear-equation inverse steps the child can already execute slowly, belongs to the parent's ears, a dumb spaced-repetition deck, or a short mixed oral. Use the model to generate the deck or the oral script, not to sit between the child and the fact. Chapter 3 §3.7 / 04-ai load-bearing finding 12. A language model that will happily hallucinate 7×8 , or the middle term of $(x + 2)(x + 3)$, is the wrong object to put between a child and an expansion.⁸⁴⁵

Detectors fail. Liang, Yuksekgonul, Mao, Wu, and Zou 2023: average false-positive rate 61.22 percent (arXiv) / 61.3 percent (*Patterns*) on human TOEFL essays. Do not use detectors to police a homeschooler's written explanations of a completing-the-square step or a residual.⁸⁴⁶ Companion-harm cases — *Garcia v. Character Technologies*, *Raine v. OpenAI* — are allegations, not verdicts. They are why a math agent should not be a companion. Not a literature on mathematics learning. Full discussion: Chapter 3 §3.6. COPPA amended Rule published 22 April 2025, general compliance 22 April 2026. A parent using an adult tool on a parent account to generate a worksheet is a different fact pattern from creating a child account on a companion app. Homeschool parents are not verified U.S. K–12 educators by default and should not assume ChatGPT for Teachers or Claude for Teachers FERPA wrappers.⁸⁴⁷ California A.B. 1159 is not law as of 30 August 2026

⁸⁴⁴Hermes Agent, <https://hermes-agent.org/>, fetched 30 August 2026: open-source MIT autonomous agent, Nous Research; self-hosted; no K–12 student-safety certification on the pages opened. Adult/developer tool, not a child-safety school product. xAI, “Grok Bot FAQ,” <https://docs.x.ai/grok-bot/faq>, fetched 30 August 2026: persistent cloud computer; “Grok Bot requires cloud data storage, so Legacy Privacy Mode is not supported”; no education SKU and no FERPA claim on the pages opened. No documented K–12 student-facing autonomous open-web agent as a school product on the pages opened 30 August 2026. El Salvador's Grok-as-tutor is a ministry deployment of Grok *the assistant*, announcement not RCT, not Grok Bot, not a U.S. homeschool SKU. Full distinctions: Chapter 3 §3.2. Hard no: the child's hands; messaging gateway; IEP/504 PDFs on a shared Grok Bot computer; Photomath logins in an agent.

⁸⁴⁵Bastani et al. 2025: GPT Base fully correct 51%; 8% arithmetic errors even when the method was right. Flash retrieval of facts and fluent algebra moves is not an LLM job. Chapter 3 §3.7 / 04-ai load-bearing finding 12: use the model to generate the deck or the oral script, not to sit between the child and the fact.

⁸⁴⁶Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou, “GPT Detectors Are Biased against Non-Native English Writers,” arXiv:2304.02819; *Patterns* 4, no. 7 (2023), <https://doi.org/10.1016/j.patter.2023.100779>. Average false-positive rate 61.22% (arXiv) / 61.3% (*Patterns*) on human TOEFL essays. Do not use detectors to police a homeschooler's written algebra explanations. Full discussion: Chapter 3 §3.6.

⁸⁴⁷Federal Trade Commission, “Children's Online Privacy Protection Rule,” 90 FR 16918 (22 April 2025), 16 CFR

(Senate third reading). Do not freeze it as a statute.

Session shape, Chapter 3 §3.8, scaled to this course. Practice suggestion, not an RCT. Unaided warm-up: one signed-fraction two-step equation, one “which family is this table?,” one expansion. Teach. Struggle. Hint not key. Unaided exit ticket: a mixed three-item check without the model, which is Bastani’s exam phase in five minutes. AP-style work later is unaided — Chapter 10 will say every FRQ on paper, timed. Even here, the exit ticket is unaided. “Finish your Algebra with Photomath so we can go to soccer” is GPT Base, the pattern with a large-sample causal harm estimate.⁸⁴⁸ If Khanmigo is in the house, expect the Tennessee pattern unless the parent requires the hint dialogue. Practice minutes predicted MAP; chat did not, descriptively. Working paper, not yet journal-peer-reviewed.

A *hypothetical*, labelled as such, can put the ban in a room. A parent whose fourteen-year-old photographs tonight’s quadratic-formula worksheet into Photomath, copies the green steps, and scores 100 percent on the page, then cannot complete the square on tomorrow’s unaided three-item check, is running GPT Base.⁸⁴⁹ The empirical pieces are Bastani’s crutch, Photomath’s scan-to-solution homepage, and Soderstrom and Bjork’s performance/learning split. The age, the topic, and the 100 percent are hypothetical. The move is not: take the phone, put the area rectangle and the formula derivation on paper, and run the unaided check.

Hermes Agent, if used, lives on the parent machine. Model split this day (Nous Portal): Hermes 4 or any strong chat model for parent explanation and hint scripts; agentic examples — Claude Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro — for the tool loop. Hermes 4 is **not recommended inside Hermes Agent** (tuned for chat, not rapid tool-calling). Hermes 4 ≠ Hermes Agent. Not a child-safe school SKU. Memory in `~/hermes/` is sensitive.⁸⁵⁰ Grok Bot, if used, is included with SuperGrok and Cursor Pro/Teams as of the 26 August 2026 expansion post. Cloud storage required; Legacy Privacy Mode unsupported; shared computer per user; no education SKU and no FERPA claim on the pages opened. Grok ≠ Grok Bot ≠ El Salvador Grok-as-tutor. Hard no: the child’s hands.⁸⁵¹

The agent does not take the test. The child does. Coordinate geometry and completing the square

312. Effective date 23 June 2025; general compliance date 22 April 2026. A parent using an adult ChatGPT / Claude / Grok / Hermes account to generate a worksheet, without creating a child profile at the operator, is a different fact pattern from a child-directed app. Still not a license to upload the child’s face, voice, or full name. Claude consumer policy is 18+; Claude for Teachers is verified U.S. K–12 educators only. Homeschool parents are not verified U.S. K–12 educators by default. California Legislature, A.B. 1159 (Addis), 2025–2026 session: Active Bill — In Floor Process, Senate third reading (Daily File 30 August 2026). **Not law.** Full discussion: Chapter 3 §3.6.

⁸⁴⁸Session shape: Chapter 3 §3.8 / 04-ai §9.4, scaled to this course. Practice suggestion, not an RCT. Grounded in Bastani’s exam phase (n. 831) and Soderstrom and Bjork (n. 783). Unaided warm-up, teach, struggle, hint not key, unaided exit ticket. AP-style work later is unaided (Chapter 10). Even here, the exit ticket is unaided. “Finish your Algebra with Photomath so we can go to soccer” is GPT Base.

⁸⁴⁹Composite *hypothetical*, labelled in the sentence that introduces it. Empirical pieces: Bastani’s crutch (n. 831); Photomath scan-to-solution homepage (n. 843); Soderstrom and Bjork performance/learning split (n. 783). The age, the topic, and the 100 percent are not a reported family.

⁸⁵⁰Hermes Agent / Hermes 4 split (n. 834, n. 844). Hermes 4 ≠ Hermes Agent. Memory in `~/hermes/` is sensitive. Not a child-safe school SKU.

⁸⁵¹Grok Bot, included with SuperGrok and Cursor Pro/Teams as of the 26 August 2026 expansion post (Chapter 3). Cloud storage required; Legacy Privacy Mode unsupported; shared computer per user; no education SKU and no FERPA claim on the pages opened. Grok ≠ Grok Bot ≠ El Salvador Grok-as-tutor. Hard no: the child’s hands.

for circles wait on this course — Chapter 8. Polynomials beyond quadratic, complex numbers, logarithms, and the unit circle wait — Chapter 9. Calculus waits on the function concept this year was supposed to finish — Chapter 10. Descriptive statistics continue — Chapter 11. The transcript title “Algebra I,” with a parenthetical if the spine was a vendor name, waits — Chapter 12. Diagnose fractions, including signed, integer operations, and the equal sign, before buying the book. A child who can operate on signed rationals, see structure in an expression, treat “=” as a relation, and assign to each input exactly one output is ready for the course Appendix A calls a more ambitious Algebra I than has generally been offered. A child who cannot is not ready, regardless of calendar grade, regardless of a publisher placement test, regardless of a chatbot that can print tonight’s completed square while the child watches. Kitchen-math is not a linear function. Photomath is not the explainer. The parent holds the key.

Chapter 8

Geometry



Figure 9: A compass, a straightedge, and a paper with a circle and a triangle.

Two documents, opposite jobs. The first is a course map. CCSSM Appendix A, published by Achieve in partnership with the Common Core writing team, names Geometry as the second course in the U.S. traditional pathway and states its purpose: “formalize and extend students’ geometric experiences from the middle grades. Students explore more complex geometric situations and

deepen their explanations of geometric relationships, moving towards formal mathematical arguments.” Transformations are emphasized early.⁸⁵² That sentence is an explicit difference from many older high-school geometry courses — including the one a parent who stopped at algebra is most likely to remember — that began with axioms and two-column proof of triangle congruence. The second document is a warning about fluent fake proofs. AlphaGeometry2, Yuri Chervonyi and colleagues, 2025, solved 84 percent of International Mathematical Olympiad geometry problems from 2000 through 2024 — 42 of 50 — as a *specialized neuro-symbolic system*, a Gemini-based language model paired with a symbolic engine. AlphaGeometry 1 had solved 54 percent.⁸⁵³ DeepMind-adjacent reporting, InfoQ, February 2025, says flagship chat models o1 and Gemini Thinking solved *none* of that IMO-geometry set.⁸⁵⁴ A kitchen-table chatbot is not AlphaGeometry. GSM8K saturation is not geometry proof. Humanity’s Last Exam, lastexam.ai fetched 30 August 2026: Gemini 3 Pro 38.3 percent, GPT-5 25.3 percent, Grok 4 24.5 percent, Claude 4.5 Sonnet 13.7 percent, GPT-4o 2.7 percent overall — not a K–12 geometry-proof score.⁸⁵⁵ A named 2026 evaluation of consumer ChatGPT on high-school two-column proofs was not opened for this project. That eval stays **UNVERIFIED**.⁸⁵⁶ Treat model-written proofs as untrusted drafts. The child writes the proof unaided. The parent, or a human solution manual, is the judge.

This chapter has to keep both documents in view. The course is six critical areas, not two-column theater. The chatbot that will write a fluent numbered argument is not the judge of that argument. A parent who thinks “geometry = two-column proofs” is using a narrower map than Appendix A publishes. A parent who thinks “the AI will do the proofs” is using a narrower map than the evaluations publish. Both corrections land here, because this is the year the parent who learned two-column will try to reproduce a high-school course that no longer opens that way, and because this is the year a language model will most confidently skip a correspondence, cite a non-theorem, or assume the conclusion, while sounding like a geometry teacher.

⁸⁵²Achieve / Common Core State Standards writing team, *Appendix A: Designing High School Mathematics Courses Based on the Common Core State Standards*, Oregon Department of Education copy, <https://www.oregon.gov/ode/educator-resources/standards/mathematics/Documents/math-appendix-a-model-course-pathways.pdf>, accessed 30 August 2026. Geometry course purpose: “formalize and extend students’ geometric experiences from the middle grades. Students explore more complex geometric situations and deepen their explanations of geometric relationships, moving towards formal mathematical arguments.” Transformations emphasized early; explicit difference from older courses that began with axioms and two-column proof of triangle congruence. Access date for all URLs in these notes: 30 August 2026.

⁸⁵³Yuri Chervonyi et al., “Gold-medalist Performance in Solving Olympiad Geometry with AlphaGeometry2,” arXiv:2502.03544 (2025), <https://arxiv.org/html/2502.03544v1>, accessed 30 August 2026. Specialized neuro-symbolic system (Gemini-based language model plus symbolic engine). **84 percent** of IMO 2000–2024 geometry problems (42 of 50) versus **54 percent** for AlphaGeometry 1; coverage of the AG language 88 percent. Not a home product and not consumer ChatGPT.

⁸⁵⁴DeepMind-adjacent reporting, InfoQ, February 2025, as used in 04-ai.md §4.3: flagship chat models o1 and Gemini Thinking solved **none** of the IMO 2000–2024 geometry set on which AlphaGeometry2 scored 84 percent. Journalism/adjacent reporting, labelled; not a claim that no chat model will ever solve an IMO geometry problem.

⁸⁵⁵Humanity’s Last Exam, <https://lastexam.ai/>, fetched 30 August 2026. Gemini 3 Pro **38.3%**, GPT-5 **25.3%**, Grok 4 **24.5%**, Claude 4.5 Sonnet **13.7%**, GPT-4o **2.7%** overall. *Nature* 649 (2026): 1139–1146. Not a K–12 geometry-proof score. Math-subset percentage not broken out on the page — **UNVERIFIED**. Calibration error still high (50–89 percent on the page).

⁸⁵⁶Named 2026 evaluation of consumer ChatGPT on high-school two-column proofs: not opened this research day. **UNVERIFIED**. Brief open question 21; 04-ai §4.3. Do not invent the eval. Treat model-written proofs as untrusted drafts until a named eval is opened.

When the course typically appears: Grade 10 after Algebra I, or Grade 9 after Grade 8 Algebra I. In the integrated pathway the same clusters are distributed across Mathematics I–III rather than concentrated in one year.⁸⁵⁷ Grade labels in this chapter are sequence anchors, not prison bars. The National Center for Education Statistics already knew the sequence was stretched. In the 2016 Parent and Family Involvement tables, Algebra I was taught *that year* to 41 percent of the home-schooled 6–8 band; arithmetic was still taught to 31 percent of the 9–12 band.⁸⁵⁸ Geometry, *ever* taught during home instruction, reached 52 percent of high-school-level homeschoolers; taught *that year*, 26 percent of 9–12. The same table puts geometry in 15 percent of the 3–5 band and 19 percent of 6–8 *that year* — the word is doing different work at different ages.⁸⁵⁹ Finding 15 of the National Mathematics Advisory Panel is still the age-lock refusal: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; claims that children of a given age cannot learn certain content because they are “too young,” “not in the appropriate stage,” or “not ready” have “consistently been shown to be wrong.”⁸⁶⁰ Accelerate a child who has Grade 8 rigid motions, similarity, and the Pythagorean theorem, and Algebra I linear equations and quadratics. Do not withhold congruence from a fourteen-year-old who has them. Do not start this course while proportional reasoning is still a cross-multiply chant, or while completing the square is still a mystery. Skipping similarity because “trigonometry is Algebra II” is the error Finding 30 does not license.⁸⁶¹ Chapter 6 was the last chance the Panel named to close the Critical Foundations before authentic algebra. Chapter 7 was Algebra I. This chapter is the second high-school course on the traditional map. Diagnose first. Then this course. Do not treat a drawing as a theorem, and do not treat a chatbot’s numbered paragraph as a proof.

The Common Core State Standards for Mathematics are treated here as a map, not as a homeschool mandate. Many homeschoolers do not use CCSSM. Many U.S. states have adopted, adapted, or replaced it. A 2026 state-by-state census of who still uses it is **UNVERIFIED** in this project; this chapter will not say “most states.”⁸⁶² What the map is good for is specifying what the mathematics

⁸⁵⁷Appendix A (n. 852): typical appearance Grade 10 after Algebra I, or Grade 9 after Grade 8 Algebra I. Integrated pathway distributes the same clusters across Mathematics I–III. “The pathways and courses are models, not mandates.”

⁸⁵⁸Jiashan Cui and Rachel Hanson, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey*, NCES 2020-001 (Washington, DC: National Center for Education Statistics, 2019), Table 9: Algebra I taught *that year* to 41 percent of the homeschooled 6–8 band; arithmetic still taught to 31 percent of 9–12. <https://nces.ed.gov/pubs2020/2020001.pdf>. Grade labels are sequence anchors, not prison bars. The 2023 First Look (NCES 2024-113) did not republish those subject tables.

⁸⁵⁹Cui and Hanson, NCES 2020-001, Table 8: geometry *ever* taught during home instruction, 52 percent of high-school-level homeschoolers. Table 9: geometry taught *that year* — 3–5, 15 percent; 6–8, 19 percent; 9–12, 26 percent. The word is doing different work at different ages. Cells flagged by the agency remain caution cells.

⁸⁶⁰National Mathematics Advisory Panel, *Foundations for Success: The Final Report of the National Mathematics Advisory Panel* (U.S. Department of Education, 2008), Finding 15, ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>, accessed 30 August 2026. What is developmentally appropriate is “largely contingent on prior opportunities to learn.” Age-lock claims (“too young,” “not in the appropriate stage,” “not ready”) have “consistently been shown to be wrong.” Brain-immaturity claims are not justified even when students possess the prerequisite knowledge. Still-cited algebra-foundations synthesis, not the science of 2026 classrooms.

⁸⁶¹NMAP, *Foundations for Success* (2008), Finding 30: mathematically gifted students with motivation “appear to be able to learn mathematics much faster... with no harm to their learning, and should be allowed to do so.” Acceleration requires prerequisites. Do not skip similarity because trigonometry “is Algebra II.”

⁸⁶²State-by-state status of CCSSM in 2026 (who still uses it, who uses “standards formerly known as,” who never did): **UNVERIFIED** in this project (01-scope open question 8). Do not say “most states” without a current count.

is: congruence via rigid motions; triangle congruence ASA, SAS, SSS from the definition of congruence in terms of those motions; similarity via dilations and AA; trigonometric ratios for acute angles; the Pythagorean theorem via similarity; coordinate geometry; circles with and without coordinates; informal arguments for circumference, area, and volume using dissection, Cavalieri's principle, and informal limits; applications of probability. The standards "do not dictate curriculum or teaching methods," and the order of topics inside a grade is not a required teaching order.⁸⁶³ This book will not dump 180 days. It will not sell Singapore, Saxon, Math-U-See, or Art of Problem Solving. Those names will appear as options, by grain and placement, labelled **vendor** where the claim is the publisher's. Appendix A's pathways and courses are models, not mandates.⁸⁶⁴ A family on the integrated path will not have a "geometry year"; the same six clusters still need to be taught.

Four lanes run through the sections. They are not four identical headings. Sections 8.1 through 8.3 explain the mathematics to the parent — what the course is, and why two-column theater is a narrower map; congruence, similarity, and right-triangle trigonometry; coordinate geometry, circles, and informal three-dimensional arguments. Section 8.4 is proof as viable arguments, not a format: Appendix A's instructional note on G.CO.9–11, CCSSM Practice 3, and a first-proof sequence labelled as an extrapolation from worked-example fading, not as a geometry-proof randomized trial. Section 8.5 plants applications of probability for Chapter 11 and refuses to treat spatial play as proof. Section 8.6 is teaching guidance for the parent whose own course was axiom-first: representations connected to the written argument, fading, generation; proportional reasoning as the load-bearing skill; Algebra I as the coordinate prerequisite; co-op as a named NCES option when the parent is the ceiling. Section 8.7 is the parent-hosted AI lane for this course: a short reminder of Chapter 3's ban versus allow, then the geometry jobs. Diagram generation is not proof. GSM8K saturation is not a proof-writing score. Ban "write the two-column proof for this figure." Allow parent-first: what would count as a viable argument; an isomorphic figure with the parent-held key; diagnose a student's gap. The child writes unaided. The parent holds the key.

Chapter 2 already locked the method this chapter will use: five interwoven strands; conceptual and procedural knowledge iterating, not a one-way street; representations connected to symbols in the same visual field, then faded; worked examples faded for novices, incorrect examples explained; blocked practice to learn execution, interleaved and spaced practice to choose a strategy and keep it — and the clean classroom randomized trials of interleaving are grade 7, not a geometry *d* this chapter will invent; word problems as schemas, not keyword hunts; fluency as flexible, accurate, efficient retrieval; explicit instruction for students already struggling, productive failure as generation-then-instruction on a new concept when priors exist — those two literatures not the same comparison; kitchen-math transfer hard. Chapter 3 already locked the AI lane: unguarded solvers raise practice scores and cut unaided exams; the model may prepare the adult and the next

CCSSM is treated as a widely used map, not a mandate.

⁸⁶³Common Core State Standards Initiative, *Common Core State Standards for Mathematics*, "How to read the grade level standards," https://www.thecorestandards.org/assets/CCSSI_Math%20Standards.pdf, accessed 30 August 2026. The standards "do not dictate curriculum or teaching methods"; the order of topics inside a grade is not a required teaching order. Map, not a homeschool mandate. Not a Common Core advocacy tract and not an anti-Common Core tract.

⁸⁶⁴Appendix A (n. 852): "The pathways and courses are models, not mandates." Four model pathways: Traditional (Algebra I, Geometry, Algebra II), Integrated (Mathematics I, II, III), and compacted versions of each.

problem; it may not do the child's problem. This chapter will not re-litigate Bastani at monograph length. One short reminder, then the objects of this course. Chapter 6 already did Grade 8 congruence via rigid motions, similarity via dilations, and the Pythagorean theorem. This chapter will not dump that band. A child who cannot move a triangle by a rotation, a reflection, or a translation, who treats similar as "looks like," who cannot write a ratio of corresponding sides, is not ready for this course regardless of calendar grade. Diagnose that. Then teach this course.

The popular story collapses those locks into slogans they do not support. Geometry is two-column proofs. Rigid motions are elementary, so skip them; the real course starts with postulates. Similarity is a middle-school leftover; trigonometry is Algebra II. A drawing that looks right is a theorem. They'll learn proof by building a shed. The AI will write the two-column. GSM8K is high, so the model can do geometry. AlphaGeometry lives in ChatGPT. The documents do not say any of those things, not as the slogans say them. This chapter prints the documents. It prints the six critical areas, the Practice 3 sentence, the 52 percent, the 84 percent as a specialized system, and the "none of that set." It leaves the open questions standing — a named 2026 consumer-ChatGPT two-column evaluation, not opened; proof, modeling, and the limit concept, thin experimental base relative to fractions and facts; Wai, Lubinski, and Benbow 2009 spatial-STEM coefficients, paper not opened, **UNVERIFIED**; Tall and Vinner 1981 concept-image quotations, not opened, **UNVERIFIED**; Appendix A's integrated Mathematics I–III topic-by-topic, only partly extracted. It does not invent a kitchen scene to make the science feel warmer. The parent who stopped at algebra is the design user. The job is to hear "they look the same" offered as SAS, and a fluent numbered paragraph that assumes the conclusion, before any tool is allowed to print a two-column the parent never checks.

A compacted path exists so that calculus is reachable in senior year. Appendix A is explicit: compacted versions of the traditional and integrated pathways complete Grade 7, Grade 8, and the first high-school course in Grades 7–8 so that a student can reach Calculus or another college-level course by senior year. "The pathways and courses are models, not mandates."⁸⁶⁵ A non-accelerated student who starts Algebra I in Grade 9 finishes Geometry in Grade 10 and Algebra II in Grade 11, and has senior year for precalculus, statistics, or a first calculus course — not both precalculus and BC unless the fourth course is overloaded.⁸⁶⁶ Compact only if the gates are open: Grade 8 rigid motions, similarity, Pythagorean theorem; Algebra I linear equations and quadratics; Grade 7 proportional reasoning. Do not skip this course because calculus is the destination. Right-triangle trigonometry here is not unit-circle trigonometry. Chapter 9 will say that. Informal-limit arguments for area and volume here are not the precise limits of Chapter 10. This chapter is where those later objects become possible, or not.

⁸⁶⁵ Appendix A (n. 852), compacted pathways: complete Grade 7 + Grade 8 + the first high-school course in Grades 7–8 so that a student can reach Calculus or another college-level course by senior year. Models, not mandates. Repeated from n. 864 because the compacted claim is load-bearing for the calculus-destination sentence.

⁸⁶⁶ Appendix A (n. 852) and 01-scope.md Calculus band: a non-accelerated student who starts Algebra I in Grade 9 finishes Algebra II in Grade 11 and has senior year for precalculus, statistics, or a first calculus course — not both precalculus and BC unless the fourth course is overloaded. Regular three-year pathway plus a fourth course reaches precalculus as a junior and calculus as a senior *only if* high-school content started in Grade 8.

8.1 What the course is: six critical areas, not two-column theater

The mathematics of this course is six critical areas, named more specifically than a slogan about “proofs.” Appendix A’s Geometry critical areas are: (1) congruence, proof, and constructions — rigid motions; triangle congruence ASA, SAS, SSS from the definition of congruence in terms of rigid motions; theorems about lines, angles, triangles, parallelograms; compass-and-straightedge constructions; (2) similarity, proof, and trigonometry — dilations; AA similarity; trigonometric ratios for acute angles; the Pythagorean theorem via similarity; Laws of Sines and Cosines as (+) standards; (3) extending to three dimensions — informal arguments for circumference, area, and volume formulas using dissection, Cavalieri, informal limits; cross-sections and solids of revolution; (4) connecting algebra and geometry through coordinates — distance, midpoint, slope criteria for parallel and perpendicular lines, equation of a parabola given focus and directrix; (5) circles with and without coordinates — inscribed angles, tangent-radius perpendicularity, arc length and radian as a unit of measure, equation of a circle; (6) applications of probability — independence, conditional probability, two-way tables, addition rule; permutations and combinations as (+).⁸⁶⁷ That catalog is the course. Two-column proof of triangle congruence is one format inside one cluster of one critical area. It is not the course.

NCTM’s *Principles and Standards for School Mathematics*, 2000, already named the through-line a parent who thinks geometry begins in tenth grade has to hear: Geometry is “a natural area of mathematics for the development of students’ reasoning and justification skills,” beginning well before high school.⁸⁶⁸ PSSM predates CCSSM. It is not “the current U.S. standard.” It remains useful as professional consensus on the strand. Chapter 4 already partitioned circles and rectangles. Chapter 5 already classified figures by properties and put the first quadrant on the page. Chapter 6 already moved triangles by rigid motions, dilated them, and proved the Pythagorean theorem informally. This course formalizes those experiences. It does not invent them. A parent whose own geometry course never mentioned a translation is not looking at a dumbed-down tenth grade. They are looking at a different opening, chosen so that congruence is a transformation you can see before it is a list of criteria you can chant.

What the course depends on is already named. Grade 8 rigid motions, similarity, and the Pythagorean theorem. Algebra I linear equations and quadratics, for coordinate geometry and for completing the square when a circle’s equation arrives. Grade 7 proportional reasoning, the load-bearing skill for similarity and for right-triangle trigonometry.⁸⁶⁹ A child who still treats 2:3 as “two and three,” who cannot see a constant of proportionality in a table and a graph through

⁸⁶⁷Appendix A (n. 852), Geometry critical areas (six): (1) congruence, proof, and constructions; (2) similarity, proof, and trigonometry; (3) extending to three dimensions; (4) connecting algebra and geometry through coordinates; (5) circles with and without coordinates; (6) applications of probability. Laws of Sines and Cosines, and permutations/combinations, are (+) standards.

⁸⁶⁸National Council of Teachers of Mathematics, *Principles and Standards for School Mathematics*, Executive Summary (Reston, VA, 2000), https://www.nctm.org/uploadedFiles/Standards_and_Positions/PSSM_ExecutiveSummary.pdf, accessed 30 August 2026. Geometry is “a natural area of mathematics for the development of students’ reasoning and justification skills,” beginning well before high school. PSSM predates CCSSM; professional consensus on the strand, not “the current U.S. standard.”

⁸⁶⁹01-scope.md Geometry band: depends on Grade 8 rigid motions, similarity, Pythagorean theorem; Algebra I linear equations and quadratics (coordinate geometry and completing the square); Grade 7 proportional reasoning as the load-bearing skill for similarity and right-triangle trigonometry.

the origin, will meet AA similarity as a slogan and SOH-CAH-TOA as a chant. Chapter 6 was that gate. Do not start this course while it is still open. What the course unlocks is also already named. Unit-circle trigonometry in Algebra II and precalculus. Analytic geometry of conics. The informal-limit arguments for area and volume that calculus will make precise.⁸⁷⁰ Skip this course, or reduce it to two-column theater, and those later objects arrive as symbol soup — or they never arrive, because the transcript never showed Geometry.

The word “geometry” is doing different work at different ages. NCES 2016, restated because the percents will otherwise be mashed: geometry *ever* taught during home instruction, 52 percent of high-school-level homeschoolers; taught *that year* to 26 percent of 9–12, 19 percent of 6–8, 15 percent of 3–5.⁸⁷¹ The 3–5 cell is not this course. It is attributes of shapes, area, angle. The 6–8 cell is not this course. It is 8.G: congruence via rotations, reflections, translations; similarity via dilations; informal arguments for triangle angle-sum; the Pythagorean theorem. The 9–12 cell *that year* is closer, and still not a completion rate. The 2012 report found 54 percent of high-school-level homeschoolers taught geometry, with calculus “not often.” Do not narrate 54 → 52 as decline without checking standard errors and questionnaire differences. Both figures say: Algebra I is common (79 percent ever-taught at high-school level in 2016; 88 percent taught basic algebra in the 2012 report), Geometry and Algebra II are not universal, calculus at home is a minority.⁸⁷² Algebra I 79, geometry 52, Algebra II 49, calculus 15 with caution. The drop from Algebra I to Geometry is this book’s high-school problem as much as the further drop to Algebra II. A kitchen that stops after Algebra I because “the state doesn’t require it” has a college problem in every named university page this chapter will print, and a mathematics problem in every later course that needs a similar triangle or a coordinate proof.

NCAA and named colleges want the title. NCAA Eligibility Center: math core courses are Algebra I or higher; Division I requires three years of such math among sixteen core courses.⁸⁷³ Geometry is inside that “or higher.” The University of Pittsburgh’s home-schooled-applicant page asks for a minimum of three years of mathematics, plus a Home School Supplemental Form and a transcript by year with grades.⁸⁷⁴ Southern Methodist University’s application guide for home-schooled and

⁸⁷⁰01-scope.md Geometry band: unlocks unit-circle trigonometry (Algebra II / Precalculus); analytic geometry of conics; informal-limit arguments for area and volume that calculus will make precise. Plant for Chapters 9 and 10; do not write those chapters here.

⁸⁷¹Cui and Hanson, NCES 2020-001, Tables 8–9 (n. 859). Geometry 52 percent ever-taught at high-school level; 26 / 19 / 15 percent taught *that year* in 9–12 / 6–8 / 3–5.

⁸⁷²Cui and Hanson, NCES 2020-001, Table 8: high-school-level homeschoolers, subjects *ever* taught — arithmetic 76; basic algebra (Algebra I) 79; geometry 52; advanced algebra (Algebra II) 49; probability 24; calculus 15 (interpret with caution). Jeremy Redford, Danielle Battle, and Stacey Bielick, *Homeschooling in the United States: 2012*, NCES 2016-096 (Washington, DC: NCES, 2016), figure 5: 88 percent taught basic algebra; 54 percent geometry; calculus “not often” (10 percent, interpret with caution). ERIC ED569947. Do not narrate 88 → 79 or 54 → 52 as decline without checking standard errors and questionnaire differences. Both figures: Algebra I is common; Geometry and Algebra II are not universal; calculus at home is a minority.

⁸⁷³NCAA Eligibility Center, Division I initial-eligibility requirements and Home School Toolkit 2025–26: math core courses are Algebra I or higher; Division I requires 3 years of such math among 16 core courses and a 2.3 core GPA. NCAA is a private association’s eligibility rules, not law, but it binds athletic scholarships. Audited, CLEP, or credit-by-exam courses are not NCAA-approved core courses.

⁸⁷⁴University of Pittsburgh, Home Schooled Applicants, admissions.pitt.edu/home-school/: minimum **3 years of mathematics** (plus 4 English, 3 science, 3 social studies, 2 arts/humanities); Home School Supplemental Form; transcript by year with grades; college courses designated and official college transcripts sent. Pattern, not a rule, for the

nontraditional students lists a minimum *suggested* three mathematics — Algebra I, Geometry, Algebra II — and wants a transcript with consistent grading and course descriptions including textbook titles.⁸⁷⁵ North Central College recommends three years of mathematics including algebra and geometry, one course beyond geometry preferred.⁸⁷⁶ Pattern, not a rule: three years of high-school math through Algebra II is the floor most of these pages name. Geometry appears by name on two of the three. Transcript the course as Geometry. Not “Math 10.” Not “Saxon Algebra 2” without a parenthetical a stranger can map onto the geometry clusters. Chapter 12 will say that as an export rule. This chapter says it as a course-identity rule. State compliance is not college evidence. Texas legally needs neither portfolio nor test for compliance; colleges and NCAA will still want a transcript that names this course.⁸⁷⁷

Saxon classic editions integrate geometry inside Algebra 1, Algebra 2, and Advanced Math; a separate Geometry course also exists. That is **vendor** / Rainbow Resource catalogue, not a finding that integration is better.⁸⁷⁸ The transcript still needs a course a stranger can map onto “Geometry.” A family that ran the classic integration and never named the geometry work on paper has done some of the mathematics and none of the export. Math-U-See has a named Geometry level — **vendor**, a topic, not a birthday.⁸⁷⁹ Art of Problem Solving has a Geometry title after Algebra — **vendor**. Teaching Textbooks, RightStart, Math Mammoth, and Life of Fred geometry placements that were not opened on a vendor primary remain **UNVERIFIED**. Do not promote those rows to handbook fact.⁸⁸⁰ No homeschool-curriculum randomized trial comparing any of these programs was opened for this project. Choose by fit — parent math confidence, student stance, special needs, budget, college target, and whether the book opens with rigid motions or with postulates — not by a phantom homeschool meta-analysis. Cathy Duffy Reviews and Rainbow Resource are **catalogue** / **reviewer**, not independent effectiveness trials.

Integrated-path families will not have a “geometry year.” Appendix A’s Integrated Mathematics I–II–III distributes the same clusters across three years rather than concentrating them. Topic-by-

three-year floor.

⁸⁷⁵Southern Methodist University, Application Guide for Homeschool, Non-Traditional Students: minimum *suggested* four units English, **three mathematics (Algebra I, Geometry, Algebra II)**, three science (two lab), three social science, two-year foreign-language sequence; transcript with consistent grading and course descriptions including textbook titles; dual-credit, AP, IB as additional measures.

⁸⁷⁶North Central College, Homeschool Applicants: transcript with course names, grades, credit, year; recommended **3 years mathematics including algebra and geometry, one course beyond geometry preferred**.

⁸⁷⁷Home School Legal Defense Association, “Texas Homeschool Laws (at a glance)” and “How to Comply with Texas’s Homeschool Law.” **Advocacy**, attorney-reviewed for members, not the Texas Education Code and not the *Leeper* opinion. Teacher qualifications: none. Notification: no. Assessment: no. Required subjects include mathematics. State compliance ≠ college evidence: colleges and NCAA still want a transcript. Confirm against the code.

⁸⁷⁸Houghton Mifflin Harcourt / Saxon Homeschool scope: geometry integrated in Algebra 1, Algebra 2, and Advanced Math in classic editions; a separate Geometry course also exists. Rainbow Resource catalogue as **vendor** / **catalogue**, not an independent effectiveness trial. Transcript still needs a course a stranger can map onto “Geometry.”

⁸⁷⁹Math-U-See / Demme Learning, curriculum sequence, <https://mathusee.com/products/math-u-see-curriculum/>, accessed 30 August 2026. **Vendor**. Named Geometry level is a topic, not a birthday. Greek letters uncoupled from calendar grade.

⁸⁸⁰Teaching Textbooks, RightStart, Math Mammoth, and Life of Fred topic placements not opened on a vendor primary: **UNVERIFIED** (01-scope curriculum-sequence table). Do not promote those rows to handbook fact. No homeschool-curriculum RCT comparing named programs was opened for this project. Cathy Duffy Reviews and Rainbow Resource are **catalogue** / **reviewer**, not independent effectiveness trials.

topic extraction of that pathway was only partly done for this project; the traditional pathway was extracted in more detail.⁸⁸¹ That remainder stays thinner. What does not stay thinner is the obligation: congruence, similarity, trigonometry, coordinates, circles, informal three-dimensional arguments, and S.CP still have to be taught. Do not skip them because the book spines say Mathematics II instead of Geometry. Do not skip them because a co-op offers Algebra II next and “we already did shapes in middle school.” The six critical areas are the course, however packaged.

Transformation-first versus axiom-first is the live debate this section will not pretend to settle. Appendix A has chosen transformations. Many homeschool reprints of older texts have not. Euclid’s *Elements* is a real book; it is not Appendix A’s Geometry. A parent who learned SAS as a postulate, then proved ASA, then spent a semester on two-column, is not wrong about the mathematics they learned. They are looking at a different instructional opening. This book should explain the difference and not pretend two-column is the only honest proof. Section 8.4 will do that work. Section 8.2 will do the mathematics of congruence from rigid motions, because that is the map this chapter is using, and because a parent who skips rigid motions “because we did that in eighth grade” is skipping the definition on which ASA, SAS, and SSS then stand.

What this course is *not*. It is not two-column theater. It is not a year of constructions with no arguments. It is not a year of arguments with no figures. It is not unit-circle trigonometry — Chapter 9. It is not Cavalieri made into a precise limit — Chapter 10. It is not AP Statistics — Chapter 11; S.CP is a plant, not a dump. It is not a spatial-IQ program. Jonathan Wai, David Lubinski, and Camilla P. Benbow, 2009, on spatial skill and later STEM achievement: paper not opened; exact coefficients **UNVERIFIED**. This chapter will not launder an unopened spatial-STEM paper into a reason to skip proof, and it will not launder kitchen building into a reason to skip proof either.⁸⁸² David Tall and Shlomo Vinner, 1981, concept image versus concept definition: not opened; quotations **UNVERIFIED**. Do not invent a teaching sequence from an unopened paper.⁸⁸³ Proof, modeling, and the limit concept have a thin experimental base relative to fractions and facts. That is an open question this chapter leaves standing, not a license to fill the gap with folklore.⁸⁸⁴

Failure points visible in this course, named so a parent can steal the diagnostic before buying the next book, are five, and they are not cute. Congruence treated as “they look the same,” with no correspondence of vertices and no rigid motion that would take one triangle onto the other. Similarity skipped because trigonometry “is Algebra II,” so SOH-CAH-TOA arrives as a chant with no ratio of corresponding sides. Coordinate geometry attempted without Algebra I’s linear equations and quadratics, so the circle’s equation is completing-the-square soup. Proof reduced to two-column format, so a valid paragraph is marked wrong and an invalid two-column is marked right. Spatial

⁸⁸¹01-scope open question 7: Integrated Math I–III topic-by-topic from Appendix A only partly extracted (traditional pathway extracted in more detail). Remainder thinner. The obligation to teach the six critical areas does not thin with the extraction.

⁸⁸²Jonathan Wai, David Lubinski, and Camilla P. Benbow (2009), spatial skill and later STEM achievement. Paper not opened; exact coefficients **UNVERIFIED**. Do not launder an unopened spatial-STEM paper into a reason to skip proof, and do not launder kitchen building into a reason to skip proof. Open question this chapter leaves standing.

⁸⁸³David Tall and Shlomo Vinner, “Concept Image and Concept Definition in Mathematics with Particular Reference to Limits and Continuity,” *Educational Studies in Mathematics* 12 (1981): 151–169. Not opened this pass; quotations **UNVERIFIED**. Do not invent a teaching sequence from an unopened paper.

⁸⁸⁴Proof, modeling, and the limit concept — thin experimental base relative to fractions and facts. Brief open question 21; 02-learning open questions. Leave standing; do not fill with folklore or with a homemade geometry-proof *d*.

play — building, making, Minecraft, a shed — treated as proof. Those five are why this chapter is this long. Algebra II’s unit circle and calculus’s informal-limit on-ramp will be symbol soup if this chapter thins.

What this section has done is place the course and kill two-column theater as the definition of it. Congruence, similarity, and right-triangle trigonometry come next, because they are the first two critical areas, and because a parent who learned axiom-first has to see why a translation is not an elementary leftover.

8.2 Congruence, similarity, and right-triangle trigonometry

Congruence is a rigid motion. That sentence is the first teaching move of this course on Appendix A’s map, and it is a sentence the parent who learned axiom-first has to be able to hear wrong. Two figures are congruent if one can be taken onto the other by a sequence of rotations, reflections, and translations — the rigid motions that preserve distance and angle.⁸⁸⁵ “They look the same” is not that sentence. A drawing that appears to match is not that sentence. SAS as a remembered postulate, with no motion named, is not yet that sentence. The child who can chant SAS and cannot say which vertex goes to which vertex, or cannot describe a rotation that would take $\triangle ABC$ onto $\triangle DEF$, has the chant and not the definition. Failure point (a) from the last section is this one. It is the load-bearing failure of congruence.

What the map actually contains is more specific than “do some triangles.” Critical area 1: congruence, proof, and constructions. Rigid motions first. Then triangle congruence ASA, SAS, SSS *from the definition of congruence in terms of rigid motions*. Then theorems about lines, angles, triangles, parallelograms. Then compass-and-straightedge constructions.⁸⁸⁶ Do not skip rigid motions because the parent’s own geometry course started with postulates. Grade 8 already used rotations, reflections, and translations as the meaning of congruence (8.G). This course uses that meaning as a definition, then derives the familiar criteria. A translation that takes A to D , followed by a rotation about D that takes the image of AB onto DE , followed by a reflection if orientation is reversed, is a proof that the triangles are congruent *if* the corresponding parts named by SAS, ASA, or SSS are given. The criteria are compressions of a motion. They are not a list of magic letter-triples. SSA is not on the list because a motion is not determined: two non-congruent triangles can share that configuration. HL for right triangles is SSS in disguise, hypotenuse and leg determining the third side by the Pythagorean theorem. AAA is similarity, not congruence — the same shape, not the same size. Those four sentences are the parent-knowledge that makes a wrong letter-triple audible. A child who writes SSA as a reason has not “forgotten the list.” They have treated a configuration that does not determine a rigid motion as if it did.

Correspondence of vertices is the notation habit that later saves the proof. $\triangle ABC \cong \triangle DEF$ names $A \leftrightarrow D, B \leftrightarrow E, C \leftrightarrow F$. Writing $\triangle ABC \cong \triangle FED$ is a different claim. A fluent paragraph that

⁸⁸⁵Common Core State Standards Initiative, *Common Core State Standards for Mathematics*, G.CO (Congruence) and Grade 8 8.G: congruence via rotations, reflections, translations — rigid motions that preserve distance and angle. Appendix A critical area 1 uses that definition as the high-school opening.

⁸⁸⁶Appendix A (n. 852), Geometry critical area 1: rigid motions; triangle congruence ASA, SAS, SSS from the definition of congruence in terms of rigid motions; theorems about lines, angles, triangles, parallelograms; compass-and-straightedge constructions. Do not skip rigid motions because the parent’s own course started with postulates.

proves the wrong correspondence has not proved the claim on the page. Chapter 2 already treated representations connected to symbols in the same visual field. Here the diagram and the written correspondence have to sit in the same visual field. A diagram with no correspondence marked is a picture. A two-column with no diagram is a format. Neither is yet an argument. Mark the vertices. Mark the given equal lengths and equal angles. Then write. Section 8.4 will say that again as a proof-format claim. This section says it as a congruence claim.

Constructions belong in the same critical area because a compass-and-straightedge construction is a rigid-motion recipe you can watch. Copying a segment is a translation of length. Copying an angle is a rigid motion of a ray. The perpendicular bisector is the set of points equidistant from two endpoints — a reflection’s fixed set. The angle bisector is a ray a reflection across which swaps the sides of the angle. A parent who treats constructions as “the drawing unit” and proofs as “the logic unit” has split a single critical area. Build the perpendicular bisector. Then prove that every point on it is equidistant from the endpoints. Then use that theorem in a later argument. Iterate. Do not withhold the construction until the theorem is written, and do not run a page of constructions with no argument attached.

Theorems about lines, angles, triangles, and parallelograms are the rest of critical area 1: vertical angles; alternate interior angles with a transversal across parallels; the triangle angle-sum; the exterior-angle theorem; base angles of an isosceles triangle; opposite sides of a parallelogram. Grade 8 already asked for informal arguments for several of these (8.G). This course asks for arguments a stranger could check. “Informal” in Grade 8 meant a diagram, a few sentences, a transformation. “Formal” here does not mean two-column. It means the reasons are named, the correspondence is named, and the conclusion is not assumed in the second line. Appendix A’s instructional note, waiting in section 8.4, will license paragraph, flow diagram, two-column, and diagram without words. The parent’s job is the validity of the underlying reasoning, not the ruling of a format.

Similarity is a dilation. Two figures are similar if one can be taken onto the other by a rigid motion and a dilation — a scaling about a point that multiplies all distances by the same positive constant, the scale factor.⁸⁸⁷ “Looks like” is not that sentence. “Same shape, different size” is closer, and still not a correspondence of vertices and a scale factor. AA similarity is the compression: two pairs of corresponding congruent angles determine a similarity of triangles, because the third angles then match and a dilation-plus-rigid-motion exists. SAS similarity and SSS similarity are the other common compressions. Do not skip similarity because trigonometry “is Algebra II.” Right-triangle trigonometry in this course *is* similarity. The trigonometric ratios for an acute angle are the constant ratios of corresponding sides in the family of similar right triangles that share that acute angle.⁸⁸⁸ SOH-CAH-TOA is a memory aid for those three ratios, not a third course. A child who can chant SOH-CAH-TOA and cannot say that $\sin \theta$ is the scale factor from the hypotenuse to the opposite side, in every right triangle with acute angle θ , has the chant and not the similarity. Failure point (b) from the last section is this one.

⁸⁸⁷CCSSM G.SRT and Appendix A critical area 2: similarity via dilations (a scaling about a point that multiplies all distances by the same positive constant) plus rigid motions; AA similarity. Grade 8 8.G already used dilations informally.

⁸⁸⁸Appendix A (n. 852), Geometry critical area 2: trigonometric ratios for acute angles as constant ratios of corresponding sides in similar right triangles that share the acute angle. Right-triangle trigonometry in this course *is* similarity. Unit-circle trigonometry is Algebra II / Precalculus (Chapter 9). Do not skip similarity because trigonometry “is Algebra II.”

Proportional reasoning is the load-bearing skill. Chapter 6 already installed a ratio as a relationship, a unit rate, a constant of proportionality k visible in a table and a graph through the origin. Similarity is that relationship on corresponding lengths. If $\triangle ABC \sim \triangle DEF$ with scale factor k , then $DE/AB = EF/BC = FD/CA = k$. Cross-multiply is a consequence of that equality of ratios, not a belief. The child who always cross-multiplies a missing-length problem, including when a unit-rate scaling would be one multiplication, has the trick from 7.RP still running. Jitendra, Star, Dupuis, and Rodriguez 2012 remains the trial this book can quote for proportion problems — seventh grade, 1,163 students, schema-based instruction versus regular instruction, immediate $g = 1.24$, six-week $g = 1.27$, no significant transfer — and it is not a geometry-proof trial.⁸⁸⁹ Do not mash it with a geometry d this chapter does not have. The kitchen how-to that survives is the same one Chapter 6 stole: identify the schema (here: corresponding sides of similar triangles), represent it with a diagram that marks the correspondence, estimate, solve, check. Vary the cover story. Never teach a magic word. “Similar” is not a keyword that means “cross-multiply the first two numbers you see.”

The Pythagorean theorem via similarity is the other load-bearing argument of critical area 2. Drop an altitude to the hypotenuse of a right triangle and you have three similar right triangles: the two smaller ones and the original. The altitude creates $a^2 = c \cdot p$ and $b^2 = c \cdot q$ with $p + q = c$, hence $a^2 + b^2 = c(p + q) = c^2$.⁸⁹⁰ That is a similarity argument, not a chant about 3-4-5. Grade 8 already used the theorem and its converse (8.G). This course asks why. Euclid’s proof with squares is real; the similarity proof is the one Appendix A names; a rearrangement of area is a dissection the next section will license. The parent who can hear a wrong answer hears a child who writes $a^2 + b^2 = c^2$ on a non-right triangle, or who treats the converse as automatic without checking the sides, or who cannot show the three similar triangles the altitude made. 3-4-5 and 5-12-13 are useful retrieve-able triples. They are not the theorem.

Trigonometric ratios for acute angles, then. Sine of an acute angle is opposite over hypotenuse. Cosine is adjacent over hypotenuse. Tangent is opposite over adjacent. Those three are the ratios this course owns. They live in similar right triangles. They do not yet live on the unit circle. Chapter 9 will put them on the unit circle, add radian measure as a completed unit, add the Pythagorean identity as a circle equation, and add the other quadrants. This chapter plants the acute-angle ratios and, in critical area 5, plants radian as a unit of arc length. Do not skip the plant. Do not pretend the plant is the tree. A student who arrives at calculus with only SOH-CAH-TOA is the common gap the scope memo named as inference from the maps, not a survey.⁸⁹¹ Hear it here: a child who can find a missing side in a 30-60-90 triangle by a chant and cannot say that $\sin 30^\circ = 1/2$ *because*

⁸⁸⁹Asha K. Jitendra, Jon R. Star, Danielle Dupuis, and Michael C. Rodriguez, “Effectiveness of Schema-Based Instruction for Improving Seventh-Grade Students’ Proportional Reasoning: A Randomized Experiment,” *Journal of Research on Educational Effectiveness* 5 (2012), <https://doi.org/10.1080/19345747.2012.725804>. Immediate $g = 1.24$, six-week retention $g = 1.27$, null transfer, $N = 1,163$, 42 classrooms. Not a geometry-proof trial. Do not quote 1.24 as a general-achievement miracle. Do not mash with Rohrer $d = 1.05$ (different design, different comparison, different outcome).

⁸⁹⁰Pythagorean theorem via similarity: Appendix A critical area 2 and CCSSM G.SRT. Drop an altitude to the hypotenuse; three similar right triangles; $a^2 = c \cdot p$ and $b^2 = c \cdot q$ with $p + q = c$, hence $a^2 + b^2 = c^2$. Grade 8 already used the theorem and its converse (8.G); this course asks why. Euclid’s squares proof is real; Appendix A names the similarity argument.

⁸⁹¹01-scope.md Calculus band, common gaps at the calculus door: “only right-triangle SOH-CAH-TOA” among them. **Inference from the maps, not a named survey.** Do not upgrade to a prevalence study.

every right triangle with a 30° angle is similar to every other, scale factor notwithstanding, has the chant. Special right triangles — 30-60-90, 45-45-90 — are retrieve-able compressions of those similarities, with scale factors $1 : \sqrt{3} : 2$ and $1 : 1 : \sqrt{2}$. Retrieve them after the similarity is visible, not instead of it.

Laws of Sines and Cosines are (+) standards.⁸⁹² Appendix A's plus marks content beyond college-and-career readiness as the Appendix defines it; they often live in a fourth course or in a richer Geometry. A parent shopping for "the whole geometry" should know they are optional on the map, not optional if the next course is a precalculus that assumes them. Do not treat their absence from a given book as a defect of the book without checking whether the family's fourth course will carry them. Do not treat their presence as proof that the book is "honors." They are the extension of right-triangle trigonometry to non-right triangles. The Law of Sines is a ratio of a side to the sine of the opposite angle, constant in a given triangle. The Law of Cosines is the Pythagorean theorem with a correction term for the included angle: $c^2 = a^2 + b^2 - 2ab \cos C$. When C is 90° , cosine is zero, and the Pythagorean theorem returns. That last sentence is why the Pythagorean theorem via similarity still has to be taught even if the Law of Cosines is coming. The special case is the structure.

Transformation-first versus axiom-first, restated as a parent choice rather than a franchise war. A transformation-first book will spend early weeks on rigid motions, then derive SAS, ASA, SSS. An axiom-first book will list postulates, including some of those criteria, then prove the rest. Both can produce viable arguments. Both can fail, the first by never writing an argument, the second by never connecting a criterion to a motion you could watch. This book uses Appendix A's opening because it is the most fully specified public map, not because Euclid is wrong. A family using a reprint of an older text is not behind. They are on a different train, often later to transformations and earlier to two-column. Say that. Do not say the reprint is "not real geometry." Do not say Appendix A "replaced proof with folding." Proof is Practice 3, next but one section. Folding, when it is a reflection you then write down, is a representation. Folding that never becomes a written reason is a spatial play, and section 8.5 will refuse to upgrade it.

Worked examples, faded, for the first pass through a congruence criterion. Chapter 2 already locked this for high-element-interactivity material. Show one fully worked argument that a given SAS configuration determines a rigid motion, then a second the child must explain, then a completion problem missing the last reason, then a conventional problem.⁸⁹³ Include at least one incorrect example: a two-column that cites SSA, or a paragraph that assumes the triangles congruent in line two in order to prove them congruent in line six. Booth, Lange, Koedinger, and Newton 2013 used incorrect examples on algebra. Applying that move to a first geometric argument is an extrapolation, not a geometry-proof randomized trial. Label it that way. Section 8.4 will say it again as the first-proof sequence. Once the child can execute a criterion, stop showing full examples. Expertise reversal is real. Then generate: the child writes the correspondence, marks the diagram,

⁸⁹²Appendix A (n. 852): Laws of Sines and Cosines are (+) standards — content beyond college-and-career readiness as the Appendix defines it. Often in a fourth course or a richer Geometry. Not a defect of a book that omits them without checking the family's fourth course; not proof that a book that includes them is "honors."

⁸⁹³Worked-example fading: John Sweller, worked-example effect for high-element-interactivity material; fading (completion problems) as the path off full examples. Chapter 2 lock. Applied here to a first congruence argument as **analogue / extrapolation**, not a geometry-proof RCT. See n. 908.

and produces a short unaided argument. Soderstrom and Bjork 2015: current performance is “often an unreliable index” of learning.⁸⁹⁴ A child who can follow a worked two-column and cannot produce a three-sentence paragraph unaided has performed, not learned.

What this section has done is put congruence on a rigid motion and similarity on a dilation, and refuse to let SOH-CAH-TOA swallow the second of those. Coordinate geometry is where Algebra I earns its keep. Circles and informal three-dimensional arguments are the other half of what two-column theater leaves out.

8.3 Coordinate geometry, circles, and informal 3-D arguments

Coordinate geometry is Algebra I sitting on the plane Chapter 5 opened in the first quadrant and Chapter 6 opened in four. Distance, midpoint, slope criteria for parallel and perpendicular lines, and the equation of a parabola given focus and directrix are critical area 4.⁸⁹⁵ They are not a Friday graphing worksheet. They are the place a congruence you cannot see in a diagram can still be proved, because the distance formula does not care how badly the figure was drawn. That sentence is why a drawing is not a theorem, and why this critical area belongs in a proof course rather than only in Algebra I.

Distance between (x_1, y_1) and (x_2, y_2) is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, which is the Pythagorean theorem on a coordinate right triangle whose legs are the run and the rise. Midpoint is the average of the coordinates, $((x_1 + x_2)/2, (y_1 + y_2)/2)$. Parallel lines have equal slope; perpendicular lines have slopes whose product is -1 , for non-vertical lines. Those three sentences require Algebra I’s linear equations and Grade 8’s slope-as-unit-rate. A child who still treats slope as “the number next to x ” and cannot compute it from two points will not prove that the diagonals of a parallelogram bisect each other by showing that the midpoints coincide. A child who cannot complete the square will not turn $x^2 + y^2 + 4x - 6y - 3 = 0$ into $(x + 2)^2 + (y - 3)^2 = 16$, a circle of radius 4 centered at $(-2, 3)$.⁸⁹⁶ Algebra I linear equations and quadratics are required for this critical area. Do not start it while they are still fragile. Do not skip Algebra I because “geometry is shapes.” Completing the square for a circle is the same completing the square Chapter 7 used for a quadratic. The representation changed. The algebra did not.

The equation of a parabola given focus and directrix is the first conic this course owns as a locus: the set of points equidistant from a point (the focus) and a line (the directrix). If the focus is $(0, p)$ and the directrix is $y = -p$, the distance definition produces $y = x^2/(4p)$, which is Algebra I’s $y = ax^2$ with a geometric reason for a . Ellipse and hyperbola, as remaining conics, typically wait for a fourth course or for Algebra II’s (+) remainder.⁸⁹⁷ Plant them. Do not dump them. Analytic

⁸⁹⁴Nicholas C. Soderstrom and Robert A. Bjork, “Learning Versus Performance: An Integrative Review,” *Perspectives on Psychological Science* 10, no. 2 (2015): 176–199, <https://doi.org/10.1177/1745691615569000>. Current performance is “often an unreliable index” of learning. People “mistakenly interpret their performance during acquisition as a reliable guide to long-term learning.” Full discussion in Chapter 3; this chapter uses the distinction.

⁸⁹⁵Appendix A (n. 852), Geometry critical area 4: distance, midpoint, slope criteria for parallel and perpendicular lines, equation of a parabola given focus and directrix. Algebra I linear equations and quadratics required.

⁸⁹⁶CCSSM G.GPE (Expressing Geometric Properties with Equations) and Appendix A critical area 5: equation of a circle; completing the square to convert a general quadratic of this form into $(x - h)^2 + (y - k)^2 = r^2$. The numerical example in the text is a worked illustration of that cluster, not a named study.

⁸⁹⁷Appendix A “Fourth Courses” column and Algebra II (+) remainder: remaining conics (ellipse / hyperbola). Plant;

geometry of conics is what this critical area unlocks; Chapter 9 and a precalculus book will carry the rest.

Circles, critical area 5, with and without coordinates. Inscribed angle: an angle with vertex on the circle, sides through two other points of the circle, measures half the intercepted arc. A central angle intercepts the same arc and measures twice the inscribed angle. Angles inscribed in the same arc are equal. An angle inscribed in a semicircle is a right angle. Tangent-radius perpendicularity: a tangent line is perpendicular to the radius at the point of contact.⁸⁹⁸ Those theorems are arguments, not observations about a drawing. A drawing will make a tangent “look perpendicular.” A proof says why, from the definition of tangent as a line that meets the circle at exactly one point, or from the radius as the shortest path from center to line. Arc length is a fraction of the circumference: $(\theta/360) \cdot 2\pi r$ in degrees. Radian as a unit of measure is the other plant: one radian is the central angle that intercepts an arc equal in length to the radius, so θ radians intercepts arc length θr . Chapter 9 will need that unit on the unit circle. This course introduces it as a circle measurement, not yet as the input of a trigonometric function of a real number. Do not skip the plant because “radians are precalculus.” Do not pretend the plant is F.TF.

The equation of a circle, $(x - h)^2 + (y - k)^2 = r^2$, is the distance formula set equal to a constant. Completing the square, already named, is how a general quadratic in x and y of this form becomes that equation. A child who expands and cannot complete, or who completes and cannot say that the left side is a distance squared, has a procedure and not a locus. Three representations again, one sitting: the definition (set of points at distance r from a center), the equation, the graph. Ask all three. Then ask the tangent-radius claim on a coordinate circle, so that perpendicular slopes have something to do.

Extending to three dimensions is critical area 3, and it is the on-ramp to calculus this course actually owns. Informal arguments for circumference, area, and volume formulas using dissection, Cavalieri’s principle, and informal limits; cross-sections and solids of revolution.⁸⁹⁹ Circumference: the limit of perimeters of inscribed regular polygons, or the “unrolled” proportion $C/D = \pi$. Area of a circle: dissect into sectors, rearrange toward a parallelogram of base πr and height r , so $A = \pi r^2$ — an informal limit, because the more sectors, the closer the rearrangement is to a rectangle. Volume of a pyramid or cone: Cavalieri, or a dissection argument that three pyramids of equal height and equal base-area fill a prism, hence $V = (1/3)Bh$. Cavalieri’s principle, informally: solids with the same cross-sectional area at every height have the same volume. A stack of cards shifted sideways has the same volume as the unshifted stack. That is the argument; it is not yet a precise integral. Chapter 10 will make the informal limit precise. This chapter plants the argument. Do not skip the plant because “volume formulas are middle school.” Grade 8 already had $V = \pi r^2 h$ for a cylinder, and $(1/3)\pi r^2 h$ for a cone, and $(4/3)\pi r^3$ for a sphere, as formulas (8.G). This course asks why, with dissection, Cavalieri, and an informal limit. A child who can retrieve $V = (1/3)Bh$ and cannot show three pyramids in a prism has the formula and not the argument.

do not dump Chapter 9.

⁸⁹⁸Appendix A (n. 852), Geometry critical area 5 / CCSSM G.C: inscribed angles; tangent-radius perpendicularity; arc length; radian as a unit of measure; equation of a circle. Radian here is a circle-measurement plant for F.TF in Algebra II, not yet trigonometric functions of a real number.

⁸⁹⁹Appendix A (n. 852), Geometry critical area 3: informal arguments for circumference, area, and volume formulas using dissection, Cavalieri’s principle, and informal limits; cross-sections and solids of revolution. On-ramp to calculus’s precise limits (Chapter 10). Grade 8 already had cylinder/cone/sphere formulas (8.G); this course asks why.

Cross-sections and solids of revolution are the other half of critical area 3. Slice a cube parallel to a face, get a square; slice on a diagonal, get a rectangle; slice through a vertex more freely, get a triangle. Revolve a rectangle about one side, get a cylinder; revolve a right triangle about a leg, get a cone. Those are the pictures calculus will later integrate. They are also the place a parent can hear a wrong answer without a two-column: “what two-dimensional figure do you get if you slice this right cylinder with a plane not parallel to the base and not through the axis?” An ellipse, typically, if the plane cuts all generators. A child who always says “circle” has a slogan from one slice.

Do not invent a teaching sequence from Tall and Vinner 1981. Concept image versus concept definition is a real distinction in the mathematics-education literature; the paper was not opened for this project; quotations stay **UNVERIFIED**.⁹⁰⁰ Informal limits in this course are Appendix A’s language, not a reconstructed Tall-and-Vinner lesson plan. The parent-facing claim that survives without that paper is simpler: the child needs a reason the area of a circle is πr^2 that is not “the formula,” and the reason this map licenses is an informal limit of a dissection. That is enough. The precise ε - δ definition is not this course. The concept-image literature, if a later edition opens it, can thicken the plant. This edition will not fake the thickening.

Do not invent Wai, Lubinski, and Benbow 2009 coefficients as a reason this critical area matters for STEM. Spatial skill and later STEM achievement is a real research program; the 2009 paper was not opened; exact coefficients stay **UNVERIFIED**.⁹⁰¹ What this chapter can say without that paper is already on the map: three-dimensional arguments, cross-sections, and solids of revolution are part of the course; they unlock calculus’s volume applications; they are not a substitute for proof. Kitchen building and making are spatial. They do not automatically produce Cavalieri. Section 8.5 will say that as NMAP Finding 26. This section says it as a critical-area claim: informal 3-D arguments are arguments. A shed is a shed.

Representations, again, because this is the section where the diagram most tempts the parent to skip the writing. Fuchs and colleagues 2021, Recommendation 3, already used in Chapters 5 and 6: choose representations that match the mathematics; connect concrete and semi-concrete to the notation in the same visual field; fade for accuracy, not for a calendar date.⁹⁰² A net of a cylinder next to the written $S = 2\pi r^2 + 2\pi rh$, in the same sitting. A stack of cards next to the sentence “same cross-section at every height, same volume.” A coordinate grid next to the distance formula, with the right triangle drawn. Carbonneau, Marley, and Selig 2013: overall d approximately 0.37 versus abstract-only, larger on retention than on transfer or justification, high guidance associated with higher retention and problem solving.⁹⁰³ That number is not “nets work.” Put the net next to the

⁹⁰⁰Tall and Vinner 1981 (n. 883). Not opened; quotations **UNVERIFIED**. Informal limits in this course are Appendix A’s language, not a reconstructed Tall-and-Vinner lesson plan.

⁹⁰¹Wai, Lubinski, and Benbow 2009 (n. 882). Not opened; exact coefficients **UNVERIFIED**. Three-dimensional arguments are on the map without that paper; they are not a substitute for proof.

⁹⁰²Lynn S. Fuchs et al., *Assisting Students Struggling with Mathematics: Intervention in the Elementary Grades*, WWC 2021006 (IES / What Works Clearinghouse, 2021), Recommendation 3: choose representations that match the mathematics; connect concrete and semi-concrete to the notation in the same visual field; fade for accuracy, not for a calendar date. Written for struggling elementary students; the representation rule does not retire in high school. Full discussion in Chapter 2.

⁹⁰³Kira J. Carbonneau, Scott C. Marley, and James P. Selig, “A Meta-Analysis of the Efficacy of Teaching Mathematics with Concrete Manipulatives,” *Journal of Educational Psychology* 105, no. 2 (2013): 380–400. Overall d approximately 0.37 versus abstract-only; larger on retention than on transfer or justification; high guidance associated with higher retention and problem solving. Not “nets work.” Moderators required. Virtual versus physical manipula-

formula. Fade when the child is accurate without it. Bring it back when a sphere's informal-limit argument needs a picture. Virtual versus physical manipulatives remain **UNVERIFIED**. GeoGebra is a representation, not a proof, and section 8.7 will say who may generate the file.

What this section has done is put Algebra I on the plane, put the circle's theorems next to its equation, and plant Cavalieri for Chapter 10 without writing Chapter 10. Proof is next: not a format, a viable argument. The parent who learned two-column needs that correction before the probability plant, because a two-column about a tangent is still a two-column, and Practice 3 is larger than that.

8.4 Proof as viable arguments, not a format

Proof readiness is not two-column format. It is CCSSM Practice 3: construct viable arguments and critique the reasoning of others.⁹⁰⁴ That practice runs every grade. It rests on NCTM's process standards — problem solving, reasoning and proof, communication, representation, connections — and on the National Research Council *Adding It Up* strand of adaptive reasoning.⁹⁰⁵ It does not begin in tenth grade. Chapter 4 already treated the equal sign as a relation, not an operator, which is a viable argument about $a + b = c$. Chapter 6 already asked for informal geometric arguments for the triangle angle-sum and for alternate interior angles. This course asks for arguments a stranger could check, in more than one format. A parent who learned that geometry is two-column is using a narrower map than Appendix A publishes. This section is that parent's correction.

Appendix A's instructional note on G.CO.9–11, quoted because the wording is the load-bearing object: “Encourage multiple ways of writing proofs, such as in narrative paragraphs, using flow diagrams, in two-column format, and using diagrams without words. Students should be encouraged to focus on the validity of the underlying reasoning while exploring a variety of formats for expressing that reasoning.”⁹⁰⁶ Four formats, one criterion. Narrative paragraph. Flow diagram. Two-column. Diagram without words. The criterion is the validity of the underlying reasoning. A valid paragraph is a proof. An invalid two-column is not. A diagram that marks a correspondence and a sequence of rigid motions, with the given parts named, can be a proof. A diagram that “looks right” is a drawing. Two-column is licensed. It is not required. It is not the definition of geometry. Do not reduce this course to two-column theater.

What a viable argument is, as Practice 3 actually writes it: the student uses stated assumptions, definitions, and previously established results in constructing arguments; makes conjectures and builds a logical progression of statements to explore the truth of those conjectures; justifies conclusions,

tives **UNVERIFIED** this pass.

⁹⁰⁴*Common Core State Standards for Mathematics*, pp. 6–8, Standards for Mathematical Practice, Practice 3: construct viable arguments and critique the reasoning of others. Runs every grade. Proof readiness is this practice, not two-column format.

⁹⁰⁵NCTM process standards (problem solving, reasoning and proof, communication, representation, connections) and National Research Council, *Adding It Up: Helping Children Learn Mathematics* (Washington, DC: National Academy Press, 2001), five strands including adaptive reasoning, p. 116 as cited inside CCSSM and NMAP. Full NRC book not re-opened this pass; cite via CCSSM's own sourcing of the eight practices.

⁹⁰⁶Appendix A (n. 852), instructional note on G.CO.9–11, quoted: “Encourage multiple ways of writing proofs, such as in narrative paragraphs, using flow diagrams, in two-column format, and using diagrams without words. Students should be encouraged to focus on the validity of the underlying reasoning while exploring a variety of formats for expressing that reasoning.” Four formats, one criterion. Wording checked against the Oregon ODE copy of Appendix A, 30 August 2026.

communicates them, and responds to the arguments of others; distinguishes correct reasoning from that which is flawed.⁹⁰⁷ Elementary students can construct arguments using concrete referents such as objects, drawings, diagrams, and actions. Later, arguments become more general and abstract. That last clause is this course. It is not a format clause. A ninth-grader who writes, in prose, “ $\triangle ABC$ and $\triangle DEF$ have $AB = DE$, $AC = DF$, and included $\angle A = \angle D$. A rigid motion takes A to D and ray AB onto ray DE . Because the included angles match, ray AC lands on ray DF . Because the sides match, C lands on F . So the triangles coincide, hence they are congruent,” has produced a viable argument. A ninth-grader who fills two columns with “Given / SAS / Therefore congruent” without a correspondence has produced a format. Grade the first one right. Grade the second one as a gap, not as a style preference.

Definitions first. A first formal proof that skips the definition has nothing to attach a reason to. Congruence as a rigid motion. Similarity as a dilation plus a rigid motion. Parallel as never meeting, or as equal corresponding angles with a transversal, once that theorem is in hand. Circle as the set of points at a fixed distance from a center. Tangent as a line meeting the circle at exactly one point. Write the definition. Then use it. A child who cites “definition of congruent” and means “they look the same” has not used a definition. A child who cites SAS without naming the correspondence has not used a criterion. The parent who stopped at algebra has to be able to hear both.

Then a worked proof with fading, then “what is wrong with this proof.” That sequence is an extrapolation from Booth, Sweller, and the worked-example literature Chapter 2 already locked, not a geometry-proof randomized trial. Label it as such.⁹⁰⁸ Proof, modeling, and the limit concept have a thin experimental base relative to fractions and facts. This chapter will not invent a d for two-column versus paragraph. What transfers from the worked-example literature is the load, not the coefficient. High-element-interactivity material — a first proof is that — is learned better from a worked example than from unguided generation, for novices; examples should be faded; incorrect examples, explained, help conceptual understanding. Booth, Lange, Koedinger, and Newton 2013, Algebra I Cognitive Tutor, incorrect examples on two-step equations: conceptual gains; the misconceptions the examples were built to hit included the equal sign as “where the answer goes” and variables as a single mystery number.⁹⁰⁹ Applying “show one fully worked argument, then a second the child must explain, then a completion missing the last reason, then a conventional problem, then an incorrect example” to G.CO.9–11 is the analogue, labelled analogue. Combined with guided practice in that paper, not as a kitchen-table Cognitive Tutor. MATHia family purchase

⁹⁰⁷CCSSM Practice 3 (n. 904), full descriptor: stated assumptions, definitions, and previously established results; conjectures and a logical progression of statements; justify, communicate, respond; distinguish correct from flawed reasoning. Elementary students may use concrete referents; later arguments become more general and abstract — this course is the later clause.

⁹⁰⁸Extrapolation from worked-example fading (Sweller) and from Julie L. Booth, Kenneth R. Koedinger, and colleagues on incorrect examples, applied to a first geometric argument. **Not a geometry-proof RCT. Label as analogue / extrapolation.** Proof-teaching experimental base is thin relative to fractions and facts (n. 884).

⁹⁰⁹Julie L. Booth, Karin E. Lange, Kenneth R. Koedinger, and Kristie J. Newton, “Using Example Problems to Improve Student Learning in Algebra: Differentiating between Correct and Incorrect Examples,” *Learning and Instruction* 25 (2013): 24–34. Algebra I Cognitive Tutor; incorrect examples helped conceptual understanding. Misconceptions the examples were built to hit included the equal sign as “where the answer goes” (Baroody and Ginsburg 1983; Kieran 1981) and variables as a single mystery number (Booth 1984; Knuth, Stephens, McNeil, and Alibali 2006; Küchemann 1978). Combined with guided practice in that paper, not as a kitchen-table Cognitive Tutor.

remains **UNVERIFIED** as of 30 August 2026; school RCTs are not homeschool results.⁹¹⁰ The move that transfers is the parent showing one fully worked SAS argument on paper — paragraph or two-column, either licensed — then a second the child must explain, then a completion, then a conventional, then: “They cited SSA. What did they think those three letters guaranteed?”

Error analysis applied to arguments is the Practice 3 half that “critique the reasoning of others” names. A drawing is not a theorem. An argument that assumes the conclusion is not a theorem. An argument that skips a correspondence is not a theorem. An argument that cites a non-theorem — “all triangles with a right angle are similar,” “vertical angles of a parallelogram are supplementary,” a made-up “angle-side-side” — is not a theorem. The parent’s job is to hear those, not to prefer a format. A *hypothetical*, labelled as such, can put the hearing in a room without inventing a reported family. A parent whose fifteen-year-old produces a numbered paragraph proving two triangles congruent by assuming a side equal that was never given, then copying a two-column from a solution manual that uses a different correspondence, is watching a validity failure and a format-worship failure, not a reason to “drill more two-column.”⁹¹¹ The empirical pieces that composite combines are in the documents: Appendix A licenses multiple formats; Practice 3 is validity; Bastani’s crutch is waiting in section 8.7 for the copied two-column. The sitting, the age, and the two papers are hypothetical. The teaching move is not. Ask which correspondence the paragraph used. Ask which given the two-column used. Ask whether either argument would survive a figure in which the unmarked side is drawn longer. Then the child writes unaided.

A drawing is not a theorem, restated because the diagram is this course’s pizza. Pies introduced part-whole in Chapter 5 and did not by themselves teach that $\frac{3}{4}$ is a number. Diagrams introduce configuration and do not by themselves teach that the configuration is necessary. A carefully drawn figure can hide a false converse: the altitude that *looks* like a median, the rectangle that *looks* like a square, the tangent that *looks* perpendicular. Coordinate geometry, last section, is one check: compute. A second figure, isomorphic in structure and different in appearance, is another: if the argument still holds, it was not riding on the drawing. Section 8.7 will ask the model for that second figure and keep the key with the parent. The child who can only “see it” on the original drawing has performed. Soderstrom and Bjork again.⁹¹²

Do not wait for two-column fluency to begin arguments, and do not withhold two-column from a child who wants it. Format is a tool. Practice 3 is the mathematics. A family using an older text that is two-column-first is not in sin; they need the validity criterion named, and they need at least one other format practiced, so that a valid paragraph is not marked wrong at a co-op or on a placement. A family using a transformation-first text that never writes a reason is in a different hole: they have motions and no argument. Both holes are this section’s. Fill them with definitions, a faded worked argument, an incorrect example, and an unaided generation. Not with a franchise

⁹¹⁰MATHia (Carnegie Learning), <https://www.carnegielearning.com/solutions/math/mathia>, fetched 30 August 2026: public page is school/district (“Request a Demo”). Direct family purchase **UNVERIFIED** as of 30 August 2026. RAND study of the blended Carnegie approach is school-side, not MATHia-alone, not homeschool. School RCTs are not homeschool results.

⁹¹¹Composite *hypothetical*, labelled in the sentence that introduces it. Empirical pieces: Appendix A licenses multiple formats (n. 906); Practice 3 is validity (n. 904); Bastani’s crutch (n. 936) for a copied two-column. The sitting, the age, and the two papers are not a reported family.

⁹¹²Soderstrom and Bjork 2015 (n. 894). A child who can only “see it” on the original drawing has performed, not learned. A second isomorphic figure is the generation-and-transfer check.

about which format is “real math.”

When the child is stuck, explicit instruction for students already struggling is still NMAP Finding 27: clear models, extensive practice, think-alouds, extensive feedback, and this “does not mean that all of a student’s mathematics instruction should be delivered in an explicit fashion.”⁹¹³ A first proof is a place a struggling student often needs the model. Productive failure — Kapur 2014, generation then instruction, greater conceptual understanding and transfer, comparable procedural knowledge — is a different comparison, on different objects, when priors exist.⁹¹⁴ Do not mash them. Do not put a child who cannot yet name SAS in front of a blank proof and call it productive. Do not withhold every generation until twenty worked two-columns have been copied. The faded sequence is the mash this book will not do: a model, then a generation. Label the sequence as extrapolation. Do not label it as a geometry RCT.

What this section has done is take two-column off the throne and put Practice 3 on it. Applications of probability sit in this course as critical area 6. Spatial play does not produce them, and does not produce proof either.

8.5 Applications of probability; spatial play is not proof

Applications of probability are the sixth critical area of Appendix A’s Geometry, not a Friday worksheet and not a reason to skip Chapter 11. Independence, conditional probability, two-way tables, the addition rule; permutations and combinations as (+).⁹¹⁵ These are Chapter 11’s Geometry stop. Plant them. Do not dump Chapter 11 here. A student who skipped the K–8 data strand — Grade 6 distributions, Grade 7 sampling and probability models, Grade 8 bivariate association — will spend AP Statistics Unit 1 retrofitting, and will spend this critical area retrofitting too. Algebra I already includes S.ID: shape, center, spread, two-way tables, linear models, correlation versus causation. This course adds S.CP: the probability rules that make a two-way table into a conditional, and independence into a check rather than a slogan.⁹¹⁶

What S.CP actually asks, as a parent-facing cluster rather than a dump. Two events A and B are independent if $P(A \text{ and } B) = P(A) \cdot P(B)$, or equivalently if $P(A | B) = P(A)$. Conditional probability $P(A | B)$ is the probability of A given that B has occurred, $P(A \text{ and } B) / P(B)$ when $P(B) \neq 0$. A two-way table is the representation: rows A and not- A , columns B and not- B , four cells of counts or relative frequencies. The addition rule: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$, so that the

⁹¹³NMAP, *Foundations for Success* (2008), Finding 27: explicit instruction (clear models, extensive practice, think-alouds, extensive feedback) has consistently positive effects for students with mathematical difficulties and for the lowest third of a typical class — and this “does not mean that all of a student’s mathematics instruction should be delivered in an explicit fashion.” Do not mash with Kapur 2014 (n. 914).

⁹¹⁴Manu Kapur, “Productive Failure in Learning Math,” *Cognitive Science* 38, no. 5 (2014): 1008–1022. Two randomized studies; comparable procedural knowledge to direct instruction first; greater conceptual understanding and transfer; number of student-generated solutions predicted outcomes. Different comparison from NMAP Finding 27. Do not mash. Do not put a child who cannot yet name SAS in front of a blank proof and call it productive.

⁹¹⁵Appendix A (n. 852), Geometry critical area 6 / CCSSM S.CP: independence, conditional probability, two-way tables, addition rule; permutations and combinations as (+). Chapter 11’s Geometry stop. Plant; do not dump Chapter 11.

⁹¹⁶01-scope.md Statistics band and Algebra I S.ID.1–3, 5–9: shape, center, spread, two-way tables, linear models, correlation versus causation. Geometry S.CP adds the probability rules. A skipped K–8 data strand is AP Statistics Unit 1 retrofit and this critical-area retrofit.

intersection is not counted twice. Mutually exclusive is the special case $P(A \text{ and } B) = 0$, which is not the same as independent. A child who treats mutually exclusive and independent as synonyms has the slogan pair this plant exists to break. Permutations and combinations as (+) are counting rules for when order matters and when it does not; they often wait for Algebra II or a richer Geometry. NCES 2016: probability ever-taught to 24 percent of high-school-level homeschoolers.⁹¹⁷ Not a statistics-course completion rate. Not this critical area done. A kitchen that never opens a two-way table in Geometry has skipped a sixth of Appendix A's course, however many two-columns it produced.

The two-way table is the representation that matches the mathematics, same sitting as the formula. 40 students: 18 take French, 15 take Spanish, 7 take both. $P(\text{French} | \text{Spanish})$ is $7/15$, not $7/40$ and not $18/40$. A child who divides by the grand total has treated a conditional as a joint. A child who adds 18 and 15 and calls it “or” has forgotten to subtract the intersection. Schema versus keywords still applies: “given” is not a magic word that always means divide by the second number you see; “and” is not always multiply. Chapter 2 already rejected keywords. Chapter 11 will do the full data-and-chance through-line, including AP Statistics' five units effective Fall 2026. This section plants the Geometry cluster and stops. College Board's AP Statistics is a different college-equivalent, non-calculus-based, official prerequisite as of the Fall 2026 CED a first-year algebra course. It does not substitute for this course, and this course does not substitute for it.⁹¹⁸

Kitchen building and making are spatial. They do not automatically produce proof. The National Mathematics Advisory Panel's Finding 26 is still the load-bearing caution: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving.⁹¹⁹ Building a shed, stacking boxes, folding paper, Minecraft, a sewing pattern, a scale model — those can motivate and can provide a second representation of a similarity or a volume the child is already being taught as an argument. They do not replace the written reason, the correspondence, or the informal Cavalieri. A child who can bisect a board and cannot prove that the perpendicular bisector is the set of equidistant points has performed, not learned. Soderstrom and Bjork: current performance is “often an unreliable index” of learning.⁹²⁰ Homeschool culture overclaims kitchen-math. Chapter 5 already refused “they'll learn fractions by cooking.” This chapter refuses “they'll learn proof by building.” NMAP Finding 26 is the opposite caution, not a vibe.

Do not treat spatial play as proof. Restate that, because the slogan is loud in unschooling literature and in well-meant maker culture, and because an unopened spatial-STEM paper will otherwise get

⁹¹⁷Cui and Hanson, NCES 2020-001, Table 8: probability ever-taught to 24 percent of high-school-level homeschoolers. Not a statistics-course completion rate. Not this critical area done.

⁹¹⁸College Board, *AP Statistics Course and Exam Description*, Effective Fall 2026, <https://apcentral.collegeboard.org/media/pdf/ap-statistics-course-and-exam-description.pdf>, accessed 30 August 2026. Equivalent to a one-semester introductory **non calculus-based** college course; prerequisite “any secondary school student who has successfully completed a **first-year algebra course**.” Not a substitute for Geometry, and Geometry is not a substitute for it. Chapter 11 gathers the through-line.

⁹¹⁹NMAP, *Foundations for Success* (2008), Finding 26: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving. Kitchen building and making are spatial; they do not automatically produce proof. Homeschool culture overclaims kitchen-math.

⁹²⁰Soderstrom and Bjork 2015 (n. 894), applied to a child who can bisect a board and cannot prove that the perpendicular bisector is the set of equidistant points.

laundered into it. Wai, Lubinski, and Benbow 2009 stays **UNVERIFIED** as to exact coefficients.⁹²¹ Spatial skill may well predict later STEM; this book has not opened the coefficients, and a prediction is not a curriculum. Holt's *Teach Your Own* and later unschooling literature describe math arising from cooking, building, games, and later self-directed study. That is a named practice. NCES 2016: 12 percent of homeschoolers "mostly or strictly informal learning." The thesis of this book — the parent understands the concept well enough to hear a wrong answer; the child does the struggle before the tool; any agent is a supplement hosted and constrained by the parent — is closer to structured and mastery than to pure unschooling. Say so without scorning the 12 percent.⁹²² Informal geometry play in grades 1–8 is NCTM's "natural area" for reasoning; it is the on-ramp, not the high-school course. Pennsylvania option 1 even names geometry among required secondary mathematics. NCAA wants Algebra I or higher as core. A shed does not transcript as Geometry.

Tips that survive evidence, gathered here so they are not scattered as slogans.

A drawing is not a theorem. Mark the correspondence. Write the reason. Check on a second figure.

Do not skip rigid motions because the parent's own course started with postulates. Congruence is defined as a rigid motion on this map. The criteria are compressions of that definition.

Do not skip similarity because trigonometry "is Algebra II." Right-triangle trigonometry in this course is similarity. Unit-circle trigonometry is Chapter 9.

Transcript the course as Geometry even if Saxon integrated it inside Algebra 1, Algebra 2, and Advanced Math. **Vendor** integration is a packaging. The export is a course title a stranger can map.⁹²³

Do not treat spatial play as proof. Finding 26. A second representation, then the argument, then retrieval. Not the shed instead.

Do not skip S.CP because "we'll do statistics later." A skipped Geometry probability cluster is Chapter 11 retrofit. Plant. Do not dump.

Do not wait for two-column fluency to count an argument as a proof. Appendix A named four formats. Practice 3 is the criterion.

When the parent is the ceiling, co-op geometry is a named option: NCES 2016, 31 percent any local homeschool group or co-op; 23 percent any tutor.⁹²⁴ Section 8.6 will put a labelled hypothetical in that room. Video curricula exist because of Maloney; keep the parent as the judge of a proof's

⁹²¹Wai, Lubinski, and Benbow 2009 (n. 882). **UNVERIFIED** as to exact coefficients. A prediction is not a curriculum. Do not treat spatial play as proof.

⁹²²Cui and Hanson, NCES 2020-001, Table 6: 77 percent "mostly or strictly formal curriculum"; 12 percent "mostly or strictly informal learning." Unschooling is a named minority, not the market majority. The thesis of this book is closer to structured/mastery; say so without scorning the 12 percent. Holt's *Teach Your Own* is a named practice, not a geometry RCT.

⁹²³Saxon classic integration (n. 878). **Vendor** packaging. Export rule: transcript the course as Geometry. Chapter 12 will say it as an export rule; this chapter says it as a course-identity rule.

⁹²⁴Cui and Hanson, NCES 2020-001, Table 3 and related: any tutor or private teacher 23 percent; any local homeschool group or co-op 31 percent. Named release valves when the parent is the ceiling, not a finding that the kitchen failed.

validity, not as the person who “explains it the way I learned it” if that way omitted transformations and treated a drawing as a theorem.

What this section has done is plant S.CP for Chapter 11 and refuse to upgrade a shed. The parent who learned two-column still needs a teaching recap: representations connected to the written argument, fading, generation. Then the model, which will write a fluent numbered argument that is not a proof.

8.6 Teaching guidance recap and the parent who learned two-column

The parent who learned axiom-first, two-column-only geometry is the design user of this recap. Maloney, Ramirez, Gunderson, Levine, and Beilock 2015 remains a school-homework study, not a parent-as-sole-teacher randomized trial: parent math anxiety plus frequent help predicted worse child math, math-specific; when help was infrequent, no such relation.⁹²⁵ A homeschool parent cannot “help less often” as a daily strategy. Prepare so you can hear a wrong correspondence. Then the child generates. “I stopped at algebra” is still the kitchen constraint; “I remember two-column and not transformations” is the geometry-specific version. Video curricula exist because of that constraint. Teaching Textbooks’ “does 100% of the teaching” is a **vendor** claim — relevant to a parent who is anxious, not a finding that the parent can stop hearing wrong arguments.⁹²⁶ Keep the parent as the judge of a proof’s validity, not as the person who “explains it the way I learned it” if that way omitted rigid motions, treated a drawing as a theorem, or marked a valid paragraph wrong for being a paragraph. Do not claim ChatGPT can “do geometry.” Do not claim AlphaGeometry2 is a home product. Those two claims wait one section. This section is the teaching recap: representations, fading, generation; diagrams connected to the written argument; proportional reasoning and Algebra I as the gates; co-op as a named release valve.

Representations first, connected to the written argument in the same visual field. Chapter 2 already locked Fuchs and colleagues 2021 Recommendation 3: choose representations that match the mathematics; connect concrete and semi-concrete to the notation in the same visual field, not “manipulatives today, symbols next month”; fade for accuracy, not for a calendar date.⁹²⁷ This course’s representations are the marked diagram, the rigid-motion sketch, the dilation with a scale factor named, the coordinate grid, the net, the two-way table, and the written argument. They belong in the same sitting. A diagram with vertices unmarked next to a two-column that cites SAS is two representations that do not touch. A GeoGebra file that moves a triangle, with no sentence

⁹²⁵Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, “Intergenerational Effects of Parents’ Math Anxiety on Children’s Math Achievement and Anxiety,” *Psychological Science* 26 (2015): 1480–1488, <https://doi.org/10.1177/0956797615592630>. School-homework study: parent math anxiety $\times$ frequent help predicted worse child math and more child math anxiety; when help was infrequent, no such relation; math-specific, not reading. Not a parent-as-sole-teacher RCT. Homeschool parent cannot “help less often” as a daily strategy. Do not treat the finding as “anxious parents should not teach.” Replication status open; no registered replication opened. Fraulein Retanal et al., *Education Sciences* 11, no. 10 (2021): 620: controlling-supportive homework-helping style partially mediated the relation in a survey of parents of 11- to 14-year-olds.

⁹²⁶Teaching Textbooks vendor claim that the program “does 100% of the teaching.” **Vendor** copy, labelled vendor. Relevant to the Maloney constraint; not a finding that the parent can stop hearing wrong arguments. Keep the parent as judge of a proof’s validity.

⁹²⁷Fuchs et al. 2021, Recommendation 3 (n. 902). Geometry-specific application: marked diagram, rigid-motion sketch, dilation with scale factor, coordinate grid, net, two-way table, and written argument in the same sitting.

saying which motion and which corresponding parts, is a movie. Put the motion next to the sentence. Put the correspondence next to both. Carbonneau $d \approx 0.37$ is versus abstract-only, under guidance, larger on retention than on transfer or justification.⁹²⁸ That number is not “diagrams work.” High guidance is the moderator this course needs: the parent pointing at A and D and saying the translation, then the child writing it. Fade when the child marks the correspondence without being told. Bring the diagram back when a circle theorem needs an auxiliary line. Virtual versus physical remain **UNVERIFIED**.

Fading next. A first formal proof is high-element-interactivity material. Worked-example fading is the analogue Chapter 2 locked and section 8.4 labelled as extrapolation, not a geometry-proof RCT: one fully worked argument, one the child explains, one completion missing the last reason, one conventional, one incorrect.⁹²⁹ Then stop showing full examples. Expertise reversal is real. Incorrect examples do the Practice 3 critique: “what is wrong with this proof?” SSA cited; conclusion assumed in line two; correspondence reversed; a drawing used as a reason (“as shown”). Booth’s algebra incorrect examples helped conceptual understanding. The geometry application is labelled analogue. Do not wait for a geometry RCT to use a wrong proof as a teaching object. Do not use *only* wrong proofs and never show a right one.

Generation after the fade. The child writes unaided. Soderstrom and Bjork: performance during acquisition is often an unreliable index of learning; conditions that boost performance now (a worked two-column at the elbow, a solver, an open diagram with all the marks already in) can fail to produce learning; conditions that depress current performance (generation, delayed feedback, a second isomorphic figure) often produce more.⁹³⁰ An unaided short argument, every session that introduced a new criterion or theorem, is Bastani’s exam phase scaled to ten minutes. Chapter 3 already locked that shape. This course’s version is: one rigid-motion image, one missing-length on similar triangles, one “is this a proof or a drawing?” The child writes. The parent holds the key. Interleaving after a short blocked introduction remains the Rohrer move, and the clean classroom $d = 1.05$ is still grade 7, $n = 140$, two-week delay, interleaved 72 percent versus blocked 38 percent, not a geometry coefficient.⁹³¹ Do not export the d . Do export the practice: mix yesterday’s SAS with last month’s inscribed-angle and last week’s distance formula, so the child has to choose a strategy, not only execute the one on the worksheet title. Rohrer, Dedrick, and Burgess 2014 treated “graph $y = mx + b$ ” and “slope from two points” as different problem kinds students fail to associate when practice is blocked. This course’s analogue, labelled analogue, is “prove congruent” versus “find

⁹²⁸Carbonneau, Marley, and Selig 2013 (n. 903). $d \approx 0.37$ versus abstract-only, under guidance. Not “diagrams work.” High guidance is the moderator this course needs.

⁹²⁹Worked-example fading plus incorrect examples (nn. 893, 908–909), restated as the teaching recap. **Extrapolation, labelled.** Not a geometry-proof RCT.

⁹³⁰Soderstrom and Bjork 2015 (n. 894). Generation, delayed feedback, and a second isomorphic figure as the geometry versions of conditions that often produce more learning while depressing current performance.

⁹³¹Doug Rohrer, Robert F. Dedrick, and Kaleena Burgess, “The Benefit of Interleaved Mathematics Practice Is Not Limited to Artificially Accumulated Skills,” *Journal of Educational Psychology* 106, no. 3 (2014). Interleaved 72 percent versus blocked 38 percent, $d = 1.05$, 95% CI [0.80, 1.30], $n = 140$, unannounced test two weeks later, grade 7. Four problem kinds included graph $y = mx + b$ and slope from two points. Clean classroom RCT of interleaving is *here* in grade 7, not a geometry d . Do not export the coefficient. Do export the practice of mixing problem kinds. Doug Rohrer, Robert F. Dedrick, Marissa K. Hartwig, and Chi-Ngai Cheung, 2020, preregistered cluster RCT, 54 seventh-grade classes, interleaved 61 percent versus blocked 38 percent, $d = 0.83$, one-month delay — a different paper, joint with spacing; do not mash.

the missing length” versus “write the circle’s equation” versus “read $P(A | B)$ off a table.” Blocked practice to learn execution. Interleaved practice to choose.

Diagrams must connect to the written argument. Restate that as the geometry-specific representation rule, because it is the rule a two-column-trained parent will skip and a transformation-trained parent will also skip, from opposite sides. The first will write reasons with no figure. The second will move a triangle and never write. Appendix A licensed “diagrams without words” as a format; that format still has to *be* an argument — vertices, given parts, a sequence of motions or a flow of implications a stranger could check — not a pretty picture. A diagram without words that is only a pretty picture is a drawing, and a drawing is not a theorem.

Proportional reasoning for similarity and right-triangle trigonometry; Algebra I for coordinates. Those two gates are the diagnostic at the door of this course, restated from section 8.1 so they survive as teaching moves. Can the child see a ratio as a relationship, find k , and mark corresponding sides? If not, Chapter 6 is not finished; SOH-CAH-TOA will be a chant. Can the child write the equation of a line, compute a slope from two points, and complete the square? If not, Chapter 7 is not finished; the circle’s equation will be soup. NMAP Finding 15 still refuses the age lock; Finding 30 still allows prepared, motivated students to move faster.⁹³² Neither finding licenses a skip of similarity because the child is “good at math,” or a skip of Algebra I because geometry is “the non-algebra year.” Diagnose. Then this course.

When the parent is the ceiling, co-op geometry is a named NCES 2016 option: 31 percent any local homeschool group or co-op; 23 percent any tutor.⁹³³ Dual enrollment is a later release valve, a college transcript NCAA and admissions offices already know how to read; it is not the typical path in the 2016 tables.⁹³⁴ A *hypothetical*, labelled as such, can put the co-op in a room. A parent of three, ages 8, 11, and 14, sits the eight-year-old on Math-U-See Beta with blocks, the eleven-year-old on Singapore 5A independently, and drives the fourteen-year-old to a Tuesday co-op geometry class.⁹³⁵ That composite is **hypothetical**. The empirical pieces it combines — multi-child household, mixed curricula, 31 percent any co-op — are NCES 2016 / 2019. The sitting, the driving, and the three named ages are not a reported family. The teaching move is not hypothetical: the parent who cannot hear a wrong SSA still has to remain the judge of the homework, or has to hire a judge (the co-op teacher as teacher of record, useful for an NCAA worksheet) rather than handing the figure to a chatbot. Co-op is a release valve, not a finding that the kitchen failed, and not a finding that the parent can stop hearing.

⁹³²NMAP Findings 15 and 30 (nn. 860–861). Diagnose proportional reasoning and Algebra I before this course. Neither finding licenses a skip of similarity or of Algebra I because the child is “good at math.”

⁹³³Cui and Hanson, NCES 2020-001 (n. 924): 31 percent any co-op; 23 percent any tutor. Repeated because the recap names the release valve.

⁹³⁴Cui and Hanson, NCES 2020-001, Table 4: 18 percent of homeschoolers also attended a school (15 percent public; 2 percent private, interpret with caution; 1 percent college, interpret with caution). Dual enrollment visible but small; not the typical path. NCAA treats a college transcript more cleanly than a parent-graded course; CLEP is not an NCAA core course (n. 873).

⁹³⁵Composite *hypothetical*, labelled in the sentence that introduces it. 03-homeschool.md §2 already labelled this composite: a parent of three (ages 8, 11, 14) sits the eight-year-old on Math-U-See Beta with blocks, the eleven-year-old on Singapore 5A independently, and drives the fourteen-year-old to a Tuesday co-op geometry class. Empirical pieces: NCES 2016 / 2019 multi-child household, mixed curricula, 31 percent any co-op. The sitting, the driving, and the three named ages are not a reported family.

A video curriculum is the other release valve, and the same constraint applies. The parent watches enough of the lesson to hear a wrong correspondence. The child does the exercises without a solver. The unaided exit ticket still happens. “The video explains it the way I learned it” is only a virtue if the way the parent learned it included a viable argument and not only a format. If the video is axiom-first and the family’s next book is transformation-first, name the difference; do not pretend they are the same course. If the video never writes a paragraph proof, add one. Format is not the course.

Do not claim ChatGPT can “do geometry.” Do not claim AlphaGeometry2 is a home product. The next section is those two sentences at full length. This recap’s last teaching move is the thesis applied to a proof: the parent understands the criterion well enough to hear a wrong reason; the child writes the argument before any tool is allowed to print one; any agent is a supplement hosted and constrained by the parent. The sources do not prove that as a causal claim about homeschool geometry. They do not contradict it. They constrain how it can be attempted. Fractions were the gate to algebra. Viable arguments are the gate through this course. The agent does not take the test.

8.7 Parent-hosted AI for geometry

One short reminder, then the objects of this course. Hamsa Bastani and colleagues, 2025: a ChatGPT-like tutor raised assisted practice grades 48 percent relative to control and then cut the unaided exam 17 percent. A guardrailed tutor that withheld the full solution raised practice 127 percent and was statistically indistinguishable from control on the unaided exam. GPT Base, asked for the answer, was fully correct 51 percent of the time. Students copied anyway.⁹³⁶ Soderstrom and Bjork 2015: current performance is “often an unreliable index” of learning.⁹³⁷ Full design, product landscape, ban-versus-allow list, six parent workflows, COPPA dates, detectors, companion-harm cases as allegations: Chapter 3. This section is a pointer, not a sequel. The workflows below are practice suggestions grounded in Bastani, Soderstrom and Bjork, and Oreopoulos and Low, not a new randomized trial of homeschool geometry AI. No year-plus kitchen-table RCT of a parent-constrained GPT-Tutor workflow was opened on 30 August 2026.⁹³⁸

GSM8K saturation is not geometry proof. GSM8K is grade-school word problems. Frontier models have long since scored very high on it; Humanity’s Last Exam’s introduction notes that popular

⁹³⁶Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman, “Generative AI without Guardrails Can Harm Learning: Evidence from High School Mathematics,” *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122, <https://doi.org/10.1073/pnas.2422633122>; author PDF https://hamsabastani.github.io/education_llm.pdf, accessed 30 August 2026. GPT Base: assisted practice +48% relative to control; unaided exam −17%. GPT Tutor: practice +127%; exam point estimate −0.004, not significant. GPT Base fully correct 51% of the time when asked for the answer; 42% logical errors, 8% arithmetic errors. Classroom-level RCT, large high school in Turkey, Fall 2023–24, ~1,000 students in grades 9–11, four 90-minute sessions, GPT-4-0613. Not a homeschool study, not GPT-5, not Khanmigo, not a geometry-proof study. Mechanism: GPT Base as a crutch. Percentages are of control grades, not Cohen’s *d*. Full discussion: Chapter 3. This chapter is a pointer, not a sequel.

⁹³⁷Soderstrom and Bjork 2015 (n. 894). Bastani’s student-perception finding is the AI-era replication of that metacognitive error: GPT Base students did not perceive that they had learned less.

⁹³⁸Parent workflows in this section are practice suggestions grounded in Bastani, Soderstrom and Bjork, and Oreopoulos and Low, not a new RCT of homeschool geometry AI. No year-plus kitchen-table RCT of a parent-constrained GPT-Tutor workflow was opened on 30 August 2026 (04-ai open questions). Label them that way. Chapter 3 §3.3–3.8 is the lock; this section is the Geometry pointer.

benchmarks like MMLU now sit “over 90 percent.” That number is not a proof-writing score. Exact GSM8K percentages for named 2026 models were not extracted this research day; treat saturation as qualitative.⁹³⁹ Humanity’s Last Exam, lastexam.ai fetched 30 August 2026: Gemini 3 Pro **38.3 percent**, GPT-5 **25.3 percent**, Grok 4 24.5 percent, Claude 4.5 Sonnet 13.7 percent, GPT-4o **2.7 percent** overall. Calibration error still high. Math-subset percentage not broken out on the page — **UNVERIFIED**. HLE is not a K–12 geometry-proof score.⁹⁴⁰ FrontierMath exists, 338 problems after the 12 June 2026 v2 correction; current leaderboard percentages for Tiers 1–3 were not extracted from the Epoch hub this day — **UNVERIFIED** as a single number.⁹⁴¹ The kitchen-table meaning of that table: even “smart” models are not expert; high confidence plus wrong is the failure mode the parent is guarding against; a fluent numbered argument is the geometry version of that failure mode.

AlphaGeometry2 is a specialized neuro-symbolic system, not ChatGPT. Chervonyi and colleagues, arXiv 2502.03544, 2025: Gemini-based language model plus a symbolic engine; **84 percent** of IMO 2000–2024 geometry problems, 42 of 50, versus 54 percent for AlphaGeometry 1; coverage of the AG language 88 percent.⁹⁴² DeepMind-adjacent reporting (InfoQ, February 2025): flagship chat models **o1 and Gemini Thinking solved none** of those IMO geometry problems.⁹⁴³ InternGeometry (arXiv 2512.10534) later reported 44 of 50 on the same set — a research agent, not a home product.⁹⁴⁴ A kitchen-table LLM asked to “prove two triangles similar” will often produce a fluent, numbered argument that skips a correspondence, cites a non-theorem, or assumes the conclusion. A named 2026 evaluation of consumer ChatGPT on high-school two-column proofs: **UNVERIFIED**. Until one is opened, this book says: treat model-written proofs as untrusted drafts; the child writes the proof unaided; the parent, or a human solution manual, is the judge.⁹⁴⁵ Do not invent that eval to make the warning feel newer. Do not claim ChatGPT can “do geometry.” Do not claim AlphaGeometry2 is a home product. A kitchen-table chatbot is not AlphaGeometry.

Ban, child-facing, during the attempt.

Paste the figure and “write the two-column proof for this figure.” That prompt is this course’s

⁹³⁹GSM8K (grade-school word problems) is saturated for frontier models; HLE site notes MMLU-class benchmarks “over 90%.” **Exact GSM8K % for a named 2026 model not extracted this day (UNVERIFIED as a single current figure)**. Saturation is qualitative. Easy word problems are not a proof-writing safety certificate. 04-ai §4.3 and load-bearing finding 11.

⁹⁴⁰Humanity’s Last Exam (n. 855). Math-subset % **UNVERIFIED**. Not a K–12 geometry-proof score. High confidence + wrong is the failure mode the parent is guarding against.

⁹⁴¹FrontierMath, Epoch AI, <https://epoch.ai/benchmarks/frontiermath-tiers-1-3-v2>, hub opened 30 August 2026: **338 problems** after **12 June 2026** v2 correction (Tiers 1–3 = 295; Tier 4 = 43). **Current accuracy % not extracted from the live plot this day (UNVERIFIED)**. Stanford HAI AI Index 2025 reported ~2% — that figure is 2025, pre-v2, likely stale. Funded in part by OpenAI (conflict noted on-page).

⁹⁴²Chervonyi et al., AlphaGeometry2 (n. 853). Specialized neuro-symbolic system, not ChatGPT. 84% / 42 of 50; AG1 54%; AG language coverage 88%.

⁹⁴³DeepMind-adjacent reporting, InfoQ, February 2025 (n. 854): o1 and Gemini Thinking solved **none** of that IMO-geometry set. A kitchen-table chatbot is not AlphaGeometry.

⁹⁴⁴InternGeometry, arXiv:2512.10534 (2025): **44/50** IMO geometry 2000–2024. A research agent, not a home product. 04-ai §4.3.

⁹⁴⁵Named 2026 consumer-ChatGPT two-column eval: **UNVERIFIED** (n. 856). Book sentence until one is opened: treat model-written proofs as untrusted drafts; the child writes unaided; the parent or a human solution manual is the judge. Do not invent the eval. Do not claim ChatGPT can “do geometry.” Do not claim AlphaGeometry2 is a home product.

GPT Base. It is the worksheet-paste from Chapter 3, wearing a compass. Photomath, Mathway, and Symbolab lead with scan-to-solution; Symbolab **vendor**-advertises geometry proofs and shape drawing. No product-specific unaided-transfer RCT was opened this research day; the transferable evidence is Bastani's crutch plus the homepages.⁹⁴⁶ Photo-to-key. "What is the answer." "Just give me the steps." Open-web agents — Hermes Agent, Grok Bot, operator-enabled ChatGPT agents — in the child's account. Companion persona, including a "geometry friend." Detectors used to police the child's written proofs: Liang, Yuksekgonul, Mao, Wu, and Zou 2023, average false-positive rate 61.22 percent (arXiv) / 61.3 percent (*Patterns*) on human TOEFL essays.⁹⁴⁷ Hard no: messaging gateway to a child's Telegram or WhatsApp; Grok Bot iOS app in the child's hands; IEP or 504 PDFs on a shared Grok Bot computer; the child's school portal or Photomath logins in an agent. El Salvador's Grok-as-tutor is a ministry deployment of Grok *the assistant*, announcement not RCT, not Grok Bot, not a U.S. homeschool SKU.⁹⁴⁸ Hermes Agent and Grok Bot can browse the open web; they are adult and developer tools, not a child-safe school SKU. No documented K–12 student-facing autonomous open-web agent as a school product on the pages opened 30 August 2026.

Allow, parent-hosted, child doing the work, parent holding the key.

Parent-first: what would count as a viable argument. Chapter 3 workflow (a): explain the concept to the *parent* first. Bastani's GPT Tutor worked in part because teachers had already written the solution and common mistakes into the hidden prompt. The parent is that teacher. Prefer a chat or reasoning model, not an open-web agent, for this job.⁹⁴⁹ Ask: "I am teaching SAS congruence from rigid motions to a homeschooler whose last math course was Algebra I. Explain to *me* as an adult who learned two-column from postulates. Give: the definition of congruence as a rigid motion, in one paragraph; why SAS determines a motion and SSA does not; what a viable argument would look like as a short paragraph, not only as two-column; the two mistakes this age makes; one tiny configuration I should be able to judge before I sit down with the child; the judgment on that

⁹⁴⁶Photomath, <https://photomath.com/>, fetched 30 August 2026: scan including word problems; "A solution will pop up for your exact problem"; Basic \$0 / Monthly \$9.99 / Annually \$69.99 (USD, this day); **vendor** "designed for learning." Mathway, <https://www.mathway.com/>: "Got a math problem? Get the steps." Symbolab, <https://www.symbolab.com/>: solver first; **vendor** geometry proofs and shape drawing. No product-specific unaided-transfer RCT opened this day. Transferable evidence is Bastani's crutch (n. 936) plus the homepages. Ban during the attempt: paste-the-figure, "write the two-column proof for this figure," photo-to-key, "what is the answer."

⁹⁴⁷Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou, "GPT Detectors Are Biased against Non-Native English Writers," arXiv:2304.02819; *Patterns* 4, no. 7 (2023), <https://doi.org/10.1016/j.patter.2023.100779>. Average false-positive rate 61.22% (arXiv) / 61.3% (*Patterns*) on human TOEFL essays. Do not use detectors to police a homeschooler's written proofs.

⁹⁴⁸xAI, "El Salvador partnership," 11 December 2025, <https://x.ai/news/el-salvador-partnership>: ministry deployment of **Grok the assistant** across 5,000+ public schools / 1 million students. Announcement, not RCT. Not Grok Bot, not a U.S. homeschool SKU. xAI, "Grok Bot FAQ," <https://docs.x.ai/grok-bot/faq>, fetched 30 August 2026: persistent cloud computer; "Grok Bot requires cloud data storage, so Legacy Privacy Mode is not supported"; one computer per user shared across Bots; no education SKU and no FERPA claim on the pages opened. Hermes Agent, <https://hermes-agent.org/>: self-hosted adult agent, open web, messaging gateways; no K–12 student-safety certification on the pages opened. No documented K–12 student-facing autonomous open-web agent as a school product on the pages opened 30 August 2026. Full distinctions: Chapter 3 §3.2. Grok ≠ Grok Bot ≠ El Salvador Grok-as-tutor. Hermes 4 ≠ Hermes Agent.

⁹⁴⁹Chapter 3 workflow (a): explain the concept to the *parent* first. Practice suggestion, not an RCT. Bastani's GPT Tutor worked in part because teachers had already written the solution and common mistakes into the hidden prompt. Prefer a chat/reasoning model, not an open-web agent, for this job.

configuration, under PARENT KEY.” Then the parent judges the tiny configuration **without** the model, then checks. The same workflow for why AA similarity is enough: similar right triangles, the trigonometric ratio as a scale factor, the wrong slogan “looks like.” The parent reads that before the lesson. The child does not.

Isomorphic figure, parent-held key. Chapter 3 workflow (c): extra practice with the answer key held by the parent. Do not generate “practice” by completing tonight’s assigned proof. That is the ban.⁹⁵⁰ Ask the model for three figures that share a structure with tonight’s type — SAS, or AA plus a missing length, or an inscribed angle — and differ in appearance, with a worked key the child never sees. The child does one unaided. The parent diagnoses. A second figure is how you test whether the argument rode on the drawing. Models hallucinate theorems and skip correspondences. The parent holds the key, which means the parent *checks* the key against a human solution manual or against their own working, because GPT Base was fully correct 51 percent of the time and a fluent wrong proof is worse than a blank page.⁹⁵¹ IXL Geometry lists 310 skills — **vendor** skill count, not a trial. Curriculum proofs remain the spine. Adaptive practice is not an agent and not a proof tutor.⁹⁵²

Diagnose a student’s gap. Chapter 3 workflow (f): photograph the child’s already-wrong work. Parent-only prompt: “Here is a student’s argument. Do not praise. Do not rewrite it as a correct two-column. Name the most likely gap in one sentence — skipped correspondence, assumed conclusion, SSA, drawing used as a reason, wrong scale factor. Give me one question I should ask the child, not the correct steps. Give me one isomorphic figure to retest after we talk.” Then the parent asks the question. The child answers the parent, not the model.⁹⁵³ For a two-column that cites SSA, the sentence is “those three pieces do not determine a rigid motion,” not “they forgot the list.” For a paragraph that uses the claim as a reason for the claim, the sentence is “assumed the conclusion.” For a missing-length that cross-multiplied the wrong sides, the sentence is “correspondence not marked,” Chapter 6’s ratio-as-a-pair-of-counts still alive inside similarity.

Diagram generation versus proof. The model may generate a *diagram description* or a TikZ or GeoGebra construction the parent checks. The model may propose auxiliary lines as *questions* — “what if you drew the altitude to the hypotenuse?” — not as a finished proof.⁹⁵⁴ That split

⁹⁵⁰Chapter 3 workflow (c): extra isomorphic practice with the answer key held by the parent. Do not generate “practice” by completing tonight’s assigned proof. That is the ban. Grounded in Bastani (practice with access $\neq$ unaided learning) and Soderstrom and Bjork (generation + delayed feedback).

⁹⁵¹Bastani et al. 2025 (n. 936): GPT Base fully correct 51%. Models hallucinate theorems and skip correspondences. The parent checks the generated key against a human solution manual or against their own working. Never let a model’s fluency stand in for a known-correct key.

⁹⁵²IXL Math, <https://www.ixl.com/math>, fetched 30 August 2026: 310 Geometry skills listed. **Vendor** skill count, not a trial. Adaptive skill practice / supplement, not a full curriculum and not an agent. Live homeschool price **UN-VERIFIED** as an invoice number. Curriculum proofs remain the spine. DreamBox, ALEKS, Khan practice paths are adaptive $\neq$ agent (Chapter 3 §3.2). MATHia family purchase **UNVERIFIED** (n. 910).

⁹⁵³Chapter 3 workflow (f): diagnose a wrong answer the child already produced. Practice suggestion, not an RCT. Name the misconception; ask a question; retest with an isomorphic item. The child answers the parent, not the model. For SSA cited as a reason, the sentence is “those three pieces do not determine a rigid motion,” not “they forgot the list.”

⁹⁵⁴04-ai.md §4.3 allow list (practice suggestion): model generates a diagram description or TikZ/GeoGebra construction the parent checks; model proposes auxiliary lines as questions (“what if you drew the altitude?”) not as a finished proof. Diagram generation versus proof is this course’s version of hint script versus answer key.

is this course's version of Chapter 3's hint script versus answer key. A GeoGebra file the parent exports, and the child then uses offline to test a conjecture, is a representation. A chatbot that writes the two-column while the child watches is GPT Base. Hermes Agent, if used, lives on the parent machine and can build a GeoGebra-able figure description as a tool-loop job. Do not put the agent on the child's phone. Model split this day (Nous Portal): Hermes 4 or any strong chat model for parent explanation and hint scripts; agentic examples — Claude Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro — for the tool loop. Hermes 4 is **not recommended inside Hermes Agent** (tuned for chat, not rapid tool-calling). Hermes 4 $\neq$ Hermes Agent. Not a child-safe school SKU. Memory in `~/hermes/` is sensitive.⁹⁵⁵ Grok Bot, if used, is included with SuperGrok and Cursor Pro/Teams as of the 26 August 2026 expansion post. Cloud storage required; Legacy Privacy Mode unsupported; shared computer per user; no education SKU and no FERPA claim on the pages opened. Grok $\neq$ Grok Bot $\neq$ El Salvador Grok-as-tutor. Hard no: the child's hands.⁹⁵⁶

Hint script, not a finished proof. Chapter 3 workflow (d): withhold the full solution; ask what they have tried; point out a named mistake; escalate information slowly.⁹⁵⁷ Level 1: "Which vertices correspond?" Level 2: "Congruence is a rigid motion; which motion would take *A* to *D*?" Level 3: "You have two sides and the included angle; SSA is not on the list, and here the angle is included — what criterion is that?" Not the two-column. The parent reads the hints aloud. Oreopoulos and Low, NBER w35620, August 2026: assignment to Khan Academy plus Khanmigo raised MAP math by 1.26 national percentile ranks per term, about 0.040 population SD per term and about 0.06 to 0.08 SD over a school year; gains resemble Khan Academy practice without AI assistance; 96 percent tried Khanmigo; the median student messaged it on only a third of practice days and in 17 percent of mistake sessions; 14.5 percent of messages contained a mathematical question or a step of reasoning.⁹⁵⁸ Do not invent standalone 0.06 as the ITT. Do not advertise Khanmigo

⁹⁵⁵Hermes Agent parent playbook: Chapter 3 §3.8 / 04-ai §9.1. Practice suggestion, not an RCT. Model split this day (Nous Portal, <https://hermes-agent.nousresearch.com/docs/integrations/nous-portal>): Hermes 4 or any strong chat model for parent explanation and hint scripts; agentic examples (Claude Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro) for the tool loop. Hermes 4 is **not recommended inside Hermes Agent** (tuned for chat, not rapid tool-calling). Hard no: messaging gateway to a child's Telegram/WhatsApp. Not a child-safe school SKU. Memory in `~/hermes/` is sensitive. Hermes 4 $\neq$ Hermes Agent.

⁹⁵⁶Grok Bot parent playbook: Chapter 3 §3.8 / 04-ai §9.2. Practice suggestion, not an RCT. Included with SuperGrok and Cursor Pro/Teams as of the 26 August 2026 expansion post (<https://x.ai/news/grok-bot-more-plans>). Cloud storage required; Legacy Privacy Mode unsupported; shared computer per user; no FERPA claim (n. 948). Hard no: child's school portal or Photomath logins; IEP/504 PDFs on the shared computer; Grok Bot iOS app in the child's hands. El Salvador is not this playbook.

⁹⁵⁷Bastani et al., author PDF, Figure 1 (GPT Tutor prompt): withhold the full solution; ask what they have tried; point out a named mistake; escalate information slowly; accept equivalent answers. Homeschool analogue is a parent-read hint script, not a child-held chatbot. Chapter 3 workflow (d).

⁹⁵⁸Philip Oreopoulos and Nina Low, "One Click Away: AI Tutoring with Khanmigo in a Two-Year School Experiment," NBER Working Paper 35620, August 2026, <https://www.nber.org/papers/w35620>. Two-year cluster RCT, 18 Tennessee middle schools. Assignment raises math achievement by 1.264 NPR per term (SE 0.597); 0.040 population SD per term; about 0.06 to 0.08 SD over a school year; implied effect of a full year of active participation 0.14 SD. "These gains resemble those from Khan Academy practice without AI assistance." 96% tried Khanmigo at least once; median student messaged it on only a third of practice days and in 17% of mistake sessions; 14.5% of messages contained a mathematical question or a step of reasoning. Do not invent standalone 0.06 as the ITT. Do not advertise Khanmigo as Bloom's two-sigma tutor. Working paper, not yet journal-peer-reviewed. Pre-registration: AEA RCT Registry AEARCTR-0013519. Whether a parent sitting next to the child and requiring a hint dialogue is a different

as Bloom’s two-sigma tutor. A homeschool parent who buys Khanmigo and walks away should expect the Tennessee pattern. A parent who sits next to the child and requires a hint dialogue after a miss is running a different intervention than that ITT — a design hypothesis, not a fitted result.

Session shape, Chapter 3 §3.8, scaled to this course. Practice suggestion, not an RCT. Unaided warm-up: one rigid-motion image (“does this sequence take $\triangle ABC$ onto $\triangle DEF$?”), one missing-length on similar triangles, one “is this a proof or a drawing?” Teach from the curriculum, with the parent already having used workflow (a) *before* sitting down. Struggle on two or three items, child writes, no phone. Hint, not key: parent uses the hint script, or Khanmigo with the parent watching, only after a miss. Hands-on or diagram if the miss was conceptual — a paper-fold reflection, a GeoGebra file the parent already checked. Exit ticket unaided: a short argument, or a second isomorphic figure, without the model. Parent grades against the parent key. Optionally later, workflow (f) on misses.⁹⁵⁹ “Finish your geometry with Photomath so we can go to soccer” is GPT Base, the pattern with a large-sample causal harm estimate. Detectors fail; do not use them on the written argument. Companion-harm cases — *Garcia v. Character Technologies*, *Raine v. OpenAI* — are allegations, not verdicts; they are why a math agent should not be a companion. COPPA amended Rule published 22 April 2025, general compliance 22 April 2026. Homeschool parents are not verified U.S. K–12 educators by default and should not assume ChatGPT for Teachers or Claude for Teachers FERPA wrappers. California A.B. 1159 is not law as of 30 August 2026 (Senate third reading).⁹⁶⁰

A *hypothetical*, labelled as such, can put the ban in a room. A parent whose fifteen-year-old photographs tonight’s SAS figure into a chatbot, copies a fluent numbered two-column, scores 100 percent on the page, and then cannot mark the correspondence on tomorrow’s unaided isomorphic figure, is running GPT Base.⁹⁶¹ The empirical pieces are Bastani’s crutch, solver homepages, the UNVERIFIED status of a consumer two-column eval, and Soderstrom and Bjork’s performance/learning split. The age, the 100 percent, and the figure are hypothetical. The move is not: take the phone, put the correspondence on paper, and run the unaided argument.

If the family only remembers one sentence from this section: the model may prepare the adult, the next isomorphic figure, a diagram description, and a question about an auxiliary line; it may not

intervention than the ITT is a design hypothesis, not a fitted result.

⁹⁵⁹Session shape: Chapter 3 §3.8 / 04-ai §9.4, scaled to this course (one rigid-motion image, one missing-length, one “is this a proof or a drawing?”). Practice suggestion, not an RCT. Grounded in Bastani’s exam phase (n. 936), Soderstrom and Bjork (n. 894), and Oreopoulos and Low’s engagement finding (n. 958). “Finish your geometry with Photomath so we can go to soccer” is GPT Base, the pattern with a large-sample causal harm estimate.

⁹⁶⁰Federal Trade Commission, “Children’s Online Privacy Protection Rule,” 90 FR 16918 (22 April 2025), Document 2025-05904, 16 CFR 312. Effective date 23 June 2025; general compliance date 22 April 2026. Full discussion: Chapter 3 §3.6. A parent using an adult tool on a parent account to generate a worksheet is a different fact pattern from creating a child account on a companion app. Homeschool parents are not verified U.S. K–12 educators by default and should not assume ChatGPT for Teachers (OpenAI, 19 November 2025) or Claude for Teachers (Anthropic, 14 July 2026) FERPA wrappers. California A.B. 1159 (Addis), 2025–2026 session: **Active Bill — In Floor Process**, Senate third reading as of 30 August 2026; last amended 21 August 2026; **not law**. Companion-harm cases — *Garcia v. Character Technologies et al.*; *Raine v. OpenAI* — are **allegations, not verdicts**. Setzer settlement terms **UNVERIFIED**. Why a math agent should not be a companion. Not a literature on mathematics learning.

⁹⁶¹Composite *hypothetical*, labelled in the sentence that introduces it. Empirical pieces: Bastani’s crutch (n. 936); solver homepages (n. 946); UNVERIFIED status of a consumer two-column eval (n. 856); Soderstrom and Bjork’s performance/learning split (n. 894). The age, the 100 percent, and the figure are not a reported sitting. The move is not: take the phone, put the correspondence on paper, and run the unaided argument.

write the child's proof. GSM8K saturation is not geometry proof. AlphaGeometry2's 84 percent is a specialized system. Chat LLMs reportedly solved none of that IMO geometry set. The child writes unaided. The parent holds the key.

Right-triangle trigonometry here is not unit-circle trigonometry. Chapter 9 is the third course, and the unit circle. Informal-limit arguments here are not yet Chapter 10's precise ones. S.CP here is not yet Chapter 11. Transcript this course as Geometry. The agent does not take the test. The child does.

Chapter 9

Algebra II, Functions, and Precalculus

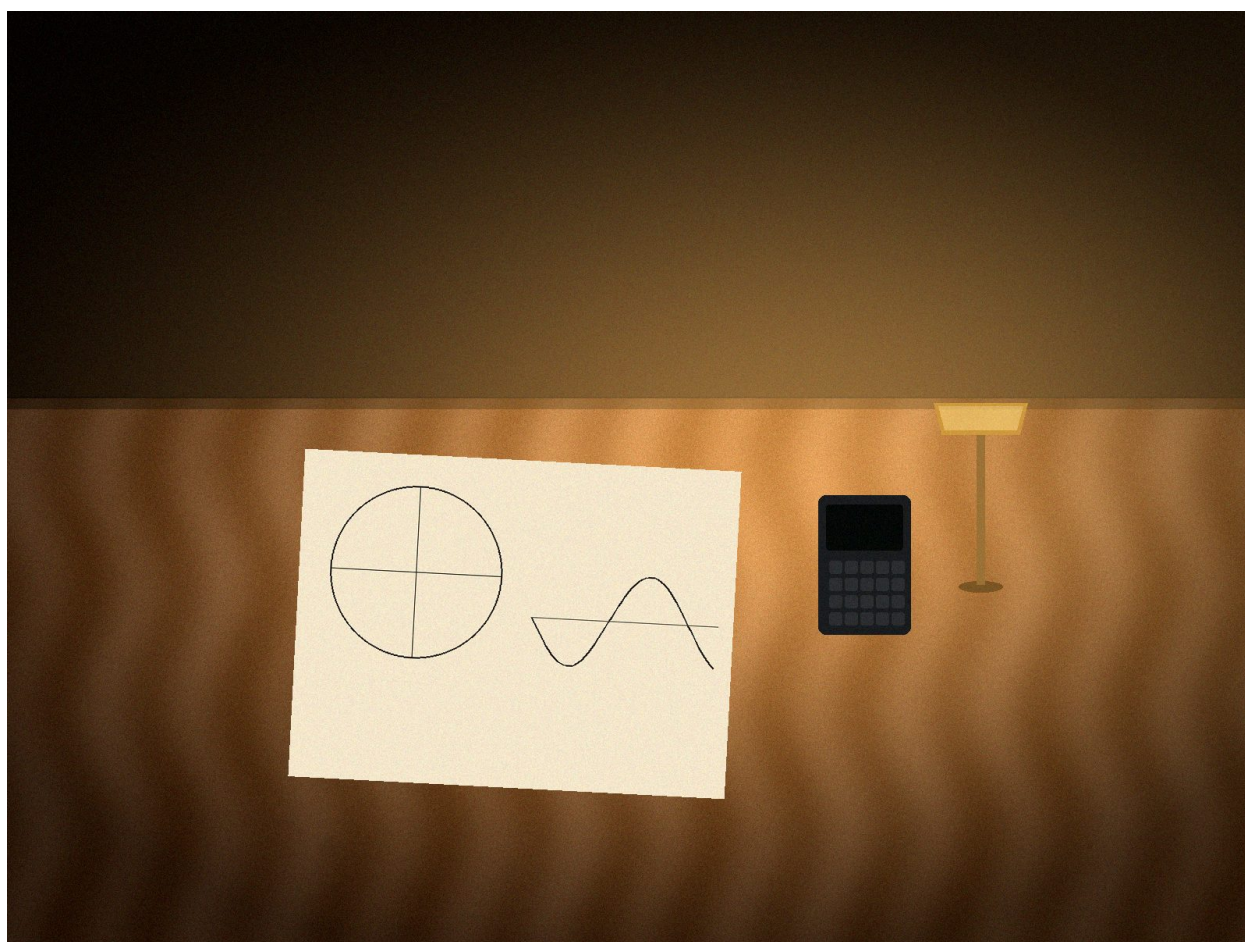


Figure 10: A paper unit circle and sine curve beside a graphing calculator with a blank dark screen.

Two percents from the same table already contain the high-school problem this chapter has to name. In the 2016 Parent and Family Involvement tables, Algebra II had ever been taught to 49 percent of high-school-level homeschoolers. Calculus had ever been taught to 15 percent — and that cell

carries the National Center for Education Statistics caution flag.⁹⁶² Those are not completion rates. They are not 2026. The 2023 First Look did not republish the subject tables. They are the last federal subject-by-band snapshot this project opened, and the drop is the book's high-school problem. Algebra I is common: 79 percent ever-taught at high-school level in the same table. Geometry is not universal: 52 percent. Algebra II sits just under half. Calculus at home is a minority. A kitchen that stops after Algebra I because "the state doesn't require it" has a college problem in every named university page Chapter 12 will print. A kitchen that transcripts "Algebra 2" after a year of leftover quadratics and a little log has a mathematics problem the National Mathematics Advisory Panel already named. This chapter is the third course on the traditional map, plus the remainder a parent shopping for a book titled Precalculus is usually buying. It is not a legally defined grade. It is not a franchise. It is school algebra, finished or not.

The catalog independent of packaging is Table 1 of the Panel's 2008 *Foundations for Success*. The Major Topics of School Algebra are: symbols and expressions — polynomial and rational expressions; arithmetic and finite geometric series; linear equations; quadratic equations; functions — linear, quadratic, polynomial, simple nonlinear, exponential, logarithmic, trigonometric, and fitting models to data; algebra of polynomials, including complex numbers and the fundamental theorem of algebra; combinatorics and finite probability.⁹⁶³ Chapter 7 was the first half of that catalog: linear, exponential, and quadratic functions, systems, descriptive statistics. This chapter is the remainder. Polynomials beyond quadratic. Rational expressions. Complex numbers. Inverse functions. Logarithms as solutions of exponentials. Trigonometric functions on the unit circle. The Algebra II statistics clusters. A homeschool Algebra II that omits functions, logarithms, and trigonometry has not covered school algebra as the Panel defined it.⁹⁶⁴ That sentence is not a culture-war claim. It is a checklist. Section 9.5 will print it as a parent sheet. Do not skip the sheet because the book's last chapter was the quadratic formula.

Achieve, in partnership with the Common Core writing team, published Appendix A as four model high-school pathways, not as a mandate. Of Algebra II the Achieve overview says the course "focuses on rational and polynomial expressions, equations, and functions, as well as a continued study of probability and statistics," with additional emphasis on modeling relative to many older Algebra II courses.⁹⁶⁵ The typical U.S. traditional organization is Algebra I, then Geometry, then Algebra II.

⁹⁶²National Center for Education Statistics, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey* (NCES 2020-001). High-school-level homeschooled students ever taught Algebra II, 49 percent; calculus, 15 percent (interpret with caution). Last opened subject tables; the 2023 First Look (NCES 2024-113) did not republish them. Not completion rates and not 2026.

⁹⁶³National Mathematics Advisory Panel, *Foundations for Success: The Final Report of the National Mathematics Advisory Panel* (U.S. Department of Education, 2008), Table 1, p. 16, ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>, accessed 30 August 2026. Major Topics of School Algebra: Symbols and Expressions (polynomial and rational expressions; arithmetic and finite geometric series); Linear Equations; Quadratic Equations; Functions (linear, quadratic, polynomial, simple nonlinear, exponential, logarithmic, trigonometric, and fitting models to data); Algebra of Polynomials (including complex numbers and the fundamental theorem of algebra); Combinatorics and Finite Probability. Catalog of "algebra through Algebra II," independent of course packaging.

⁹⁶⁴NMAP, *Foundations for Success* (2008), Table 1 (n. 963). A homeschool Algebra II that omits functions, logarithms, and trigonometry has not covered school algebra as the Panel defined it. Restated as a parent checklist in §9.5.

⁹⁶⁵Achieve / CCSS writing team, *Appendix A: Designing High School Mathematics Courses Based on the Common Core State Standards*, Oregon ODE copy, <https://www.oregon.gov/ode/educator->

An international-style integrated sequence of Mathematics I–II–III also exists. Compacted versions of each complete Grade 7, Grade 8, and the first high-school course in Grades 7–8 so that a student can reach Calculus or another college-level course by senior year. “The pathways and courses are models, not mandates.”⁹⁶⁶ This chapter is the traditional third course. Integrated Mathematics III is the alternative packaging of much of this content plus remaining geometry and statistics clusters. Name it. Do not write three integrated chapters. Appendix A’s integrated pathway was only partly extracted for this project; topic-by-topic Mathematics III remains **UNVERIFIED** pending a re-opening of Appendix A pages 49–90.⁹⁶⁷ A family on that train will not have a year titled Algebra II. They will still need the objects this chapter names.

When the course typically appears is a calendar fact, not a developmental law. Grade 11 in the non-compacted traditional pathway, after Geometry. Grade 10 after a Grade 8 Algebra I and a Grade 9 Geometry. Some students sit it earlier. Finding 15 of the Panel is still the age-lock refusal: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; claims that children of a given age cannot learn certain content because they are “too young,” “not in the appropriate stage,” or “not ready” have “consistently been shown to be wrong.”⁹⁶⁸ Accelerate a child who has Algebra I’s function fluency and Geometry’s right-triangle trigonometry. Do not withhold authentic Algebra II from a fifteen-year-old who has them. Do not start the course to keep up with public-school Grade 11 if the gates are closed. A compacted path to calculus in senior year exists so that the destination is reachable. Appendix A is explicit: the regular three-year pathway plus a fourth course reaches precalculus as a junior and calculus as a senior *only if* high-school content started in Grade 8.⁹⁶⁹ A non-accelerated student who starts Algebra I in Grade 9 finishes Algebra II in Grade 11 and has senior year for precalculus, statistics, or a first calculus course — not both precalculus and BC unless the fourth course is overloaded. That sentence is a pathway fact. It is not a mandate to compact. Compact only if the gates are open. Do not skip this course because calculus is the destination. Chapter 10 will say that again. This chapter is where the destination becomes possible, or not.

What a parent buying a book titled Precalculus is actually buying is a publisher’s own scope, not a legal course. Appendix A’s “Fourth Courses” column holds most of the (+) standards not required

resources/standards/mathematics/Documents/math-appendix-a-model-course-pathways.pdf, accessed 30 August 2026. Algebra II “focuses on rational and polynomial expressions, equations, and functions, as well as a continued study of probability and statistics,” with additional modeling emphasis relative to many older Algebra II courses (Achieve overview of the pathways). Map, not a mandate.

⁹⁶⁶Appendix A (n. 965). Four model pathways: Traditional (Algebra I, Geometry, Algebra II); Integrated (Mathematics I, II, III); compacted versions of each that complete Grade 7 + Grade 8 + the first high-school course in Grades 7–8 so that a student can reach Calculus or another college-level course by senior year. “The pathways and courses are models, not mandates.”

⁹⁶⁷Appendix A integrated pathway, pages 49–90: only partly extracted for this project (01-scope open question 7). Topic-by-topic Mathematics III remains **UNVERIFIED**. Name the pathway; do not write three integrated chapters.

⁹⁶⁸NMAP, *Foundations for Success* (2008), Finding 15. What is developmentally appropriate is “largely contingent on prior opportunities to learn.” Age-lock claims (“too young,” “not in the appropriate stage,” “not ready”) have “consistently been shown to be wrong.” Brain-immaturity claims are not justified even when students possess the prerequisite knowledge.

⁹⁶⁹Appendix A (n. 965): the regular three-year pathway plus a fourth course reaches precalculus as a junior and calculus as a senior *only if* high-school content started in Grade 8. A non-accelerated student who starts Algebra I in Grade 9 finishes Algebra II in Grade 11 and has senior year for precalculus, statistics, or a first calculus course — not both precalculus and BC unless the fourth course is overloaded. Model, not a mandate to compact.

for college-and-career readiness: vectors and matrices; remaining complex-plane work; further inverse-function work; further trigonometry; remaining conics; further probability and expected value.⁹⁷⁰ A parent shopping for “precalculus” is usually buying this remainder plus a tightening of Algebra II functions. College Board’s AP Precalculus, first widely offered 2023–24, with content unchanged in the 2026–27 clarifications, is one named fourth-course object: “AP Precalculus prepares students for other college-level mathematics and science courses. Through regular practice, students build deep mastery of modeling and functions.” The official course page, refetched 30 August 2026, publishes four units with MC weights: Unit 1 Polynomial and Rational Functions 30%–40%; Unit 2 Exponential and Logarithmic Functions 25%–40%; Unit 3 Trigonometric and Polar Functions 30%–35%; Unit 4 Functions Involving Parameters, Vectors, and Matrices not assessed. Units 1–3 are on the exam. Topic lists below those titles stay **UNVERIFIED**.⁹⁷¹ Do not dump Chapter 10’s Calculus AB and BC unit lists here. Precalculus is not a legally defined course. Algebra II, as Appendix A and the Panel write it, is a named object. Keep the split.

Four lanes run through the sections. They are not four identical headings. Sections 9.1 through 9.5 explain the mathematics to the parent — what Algebra II is and what a precalculus title is buying; polynomials, rationals, and complex numbers; inverse, exponential, and logarithmic functions; unit-circle trigonometry; the Algebra II statistics clusters and the Panel’s finished-algebra checklist, planted for Chapter 11. Section 9.6 is teaching guidance: the gates (fractions still, functions from 8.F and Algebra I, Geometry’s right-triangle trigonometry); worked-example fading for the first pass through a new symbol-heavy procedure; interleaving of function types, labelled as transfer from Rohrer’s grade-7 classroom trials, not as a new Algebra II *d*; release valves when the parent is the ceiling — tutor, co-op, dual-enrollment College Algebra or Precalculus — planted for Chapter 12. Section 9.7 is the parent-hosted AI lane: a parent-first explanation of why $\ln$ and $\log_{10}$ differ and why the unit circle is the right home for trigonometry; isomorphic items, parent-held key; Wolfram|Alpha as a computer-algebra check after the child has attempted, not as an explainer; ban solver-for-the-child. Point to Chapter 3. Unaided exit tickets. AP Precalculus free response, if sat, is unaided.

Chapter 2 already locked the method this chapter will use: five interwoven strands; conceptual and procedural knowledge iterating, not a one-way street; worked examples faded for novices, incorrect examples explained; blocked practice to learn execution, interleaved practice to choose a strategy — and the clean classroom randomized trials of interleaving are grade 7, not an Algebra II effect size this chapter will invent; word problems as schemas, not keywords; explicit instruction for students already struggling, productive failure as generation-then-instruction on a new concept when priors

⁹⁷⁰Appendix A, “Fourth Courses” column: most (+) standards not required for college-and-career readiness — N.VM vectors and matrices; remaining complex-plane work; further inverse functions; F.TF.3, 4, 6, 7, 9; remaining conics (ellipse/hyperbola); S.MD expected value. A parent shopping for “precalculus” is usually buying this remainder plus a tightening of Algebra II functions. Precalculus is not a legally defined course.

⁹⁷¹College Board, AP Precalculus course page, <https://apcentral.collegeboard.org/courses/ap-precalculus>, accessed 30 August 2026. First widely offered 2023–24; 2026–27 clarifications: “Course content has not changed.” “AP Precalculus prepares students for other college-level mathematics and science courses. Through regular practice, students build deep mastery of modeling and functions.” Course page unit table (MC weighting): Unit 1 Polynomial and Rational Functions 30%–40%; Unit 2 Exponential and Logarithmic Functions 25%–40%; Unit 3 Trigonometric and Polar Functions 30%–35%; Unit 4 Functions Involving Parameters, Vectors, and Matrices not assessed. Units 1–3 assessed; Unit 4 local/state. Exam format changes (MC counts, FRQ 1–2, section timing) take effect May 2027. Topic lists below those four titles remain **UNVERIFIED** (full CED PDF not opened). Do not invent a topic catalog.

exist — those two literatures not the same comparison; kitchen-math transfer hard. Chapter 3 already locked the AI lane: unguarded solvers raise practice scores and cut unaided exams; the model may prepare the adult and the next problem; it may not do the child’s problem. This chapter will not re-litigate Bastani at monograph length. One short reminder, then the objects of Algebra II. Chapter 5 already did the fractions gate. Chapter 6 already planted 8.F. Chapter 7 already did linear, exponential, and quadratic functions. Chapter 8 already did right-triangle trigonometry and planted the unit circle. This chapter will not rewrite them. It will use them as a door. Do not mash the Panel’s Finding 27 with Kapur 2014. Do not sell Saxon or Art of Problem Solving. Do not dump Chapter 10’s eight AB units. Do not dump Chapter 11’s five AP Statistics units. Plant both. No homeschool Algebra II randomized trial was opened for this project. Do not invent one.

The popular story collapses those locks into slogans they do not support. Algebra II is eleventh grade. Precalculus is a legal course. Functions are $y = mx + b$, still. Logarithms are a calculator button. Trigonometry is SOH-CAH-TOA from geometry, so skip the unit circle. Complex numbers are a curiosity. Statistics can wait for the AP year. The video curriculum does 100 percent of the teaching, so the parent can stop hearing a wrong inverse. Photomath will tutor tonight’s identities. The documents do not say any of those things, not as the slogans say them. This chapter prints the documents. It prints the objects, the sample sizes, and the nulls. It leaves the open questions standing — AP Precalculus topic lists below the four official unit titles **UNVERIFIED**; Teaching Textbooks / Life of Fred / Math Mammoth topic placements **UNVERIFIED**; Adelman 1999 *Answers in the Toolbox* cited by the Panel for the Algebra II–college-graduation correlation, primary not re-opened this pass; college-credit comparison of AP versus dual enrollment versus CLEP at scale, not found as a single national experiment; when to send the child out of the kitchen, a household decision constrained by Maloney, not a fitted age; Integrated Mathematics III topic-by-topic **UNVERIFIED**. It does not invent a kitchen scene to make the science feel warmer. The parent who stopped at algebra is still the design user. The job is to hear $\log(ab)$ written as $\log a + \log b$ with the bases mismatched, and $\sin^2\theta + \cos^2\theta$ written as $\sin(2\theta)$, before any tool is allowed to print a simplified identity the parent never sees.

The function concept from 8.F and from Algebra I’s F-IF must be solid before this course earns its name. A function assigns to each input exactly one output. The student must move among formula, graph, table, and verbal. Without that concept, calculus’s $f'(x)$ has nothing to attach to.⁹⁷² That sentence is a plant for Chapter 10. It is not a scare. Calculus is a new operation on functions the student already knows. A student with a shaky function concept will experience calculus as symbol soup. Diagnose at this door. Do not buy a book titled Precalculus to hide a missing 8.F.

9.1 What Algebra II is, and what “precalculus” is buying

Algebra II, as Appendix A writes it, is the third course in the traditional pathway. The fundamental purpose, as the Achieve overview already quoted, is rational and polynomial expressions, equations, and functions, plus a continued study of probability and statistics, with more modeling than many

⁹⁷²01-scope load-bearing 13 and Algebra I / Grade 8 unlocks: without the function concept (8.F, then F-IF), calculus’s $f'(x)$ has nothing to attach to. Plant for Chapter 10. Common calculus-door gaps named in 01-scope as **inference from the maps, not a survey**.

older Algebra II courses carried.⁹⁷³ That is the object. It is not a birthday. It is not a publisher's grade. It is not "the year after Geometry" as a franchise slogan. It is a model, not a law. The pathways and courses are models, not mandates. The typical U.S. traditional organization places it after Geometry. The integrated pathway distributes the same clusters across Mathematics I–II–III. Compacted versions exist so calculus is reachable in senior year. None of those calendars is the mathematics.

The content catalog, with Appendix A's cluster homes, is the parent checklist for the *third* course, not a 180-day plan. Complex numbers: N.CN.1–2, and use in polynomial identities and equations — N.CN.7, with N.CN.8 and N.CN.9 as (+) standards. Polynomial and rational expressions beyond quadratic: A.SSE; A.APR.2–3, zeros and factors, the remainder theorem; A.APR.4 and A.APR.5 as (+); A.APR.6, rewrite rational expressions. Equations: radical and rational, A.REI.2; graphs combining polynomial, rational, radical, absolute value, exponential, logarithmic, A.REI.11. Functions: polynomial, rational, radical, exponential, logarithmic, trigonometric; inverse functions; transformations across function types. Trigonometric functions: F.TF.1–2, unit circle and radian; F.TF.5, model periodic phenomena; F.TF.8, Pythagorean identity. Exponential and log: F.LE.4, logarithms as solutions of exponentials. Statistics: S.ID.4, normal distribution; S.IC, making inferences and justifying conclusions from surveys, experiments, observational studies; probability rules continued.⁹⁷⁴ That is a year. It is not a lot because a publisher padded a catalog. It is a lot because the course finishes school algebra as the Panel defined it, and because a kitchen that stops at leftover quadratics has not finished.

What is left for a "fourth course" is a different column. Appendix A's Fourth Courses hold most of the (+) standards not required for college-and-career readiness: N.VM, vectors and matrices; remaining complex-plane work; further inverse-function work; further trigonometry — F.TF.3, 4, 6, 7, 9; remaining conics, ellipse and hyperbola; further probability and expected value, S.MD.⁹⁷⁵ A parent shopping for "precalculus" is usually buying this remainder plus a tightening of Algebra II functions. Polar coordinates and parametrics sit in that remainder, or in a calculus BC year. They are not a legal requirement of a course titled Algebra II. They are also not optional extras if the destination is BC. Keep the split. Do not dump Chapter 10's Units 9–10 here. Do not pretend a book titled Precalculus has a statutory definition.

AP Precalculus is one named fourth-course object, not the definition of the remainder. The course-page facts already named in the opening stand: modeling and functions; four official units with published MC weights; Units 1–3 assessed; Unit 4 (parameters, vectors, matrices) not on the exam; exam format changes May 2027; "Course content has not changed." Topic lists below those four

⁹⁷³Appendix A Algebra II purpose (n. 965), restated at the section opening so the object is in view before the cluster list.

⁹⁷⁴Appendix A traditional-pathway cluster chart for Algebra II, as extracted in 01-scope: N.CN.1–2 (and use in polynomial identities and equations, N.CN.7, +8, +9); A.SSE; A.APR.2–3 (zeros and factors; remainder theorem), A.APR.4 (+5), A.APR.6 (rewrite rational expressions); A.REI.2 radical and rational; A.REI.11 graphs combining function types; polynomial, rational, radical, exponential, logarithmic, trigonometric functions; inverse functions; F.TF.1–2 unit circle and radian, F.TF.5 periodic, F.TF.8 Pythagorean identity; F.LE.4 logarithms as solutions of exponentials; S.ID.4 normal; S.IC inference. Map, not a 180-day plan.

⁹⁷⁵Appendix A Fourth Courses (n. 970). Polar/parametrics as (+) remainder or as AP Calculus BC Unit 9 — plant for Chapter 10; do not dump BC Units 9–10 here.

titles stay **UNVERIFIED**.⁹⁷⁶ Do not invent a topic catalog under the titles. A family sitting AP Precalculus is sitting a College Board course with an exam. A family running a non-AP fourth course from a textbook is running a tightening of Algebra II functions plus some of the (+) remainder. Those are different objects that share a lot of mathematics. Credit and placement are set by each college. College Board itself says to search each college's policy. Do not claim AP is always more widely accepted than CLEP or dual enrollment as a universal fact.⁹⁷⁷ Dual-enrollment College Algebra is a third object: a college transcript NCAA and admissions offices already know how to read. Plant for Chapter 12. Do not dump the Core-Course Worksheet.

The live debate this section will not pretend to settle is whether a homeschool “Algebra 2” that is mostly quadratics plus a little log has earned the transcript title. The Panel's Table 1 says no. Functions through trigonometric, algebra of polynomials including complex numbers and the fundamental theorem of algebra, logarithms, combinatorics and finite probability — those are in the catalog of school algebra, independent of course packaging.⁹⁷⁸ A book that never reaches the unit circle, never treats a logarithm as the solution of an exponential, and never writes a complex number as $a + bi$, has not covered the catalog. Transcript the work that was done. Do not upgrade a quadratic-review year into Algebra II because the cover said so. Do not withhold the title from a year that did the catalog because the publisher used a different spine. The title is an export object. Chapter 12 will say that as a rule. This chapter says it as a mathematics rule.

Clifford Adelman, *Answers in the Toolbox*, 1999, is cited by the Panel for the Algebra II–college-graduation correlation.⁹⁷⁹ The primary was not re-opened for this project. Cite it as the Panel's citation, not as a re-fetched finding, and not as a homeschool experiment. The Panel used it. This book will not launder an unopened 1999 report into a claim that finishing a kitchen Algebra II *causes* a bachelor's degree. Cardus 2023 already refused a causal reading of BA-rate gaps. Do not invent a new one from Adelman-via-NMAP. What survives is narrower and more useful: named college pages want the title. Southern Methodist University's application guide for home-schooled and nontraditional students lists a minimum *suggested* three mathematics — Algebra I, Geometry, Algebra II.⁹⁸⁰ The University of Pittsburgh asks for a minimum of three years of mathematics. North Central College recommends three years including algebra and geometry, one course beyond geometry preferred.⁹⁸¹ Pattern, not a rule: three years of high-school math through Algebra II is

⁹⁷⁶College Board AP Precalculus course page (n. 971). Official unit names and MC weights now taken from the course-page table. Full CED PDF not opened this pass. Topic lists below those four titles remain **UNVERIFIED**.

⁹⁷⁷College Board, including the CLEP/AP comparison page, <https://clep.collegeboard.org/for-high-schools/comparing-clep-and-ap>, accessed 30 August 2026: search each college's policy. Do not claim AP is always more widely accepted than CLEP or dual enrollment as a universal fact. Secondary journalism that ranks AP above CLEP for selective colleges is labelled journalism.

⁹⁷⁸NMAP Table 1 (n. 963). Live debate: whether a homeschool “Algebra 2” that is mostly quadratics plus a little log has earned the transcript title — the Panel's catalog says no.

⁹⁷⁹Clifford Adelman, *Answers in the Toolbox* (1999), cited by NMAP for the Algebra II–college-graduation correlation. Primary not re-opened this pass (01-scope working source 38). Cite as the Panel's citation, not as a re-fetched finding and not as a homeschool experiment. Do not upgrade into a causal claim that finishing kitchen Algebra II produces a bachelor's degree.

⁹⁸⁰Southern Methodist University, application guide for home-schooled and nontraditional students: minimum *suggested* three mathematics (Algebra I, Geometry, Algebra II); transcript with consistent grading and course descriptions including textbook titles. Pattern, not a rule. Full discussion: Chapter 12.

⁹⁸¹University of Pittsburgh, home-schooled-applicant page: minimum 3 years of mathematics; Home School Supplemental Form; transcript by year with grades. North Central College: recommended 3 years mathematics including

the floor most of these pages name. Selective STEM programs will want precalculus or calculus plus an external signal — AP, dual enrollment, SAT math. The kitchen that stops at Algebra I has a college problem in every named example. The kitchen that transcribes Algebra II without the Panel’s catalog has a different problem: a title a stranger cannot trust.

NCAA Eligibility Center: math core courses are Algebra I or higher. Division I requires three years of such math among sixteen core courses.⁹⁸² Algebra II is inside that “or higher.” Transcript the course as Algebra II, not “Math 11,” not “Saxon Algebra 2” without a parenthetical a stranger can map onto the third-course clusters. A family that ran Saxon’s classic integration — geometry inside Algebra 1, Algebra 2, and Advanced Math — still needs a course a stranger can map onto Algebra II, and a separate mapping onto Geometry. That is **vendor** / Rainbow Resource catalogue, not a finding that integration is better.⁹⁸³ Math-U-See has a named Algebra 2 level after Geometry, then PreCalculus, then Calculus — **vendor** / Demme Learning, a topic ladder, not a birthday.⁹⁸⁴ Art of Problem Solving has Algebra, then Intermediate Algebra, then Precalculus titles in the post-Beast Academy sequence — **vendor**. Teaching Textbooks, RightStart, Math Mammoth, and Life of Fred Algebra II / precalculus topic placements that were not opened on a vendor primary remain **UNVERIFIED**. Do not promote those rows to handbook fact.⁹⁸⁵ No homeschool-curriculum randomized trial comparing any of these programs was opened for this project. Choose by fit — parent math confidence, student stance, special needs, budget, college target — not by a phantom home-school meta-analysis. Cathy Duffy Reviews and Rainbow Resource remain **catalogue** / **reviewer**, not independent effectiveness trials.

Integrated Mathematics III is the other train. Appendix A publishes it as the third year of a model pathway that distributes the same high-school clusters across three years rather than concentrating algebra in two and geometry in one. A family on that train will not have a year titled Algebra II. They will still need polynomials beyond quadratic, rationals, complex numbers, inverse functions, exponential and log, unit-circle trigonometry, and the S.ID.4 / S.IC clusters this chapter names. They will also need remaining geometry clusters this chapter will not write. Name the pathway. Transcript the course as Integrated Mathematics III, or Mathematics III, with a parenthetical a stranger can map onto Algebra II plus remaining geometry — a plant for Chapter 12,

algebra and geometry, one course beyond geometry preferred. Pattern, not a rule: three years through Algebra II is the floor most of these named pages want. Selective STEM programs will want precalculus or calculus plus an external signal.

⁹⁸²NCAA Eligibility Center: math core courses are Algebra I or higher. Division I: 16 core courses including 3 years math (Algebra I or higher). Division II: 2 years math (Algebra I or higher). Algebra II is inside “or higher.” Transcript the course as Algebra II, not “Math 11.” Plant for Chapter 12. NCAA is a private association’s eligibility rules, not law.

⁹⁸³Houghton Mifflin Harcourt / Rainbow Resource catalogue: Saxon classic editions integrate geometry in Algebra 1, Algebra 2, and Advanced Math; a separate Geometry course also exists. **Vendor** / catalogue, not a finding that integration is better. The transcript still needs a course a stranger can map onto Algebra II and onto Geometry.

⁹⁸⁴Math-U-See / Demme Learning, curriculum sequence, <https://mathusee.com/products/math-u-see-curriculum/>, accessed 30 August 2026. Named Algebra 2 level after Geometry; then PreCalculus; then Calculus. **Vendor**. A topic ladder, not a birthday.

⁹⁸⁵Teaching Textbooks, RightStart, Math Mammoth, and Life of Fred Algebra II / precalculus topic placements: vendor primaries not opened this pass (01-scope open question 5). **UNVERIFIED**. Do not promote those rows to handbook fact. Art of Problem Solving post-Beast Academy Algebra / Intermediate Algebra / Precalculus titles: **vendor**. No homeschool-curriculum RCT comparing these was opened. Cathy Duffy Reviews and Rainbow Resource: **catalogue** / **reviewer**, not independent effectiveness trials.

not a dump of the NCAA worksheet.⁹⁸⁶ Do not write three integrated chapters. Topic-by-topic Mathematics III from Appendix A pages 49–90 remains **UNVERIFIED**. If the family’s book is integrated, map tonight’s lesson onto the objects in sections 9.2 through 9.5 and onto Chapter 8’s remaining geometry objects. The mathematics does not change with the cover.

A compacted path through Grades 7–8 exists so that this course can sit in Grade 10, a fourth course in Grade 11, and calculus in Grade 12. Appendix A’s compacted models, already named in the opening, are models, not mandates.⁹⁸⁷ The Panel recommended that all *prepared* students have access to an authentic algebra course by Grade 8.⁹⁸⁸ Authentic Grade 8 Algebra I is Chapter 7 sitting early. It is not a license to skip this course’s logarithms and unit circle because the calendar is crowded. Compact if the gates are open. Do not delay this course because the parent last passed Algebra I in 1998. That last sentence is a release-valve problem. Section 9.6 will name the valves.

Singapore Math Inc., **vendor**, tells parents that after Dimensions 8 “students may move on to Geometry and Algebra 2 programs.” Dimensions 7 and 8 together cover pre-algebra and Algebra 1 with some geometry; they are not this course.⁹⁸⁹ Do not treat a finished Grade 8 book, of any franchise, as Algebra II. The Panel’s catalog is the test.

What the course depends on is already named, and this chapter will not dump Chapters 5 through 8 to say it again. Algebra I function fluency — especially linear, exponential, and quadratic — and Geometry’s right-triangle trigonometry and similarity.⁹⁹⁰ Polynomial long division and the remainder theorem still need fraction fluency. The gate from Chapter 5 is not closed by an Algebra I credit. A student who cannot operate on rationals, cannot move among formula, graph, table, and verbal for a function, or treats SOH-CAH-TOA as the whole of trigonometry, is not ready, regardless of calendar grade. What the course unlocks is Calculus — limits of functions, derivatives of the elementary functions, the chain rule — and AP Statistics, whose official prerequisite as of the Fall 2026 CED is first-year algebra only, with Algebra II still useful.⁹⁹¹ Statistics does not substitute for calculus as a mathematics sequence. Calculus does not substitute for statistics as a data-reasoning course. Chapter 11 will hold that split. This chapter plants the Algebra II clusters so they are not skipped because “stats is later.”

⁹⁸⁶Appendix A Integrated Mathematics III (n. 967). Transcript as Integrated Mathematics III or Mathematics III, with a parenthetical a stranger can map onto Algebra II plus remaining geometry. Plant for Chapter 12, not a dump of the Core-Course Worksheet.

⁹⁸⁷Appendix A compacted pathways (nn. 966, 969). Restated so the destination cannot overwrite the gates.

⁹⁸⁸NMAP, *Foundations for Success* (2008): all *prepared* students should have access to an authentic algebra course by Grade 8. “Authentic” means the Major Topics of School Algebra (Table 1), not a course that is algebra in name only.

⁹⁸⁹Singapore Math Inc., School FAQ, <https://www.singaporemath.com/pages/school-faq>, accessed 30 August 2026. After Grade 8 “students may move on to Geometry and Algebra 2 programs.” Dimensions 7 and 8 together cover pre-algebra and Algebra 1 with some geometry. **Vendor**. Not this course.

⁹⁹⁰01-scope Algebra II “Depends on”: Algebra I function fluency (linear, exponential, quadratic); Geometry right-triangle trigonometry and similarity. Polynomial long division and the remainder theorem still need fraction fluency. Unlock: Calculus (limits of functions, derivatives of the elementary functions, the chain rule) and AP Statistics (official prereq as of Fall 2026 CED: Algebra I only; Algebra II still useful).

⁹⁹¹College Board, *AP Statistics Course and Exam Description*, Effective Fall 2026, <https://apcentral.collegeboard.org/media/pdf/ap-statistics-course-and-exam-description.pdf>, accessed 30 August 2026. Non-calculus-based; prerequisite “any secondary school student who has successfully completed a first-year algebra course.” Algebra II still useful. Statistics does not substitute for calculus in a STEM calculus sequence; calculus does not substitute for statistics as a data-reasoning course. Plant for Chapter 11.

Common gaps when a homeschooler arrives underprepared, named as **inference from the maps, not a survey**, because this project opened no nationally representative Algebra II-teacher survey analogous to the Panel’s 2007 Algebra I survey of 743 teachers.⁹⁹² Weak function concept: cannot move among graph, table, formula, verbal. Chain-rule algebra later will fail because composition was never seen as a structure. Trigonometry of the unit circle never learned — only right-triangle SOH-CAH-TOA. No fluency with exponential and log. Those are the calculus-door gaps the 01-scope memo labelled inference. Diagnose them here, while there is still a year to close them. Do not wait for a book titled Calculus to discover them.

The Common Core State Standards for Mathematics are treated here as a map, not as a homeschool mandate. Many homeschoolers do not use CCSSM. Many U.S. states have adopted, adapted, or replaced it. A 2026 state-by-state census of who still uses it is **UNVERIFIED** in this project; this chapter will not say “most states.”⁹⁹³ What the map is good for is specifying what the mathematics is. The standards “do not dictate curriculum or teaching methods,” and the order of topics inside a grade or course is not a required teaching order.⁹⁹⁴ This book will not dump 180 days. It will not sell Singapore, Saxon, Math-U-See, or Art of Problem Solving. Those names appear as options, by grain and placement, labelled **vendor** where the claim is the publisher’s.

What this section has done is split the third course from the fourth course, refuse the calendar, and install the title as an export object that has to earn the Panel’s catalog. Polynomials, rationals, and complex numbers come next, because they are the algebra-of-polynomials remainder Chapter 7 deferred, and because a parent who stopped at algebra often remembers FOIL and does not remember why a remainder of 4 at $x = 1$ means $f(1) = 4$.

9.2 Polynomials, rationals, complex numbers

The algebra of polynomials is the Panel’s name for the remainder Chapter 7 deferred: roots, complex numbers, the fundamental theorem of algebra, and, in the Panel’s catalog, the binomial theorem and induction.⁹⁹⁵ Algebra I already did arithmetic on linear and quadratic polynomials, A.APR.1, and it recognized complex solutions of quadratics without developing them. This course develops them. This year’s object is structure in expressions that are no longer quadratic, a remainder that is a value of a function, a rational expression rewritten by division, and a number system in which every polynomial of degree n has n roots, counting multiplicity, once the coefficients are allowed to be complex.

⁹⁹²Common calculus-door gaps (weak function concept; chain-rule algebra; only right-triangle SOH-CAH-TOA; no fluency with exponential/log): 01-scope, labelled **inference from the maps, not a survey**. No nationally representative Algebra II-teacher survey analogous to NMAP’s 2007 Algebra I survey of 743 teachers was opened. Do not write “teachers today say.”

⁹⁹³2026 state-by-state census of who still uses CCSSM, who uses “standards formerly known as,” and who never did: **UNVERIFIED** (01-scope open question 8). Do not say “most states.”

⁹⁹⁴Common Core State Standards Initiative, *Common Core State Standards for Mathematics*, https://www.thecorestandards.org/assets/CCSSI_Math%20Standards.pdf, accessed 30 August 2026. “How to read the grade level standards”: the standards “do not dictate curriculum or teaching methods”; order of topics inside a grade is not a required teaching order. Map, not a homeschool mandate.

⁹⁹⁵NMAP Table 1, algebra of polynomials, including complex numbers and the fundamental theorem of algebra; binomial theorem and induction named in the Panel’s catalog. Appendix A places much of this in Algebra II, with some as (+).

Complex numbers first, because they are the extension that makes the fundamental theorem possible, and because they are the object a kitchen most often skips as a curiosity. A complex number is a number of the form $a + bi$, where a and b are real and $i^2 = -1$. N.CN.1–2: know there is a complex number i such that $i^2 = -1$, and every complex number has the form $a + bi$ with a and b real; use the relation $i^2 = -1$ and the commutative, associative, and distributive properties to add, subtract, and multiply complex numbers.⁹⁹⁶ That last clause is algebra as generalized arithmetic, Chapter 7’s first teaching move, now in a new system. $(2 + 3i) + (1 - i)$ is $3 + 2i$, because real parts add and imaginary parts add. $(2 + 3i)(1 - i)$ is $2 - 2i + 3i - 3i^2$, and $-3i^2$ is $-3(-1) = 3$, so the product is $5 + i$. The parent who hears “just FOIL it” and cannot say that i^2 was replaced by -1 because that is the defining relation has a procedure and not the system. Iterate. A short conceptual pass: we needed a number whose square is -1 so that $x^2 + 1 = 0$ has a solution; we write it i ; the plane of pairs (a, b) is the new number line, two-dimensional. Then a handful of sums and products. Then a return to the relation. Then more. Do not withhold practice for weeks. Do not run pages of $(a + bi)(c + di)$ with no why.

Use in polynomial identities and equations is N.CN.7, with (+) extensions in N.CN.8–9. Solve quadratic equations with real coefficients that have complex solutions. $x^2 + 1 = 0$ has solutions i and $-i$. $x^2 - 2x + 2 = 0$ has discriminant $4 - 8 = -4$, so the quadratic formula produces $1 \pm i$. Chapter 7 recognized that the discriminant can be negative. This course writes the solutions. The fundamental theorem of algebra, in the form a high-school course can use, is the claim that a polynomial of degree n with complex coefficients has n roots in the complex numbers, counting multiplicity. Appendix A places the theorem in the Algebra II neighborhood, often as (+) or as a statement to know rather than a proof to produce.⁹⁹⁷ Do not pretend a kitchen will prove it. Do pretend a kitchen can use it: $x^3 - 1 = 0$ has three roots, one real and two complex; $x^2 + 1$ is irreducible over the reals and factors as $(x - i)(x + i)$ over the complexes. A child who can recast “no real roots” as “two complex roots” has the theorem as a counting principle, not as a slogan.

Polynomials beyond quadratic, A.APR.2–3. The remainder theorem: for a polynomial $p(x)$ and a number a , the remainder on division by $x - a$ is $p(a)$.⁹⁹⁸ That sentence is the one the parent who stopped at algebra has to be able to hear wrong. Polynomial long division of $p(x) = x^3 - 1$ by $x - 1$ produces quotient $x^2 + x + 1$ and remainder 0, and $p(1) = 0$, which is why $x - 1$ is a factor. Division of $x^3 - 1$ by $x - 2$ produces some quotient and remainder $p(2) = 7$. The remainder is the value. Synthetic division is a compression of the same arithmetic, not a different religion. A child who can run synthetic division and cannot say that the last number is $p(a)$ has a procedure. A child who can evaluate $p(2)$ by substituting and cannot see that the same number is the remainder has a different procedure. Compare them. Rittle-Johnson and Star’s comparison-of-methods papers support the flexibility move; those 2007 and 2009 papers were not opened in full for this project, and their exact d values remain **UNVERIFIED**.⁹⁹⁹ The qualitative finding is enough for a kitchen

⁹⁹⁶CCSSM N.CN.1–2, as placed in Appendix A Algebra II: know there is a complex number i such that $i^2 = -1$, and every complex number has the form $a + bi$ with a and b real; use $i^2 = -1$ and the commutative, associative, and distributive properties to add, subtract, and multiply complex numbers.

⁹⁹⁷Fundamental theorem of algebra: a polynomial of degree n has n roots in the complex numbers, counting multiplicity. Appendix A places the theorem in the Algebra II neighborhood, often as (+) or as a statement to know rather than a proof to produce. Do not pretend a kitchen will prove it. Do use it as a counting principle.

⁹⁹⁸CCSSM A.APR.2, Appendix A Algebra II: remainder theorem — for a polynomial $p(x)$ and a number a , the remainder on division by $x - a$ is $p(a)$.

⁹⁹⁹Bethany Rittle-Johnson and Jon R. Star, comparison-of-methods papers (2007, 2009). Comparison improves

rule: after the child can execute one method, put a second method next to it and ask which is better for this problem, and why.

Zeros and factors, A.APR.3: graph polynomial functions, identifying zeros when suitable factorizations are available, and showing end behavior.¹⁰⁰⁰ If $p(a) = 0$, then $x - a$ is a factor. If $x - a$ is a factor, then $p(a) = 0$. That biconditional is the remainder theorem plus the factor theorem, which is the remainder-zero case. A cubic that factors as $(x - 1)(x - 2)(x + 3)$ has zeros 1, 2, and -3 , and the graph crosses the x -axis at those three points. End behavior of an odd-degree polynomial with positive leading coefficient: down on the left, up on the right. Even degree, positive leading coefficient: up on both sides. Those are compressions of “the leading term dominates for large $|x|$,” not decorations. A.REI.11 asks students to explain why the x -coordinates of the points where the graphs of $y = f(x)$ and $y = g(x)$ intersect are the solutions of $f(x) = g(x)$, and to find the solutions approximately, including intersections of polynomial, rational, radical, absolute value, exponential, and logarithmic graphs.¹⁰⁰¹ Three representations in one sitting: table, graph, formula. Do not introduce this year’s functions as “ $y = mx + b$ ” only, so the general “each input exactly one output” idea never forms. Chapter 6’s 8.F was the correspondence. Chapter 7’s F-IF was the notation. This course’s polynomial, rational, radical, exponential, logarithmic, and trigonometric families are the objects calculus will differentiate. If the student cannot move among representations for these families, Chapter 10 will be symbol soup. Diagnose. Plant.

Rational expressions, A.APR.6: rewrite simple rational expressions in different forms; write $a(x)/b(x)$ in the form $q(x) + r(x)/b(x)$, where $a(x)$, $b(x)$, $q(x)$, and $r(x)$ are polynomials with the degree of r less than the degree of b , using inspection, long division, or a computer algebra system.¹⁰⁰² The last clause is Appendix A’s permission to use a CAS as a rewrite tool. Section 9.7 will keep Wolfram|Alpha on the parent’s side of the table, as a check after the child has attempted the division by hand. The rewrite itself is the same move as $7 \div 3 = 2$ remainder 1, written $2 + 1/3$, now with polynomials. $(x^3 - 1)/(x - 1) = x^2 + x + 1$, remainder 0. $(x^2 + 3x + 1)/(x + 1) = x + 2$ remainder -1 , so $x + 2 - 1/(x + 1)$. A child who “cancels” x in $(x^2 + x)/(x + 1)$ to get x , or who cancels the x in $(x + 2)/(x + 3)$ to get $2/3$, has treated a rational expression as a pair of decorations to be crossed out rather than as a quotient of two numbers that happen to be written with a letter. That error is the incorrect example this year’s fading sequence has to name. “Cancel everything that looks alike” is the cousin of Ni and Zhou’s whole-number bias: adding the counts you see. Here the child is cancelling the letters they see. Name the misconception. Then an isomorphic item. Then the child generates.

Radical and rational equations, A.REI.2: solve simple rational and radical equations in one variable,

flexibility; exact d values **UNVERIFIED** this pass (02-learning; outline do-not-invent). Qualitative finding is enough for a kitchen rule: after the child can execute one method, put a second next to it.

¹⁰⁰⁰CCSSM A.APR.3, Appendix A Algebra II: identify zeros of polynomials when suitable factorizations are available; use the zeros to construct a rough graph including end behavior.

¹⁰⁰¹CCSSM A.REI.11, Appendix A Algebra II: explain why the x -coordinates of the points where the graphs of $y = f(x)$ and $y = g(x)$ intersect are the solutions of $f(x) = g(x)$; find solutions approximately, including intersections involving polynomial, rational, radical, absolute value, exponential, and logarithmic functions.

¹⁰⁰²CCSSM A.APR.6, Appendix A Algebra II: rewrite simple rational expressions; write $a(x)/b(x)$ in the form $q(x) + r(x)/b(x)$ using inspection, long division, or, as the standard permits, a computer algebra system. Section 9.7 keeps the CAS on the parent’s side, after the child has attempted.

and give examples showing how extraneous solutions may arise.¹⁰⁰³ $\sqrt{x+3} = x-3$, squared, becomes $x+3 = (x-3)^2$. One of the candidates will fail when substituted back, because squaring is not a reversible step on the whole real line in the same way adding 3 is. A rational equation $(x+1)/(x-2) = 3$ produces a candidate that must be checked against the excluded value $x=2$. Solving is still asking which numbers, substituted for the letter, make the two sides name the same quantity. The operational equal sign, which wanted a number on the right, still has nowhere to go. Checking is not a courtesy. Checking is the definition. A child who can run the algebra and will not substitute back has a dance.

Polynomial long division and the remainder theorem still need fraction fluency. Restate the gate, because this is the year a signed-fraction coefficient inside a cubic will turn the page into soup. $(2/3)x^3 - (1/2)x + 1/4$, divided by $x - 1/2$, is not a new algebra topic. It is Chapter 5's fractions as numbers, Chapter 6's signed rationals, and Chapter 7's structure in expressions, unfinished, arriving as symbol soup. The Panel's 2007 survey of 743 Algebra I teachers named rational numbers as the top incoming gap.¹⁰⁰⁴ There is no 2026 Algebra II replication in the sources this project opened. Do not write "teachers today say." Do write: the arithmetic did not retire because the cover said Algebra II. If those operations are still fragile, the Algebra II book is early. Diagnose. Then Chapter 5 and 6's bands. Then this course.

Function fluency from Algebra I — linear, exponential, quadratic — is the other gate. A polynomial function is a new family built from the same function concept. $p(x) = x^3 - x$ assigns to each input exactly one output. Domain, for a polynomial with real coefficients, is all reals. A rational function $r(x) = 1/(x-2)$ has a domain that excludes 2. End behavior, zeros, holes, vertical asymptotes — those are features of graphs the student already learned to read in F-IF, now on new families. A child who can compute $p(3)$ and cannot say what p is has a chant. A child who can sketch $y = 1/x$ from memory and cannot say why $x = 0$ is excluded has a picture and not a domain. Iterate. A short conceptual pass: each family is a function; the differences are algebraic structure and graphical features. Then a handful of graphs. Then a return to "each input exactly one output." Then more.

The binomial theorem and induction sit in the Panel's algebra-of-polynomials row. Appendix A places the binomial theorem as A.APR.5, a (+) standard: know and apply it for expansion of $(x+y)^n$ in powers of x and y for a positive integer n .¹⁰⁰⁵ A kitchen that never writes $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$ from Pascal's triangle, or from choosing k of the y 's, has omitted a named topic. It has not therefore failed the whole course. Mark it. Mathematical induction as a proof method is thinner in Appendix A's Algebra II cluster list than in the Panel's catalog; do not invent an Appendix A induction standard this project did not extract. Name induction as a Panel topic. Teach a first inductive argument on a sum formula if the book has one. Do not pretend a kitchen will finish a foundations-of-mathematics course.

¹⁰⁰³CCSSM A.REI.2, Appendix A Algebra II: solve simple rational and radical equations in one variable, and give examples showing how extraneous solutions may arise.

¹⁰⁰⁴NMAP, *Foundations for Success* (2008), nationally representative survey of Algebra I teachers, 743 teachers, 2007: poorest incoming preparation in "rational numbers and operations involving fractions and decimals," alongside word problems and study habits. 2008 data, not a 2026 Algebra II census. Do not write "teachers today say."

¹⁰⁰⁵CCSSM A.APR.5 (+): know and apply the binomial theorem for the expansion of $(x+y)^n$ in powers of x and y for a positive integer n . Panel catalog includes it; Appendix A marks it (+). A kitchen that omits it has a hole, not an automatic course failure.

Series — arithmetic and finite geometric — sit in the Panel’s symbols-and-expressions row, and Appendix A already placed arithmetic and geometric sequences in Algebra I’s F.BF.¹⁰⁰⁶ Finite geometric series, the sum $1 + r + r^2 + \dots + r^{n-1} = (1 - r^n)/(1 - r)$ for $r \neq 1$, is the compression this course can finish. Infinite series wait for Calculus BC Unit 10. Do not dump Chapter 10 here.

Teaching move, planted for section 9.6: worked-example fading for the first pass through a new symbol-heavy procedure. Sweller and Cooper 1985, algebra novices.¹⁰⁰⁷ Renkl and Atkinson, backward fading.¹⁰⁰⁸ Booth, Lange, Koedinger, and Newton 2013, incorrect examples especially useful for conceptual understanding.¹⁰⁰⁹ Those papers are Algebra I and earlier. This chapter will not invent an Algebra II *d*. It will steal the move. First pass through polynomial long division: one fully worked division, then a second the child must explain, then a completion problem missing the last line, then a conventional problem. Incorrect examples of dropped negative signs — Vlassis 2004 as cited by Booth — and of “cancel everything that looks alike” in a rational expression. The child names the error. Then an isomorphic item. Then the child generates. Do not mash Kapur 2014 with NMAP Finding 27.¹⁰¹⁰ A child already lost inside long division needs a worked example, not a week of unguided discovery.

A *hypothetical*, labelled as such, can put the remainder theorem in a room without inventing a reported family. A parent whose sixteen-year-old divided $x^3 - 1$ by $x - 2$, wrote remainder 0 because “it came out even, I think,” and then, asked for $p(2)$, substituted and got 7, is watching a procedure and a substitution that have not yet met.¹⁰¹¹ The empirical pieces that composite combines are in the documents: the remainder theorem is A.APR.2; Booth’s incorrect-example move names the gap; fraction fluency still sits under the arithmetic. The sitting, the age, and the two marks are hypothetical. The teaching move is not. Write $p(2)$ next to the remainder line. Ask why those two

¹⁰⁰⁶NMAP Table 1, symbols and expressions: arithmetic and finite geometric series. Appendix A placed arithmetic and geometric sequences in Algebra I F.BF. Finite geometric sum is the Algebra II compression; infinite series wait for AP Calculus BC Unit 10. Do not dump Chapter 10.

¹⁰⁰⁷John Sweller and Graham A. Cooper, “The Use of Worked Examples as a Substitute for Problem Solving in Learning Algebra,” *Cognition and Instruction* 2, no. 1 (1985): 59–89. For novices, studying a fully worked solution produces more learning than unguided problem solving (cognitive-load account). Algebra I and earlier; steal the move, do not invent an Algebra II *d*.

¹⁰⁰⁸Alexander Renkl and Robert K. Atkinson, guidance fading / backward fading (Renkl and Atkinson 2003; Renkl, Atkinson, Maier, and Staley 2002; Atkinson, Derry, Renkl, and Wortham 2000): complete example, then completion problems, fade from the end first, then conventional problems. Chapter 2 locked the method.

¹⁰⁰⁹Julie L. Booth, Karin E. Lange, Kenneth R. Koedinger, and Kristie J. Newton, “Using Example Problems to Improve Student Learning in Algebra: Differentiating between Correct and Incorrect Examples,” *Learning and Instruction* 25 (2013): 24–34. Explaining incorrect examples, alone or mixed with correct ones, especially useful for conceptual understanding of two-step linear equations, combined with guided practice in the Algebra I Cognitive Tutor. Misconceptions the examples were built to hit include minus signs as “subtract” rather than as modifiers of terms (Vlassis 2004). Algebra I Cognitive Tutor, not an Algebra II RCT.

¹⁰¹⁰NMAP Finding 27: explicit instruction (clear models, extensive practice, think-alouds, extensive feedback) has consistently positive effects for students with mathematical difficulties and for the lowest third of a typical class — and this “does not mean that all of a student’s mathematics instruction should be delivered in an explicit fashion.” Manu Kapur, “Productive Failure in Learning Math,” *Cognitive Science* (2014): generation-then-instruction on a new concept when priors exist; greater conceptual understanding and transfer, comparable procedures. Different comparisons. Do not mash. Chapter 2 locked the refusal.

¹⁰¹¹Composite *hypothetical*, labelled in the sentence that introduces it. Empirical pieces: A.APR.2 remainder theorem; Booth incorrect-example move; fraction fluency still under the arithmetic. Sitting, age, and the two marks are hypothetical.

numbers must be the same. Then an isomorphic cubic. Then the child generates.

What this section has done is name complex numbers as a system, the remainder theorem as a value of a function, rationals as quotients rather than decorations, and the fraction gate as still open. Inverse, exponential, and logarithmic functions come next, because they are the functions calculus will differentiate, and because a logarithm is not a button.

9.3 Inverse, exponential, and logarithmic functions

A logarithm is not a button. An inverse is not the inv key. Those two sentences are the first teaching moves of this section, and they are the sentences the parent who stopped at algebra has to be able to hear wrong. F.LE.4: understand the inverse relationship between exponents and logarithms, and use it to solve problems involving logarithms and exponents; for exponential models, express as a logarithm the solution to $ab^{ct} = d$ where a , c , and d are numbers and the base b is 2, 10, or e ; evaluate the logarithm using technology.¹⁰¹² That is Appendix A's Algebra II home for logarithms as solutions of exponentials. The button is technology for evaluating. The mathematics is the inverse relationship. A child who can press LOG on 100 and read 2, and cannot say that $\log_{10} 100 = 2$ because $10^2 = 100$, has a button. A child who can recast $2^x = 8$ as $x = 3$, and cannot recast $2^x = 10$ as $x = \log_2 10$, has a special case and not the inverse.

Inverse functions first, because they are the structure logarithms sit on, and because they are the structure calculus will differentiate. A function f assigns to each input exactly one output. An inverse function f^{-1} , when it exists, assigns to each output of f the unique input that produced it. $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$, on the appropriate domains. The superscript -1 is not an exponent. f^{-1} is not $1/f$. That confusion is the one this year's fading sequence has to name, because the same superscript on a number *means* a reciprocal, and the same superscript on a function *names* an inverse. Notation is not decoration. A child who writes $\sin^{-1}(x)$ as $1/\sin(x)$ has read the superscript as arithmetic. A child who writes $1/f(x)$ as $f^{-1}(x)$ has done the same in the other direction. Hear it.

Not every function has an inverse that is a function. $f(x) = x^2$, from reals to reals, sends both 2 and -2 to 4. Restrict the domain to $x \geq 0$, and the inverse is the principal square root. Horizontal-line test: if a horizontal line hits the graph more than once, the function is not one-to-one, and no inverse function exists on that domain. The parent who stopped at algebra often remembers “swap x and y and solve.” That is a procedure for finding a formula, when a formula exists. The concept is the correspondence run backwards, when the correspondence is one-to-one. Graph of f^{-1} is the graph of f reflected across $y = x$. Three representations in one sitting: table, graph, formula. $f(2) = 5$, so $f^{-1}(5) = 2$. The table of the inverse is the table of f with the columns swapped. Do not introduce inverses as a button. Do not skip the table.

Exponential functions, already in Algebra I as a family compared with linear and quadratic, now as the functions whose inverses are logarithms. $f(x) = b^x$, $b > 0$, $b \neq 1$. Domain all reals. Range positive reals. $f(0) = 1$. Passing through $(0, 1)$ is a feature, not a coincidence: $b^0 = 1$. Growth

¹⁰¹²CCSSM F.LE.4, Appendix A Algebra II: inverse relationship between exponents and logarithms; for exponential models, express as a logarithm the solution to $ab^{ct} = d$ where a , c , and d are numbers and the base b is 2, 10, or e ; evaluate the logarithm using technology. The button is technology for evaluating. The mathematics is the inverse relationship.

when $b > 1$; decay when $0 < b < 1$. The parent who hears “exponential means it grows fast” has a slogan that fails for $f(x) = (1/2)^x$. Interleave. Graph this: is it polynomial, rational, exponential? A cubic goes to $\pm\infty$ on one side and the other depending on degree and leading coefficient. An exponential b^x with $b > 1$ goes to 0 on the left and to ∞ on the right. A rational with a horizontal asymptote goes to a constant. Those are different end behaviors. Rohrer, Dedrick, and Burgess, 2014, is a grade-7 classroom crossover on interleaved versus blocked practice — interleaved 72 percent versus blocked 38 percent, $d = 1.05$, $n = 140$, unannounced test two weeks later — and Rohrer, Dedrick, Hartwig, and Cheung, 2020, is a preregistered cluster RCT, 54 seventh-grade classes, interleaved 61 percent versus blocked 38 percent, $d = 0.83$, unannounced test one month later.¹⁰¹³ Those are grade 7. This chapter will not export the d to Algebra II. Label the move as transfer: after a short blocked introduction of a new family, mix yesterday’s polynomial graphs with last month’s exponentials and this week’s rationals. The clean classroom evidence is middle school. The kitchen rule is the same rule Chapter 2 locked. Do not mash Rohrer 2014 with Rohrer 2020 with Dunlosky “high utility” for spacing. Interleaving classroom d s are joint with spacing. Almost all clean classroom RCTs are grade 7.

Logarithmic functions are the inverses of exponential functions. $y = \log_b x$ means $b^y = x$. Domain positive reals. Range all reals. The graph of $y = \log_b x$ is the graph of $y = b^x$ reflected across $y = x$. $\log_b 1 = 0$, because $b^0 = 1$. $\log_b b = 1$, because $b^1 = b$. Change of base: $\log_b x = \log_k x / \log_k b$ for any valid k . The two bases a parent must be able to tell apart, as a parent-first object planted for section 9.7, are $\log_{10}$ — common log, the LOG button on many calculators, the base of scientific notation and of the pH scale as a context, not as a substitute for the definition — and $\ln$, $\log_e$, natural log, the base that calculus will prefer because the derivative of e^x is e^x .¹⁰¹⁴ Why they differ is not “one is for science and one is for calculus” as a slogan. They differ because they are logarithms in different bases: $\log_{10} 100 = 2$, $\ln 100 = \ln(10^2) = 2 \ln 10$, and $\ln 10$ is not 1. A child who treats $\ln$ and $\log$ as synonyms, or who writes $\ln(ab) = \ln a \cdot \ln b$, or who writes $\log(a + b) = \log a + \log b$, has smashed a function that turns products into sums with a function that does not. The product rule: $\log_b(xy) = \log_b x + \log_b y$, same base. The quotient rule: $\log_b(x/y) = \log_b x - \log_b y$. The power rule: $\log_b(x^p) = p \log_b x$. Those three are compressions of exponent rules, because a logarithm is an exponent. Derive them from $b^{\log_b x} = x$. Do not present them as a list to be laminated.

Solving $ab^{ct} = d$ as a logarithm is F.LE.4’s modeling move. $3 \cdot 2^{0.5t} = 24$ becomes $2^{0.5t} = 8$, so $0.5t = \log_2 8 = 3$, so $t = 6$, when the numbers cooperate; or $0.5t = \ln 8 / \ln 2$ when they do not. Compound interest, population, radioactive decay — Appendix A wants modeling, more of it than many older Algebra II courses carried. NMAP Finding 26 still applies: real-world contexts help similar real-world items; they do not, in the Panel’s small high-quality set, improve computa-

¹⁰¹³Doug Rohrer, Robert F. Dedrick, and Kaleena Burgess, “The Benefit of Interleaved Mathematics Practice Is Not Limited to Superficially Similar Kinds of Problems,” *Psychonomic Bulletin and Review* 21 (2014): 1323–1330. Grade-7 classroom crossover, $n = 140$, unannounced test two weeks later, interleaved 72% vs blocked 38%, $d = 1.05$, 95% CI [0.80, 1.30]. Doug Rohrer, Robert F. Dedrick, Kelli Taylor Hartwig, and Christina Cheung, 2020: preregistered cluster RCT, 54 seventh-grade classes, four months of assignments, unannounced test one month later, interleaved 61% vs blocked 38%, $d = 0.83$. Authors: interleaving also spaces same-kind problems; do not treat the d as “pure interleaving.” Almost all clean classroom RCTs are grade 7. Label Algebra II use as transfer. Do not mash 1.05 with 0.83 with Dunlosky “high utility” for spacing.

¹⁰¹⁴Parent-first object for §9.7: why $\ln$ ($\log_e$) and $\log_{10}$ differ. Different bases; calculus prefers e because the derivative of e^x is e^x . Not a slogan that “one is for science and one is for calculus.”

tion, simple word problems, or equation solving.¹⁰¹⁵ A compound-interest problem is a legitimate exponential model. It is not a substitute for the inverse relationship. Kitchen-math transfer is hard. Do not claim the family savings account produces $\ln$.

A common calculus-door gap, named as **inference from the maps, not a survey**: no fluency with exponential and log.¹⁰¹⁶ CLEP Calculus assumes algebra, geometry, trigonometry, and analytic geometry. AP Calculus, in the official CED prerequisite paragraph now extracted, requires the equivalent of four years of college-bound secondary mathematics — algebra, geometry, trigonometry, analytic geometry, and elementary functions, including linear, polynomial, rational, exponential, logarithmic, trigonometric, inverse trigonometric, and piecewise-defined — and familiarity with the properties, composition, algebra, and graphs of functions.¹⁰¹⁷ What the map still requires is the same: if this course never treats a logarithm as the solution of an exponential, Chapter 10's derivatives of e^x and $\ln x$ have nothing to attach to. Do not omit logarithms and still call it Algebra II in the Panel's sense. The functions list in Table 1 runs through logarithmic and trigonometric. A year of leftover quadratics plus a little log, where “a little log” means the LOG button, has not covered the catalog.

The function concept from 8.F and Algebra I must be solid. Restate the diagnostic, because this is the year a shaky correspondence becomes four families the child cannot tell apart. Can the student, given a graph, a table, a formula, and a verbal description, say whether they name the same function? Can they find an inverse from a table by swapping columns, and say when that swap fails to be a function? Can they distinguish $f(x + 1)$ from $f(x) + 1$, still, now on $f(x) = 2^x$? $f(x + 1) = 2^{x+1} = 2 \cdot 2^x$, a vertical stretch that happens to look like a horizontal shift for this family because of the exponent rules — a feature, not a contradiction, and a trap. Hear the trap. Diagnose. If the correspondence is still “each input one output” as a chant with no move among representations, this course is early. Chapter 6 planted 8.F. Chapter 7 formalized F-IF. This chapter will not rewrite them. It will use them. Plant for Chapter 10: calculus is a new operation on functions they already know. A student who cannot say what f is will experience f' as symbol soup.

Interleave function types that share a surface. Graph this: is it polynomial, rational, exponential, logarithmic? Compare methods for solving $b^x = d$: guess-and-check with integer powers; rewrite as a logarithm; change of base and a calculator. After the child can execute one method, put a second next to it. Do not run twenty identical log-law items in a block and then twenty identical exponential-solve items in a block and call the week “functions.” That is blocked practice, which teaches execution. Choosing a strategy is the interleaved job. Label the d as grade 7. Steal the

¹⁰¹⁵NMAP Finding 26: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving. Kitchen-math transfer is hard. A compound-interest problem is a legitimate exponential model, not a substitute for the inverse relationship.

¹⁰¹⁶01-scope calculus-door gaps, labelled **inference from the maps, not a survey**: no fluency with exponential/log.

¹⁰¹⁷CLEP Calculus assumes algebra, geometry, trigonometry, and analytic geometry (College Board CLEP Calculus page, accessed 30 August 2026). AP Calculus AB and BC CED, front-matter Prerequisites, extracted 30 August 2026: four years of college-bound secondary mathematics; algebra, geometry, trigonometry, analytic geometry, and elementary functions including inverse trigonometric and piecewise-defined; properties, composition, algebra, and graphs of functions; language of functions; unit-circle sine and cosine at 0, $\pi/6$, $\pi/4$, $\pi/3$, $\pi/2$ and multiples. BC additional: basic familiarity with sequences and series and some exposure to parametric and polar equations. Source: <https://apcentral.collegeboard.org/media/pdf/ap-calculus-ab-and-bc-course-and-exam-description.pdf>. Unit-guide topic rows remain **UNVERIFIED**.

assignment shape.

What this section has done is name inverses as correspondences run backwards, logarithms as exponents, and $\ln$ versus $\log_{10}$ as a parent-first object. Unit-circle trigonometry comes next, because SOH-CAH-TOA only is a gap at the calculus door, and because a parent who learned right-triangle ratios in geometry will try to skip this section as already done.

9.4 Unit-circle trigonometry

Right-triangle trigonometry is the gate, not the destination. Chapter 8 already taught trigonometric ratios for acute angles — sine, cosine, tangent as ratios of sides in a right triangle, SOH-CAH-TOA as a mnemonic for that special case, the Pythagorean theorem via similarity, and, as (+) standards, the Laws of Sines and Cosines.¹⁰¹⁸ That work is load-bearing. Proportional reasoning from Grade 7 is load-bearing for it. Skipping similarity because “trigonometry is Algebra II” was the error Chapter 8 refused. This section refuses the complementary error: treating SOH-CAH-TOA as the whole of trigonometry, and skipping the unit circle because “we did trig in geometry.” Calculus differentiates sine and cosine as functions of a real variable, in radians, defined for all reals, periodic. A ratio of two sides of an acute triangle is not that object. It is the seed of that object. F.TF.1–2, F.TF.5, and F.TF.8 are Algebra II. The rest of F.TF sits in the fourth-course remainder. Do not skip this because “trig is precalculus.”

F.TF.1: understand radian measure of an angle as the length of the arc on the unit circle subtended by the angle.¹⁰¹⁹ A unit circle has radius 1. An angle of one radian subtends an arc of length 1. A full turn is 2π radians, because the circumference is 2π . 180 degrees is π radians. 90 degrees is $\pi/2$. 30 degrees is $\pi/6$. 45 is $\pi/4$. 60 is $\pi/3$. Those conversions are fluent facts this year, analogous to multiplication facts in Grade 3: they free working memory for the functions. They are not the concept. The concept is that the real number θ is the directed arc length, and that sine and cosine of θ are the coordinates of the point you land on after walking that far from (1, 0) along the circle. Degrees are a different unit. Calculators have a mode. A child who computes $\sin 90$ in degree mode and gets 1, then computes $\sin 90$ in radian mode and gets a number near 0.89, and cannot say why, has a mode error and not a concept error — unless they also cannot say what 90 radians is. Hear both.

F.TF.2: explain how the unit circle in the coordinate plane enables the extension of trigonometric functions to all real numbers, interpreted as radian measures of angles traversed counterclockwise from the positive x -axis.¹⁰²⁰ $(\cos \theta, \sin \theta)$ is the point. Cosine is the x -coordinate. Sine is the y -coordinate. Tangent is sine over cosine, when cosine is not zero. That is the definition this course needs. SOH-CAH-TOA is the first-quadrant, acute-angle special case, recovered when the terminal ray cuts the unit circle in the first quadrant and you drop a perpendicular: the opposite side is $\sin$

¹⁰¹⁸Appendix A Geometry critical area 2: trigonometric ratios for acute angles; Pythagorean theorem via similarity; Laws of Sines and Cosines as (+) standards. Chapter 8. Right-triangle trigonometry is the gate to this section, not the destination.

¹⁰¹⁹CCSSM F.TF.1, Appendix A Algebra II: understand radian measure of an angle as the length of the arc on the unit circle subtended by the angle.

¹⁰²⁰CCSSM F.TF.2, Appendix A Algebra II: explain how the unit circle in the coordinate plane enables the extension of trigonometric functions to all real numbers, interpreted as radian measures of angles traversed counterclockwise from the positive x -axis. $(\cos \theta, \sin \theta)$ is the point.

θ , the adjacent is $\cos \theta$, the hypotenuse is 1. The mnemonic did not become false. It became a special case. A child who can chant SOH-CAH-TOA and cannot place $(\cos 2\pi/3, \sin 2\pi/3)$ in the second quadrant as $(-1/2, \sqrt{3}/2)$ has the special case and not the function. A child who can recast a reference angle and a sign chart as “ASTC” or “all students take calculus” and cannot say that cosine is negative in the second quadrant because the x -coordinate is negative has a chant.

The parent-first object, planted for section 9.7, is why the unit circle is the right home for trigonometry. Two reasons, neither of them a kitchen scene. First: a function of a real variable needs a real input. Arc length on a circle of radius 1 gives every real number a geometric meaning, including numbers larger than 2π and negative numbers, clockwise. Sine and cosine become functions $\mathbb{R} \rightarrow \mathbb{R}$, periodic with period 2π . Second: the coordinates are the values. There is nothing else to remember, once the point is placed, except the special-angle coordinates the child should retrieve: $0, \pi/6, \pi/4, \pi/3, \pi/2$, and their cousins in the other quadrants. A language model can generate a description of a GeoGebra figure — unit circle, ray at θ , point $(\cos \theta, \sin \theta)$, perpendicular to the x -axis. Hermes Agent on the parent machine can build that description. The child draws, or opens GeoGebra under the parent’s eye. Do not put the agent on the child’s phone.¹⁰²¹ The mathematics happens in the coordinates, not in the chat.

F.TF.8: prove the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$ and use it to find $\sin \theta$, $\cos \theta$, or $\tan \theta$, given $\sin \theta$, $\cos \theta$, or $\tan \theta$, and the quadrant.¹⁰²² The identity is the equation of the unit circle, $x^2 + y^2 = 1$, with $x = \cos \theta$ and $y = \sin \theta$. It is not a new fact. It is the circle. A child who writes $\sin^2\theta + \cos^2\theta = \sin(2\theta)$, or who writes $\sin^2\theta + \cos^2\theta = 1$ as $\sin \theta + \cos \theta = 1$, has smashed a square of a function with a function of a double, or has dropped the squares. Those are this year’s incorrect examples. Name them. Then an isomorphic item: given $\sin \theta = 3/5$ and θ in quadrant II, find $\cos \theta$. Cosine is negative in II; $\cos^2\theta = 1 - 9/25 = 16/25$; $\cos \theta = -4/5$. Fraction fluency, still. The 3-4-5 triangle from geometry is sitting inside the unit circle, scaled. Similarity did not retire.

F.TF.5: choose trigonometric functions to model periodic phenomena with specified amplitude, frequency, and midline.¹⁰²³ $y = A \sin(B(x - C)) + D$, or cosine. Amplitude $|A|$. Period $2\pi/|B|$. Phase shift C . Midline $y = D$. Ferris wheels, tides, temperature over a year — Appendix A wants modeling. Finding 26 still applies. A tide table is a legitimate periodic model. It is not a substitute for the unit-circle definition. Do not claim the family beach trip produces radians.

A common calculus-door gap, named as **inference from the maps, not a survey**: trigonometry of the unit circle never learned — only right-triangle SOH-CAH-TOA.¹⁰²⁴ Derivatives of sine and cosine are limits of difference quotients of functions defined on the reals. A student who only knows an acute ratio has no function to differentiate when the input is $2\pi/3$, or 7, or $-\pi$. Diagnose at this door. Can the student place $(\cos \theta, \sin \theta)$ for the special angles in all four quadrants, in radians,

¹⁰²¹Chapter 3 / 04-ai: Hermes Agent on the parent machine; do not put the agent on the child’s phone. Browser only on sites the parent names (curriculum publisher, Khan, Wolfram, GeoGebra). Not a child-safe school SKU. GeoGebra-able figure description is materials generation, Chapter 3 workflow (e).

¹⁰²²CCSSM F.TF.8, Appendix A Algebra II: prove the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$ and use it to find $\sin \theta$, $\cos \theta$, or $\tan \theta$ given one of them and the quadrant. The identity is the equation of the unit circle.

¹⁰²³CCSSM F.TF.5, Appendix A Algebra II: choose trigonometric functions to model periodic phenomena with specified amplitude, frequency, and midline. Finding 26 still applies (n. 1015).

¹⁰²⁴01-scope calculus-door gaps, labelled **inference from the maps, not a survey**: trigonometry of the unit circle never learned — only right-triangle SOH-CAH-TOA.

without a calculator? Can they use $\sin^2\theta + \cos^2\theta = 1$ to find a missing value given a quadrant? Can they sketch one period of sine and of cosine from the circle, not from a memorized wavy picture? If not, the precalculus book is early, or this course is unfinished. Do not wait for Chapter 10 to discover the gap.

What sits in the fourth-course remainder, so a parent shopping for “precalculus” knows what they are buying: F.TF.3, special angles such as $\pi/3$, $\pi/4$, $\pi/6$ — which this section already treated as Algebra II fluency, and which Appendix A also lists as (+) in some pathway charts; F.TF.4, even and odd, periodicity, using those to fill a unit circle from a first-quadrant scrap; F.TF.6, trigonometric functions restricted to a domain on which they are always increasing or always decreasing, inverse trigonometric functions; F.TF.7, inverse trig to solve equations, technology; F.TF.9, addition and subtraction formulas, (+).¹⁰²⁵ Inverse trigonometric functions are the inverse-function idea of section 9.3, now on sine, cosine, and tangent with restricted domains. Arcsin takes $[-1, 1]$ to $[-\pi/2, \pi/2]$. A child who writes $\sin^{-1}$ as $1/\sin$ has the superscript error already named. Vectors and matrices, N.VM; remaining conics, ellipse and hyperbola; polar coordinates and parametrics as (+) — those are further remainder. Polar and parametric calculus lives in AP Calculus BC Unit 9. Do not dump that unit here. Do plant: a student who never leaves rectangular coordinates will meet polar as a new language in BC, or will not sit BC.

Laws of Sines and Cosines were Geometry (+) standards. A kitchen that did them in the geometry year has them. A kitchen that did not can do them now, as applications of right-triangle and general-triangle trigonometry. They are not unit-circle trigonometry. They do not replace it.

Radian measure already appeared in Chapter 8 as a unit of measure for arc length on a circle, not necessarily a unit circle.¹⁰²⁶ This course puts that unit on the unit circle and names the coordinates as cosine and sine. The informal-limit arguments for circumference and area in Geometry are the on-ramp to calculus’s precise ones. The unit circle is the on-ramp to the derivatives of the trigonometric functions. Skip either, and Chapter 10 arrives as soup.

Interleave. After a short blocked pass on placing special angles, mix: one inverse-function item from 9.3, one remainder-theorem item from 9.2, one “which function family is this graph,” one unit-circle coordinate, one Pythagorean-identity recovery. That mixed page is the Rohrer assignment shape, in a kitchen, labelled as a practice suggestion transferred from grade 7, not as $d = 1.05$ sitting on the table.¹⁰²⁷ Rohrer’s teachers implemented without training in 2020. The parent can implement without a product.

What this section has done is extend acute ratios to functions of a real variable on the unit circle, name the Pythagorean identity as the circle, and refuse the skip. Algebra II still has statistics clusters, and the Panel still has a finished-algebra checklist. Those come next, because a kitchen that treats stats as “the AP year” will skip S.ID.4 and S.IC, and because a kitchen that treats a quadratic-review year as Algebra II needs a sheet that says no.

¹⁰²⁵Appendix A Fourth Courses: F.TF.3, 4, 6, 7, 9; N.VM; remaining conics; polar/parametrics as (+). Inverse trigonometric functions are the inverse-function idea of §9.3 on restricted domains. Polar/parametric calculus is AP Calculus BC Unit 9 — plant for Chapter 10; do not dump it.

¹⁰²⁶Appendix A Geometry: arc length and radian as a unit of measure (circles critical area). This course puts that unit on the unit circle and names the coordinates as cosine and sine.

¹⁰²⁷Rohrer assignment shape (n. 1013), labelled as a practice suggestion transferred from grade 7, not as $d = 1.05$ sitting on the table. Rohrer 2020: teachers implemented without training.

9.5 Algebra II stats clusters and the “have we finished?” checklist

Algebra II still has statistics clusters. They are not AP Statistics. They are not a reason to skip Chapter 11. They are S.ID.4 and S.IC, and a kitchen that waits for “the AP year” will skip them, just as a kitchen that skips Grade 6 distributions, Grade 7 sampling, Algebra I’s S.ID, and Geometry’s S.CP will spend AP Statistics Unit 1 retrofitting.¹⁰²⁸ This section plants Chapter 11. It does not write it.

S.ID.4: use the mean and standard deviation of a data set to fit it to a normal distribution and to estimate population percentages; recognize that there are data sets for which such a procedure is not appropriate; use calculators, spreadsheets, and tables to estimate areas under the normal curve.¹⁰²⁹ That is the Algebra II home for the normal. Mean and standard deviation as parameters of a bell-shaped model. The empirical rule as a compression: about 68 percent within one standard deviation of the mean, about 95 percent within two, about 99.7 percent within three, *when the normal is a reasonable fit*. The last clause is the one the parent has to hear. Not every data set is normal. A bimodal distribution, a heavy skew, a ceiling effect — the procedure is not appropriate. Algebra I already taught shape, center, spread, and “correlation is not causation.” This cluster adds a specific model and a caution about using it. It does not add inference from a random sample to a population, not yet. That is S.IC.

S.IC: make inferences and justify conclusions from sample surveys, experiments, and observational studies.¹⁰³⁰ The cluster includes: understand statistics as a process for making inferences about population parameters based on a random sample; decide if a specified model is consistent with results from a given data-generating process, via simulation; recognize the purposes of and differences among sample surveys, experiments, and observational studies, and explain how randomization relates to each; use data from a sample survey to estimate a population mean or proportion, and develop a margin of error through simulation; use data from a randomized experiment to compare two treatments, and use simulations to decide if differences between parameters are significant; evaluate reports based on data. That is a lot of statistics. It is still not AP Statistics. College Board’s AP Statistics, effective Fall 2026, is a one-semester-college-equivalent, **non-calculus-based** introductory statistics course whose only published prerequisite is successful completion of a first-year algebra course.¹⁰³¹ Five units, first revised exam May 2027: exploring one-variable data and col-

¹⁰²⁸01-scope Statistics band and brief Chapter 11: CCSSM 6.SP, 7.SP, 8.SP; Algebra I S.ID.1–3, 5–9; Geometry S.CP; Algebra II S.ID.4 and S.IC. A skipped K–8 / Algebra I / Geometry / Algebra II data strand is AP Statistics Unit 1 retrofit. Plant Chapter 11. Do not skip S.ID.4 and S.IC because “stats is later.” Do not treat AP Statistics as this course.

¹⁰²⁹CCSSM S.ID.4, Appendix A Algebra II: use the mean and standard deviation of a data set to fit it to a normal distribution and to estimate population percentages; recognize that there are data sets for which such a procedure is not appropriate; use calculators, spreadsheets, and tables to estimate areas under the normal curve.

¹⁰³⁰CCSSM S.IC, Appendix A Algebra II: make inferences and justify conclusions from sample surveys, experiments, and observational studies — including statistics as a process for inferences about population parameters based on a random sample; simulation; purposes of and differences among surveys, experiments, and observational studies; estimate a population mean or proportion from a sample survey with margin of error through simulation; compare two treatments from a randomized experiment; evaluate reports based on data.

¹⁰³¹College Board, *AP Statistics CED*, Effective Fall 2026 (n. 991). Five units, first revised exam May 2027, MC weights: Unit 1 Exploring One-Variable Data and Collecting Data 20–30%; Unit 2 Probability, Random Variables, and Probability Distributions 15–25%; Unit 3 Inference for Categorical Data: Proportions 15–25%; Unit 4 Inference for Quantitative Data: Means 10–20%; Unit 5 Regression Analysis 10–20%. Printed here only to refuse dumping them as Algebra II. Prerequisite: first-year algebra.

lecting data; probability, random variables, and probability distributions; inference for categorical data, proportions; inference for quantitative data, means; regression analysis. This chapter will not dump those unit weights as if they were Algebra II. Algebra II's S.IC is the high-school inference on-ramp: randomization, simulation, margin of error, the survey / experiment / observational-study split. AP Statistics is a different college-equivalent. A student who did S.IC will spend less of Unit 1 retrofitting. A student who skipped S.IC because "we'll do AP Stats later" will retrofit. Plant Chapter 11. Do not skip the plant.

AP Statistics after Algebra I is a legitimate senior elective for a non-STEM path and a legitimate *add-on* for a STEM path. It is a weak stealth substitute for Calculus I if the destination is Physics C or a calculus-based engineering sequence.¹⁰³² College Board's revised course framework encourages students with appropriate background to take **both** AP Statistics and AP Calculus, and notes that students interested in data science are encouraged to take AP Statistics and an AP Computer Science course. It is not published as a substitute for calculus in STEM calculus sequences. Journalism claiming five content areas were "cut" from AP Statistics is labelled journalism and is not used as a primary for what remains. Use the CED. Homeschool-specific AP Statistics participation rates were not in the sources this project opened. That rate stays **UNVERIFIED**. Do not invent one. GAISE II, the American Statistical Association's guidelines, is cited inside the CED as the alignment for statistical practices; the ASA primary was not separately opened this pass. Page-level GAISE II claims stay **UNVERIFIED**; treat as spirit, not as quoted recommendations.¹⁰³³

NCES 2016: probability ever-taught to 24 percent of high-school-level homeschoolers.¹⁰³⁴ That is not a stats-course completion rate. Algebra II ever-taught, 49 percent; calculus, 15 percent with caution. The drop from Algebra II to calculus is this book's high-school problem, restated because the checklist that follows is how a parent knows whether the 49 percent was the catalog or a title. Combinatorics and finite probability sit in the Panel's last Major Topic. Geometry already placed S.CP: independence, conditional probability, two-way tables, addition rule; permutations and combinations as (+). This course continues probability rules and can finish a finite-probability pass the geometry year did not. Do not dump Chapter 11's Unit 2. Do not skip finite probability because "stats is later."

The Panel's Major Topics of School Algebra, restated as a parent checklist for "have we finished Algebra II?" — stealable, not a 180-day plan, independent of course packaging.¹⁰³⁵

Symbols and expressions. Polynomial expressions, including arithmetic on polynomials beyond quadratic. Rational expressions, including rewrite by division. Arithmetic series and finite geometric series. A kitchen that never rewrote a rational and never summed a finite geometric series has

¹⁰³²College Board, AP Statistics Revised Course Framework, <https://apcentral.collegeboard.org/media/pdf/ap-statistics-revised-course-framework.pdf>, partially opened 30 August 2026: encourages both AP Statistics and AP Calculus when background allows; data-science-interested students also encouraged to take AP Computer Science. Not published as a calculus substitute. Journalism claiming five content areas were "cut" is labelled journalism; use the CED.

¹⁰³³American Statistical Association GAISE II: cited inside the AP Statistics CED as the alignment for statistical practices; ASA primary not separately opened this pass. Page-level GAISE II claims **UNVERIFIED**. Homeschool-specific AP Statistics participation rate: **UNVERIFIED**; do not invent one.

¹⁰³⁴NCES PFI 2016: probability ever-taught to 24 percent of high-school-level homeschoolers. Not a stats-course completion rate. Algebra II 49 percent; calculus 15 percent with caution (n. 962).

¹⁰³⁵NMAP Table 1 (n. 963), restated as a stealable parent checklist. Independent of course packaging.

holes. Mark them.

Linear equations and systems. Already Algebra I. If they are still fragile, this course is early, not finished.

Quadratic equations. Factor, complete the square, quadratic formula. Real solutions in Algebra I; complex solutions in this course. A kitchen that never wrote $1 \pm i$ has not finished the quadratic.

Functions. Linear, quadratic, polynomial, simple nonlinear, exponential, logarithmic, trigonometric, and fitting models to data. Move among formula, graph, table, verbal. Inverse functions as correspondences, not buttons. A kitchen that never treated a logarithm as the solution of an exponential, and never placed $(\cos \theta, \sin \theta)$ on the unit circle, has not covered school algebra as the Panel defined it. Do not omit those and still call it Algebra II.

Algebra of polynomials. Roots, complex numbers, the fundamental theorem of algebra as a counting principle, binomial theorem, induction. A kitchen that never wrote $a + bi$ has not finished this row.

Combinatorics and finite probability. Counting, independent events, conditional probability continued from Geometry S.CP. A kitchen that never distinguished a survey from an experiment has not finished S.IC.

That is the sheet. A homeschool Algebra II that omits functions, logarithms, and trigonometry has not covered school algebra as the Panel defined it. Adelman 1999, cited by the Panel for the Algebra II–college-graduation correlation, was not re-opened; do not upgrade the citation into a kitchen experiment.¹⁰³⁶ Named college pages want the title. The title has to earn the sheet. Transcript the work that was done. If the year was leftover quadratics plus a little log, transcript a more honest name, or finish the catalog.

What the finished course unlocks is Calculus: limits of functions, derivatives of the elementary functions — polynomial, rational, exponential, logarithmic, trigonometric — and the chain rule, which is a derivative of a composition, which is a structure this course’s inverse and exponential work should have made visible.¹⁰³⁷ It also unlocks AP Statistics in the weaker sense that Algebra II is useful; the official prerequisite as of Fall 2026 is Algebra I only. It does not unlock Physics C or a calculus-based engineering sequence by itself. Precalculus, as a fourth course, is the usual last high-school course before AB or BC. A non-accelerated student who starts Algebra I in Grade 9 has senior year for one of those, not both a full precalculus and BC unless the year is overloaded. Compacted path to calculus senior year is Appendix A’s model, not a mandate.

A “have we finished?” sitting, stealable. Not a publisher placement test. Can the student divide a cubic by a linear factor and say that the remainder is $p(a)$? Write a complex solution of a quadratic? Find an inverse from a table and say when the swap fails? Solve $ab^{\wedge}ct = d$ as a logarithm, and say why $\ln$ and $\log_{10}$ differ? Place $(\cos 2\pi/3, \sin 2\pi/3)$ without a calculator? Use $\sin^2\theta + \cos^2\theta = 1$ in quadrant II? Sketch one period of sine from the circle? Say whether a normal model is appropriate, and whether a study was a survey, an experiment, or observational? If the answers are yes, the

¹⁰³⁶Adelman 1999 via NMAP only (n. 979). Do not upgrade.

¹⁰³⁷01-scope Algebra II unlocks: Calculus — limits of functions, derivatives of the elementary functions, the chain rule. Chain rule is a derivative of a composition; composition as structure is this course’s inverse and exponential work.

catalog is in reach, and a fourth course or a first calculus course is a conversation, not a leap. If the answers are no, buying a book titled Precalculus will not close the holes. Diagnose. Then this course, finished. Then Chapter 10 or Chapter 11, as the destination requires.

What this section has done is plant Chapter 11, refuse AP Statistics as this course, and print the Panel's catalog as a sheet a parent can steal. Teaching guidance comes next: the gates, the interleaving labelled as transfer, and the release valves when the parent is the ceiling.

9.6 Teaching guidance: gates, interleaving, release valves

The parent is still the design user who stopped at algebra. The job has not changed: understand the concept well enough to hear a wrong answer; let the child do the struggle before the tool; host and constrain any agent. What has changed is the altitude. Inverse functions, log laws, and the unit circle are the year many kitchens discover that the parent is the ceiling. That discovery is not a failure of the kitchen. It is a planning fact. NCES 2016 already counted the valves: 23 percent of homeschoolers had any instruction by a tutor; 31 percent had any from a local homeschool group or co-op.¹⁰³⁸ Video and self-teaching curricula exist because of Maloney. Dual-enrollment College Algebra and Precalculus exist because a community-college transcript is an object NCAA and admissions offices already know how to read. This section names the gates, steals the teaching moves Chapter 2 already locked, labels the interleaving evidence as grade 7, and then names the valves. It does not promise that a parent who last passed Algebra I in 1998 must explain the unit circle unaided until June.

Function fluency from Algebra I and Geometry's right-triangle trigonometry are the gate. Polynomial long division still needs fraction fluency. Three diagnostics, stealable, content not birthday.

First: fractions, including signed, still. Can the child operate on $(2/3)x^3 - (1/2)x + 1/4$? Place $-3/4$ and $2/3$ on the same line? The Panel named fractions, including decimals, percents, and negative fractions, as "the most important foundational skill not currently developed among American students."¹⁰³⁹ Siegler, Duncan, Davis-Kean, Duckworth, Claessens, Engel, Susperreguy, and Chen, 2012: elementary fraction and division knowledge uniquely predicted high-school algebra five or six years later, after controls — a 1-SD increase in age-10 fractions knowledge uniquely associated with a 0.15-SD (United Kingdom) / 0.17-SD (United States) increase in high-school algebra.¹⁰⁴⁰ Prediction, not a curriculum RCT. Do not upgrade it. Do not skip the gate because an Algebra II book is on the shelf. The 743-teacher Algebra I survey's top content gap was rational numbers.

¹⁰³⁸NCES PFI 2016 (NCES 2020-001): 23 percent of homeschoolers had any instruction by a tutor; 31 percent had any from a local homeschool group or co-op. 03-homeschool §5.2: these are the release valves for Algebra II–calculus.

¹⁰³⁹NMAP, *Foundations for Success* (2008), Executive Summary findings 4 and 12, and Chapter 4: "knowledge of fractions is the most important foundational skill not currently developed among American students." Critical Foundation includes decimals, percents, and negative fractions. Prefer the Panel's sentence.

¹⁰⁴⁰Robert S. Siegler, Greg J. Duncan, Pamela E. Davis-Kean, Kathryn Duckworth, Amy Claessens, Mimi Engel, Maria Ines Susperreguy, and Meichu Chen, "Early Predictors of High School Mathematics Achievement," *Psychological Science* 23, no. 7 (2012): 691–697, <https://doi.org/10.1177/0956797612440101>; author PDF <https://sieglertc.columbia.edu/wp-content/uploads/2019/02/Siegler-et-al-PsySci12.pdf>, accessed 30 August 2026. British Cohort Study $N = 3,677$; U.S. PSID-CDS $N = 599$. A 1-SD increase in age-10 fractions knowledge uniquely associated with 0.15-SD (UK) / 0.17-SD (US) increase in high-school algebra after other arithmetic, IQ or working memory, and family background. Prediction, not a curriculum RCT.

There is no 2026 Algebra II replication in the sources opened. The arithmetic did not retire.

Second: the function concept from 8.F and Algebra I's F-IF. Can the student move among formula, graph, table, and verbal? Assign to each input exactly one output, and say when a graph fails that test? Distinguish $f(x + 1)$ from $f(x) + 1$, now on exponentials and on sine? Without that concept, calculus's $f'(x)$ has nothing to attach to. Diagnose here. Plant Chapter 10. A student who treats a function as “the equation” will experience this year's six families as six chants.

Third: Geometry's right-triangle trigonometry and similarity. Can the student write a trigonometric ratio for an acute angle as a ratio of corresponding sides, not as a button? Place a reference triangle inside a unit circle and say why cosine is the x-coordinate? If SOH-CAH-TOA is the whole of trigonometry, this course's F.TF.1–2 is not a review. It is the first time sine is a function of a real variable. Do not skip it because Chapter 8 used the mnemonic.

Worked-example fading for the first pass through a new symbol-heavy procedure. Sweller and Cooper 1985, algebra novices. Renkl and Atkinson, backward fading. Booth, Lange, Koedinger, and Newton 2013, incorrect examples especially useful for conceptual understanding of two-step linear equations in the Algebra I Cognitive Tutor.¹⁰⁴¹ Steal the sequence, labelled as transfer, not as an Algebra II trial. First pass through a log law, a remainder-theorem division, a unit-circle evaluation, an inverse-function swap: one fully worked example the parent already understands; a second the child must explain; a completion problem missing the last step; a conventional problem. Incorrect examples: dropped negative signs (Vlassis 2004 as cited by Booth); “cancel everything that looks alike” in a rational; $\log(a + b)$ written as $\log a + \log b$; $\sin^{-1}$ written as $1/\sin$; $\sin^2\theta + \cos^2\theta$ written as $\sin(2\theta)$; $\ln$ and $\log_{10}$ treated as synonyms. The child names the error. Then an isomorphic item. Then the child generates. Self-explanation makes a worked example more than a script to copy; Dunlosky, Rawson, Marsh, Nathan, and Willingham 2013 rate self-explanation as moderate utility, not as high as practice testing and distributed practice.¹⁰⁴² Do not start a brand-new identity with “see if you can figure it out” as the only move if the child is a novice in that identity. Kapur 2014 is a different comparison. NMAP Finding 27 is a different comparison. Do not mash them. A child already lost inside the unit circle needs a worked placement of $(\cos 2\pi/3, \sin 2\pi/3)$, not a week of unguided discovery. A child who has priors — right-triangle trig, the function concept, the circle's equation — can spend twenty minutes trying to extend sine past 90 degrees with what they already know, then be taught. That is productive failure, designed. It is not leaving them stuck.

Interleave function types that share a surface. Graph this: is it polynomial, rational, exponential, logarithmic, trigonometric? Mix yesterday's remainder theorem with last month's inverse and this week's Pythagorean identity. Section 9.3 already printed Rohrer 2014 and Rohrer 2020 as grade-7 classroom trials, $d = 1.05$ and $d = 0.83$, joint with spacing.¹⁰⁴³ Almost all clean classroom RCTs

¹⁰⁴¹Sweller and Cooper 1985 (n. 1007); Renkl and Atkinson fading (n. 1008); Booth et al. 2013 (n. 1009). Steal the sequence, labelled as transfer, not as an Algebra II trial.

¹⁰⁴²John Dunlosky, Katherine A. Rawson, Elizabeth J. Marsh, Mitchell J. Nathan, and Daniel T. Willingham, “Improving Students' Learning with Effective Learning Techniques,” *Psychological Science in the Public Interest* 14, no. 1 (2013). Practice testing and distributed practice rated high utility; interleaved practice moderate (less classroom generality at the time of that review); self-explanation moderate utility. Do not mash with Rohrer classroom d s.

¹⁰⁴³Rohrer 2014 and Rohrer 2020 (n. 1013), restated in §9.6 so the grade-7 label sits next to the kitchen assignment. Do not export the d to Algebra II.

of interleaving in mathematics are grade 7. This chapter will not export the d to Algebra II. Label the move as transfer. After a short blocked introduction of a new skill — enough to execute — mix kinds. Rohrer’s teachers implemented without training in 2020. The parent can implement without a product. Do not mash 1.05 with 0.83 with Dunlosky “high utility” for spacing.

Compare methods. After the child can execute one method, put a second next to it and ask which is better for this problem, and why. Rittle-Johnson and Star 2007 and 2009: comparison improves flexibility; exact d **UNVERIFIED** this pass.¹⁰⁴⁴ The qualitative finding is enough. Factor versus remainder theorem versus graph to find a zero. Change of base versus a same-base rewrite to solve an exponential. Reference angle plus sign versus placing the point on the circle. Two representations in the same visual field, then faded. Carbonneau, Marley, and Selig 2013: manipulatives versus abstract-only, overall mean d approximately 0.37, small-to-moderate, under guidance, then a crutch if not connected to symbols and faded.¹⁰⁴⁵ A unit-circle drawing is a representation, not a toy. Fade when the child can retrieve $(\cos \pi/3, \sin \pi/3)$ without it. Revisit the circle when the retrieval fails.

Conceptual and procedural knowledge iterate. Rittle-Johnson, Schneider, and Star 2015: “It is a myth that it is a one-way street from conceptual knowledge to procedural knowledge.”¹⁰⁴⁶ NRC *Adding It Up*, 2001, p. 116: five strands, “interwoven and interdependent.” NMAP Finding 10: ranking conceptual understanding, computational fluency, and problem solving is “misguided.”¹⁰⁴⁷ Do not wait for full understanding of the unit circle before any special-angle retrieval. Do not drill twenty identities with zero talk about the circle. Iterate. NCTM 2014 “fluency follows understanding” is policy, not an experiment. Finding 23: high-quality research does not support exclusive use of either all-teacher-directed or all-student-centered instruction.

Kitchen-math transfer is hard. Finding 26: real-world contexts help similar real-world items, not computation, simple word problems, or equation solving.¹⁰⁴⁸ A compound-interest problem is a legitimate exponential model. A Ferris-wheel problem is a legitimate periodic model. Neither replaces the inverse relationship or the unit-circle definition. A child who can double a recipe and cannot place $3/2$ on a line has performed, not learned — Soderstrom and Bjork 2015, current

¹⁰⁴⁴Rittle-Johnson and Star 2007, 2009: comparison improves flexibility; exact d **UNVERIFIED** this pass (n. 999).

¹⁰⁴⁵Kira J. Carbonneau, Scott C. Marley, and James P. Selig, “A Meta-Analysis of the Efficacy of Teaching Mathematics with Concrete Manipulatives,” *Journal of Educational Psychology* 105 (2013). 55 studies, $N = 7,237$, kindergarten through college, manipulatives vs abstract symbols only. Overall mean d approximately 0.37 (small-to-moderate) favoring manipulatives, under guidance; a crutch if not connected to symbols and faded. A unit-circle drawing is a representation, not a toy.

¹⁰⁴⁶Bethany Rittle-Johnson, Michael Schneider, and Jon R. Star, “Not a One-Way Street: Bidirectional Relations between Procedural and Conceptual Knowledge of Mathematics,” *Educational Psychology Review* 27, no. 4 (2015): 587–597. “It is a myth that it is a one-way street from conceptual knowledge to procedural knowledge.” Verify wording against the article before typesetting an epigraph; eligible as Chapter 2’s load-bearing sentence, used here as method.

¹⁰⁴⁷National Research Council, Jeremy Kilpatrick, Jane Swafford, and Bradford Findell, eds., *Adding It Up: Helping Children Learn Mathematics* (Washington, DC: National Academy Press, 2001), p. 116: five strands “interwoven and interdependent.” NMAP Finding 10: the curriculum “must simultaneously develop conceptual understanding, computational fluency, and problem-solving skills. Debates regarding the relative importance of these aspects of mathematical knowledge are misguided.” NCTM 2014 “fluency follows understanding” is policy, not an experiment. NMAP Finding 23: high-quality research does not support exclusive use of either all-teacher-directed or all-student-centered instruction.

¹⁰⁴⁸NMAP Finding 26 (n. 1015), restated so kitchen Ferris-wheel and compound-interest problems cannot replace the unit-circle definition or the inverse relationship.

performance “often an unreliable index” of learning.¹⁰⁴⁹ A child who can watch a tide app and cannot place $(\cos \theta, \sin \theta)$ has done the same at a higher altitude.

When the parent is the ceiling, this is the band where tutor, co-op, and dual-enrollment College Algebra or Precalculus become the release valve.¹⁰⁵⁰ Practical notes from named sources, not a directory. Co-op math classes create a teacher of record useful for NCAA worksheets and transcripts. Community-college dual enrollment produces an official transcript NCAA and admissions offices already know how to read. Public-school à la carte access is **state-law dependent**; HSLDA “public school access” pages exist per state and were not opened in full here — **advocacy**, not a statute.¹⁰⁵¹ Umbrella schools can issue transcripts; they do not magically teach the unit circle. Dual enrollment: NCES 2020-125 is **public-school** supply — 82 percent of public schools with any of grades 9–12 offered dual or concurrent enrollment in 2017–18.¹⁰⁵² IPEDS 2022–23, as reported on the IES project page and in Community College Daily journalism: nearly 2.5 million dual-enrolled students.¹⁰⁵³ Homeschool-specific dual-enrollment counts: **not in these sources**. Do not invent a homeschool dual-enrollment rate. IES dual-enrollment outcomes study first report expected 2027. Cost ranges from free, in some state programs, to ordinary community-college tuition. Keep the syllabus. Credit is set by each receiving college. Do not promise it is automatic.

Video curricula exist because of Maloney, Ramirez, Gunderson, Levine, and Beilock, 2015: first- and second-graders of more math-anxious parents learned less math over the school year and had more math anxiety at year’s end — but only if those parents reported frequent homework help. When help was infrequent, parent anxiety was unrelated to child outcomes. The effect was specific to math, not reading.¹⁰⁵⁴ That is a school-homework study, not a parent-as-sole-teacher RCT. A

¹⁰⁴⁹Nicholas C. Soderstrom and Robert A. Bjork, “Learning versus Performance: An Integrative Review,” *Perspectives on Psychological Science* 10, no. 2 (2015): 176–199. Current performance is “often an unreliable index” of learning. Eligible quote; Chapter 3 locked the split. A child who can watch a tide app and cannot place $(\cos \theta, \sin \theta)$ has performed, not learned.

¹⁰⁵⁰03-homeschool §5.2, “When the parent is the ceiling: tutor, co-op, community college.” NCES 2016: 23 percent any tutor; 31 percent any co-op; a small share of 9–12 also in college classes. Co-op math classes create a teacher of record useful for NCAA worksheets. Community-college dual enrollment produces an official transcript NCAA and admissions offices already know how to read. Umbrella schools can issue transcripts; they do not magically teach calculus.

¹⁰⁵¹HSLDA “public school access” pages: **advocacy**, attorney-reviewed for members, not statutes. Public-school à la carte access is state-law dependent. Pages not opened in full this pass.

¹⁰⁵²NCES, *Dual or Concurrent Enrollment in Public Schools in the United States*, NCES 2020-125 (December 2020), <https://nces.ed.gov/pubs2020/2020125.pdf>, accessed 30 August 2026. From the 2017–18 National Teacher and Principal Survey: 82 percent of public schools with any of grades 9–12 offered dual or concurrent enrollment. **Public-school** supply, not a homeschool participation rate.

¹⁰⁵³IES/NCEE project page, Outcomes Associated with Dual Enrollment Programs; Community College Daily / AACC (journalism), “DataPoints: New data on dual enrollment” (August 2024), reporting IPEDS 2022–23: nearly 2.5 million dual-enrolled students. Journalism labelled journalism for the CCD article; the IES page is a federal project description. Homeschool-specific dual-enrollment counts: **not in these sources**. IES dual-enrollment outcomes study first report expected 2027.

¹⁰⁵⁴Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, “Intergenerational Effects of Parents’ Math Anxiety on Children’s Math Achievement and Anxiety,” *Psychological Science* 26, no. 9 (2015): 1480–1488. First- and second-graders of more math-anxious parents learned less math over the school year and had more math anxiety at year’s end — only if those parents reported frequent homework help. When help was infrequent, parent anxiety was unrelated to child outcomes. Math-specific, not reading. School-homework study, not a parent-as-sole-teacher RCT. A homeschool parent cannot “help less often” as a daily strategy. Do not treat as

homeschool parent cannot “help less often” as a daily strategy. Do not treat the finding as “anxious parents should not teach.” Replication status open. The constructive response the literature supports: the parent learns the day’s concept well enough to stay out of the child’s working memory; the child generates; the parent does not narrate worry. When the parent is the anxiety vector, shift the instructional voice to a well-designed curriculum video, a co-op teacher, a tutor, or a community-college instructor — and keep the parent as supervisor who can still hear a wrong inverse, not as the person who “explains it the way I learned it.” Teaching Textbooks, **vendor**, “does 100% of the teaching” and “100% of the grading” — a vendor claim, relevant to the parent who stopped at algebra, not a finding that the parent can stop hearing a wrong inverse.¹⁰⁵⁵ CTCMath, **vendor**. Khan Academy, free map, not a researched curriculum outcome. Topic placements for Teaching Textbooks, Life of Fred, and Math Mammoth that were not opened on vendor primaries remain **UNVERIFIED**. Do not promote those rows to handbook fact.

A *hypothetical*, labelled as such, built from NCES pieces, not a case study: a parent who last passed Algebra I in 1998 hires a weekly tutor for Algebra II, or enrolls the student in College Algebra at the community college, while remaining the homeschool administrator of record.¹⁰⁵⁶ That is a normal high-school plan, not a failure of the kitchen. The empirical pieces are the 23 percent tutor cell, the 31 percent co-op cell, a small share of 9–12 also in college classes, and Maloney as the reason video curricula exist. The year 1998, the weekly cadence, and the sitting are hypothetical. The move is not. Stay the administrator. Do not disappear. Hear the wrong inverse on the unaided exit ticket even if someone else explained the unit circle on Tuesday.

Do not promise that AP self-study yields a seat in May. College Board’s Homeschoolers’ Guide, Jacob Mishook, 29 June 2026: homeschoolers may self-study; a formal AP course is recommended not required; parents **cannot order exams**; find a school via the AP Course Ledger; the AP coordinator enrolls the student in an “exam only” My AP section; mid-November order deadline.¹⁰⁵⁷ Plant for Chapter 12. Do not invent a registration recipe beyond that guide. AP Precalculus, if sat, is one named fourth-course object. Credit is set by each college. NCAA: audited, CLEP, or credit-by-exam courses are **not** NCAA-approved core courses. A kitchen Algebra II taught from a textbook can count if the Core-Course Worksheet is honest. Dual-enrollment College Algebra is a college transcript NCAA already knows how to read.¹⁰⁵⁸ Math core is Algebra I or higher. Do not

“anxious parents should not teach.” Replication status open. Video curricula exist because of this constraint; keep the parent as supervisor who can still hear a wrong inverse.

¹⁰⁵⁵Teaching Textbooks, teachingtextbooks.com, accessed 30 August 2026: “does 100% of the teaching” and “100% of the grading.” **Vendor** claim, not an effectiveness trial. Relevant to the parent who stopped at algebra; not a finding that the parent can stop hearing a wrong inverse. CTCMath **vendor**. Khan Academy: free map, not a researched curriculum outcome.

¹⁰⁵⁶Composite *hypothetical*, labelled in the sentence that introduces it, built from NCES pieces (n. 1038) and Maloney (n. 1054), not a case study. Year 1998, weekly cadence, and sitting are hypothetical. 03-homeschool §5.2 already labelled this composite.

¹⁰⁵⁷College Board, Homeschoolers’ Guide (Jacob Mishook, 29 June 2026, All Access): homeschoolers may self-study AP; a formal AP course is recommended not required; parents **cannot order exams**; find a school via the AP Course Ledger; AP coordinator enrolls the student in an “exam only” My AP section; mid-November order deadline. Do not invent a registration recipe beyond that guide. Plant for Chapter 12. Do not promise that AP self-study yields a May seat.

¹⁰⁵⁸NCAA Home School Toolkit 2025–26 and Core-Course Worksheet (July 2025): audited, CLEP, or credit-by-exam courses are **not** NCAA-approved core courses. A kitchen Algebra II taught from a textbook can count if the worksheet is honest and college-preparatory. Dual-enrollment College Algebra is a college transcript NCAA already knows how

promise NCAA eligibility from a course titled “fun with numbers.”

No homeschool-curriculum RCT was opened. Choose by fit: parent math confidence, student stance, special needs, budget, college target, and whether the parent will remain supervisor. Cathy Duffy Reviews and Rainbow Resource are **catalogue / reviewer**, not independent effectiveness trials. Saxon Algebra 2, Math-U-See Algebra 2, AoPS Intermediate Algebra, then a fourth-course title — **vendor** names, grain and placement, not a ranking.

What this section has done is name the gates, steal fading and interleaving with the grade-7 label, and install the release valves as a normal high-school plan. Wolfram checks identities; the model explains to the parent; the child writes unaided. That split is the last section.

9.7 Parent-hosted AI for Algebra II and precalculus

Chapter 3 already locked the kitchen condition. Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman, 2025: GPT Base raised assisted-practice grades 48 percent relative to control and then cut unaided-exam grades 17 percent; GPT Tutor raised practice 127 percent and was statistically indistinguishable from control on the exam.¹⁰⁵⁹ Unguarded solvers raise practice scores and cut unaided exams. The model may prepare the adult and the next problem; it may not do the child’s problem. This chapter will not re-litigate Bastani at monograph length. One short reminder, then the Algebra II jobs. Parent workflows here are **practice suggestions, not an RCT**. No year-plus kitchen GPT-Tutor trial exists in the literature opened on 30 August 2026. Bastani is Turkish high school. Oreopoulos and Low is Tennessee middle school with a teacher present.

Allow, matching Chapter 3’s six workflows, scaled to this course.

Parent-first explanation of why $\ln$ and $\log_{10}$ differ, and why the unit circle is the right home for trigonometry. Chapter 3 workflow (a): the parent, adult account, child not in the room, asks a reasoning chat model — not an open-web agent — to explain the idea, the notation, two mistakes this age makes, and one tiny problem the parent should solve before sitting down. Then the parent solves without the model and checks.¹⁰⁶⁰ Why $\ln$ and $\log_{10}$ differ: different bases; $\log_{10} 100 = 2$ because $10^2 = 100$; $\ln 100 = 2 \ln 10$; $\ln 10$ is not 1; calculus will prefer e because the derivative of e^x is e^x . Why the unit circle: sine and cosine as functions of a real variable need a real input; arc length on radius 1 gives every real a geometric meaning; $(\cos \theta, \sin \theta)$ are the coordinates. Two mistakes: treating $\ln$ and $\log$ as synonyms; writing $\sin^{-1}$ as $1/\sin$. One tiny problem: place $(\cos 2\pi/3, \sin 2\pi/3)$ and say why cosine is negative. The parent does that problem on paper. Then the

to read. Math core = Algebra I or higher. Do not promise NCAA eligibility from a course titled “fun with numbers.”

¹⁰⁵⁹Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman, “Generative AI without Guardrails Can Harm Learning: Evidence from High School Mathematics,” *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122. GPT Base: assisted practice +48% relative to control; unaided exam −17%. GPT Tutor: practice +127%; exam statistically indistinguishable from control (point estimate −0.004). Turkish high school, grades 9–11, four sessions, GPT-4-0613. Percentages of control *grades*, not Cohen’s *d*. Not a homeschool RCT. Chapter 3 locked the object.

¹⁰⁶⁰Chapter 3 §3.4 workflow (a): parent-first explanation, adult account, child not present; idea, notation, two mistakes, one tiny problem the parent solves without the model. Bastani GPT Tutor worked in part because teachers had written the solution and common mistakes into the prompt. The parent is that teacher. Prefer a reasoning chat model, not an open-web agent. Practice suggestion, not an RCT.

lesson. Bastani’s GPT Tutor worked in part because *teachers* had already written the solution and the common mistakes into the prompt. The parent is that teacher. Prefer a reasoning chat model. Hermes 4 as chat is a candidate; Hermes 4 is **not recommended inside Hermes Agent** (tuned for chat, not rapid tool-calling). Hermes 4 $\neq$ Hermes Agent.¹⁰⁶¹

Lesson outline, Chapter 3 workflow (b): hook with a physical object or a drawn circle; one worked example the parent will write by hand; three student attempts; one exit problem; answers only in a marked PARENT KEY. Analogues — Claude for Teachers, ChatGPT for Teachers — are **vendor educator features**, 18+ / verified educators, not homeschool RCTs and not FERPA wrappers a homeschool parent should assume.¹⁰⁶² The outline is a map, not a franchise.

Isomorphic practice, key held by the parent. Chapter 3 workflow (c): five to eight isomorphic items. Grounded in Bastani (practice with access is not unaided learning), Soderstrom and Bjork (generation plus delayed feedback), and Oreopoulos and Low (practice minutes predicted MAP; chat did not, descriptively).¹⁰⁶³ Do not generate “practice” by photographing tonight’s assigned homework. Do not let the model print tonight’s Saxon or AoPS set with answers in the student file. Parent holds the key. Child does the page. Oreopoulos and Low, NBER w35620, August 2026, as Chapter 3 already reported: about 0.040 population SD per term, about 0.06–0.08 SD over a school year; gains resemble Khan without AI. Do not invent standalone 0.06 as ITT. Do not advertise Khanmigo as Bloom’s two-sigma tutor. If Khanmigo is in the house, expect the Tennessee pattern unless the parent requires the hint dialogue.¹⁰⁶⁴ Working paper, not yet journal-peer-reviewed.

Hint script, not an answer key for the child. Chapter 3 workflow (d): Bastani’s GPT Tutor analogue. Withhold the full solution; ask what they have tried; point out a named mistake; escalate slowly.¹⁰⁶⁵ Parent reads the hints aloud. That is closer to human tutoring than handing the child a solver and leaving the room. “You treated $\ln$ and $\log_{10}$ as the same function. What is $\log_{10} 100$? What is $\ln 100$ in terms of $\ln 10$?” Then wait. Then a change-of-base prompt. Not the value.

Hands-on as materials generation, Chapter 3 workflow (e): the model describes a unit circle, a ray,

¹⁰⁶¹Nous Portal documentation, accessed 30 August 2026: Hermes 4 is a chat/reasoning model family, **not recommended for use inside Hermes Agent** (tuned for chat, not rapid tool-calling). Hermes 4 $\neq$ Hermes Agent. Hermes Agent is an open-source autonomous agent (hermes-agent.org), MIT license, parent-machine tool, not a child-safety school product. Chapter 3 §3.2.

¹⁰⁶²ChatGPT for Teachers / Claude for Teachers: **vendor educator features**, 18+ / verified educators, not homeschool SKUs and not FERPA wrappers a homeschool parent should assume. Chapter 3. The outline is a map, not a franchise.

¹⁰⁶³Bastani 2025 (n. 1059): practice with access $\neq$ unaided learning. Soderstrom and Bjork 2015 (n. 1049): generation + delayed feedback. Philip Oreopoulos and Daniel Low, NBER working paper w35620 (August 2026): practice minutes predicted MAP; chat did not, descriptively. Working paper, not yet journal-peer-reviewed. Isomorphic items, parent-held key. Do not photograph tonight’s assigned homework into a model.

¹⁰⁶⁴Oreopoulos and Low, NBER w35620 (August 2026), two-year cluster RCT in 18 Tennessee middle schools: assignment to Khan Academy + Khanmigo raised MAP math by 1.26 national percentile ranks per term (SE 0.60), about 0.040 population SD per term and about 0.06–0.08 SD over a school year; implied effect of a full year of active participation in year 2 is 0.14 SD. Authors: “These gains resemble those from Khan Academy practice without AI assistance.” 96% of students tried Khanmigo; median student messaged it on only a third of practice days and in 17% of mistake sessions; 14.5% of messages contained a mathematical question or a step of reasoning. Do not invent standalone 0.06 as ITT. Do not advertise Khanmigo as Bloom’s 2-sigma tutor. Working paper.

¹⁰⁶⁵Bastani GPT Tutor hidden prompt (Figure 1 in the paper): withhold the full solution; ask what they have tried; point out a named mistake; escalate slowly. Homeschool analogue: parent-read hint script, not a child-held chatbot. Chapter 3 workflow (d).

a point, a perpendicular; the child draws, or the parent pastes a GeoGebra-able description into GeoGebra. Hermes Agent on the parent machine can build that description. Browser *only* on sites the parent names — curriculum publisher, Khan, Wolfram, GeoGebra — not open-ended “go find homework answers.”¹⁰⁶⁶ Do not put the agent on the child’s phone. Carbonneau $d \approx 0.37$ is versus abstract-only, under guidance. Virtual versus physical manipulatives remain **UNVERIFIED**.

Diagnose work the child already produced. Chapter 3 workflow (f): photograph after the attempt. Parent-only: name the misconception; one question to ask the child, not the correct steps; one isomorphic retest. The child answers the parent, not the model.¹⁰⁶⁷ For $\log(a + b) = \log a + \log b$, the sentence is “smashing a product rule onto a sum,” not “they forgot the law.” For $(\cos 2\pi/3, \sin 2\pi/3)$ placed in quadrant I, the sentence is “reference angle without the sign of the coordinate.” For a remainder of 0 next to $p(2) = 7$, the sentence is “division and substitution have not met.” No homeschool diagnosis RCT. MATHia / Carnegie **vendor** claim that their ITS recognizes *why* students err is vendor, plus Pane et al. 2014 Cognitive Tutor cited inside Oreopoulos and Low — school ITS, not kitchen.

Wolfram|Alpha is the right **check** for an identity or an inverse, after the child has attempted. Opened this research day: “Compute verified, expert-level answers using Wolfram’s breakthrough algorithms and knowledgebase.” Built on Wolfram Language. Symbolic, not next-token math.¹⁰⁶⁸ A language model is the right **explainer for the parent** — why the product rule for logs follows from exponent rules; why the Pythagorean identity is the unit circle. Mixing the two without a label is how hallucinated identities survive. Plant for Chapter 10’s stronger version of the same split: Wolfram checks an antiderivative; the model explains to the parent why this u ; the child writes the FRQ unaided. Never let a model’s fluency stand in for a known-correct key — textbook, Wolfram, parent-worked solution. Step-by-step on Wolfram Pro is **vendor**; Pro pricing was **not captured as a live number** on the homepage this day — **UNVERIFIED** current Pro price.¹⁰⁶⁹ Symbolic algebra — Symbolab, Wolfram, a CAS — is appropriate **after** the child can execute a method by hand, as a check, analogous to a calculator after fluency, not instead of it. Soderstrom and Bjork: removing the desirable difficulty of generation depresses learning.

Ban. Paste-the-precalculus-test. Photo-to-key. “What is the answer.” Photomath, Mathway, and Symbolab lead with scan-to-solution; no product-specific unaided-transfer RCT was opened this

¹⁰⁶⁶Chapter 3 / 04-ai browser constraint: sites the parent names (curriculum publisher, Khan, Wolfram, GeoGebra), not open-ended “go find homework answers.” Hermes Agent on the parent machine. Carbonneau $d \approx 0.37$ (n. 1045) versus abstract-only, under guidance. Virtual versus physical manipulatives **UNVERIFIED**.

¹⁰⁶⁷Chapter 3 workflow (f): diagnose work the child already produced. Name the misconception; one question; isomorphic retest. Child answers the parent, not the model. No homeschool diagnosis RCT. MATHia/Carnegie **vendor** claim plus Pane et al. 2014 Cognitive Tutor cited inside Oreopoulos/Low — school ITS, not kitchen. MATHia family purchase **UNVERIFIED**.

¹⁰⁶⁸Wolfram|Alpha, <https://www.wolframalpha.com/>, accessed 30 August 2026: “Compute verified, expert-level answers using Wolfram’s breakthrough algorithms and knowledgebase.” Built on Wolfram Language. Symbolic/computational engine, not an LLM. The honest **check** for an identity or an inverse after the child has attempted. A language model is the explainer for the parent. Mixing the two without a label is how hallucinated identities survive. Plant for Chapter 10.

¹⁰⁶⁹Wolfram|Alpha Pro step-by-step: **vendor**. Pro pricing **not captured as a live number** on the homepage this access day — **UNVERIFIED** current Pro price. Symbolic algebra (Symbolab, Wolfram, a CAS) is appropriate after the child can execute a method by hand, as a check — analogous to a calculator after fluency. Soderstrom and Bjork (n. 1049): removing generation depresses learning.

research day; the transferable evidence is Bastani's crutch plus the homepages.¹⁰⁷⁰ Open-web agent in the child's hands. Companion persona. Hermes Agent and Grok Bot can browse the open web; they are adult and developer tools, not a child-safe school SKU. No documented K–12 student-facing autonomous open-web agent as a school product on the pages opened 30 August 2026.¹⁰⁷¹ Hard no: messaging gateway to a child's Telegram or WhatsApp; Grok Bot iOS app in the child's hands; IEP or 504 PDFs on a shared Grok Bot computer; the child's school portal or Photomath logins in an agent. El Salvador's Grok-as-tutor is a ministry deployment of Grok *the assistant*, announcement not RCT, not Grok Bot, not a U.S. homeschool SKU. Grok $\neq$ Grok Bot $\neq$ El Salvador Grok-as-tutor. Grok Bot requires cloud data storage; Legacy Privacy Mode is not supported; no education SKU and no FERPA claim on the pages opened.¹⁰⁷² Adaptive is not an agent. IXL and Khan sequence items; they do not replace the parent hearing a wrong inverse.

Facts and fluent algebra are not an LLM job. GPT Base was fully correct 51 percent of the time, with 8 percent arithmetic errors even when the method was right.¹⁰⁷³ A language model that will happily hallucinate 7×8 , or $\sin^2\theta + \cos^2\theta = \sin(2\theta)$, is the wrong object to put between a child and an identity. Special-angle coordinates, log laws as retrieved compressions, $i^2 = -1$ — those belong to the parent's ears, a dumb spaced-repetition deck, or a short mixed oral. Use the model to generate the deck or the oral script, not to sit between the child and the fact.

AP Precalculus free response, if sat, is unaided. College Board still requires unaided work in the exam room.¹⁰⁷⁴ Every FRQ the child will meet must be written without a model, on paper, timed. The model may afterwards, parent account, mark against a rubric the parent pasted from an official FRQ — and the parent still samples the marking. Ban paste-the-FRQ. Do not promise that AP self-study yields a May seat; parents cannot order exams. Detectors fail. Liang, Yuksekgonul, Mao, Wu, and Zou 2023: 61.22 percent (arXiv) / 61.3 percent (*Patterns*) false positive on human TOEFL essays. Do not use detectors to police a written log-law or unit-circle argument.¹⁰⁷⁵ Companion-

¹⁰⁷⁰Photomath, Mathway, Symbolab: homepages lead with scan-to-solution. No product-specific unaided-transfer RCT opened 30 August 2026. Transferable evidence: Bastani's crutch (n. 1059) plus the homepages. Analogy of mechanism, not a Photomath-named RCT. Ban paste-the-prec calculus-test and photo-to-key.

¹⁰⁷¹Negative finding, pages opened 30 August 2026, not an exhaustive census: no documented K–12 student-facing autonomous open-web agent sold as a school product. Hermes Agent and Grok Bot are adult/developer tools, not child-safe school SKUs. Hard no: messaging gateway to a child's Telegram or WhatsApp; Grok Bot iOS app in the child's hands; IEP/504 PDFs on a shared Grok Bot computer.

¹⁰⁷²Grok Bot FAQ, fetched 30 August 2026: persistent cloud computer; cloud data storage required; Legacy Privacy Mode not supported; every Bot on the account shares one computer; no education SKU and no FERPA claim on the pages opened. xAI news post 11 December 2025: El Salvador ministry partnership to deploy Grok — the assistant — across more than 5,000 public schools. Announcement, not RCT; not Grok Bot; not a U.S. homeschool SKU. Grok $\neq$ Grok Bot $\neq$ El Salvador Grok-as-tutor. Independent evaluation **UNVERIFIED**.

¹⁰⁷³Bastani 2025 (n. 1059): GPT Base, asked "What is the answer?", fully correct only 51 percent of the time on the practice set; 42 percent logical errors and 8 percent arithmetic errors even when the method was right. Facts and fluent algebra are not an LLM job. Chapter 3 §3.7.

¹⁰⁷⁴College Board AP exams: unaided work in the exam room. AP Precalculus free response, if sat, is unaided. Ban paste-the-FRQ. The model may afterwards (parent account) mark against a rubric the parent pasted from an official FRQ; the parent still samples the marking. Practice suggestion, not an RCT.

¹⁰⁷⁵Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou, "GPT Detectors Are Biased against Non-Native English Writers," arXiv 2023 / *Patterns*: average false-positive rate 61.22% (arXiv) / 61.3% (*Patterns*) on human TOEFL essays. Do not use detectors to police written math explanations. Companion-harm cases — *Garcia v. Character Technologies*, *Raine v. OpenAI* — are **allegations, not verdicts**. COPPA amended Rule published 22 April 2025, general compliance 22 April 2026. California A.B. 1159: Senate third reading as of 30 August 2026; not law.

harm cases are allegations, not verdicts; they are why a math agent should not be a companion. COPPA general compliance 22 April 2026. A parent generating a worksheet on an adult account is a different fact pattern from a child account on a companion app. California A.B. 1159 is not law as of 30 August 2026. Do not freeze it as a statute.

Session shape, Chapter 3 §3.8, scaled to this course. Practice suggestion, not an RCT. Unaided warm-up: one inverse, one unit-circle value, one “which function family?” Teach. Struggle. Hint not key. Unaided exit ticket: a mixed three-item check without the model, which is Bastani’s exam phase in five minutes.¹⁰⁷⁶ “Finish your precalculus with Photomath so we can go to soccer” is GPT Base, the pattern with a large-sample causal harm estimate. If the family only remembers one sentence from this section, Chapter 3 already wrote it: the model may prepare the adult and the next problem; it may not do the child’s problem. The Algebra II version is narrower. The model may explain to the parent why $\ln$ and $\log_{10}$ differ. It may not print the change-of-base path onto the child’s page. Wolfram may check an identity after the attempt. The child writes unaided.

A *hypothetical*, labelled as such, can put the ban in a room. A parent whose sixteen-year-old photographs tonight’s identities worksheet into Photomath, copies the green steps, and scores 100 percent on the page, then cannot place $(\cos 2\pi/3, \sin 2\pi/3)$ on tomorrow’s unaided three-item check, is running GPT Base.¹⁰⁷⁷ The empirical pieces are Bastani’s crutch, Photomath’s scan-to-solution homepage, and Soderstrom and Bjork’s performance/learning split. The age, the domain, and the 100 percent are hypothetical. The move is not: take the phone, put the circle on paper, and run the unaided check.

A solid function concept is the calculus door. A shaky one is symbol soup. Diagnose the correspondence, the inverse, the exponential, the log, the unit circle, before buying a book titled Calculus. Chapter 10 is a new operation on functions they already know. This chapter was the year those functions had to become known, or not. The agent does not take the test. The child does.

Do not freeze it as a statute. Chapter 3 §3.6.

¹⁰⁷⁶Chapter 3 §3.8 session shape, scaled to this course: unaided warm-up (one inverse, one unit-circle value, one “which function family?”); teach; struggle; hint not key; unaided exit ticket. Bastani’s exam phase in five minutes. Practice suggestion, not an RCT. “Finish your precalculus with Photomath so we can go to soccer” is GPT Base.

¹⁰⁷⁷Composite *hypothetical*, labelled in the sentence that introduces it. Empirical pieces: Bastani’s crutch (n. 1059); Photomath scan-to-solution homepage (n. 1070); Soderstrom and Bjork performance/learning split (n. 1049). Age, domain, and 100 percent are hypothetical. The move is not: take the phone, put the circle on paper, run the unaided check.

Chapter 10

Calculus: AB, BC, CLEP, and the Function Concept

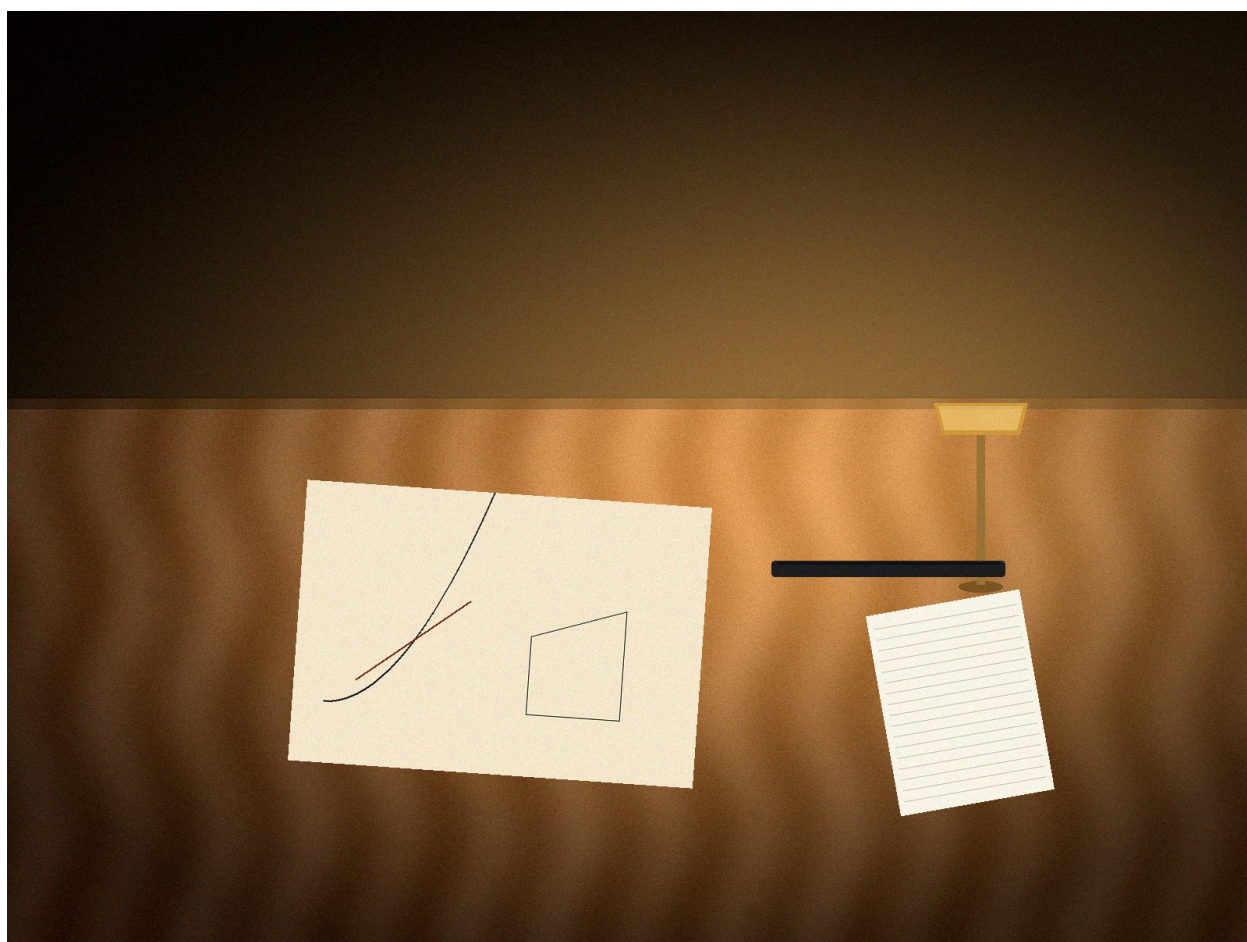


Figure 11: A paper sketch of a tangent to a curve, a closed laptop, and a paper free-response booklet.

Two documents, not a genius myth, already contain the argument this chapter has to keep in view. The first is CCSSM Appendix A, published by Achieve in partnership with the Common Core writing team: the regular three-year high-school pathway plus a fourth course reaches precalculus as a junior and calculus as a senior *only if* high-school content started in Grade 8. Compacted

versions of the traditional and integrated pathways complete Grade 7, Grade 8, and the first high-school course in Grades 7–8 so that a student can reach Calculus or another college-level course by senior year. “The pathways and courses are models, not mandates.”¹⁰⁷⁸ The second is College Board’s AP Calculus AB course page, opened 30 August 2026: eight units, Limits and Continuity through Applications of Integration, with published multiple-choice weightings, and a 2026–27 clarification that does **not** change course content.¹⁰⁷⁹ Those two sentences are not a placement slogan. They are a refusal. Calculus in senior year is a pathway fact, not a birthday. Eight units is a map of what a first college-facing calculus course actually contains, not a claim that a parent who last passed Algebra I in 1998 casually teaches BC from a kitchen table. Compact only if the gates are open. Do not compact because a neighbor’s child is already in a public-school AB classroom.

The National Center for Education Statistics already knew the destination is a minority. In the 2016 Parent and Family Involvement tables, calculus was ever-taught to 15 percent of high-school-level homeschoolers — interpret with caution — and taught *that year* to 13 percent of the 9–12 band.¹⁰⁸⁰ Those figures are *high-school-level ever-taught calculus during home instruction*, not college calculus and not a completion rate. They are 2016, not 2026. The 2023 First Look did not republish the subject tables. No named nationally representative source opened for this project gives the share of homeschool graduates who complete a college calculus course or who major in STEM.¹⁰⁸¹ Do not treat 15 percent as a plan, a percentile, or a proof that “homeschoolers do calculus.” Promise a path, not a headcount.

What the mathematics actually is, named more specifically than a slogan about “hard algebra,” is the mathematics of instantaneous change and accumulation: limits, derivatives, integrals, and, in a second course, infinite series and the calculus of parametric, polar, and vector-valued functions.¹⁰⁸² It is not hard algebra. A student with shaky algebra will experience calculus as symbol soup. A student with a solid function concept will experience it as a new operation on functions they already

¹⁰⁷⁸Achieve / CCSS writing team, *Appendix A: Designing High School Mathematics Courses Based on the Common Core State Standards*, Oregon ODE copy, <https://www.oregon.gov/ode/educator-resources/standards/mathematics/Documents/math-appendix-a-model-course-pathways.pdf>, accessed 30 August 2026. Regular three-year pathway plus a fourth course reaches precalculus as a junior and calculus as a senior only if high-school content started in Grade 8. Compacted traditional and integrated pathways complete Grade 7 + Grade 8 + the first high-school course in Grades 7–8 so that a student can reach Calculus or another college-level course by senior year. “The pathways and courses are models, not mandates.”

¹⁰⁷⁹College Board, AP Calculus AB course page, <https://apcentral.collegeboard.org/courses/ap-calculus-ab>, accessed 30 August 2026. Eight units with MC weightings as tabled in §10.3. *AP Calculus AB and BC Course and Exam Description Clarifications and Corrections, Effective Fall 2026*, <https://apcentral.collegeboard.org/media/pdf/ap-calculus-ab-bc-course-and-exam-description-clarifications.pdf>, accessed 30 August 2026: “Course content has not changed.”

¹⁰⁸⁰National Center for Education Statistics, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey* (NCES 2020-001), Table 8 (ever-taught at high-school level during home instruction) and Table 9 (subject taught *that year* by grade band). Calculus ever-taught 15 percent of high-school-level homeschoolers — interpret with caution — and taught *that year* to 13 percent of 9–12. Last opened subject tables; NHES:2023 did not republish them. Not a completion rate and not college calculus.

¹⁰⁸¹No named nationally representative source opened for this project (01-scope; 03-homeschool; brief open question 5) gives the share of homeschool graduates who complete a college calculus course or who major in STEM. Cardus Education Survey 2023 reports bachelor’s-attainment gaps and does not report SAT math, calculus, or STEM-major rates. Do not close the hole with analogy.

¹⁰⁸²01-scope, Calculus band: the mathematics of instantaneous change and accumulation — limits, derivatives, integrals, and, in a second course, infinite series and the calculus of parametric, polar, and vector-valued functions. Not “hard algebra.”

know. That sentence is the door. Chapter 6 planted 8.F — a function assigns to each input exactly one output — as the first Functions domain in K–8. Chapter 7 planted F-IF: function notation, domain and range, key features, average rate of change, graphs of linear, quadratic, exponential, absolute-value, step, and piecewise functions. Chapter 9 planted the elementary functions calculus will differentiate and integrate: polynomial, rational, radical, exponential, logarithmic, trigonometric, including inverse trigonometric, and the unit circle as the right home for trigonometry. This chapter does not start calculus on a fragile function concept. Diagnose first. Then the eight units. Then, if and only if those eight are honest, Units 9 and 10.

The Common Core State Standards for Mathematics are treated here as a map, not as a homeschool mandate. Many homeschoolers do not use CCSSM. Many U.S. states have adopted, adapted, or replaced it. A 2026 state-by-state census of who still uses it is **UNVERIFIED** in this project; this chapter will not say “most states.”¹⁰⁸³ What the map is good for is specifying what the mathematics *depends on*: the function concept; algebraic manipulation of compositions; unit-circle trigonometry; fluency with exponential and logarithmic functions; analytic geometry. CLEP states that assumption explicitly. AP Calculus does too. The official CED prerequisite paragraph, extracted 30 August 2026 from the 7.63 MB AP Calculus AB and BC Course and Exam Description, now sits on this page: before studying calculus, all students should complete the equivalent of four years of college-bound secondary mathematics — algebra, geometry, trigonometry, analytic geometry, and elementary functions (linear, polynomial, rational, exponential, logarithmic, trigonometric, inverse trigonometric, and piecewise-defined) — and must be familiar with the properties, composition, algebra, and graphs of functions.¹⁰⁸⁴ The eight unit titles and their multiple-choice weights come from the course pages. Topic lists below those titles stay off the page. Do not invent a CED catalog in order to sound official.

Four lanes run through the sections. They are not four identical headings. Sections 10.1 through 10.3 explain the mathematics to the parent: the function concept at the door; limits, derivatives, integrals, and the Fundamental Theorem, written for an adult who stopped at algebra, not as a textbook dump; then three college-facing objects — AP Calculus AB, AP Calculus BC, and CLEP Calculus — overlapping mathematics, different assessments. Section 10.4 is pathways, credit, NCAA, and AP logistics, enough to teach the parent how the course choice and the seat interact; the full records chapter is Chapter 12. Section 10.5 is teaching guidance: multiple representations in the same sitting; worked examples faded for the first pass through a new technique; every AP free-response question written unaided, on paper, timed. Section 10.6 is the tips that survive: objects, titles, no

¹⁰⁸³2026 state-by-state census of who still uses CCSSM, who uses “standards formerly known as,” and who never did: **UNVERIFIED** (01-scope open question 8). Do not say “most states.” Common Core State Standards Initiative, *Common Core State Standards for Mathematics*, https://www.thecorestandards.org/assets/CCSSI_Math%20Standards.pdf, accessed 30 August 2026, treated throughout as a map, not a homeschool mandate.

¹⁰⁸⁴CLEP states the assumption explicitly: algebra, geometry, trigonometry, and analytic geometry (n. 1094). AP Calculus official prerequisite paragraph now extracted from College Board, AP Calculus AB and BC Course and Exam Description, <https://apcentral.collegeboard.org/media/pdf/ap-calculus-ab-and-bc-course-and-exam-description.pdf>, accessed 30 August 2026, front-matter Prerequisites: four years of college-bound secondary mathematics; algebra, geometry, trigonometry, analytic geometry, and elementary functions including inverse trigonometric and piecewise-defined; properties, composition, algebra, and graphs of functions; language of functions; unit-circle sine and cosine and values at 0, $\pi/6$, $\pi/4$, $\pi/3$, $\pi/2$ and multiples; BC additional expectation of basic familiarity with sequences and series and some exposure to parametric and polar equations. Unit-guide topic rows below the eight and ten titles remain **UNVERIFIED**.

overload fantasy. Section 10.7 is the parent-hosted AI lane: Wolfram|Alpha is a computer-algebra check, not an explainer; a language model is the explainer for the parent; ban “do this FRQ”; allow parent-first, isomorphic practice with a parent-held key, and diagnosis of a wrong derivative. The child does the work. The parent holds the key.

Chapter 2 already locked the method this chapter will use: five interwoven strands; conceptual and procedural knowledge iterating, not a one-way street; worked examples faded for novices, incorrect examples explained; blocked practice to learn execution, interleaved and spaced practice to choose a strategy and keep it — and the clean classroom randomized trials of interleaving are grade 7, not calculus; do not import Rohrer’s $d = 1.05$ onto a free-response question. Chapter 3 already locked the AI lane: unguarded solvers raise practice scores and cut unaided exams; the model may prepare the adult and the next problem; it may not do the child’s problem. This chapter will not re-litigate Bastani at monograph length. One short reminder, then the objects of this course. Chapter 9 already did the elementary functions. This chapter will not dump it. A child who cannot move among formula, graph, table, and verbal for a function they already “finished,” who treats composition as two separate problems, who has only right-triangle SOH-CAH-TOA and no unit circle, is not ready for a limit of a difference quotient, regardless of a book titled Precalculus on the shelf.

The popular story collapses those locks into slogans they do not support. Calculus is hard algebra. Finish the precalculus book and you are ready. AB and BC are the same course with different letters. CLEP is a cheaper AP. Statistics can substitute if the child is “not a math person.” The parent who stopped at algebra can casually teach BC because the video curriculum does 100 percent of the teaching. The AI will do this FRQ. The documents do not say any of those things, not as the slogans say them. This chapter prints the documents. It prints the unit titles, the weights, the ACE recommendation, the NCAA exclusion, the May 2027 timing change, and the UNVERIFIED flags. It leaves the open questions standing — the full AP Calculus CED topic-level tables below the eight and ten unit titles; Tall and Vinner 1981 on concept image versus concept definition, not opened; what non-AP homeschool calculus books actually assign; homeschool-specific dual-enrollment counts; STEM-major and college-calculus rates for homeschool graduates; the IES dual-enrollment outcomes report expected in 2027. It does not invent a kitchen scene to make the science feel warmer. The parent who stopped at algebra is still the design user. The job is to hear a derivative treated as “the slope formula,” and a limit treated as “plug it in,” before any tool is allowed to print a scored free-response the parent never sees.

A compacted path through Grades 7–8 exists so that calculus is reachable in senior year. That sentence is a model, not a mandate. A non-accelerated student who starts Algebra I in Grade 9 finishes Algebra II in Grade 11 and has senior year for precalculus, statistics, or a first calculus course — not both precalculus and BC unless the fourth course is overloaded.¹⁰⁸⁵ Overload is not a plan. Statistics is a different college-equivalent. Chapter 11 is that chapter. It will not be written here. Calculus does not substitute for statistics as a data-reasoning course. Statistics does not substitute for calculus in a calculus-based STEM sequence. College Board’s revised AP Statistics framework encourages taking both when the student has the background. That is the honest STEM

¹⁰⁸⁵Appendix A (n. 1078). A non-accelerated student who starts Algebra I in Grade 9 finishes Algebra II in Grade 11 and has senior year for precalculus, statistics, or a first calculus course — not both precalculus and BC unless the fourth course is overloaded. Overload is not a plan. Compacted 7–8 is a model, not a mandate.

advice. This chapter is the calculus object. Plant Chapter 11. Do not dump it.

The thesis of the book still holds, and it is still not a causal claim about homeschool calculus. A parent who is not a math specialist can take a child from first-grade number sense through high-school calculus if three things hold: the parent understands the concept well enough to hear a wrong answer — “the limit is the value you get when you plug it in,” “the derivative is the slope formula,” “the integral is the antiderivative chant,” “the Fundamental Theorem is two formulas I memorized”; the child does the struggle before the tool; and any agent is a supplement hosted and constrained by the parent, not a substitute for the child’s work and not a school-safe product by default.¹⁰⁸⁶ The sources do not prove that thesis as a causal claim about kitchen-table calculus. They do not contradict it. They constrain how it can be attempted. Unguarded solvers raise practice scores and cut unaided exams. No homeschool calculus randomized trial exists in the literature opened on 30 August 2026. This chapter will not invent one. Do not claim a parent who stopped at algebra casually teaches BC. When the parent is the ceiling, dual-enrollment Calculus I is the cleanest transcript a stranger already knows how to read. Chapter 12 will say that as an export rule. This chapter says it as a teaching rule.

10.1 The function concept at the door

Calculus is a new operation on functions they already know. That sentence is the first teaching move of this chapter, and it is a sentence the parent who stopped at algebra has to be able to hear wrong. If f is a function that assigns to each input exactly one output, then f' is another function — the derivative — that assigns to each input the instantaneous rate at which f is changing there, when that rate exists. The integral, in the accumulation sense, is another operation on f : it builds a new function that records how much f has piled up. The Fundamental Theorem of Calculus is the meeting of those two operations. None of that attaches if f itself is still a slogan. A child who can differentiate a polynomial by a chant and cannot say, of the same function, what the graph is doing, what the table is doing, and what a verbal description would have to say, has the chant and not the object. Do not start this chapter on that chant.

The door diagnostic is three questions, not a book title. Can the student move among formula, graph, table, and verbal for a function already on the page? Can the student manipulate compositions algebraically — see $f(g(x))$ as a structure, not as two separate problems? Does the student have unit-circle trigonometry, not only right-triangle SOH-CAH-TOA? Those three are the load-bearing questions. “Did we finish a book titled Precalculus” is not among them.¹⁰⁸⁷ Precalculus is not a legally defined course. Appendix A’s “Fourth Courses” column holds most of the (+) standards not required for college-and-career readiness; a parent shopping for “precalculus” is usually buying that remainder plus a tightening of Algebra II functions. Chapter 9 already said so. A finished book with that word on the spine is evidence of a purchase, not of a function concept.

Common Core’s 8.F is the first Functions domain in K–8, and it is still the right first sentence: a

¹⁰⁸⁶Book thesis, FRAME.md and outline, 30 August 2026. The sources do not prove the thesis as a causal claim about homeschool calculus. They do not contradict it. They constrain how it can be attempted.

¹⁰⁸⁷01-scope, calculus-door diagnostic: move among formula, graph, table, verbal; algebraic manipulation of compositions; unit-circle trigonometry — not “did we finish a book titled Precalculus.” Precalculus is not a legally defined course (Appendix A “Fourth Courses” column; Chapter 9).

function assigns to each input exactly one output. Compare functions represented in different ways. Linear versus nonlinear. Interpret $y = mx + b$ as defining a linear function, m as rate of change, b as initial value. Sketch a graph from a qualitative description.¹⁰⁸⁸ Do not introduce functions as “ $y = mx + b$ ” only. Chapter 6 already refused that collapse. Rohrer, Dedrick, and Burgess 2014 treated “graph $y = mx + b$ ” and “slope from two points” as different problem kinds that students fail to associate when practice is blocked. That trial is grade 7. The coefficient stays in grade 7. The association failure does not. A child who can write $y = 2x + 3$ and cannot read a table that is the same function, or a graph that is the same function, or a sentence that is the same function, does not have 8.F. Calculus’s $f'(x)$ has nothing to attach to.

Algebra I, on Appendix A’s traditional pathway, formalizes that concept: function notation, domain and range, key features of graphs, average rate of change over an interval, and the families linear, exponential, and quadratic, plus absolute value, step, and piecewise.¹⁰⁸⁹ Average rate of change is already a difference quotient without the limit. The parent who can hear a wrong answer hears “the slope is 2” on a curve as a confusion between the average rate on an interval and the instantaneous rate at a point. That confusion is this chapter’s object. It is not a new object. It is 8.F and F-IF, taken to a limit.

Algebra II and the fourth course supply the rest of the elementary functions. Polynomial, rational, radical. Exponential and logarithmic, with logarithms as solutions of exponentials. Trigonometric functions on the unit circle, radian measure, the Pythagorean identity, modeling periodic phenomena, inverse trigonometric functions as far as the course takes them.¹⁰⁹⁰ NMAP’s Major Topics of School Algebra already named the catalog independent of course packaging: linear, quadratic, polynomial, simple nonlinear, exponential, logarithmic, trigonometric, and fitting models to data. A homeschool Algebra II that omits functions, logarithms, and trigonometry has not covered school algebra as the Panel defined it. Chapter 9 already used that checklist. This chapter inherits it as a door, not as a dump. Unit-circle trigonometry is not optional if calculus is the destination. SOH-CAH-TOA only is a common gap at this door.

The common gaps, named in the scope memo and inherited here, are five, and they are labelled **inference from the maps, not a survey**. No named nationally representative study of homeschool calculus arrivals was opened for this project. The list is what the topic maps imply a student must already be able to do, not a 2026 kitchen census.¹⁰⁹¹ (a) Weak function concept: cannot move

¹⁰⁸⁸CCSSM 8.F: first Functions domain in K–8. A function assigns to each input exactly one output; compare representations; linear versus nonlinear; interpret $y = mx + b$ as a linear function, m as rate of change, b as initial value; sketch from a qualitative description. Planted in Chapter 6. Without this concept, calculus’s $f'(x)$ has nothing to attach to.

¹⁰⁸⁹Appendix A Algebra I critical areas and F-IF clusters: function concept and notation; interpret key features; average rate of change; graph linear, quadratic, exponential, absolute value, step, piecewise. Oregon ODE copy (n. 1078). Average rate of change is already a difference quotient without the limit.

¹⁰⁹⁰Appendix A Algebra II and Fourth Courses; National Mathematics Advisory Panel, *Foundations for Success: The Final Report* (U.S. Department of Education, 2008), Table 1, Major Topics of School Algebra, ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>, accessed 30 August 2026. Functions named: linear, quadratic, polynomial, simple nonlinear, exponential, logarithmic, trigonometric, and fitting models. A homeschool Algebra II that omits functions, logarithms, and trigonometry has not covered school algebra as the Panel defined it. Chapter 9.

¹⁰⁹¹Common gaps at the calculus door: 01-scope Calculus band, labelled **inference from the maps, not a survey**. No named nationally representative study of homeschool calculus arrivals was opened. Do not upgrade the list into a 2026 kitchen census.

among graph, table, formula, verbal. (b) Chain-rule algebra: composition not seen as a structure. (c) Trigonometry of the unit circle never learned; only right-triangle SOH-CAH-TOA. (d) No fluency with exponential and logarithmic functions. (e) Treating a derivative as “the slope formula” rather than as a limit of average rates. Those five are the diagnosis list a parent can steal before buying an AB book, a BC book, a CLEP packet, or a dual-enrollment seat. They are also why this section is this long. The eight units will be symbol soup if this door stays shut.

A *hypothetical*, labelled as such, can put the door in a room. A parent whose sixteen-year-old has a book titled Precalculus with the last chapter checked, who can differentiate x^3 by a chant, and who, shown a table of a function and asked for the average rate of change on an interval, reaches for “the slope formula” as if the table were two points on a line and nothing else, is not ready for Unit 1.¹⁰⁹² The empirical pieces are 8.F, F-IF’s average rate of change, and the maps’ demand that a function survive four representations. The age, the chant, and the checked chapter are hypothetical. The move is not: open the limits chapter. The move is: put one function on the table in four representations and stay there until the child can travel.

Limit as concept-image versus concept-definition is a named literature. Tall and Vinner 1981 is the paper the research stream marked and did not open. This chapter will not invent a teaching sequence from an unopened paper. It will not quote a concept-image sentence as if the PDF had been on the desk. The flag stays **UNVERIFIED**.¹⁰⁹³ What the parent can still hear, without that paper, is a wrong answer in ordinary language. “The limit is what you get when you plug it in” is wrong when plugging in is undefined, and it is incomplete even when plugging in works, because the limit is about the values *approaching*, not about the value *at*. “The limit is what the function wants to be” is a metaphor a child may use on the way to a definition; it is not a definition. Section 10.2 will give the parent the object. This section only needs the parent to refuse a start that skips the function the limit is *of*.

CLEP’s published assumption is the same door in vendor-neutral language. College Board: the Calculus exam covers skills and concepts usually taught in a one-semester college course in calculus, assumed after algebra, geometry, trigonometry, and analytic geometry.¹⁰⁹⁴ AP Calculus, via the course pages, is an introductory college-level calculus course. The official CED prerequisite paragraph, now extracted, is more specific than that CLEP list and is not invented here: four years of college-bound secondary mathematics; algebra, geometry, trigonometry, analytic geometry, and elementary functions including inverse trigonometric and piecewise-defined; properties, composition, algebra, and graphs of functions; the language of functions (domain and range, odd and even, periodic, symmetry, zeros, intercepts, increasing and decreasing); sine and cosine defined from the unit circle, with values at 0 , $\pi/6$, $\pi/4$, $\pi/3$, $\pi/2$ and their multiples. Students who take AP Calculus BC are expected to have basic familiarity with sequences and series, as well as some exposure to

¹⁰⁹²Hypothetical labelled in the sentence that introduces it. Empirical pieces: 8.F (n. 1088); F-IF average rate of change (n. 1089); four-representation demand (n. 1087). The age, the chant, and the checked chapter are hypothetical.

¹⁰⁹³David Tall and Shlomo Vinner, “Concept Image and Concept Definition in Mathematics with Particular Reference to Limits and Continuity,” *Educational Studies in Mathematics* 12 (1981): 151–169. Paper not opened this project (outline open question 21). Do not invent a teaching sequence or quote a concept-image sentence as if the PDF had been read. **UNVERIFIED**.

¹⁰⁹⁴College Board, CLEP Calculus exam, <https://clep.collegeboard.org/clep-exams/calculus>, accessed 30 August 2026. Assumed after algebra, geometry, trigonometry, and analytic geometry. “The Calculus exam covers skills and concepts that are usually taught in a one-semester college course in calculus.”

parametric and polar equations. A kitchen that is still teaching the unit circle inside week three of AB is not “behind.” It is doing the door. A kitchen that skipped the unit circle because the AB book starts with limits is doing the soup.

What this section has done is refuse a start. Limits come next, because they are the first new object, and because a limit treated as plugging in is the misconception Unit 1 is built to break. The parent who stopped at algebra does not need a semester of real analysis. The parent needs to hear “plug it in” as a special case that sometimes works, and to know why a hole in a graph can still have a limit. Then derivatives. Then integrals. Then the theorem that makes those two operations one story.

10.2 Limits, derivatives, integrals, and the Fundamental Theorem

A limit is not “plugging in.” That sentence is the first object of this section, and it is the sentence the parent who stopped at algebra has to steal before any derivative chant is allowed on the page. The limit of $f(x)$ as x approaches a is the value that $f(x)$ gets arbitrarily close to as x gets arbitrarily close to a , from both sides if the two-sided limit is the one being claimed, when that value exists. The function does not have to be defined at a . The function does not have to equal its limit at a . Continuity is the extra claim that the function is defined at a and that the limit equals the value. A hole in a graph can have a limit. A jump cannot have a two-sided limit. A vertical asymptote is a different failure: the values do not approach a finite number.¹⁰⁹⁵

Plugging in is a *method* that works when the function is continuous at the point, which polynomials are, and which rational functions are wherever the denominator is not zero. It is not the definition. The child who always plugs in, including when plugging in is undefined, and who then writes “does not exist” without looking at the values on either side, has a method and not an object. The child who cancels a common factor, plugs in, and cannot say why that cancellation was legal — why the new function agrees with the old one everywhere except at the hole — has a trick. Unit 1 of AB is Limits and Continuity, 10–15 percent of the multiple-choice exam.¹⁰⁹⁶ The parent who can hear a wrong answer hears “the limit is 5 because I plugged in 2” as incomplete until the child can also say what the nearby values are doing, and hears “the limit does not exist because I got 0/0” as a confusion between an indeterminate *form* and a limit that may still exist after algebra.

One-sided limits are the same idea from one side. The two-sided limit exists only if the left-hand and right-hand limits exist and agree. Infinite limits and limits at infinity are the language of asymptotes: the values grow without bound, or they settle toward a horizontal line. This is not a catalog of every limit law. The parent does not need the ϵ – δ definition to hear a wrong answer at the table. The parent needs the picture: a function approaching a height, a function approaching from one side only, a function blowing up, a function settling as x runs off the page. Multiple representations, already the door, are now the daily practice. A formula, a graph, a table of nearby values, a sentence. Same sitting. The child who can compute a limit from a formula and cannot read the same limit off a graph does not have the object. College Board’s second mathematical practice for this course is

¹⁰⁹⁵Limit as approach, not as the value at the point; continuity as the extra claim that the function is defined at the point and the limit equals the value. Parent-facing object, not a textbook dump and not an ϵ – δ course. AB Unit 1 title from the course page (n. 1079).

¹⁰⁹⁶College Board, AP Calculus AB course page (n. 1079): Unit 1 Limits and Continuity, 10%–15% MC.

exactly that translation.¹⁰⁹⁷

A derivative is not “the slope formula.” The slope formula, $(y_2 - y_1)/(x_2 - x_1)$, is the average rate of change of a function on an interval. Algebra I already named that object as average rate of change. The derivative at a point is the *limit* of those average rates as the interval shrinks to the point, when the limit exists. In symbols the parent can steal: $f'(a)$ is the limit as h approaches 0 of $[f(a + h) - f(a)] / h$. That quotient is a slope. The limit of that quotient is an instantaneous rate. The child who writes “the derivative is the slope” and means the slope of a line through two points has the Algebra I object. The child who writes the same sentence and means the slope of the tangent line, obtained as a limit of secant slopes, has the calculus object. Those are not the same sentence.¹⁰⁹⁸

Differentiability is a stronger claim than continuity. A corner, a cusp, a vertical tangent, a jump — the derivative fails for different reasons. A function can be continuous at a point and not differentiable there. The absolute-value function at zero is the usual picture: the graph is connected; the secant slopes from the left approach -1 and from the right approach 1 ; the two-sided limit of the difference quotient does not exist. The parent who can hear a wrong answer hears “it’s continuous so you can differentiate it” as the collapse of those two claims.

Then the derivative as a *function*. Once the derivative exists on an interval, f' is itself a function, and the child’s job is to move among f and f' in the same four representations. Where f is increasing, f' is positive. Where f has a horizontal tangent, f' is zero — or undefined, if the tangent is vertical or the corner refuses a tangent. Where f is concave up, f' is increasing, and f'' , the derivative of the derivative, is positive when it exists. Those readings are Unit 5’s analytical applications, 15–20 percent of AB multiple choice, the heaviest analytical slice.¹⁰⁹⁹ They are not a later luxury. They are why the derivative had to be a function and not a slogan about slope.

The computational layer is real, and it is not the object. Power rule, product rule, quotient rule, chain rule. Derivatives of the elementary functions: polynomials, exponentials, logarithms, sine and cosine and the rest of the trigonometric six, inverse trigonometric functions as far as the course takes them. Implicit differentiation, because not every relation arrives as $y = f(x)$. Inverse functions, because the derivative of an inverse is the reciprocal of the derivative of the original, at the matching point, when those derivatives exist and are nonzero. The chain rule is the load-bearing computation, and it is the algebra of composition from the door. The child who cannot see $\sin(3x^2)$ as a composition will chant a product or a power and get a fluent wrong derivative. Unit 2 of AB is the definition and the fundamental properties, 10–15 percent. Unit 3 is composite, implicit, and inverse functions, 5–10 percent.¹¹⁰⁰ Do not let Unit 2’s chants skip Unit 3’s structure. Do not let Unit 3’s structure skip the door.

Contextual applications are the derivative used as a rate. Related rates: two changing quantities tied by an equation, differentiated with respect to time. Linearization and local approximation:

¹⁰⁹⁷ College Board, AP Calculus AB course page (n. 1079): mathematical practice (2) translate mathematical information across representations. Assessed skill, not a franchise slogan.

¹⁰⁹⁸ Derivative as the limit of average rates; average rate of change already named in Algebra I F-IF (n. 1089). “The derivative is the slope formula” collapses the Algebra I object with the calculus object.

¹⁰⁹⁹ College Board, AP Calculus AB course page (n. 1079): Unit 5 Analytical Applications of Differentiation, 15%–20% MC.

¹¹⁰⁰ College Board, AP Calculus AB course page (n. 1079): Unit 2, 10%–15% MC; Unit 3 Composite, Implicit, and Inverse Functions, 5%–10% MC.

the tangent line as a stand-in for the function near the point. L'Hospital's Rule: an indeterminate quotient $0/0$ or ∞/∞ may be replaced, under hypotheses the book will name, by the quotient of the derivatives. The Mean Value Theorem: on a closed interval, if f is continuous on the closed interval and differentiable on the open interval, there is at least one point where the instantaneous rate equals the average rate. The parent who stopped at algebra does not need a proof of the Mean Value Theorem this afternoon. The parent needs to hear “the average speed was 50, so at every moment the speed was 50” as false, and “the average speed was 50, so at some moment the instantaneous speed was 50” as the theorem, when the hypotheses hold.¹¹⁰¹ Unit 4 is those contextual applications, 10–15 percent. Kitchen contexts help similar contexts. NMAP Finding 26 still holds: real-world contexts improve performance on similar real-world items; they do not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving. A related-rates ladder problem is a second representation of implicit differentiation, not a replacement for it.¹¹⁰²

Analytical applications are the first- and second-derivative tests, extrema, concavity, points of inflection, curve sketching from f' and f'' , optimization. Extreme Value Theorem: a continuous function on a closed, bounded interval attains a maximum and a minimum. The 2026–27 clarification adjusted the wording of that theorem in Unit 5; it did not change the content.¹¹⁰³ The parent's ear is the same either wording: on a closed interval, look at critical points *and* endpoints; on an open interval, a continuous function may have no maximum. Optimization is not a new derivative. It is the derivative used to find a best value, plus the discipline of saying what the variables are and what is being optimized. A child who can differentiate and cannot say what is being maximized has the chant.

An integral, in the accumulation sense, is not “the antiderivative chant.” The definite integral of f from a to b is a limit of Riemann sums: slice the interval, take a sample height on each slice, multiply height by width, add. As the slices get thinner, when the limit exists, that sum approaches a number. If f is positive, that number is an area. If f changes sign, that number is a net accumulation — area above the axis minus area below. If f is a rate, that number is the net change in the quantity whose rate is f . Grade 3 already had area as a product of side lengths and as additive. Grade 5 already had volume of a right rectangular prism. Geometry already had informal arguments for circumference, area, and volume using dissection, Cavalieri, informal limits.¹¹⁰⁴ Those informal-limit arguments become precise here. They do not become optional. A child who has never seen a Riemann rectangle will experience “the integral is the antiderivative” as a chant that happens to get area problems right. Unit 6 is Integration and Accumulation of Change, 15–20 percent of AB multiple choice, tied with Unit 5 for the heaviest slice.¹¹⁰⁵

¹¹⁰¹ Mean Value Theorem as named on the CLEP Calculus page (n. 1094) and as an AB contextual/analytical object. Parent ear, not a proof. Do not invent CED topic rows under Unit 4 (n. 1111).

¹¹⁰² NMAP, *Foundations for Success* (2008), Finding 26: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving. Kitchen-math transfer is hard. Full discussion in Chapter 2.

¹¹⁰³ College Board clarifications PDF (n. 1079): Extreme Value Theorem wording, Unit 5 EK FUN-1.C.1. Content unchanged.

¹¹⁰⁴ CCSSM 3.MD area; 5.MD volume of right rectangular prisms; Appendix A Geometry critical area (3): informal arguments for circumference, area, and volume using dissection, Cavalieri, informal limits. Planted in Chapters 5 and 8. Informal-limit arguments become precise here.

¹¹⁰⁵ College Board, AP Calculus AB course page (n. 1079): Unit 6 Integration and Accumulation of Change, 15%–20%

Antiderivatives are the other face of the same operation. An antiderivative of f is a function F whose derivative is f . The indefinite integral is the family of those antiderivatives, plus a constant, because constants vanish when differentiated. Substitution is the chain rule run backward. The computational layer is real again, and it is still not the object. A child who can antidifferentiate x^2 and cannot say what a definite integral of a rate means has the chant.

The Fundamental Theorem of Calculus is the meeting of those two operations, and it is the sentence this section exists to give the parent in two representations. First representation, evaluation: if F is an antiderivative of f , then the definite integral of f from a to b equals $F(b) - F(a)$, when the hypotheses the book names hold. The accumulated change equals the net change of an antiderivative. Second representation, derivative of an accumulator: if $F(x)$ is defined as the integral of f from a fixed a to a variable x , then $F'(x) = f(x)$, again under hypotheses. The rate of change of the accumulated amount is the original rate. Those are not two unrelated formulas. They are one theorem, read as evaluation and read as a derivative.¹¹⁰⁶ The parent who can hear a wrong answer hears “the Fundamental Theorem is $F(b) - F(a)$ ” as half the theorem, and hears “integration is the opposite of differentiation, so I just write the antiderivative” as a slogan that hides both the Riemann-sum meaning and the accumulator-as-function meaning.

Two representations in the same sitting, because that is the door applied to the theorem. A graph of f , shaded from a to x , with $F(x)$ named as that signed area, and F' read as the height of f . A formula, $F(x) = \int$ from 1 to x of $(3t^2 - 2) dt$, and $F'(x) = 3x^2 - 2$, with the chain rule waiting if the upper limit is not x but x^2 . A table of values of f , a Riemann-sum estimate of an accumulation, and then an antiderivative evaluation as a check, not as a replacement. A sentence: “the rate is f ; the amount added between noon and three is the integral; the function that records how much has been added since noon has derivative f .” The child who can evaluate $F(3) - F(1)$ and cannot point at the shaded region, or cannot say what F' is, has one representation. College Board will ask for the others. So should the kitchen.

Differential equations of the form $dy/dx = ky$ live in AB as Unit 7, 5–10 percent: a rate proportional to the current amount, exponential growth and decay, slope fields as a picture of a differential equation, particular solutions given an initial condition.¹¹⁰⁷ The 2026–27 clarification notes that there may be infinitely many solutions to a differential equation; it does not change the content. The parent’s ear: a slope field is a picture of tiny slopes; a solution curve is a graph that follows those slopes; $y = Ce^{kt}$ is a family; an initial condition picks one curve. Do not treat Unit 7 as a later course. Do not treat it as the whole of differential equations. It is the AB slice.

Applications of integration are Unit 8, 10–15 percent: area between curves, volume by slicing and by washers or disks, average value of a function, accumulation in context, arc length as far as AB takes it.¹¹⁰⁸ Volume by slicing is Grade 5’s $V = B \times h$ made precise: the base can change, the limit

MC.

¹¹⁰⁶Fundamental Theorem as evaluation and as derivative of an accumulator. Parent-facing two-representation demand, aligned to mathematical practice (2) (n. 1097). Not a textbook proof.

¹¹⁰⁷College Board, AP Calculus AB course page (n. 1079): Unit 7 Differential Equations, 5%–10% MC. Form $dy/dx = ky$ named in 01-scope and on the CLEP page (n. 1094). Clarifications PDF (n. 1079): “there may be infinitely many solutions to a differential equation,” Unit 7 EK FUN-7.B.2. Content unchanged.

¹¹⁰⁸College Board, AP Calculus AB course page (n. 1079): Unit 8 Applications of Integration, 10%–15% MC. Topic rows below the title **UNVERIFIED** (n. 1111). Area, volume, average value named at unit-title grain only.

of the slices is the integral. Cavalieri's informal argument from Geometry is the ancestor. A child who never saw a slice will experience a washer integral as a formula to memorize. This is not a 180-day plan and not a dump of every application the course page names. It is the parent-facing object: the integral as accumulation, used on area, volume, and average value, sitting on the Fundamental Theorem.

What a parent who stopped at algebra has to be able to hear, stolen as a short list.

"The limit is the value at the point." Wrong when the point is a hole; incomplete even when the function is continuous, because the limit is about approach.

"The derivative is the slope formula." Wrong as a definition. The slope formula is an average; the derivative is a limit of averages.

"It's continuous, so I can differentiate it." Continuity is necessary for differentiability on an interval in the usual school telling; it is not sufficient. Corners, cusps, vertical tangents.

"The integral is the antiderivative." The definite integral is a limit of sums; the antiderivative is the other face; the Fundamental Theorem is why those two faces meet.

"The Fundamental Theorem is $F(b) - F(a)$." Half. The other half is that the derivative of the accumulator is the original function.

"I don't need the graph; I have the formula." College Board's second practice is translation across representations. A kitchen that only computes will meet FRQs that only interpret.

This is not hard algebra. It is a new operation on functions they already know, plus the algebra of those functions under composition, plus the unit circle. Informal-limit arguments in Geometry become precise here. Area and volume become integrals. Average rate of change becomes a derivative. The parent does not have to re-learn a semester of college calculus tonight. The parent has to hear those six sentences as wrong, and to know which object each sentence collapsed. Section 10.3 names the three exams that assess versions of this mathematics. They are different objects. The mathematics in this section is what they overlap on.

10.3 AB is eight units; BC adds two; CLEP is another exam

Three college-facing objects share overlapping mathematics and are not the same assessment. AP Calculus AB is eight units, limits through applications of integration, including simple differential equations of the form $dy/dx = ky$. AP Calculus BC is those eight plus Unit 9, parametric equations, polar coordinates, and vector-valued functions, and Unit 10, infinite sequences and series. CLEP Calculus is a one-semester college-calculus exam, primarily multiple-choice, about 60 percent limits and differential calculus and 40 percent integral calculus on one College Board statement, and about 10 percent limits, 50 percent differential, 40 percent integral on another. Students who take BC receive an AB subscore. CLEP does not produce an AP score. A non-AP home course can match AB content without matching AP exam format. It cannot honestly be called "the same as BC" without Units 9 and 10.¹¹⁰⁹

¹¹⁰⁹College Board AB and BC course pages (n. 1079; n. 1113); CLEP Calculus page (n. 1094); College Board, "Comparing CLEP and AP," <https://clep.collegeboard.org/for-high-schools/comparing-clep-and-ap>, accessed 30 Au-

The eight AB units, with multiple-choice exam weighting as the College Board course page publishes them, accessed 30 August 2026, are a map of what the first college-facing course actually contains, not a 180-day plan.

Unit	Topic	MC weighting
1	Limits and Continuity	10%–15%
2	Differentiation: Definition and Fundamental Properties	10%–15%
3	Differentiation: Composite, Implicit, and Inverse Functions	5%–10%
4	Contextual Applications of Differentiation	10%–15%
5	Analytical Applications of Differentiation	15%–20%
6	Integration and Accumulation of Change	15%–20%
7	Differential Equations	5%–10%
8	Applications of Integration	10%–15%

Those weights are from the AB course page. They are not journalism. They are not a claim about how a kitchen should budget weeks. Units 5 and 6 are the heaviest multiple-choice slices for a reason section 10.2 already named: analytical use of the derivative, and accumulation, are the load-bearing work, not a warm-up before “the real formulas.”¹¹¹⁰ Topic lists *below* those eight titles — the CED’s skill-by-skill rows — were not fully extracted from the 7.63 MB PDF. They stay **UNVERIFIED**. This chapter will not invent a topic catalog under Unit 4 or Unit 8 in order to look complete.¹¹¹¹

Mathematical practices, as the same course page names them, are four: (1) determine expressions and values using procedures and rules; (2) translate mathematical information across representations; (3) justify reasoning and solutions; (4) use correct notation and conventions, free-response only.¹¹¹² Practice 2 is the door, now as an assessed skill. Practice 3 is why a chant is not enough. Practice 4 is why the child writes on paper, in the notation the course uses, without a model inventing a symbol. A kitchen that only drills procedures is practicing one of four. The exam will not be that kind.

AP Calculus BC is AB’s eight units plus two.

gust 2026. Three objects, overlapping mathematics, different assessments. Students who take BC receive an AB subscore.

¹¹¹⁰Weights from the AB course page (n. 1079), accessed 30 August 2026. Not a 180-day budget.

¹¹¹¹AP Calculus full CED, 7.63 MB, not fully extracted (01-scope open question 1; outline open question 10). Unit-level topic lists below the eight and ten titles **UNVERIFIED**. Do not invent a topic catalog.

¹¹¹²College Board, AP Calculus AB course page (n. 1079): four mathematical practices. Practice (4), notation and conventions, FRQ only.

Unit	Topic	MC weighting (BC)
9	Parametric Equations, Polar Coordinates, and Vector-Valued Functions	10%–15%
10	Infinite Sequences and Series	15%–20%

On BC, the AB units are reweighted: Units 1–4 each drop to 5–10 percent multiple choice; Unit 5 drops to 10–15 percent; Unit 6 stays 15–20 percent; Units 7–8 drop to 5–10 percent.¹¹¹³ College Board: BC is an introductory college-level calculus course covering the same big ideas as AB plus the additional units. Students who take BC receive an AB subscore. That subscore is the honest sentence for a parent who is afraid BC will “waste” an AB score. It is not a reason to sit BC without Units 9 and 10. Parametric, polar, and vector-valued functions are a different representation of motion and of curves that are not graphs of $y = f(x)$. Series are a different representation of functions: a power series as a function, Taylor and Maclaurin polynomials as approximations, tests for convergence as the discipline of saying whether an infinite sum has a sum. Those are not “harder AB.” They are more mathematics. A kitchen that has not taught them should not transcript BC.

The 2026–27 clarifications do not change course content. What they do change, as published, is wording: Extreme Value Theorem wording in Unit 5 (EK FUN-1.C.1), and the note that there may be infinitely many solutions to a differential equation (Unit 7 EK FUN-7.B.2). “Course content has not changed.”¹¹¹⁴ Exam multiple-choice timing changes take effect with the May 2027 exams, both AB and BC: 42 multiple-choice questions in 100 minutes, 50 percent of the score — Part A, 29 questions, 62 minutes, no calculator; Part B, 13 questions, 38 minutes, graphing calculator required. Free response: 6 questions, 1 hour 30 minutes, 50 percent. Hybrid digital: multiple-choice and free-response viewed in Bluebook; free-response handwritten in paper booklets.¹¹¹⁵ Those are logistics. They are not a reason to change what is taught. They are a reason to practice the 2027 timing if the child will sit in May 2027 or after. A child who sat in May 2026 used the previous timing; that sitting is already past on this manuscript’s research day of 30 August 2026. A parent reading later should re-check the live exam page rather than treat this paragraph as a 2028 fact.

CLEP Calculus is a different object. College Board: “The Calculus exam covers skills and concepts that are usually taught in a one-semester college course in calculus.” Forty-four questions, about 90 minutes, two sections — one with no calculator, one with an online TI-84 Plus CE. Content, on one College Board statement, is approximately 10 percent limits, 50 percent differential calculus, and 40 percent integral calculus. On another statement on the same family of pages, the exam is “approximately 60 percent limits and differential calculus and 40 percent integral calculus.”

¹¹¹³College Board, AP Calculus BC course page, <https://apcentral.collegeboard.org/courses/ap-calculus-bc>, and BC exam page, <https://apcentral.collegeboard.org/courses/ap-calculus-bc/exam>, accessed 30 August 2026. Units 9–10 and BC reweighting of Units 1–8 as tabled in the body. AB subscore.

¹¹¹⁴Clarifications PDF (n. 1079). EVT wording; infinitely many solutions to a differential equation. “Course content has not changed.”

¹¹¹⁵College Board BC exam page (n. 1113) and AB course page (n. 1079): effective May 2027, both AB and BC — 42 MC, 1 hour 40 minutes, 50% (Part A 29 / 62 minutes, no calculator; Part B 13 / 38 minutes, graphing calculator required); FRQ 6 questions, 1 hour 30 minutes, 50%; hybrid digital (MC and FRQ viewed in Bluebook; FRQ handwritten in paper booklets). Re-check the live exam page after this research day (30 August 2026).

Both statements were opened 30 August 2026. This chapter prints both. It does not pick one and hide the other.¹¹¹⁶ Topics the page names include definitions of the derivative, chain rule, implicit and inverse derivatives, Mean Value Theorem, L'Hospital's Rule, curve sketching, related rates, antiderivatives, substitution, $y' = ky$, Riemann sums, Fundamental Theorem, average value, area. That list is the page's list. It is not a CED. It is not AB's eight units with weights. Series, polar, parametric, and vector calculus are not this exam.

ACE recommends a credit-granting score of 50 for 4 semester hours. Each institution sets its own credit-granting policy. The list price on the Calculus page as of the 30 August 2026 fetch was \$97.¹¹¹⁷ College Board's comparison page: CLEP is prepared for independently; AP follows a designed high-school course. CLEP is primarily multiple-choice. AP includes free-response and is taken after, or alongside, a designed high-school course. CLEP is used for credit toward a two- or four-year degree and for placing out of introductory courses; AP is also used in admission.¹¹¹⁸ Do not claim that AP is always more widely accepted than CLEP as a universal fact. College Board itself says to search each college's policy. Secondary journalism that ranks AP above CLEP for selective colleges is labelled journalism. This chapter will not launder it into a rule.

CLEP is not AP Calculus AB. CLEP is not AP Calculus BC. The overlap is real: limits, derivatives, integrals, the Fundamental Theorem, $y' = ky$, area. The assessments are not. A child who has written six unaided free-response questions under timing has practiced a skill CLEP will not score and AB will. A child who has sat a 44-question multiple-choice exam in 90 minutes has practiced a skill AB's 42-plus-6 structure is not. Credit is set by each college in both cases. A CLEP 50 is not an AP 4. An AP 3 is not a CLEP 50. Search the receiving institution.

A non-AP home course is a fourth object, not a fifth exam. The mathematics can match AB's eight units without matching Bluebook, without matching a graphing-calculator policy, and without matching a May sitting. What the parent cannot drop, if the goal is college Calculus I, is the function concept, algebraic fluency, and trigonometry of the unit circle. What the parent can drop is AP-specific pacing, calculator policy, and free-response format. What the parent cannot honestly write on a transcript is "AP Calculus BC" for a course that never reached series, or "the same as BC" for a course that matched AB and stopped. Whether named homeschool calculus titles actually assign AB content — Saxon Calculus second edition's table of contents; Art of Problem Solving Calculus; a community-college Calculus I catalog sample — was **not opened** in this project. That gap stays open. Do not promote a vendor row to handbook fact.¹¹¹⁹ Math-U-See's named Calculus level,

¹¹¹⁶CLEP Calculus page (n. 1094). 44 questions, ~90 minutes, two sections (no calculator / online TI-84 Plus CE). One statement: ~10% limits, ~50% differential, ~40% integral. Another: "approximately 60% limits and differential calculus and 40% integral calculus." Both opened 30 August 2026. Print both.

¹¹¹⁷CLEP Calculus page (n. 1094): ACE recommended credit-granting score 50 for 4 semester hours; "each institution reserves the right to set its own credit-granting policy." List price \$97 as of the 30 August 2026 fetch.

¹¹¹⁸College Board, "Comparing CLEP and AP" (n. 1109). CLEP prepared for independently; AP follows a designed high-school course. CLEP primarily multiple-choice; AP includes free-response. Do not claim AP is always more widely accepted than CLEP; College Board says to search each college's policy. Secondary journalism that ranks AP above CLEP for selective colleges is labelled journalism.

¹¹¹⁹Non-AP homeschool calculus tables of contents — Saxon Calculus 2nd ed.; Art of Problem Solving Calculus; a community-college Calculus I catalog sample — not opened (01-scope open question 12). Math-U-See Calculus level one-line, **vendor**, Demme Learning sequence page, <https://mathusee.com/products/math-u-see-curriculum/>, accessed 30 August 2026. Khan Academy AP Calculus AB/BC courses exist as a free map; live course-list detail **UNVERIFIED-in-detail** (01-scope). Do not promote closed vendor rows to handbook fact.

vendor, lists derivatives, integrals, applications, and differential equations in a one-line description. That is a vendor sentence, not a CED alignment. Khan Academy offers AP Calculus AB and BC courses as a free map; useful as practice, not as a researched curriculum outcome.

The live debate this section will not pretend to settle belongs on the page so it is not settled by a franchise. AB versus BC versus non-AP versus CLEP versus dual-enrollment Calculus I — different objects, overlapping mathematics, college-by-college credit. A student with a solid function concept and a year for eight units is looking at AB or a non-AP course that matches those units, or at dual-enrollment Calculus I. A student who has those eight and a second year, or a compacted year that honestly includes Units 9 and 10, is looking at BC. A student who needs a one-semester college-calculus credit and whose receiving colleges take CLEP is looking at CLEP, knowing it is not an NCAA core course — plant Chapter 12; do not dump the worksheet. A student whose parent is the ceiling is looking at dual enrollment, because the community-college transcript is an object NCAA and admissions offices already know how to read. Statistics is not on that list as a substitute. Chapter 11 is that refusal.

What this section has done is separate the objects. Pathways and credit come next, because Appendix A is explicit about when senior-year calculus is reachable, and because none of these scores is automatic credit.

10.4 Pathways, credit, NCAA, and AP logistics

Credit is not automatic. That sentence is the first teaching move of this section, and it applies to every object section 10.3 named. AP credit is not automatic. CLEP credit is not automatic. Dual-enrollment credit is not automatic. Each receiving college sets policy. College Board says to search each college's policy. This chapter will not invent a ranking.¹¹²⁰ NCAA is a private association's eligibility rules, not law, but it binds athletic scholarships, and it does not take CLEP as a core course. AP seats are school-controlled. Parents cannot order exams. Those are the load-bearing facts. The full records chapter is Chapter 12. This section is the slice that changes which course the kitchen should attempt.

Appendix A is explicit about the calendar. The regular three-year pathway — Algebra I, Geometry, Algebra II, or Integrated Mathematics I–II–III — plus a fourth course reaches precalculus as a junior and calculus as a senior *only if* high-school content started in Grade 8. Compacted versions of each pathway complete Grade 7, Grade 8, and the first high-school course in Grades 7–8 so that a student can reach Calculus or another college-level course by senior year. “The pathways and courses are models, not mandates.”¹¹²¹ A non-accelerated student who starts Algebra I in Grade 9 finishes Algebra II in Grade 11 and has senior year for precalculus, statistics, or a first calculus course — not both precalculus and BC unless the fourth course is overloaded. Overload is not a plan. Compacted 7–8 is a model so calculus is reachable senior year. It is not a mandate to compact. Compact only if the gates are open: fractions including signed fractions, integer operations, the equal sign as a relation, the function concept, unit-circle trigonometry. NMAP recommended that all *prepared* students have access to an authentic algebra course by Grade 8; “authentic” means

¹¹²⁰Each receiving college sets AP, CLEP, and dual-enrollment credit policy. College Board credit-policy search, referenced from the AP course pages. Do not invent a ranking.

¹¹²¹Appendix A (n. 1078). Models, not mandates.

the Major Topics of School Algebra, not a course that is algebra in name only.¹¹²² Finding 15 still refuses the age lock. Finding 30 still allows prepared, motivated students to move faster, with no harm to their learning. Neither finding licenses a skip of the function concept because a child is “good at math.”

Grade labels remain sequence anchors, not prison bars. NCES 2016: Algebra I was taught *that year* to 41 percent of the homeschooled 6–8 band; arithmetic was still taught to 31 percent of the 9–12 band. High-school ever-taught: Algebra I 79 percent, geometry 52, Algebra II 49, calculus 15 with the caution this chapter already installed.¹¹²³ Taking Algebra I in Grade 8 *and* listing it as a high-school credit is a record-keeping choice with NCAA implications. Pre-ninth coursework rules live in the current Home School Toolkit; read it; do not invent a rule from a blog. Chapter 12 will walk the worksheet. This chapter only needs the parent to see that a compacted path is how senior-year calculus becomes possible on a CCSSM-aligned map, and that a child still in signed fractions at calendar grade 9 is not “behind a law.” That child is at a gate.

AP logistics, from College Board’s Homeschoolers’ Guide (Jacob Mishook, 29 June 2026), are narrower than internet advice and more useful. Homeschoolers may self-study; a formal AP course is recommended, not required. Parents **cannot order exams**. Find a school via the AP Course Ledger. The AP coordinator enrolls the student in an “exam only” My AP section. Mid-November order deadline. Homeschool educators may submit courses to the AP Course Audit to become authorized providers.¹¹²⁴ Sit at an authorized school. Seats are school-controlled. Do not promise that AP self-study yields a May seat. Do not invent a 2026 homeschool ordering process beyond that guide. Homeschool-specific access mechanics — which schools accept exam-only homeschoolers in which counties, what a coordinator may charge, whether a virtual school can order — remain **UNVERIFIED** as a registration recipe in this project. The guide is the recipe. Stop there.¹¹²⁵ SAT registration, for the parent who needs the adjacent object, uses homeschool high-school code 970000. PSAT is an “away student” arranged at a school at least four weeks ahead. Those are College Board assessment facts, not calculus facts. They sit here only so the November AP deadline is not confused with an SAT Saturday.

A course labelled “AP” on a homeschool transcript is a different fact from a May score. The AP

¹¹²²NMAP, *Foundations for Success* (2008), recommendation that all *prepared* students have access to an authentic algebra course by Grade 8; “authentic” means the Major Topics of School Algebra (n. 1090). Finding 15: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; age-lock claims have “consistently been shown to be wrong.” Finding 30: prepared, motivated students “appear to be able to learn mathematics much faster... with no harm to their learning, and should be allowed to do so.” Neither finding licenses a skip of the function concept.

¹¹²³NCES 2020-001, Tables 8 and 9 (n. 1080). Algebra I taught that year: 41 percent of homeschooled 6–8; arithmetic still taught to 31 percent of 9–12. Ever-taught at high-school level: Algebra I 79 percent, geometry 52, Algebra II 49, calculus 15 (caution). Coverage, not quality.

¹¹²⁴College Board, “The Homeschoolers’ Guide to College Board Assessments,” Jacob Mishook, 29 June 2026. AP: parents cannot order exams; find a school via the AP Course Ledger; coordinator enrolls “exam only”; mid-November order deadline; self-study allowed; a formal AP course recommended not required; homeschool educators may submit courses to the AP Course Audit. SAT homeschool high-school code 970000; PSAT as “away student” arranged at a school at least four weeks ahead.

¹¹²⁵Homeschool-specific AP exam-access mechanics beyond the 29 June 2026 guide — which schools accept exam-only homeschoolers, fees, virtual-school ordering — **UNVERIFIED** as a registration recipe (01-scope open question 6; outline open question 16). Do not invent a recipe. Seats are school-controlled.

Course Audit is how a school, including a homeschool educator who submits, becomes authorized to use the AP designation. A family that self-studies from an AB book and sits the exam as “exam only” has a score, if a seat was found, and a course that should be transcribed honestly — Calculus, or AP Calculus AB if the Audit was done, not “AP Calculus BC” for eight units. Chapter 12’s title rule: a stranger should be able to map the line onto Algebra I, Geometry, Algebra II, Precalculus, Calculus, Statistics. This chapter plants that rule. It does not write the transcript template.

CLEP logistics are the other College Board door. Register via My CLEP. No school enrollment required. Year-round testing. Thirty-four exams, including College Algebra and Calculus. Modern States is named on College Board pages as a partnership for free prep and an exam voucher; treat that as College Board’s named partnership, not as this book’s endorsement of a vendor.¹¹²⁶ ~2,900 colleges grant CLEP credit, College Board’s figure as of the pages opened. Each of those colleges still sets its own score and hour policy. ACE’s 50 / 4 semester hours is a recommendation, not a transcript line.

NCAA, from the Home School Toolkit 2025–26 and the Division I initial-eligibility page, not from blogs. A core course is college-preparatory high-school work in English, math (Algebra I or higher), science, social science, world language, comparative religion, or philosophy. Division I: 16 core courses including **3 years math (Algebra I or higher)**; 10 core courses, 7 in English, math, or science, before the seventh semester; minimum **2.3** core GPA. Division II: 16 core including **2 years math (Algebra I or higher)**; minimum **2.2** core GPA. Homeschool packet: transcript, Administrator and Accordance Statement, **one Core-Course Worksheet per core course**. Audited, CLEP, or credit-by-exam courses are **not** NCAA-approved core courses. Evaluation occurs only after an NCAA school requests it.¹¹²⁷ Book implication, stolen as a sentence: a kitchen calculus course taught from a textbook can count if the Core-Course Worksheet is honest and the course is college-preparatory. A CLEP Calculus pass cannot substitute for that core course. Dual-enrollment Calculus I from the community college is a college transcript, which NCAA already knows how to read, and which it treats differently — and often more cleanly — than a parent-graded course. Do not promise NCAA eligibility from CLEP, from an audited sit-in, or from a course titled “fun with numbers.” Math core starts at Algebra I. Calculus sits above that floor. CLEP sits outside it.

Dual enrollment is “enrollment in a class that offers both high school and college credit,” as NCES defines it. NCES Data Point 2020-125, December 2020, from the 2017–18 National Teacher and Principal Survey: **82 percent** of public schools with any of grades 9–12 offered dual or concur-

¹¹²⁶ CLEP Calculus page (n. 1094) and Homeschoolers’ Guide (n. 1124): register via My CLEP; no school enrollment required; 34 exams; Modern States named as a partnership for free prep and an exam voucher — College Board’s named partnership, not this book’s endorsement. ~2,900 colleges grant CLEP credit, College Board’s figure on the pages opened 30 August 2026; each still sets policy.

¹¹²⁷ NCAA Division I initial-eligibility requirements; *Home School Toolkit 2025–26*; Core-Course Worksheet (July 2025). Core course = college-preparatory high-school work in English, math (Algebra I or higher), science, social science, world language, comparative religion, or philosophy. Division I: 16 core including 3 years math (Algebra I or higher); 10 core (7 in English/math/science) before the seventh semester; minimum 2.3 core GPA. Division II: 16 core including 2 years math (Algebra I or higher); minimum 2.2 core GPA. Homeschool packet: transcript, Administrator and Accordance Statement, one Core-Course Worksheet per core course. Audited, CLEP, or credit-by-exam courses are **not** NCAA-approved core courses. Evaluation only after an NCAA school requests it. A kitchen calculus course taught from a textbook can count if the worksheet is honest. A CLEP pass cannot substitute for the core course. Dual-enrollment Calculus I is a college transcript NCAA already knows how to read. NCAA is a private association’s eligibility rules, not law.

rent enrollment. Among those schools, funding was provided most often by the school, district, or state (78 percent), then family or student (42 percent). Schools with 75 percent or more of students eligible for free or reduced-price lunch were less likely to offer it (71 percent versus 84–93 percent).¹¹²⁸ That is **public-school supply**, not a homeschool participation rate. IPEDS’s first collection, 2022–23, reported on IES project pages and in Community College Daily journalism: nearly **2.5 million** dual-enrolled students; about one-third of all 11th and 12th grade public high-school enrollments; 73.5 percent (1.8 million) at public community colleges. Journalism labelled journalism for the CCD article; the IES page is a federal project description.¹¹²⁹ Homeschool-specific dual-enrollment counts are **not in these sources**. The IES dual-enrollment outcomes study’s first report is expected in 2027. This chapter will not invent a homeschool dual-enrollment rate, and it will not invent a finding that dual-enrollment Calculus I “works better than AP” as a national experiment. Admissions offices vary on whether dual-enrollment calculus is “as rigorous as AP.” Named-university pages in the research — SMU among them — say dual-credit, AP, IB, and historical SAT Subject scores help them measure ability when the homeschool transcript is otherwise opaque. Pattern, not a rule: three years of high-school mathematics through Algebra II is the floor most of the named pages want; selective STEM programs will want precalculus or calculus **plus** an external signal — AP, dual enrollment, SAT math. The kitchen that stops at Algebra I because “the state doesn’t require it” has a college problem in every named example in Chapter 12, whether or not a calculus book is on the shelf.¹¹³⁰

Whether statistics can substitute is the last pathway question this section will answer, and the answer is no for a calculus-based college STEM sequence. AP Statistics, effective Fall 2026, is a one-semester, introductory, **non calculus-based** college course in statistics. The published prerequisite is successful completion of a first-year algebra course. College Board’s revised framework encourages taking both AP Statistics and AP Calculus when the student has the background. It is not published as a substitute for calculus in STEM calculus sequences.¹¹³¹ A parent choosing AP

¹¹²⁸NCES, *Dual or Concurrent Enrollment in Public Schools in the United States*, NCES 2020-125 (December 2020), <https://nces.ed.gov/pubs2020/2020125.pdf>, accessed 30 August 2026. 82 percent of public schools with any of grades 9–12 offered dual or concurrent enrollment, 2017–18 National Teacher and Principal Survey. Funding: school/district/state 78 percent; family/student 42 percent. Schools with ≥ 75 percent free/reduced-price lunch less likely (71 percent vs 84–93 percent). **Public-school supply**, not a homeschool participation rate.

¹¹²⁹IES/NCEE, Outcomes Associated with Dual Enrollment Programs, project page (first report expected 2027), <https://ies.ed.gov/use-work/evaluations/outcomes-associated-dual-enrollment-programs>, accessed 30 August 2026. Community College Daily / AACC, “DataPoints: New data on dual enrollment” (August 2024), reporting IPEDS 2022–23: nearly 2.5 million dual-enrolled students; about one-third of 11th and 12th grade public high-school enrollments; 73.5 percent (1.8 million) at public community colleges — **journalism** for the CCD article; IES page is a federal project description. Homeschool-specific dual-enrollment counts: **not in these sources**.

¹¹³⁰Named university pages in 03-homeschool (Pitt: 3 years mathematics; SMU: suggested Algebra I, Geometry, Algebra II; North Central: 3 years including algebra and geometry, one course beyond geometry preferred). Pattern, not a rule: three years through Algebra II as a floor; selective STEM programs want precalculus or calculus plus an external signal. Plant for Chapter 12. Do not dump the transcript template.

¹¹³¹College Board, *AP Statistics Course and Exam Description*, Effective Fall 2026, <https://apcentral.collegeboard.org/media/pdf/ap-statistics-course-and-exam-description.pdf>, accessed 30 August 2026: “equivalent to a one-semester, introductory, **non calculus-based** college course in statistics”; prerequisite “any secondary school student who has successfully completed a **first-year algebra course**.” Revised course framework, <https://apcentral.collegeboard.org/media/pdf/ap-statistics-revised-course-framework.pdf>, partially opened 30 August 2026: encourages both AP Statistics and AP Calculus when background allows. Not published as a calculus substitute. Plant for Chapter 11. Do not write Chapter 11 here.

Statistics *instead of* precalculus or calculus is choosing a different college-equivalent. That is a legitimate choice for a non-STEM path. It is a weak stealth substitute for Calculus I if the destination is Physics C or a calculus-based engineering sequence. Chapter 11 is that chapter. Plant it. Do not write it.

What this section has done is put the course choice inside a pathway and a paperwork constraint. Teaching the course at home comes next: representations, fading, unaided paper. The seat and the score do not teach the derivative.

10.5 Teaching guidance: representations, fading, unaided paper

The first pass through a new technique is a worked example, faded, then the child generates. That sentence is Chapter 2's lock applied to a difference quotient, and it is the sentence this section will not upgrade into a calculus RCT. Sweller and Cooper 1985 is the worked-example effect for novices in algebra. Renkl and Atkinson 2003 is guidance fading. Kalyuga, Ayres, Chandler, and Sweller 2003 is the expertise-reversal effect: the same worked example that helps a novice can hurt a relative expert.¹¹³² Full discussion in Chapter 2. This chapter applies the sequence. First: a complete solution of one limit, one derivative of a composition, one definite-integral interpretation, visible, with the reason for each line. Then: the same structure with a step missing, the child filling the step. Then: the child generates. Do not start the first chain-rule day with six unworked compositions. Do not keep writing every line for a child who can already execute. Fade for accuracy, not for a calendar date.

Incorrect examples belong in the same first pass. Booth, Lange, Koedinger, and Newton 2013 is an Algebra I Cognitive Tutor trial: explaining incorrect examples, alone or with correct ones, was especially useful for conceptual understanding of two-step linear equations. The misconceptions that paper was built to hit — equal sign as “where the answer goes,” minus as “subtract” rather than as a modifier of terms, variable as a single mystery number — are not calculus misconceptions. The *method* transfers as a method, labelled as such. The coefficient does not.¹¹³³ Build one wrong derivative: the child who differentiates $\sin(3x^2)$ as $\cos(3x^2)$, missing the chain rule. Build one wrong limit: “does not exist because I got 0/0.” Build one wrong integral: the antiderivative chant with no $+ C$ on an indefinite integral, or, worse, $F(b) - F(a)$ computed on a function that is not an antiderivative of the integrand. The child explains why the line is wrong. Then generates a correct one. The parent who stopped at algebra can hear those three errors without being able to teach BC. Hearing is the thesis's first condition. Teaching BC is a different job.

Move among formula, graph, table, and verbal in the same sitting. That is the door diagnostic, now

¹¹³²John Sweller and Graham A. Cooper, “The Use of Worked Examples as a Substitute for Problem Solving in Learning Algebra,” *Cognition and Instruction* 2, no. 1 (1985): 59–89. Alexander Renkl and Robert K. Atkinson, “Structuring the Transition from Example Study to Problem Solving in Cognitive Skill Acquisition: A Cognitive Load Perspective,” *Educational Psychologist* 38 (2003): 15–22. Slava Kalyuga, Paul Ayres, Paul Chandler, and John Sweller, “The Expertise Reversal Effect,” *Educational Psychologist* 38 (2003): 23–31. Full discussion in Chapter 2. This chapter applies the sequence to a first pass through a new calculus technique. Not a calculus RCT.

¹¹³³Julie L. Booth, Karin E. Lange, Kenneth R. Koedinger, and Kristie J. Newton, “Using Example Problems to Improve Student Learning in Algebra: Differentiating between Correct and Incorrect Examples,” *Learning and Instruction* 25 (2013): 24–34, <https://files.eric.ed.gov/fulltext/ED543090.pdf>, accessed 30 August 2026. Algebra I Cognitive Tutor; misconceptions named in Chapter 7. Method transfers as a method, labelled as such. The coefficient does not.

as daily practice. College Board's second mathematical practice is the assessed version of the same demand. A session that only computes will produce a child who can differentiate x^3 and cannot read f' from a graph of f , or who can evaluate $F(3) - F(1)$ and cannot point at a shaded accumulation. Interleave representations the way Chapter 6 interleaved a ratio table, a graph through the origin, and $y = kx$. Do not wait for "graph week." A *hypothetical*, labelled as such: a parent who runs a page of power-rule items on Monday, a page of graph-reading on Tuesday, and a page of verbal related-rates on Wednesday, and never puts the same function in all three representations on one day, should not be surprised when Thursday's mixed three-item check falls apart.¹¹³⁴ The empirical piece is Practice 2 plus the door. The days of the week are hypothetical.

A non-AP home course can match AB content without matching AP exam format. It should still match the representations. What it can drop: Bluebook, a May sitting, a graphing-calculator split that the family is not sitting. What it should not drop: the limit as approach, the derivative as a limit of average rates, the integral as accumulation, the Fundamental Theorem in two representations, $dy/dx = ky$, area and volume as integrals. What it cannot honestly add on the transcript is Units 9 and 10 it did not teach. Practice the format the child will actually meet. If the child will sit AB, write free-response on paper, timed, unaided. If the child will sit CLEP, practice 44 multiple-choice items in about 90 minutes, two calculator policies. If the child will sit dual-enrollment Calculus I, practice what that instructor grades. Do not practice a format from a franchise blog.

Interleaving in high-school calculus is a practice suggestion with a hard caveat. Rohrer, Dedrick, and Burgess 2014: grade-7 classroom crossover, $n = 140$, unannounced test two weeks later, interleaved 72 percent versus blocked 38 percent, $d = 1.05$. Rohrer 2020: 61 percent versus 38 percent, $d = 0.83$, 54 classes, joint with spacing. Almost all clean classroom randomized trials of interleaving in mathematics are grade 7. Do not import $d = 1.05$ onto an AP free-response question. Do not mash the 2014 d with the 2020 d with Dunlosky's "high utility" label for spacing.¹¹³⁵ What survives as a kitchen move, labelled as transfer of a finding and not as a coefficient sitting on the table: after a short blocked introduction to a new technique — three chain-rule items, not thirty — mix yesterday's chain rule with last week's implicit derivative and last month's limit that is not a derivative. The child has to *choose* a strategy, not only execute one. Blocked practice teaches execution. Mixed practice teaches choice. The d stays in grade 7.

When the parent is the ceiling, dual-enrollment Calculus I is the cleanest transcript. That sentence is a teaching rule, not a failure of the kitchen. NCES 2016: any tutor 23 percent; any local homeschool group or co-op 31 percent. Those are release valves, not a confession.¹¹³⁶ Maloney, Ramirez, Gunderson, Levine, and Beilock 2015 is still the school-homework study this book will not upgrade:

¹¹³⁴Hypothetical labelled in the sentence that introduces it. Empirical pieces: College Board practice (2) (n. 1097); door diagnostic (n. 1087). The days of the week are hypothetical.

¹¹³⁵Doug Rohrer, Robert F. Dedrick, and Kaleena Burgess, "The Benefit of Interleaved Mathematics Practice Is Not Limited to Superficially Similar Kinds of Problems," *Psychonomic Bulletin and Review* 21 (2014): 1323–1330, <https://files.eric.ed.gov/fulltext/ED548041.pdf>, accessed 30 August 2026. Grade-7 classroom crossover, $n = 140$, unannounced test two weeks later, interleaved 72% vs blocked 38%, $d = 1.05$, 95% CI [0.80, 1.30]. Doug Rohrer, Robert F. Dedrick, and Pooja K. Agarwal, 2020 classroom study: 61% vs 38%, $d = 0.83$, 54 classes, joint with spacing. Almost all clean classroom RCTs of interleaving in mathematics are grade 7. Do not import $d = 1.05$ onto an AP FRQ. Do not mash the 2014 d with the 2020 d with Dunlosky "high utility" for spacing. Full objects in Chapter 2 and Chapter 6.

¹¹³⁶NCES 2020-001: any tutor 23 percent; any local homeschool group or co-op 31 percent. Release valves, not a failure of the kitchen.

parent math anxiety times frequent help predicted worse child math and more child math anxiety; when help was infrequent, no such relation; math-specific, not reading. Not a parent-as-sole-teacher RCT. A homeschool parent cannot “help less often” as a daily strategy. Do not treat the finding as “anxious parents should not teach.” Replication status open.¹¹³⁷ Video curricula exist because of Maloney. Keep the parent as the supervisor who can still hear a wrong derivative, not as the person who “explains it the way I learned it in 1998.” Teaching Textbooks **vendor** “does 100% of the teaching” is a vendor claim, relevant to Maloney, not a finding that the parent can stop hearing wrong answers. Do not claim a parent who stopped at algebra casually teaches BC. BC is eight units plus parametric, polar, vector, and series. The design user can learn to hear a wrong chain rule. The design user should not be told that hearing is the same as teaching Unit 10.

Do not let a language model be the computer-algebra system. Wolfram|Alpha is the check: verified algorithms, not next-token math. A language model is the explainer for the parent. Section 10.7 is the split. This section only needs the teaching reason: a fluent wrong antiderivative, copied from a chat window, is worse than a blank page, because the child will believe the fluency. Bastani’s GPT Base was fully correct 51 percent of the time on high-school items, with 8 percent arithmetic errors even when the method was right. Facts and fluent algebra are not an LLM job.¹¹³⁸ The child executes the derivative by hand. Then, after the attempt, Wolfram checks the result. Then the parent, if needed, asks the model why the chain rule wanted that inner derivative — parent account, child not present, or child present only after the attempt.

Every AP free-response question the child will meet must be written without a model, on paper, timed. College Board still requires unaided work in the exam room. Hybrid digital, May 2027 and after: the prompt is on a screen; the writing is in a paper booklet. Practice that split if that is the sitting. Practice suggestion, not an RCT: a timed unaided FRQ slice is Bastani’s exam phase applied to this course. The model may afterwards, on the parent account, mark against a rubric the parent pasted from an official free-response scoring guideline — and the parent still samples the marking, because models hallucinate theorems and drop negative signs.¹¹³⁹ Ban “do this FRQ.” That prompt, child-facing, is paste-the-worksheet in a new font. Section 10.7 will say it again as an AI rule. This section says it as a teaching rule. The child writes. The parent holds the key.

Session shape, Chapter 3 §3.8, scaled to this course. Practice suggestion, not an RCT. Unaided

¹¹³⁷Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, “Intergenerational Effects of Parents’ Math Anxiety on Children’s Math Achievement and Anxiety,” *Psychological Science* 26 (2015): 1480–1488, <https://doi.org/10.1177/0956797615592630>. School-homework study: parent math anxiety $\times$ frequent help predicted worse child math and more child math anxiety; when help was infrequent, no such relation; math-specific, not reading. Not a parent-as-sole-teacher RCT. Homeschool parent cannot “help less often” as a daily strategy. Do not treat the finding as “anxious parents should not teach.” Replication status open. Teaching Textbooks **vendor** “does 100% of the teaching” is a vendor claim, relevant to Maloney, not a finding that the parent can stop hearing wrong answers.

¹¹³⁸Bastani et al. 2025 (n. 1145): GPT Base fully correct 51%; 8% arithmetic errors even when the method was right. Wolfram|Alpha, <https://www.wolframalpha.com/>, fetched 30 August 2026: “verified, expert-level answers using Wolfram’s breakthrough algorithms and knowledgebase,” built on Wolfram Language. Computational engine, not an LLM. Pro step-by-step is vendor; current Pro pricing **UNVERIFIED** this day. A language model is the explainer for the parent. Do not let a language model be the CAS.

¹¹³⁹College Board still requires unaided student work in the AP exam room (n. 1115). Practice suggestion, not an RCT: every FRQ written without a model, on paper, timed; model may afterwards, parent account, mark against a parent-pasted official rubric; parent still samples the marking. Ban “do this FRQ.” Chapter 3 §3.7.

warm-up: one limit, one derivative of a composition, one definite-integral interpretation. Teach from the family's book, the parent already having used a parent-first explanation before sitting down. Struggle. Hint not key. Timed unaided FRQ slice or a mixed three-item check without the model. "Finish your calculus with Photomath so we can go to soccer" is GPT Base, the pattern with a large-sample causal harm estimate.¹¹⁴⁰ If Khanmigo is in the house, expect the Tennessee pattern unless the parent requires the hint dialogue. Oreopoulos and Low, NBER w35620, August 2026: intention-to-treat 0.040 population SD per term; do not invent a standalone 0.06 as ITT. Gains resemble Khan Academy practice without AI assistance. Working paper, not yet journal-peer-reviewed.¹¹⁴¹

What this section has done is put the method on the objects of section 10.2. Tips come next, condensed, stealable, then the AI split that calculus makes unavoidable.

10.6 Tips that survive: objects, titles, no overload fantasy

Stealable sentences, not a second teaching section.

Do not start calculus on a fragile function concept. Diagnose four representations, compositions, and the unit circle before opening Unit 1. A book titled Precalculus is a purchase, not a diagnosis.

Do not call a non-AP home course "the same as BC" without Units 9 and 10. AB is eight units. BC is those eight plus parametric, polar, and vector-valued functions and infinite sequences and series. Matching AB content is honest. Matching AB content and writing BC on the transcript is not.

Do not treat the 2016 15 percent ever-taught figure as a completion rate, as college calculus, or as a 2026 headcount. Interpret with caution. No named nationally representative source gives STEM-major or college-calculus completion rates for homeschool graduates.

A non-accelerated student who starts Algebra I in Grade 9 has senior year for precalculus, statistics, or a first calculus course — not both precalculus and BC unless the fourth course is overloaded. Compacted 7–8 is a model so calculus is reachable senior year. It is not a mandate. Overload is not a plan.

Transcript the course as Calculus, or AP Calculus AB, or AP Calculus BC, or dual-enrollment Calculus I, not "Math 12," and not "Saxon Calculus" without a parenthetical a stranger can map. Chapter 12 is the title rule. This chapter plants it.¹¹⁴²

¹¹⁴⁰Session shape: Chapter 3 §3.8 / 04-ai §9.4, scaled to this course. Practice suggestion, not an RCT. Grounded in Bastani's exam phase (n. 1145) and Soderstrom and Bjork (n. 1144).

¹¹⁴¹Philip Oreopoulos and Corinne Low, "Making a Tutor: How to Design and Evaluate an AI Tutor for Mathematics," NBER Working Paper 35620 (August 2026). Two-year cluster RCT, 18 Tennessee middle schools. Assignment to Khan Academy + Khanmigo: 1.26 national percentile ranks per term (SE 0.60); about **0.040 population SD per term**; about 0.06–0.08 SD over a school year. Authors: "These gains resemble those from Khan Academy practice without AI assistance." 96% tried Khanmigo; median messaged on a third of practice days and in 17% of mistake sessions; 14.5% of messages contained a mathematical question or a step of reasoning. Do not invent standalone 0.06 as ITT. Working paper, not yet journal-peer-reviewed. Full discussion: Chapter 3 §3.5.

¹¹⁴²Course-title rule planted for Chapter 12: a stranger should map the line onto Algebra I, Geometry, Algebra II, Precalculus, Calculus, Statistics — not "Math 12." NCAA example transcript and named college pages (n. 1127, n. 1130). Do not dump the template.

AP credit is not automatic. CLEP credit is not automatic. Dual-enrollment credit is not automatic. Each college sets policy. Do not claim AP is always more widely accepted than CLEP.

CLEP is not AB. CLEP is not BC. CLEP is not an NCAA core course. A kitchen calculus course taught from a textbook can count for NCAA if the Core-Course Worksheet is honest. Dual-enrollment Calculus I is a college transcript NCAA already knows how to read. Plant Chapter 12. Do not dump the packet.

AP seats are school-controlled. Parents cannot order exams. Self-study is allowed; a formal AP course is recommended, not required. Find a school via the AP Course Ledger. Mid-November order deadline. Do not invent a registration recipe beyond the 29 June 2026 Homeschoolers' Guide.

Do not invent CED topic lists below the eight and ten unit titles. The full AP Calculus CED was not fully extracted. Topic-level tables stay **UNVERIFIED**.

Do not invent a teaching sequence from Tall and Vinner 1981. The paper was not opened.

Do not claim a parent who stopped at algebra casually teaches BC. The design user can learn to hear a wrong derivative. Hearing is not Unit 10. When the parent is the ceiling, hire a tutor, join a co-op, or enroll in dual-enrollment Calculus I. Those are release valves. They are not a confession.

Statistics does not substitute for this chapter in a calculus-based STEM sequence. Calculus does not substitute for statistics as a data-reasoning course. Chapter 11 is that refusal.

Do not let a language model be the CAS. Wolfram checks. The model explains to the parent. The child writes the FRQ unaided.

The AP Calculus full CED topic-level tables, the official prerequisite sentence, non-AP homeschool calculus tables of contents, homeschool-specific dual-enrollment counts, and STEM-major rates remain open questions. Do not close them with a kitchen scene.

10.7 Parent-hosted AI for calculus

Wolfram|Alpha is the right **check** for an antiderivative, a limit, a series test — verified algorithms, not next-token math. A language model is the right **explainer for the parent** — why integration by parts, why this u , why the chain rule wanted the inner derivative. Mixing the two without a label is how a hallucinated integral survives a parent's glance.¹¹⁴³ That split is Chapter 3 §3.7 applied to this course, and it is the sentence this section exists to make unavoidable. Calculus is the first kitchen course in which a fluent wrong answer looks like a textbook. The child who copies a chat-window antiderivative that dropped a factor, or a series test that named the wrong test, has Bastani's crutch plus a typeset identity. The parent who treats that fluency as a check has used a language model as a computer-algebra system. Do not.

Chapter 3 already printed the kitchen condition. This section will not rewrite Bastani, Soderstrom and Bjork, or the ban-versus-allow list. It will reuse them on limits, derivatives, and integrals,

¹¹⁴³Wolfram|Alpha (n. 1138); Chapter 3 §3.7; 04-ai §4.4. Mixing a language model and a CAS without a label is how a hallucinated integral survives.

labelled as practice suggestions, not as a homeschool RCT, which still does not exist.¹¹⁴⁴ One sentence if the family remembers only one: the model may prepare the adult and the next problem; it may not do the child's problem. The derivative is the child's. A CAS check comes after.

Bastani, Bastani, Sungu, Ge, Kabakcı, and Mariman, *PNAS* 2025: GPT Base raised assisted practice grades 48 percent relative to control, then cut unaided exam grades 17 percent. GPT Tutor raised practice 127 percent and was statistically indistinguishable from control on the unaided exam (point estimate -0.004). GPT Base, asked for the answer, was fully correct 51 percent of the time, with 8 percent arithmetic errors even when the method was right. Students copied anyway. They did not perceive that they had learned less. Classroom-level RCT, large high school in Turkey, roughly a thousand students in grades 9–11, four ninety-minute sessions, GPT-4-0613. Not a homeschool study. Not GPT-5. Not Khanmigo. Percentages are of control grades, not Cohen's d .¹¹⁴⁵ Full discussion: Chapter 3. The mechanism is the transferable object: access to the solution during acquisition, copying, a metacognitive miss, a closed test. Photomath, Mathway, Symbolab, and consumer ChatGPT are the same mechanism as GPT Base unless the parent constrains them. That sentence is an analogy of mechanism, not a Photomath-named RCT. No product-specific unaided-transfer trial of those solvers was opened on 30 August 2026.

Allow, parent-hosted, pointing at Chapter 3's workflows.

(a) Explain the concept to the parent first. What the Fundamental Theorem says in two representations. Why a limit is not plugging in. Why a derivative is a limit of average rates and not “the slope formula.” Why the chain rule is composition, not a product. Why $y' = ky$ is a family of exponentials. Parent account, no child present. The prompt pattern Chapter 3 already stole: explain it to *me* as an adult who last took math in [year]; the idea in one paragraph; the notation; the two mistakes this age makes; one tiny problem I should solve before I sit down; the answer under a parent key. Then the parent solves the tiny problem without the model, then checks. The check on a computation is Wolfram|Alpha, labelled as the CAS, or the textbook's known-correct key. The check on an explanation is whether the parent can now hear a wrong sentence at the table. A *hypothetical* parent-first prompt, labelled as such: “I last took algebra in 1998. In two short paragraphs, what does the Fundamental Theorem of Calculus say as an evaluation, $F(b) - F(a)$, and as a derivative of an accumulator? One graph I should be able to point at. One formula. The two mistakes a first-year student makes. One tiny problem I should do on paper. Put the answer

¹¹⁴⁴Parent workflows in this section are practice suggestions grounded in Bastani (n. 1145), Soderstrom and Bjork, and Oreopoulos and Low (n. 1141), not a new RCT of homeschool AI. No year-plus kitchen-table RCT of a parent-constrained GPT-Tutor workflow was opened on 30 August 2026. Nicholas C. Soderstrom and Robert A. Bjork, “Learning Versus Performance: An Integrative Review,” *Perspectives on Psychological Science* 10, no. 2 (2015), <https://doi.org/10.1177/1745691615569000>. Current performance is “often an unreliable index” of learning. Full discussion: Chapter 3.

¹¹⁴⁵Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman, “Generative AI without Guardrails Can Harm Learning: Evidence from High School Mathematics,” *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122, <https://doi.org/10.1073/pnas.2422633122>; author PDF https://hamsabastani.github.io/education_llm.pdf, accessed 30 August 2026. GPT Base: assisted practice +48% relative to control; unaided exam -17% . GPT Tutor: practice +127%; exam point estimate -0.004 , not significant. GPT Base fully correct 51% of the time when asked for the answer; 8% arithmetic errors even when the method was right. Design: classroom-level RCT, large high school in Turkey, Fall 2023–24, ~1,000 students in grades 9–11, four 90-minute sessions, GPT-4-0613. Not a homeschool study, not GPT-5, not Khanmigo. Percentages are of control grades, not Cohen's d . Full discussion: Chapter 3.

below a line marked PARENT KEY.”¹¹⁴⁶ The year and the 1998 are hypothetical. The split — two representations, parent solves, then checks — is the workflow.

(c) Extra isomorphic items with the parent-held key. Same structure, new numbers, new cover story. Two pages: student sheet and parent key. Five to eight items, not a completed copy of tonight’s assigned FRQ. Isomorphic here means a new chain-rule derivative after last night’s chain-rule derivative, or a new accumulator $F(x) = \int$ from a to $g(x)$ after yesterday’s $F(x) = \int$ from a to x — not a packet that mixes every unit on day two of Unit 2. Mix later, as section 10.5 already said, without importing $d = 1.05$. The child never sees the key. Sample the set. Work one item before the child sees the sheet. Bastani’s 51 percent and 8 percent arithmetic-error figures still apply. A fluent wrong key is worse than no extra practice. After the child attempts, Wolfram may check a derivative or an antiderivative. The language model does not.¹¹⁴⁷

(d) Hint script, not an answer key for the child. Level 1: a question only — “Is this a composition?” “Are you looking at an average rate or an instantaneous one?” “Is this a net accumulation or an area that cannot be negative?” Level 2: name the concept without numbers — “the chain rule is the derivative of the outside, evaluated at the inside, times the derivative of the inside.” Level 3: the next step, still not the final expression. One likely mistake and what to say if it appears. Parent reads the hints aloud. Do not put the final answer anywhere the child might see. Bastani’s GPT Tutor hidden prompt withheld the full solution, named common mistakes, and escalated slowly. The homeschool analogue is a parent-read script, not a child-held chatbot.¹¹⁴⁸

(f) Diagnose a child’s already-wrong derivative. Photograph the page *after* the attempt. Parent-only prompt: name the most likely misconception in one sentence; one question I should ask the child, not the correct steps; one isomorphic item to retest after we talk. The child answers the parent, not the model. The misconceptions this chapter exists to hit: “the derivative is the slope formula”; the chain rule dropped; $\sin(3x^2)$ differentiated as a product; a limit declared nonexistent because the form was $0/0$; $F(b) - F(a)$ computed on a function that is not an antiderivative; a series test named that does not apply, if the child is in Unit 10. Models hallucinate theorems and drop negative signs. The parent who can hear a wrong answer can also hear a fake quotient rule.¹¹⁴⁹

Afterwards, parent account, mark an FRQ against a rubric the parent pasted from an official free-response scoring guideline. Practice suggestion, not an RCT. The child has already written the question unaided, on paper, timed. The model sees the child’s work and the official rubric, not the live May prompt the night before the exam. The parent still samples the marking. Models award points for a theorem that was not used, miss a dropped minus, and invent a justification that reads like a scoring note. Sampling is not optional. This is the one “afterward” use that Chapter 3

¹¹⁴⁶Hypothetical labelled in the sentence that introduces it. Workflow (a) is Chapter 3 §3.4, practice suggestion, not an RCT. Prefer a chat/reasoning model, not an open-web agent, for this job.

¹¹⁴⁷Chapter 3 workflow (c): extra isomorphic practice with the answer key held by the parent. Do not generate “practice” by completing tonight’s assigned FRQ. That is the ban. Bastani 51% / 8% (n. 1145). Wolfram as the computational check after the attempt (n. 1138).

¹¹⁴⁸Bastani et al., author PDF, Figure 1 (GPT Tutor prompt): withhold the full solution; ask what they have tried; point out a named mistake; escalate information slowly; accept equivalent answers. Homeschool analogue is a parent-read hint script, not a child-held chatbot. Chapter 3 workflow (d).

¹¹⁴⁹Chapter 3 workflow (f): diagnose a wrong answer the child already produced. Practice suggestion, not an RCT. Name the misconception; ask a question; retest with an isomorphic item. The child answers the parent, not the model. Models hallucinate theorems and drop negative signs.

already allowed. It is not “do this FRQ.”¹¹⁵⁰

(e) Hands-on or figure-description workflow, used sparingly. The model may describe a GeoGebra-able figure — a slope field, a Riemann-sum picture, a washer — for the parent to build or for the child to draw. The child draws. The model does not shade. Carbonneau, Marley, and Selig’s $d \approx 0.37$ is manipulatives versus abstract-only, under guidance, and it is not a calculus trial; do not quote it as “slope-field apps work.” Virtual versus physical manipulatives in this course: **UNVERIFIED**.¹¹⁵¹ Hermes Agent on the parent machine can turn a lesson note into a figure description plus a student sheet and a parent key. Do not put the agent on the child’s phone.

Ban, child-facing, during the attempt.

“Do this FRQ.” Ban that prompt. Ban paste-the-FRQ. Ban photograph-tonight’s-worksheet-to-key. Photomath, Mathway, and Symbolab list calculus among subjects. Vendor “designed for learning” is vendor. The mechanism is GPT Base.¹¹⁵² College Board still requires unaided work in the exam room. Every FRQ the child will meet is written without a model, on paper, timed. Paste-the-FRQ is Chapter 3’s paste-the-worksheet in a new font.

The model as CAS. Ban that use. A chat window that prints an antiderivative is not Wolfram. A chat window that prints a series-convergence verdict is not a verified test. Label the tool. If the job is a check, use Wolfram|Alpha — or the textbook key, or a teacher of record. If the job is an explanation for the adult, use a language model on the parent account, then verify any integral the model writes against Wolfram before the explanation is trusted. Mixing them without a label is how hallucinated integrals survive.

The model between the child and fluent algebra. Ban that use. Facts and fluent algebra are not an LLM job. The chain rule’s inner derivative is algebra. A sign error in a difference quotient is arithmetic. GPT Base’s 8 percent arithmetic-error figure, even when the method was right, is the reason. Use the model to generate a deck of “which rule?” prompts or an oral script of unit-circle values, not to sit between the child and $d/dx[\sin(3x^2)]$.¹¹⁵³

Companion persona. Open-web agent in the child’s hands. Detectors as police on the child’s written justifications. Chapter 3 §3.3 and §3.6. Liang, Yuksekogonul, Mao, Wu, and Zou 2023: average false-positive rate 61.22 percent (arXiv) / 61.3 percent (*Patterns*) on human TOEFL essays. A

¹¹⁵⁰Afterwards-marking against a parent-pasted official FRQ rubric: Chapter 3 §3.7. Practice suggestion, not an RCT. Parent still samples. Ban “do this FRQ.”

¹¹⁵¹Kira J. Carbonneau, Scott C. Marley, and James P. Selig, “A Meta-Analysis of the Efficacy of Teaching Mathematics with Concrete Manipulatives,” *Journal of Educational Psychology* 105, no. 2 (2013): 380–400. 55 studies, $N = 7,237$, K through college, manipulatives vs abstract-only. Overall mean $d \approx 0.37$. Do not quote as “slope-field apps work.” Virtual vs physical manipulatives in calculus **UNVERIFIED**. Hermes Agent on the parent machine for a GeoGebra-able figure description: Chapter 3 §3.8 playbook; not a child-facing agent.

¹¹⁵²Photomath, <https://photomath.com/>, fetched 30 August 2026: scan including word problems; “A solution will pop up for your exact problem”; Basic \$0 / Monthly \$9.99 / Annually \$69.99 (USD, this day); **vendor** “designed for learning.” Mathway, <https://www.mathway.com/>: “Got a math problem? Get the steps.” Symbolab, <https://www.symbolab.com/>: solver first. Coverage through calculus listed. No product-specific unaided-transfer RCT opened this day. Transferable evidence is Bastani’s crutch (n. 1145) plus the homepages. Ban during the attempt: paste-the-FRQ, photo-to-key, “do this FRQ.”

¹¹⁵³Bastani et al. 2025 (n. 1145): 8% arithmetic errors even when the method was right. Facts and fluent algebra are not an LLM job. Chapter 3 §3.7 / 04-ai load-bearing finding 12: use the model to generate the deck or the oral script, not to sit between the child and the fact or the inner derivative.

homeschooler who writes a careful paragraph about why a limit fails to exist is exactly the writer those detectors smear.¹¹⁵⁴ *Garcia v. Character Technologies* and *Raine v. OpenAI* are allegations, not verdicts. They are why a math agent should not be a companion. Not a literature on mathematics learning. COPPA amended Rule published 22 April 2025, general compliance 22 April 2026. Homeschool parents are not verified U.S. K–12 educators by default and should not assume ChatGPT for Teachers or Claude for Teachers FERPA wrappers. California A.B. 1159 is not law as of 30 August 2026 (Senate third reading). Do not freeze it as a statute.¹¹⁵⁵

Session shape, Chapter 3 §3.8, scaled to this course. Practice suggestion, not an RCT. Unaided warm-up: one limit, one derivative of a composition, one definite-integral interpretation. Teach. Struggle. Hint not key. Timed unaided FRQ slice. The model, if used at all during the hour, is a hint script the parent already generated and is reading, or it is Wolfram after the attempt, labelled. “Finish your calculus with ChatGPT so we can go to soccer” is GPT Base.¹¹⁵⁶ If Khanmigo is in the house, expect the Tennessee pattern unless the parent requires the hint dialogue: 96 percent tried it at least once; the median student messaged it on only a third of practice days and in 17 percent of mistake sessions; 14.5 percent of messages contained a mathematical question or a step of reasoning. Practice minutes predicted MAP; chat did not, descriptively.¹¹⁵⁷

A *hypothetical*, labelled as such, can put the ban in a room. A parent whose seventeen-year-old pastes an official AB free-response question into a chatbot with “do this FRQ,” copies a fluent scoring-note paragraph, and scores 100 percent on the page, then cannot say on tomorrow’s unaided three-item check what the Fundamental Theorem says as a derivative of an accumulator, is running GPT Base.¹¹⁵⁸ The empirical pieces are Bastani’s crutch, Chapter 3’s FRQ-unaided rule, and Soderstrom and Bjork’s performance/learning split. The age, the paste, and the 100 percent are hypothetical. The move is not: take the paste, put the same question on paper, demand a written solution the child produces, and run the unaided check.

Hermes Agent, if used, lives on the parent machine. Model split this day (Nous Portal): Hermes 4 or any strong chat model for parent explanation and hint scripts; agentic examples — Claude Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro — for the tool loop that turns a lesson note into a student sheet, a parent key, and a GeoGebra-able figure description. Hermes 4 is **not recommended inside Hermes Agent**. Hermes 4 ≠ Hermes Agent. Not a child-safe school SKU. Memory in `~/hermes/` is sensitive. Hard no: messaging gateway to a child’s Telegram or

¹¹⁵⁴Weixin Liang, Mert Yuksekogonul, Yining Mao, Eric Wu, and James Zou, “GPT Detectors Are Biased against Non-Native English Writers,” arXiv:2304.02819; *Patterns* 4, no. 7 (2023), <https://doi.org/10.1016/j.patter.2023.100779>. Average false-positive rate 61.22% (arXiv) / 61.3% (*Patterns*) on human TOEFL essays. Do not use detectors to police a homeschooler’s written calculus justifications.

¹¹⁵⁵*Garcia v. Character Technologies et al.*, complaint, M.D. Fla., via CourtListener/RECAP: allegations, not a verdict. *Raine v. OpenAI*, complaint: allegations, not a verdict. Why a math agent should not be a companion. Not a literature on mathematics learning. Full discussion: Chapter 3 §3.6. Federal Trade Commission, “Children’s Online Privacy Protection Rule,” 90 FR 16918 (22 April 2025), Document 2025-05904, 16 CFR 312. Effective date 23 June 2025; general compliance date 22 April 2026. Homeschool parents are not verified U.S. K–12 educators by default. California A.B. 1159 is not law as of 30 August 2026 (Senate third reading).

¹¹⁵⁶Session shape: Chapter 3 §3.8, scaled to this course (n. 1140). Practice suggestion, not an RCT. “Finish your calculus with ChatGPT so we can go to soccer” is GPT Base.

¹¹⁵⁷Oreopoulos and Low, NBER w35620 (n. 1141). Tennessee pattern. Working paper, not yet journal-peer-reviewed.

¹¹⁵⁸Hypothetical labelled in the sentence that introduces it. Empirical pieces: Bastani’s crutch (n. 1145); Chapter 3 FRQ-unaided rule (n. 1139); Soderstrom and Bjork (n. 1144). The age, the paste, and the 100 percent are hypothetical.

WhatsApp.¹¹⁵⁹ Grok Bot, if used, is included with SuperGrok and Cursor Pro/Teams as of the 26 August 2026 expansion post. Cloud storage required; Legacy Privacy Mode unsupported; shared computer per user; no education SKU and no FERPA claim on the pages opened. Grok ≠ Grok Bot ≠ El Salvador Grok-as-tutor. Hard no: the child's hands; an IEP PDF on the shared computer; signing the Bot into a school portal or a Photomath login.¹¹⁶⁰

The open questions this chapter leaves standing are the ones a sequel would fill with analogy. AP Calculus full CED topic-level tables below the eight and ten unit titles: **UNVERIFIED** (front-matter prerequisite paragraph now extracted; unit-guide rows not). Non-AP homeschool calculus tables of contents — Saxon Calculus second edition, Art of Problem Solving Calculus, a community-college Calculus I catalog sample — not opened. Homeschool-specific dual-enrollment counts and AP exam-access mechanics beyond the 29 June 2026 guide: not in these sources. STEM-major and college-calculus completion rates for homeschool graduates: no named nationally representative source. IES dual-enrollment outcomes study first report expected 2027. Limit as concept-image versus concept-definition (Tall and Vinner 1981): not opened; do not invent a teaching sequence. No homeschool calculus RCT. No year-plus kitchen GPT-Tutor RCT.¹¹⁶¹ Hold those holes. Do not close them with a kitchen scene.

Statistics is a different college-equivalent. It does not substitute for this chapter in STEM. Chapter 11 is that through-line, from Grade 6 distributions to a non-calculus introductory course whose official prerequisite, as of the Fall 2026 CED, is first-year algebra. College Board encourages taking both when the background allows. This chapter was the calculus object: a new operation on functions they already know; eight units, or those eight plus two; a one-semester CLEP exam that is neither AB nor BC; a pathway that reaches senior-year calculus only if high-school content started in Grade 8, as a model, not a mandate; a seat a parent cannot order; a free-response the child writes unaided. Everything Chapters 4 through 11 taught still has to leave the house as a record a stranger can read. Course titles. Grades. Credits. An AP score, or a dual-enrollment transcript, or a CLEP score the receiving institution actually takes — different objects, overlapping destinations, no automatic credit. CLEP is not an NCAA core course. There is no national homeschool diploma.

¹¹⁵⁹Hermes Agent, <https://hermes-agent.org/>, fetched 30 August 2026: self-hosted adult agent, open web, messaging gateways; no K–12 student-safety certification on the pages opened. Nous Portal this day: Hermes 4 **not recommended inside Hermes Agent** (tuned for chat, not rapid tool-calling). Hermes 4 ≠ Hermes Agent. Agentic Portal examples this day: Claude Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro. Hard no: messaging gateway to a child's Telegram/WhatsApp. Memory in `~/hermes/` is sensitive. Full distinctions: Chapter 3 §3.2 and §3.8. Practice suggestion, not an RCT.

¹¹⁶⁰xAI, “Grok Bot FAQ,” <https://docs.x.ai/grok-bot/faq>, fetched 30 August 2026: persistent cloud computer; “Grok Bot requires cloud data storage, so Legacy Privacy Mode is not supported”; one computer per user shared across Bots; no education SKU and no FERPA claim on the pages opened. Included with SuperGrok and Cursor Pro/Teams as of the 26 August 2026 expansion post. Grok ≠ Grok Bot ≠ El Salvador Grok-as-tutor (xAI news, 11 December 2025: ministry deployment of Grok *the assistant*, announcement not RCT, not a U.S. homeschool SKU). Hard no: the child's hands; IEP/504 PDFs on the shared computer; school-portal or Photomath logins. Full distinctions: Chapter 3 §3.2 and §3.8.

¹¹⁶¹Open questions left standing: AP Calculus full CED topic-level tables and official prerequisite paragraph (01-scope open question 1; outline open question 10); non-AP homeschool calculus TOCs (01-scope open question 12); homeschool dual-enrollment counts and AP exam access beyond the 29 June 2026 guide (open question 16); STEM-major / college-calculus rates (open question 5); IES dual-enrollment outcomes report expected 2027 (open question 23); Tall and Vinner 1981 not opened (open question 21). No homeschool calculus RCT. No year-plus kitchen GPT-Tutor RCT. Do not close with analogy.

State compliance is not college evidence. Chapter 12 is that export layer. The child writes the FRQ. The parent holds the key. The agent does not take the test.

Chapter 11

Data and Chance Across the Years



Figure 12: A scatterplot on paper, a pair of dice, and a sketched histogram.

Two sentences on College Board pages already contain the argument this chapter has to keep in view. AP Statistics, in the Course and Exam Description effective Fall 2026, is “equivalent to a one-semester, introductory, **non calculus-based** college course in statistics.” The only published prerequisite is that “any secondary school student who has successfully completed a **first-year**

algebra course” may take it.¹¹⁶² Those two sentences are not a placement slogan. They are a refusal. Statistics is a different discipline from calculus. Calculus is the mathematics of instantaneous change and accumulation: limits, derivatives, integrals, and, in a second course, series. Statistics is formulating questions, collecting data, analyzing data, and interpreting results, with variability as the central idea rather than “the average.” A parent who stuffed this chapter into Chapter 10 because both are “senior electives” would have treated two college-equivalents as one. College Board does not. This book will not.

The former second-year-algebra prerequisite is gone in this CED. That is a real change, and it is easy to misread. It does not mean AP Statistics is stealth Algebra II. It does not mean AP Statistics is stealth calculus. It does not mean a student who has finished Algebra I and nothing else is finished with high-school mathematics. It means the official gate for this particular college-equivalent is first-year algebra — linear equations, interpreting slope and intercept, basic function notation — plus comfort with fractions, percents, and inequality language.¹¹⁶³ Calculus is not on the list. Chapter 10 is the calculus chapter. This chapter exists because College Board treats statistics as a different college-equivalent, because the Common Core State Standards for Mathematics already placed a Statistics and Probability domain from Grade 6, and because Algebra I already includes the S.ID clusters a kitchen is tempted to skip on the theory that “stats is later.” Later is here. Skipping the K–8 data strand is AP Statistics Unit 1 retrofit. Skipping S.ID in Algebra I is the same retrofit, one year later, with a scatter plot and no vocabulary for association.¹¹⁶⁴

The revised course framework, partially opened on 30 August 2026, encourages students who have the background to take **both** AP Statistics and AP Calculus, and notes that students interested in data science are also encouraged to take AP Statistics and an AP Computer Science course. It is not published as a substitute for calculus in STEM calculus sequences.¹¹⁶⁵ A parent choosing AP Statistics *instead of* precalculus or calculus is choosing a different college-equivalent. That is a legitimate choice for a non-STEM path. It is a weak stealth substitute for Calculus I if the destination is Physics C or a calculus-based engineering sequence. College Board’s encouragement of both is the honest STEM advice. Chapter 10 will not be dumped here to make that sentence longer. Chapter 12 will want the course titled Statistics or AP Statistics on a transcript a stranger can read, and will want the AP seat ordered through a school, because parents cannot order AP exams. Those two plants stay plants.

¹¹⁶²College Board, *AP Statistics Course and Exam Description*, Effective Fall 2026, <https://apcentral.collegeboard.org/media/pdf/ap-statistics-course-and-exam-description.pdf>, accessed 30 August 2026. “Equivalent to a one-semester, introductory, **non calculus-based** college course in statistics.” Prerequisite: “any secondary school student who has successfully completed a **first-year algebra course**.” The former second-year-algebra prerequisite is gone in this CED. First revised exam May 2027.

¹¹⁶³Same CED (n. 1162). Depends on: first-year algebra (linear equations, interpreting slope and intercept, basic function notation); comfort with fractions, percents, and inequality language. Not calculus. 01-scope Statistics band.

¹¹⁶⁴Common Core State Standards Initiative, *Common Core State Standards for Mathematics*, https://www.thecorestandards.org/assets/CCSSI_Math%20Standards.pdf, accessed 30 August 2026, Grades 6–8 Statistics and Probability domains (6.SP, 7.SP, 8.SP) and high-school S.ID. A skipped K–8 data strand is AP Statistics Unit 1 retrofit. Algebra I S.ID is the on-ramp, not a reason to skip Chapter 11 and not a reason to skip the Algebra I clusters. Plant from Chapters 6 and 7.

¹¹⁶⁵College Board, *AP Statistics Revised Course Framework*, <https://apcentral.collegeboard.org/media/pdf/ap-statistics-revised-course-framework.pdf>, partially opened 30 August 2026. Encourages students with appropriate background to take **both** AP Statistics and AP Calculus; students interested in data science also encouraged to take AP Statistics and an AP Computer Science course. Not published as a substitute for calculus in STEM calculus sequences.

Four lanes run through the sections. They are not four identical headings. Sections 11.1 through 11.4 explain the mathematics to the parent: why two college-equivalents cannot share a chapter; the K–8 spine of distributions, samples, and bivariate association, with informal data work earlier than Grade 6; the high-school clusters already sitting in Algebra I, Geometry, and Algebra II if those courses were honest; the five AP Statistics units of the Fall 2026 CED, with the four statistical practices College Board aligns to the American Statistical Association’s GAISE II. Section 11.5 is teaching guidance: a question that has variability, not a worksheet of mean, median, and mode with no data story; design rather than a slogan about correlation; schemas for chance rather than keyword hunts; simulation as a representation that has to connect to a written claim. Section 11.6 is the tips that survive: add-on versus substitute; do not skip K–8; do not treat journalism about “cuts” as the CED; transcript the course as Statistics. Section 11.7 is the parent-hosted AI lane: Chapter 3’s ban versus allow, reused on data and chance. Generate a dataset and a wrong conclusion for the child to attack. The parent holds the key. Ban “analyze this and write the AP paragraph.” Models hallucinate statistics. The child writes unaided.

The Common Core State Standards for Mathematics are treated here as a map, not as a homeschool mandate. Many homeschoolers do not use CCSS. Many U.S. states have adopted, adapted, or replaced it. A 2026 state-by-state census of who still uses it is **UNVERIFIED** in this project; this chapter will not say “most states.”¹¹⁶⁶ What the map is good for is specifying what the mathematics is: a statistical question as one whose answer varies; a distribution with shape, center, and spread; a sample as not a population; an association as not a function until someone specifies a rule, and not causation ever by itself; independence and conditional probability in a two-way table; a normal model as a later high-school object; inference from surveys, experiments, and observational studies as a still later one. The standards “do not dictate curriculum or teaching methods,” and the order of topics inside a grade is not a required teaching order.¹¹⁶⁷ This book will not dump 180 days. It will not sell a statistics franchise. NCTM’s *Principles and Standards* already named Data Analysis and Probability as one of five content standards across four grade bands; the 2006 *Curriculum Focal Points* put analyzing and summarizing data sets among Grade 8’s three most important topics.¹¹⁶⁸ Those documents predate CCSSM. They are not “the current U.S. standard.” They remain useful as professional consensus that data is a strand, not a senior afterthought.

The National Center for Education Statistics already knew the strand is easy to drop. In the 2016 Parent and Family Involvement tables, probability was ever-taught to 24 percent of high-school-level homeschoolers.¹¹⁶⁹ That is not a statistics-course completion rate. It is not AP Statistics

¹¹⁶⁶2026 state-by-state census of who still uses CCSSM, who uses “standards formerly known as,” and who never did: **UNVERIFIED** (01-scope open question 8). Do not say “most states.”

¹¹⁶⁷CCSSM, “How to read the grade level standards”: the standards “do not dictate curriculum or teaching methods”; order of topics inside a grade is not a required teaching order. Same PDF as n. 1164. Map, not a homeschool mandate.

¹¹⁶⁸National Council of Teachers of Mathematics, *Principles and Standards for School Mathematics* Executive Summary (2000), https://www.nctm.org/uploadedFiles/Standards_and_Positions/PSSM_ExecutiveSummary.pdf, accessed 30 August 2026. Five content standards including Data Analysis and Probability; five process standards; four grade bands (pre-K–2, 3–5, 6–8, 9–12). NCTM, *Curriculum Focal Points for Prekindergarten through Grade 8 Mathematics* (2006): Grade 8 focal points include analyzing and summarizing data sets, from a publicly posted excerpt, https://www.platonicsolids.info/focal_points_by_grade.pdf, accessed 30 August 2026 — **secondary copy of a primary**; full NCTM book paywalled. Prefer a library copy. PSSM and Focal Points predate CCSSM; they are not “the current U.S. standard.”

¹¹⁶⁹National Center for Education Statistics, *Homeschooling in the United States: Results from the 2012 and 2016*

participation. Whether homeschoolers actually sit AP Statistics at any measurable rate is **UNVERIFIED**.¹¹⁷⁰ The 24 percent is a reminder that chance and data are the units a kitchen can skip because the algebra chapters look like “the real math.” Algebra I 79 percent ever-taught, geometry 52, Algebra II 49, calculus 15 with the caution Chapter 1 already installed: those are *high-school-level ever-taught during home instruction* figures from 2016, not 2026, and not completion rates.¹¹⁷¹ Probability at 24 percent sits below all of them. This chapter is the refusal to treat that drop as a plan.

GAISE, the American Statistical Association’s Guidelines for Assessment and Instruction in Statistics Education, frames school statistics as formulate questions, collect, analyze, interpret, with variability as the central idea. College Board’s four statistical practices are aligned to GAISE II. The ASA primary was not separately opened for this project; existence and alignment are confirmed inside the CED. Page-level claims stay **UNVERIFIED**. Treat the spirit, not a quoted recommendation list.¹¹⁷² This chapter will not invent a GAISE page number in order to sound official. It will say: a question that has variability is the teaching object; a mean with no data story is not.

The thesis of the book still holds, and it is still not a causal claim about homeschool statistics. A parent who is not a math specialist can take a child from first-grade category counts through a non-calculus college-equivalent in statistics if three things hold: the parent understands the concept well enough to hear a wrong answer — “the average is 12, so they are all about 12,” “the correlation is 0.9 so X caused Y ,” “I simulated it, so it is true”; the child does the struggle before the tool; and any agent is a supplement hosted and constrained by the parent, not a substitute for the child’s work and not a school-safe product by default. The sources do not prove that thesis as a causal claim about kitchen-table statistics. They do not contradict it. They constrain how it can be attempted. Unguarded solvers raise practice scores and cut unaided exams. No homeschool AP Statistics randomized trial exists in the literature opened on 30 August 2026. This chapter will not invent one. Journalism claiming five content areas were “cut” from AP Statistics is labelled journalism and is not used as a primary for what remains. Use the CED.¹¹⁷³

Parent and Family Involvement Survey (NCES 2020-001). Probability ever-taught to 24 percent of high-school-level homeschoolers. Not a statistics-course completion rate and not an AP Statistics participation rate. Last opened subject tables of this kind; NHES:2023 did not republish them.

¹¹⁷⁰Homeschool AP Statistics participation rate: **UNVERIFIED**. No named nationally representative source in the memos opened 30 August 2026. Do not invent one.

¹¹⁷¹NCES 2020-001, high-school-level ever-taught during home instruction: Algebra I 79 percent, geometry 52, Algebra II 49, calculus 15 (caution: not a completion rate, not college calculus, not 2026). Probability 24 percent (n. 1169) sits below all of them. The 2012 report’s 88 percent “basic algebra” figure is a different questionnaire; do not narrate 88 → 79 as decline without checking standard errors. NHES:2023 First Look did not republish the subject tables.

¹¹⁷²American Statistical Association, *Guidelines for Assessment and Instruction in Statistics Education* (Pre-K–12 GAISE I 2005, rev. 2007; Pre-K–12 GAISE II 2020; 2016 is the college GAISE report). Cited inside the AP Statistics CED (n. 1162) as the alignment for the four statistical practices. ASA primary not separately opened this pass; page-level claims **UNVERIFIED**. Treat as spirit — formulate questions, collect, analyze, interpret; variability as the central idea — not as quoted recommendations.

¹¹⁷³Journalism claiming five content areas were “cut” from AP Statistics is labelled **journalism** and is not used as a primary for what remains. Use the CED (n. 1162).

11.1 Why a separate chapter: two college-equivalents

The popular story collapses two senior years into one. Take AP Calculus if the child is “good at math.” Take AP Statistics if the child is “not a math person,” or if the family wants something that “uses real data,” or if Algebra II felt like enough. The documents do not say that. They say there are two college-equivalents, with different objects, different prerequisites, and different destinations. Stuffing statistics into the calculus chapter would have been a convenience for the table of contents and a contradiction of the correction this book already locked: statistics is not stealth calculus, and calculus is not stealth statistics.¹¹⁷⁴

AP Calculus AB is eight units, limits through applications of integration. AP Calculus BC is those eight plus parametric, polar, and vector-valued functions and infinite sequences and series. The 2026–27 clarifications do not change that content. Those courses assume precalculus-level functions. CLEP Calculus is a different object again: a one-semester college-calculus exam, about 60 percent limits and differential calculus and 40 percent integral calculus. Chapter 10 is the place for those distinctions. They are named here only so the parent cannot treat “college math” as a single elective. AP Statistics is none of those objects. It is a one-semester, introductory, non-calculus-based college course in statistics. The Fall 2026 CED organizes it into five units, not the former nine. First revised exam: May 2027.¹¹⁷⁵

The five units, with multiple-choice exam weighting as the CED publishes them, are a map of what the college-equivalent actually contains, not a 180-day plan.

Unit	Title	MC weighting
1	Exploring One-Variable Data and Collecting Data	20–30%
2	Probability, Random Variables, and Probability Distributions	15–25%
3	Inference for Categorical Data: Proportions	15–25%
4	Inference for Quantitative Data: Means	10–20%
5	Regression Analysis	10–20%

Those weights are from the CED as extracted into this project’s scope memo on 30 August 2026. They are not journalism. They are not a claim about how a kitchen should budget weeks. Unit 1 is

¹¹⁷⁴01-scope load-bearing 11: statistics does not substitute for calculus as a mathematics sequence, and calculus does not substitute for statistics as a data-reasoning course. AP Statistics is explicitly “non calculus-based.” AP Calculus is about change, limits, and analysis of functions. Two college-equivalents, not an equivalent.

¹¹⁷⁵College Board AP Calculus AB and BC course pages, <https://apcentral.collegeboard.org/courses/ap-calculus-ab> and <https://apcentral.collegeboard.org/courses/ap-calculus-bc>, accessed 30 August 2026. AB: eight units. BC: those eight plus Units 9–10. 2026–27 clarifications do not change course content. CLEP Calculus: <https://clep.collegeboard.org/clep-exams/calculus>, accessed 30 August 2026 — a different object (one-semester college calculus exam; about 60 percent limits and differential, 40 percent integral). Named here only as contrast. Chapter 10 is the calculus chapter; do not dump it. AP Statistics five units, not the former nine: CED (n. 1162); first revised exam May 2027.

the heaviest multiple-choice slice for a reason the rest of this chapter will keep repeating: exploring data and collecting it — census versus sample, experiment versus observational study, bias, random assignment versus random sampling — is the load-bearing work, not a warm-up before “the real formulas.”¹¹⁷⁶ Section 11.4 will walk the units. This section only needs the parent to see that the course is five units of data, chance, and inference, not a stealth derivative.

The prerequisite sentence does work the old second-year-algebra gate used to do by exclusion. A student who has completed a first-year algebra course may take AP Statistics. Linear equations. Slope and intercept in context. Basic function notation. Fractions, percents, inequalities. Not a second algebra. Not precalculus. Not a limit. Algebra II remains useful: the normal model (S.ID.4) and making inferences from surveys, experiments, and observational studies (S.IC) live there on the Common Core map, and a student who has seen them will spend less of Unit 1 and Unit 4 retrofitting. Useful is not required. The official gate is Algebra I.¹¹⁷⁷ A kitchen that hears “you can take AP Stats after Algebra I” as “we can stop mathematics at Algebra I, and statistics will cover the rest” has misread the gate as a ceiling. Chapter 12’s named university pages want, as a pattern not a rule, three years of high-school mathematics through Algebra II as a floor. Selective STEM programs will want precalculus or calculus plus an external signal. The kitchen that stops at Algebra I because “the state doesn’t require it” has a college problem in every named example in Chapter 12, whether or not AP Statistics is on the transcript.¹¹⁷⁸ That sentence is a plant. It is not a dump of NCAA worksheets.

The live debate this section will not pretend to settle belongs on the page so a franchise cannot settle it. Statistics as a senior *instead of* precalculus or calculus, versus statistics as an *add-on*. College Board’s revised framework encourages both when the student has the background. That is the honest STEM advice, and it is a sequence fact about two courses, not a survey of what kitchens actually choose. Whether stats-as-substitute is actually chosen in homeschool kitchens is **not in these sources**. Whether homeschoolers sit AP Statistics at any measurable rate is **UNVERIFIED**. This chapter will not invent a participation rate in order to scold or to cheer.¹¹⁷⁹ Data-science-interested students are also encouraged, in the same framework, to take AP Statistics and an AP Computer Science course. That is a third college-equivalent, not a rewrite of this one. This book is a mathematics handbook. It will not write an AP Computer Science chapter in order to sound current.

Two destinations, named as sequence facts, not as a survey of majors. Physics C is a calculus-

¹¹⁷⁶CED (n. 1162), unit MC weights as extracted into 01-scope on 30 August 2026: Unit 1, 20–30%; Unit 2, 15–25%; Unit 3, 15–25%; Unit 4, 10–20%; Unit 5, 10–20%. Weights are from the CED, not journalism. They are not a kitchen week-budget.

¹¹⁷⁷CED (n. 1162) prerequisite sentence. Algebra II remains useful: S.ID.4 (normal) and S.IC (inference from surveys, experiments, observational studies) live there on the Appendix A traditional pathway. Useful is not required. The official gate is Algebra I.

¹¹⁷⁸Plant for Chapter 12. Named university pages in the brief (Pitt: minimum 3 years of mathematics; SMU: suggested Algebra I, Geometry, Algebra II; North Central: 3 years including algebra and geometry, one course beyond geometry preferred) want a transcript with course titles, grades, and credits. Pattern, not a rule: three years through Algebra II is the floor those pages tend to name. Selective STEM programs will want precalculus or calculus plus an external signal. The kitchen that stops at Algebra I because “the state doesn’t require it” has a college problem in every named example, whether or not AP Statistics is on the transcript. Do not dump NCAA worksheets here.

¹¹⁷⁹Whether stats-as-substitute is actually chosen in homeschool kitchens: **not in these sources**. Homeschool AP Statistics participation: **UNVERIFIED** (n. 1170). Do not invent a rate in order to scold or to cheer.

based AP physics course. A calculus-based engineering sequence in college assumes Calculus I, then usually Calculus II, with physics that uses derivatives and integrals. AP Statistics does not unlock those. It unlocks, where the receiving institution accepts the score, credit or placement in an introductory statistics course, and it unlocks data literacy for social science, biology, and civic life.¹¹⁸⁰ Credit is not automatic. Chapter 12 will say that about AP, CLEP, and dual enrollment together. This chapter says it about this exam: a 5 is not a registrar's stamp. Search the college.

Journalism about the 2026 revision is a separate object from the CED. Pieces that claimed five content areas were “cut” from AP Statistics circulated around the revision. This project labels that journalism and does not use it as a primary for what remains. The primary is the CED: five units, the weights in the table, the four practices, the Unit 1 design spine section 11.4 will list. A parent shopping by headline will buy a rumor. A parent who opens the CED will buy a course.¹¹⁸¹

Why a separate chapter, then, in one paragraph the design user can steal. Because College Board publishes two college-equivalents, not one “hard math” and one “easy math.” Because the statistics equivalent is non-calculus-based and opens after Algebra I, which is a fact about the course, not a license to skip Algebra II, precalculus, or calculus if those are the destination. Because CCSSM already placed the strand from Grade 6, and Algebra I, Geometry, and Algebra II already contain the clusters. Because NCES 2016 shows probability ever-taught to about a quarter of high-school-level homeschooolers, which is how a strand dies. Because a skipped K–8 data strand is Unit 1 retrofit, and retrofit is not a pedagogy. Because the agent will happily compute a p -value the child has not set up, and Chapter 3 already forbade the crutch. Because everything this chapter teaches still has to leave the house as a record a stranger can read — Chapter 12, not a footnote.

What this section has done is split the two college-equivalents and refuse the stuffing. The K–8 spine is already on the map. Skipping it is the retrofit. That spine comes next.

11.2 The K–8 spine: distributions, samples, bivariate association

Statistical thinking does not begin in AP Statistics, and it does not begin in Algebra I. The Common Core map places a Statistics and Probability domain in Grade 6, Grade 7, and Grade 8. Informal data work sits earlier: Grade 1 organizes, represents, and interprets data with up to three categories; Grade 2 adds picture graphs and bar graphs with a single-unit scale and up to four categories, and line plots of measurement data; Grades 3–5 scale the graphs and put fractions on the line plot — halves and fourths of an inch in Grade 3, halves, fourths, and eighths in Grades 4 and 5, with fraction operations on the Grade 5 data.¹¹⁸² That earlier work is not a mean. It is a category count the child made, or a measurement the child plotted. Chapter 4 already planted three categories and

¹¹⁸⁰Unlocks: college intro-stats credit or placement where the institution accepts the AP score; data literacy for social science, biology, and civic life. Does not unlock Physics C or a calculus-based engineering sequence. Sequence facts, not a survey of majors. AP credit is not automatic; each receiving college sets policy (College Board AP Credit Policy Search, referenced from AP course pages).

¹¹⁸¹Journalism-about-cuts labelled journalism (n. 1173). Primary for what remains: the CED (n. 1162).

¹¹⁸²CCSSM 1.MD: data with up to three categories. CCSSM 2.MD: picture graphs and bar graphs, single-unit scale, up to four categories; line plots of measurement data. CCSSM 3.MD: scaled picture/bar graphs; line plots with halves and fourths of an inch. CCSSM 4.MD and 5.MD: line plots in 1/2, 1/4, 1/8 units; Grade 5 fraction operations on the data. Informal earlier than the Grade 6 SP domain. Chapter 4 planted three categories and a bar; Chapter 5 used line plots as a place fractions live. Not a mean.

a bar. Chapter 5 already used line plots as a place fractions live. This section will not dump those chapters. It will say: the spine starts as a representation of a question the child can see, years before anyone says *median*.

NCTM's 2000 *Principles and Standards* had already named Data Analysis and Probability as one of five content standards across four grade bands — pre-K–2, 3–5, 6–8, 9–12 — not thirteen separate grade lists.¹¹⁸³ The 2006 *Curriculum Focal Points*, from the publicly posted excerpt this project used as a **secondary copy of a primary** (the full NCTM book is paywalled), put analyzing and summarizing data sets among Grade 8's three most important mathematical topics, alongside linear functions and systems and two- and three-dimensional space including the Pythagorean theorem.¹¹⁸⁴ Prefer a library copy before treating the excerpt as the official book. The consensus is still the one a kitchen that “saves stats for high school” has to hear: data is a strand in the middle grades, not an elective the Algebra I book will magically supply.

Grade 6 is the first year the map names statistical thinking as a critical area. The official PDF's four Grade 6 critical areas are connecting ratio and rate to whole-number multiplication and division; completing division of fractions and extending number to the rationals, including negatives; writing, interpreting, and using expressions and equations; and **developing understanding of statistical thinking**.¹¹⁸⁵ The Statistics and Probability domain, 6.SP, is specific. Statistical variability. Statistical questions. Distributions: center (median, mean), spread (interquartile range, mean absolute deviation), shape (clusters, peaks, gaps, symmetry). Summarize numerical data sets in context.¹¹⁸⁶ Mean, median, and mode as a triad with no data story is not this domain. A question that has variability — how long did these sunflower seedlings grow in a week? — is.

A statistical question, as the parent who stopped at algebra has to be able to hear, is a question whose answer varies. “How tall is this sunflower?” is not a statistical question. It has one answer. “How tall are the sunflowers in this row?” is. The answers are a distribution. The distribution has a shape: clustered? peaked? gapped? roughly symmetric, or pulled to one side? It has a center: a median that splits the ordered list in half; a mean that is the balance point, the total divided by the count. It has a spread: a range that is only two points and is therefore fragile; an interquartile range that is the middle half; a mean absolute deviation that asks how far, on average, the points sit from the mean. Those objects are Grade 6. They are not AP Statistics. A child who can compute a mean from a tidy list and cannot say that the mean is pulled by a long tail, or that two classes can share a mean and not share a spread, has the computation and not the distribution. The parent who can hear a wrong answer hears “they are all about 12” when the list is 2, 3, 4, 12, 40, and asks to see

¹¹⁸³NCTM PSSM Executive Summary (n. 1168). Data Analysis and Probability is one of five content standards across four grade bands, not a senior afterthought.

¹¹⁸⁴NCTM Focal Points 2006 (n. 1168), Grade 8: linear functions and systems of linear equations; analyzing two- and three-dimensional space using distance and angle (including the Pythagorean theorem); analyzing and summarizing data sets. **Secondary copy of a primary.** Programs offering algebra in Grade 8 should still not omit Grade 7–8 focal-point content; it can be moved, but not dropped.

¹¹⁸⁵CCSSM Grade 6 overview / four critical areas, official PDF (n. 1164): connecting ratio and rate; completing division of fractions and extending to rationals including negatives; expressions and equations; developing understanding of statistical thinking. Also area, surface area, volume.

¹¹⁸⁶CCSSM 6.SP: statistical variability; statistical questions; distributions — center (median, mean), spread (IQR, MAD), shape (clusters, peaks, gaps, symmetry); summarize numerical data sets in context. Not a worksheet of mean/median/mode with no data story.

the shape.

Grade 7's fourth critical area is drawing inferences about populations based on samples.¹¹⁸⁷ 7.SP: random sampling to draw inferences about a population; informal comparative inferences about two populations; chance processes and probability models — sample spaces, compound events, simulations.¹¹⁸⁸ A sample is not a population. A voluntary-response sample is a biased sample. Those two sentences are Unit 1 of AP Statistics, years early, and they are 7.SP. Do not skip them because “we’ll do AP Stats later.” Informal comparative inferences are the Grade 7 version of “do these two groups look different,” without a two-sample *t*-procedure. Overlap of the two distributions, difference of centers relative to spread, a skeptical eye on sample size: that is the work. It is not a *p*-value. Do not import one.

Chance processes in Grade 7 are the first formal probability models on the map. A sample space is the set of possible outcomes. A probability is a number between 0 and 1 assigned to an outcome or an event, with the whole sample space totaling 1. Compound events are “and,” “or,” “not,” read from a list, a tree, a table, or a simulation. A simulation is a representation. It is not a proof and it is not a vibe. The child who runs a bag-draw fifty times and says “I simulated it, so it is true” has performed a ritual. The child who writes “in 50 trials I got both red 18 times, so I estimate $P(\text{both red}) \approx 18/50$, which is near the $3/8$ I counted from the sample space” has connected the representation to a written claim. Section 11.5 will make that connection a teaching rule. Grade 7 already requires the objects.

Grade 8's first critical area includes modeling an association in bivariate data with a linear equation.¹¹⁸⁹ 8.SP: bivariate data — scatter plots; informal linear models; clustering, outliers, positive and negative association; intercept, slope, and frequency tables of bivariate categorical data.¹¹⁹⁰ An association is not a function until someone specifies a rule, and it is not causation ever, by itself. Chapter 6 already planted that sentence next to 8.F, because a scatter that “looks like a line” is the confusion a child will bring into Algebra I. Algebra I's S.ID will add residuals and correlation-versus-causation as named clusters. AP Statistics Unit 5 will make regression a course. This is the scatter, the “looks like a line,” the informal $y = mx + b$ fitted by eye. Statistical thinking remains a question that has variability, not a worksheet of mean, median, and mode.

What the three years depend on is not mysterious, and it is not calculus. Grade 6 distributions need the number line, ordered lists, and fraction sense enough to place a median between two values. Grade 7 sampling needs ratio and percent, because a sample of 30 from 300 is 10 percent, and a biased 10 percent is still biased. Grade 7 probability needs fraction operations, because $3/8$

¹¹⁸⁷Common Core State Standards Initiative, Grade 7 Introduction, <https://www.thecorestandards.org/Math/Content/7/introduction/>, accessed 30 August 2026. Four critical areas; the fourth is drawing inferences about populations based on samples.

¹¹⁸⁸CCSSM 7.SP: random sampling to draw inferences about a population; informal comparative inferences about two populations; chance processes and probability models (sample spaces, compound events, simulations). A sample is not a population. A voluntary-response sample is a biased sample. Those two sentences are Unit 1 of AP Statistics, years early.

¹¹⁸⁹Common Core State Standards Initiative, Grade 8 Introduction, <https://www.thecorestandards.org/Math/Content/8/introduction/>, accessed 30 August 2026. Three critical areas; the first includes modeling an association in bivariate data with a linear equation.

¹¹⁹⁰CCSSM 8.SP: bivariate data — scatter plots; informal linear models; clustering, outliers, positive/negative association; intercept, slope, and frequency tables of bivariate categorical data. An association is not a function until someone specifies a rule, and is not causation by itself. Plant for Algebra I S.ID and AP Statistics Unit 5.

of a sample space is a fraction, and compound events are products and sums of fractions under independence and disjointness the child is only beginning to name. Grade 8 bivariate work needs 8.EE's slope and 8.F's function concept as neighbors, not as synonyms. A skipped fractions gate from Chapter 5 will show up here as a child who cannot find a median in an even list, cannot compute a relative frequency, and cannot read 0.3 as a probability. Diagnose that. Then this spine. Do not start AP Statistics here as if the course were finished.

What the three years unlock is Algebra I's S.ID, Geometry's S.CP, and, later, AP Statistics Unit 1 as a course rather than as a retrofit. A skipped K–8 data strand is Unit 1 retrofit. NCES 2016: probability ever-taught to 24 percent of high-school-level homeschoolers. Not a statistics-course completion rate. A reminder that the data strand is easy to drop.¹¹⁹¹ NCTM Focal Points already warned that programs offering algebra in Grade 8 should still not omit Grade 7–8 focal-point content; it can be moved into Grade 6–7 or into high school, but not dropped.¹¹⁹² Data is in that warning. Similarity is in that warning. A family that “starts Algebra I early” and drops 7.SP and 8.SP has dropped the on-ramp this chapter exists to keep.

GAISE spirit, restated without a page quote. Formulate a question that has variability. Collect data with a design the parent can name — a census of the household's seedlings is a census; a survey of the co-op's email list is a voluntary-response sample. Analyze: a representation, a center, a spread, a shape, later a comparison, later a scatter. Interpret in context: the number is about these plants, this week, this sample, not about “sunflowers” as a law of nature. Page-level GAISE II claims remain **UNVERIFIED**. The spirit is the teaching object. A worksheet of already-tallied bars with no question is not.¹¹⁹³

A *hypothetical*, labelled as such, can put the skip in a room. A parent whose eleven-year-old's Grade 6 book has a statistics chapter after the ratio chapters, who skips it because “the algebra is the real math,” and whose fourteen-year-old then meets a scatter plot in Algebra I with no word for association, is meeting a hole the map already named.¹¹⁹⁴ The empirical pieces are 6.SP, 8.SP, and S.ID. The skipping and the sitting are hypothetical. The move is not: ask a question that has variability in Grade 6, sample in Grade 7, scatter in Grade 8, and do not wait for a course titled Statistics.

What this section has done is place the spine and refuse the later. High-school clusters already taught, if Algebra I through Algebra II and Geometry were honest, come next. This chapter is the through-line. Those clusters are not a reason to skip it, and it is not a reason to skip them.

11.3 Algebra I S.ID, Geometry S.CP, Algebra II S.ID.4 and S.IC

Do not skip these clusters in Chapters 7 through 9 because “stats is Chapter 11.” This chapter is the through-line and the AP course. Those clusters are the on-ramp, the chance year, and the inference

¹¹⁹¹NCES 24 percent (n. 1169). A skipped K–8 data strand is Unit 1 retrofit.

¹¹⁹²NCTM Focal Points warning (n. 1184). Data is in that warning. Similarity is in that warning. A family that “starts Algebra I early” and drops 7.SP and 8.SP has dropped the on-ramp.

¹¹⁹³GAISE spirit (n. 1172). Page-level **UNVERIFIED**. A worksheet of already-tallied bars with no question is not the spirit.

¹¹⁹⁴Composite *hypothetical*, labelled as such in the sentence that introduces it. Empirical pieces: 6.SP, 8.SP, S.ID. The skipping and the sitting are hypothetical.

preview. A kitchen that saved every data lesson for a senior elective will arrive at AP Statistics Unit 1 having never asked a question that has variability, and will arrive at Geometry’s conditional probability having never built a two-way table. The map already placed the work. Teach it in the year it sits. Point here for the through-line. Do not dump the AP course into Algebra I, and do not dump Algebra I’s S.ID into this section as if Chapter 7 had not already named it.¹¹⁹⁵

Algebra I, S.ID.1–3 and 5–9. Appendix A’s third critical area for Algebra I is descriptive statistics: shape, center, spread; two-way tables; linear models, residuals, correlation versus causation.¹¹⁹⁶ That is not a sidebar. It is one of five critical areas of the course Appendix A calls a more ambitious Algebra I than has generally been offered. Chapter 7 §7.4 is the teaching home. This section is the parent’s reminder of what the clusters actually contain, at enough grain to hear a wrong answer.

S.ID.1–3: represent data with plots on the real number line — dot plots, histograms, box plots; compare center (median, mean) and spread (interquartile range, standard deviation) of two or more data sets, using shape, outliers, and the context; interpret differences in shape, center, and spread in the context of the data, accounting for possible effects of extreme data points.¹¹⁹⁷ Grade 6 had mean, median, IQR, and mean absolute deviation. Algebra I adds the standard deviation as a named measure of spread, the box plot as a display of the five-number summary, and the comparison of two or more distributions as a requirement rather than a Grade 7 informal look. A child who can compute a mean and cannot say why a skewed distribution might want the median, or why an outlier pulls the mean and the standard deviation harder than it pulls the median and the IQR, has the formula and not the cluster. Hear “they are all about 12” again. Ask for the shape.

S.ID.5: two-way frequency tables for two categorical variables; relative frequencies; possible association. S.ID.6: represent two quantitative variables on a scatter plot; describe how the variables are related; fit a function to the data, informally; informally assess the fit by plotting and analyzing residuals; fit a linear function for a scatter plot that suggests a linear association. S.ID.7: interpret the slope and the intercept of a linear model in the context of the data. S.ID.8: compute (using technology) and interpret the correlation coefficient of a linear fit. S.ID.9: distinguish between correlation and causation.¹¹⁹⁸ A residual is the difference between an observed y and the y the line predicts. A residual plot that looks random is consistent with a linear model being reasonable. A residual plot that curves is a reason to doubt the linear family. Correlation r is a number between

¹¹⁹⁵Brief Chapter 11 do-not-overclaim: do not skip Algebra I S.ID, Geometry S.CP, or Algebra II S.IC because “stats is Chapter 11.” Chapters 7–9 plants. This chapter is the through-line and the AP course.

¹¹⁹⁶Achieve / CCSS writing team, *Appendix A: Designing High School Mathematics Courses Based on the Common Core State Standards*, Oregon ODE copy, <https://www.oregon.gov/ode/educator-resources/standards/mathematics/Documents/math-appendix-a-model-course-pathways.pdf>, accessed 30 August 2026. Algebra I critical area (3): descriptive statistics — shape, center, spread; two-way tables; linear models, residuals, correlation versus causation. “The pathways and courses are models, not mandates.”

¹¹⁹⁷CCSSM S.ID.1–3 (high-school Statistics and Probability conceptual category), as limited to Algebra I in Appendix A (n. 1196): plots on the real number line (dot plots, histograms, box plots); compare center (median, mean) and spread (IQR, standard deviation) of two or more data sets; interpret differences in shape, center, and spread in context, accounting for extreme data points. Grade 6 had mean, median, IQR, MAD; Algebra I adds standard deviation, box plots, and required comparison of distributions.

¹¹⁹⁸CCSSM S.ID.5–9, Algebra I limits in Appendix A: two-way frequency tables (S.ID.5); scatter plots, informal fit, residuals, linear fit (S.ID.6); interpret slope and intercept of a linear model in context (S.ID.7); correlation coefficient using technology (S.ID.8); distinguish correlation and causation (S.ID.9). $r = 0.9$ does not mean X causes Y . Random assignment versus random sampling is the Unit 1 design distinction; Algebra I already says the slogan.

-1 and 1 that measures the strength and direction of a linear association. $r = 0.9$ does not mean X causes Y . Hear a child say “the correlation is 0.9 so X causes Y ” and the sentence is the slogan, not “they computed r wrong.” Random assignment versus random sampling is the design distinction AP Statistics Unit 1 will make precise. This course already says the slogan. Chapter 7 already said it. This chapter gathers it so the parent cannot treat S.ID.9 as a poster.

S.ID.4, the normal distribution, is **not** an Algebra I cluster on the Appendix A traditional-pathway chart. It is Algebra II. Do not steal it into this year because a vendor book printed a bell curve. Do not skip it in Algebra II because “we’ll do AP Stats later.”¹¹⁹⁹

Geometry, S.CP. Appendix A’s sixth critical area for Geometry is applications of probability: independence, conditional probability, two-way tables, the addition rule; permutations and combinations as (+) standards.¹²⁰⁰ Chapter 8 is the course. This section is the chance stop, not a reason to skip Chapter 11, and not a reason to skip S.CP because “stats is Chapter 11.” Conditional probability is $P(A \text{ given } B)$, read from a two-way table as the fraction of B that is also A , not as a keyword hunt for the word *given*. Independence is $P(A \text{ and } B) = P(A) \cdot P(B)$, or equivalently $P(A \text{ given } B) = P(A)$: knowing B happened does not change the probability of A . The addition rule is $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$, which is inclusion-exclusion on a table, not a chant. Two-way tables are the representation that makes all three visible. A child who can recite “independent means multiply” and cannot read a table to check whether the product matches the joint relative frequency has the chant and not the cluster.

Permutations and combinations are (+) on this map: additional mathematics for students who will continue past college-and-career readiness, not a required Geometry object. NMAP’s Major Topics of School Algebra include combinatorics and finite probability as a topic of school algebra through Algebra II, not as a Geometry mandate.¹²⁰¹ A homeschool Geometry that omits S.CP because the parent remembers high-school geometry as two-column proofs of triangles has omitted a critical area Appendix A published. A homeschool Geometry that dumps AP Statistics Unit 2 into the last month of the year has not taught Geometry. Teach independence, conditionals, and tables. Point here for the through-line.

Algebra II, S.ID.4 and S.IC. Appendix A’s Algebra II is where the normal distribution and inference live on the traditional pathway. S.ID.4: use the mean and standard deviation of a data set to fit it to a normal distribution and to estimate population percentages; recognize that there are data sets for which such a procedure is not reasonable; use calculators, spreadsheets, and tables to estimate areas under the normal curve.¹²⁰² S.IC: make inferences and justify conclusions from

¹¹⁹⁹CCSSM S.ID.4 (normal distribution) is an Algebra II cluster on the Appendix A traditional-pathway chart, not Algebra I. Do not steal it into Algebra I because a vendor book printed a bell curve. Do not skip it in Algebra II because “we’ll do AP Stats later.”

¹²⁰⁰Appendix A (n. 1196), Geometry critical area (6): applications of probability — independence, conditional probability, two-way tables, addition rule; permutations/combinations as (+) standards. Chapter 8 is the course. This chapter is the chance stop.

¹²⁰¹National Mathematics Advisory Panel, *Foundations for Success: The Final Report* (U.S. Department of Education, 2008), ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>, accessed 30 August 2026, Table 1, Major Topics of School Algebra: combinatorics and finite probability is a topic of school algebra through Algebra II, not a Geometry mandate. (+) standards in Appendix A are additional mathematics, not required for college-and-career readiness as the map defines it.

¹²⁰²CCSSM S.ID.4, Algebra II in Appendix A: use mean and standard deviation to fit a normal distribution and

sample surveys, experiments, and observational studies — including recognizing the purposes of and differences among those designs; using data from a sample survey to estimate a population mean or proportion and understanding the margin of error; using data from a randomized experiment to compare two treatments; evaluating reports.¹²⁰³ Chapter 9 is the course. This section is the inference preview, not AP Statistics Units 3 and 4 dumped into Algebra II, and not a reason to skip Algebra II's clusters because a senior elective is on the calendar.

A normal model is a special shape, not a law of nature. Many distributions are not normal. S.ID.4 already says to recognize data sets for which the fit is not reasonable. A child who draws a bell over every histogram has a doodle, not a model. The 68–95–99.7 rule, if the family's book uses it, is a property of *this* model, not a property of data. AP Statistics will make z-scores and the normal model a working tool in Units 1 and 2. Algebra II is where the shape is named. Do not skip the naming.

S.IC is the design distinction Unit 1 will demand at AP grain. A sample survey estimates a population quantity. A randomized experiment, if well designed, can support a causal comparison of treatments. An observational study can show association and cannot, by itself, justify a causal claim. Margin of error is the honest width around a sample estimate, not a confession of incompetence. Evaluating reports is reading a study as a design, not as a headline. Those objects are Algebra II. They are also the spine of AP Statistics Unit 1. A student who has seen them in Algebra II will still have a course's worth of procedures in Units 3–5. A student who has not seen them will spend Unit 1 learning that a survey is not an experiment. Either path is teachable. The second is retrofit.

Depends on. First-year algebra: linear equations, interpreting slope and intercept, basic function notation. Comfort with fractions, percents, and inequality language. Not calculus.¹²⁰⁴ A child who cannot read $y = 2.4x + 11$ as “about 2.4 units of y per one unit of x , starting at 11 when x is 0” is not ready to interpret a regression slope. A child who cannot treat 0.23 as a relative frequency and as a probability is not ready for Unit 2. Diagnose the Algebra I objects. Then this course, or then the Algebra II clusters, or then AP Statistics. Do not diagnose calculus.

Unlocks. College introductory-statistics credit or placement where the institution accepts the AP score. Data literacy for social science, biology, and civic life. Does not unlock Physics C. Does not unlock a calculus-based engineering sequence.¹²⁰⁵ Those last two sentences are sequence facts. They are the plant for Chapter 10: statistics is not a STEM-calculus substitute. Chapter 10 is the calculus chapter. This chapter will not dump it. A parent choosing AP Statistics as a senior elective after Algebra I has chosen a legitimate college-equivalent. A parent choosing it *instead of* the precalculus-to-calculus path, with Physics C or engineering as the destination, has chosen a

estimate population percentages; recognize data sets for which the procedure is not reasonable; use technology and tables to estimate areas under the normal curve. A normal model is a special shape, not a law of nature.

¹²⁰³CCSSM S.IC, Algebra II in Appendix A: make inferences and justify conclusions from sample surveys, experiments, and observational studies — purposes and differences among those designs; estimate a population mean or proportion from a sample survey, including margin of error; compare two treatments from a randomized experiment; evaluate reports. Design distinction Unit 1 will demand at AP grain.

¹²⁰⁴CED prerequisite and 01-scope Statistics “Depends on.” Not calculus. Diagnose Algebra I objects (linear equations, slope and intercept in context, basic function notation; fractions, percents, inequalities) before AP Statistics, not a limit.

¹²⁰⁵01-scope Statistics “Unlocks.” Physics C and calculus-based engineering named as sequence facts. Plant for Chapter 10: statistics is not a STEM-calculus substitute. Do not dump Chapter 10.

different object than the destination requires. Take both if the student has the background and the destination wants both. College Board already said that.

Integrated Mathematics I–III packages the same clusters across three years rather than concentrating them in Algebra I, Geometry, and Algebra II. Topic-by-topic Mathematics I from Appendix A pages 49–90 remains **UNVERIFIED** in this project.¹²⁰⁶ If the family’s book is integrated, map tonight’s lesson onto S.ID, S.CP, or S.IC as the object, and onto this chapter as the through-line. The mathematics does not change with the cover. Transcript the course as Integrated Mathematics I, II, or III, with a parenthetical a stranger can map — a plant for Chapter 12.

What this section has done is gather the high-school clusters so they cannot be skipped, and so they cannot replace the AP course. The AP course, five units, first revised exam May 2027, comes next.

11.4 AP Statistics five units (Fall 2026 CED)

The college-equivalent, explained to the parent who stopped at algebra, is five units and four practices, not nine old units and a rumor about cuts. Effective Fall 2026. First revised exam May 2027. Non-calculus-based. Prerequisite: successful completion of a first-year algebra course.¹²⁰⁷ Journalism about what was “cut” is not this section. This section is the CED.

Four statistical practices, which College Board aligns to ASA GAISE II: formulate questions; collect data; analyze data; interpret results.¹²⁰⁸ GAISE II page-level claims stay **UNVERIFIED**. The four verbs are the course’s job description. A unit that is only “compute r ” has skipped formulate, collect, and interpret. A unit that is only “we surveyed the family” has skipped analyze. Variability is the central idea, not “the average.” Correlation is not causation — Algebra I already says so. This course’s Unit 1 makes the design distinction the slogan does not: random assignment versus random sampling.

Unit 1. Exploring One-Variable Data and Collecting Data. 20–30 percent of multiple-choice.¹²⁰⁹ This is the heaviest slice, and it is the retrofit trap. A student who skipped 6.SP, 7.SP, 8.SP, and S.ID will spend this unit learning vocabulary the map already placed. A student who did those years will spend this unit tightening design. The topic spine, from the CED unit guide as extracted into the scope memo, is a parent checklist, not a 180-day plan.¹²¹⁰

¹²⁰⁶Integrated Mathematics I–III topic-by-topic from Appendix A pp. 49–90: **UNVERIFIED** (01-scope open question 7). If the family’s book is integrated, map tonight’s lesson onto S.ID, S.CP, or S.IC. Transcript as Integrated Mathematics I/II/III with a parenthetical a stranger can map (Chapter 12).

¹²⁰⁷CED (n. 1162): five units, effective Fall 2026, first revised exam May 2027, non-calculus-based, first-year-algebra prerequisite.

¹²⁰⁸CED (n. 1162): four statistical practices — Formulate Questions; Collect Data; Analyze Data; Interpret Results — aligned to ASA GAISE II. GAISE II page-level **UNVERIFIED** (n. 1172).

¹²⁰⁹CED (n. 1162), Unit 1 title and MC weighting 20–30%. Heaviest multiple-choice slice. Exploring data and collecting it is load-bearing work, not a warm-up before “the real formulas.”

¹²¹⁰CED unit guide, Unit 1 topic spine as extracted into 01-scope on 30 August 2026: investigative questions; variables (categorical vs quantitative; discrete vs continuous; parameter vs statistic); frequency/relative-frequency tables; bar charts and pie charts; histograms, stem-and-leaf, dotplots; shape/center/variability/outliers; mean, median, range, IQR, standard deviation, percentiles; 1.5×IQR and 2-SD outlier rules; five-number summary and boxplots; comparisons; z-scores; census vs sample; experiment vs observational study (prospective, retrospective, survey); confounding; random sampling methods (SRS, stratified, cluster, systematic); bias (voluntary response, undercoverage, nonresponse, response); experimental design (comparison, random assignment, replication, control; placebo, blinding; completely

Investigative questions. Variables: categorical versus quantitative; discrete versus continuous; parameter versus statistic. A parameter describes a population. A statistic describes a sample. Confusing the two is how a sample mean becomes “the true average.” Frequency and relative-frequency tables. Bar charts and pie charts for categorical data. Histograms, stem-and-leaf plots, and dot plots for quantitative data. Shape, center, variability, outliers. Mean, median, range, IQR, standard deviation, percentiles. The $1.5 \times \text{IQR}$ and 2-standard-deviation outlier rules. Five-number summary and box plots. Comparisons of distributions. z-scores. Census versus sample. Experiment versus observational study, including prospective, retrospective, and survey. Confounding. Random sampling methods: simple random sample, stratified, cluster, systematic. Bias: voluntary response, undercoverage, nonresponse, response. Experimental design: comparison, random assignment, replication, control; placebo, blinding; completely randomized, randomized block, matched pairs. Generalization versus causal inference.

That last pair is the load-bearing distinction of the unit, and of the course. Random *sampling* is how a sample can represent a population, and so how a statistic can be generalized to that population, with honest uncertainty. Random *assignment* is how treatments can be compared causally, inside the experimental units you actually have. A voluntary-response sample of homeschool parents on a forum is not a simple random sample of U.S. homeschoolers. An observational comparison of two curricula the families already chose is not a randomized experiment. Confounding is the unnamed third variable that travels with the treatment. The parent who can hear a wrong answer hears “we asked the group, so this is what homeschoolers think,” and asks who is missing. Hear “the kids who used the app scored higher, so the app caused it,” and ask who chose the app.

Unit 2. Probability, Random Variables, and Probability Distributions. 15–25 percent of multiple-choice.¹²¹¹ Geometry’s S.CP is the on-ramp: sample spaces, two-way tables, independence, conditionals, the addition rule. This unit makes those into random variables and named distributions. A random variable is a numerical outcome of a chance process. A probability distribution lists the values and the probabilities, or describes them with a model. Discrete versus continuous again, now as the type of random variable. Expected value is a long-run mean, not a promise about the next trial. Named models the parent will see in a typical book include the binomial and the normal; this chapter will not invent a complete catalog of which named models the CED requires beyond what the unit title and the Unit 1 spine already named. The normal model, Algebra II’s S.ID.4, becomes a working approximation for some sampling distributions later. Unit 2 is where probability stops being a Geometry chapter and becomes the language inference will speak.

Unit 3. Inference for Categorical Data: Proportions. 15–25 percent of multiple-choice.¹²¹² A population proportion p . A sample proportion $\hat{p}$. A sampling distribution of $\hat{p}$ that, under conditions the child has to check, is approximately normal. A confidence interval as an interval of plausible values for p , not as a probability that a fixed interval “contains the truth” in the way a

randomized, randomized block, matched pairs); generalization vs causal inference. Parent checklist, not a 180-day plan. Do not invent additional CED topic lists below this spine.

¹²¹¹CED (n. 1162), Unit 2 title and MC weighting 15–25%. Geometry S.CP is the on-ramp. This chapter does not invent a complete catalog of which named probability models the CED requires beyond the unit title and the Unit 1 spine.

¹²¹²CED (n. 1162), Unit 3 title and MC weighting 15–25%. First inference procedure at AP grain. A small p -value is evidence against a specified null, in a specified design; it is not a proof and not a cause.

child first wants to say it. A significance test as a measurement of how surprising the sample is if a hypothesized p were true. Conditions. A conclusion in context. This is the first inference procedure at AP grain. Worked-example fading belongs here, in section 11.5, because the procedure is high-element-interactivity for a novice: hypotheses, conditions, mechanism, numbers, sentence. The incorrect example this course exists to hit is “ p is small, so the null is false and X caused Y .” A small p -value is evidence against a specified null, in a specified design. It is not a proof. It is not a cause. If the design was an observational study, a small p -value still does not license a causal sentence.

Unit 4. Inference for Quantitative Data: Means. 10–20 percent of multiple-choice.¹²¹³ The same logic as Unit 3, now for a mean. A population mean μ . A sample mean $\bar{x}$. A t -procedure rather than a z -procedure when the population standard deviation is not known, which is the usual case. One sample, two samples, paired data as different designs, not as a keyword for “two numbers.” Conditions again. Conclusion in context again. The parent who stopped at algebra does not need to become a statistician overnight. The parent needs to hear three wrong answers: using a proportions procedure on a quantitative mean; treating paired data as two independent samples; writing a causal sentence from a t -test on observational data. Interleaving of inference procedures, after a short blocked introduction to each, is the practice shape. Rohrer, Dedrick, and Burgess’s $d = 1.05$ is still a grade-7 algebra result on four problem kinds. Do not import it as this course’s effect size. Do import the shape: blocked to learn execution, mixed to choose the procedure.¹²¹⁴

Unit 5. Regression Analysis. 10–20 percent of multiple-choice.¹²¹⁵ Algebra I’s S.ID.6–9 are the on-ramp: scatter, linear fit, residuals, r , correlation versus causation. This unit makes inference on the slope, conditions for residual analysis at AP grain, and the written claim that a linear model is or is not reasonable for *these* data. $r = 0.9$ is still not a cause. A residual plot that fans is a reason to doubt constant variance, which is a condition, not a vibe. Extrapolation is a prediction outside the data’s x -range, and it is a claim the course wants the child to refuse or to qualify. Technology computes. The child interprets. A kitchen that lets the calculator or the model print $\hat{y} = 2.4x + 11$ and then stops has stopped before the unit’s job.

Exam logistics beyond the five-unit weights were not fully extracted as a 2026–27 free-response timing table in the pass that opened the CED for this project. Do not invent a minutes-and-questions recipe from memory of an older CED. College Board’s homeschool exam-ordering process is Chapter 12: parents cannot order exams; find a participating school via the AP Course Ledger; the coordinator enrolls the student as “exam only”; mid-November order deadline on the 29 June 2026 homeschool guide.¹²¹⁶ Self-study is allowed. A formal AP course is recommended, not required.

¹²¹³CED (n. 1162), Unit 4 title and MC weighting 10–20%. Same logic as Unit 3, for a mean. Paired versus two-sample is a design distinction, not a keyword.

¹²¹⁴Doug Rohrer, Robert F. Dedrick, and Kaleena Burgess, “The Benefit of Interleaved Mathematics Practice Is Not Limited to Superficially Similar Kinds of Problems,” *Psychonomic Bulletin and Review* 21 (2014): 1323–1330, <https://files.eric.ed.gov/fulltext/ED548041.pdf>, accessed 30 August 2026. Grade-7 classroom crossover, $n = 140$, unannounced test two weeks later, interleaved 72% vs blocked 38%, $d = 1.05$, 95% CI [0.80, 1.30]. Four problem kinds, none of them hypothesis tests. Do not import this d as an AP Statistics result. Do import blocked-to-learn-execution, mixed-to-choose-the-procedure. Full objects in Chapter 2.

¹²¹⁵CED (n. 1162), Unit 5 title and MC weighting 10–20%. Algebra I S.ID.6–9 are the on-ramp. Technology computes; the child interprets. $r = 0.9$ is still not a cause.

¹²¹⁶Exam minutes-and-questions recipe beyond the five-unit MC weights: not fully extracted as a 2026–27 free-

Seats are school-controlled. This chapter will not invent a registration recipe beyond that guide. Credit is set by each college.

What this section has done is put five units and four practices on the table so the parent can hear a wrong answer in each. Teaching is not a unit table. Teaching is a question that has variability, a design, a schema for chance, and a simulation that earns a sentence. That guidance comes next.

11.5 Teaching guidance: variability, design, schemas for chance

A question that has variability is the first teaching move, and it is a move the parent who stopped at algebra has to be able to hear missing. “Find the mean, median, and mode of 3, 5, 5, 7, 12” is a worksheet. It has no question. It has no variability in the GAISE sense, because nothing was asked whose answer varies; a list was handed over. “How long do our sunflower seedlings grow in a week?” is a question. The answers will differ. The child will have to decide what to measure, on which plants, with which ruler, on which day. The distribution is then about those decisions as much as it is about sunflowers. Compute the mean after the question, not instead of it.¹²¹⁷ GAISE page-level claims remain **UNVERIFIED**. The spirit is the move: formulate, collect, analyze, interpret. A triad of averages with no story is not the move.

Design is the second move, and it is the move a slogan cannot replace. Correlation is not causation. Algebra I S.ID.9 already says so. AP Statistics Unit 1 makes the distinction the slogan does not. Random *sampling* licenses generalization to a population, with uncertainty, when the sample is actually random in a named way. Random *assignment* licenses causal comparison of treatments, inside the units you assigned, when the assignment is actually random and the rest of the design — comparison, control, replication — is actually there. A kitchen survey of “who in this co-op likes the new book” is real data. It is a voluntary-response sample of a convenience population. It does not license “homeschoolers prefer.” A family that tries two breakfasts on alternate weeks and measures morning-work time has an informal experiment with confounded days, a tiny n , and no blinding. It does not license “eggs cause focus.” Use the data. Name the design. Do not crown it.¹²¹⁸

Kitchen surveys and kitchen experiments are the homeschool temptation that Finding 26 already answered in another key. The National Mathematics Advisory Panel’s Finding 26: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the small set of high-quality studies the Panel admitted, improve computation, simple word problems, or equation solving.¹²¹⁹ The analogue for this chapter is not a new Panel finding. It is the same trans-

response timing table in the pass that opened the CED. Do not invent one from an older CED. College Board Home-schoolers’ Guide (Jacob Mishook, 29 June 2026): parents **cannot order** AP exams; find a school via the AP Course Ledger; coordinator enrolls “exam only”; mid-November order deadline; self-study allowed; a formal AP course is recommended not required. Do not invent a registration recipe beyond that guide. Plant for Chapter 12.

¹²¹⁷Teaching object: a question that has variability, not a mean/median/mode worksheet with no data story. GAISE spirit (n. 1172). Sunflower-seedling example is the brief’s illustration, not a reported kitchen.

¹²¹⁸CED Unit 1 design spine (n. 1210): random sampling vs random assignment; generalization vs causal inference; confounding. Algebra I S.ID.9 already says correlation is not causation; Unit 1 makes the design distinction the slogan does not.

¹²¹⁹NMAP, *Foundations for Success* (2008), Finding 26: teaching with “real-world” contexts improves performance on similar real-world items; it does not, in the Panel’s small high-quality set, improve computation, simple word

fer caution, applied. Kitchen surveys are real data. They do not by themselves teach inference. They help on similar kitchen-survey items. They do not replace a worked example of a confidence interval, a residual plot, or a two-way table the child has to read. Cooking can be a second representation of a fraction already being taught as a number; it is not the number line. A household height-versus-age scatter can be a second representation of a linear association already being taught as a fit; it is not S.ID.9, and it is not Unit 5. Informal activity is not a substitute for the trained objects. Transfer from kitchen data to formal inference, measured rather than assumed, remains an open question this book will not fill with analogy.

Schemas, not keywords, still apply to probability word problems. Chapter 2 already locked the principle: word problems are learned as schemas, not as keyword hunts. Jitendra, Star, Dupuis, and Rodriguez 2012 is a seventh-grade *proportion* trial — $g = 1.24$ immediate, $g = 1.27$ at six weeks, null transfer — not a probability trial. IES 2021 Recommendation 5, strong evidence, 18 studies, and the 2009 RtI guide both reject keyword strategies.¹²²⁰ This chapter will not invent a homeschool probability-schema RCT, and it will not quote the 1.24 as if it had been measured on compound events. What transfers is the teaching move. Identify the chance structure. Represent it. Solve. Check. Chance structures the parent can name without a new experiment: a single-stage sample space (one spin, one draw with a list of outcomes); a compound event with replacement or independence (multiply along a tree if independence holds); a compound event without replacement (condition along the tree); a two-way table (read joint, marginal, and conditional relative frequencies from the cells); a simulation (repeat a chance process to estimate a probability). Keywords fail here the way *all*, *left*, and *total* fail in the additive cases. *And* does not always mean multiply. *Or* does not always mean add. *Given* does not always sit next to the condition the child should use. *At least* is a complement in one structure and a slog in another. Teach the structure. Vary the cover story. Never teach a magic word.

Simulation is a representation that must connect to a written claim. Grade 7 already named simulations as a way to approximate probabilities of compound events. AP Statistics will use simulation again for sampling distributions the child cannot yet derive. The teaching rule is the same at both grains. A simulation without a sentence is a ritual. The child writes, before running it, what the chance process is, what counts as a success, how many trials, and what the estimate will be an estimate *of*. After running it, the child writes the count, the estimate, and a comparison to any theoretical value the sample space already gave. “I ran it 50 times” is not a claim. “About 18/50 of the trials were both red, so I estimate $P(\text{both red}) \approx 0.36$, near the $3/8$ from the table” is a claim. The

problems, or equation solving. Analogue for this chapter, labelled as analogue: kitchen surveys are real data; they do not by themselves teach inference. Not a new Panel finding. Confrey, Maloney, and Nguyen (2008) published a methodological critique of the Panel; the critique is real; the Panel is still the synthesis later documents cite.

¹²²⁰Asha K. Jitendra, Jon R. Star, Danielle Dupuis, and Michael C. Rodriguez, “Effectiveness of Schema-Based Instruction for Improving Seventh-Grade Students’ Proportional Reasoning: A Randomized Experiment,” *Journal of Research on Educational Effectiveness* 5 (2012): seventh-grade RCT, 1,163 students, 42 classrooms, proportion problems, immediate $g = 1.24$, six-week retention $g = 1.27$, no significant transfer. **Not a probability trial.** Do not quote 1.24 as if it had been measured on compound events. Lynn S. Fuchs et al., *Assisting Students Struggling with Mathematics: Intervention in the Elementary Grades*, WWC 2021006 (IES, 2021), Recommendation 5, strong evidence, 18 studies; Russell Gersten et al., *Assisting Students Struggling with Mathematics: Response to Intervention (RtI) for Elementary and Middle Schools*, NCEE 2009-4060 (IES, 2009): both reject keyword strategies. What transfers is the teaching move (identify the chance structure; represent; solve; check), not a new coefficient. No homeschool probability-schema RCT was opened.

representation earned the sentence. A language model that offers to “run a simulation” and then prints a fluent paragraph has stolen both the representation and the sentence. Section 11.7 will ban that theft. Here the ban is pedagogical: the child runs it, or draws the random digits, or uses a table in the book, and then writes.

Worked-example fading for the first hypothesis-test procedure, because the procedure is new and high in element interactivity. Sweller and Cooper 1985: for novices, studying a worked solution outperforms unguided problem solving. Renkl and Atkinson: fade from the end. Booth, Lange, Koedinger, and Newton 2013: explaining incorrect examples helps conceptions, in their case for two-step linear equations in the Algebra I Cognitive Tutor.¹²²¹ Those papers are not AP Statistics trials. They are the method Chapter 2 already locked. Apply it here and label the application. First test: a complete worked example the parent already understands — hypotheses, conditions, mechanism, numbers, sentence in context. Second: an example the child must explain, including one incorrect example whose error is the error this course exists to hit. “ p is small, so the null is false and X caused Y ” is that error. The null is a specified hypothesis about a parameter. A small p -value is evidence against it, under the design you actually ran. It does not prove the null false in the logical sense the child wants. It does not name a cause if the design was observational. The child names the error. Then a completion problem with the last sentence blank. Then a conventional problem. Fade. Do not assign twenty tests on day one and call the struggle productive. Productive failure, Kapur 2014, is generation-then-instruction on a new concept when priors exist. It is not NMAP Finding 27, which is explicit instruction for students already struggling. Do not mash them. A first t -procedure is a novice object. Fade it.¹²²²

Incorrect examples belong in design as much as in computation. A voluntary-response sample labelled “SRS.” A randomized experiment that assigned by who arrived first. A two-way table whose conditionals were read from the wrong margin. A regression whose residual plot curves and whose writer still claims the linear model is appropriate. A confidence interval interpreted as “there is a 95 percent chance that *this* interval contains p ,” when the course’s language is that 95 percent of intervals constructed this way would capture p . The parent does not need a graduate course to hear those. The parent needs one correct worked example and one incorrect one per new object, and the ear to ask the child to name the error.

Interleaving of inference procedures after a short blocked introduction is the practice shape, not an imported effect size. Rohrer, Dedrick, and Burgess 2014: grade-7 classroom crossover, $n = 140$, unannounced test two weeks later, interleaved 72 percent versus blocked 38 percent, $d = 1.05$, on

¹²²¹John Sweller and Graham A. Cooper, “The Use of Worked Examples as a Substitute for Problem Solving in Learning Algebra,” *Cognition and Instruction* 2, no. 1 (1985): 59–89. Alexander Renkl, Robert K. Atkinson, and colleagues on guidance fading / backward fading. Julie L. Booth, Kenneth R. Koedinger, et al., “Using Example Problems to Improve Student Learning in Algebra: Differentiating between Correct and Incorrect Examples,” *Learning and Instruction* 25 (2013) — Booth, Lange, Koedinger, and Newton 2013 as cited in Chapter 2: explaining incorrect examples (alone or with correct ones) especially useful for conceptual understanding of two-step linear equations, combined with guided practice in the Algebra I Cognitive Tutor. **Not AP Statistics trials.** Application of Chapter 2’s method, labelled as application.

¹²²²Manu Kapur, “Productive Failure in Learning Math,” *Cognitive Science* 38 (2014): generation-then-instruction on a new concept when priors exist. NMAP Finding 27: explicit instruction for students already struggling. Different comparisons; do not mash. A first hypothesis-test procedure is a novice object; fade it (n. 1221).

four problem kinds that were not hypothesis tests.¹²²³ Rohrer, Dedrick, Hartwig, and Cheung 2020: 61 percent versus 38 percent, $d = 0.83$, 54 classes, joint with spacing. Almost all clean classroom randomized trials of interleaving in mathematics are grade 7. Do not export $d = 1.05$ to AP Statistics. Do export the discrimination problem the 2014 materials already showed: students fail to associate “this story to this strategy” when practice is blocked. Unit 3 versus Unit 4 versus a two-sample versus a paired versus a chi-square the family’s book may include is exactly that discrimination problem. Blocked to learn execution of one procedure. Then mixed: which procedure, which conditions, which sentence. A parent whose purchased book is almost entirely blocked — most are, Rohrer, Dedrick, and Hartwig 2020, “The scarcity of interleaved practice in mathematics textbooks” — can still mix yesterday’s proportion interval with last week’s one-sample mean test on the same page.¹²²⁴ The book does not have to do it. The parent can.

Representations that scale. A two-way table is the representation for conditionals, independence, and the addition rule, from Geometry S.CP through Unit 2 and into Unit 3’s χ^2 if the book goes there. A dot plot or a box plot is the representation for shape, center, and spread, from Grade 6 through Unit 1 and Unit 4. A scatter plot plus a residual plot is the representation for association, from Grade 8 through S.ID through Unit 5. A tree is the representation for staged chance. A simulation is the representation for “I cannot list the sample space, but I can estimate.” Do not let a calculator or a model replace the representation before the child can produce it. Technology computes a standard deviation the child already understands as spread. It does not introduce spread.

The parent who stopped at algebra is the design user here as everywhere. Maloney, Ramirez, Gunderson, Levine, and Beilock 2015 is still a school-homework study: parent math anxiety plus frequent help predicted worse child math, math-specific. It is not a parent-as-sole-teacher RCT.¹²²⁵ The constructive response Chapter 2 already stole still holds. Prepare so that you can hear a wrong answer. Then do not give the next step. The wrong answers in this chapter are design sentences and chance structures, not only arithmetic. “The average is 12, so they are all about 12.” “We asked the group.” “ $r = 0.9$, so it caused it.” “ p is small, so the null is false and the treatment worked.” “I simulated it, so it is true.” Hear those. Ask for the shape, the sample, the design, the residual, the sentence. The model may prepare the adult to hear them. It may not write the AP paragraph.

What this section has done is lock the teaching objects: a question that has variability, a named design, a chance schema rather than a keyword, a simulation that earns a sentence, a faded first test, an incorrect causal p -value, mixed practice after a short block, without importing Rohrer’s d . Tips that survive — add-on versus substitute, do not skip K–8, title the course — come next.

¹²²³Rohrer, Dedrick, and Burgess 2014 (n. 1214). Do not export $d = 1.05$ to AP Statistics.

¹²²⁴Doug Rohrer, Robert F. Dedrick, Marissa K. Hartwig, and Chi-Ngai Cheung, “A Randomized Controlled Trial of Interleaved Mathematics Practice,” *Journal of Educational Psychology* 112, no. 1 (2020): 40–52, 61% vs 38%, $d = 0.83$, 54 seventh-grade classes, one-month delayed unannounced test; interleaved assignments also space same-kind items, so d is a joint effect. Doug Rohrer, Robert F. Dedrick, and Marissa K. Hartwig, “The Scarcity of Interleaved Practice in Mathematics Textbooks,” *Educational Psychology Review* 32 (2020): 873–83. A parent whose book is blocked can still mix yesterday’s proportion interval with last week’s one-sample mean test. Brunmair and Richter interleaving g for math: **UNVERIFIED** (not opened).

¹²²⁵Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, “Intergenerational Effects of Parents’ Math Anxiety on Children’s Math Achievement and Anxiety,” *Psychological Science* 26, no. 9 (2015). School-homework study: parent math anxiety plus frequent help predicted worse child math, math-specific. **Not** a parent-as-sole-teacher RCT. Constructive response as in Chapter 2: prepare so you can hear a wrong answer; then do not give the next step.

11.6 Tips that survive: add-on vs substitute; do not skip K–8

Steal these as sentences, not as a franchise.

AP Statistics after Algebra I is a legitimate senior elective for a non-STEM path and a legitimate add-on for a STEM path. College Board’s revised framework encourages taking both AP Statistics and AP Calculus when the student has the background. Data-science-interested students are also encouraged to take AP Statistics and an AP Computer Science course. It is not published as a substitute for calculus in STEM calculus sequences.¹²²⁶ A parent choosing statistics as the senior college-equivalent, after Algebra II, with no plan for Physics C or a calculus-based engineering sequence, has chosen a real object. A parent choosing it *and* calculus has chosen the pair College Board already named.

It is a weak stealth substitute for Calculus I if the destination is Physics C or a calculus-based engineering sequence. Those destinations are calculus-based. This course is non-calculus-based. Chapter 10 is the calculus chapter. This chapter plants the refusal and does not dump eight AP Calculus units to make the refusal louder.¹²²⁷ Take both if the student has the background and the destination wants both. Take statistics alone if the destination is data literacy, social science, biology, or an introductory-stats credit the receiving college actually grants. Do not take statistics alone and call it the STEM math sequence.

Do not skip K–8 data because “we’ll do AP Stats later.” Grade 6 distributions, Grade 7 sampling and probability models, Grade 8 bivariate association are the spine. Informal category counts and line plots sit earlier, in Grades 1–5. A skipped K–8 data strand is Unit 1 retrofit. NCES 2016: probability ever-taught to 24 percent of high-school-level homeschoolers. That is not a completion rate. It is how a strand dies.¹²²⁸ NCTM Focal Points: do not omit Grade 7–8 focal-point content if algebra is offered in Grade 8. Data is in that warning.

Do not skip Algebra I S.ID, Geometry S.CP, or Algebra II S.ID.4 and S.IC because “stats is Chapter 11.” Chapter 7, Chapter 8, and Chapter 9 already have the clusters. This chapter is the through-line. Teach the clusters in the year they sit. Point here for the AP course and for the years that came before.

Do not treat AP Statistics as stealth calculus. Do not treat calculus as stealth statistics. Two college-equivalents. Different objects. Different prerequisites. Different destinations. The Fall 2026 CED’s Algebra I prerequisite is a fact about this course, not a ceiling on high-school mathematics, and not a claim that statistics covers limits.

Do not treat AP Statistics as stealth Algebra II. The official prerequisite is first-year algebra. Algebra II remains useful. Useful is not required. A homeschool Algebra II that omits functions, logarithms, and trigonometry has not covered school algebra as NMAP defined it, whether or not AP Statistics is on the transcript.¹²²⁹

¹²²⁶Revised course framework (n. 1165). Legitimate senior elective for a non-STEM path; legitimate add-on for a STEM path. Not published as a calculus substitute.

¹²²⁷Plant for Chapter 10. Physics C and calculus-based engineering sequences are calculus-based destinations. AP Statistics is non-calculus-based (n. 1162). Do not dump AB/BC unit tables here.

¹²²⁸6.SP / 7.SP / 8.SP (nn. 1186, 1188, 1190); Grades 1–5 data (n. 1182); NCES 24 percent (n. 1169). Do not skip K–8 data because “we’ll do AP Stats later.”

¹²²⁹NMAP Table 1 (n. 1201), restated as a parent checklist in Chapter 9: a homeschool Algebra II that omits functions,

Do not use journalism about “cuts” as the CED. Five units, the weights in section 11.4, the four practices, the Unit 1 design spine. Use the CED. Label journalism as journalism.¹²³⁰

Do not invent a homeschool AP Statistics participation rate. Whether homeschoolers sit this exam at any measurable rate is **UNVERIFIED**. NCES 24 percent is probability ever-taught, not this exam.¹²³¹

GAISE II was not opened in full. Treat the spirit — formulate, collect, analyze, interpret; variability as the central idea — not a quoted page. Page-level claims stay **UNVERIFIED**.¹²³²

Transcript the course as Statistics or AP Statistics, with a parenthetical a stranger can map if the family’s book used a publisher title. Not “Math 12.” Not “Saxon something” without the parenthetical. Not “data fun.” Chapter 12 is the export layer: no national homeschool diploma; state compliance is not college evidence; course titles a stranger can map onto Algebra I, Geometry, Algebra II, Precalculus, Calculus, Statistics. Dual-enrollment statistics, if taken, listed and backed by the college transcript. AP seats are school-controlled; parents cannot order exams; the 29 June 2026 College Board homeschool guide is the recipe this book will not invent beyond.¹²³³ Plant that. Do not dump the NCAA worksheet here.

Simulation must connect to a written claim. A run count is not a claim. The child writes what was estimated, from what process, in how many trials, compared with what. Section 11.5 is the teaching rule. This bullet is the reminder when a calculator or a model offers to “simulate it for you.”

Kitchen surveys are real data. They do not by themselves teach inference. Finding 26 analogue: similar-context help, not transfer to a residual plot or a two-sample t the child has not been taught to run. Use the household scatter as a second representation. Do not crown it as Unit 5.

Technology computes after the child has chosen the procedure. A graphing calculator, a spreadsheet, Wolfram|Alpha, or a statistics package can check a standard deviation, a t -interval, or a least-squares slope *after* the child has named the procedure, the conditions, and the sentence. It is a check, not an explainer. Chapter 3 already said that about Wolfram and a CAS. The analogue here is the same object with a different button.¹²³⁴

logarithms, and trigonometry has not covered school algebra as NMAP defined it. AP Statistics’ Algebra I prerequisite does not make AP Statistics stealth Algebra II.

¹²³⁰Journalism labelled journalism (n. 1173). Use the CED.

¹²³¹Homeschool AP Statistics participation **UNVERIFIED** (n. 1170). NCES 24 percent is probability ever-taught, not this exam.

¹²³²GAISE II not opened in full (n. 1172). Page-level **UNVERIFIED**.

¹²³³Plant for Chapter 12. Transcript the course as Statistics or AP Statistics, with a parenthetical a stranger can map if the family’s book used a publisher title. No national homeschool diploma. State compliance is not college evidence. Dual-enrollment statistics, if taken, listed and backed by the college transcript. AP ordering: Homeschoolers’ Guide 29 June 2026 (n. 1216). NCAA: math core = Algebra I or higher; audited, CLEP, or credit-by-exam courses are not NCAA-approved core courses (NCAA Home School Toolkit 2025–26, as in the brief). Do not dump the worksheet here.

¹²³⁴Wolfram Research, Wolfram|Alpha, <https://www.wolframalpha.com/>, fetched 30 August 2026: computational engine on Wolfram Language, not an LLM. Check, not explainer. Chapter 3 §3.7 analogue: a statistics package or Wolfram may check a calculation *after* the child has chosen the procedure. Pro step-by-step pricing **UNVERIFIED** as a live number on the homepage this day.

The first hypothesis test is a faded worked example, then an incorrect causal p -value, then a completion, then a mixed page. Do not assign a packet of tests on day one. Do not import Rohrer's $d = 1.05$ as this course's result. Do import blocked-then-mixed.

Unaided exit tickets still apply. Bastani's exam phase is five minutes with the phone in the other room: one "census or sample," one correlation-versus-causation, one "which procedure." Performance on a page the model helped finish is not learning. Chapter 3. This chapter will not re-litigate the 48 percent / -17 percent at monograph length.¹²³⁵

The kitchen that stops at Algebra I because "the state doesn't require it" has a college problem in every named university page in Chapter 12, whether or not AP Statistics is on the transcript. Three years through Algebra II is the floor those pages tend to name, as a pattern, not a law. Statistics does not erase that floor.¹²³⁶

What this section has done is print the stealable sentences. The model will happily compute a p -value. That is the problem. Parent-hosted AI, with the child doing the work and the parent holding the key, is the last lane.

11.7 Parent-hosted AI for data and chance

Chapter 3 already printed the kitchen condition. This section will not rewrite Bastani, Soderstrom and Bjork, or the ban-versus-allow list. It will reuse them on data and chance, labelled as practice suggestions, not as a homeschool RCT, which still does not exist.¹²³⁷ One sentence if the family remembers only one: the model may prepare the adult and the next problem; it may not do the child's problem. Procedure choice is the child's. A calculation check comes after.

Allow, parent-hosted, pointing at Chapter 3's workflows.

(a) Explain the concept to the parent first. Why a voluntary-response sample is biased. Why random assignment is not random sampling. Why a small p -value is not a cause. Why $r = 0.9$ is not X caused Y . Why a simulation estimate is not a proof. Parent account, no child present. The prompt pattern Chapter 3 already stole: explain it to *me* as an adult who last took math in [year]; the idea in one paragraph; the notation; the two mistakes this age makes; one tiny problem I should solve before I sit down; the answer under a parent key. Then the parent solves the tiny problem without the model, then checks. The check can be the textbook, a known-correct key, or Wolfram|Alpha

¹²³⁵Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman, "Generative AI without guardrails can harm learning: Evidence from high school mathematics," *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122. Assisted practice: GPT Base +48 percent relative to control; unaided exam: -17 percent. GPT Tutor: +127 percent practice / ~0 vs control on the exam. Percentages of control grades, not Cohen's d . Not a homeschool study. Full discussion: Chapter 3. Nicholas C. Soderstrom and Robert A. Bjork, "Learning Versus Performance," *Perspectives on Psychological Science* 10, no. 2 (2015): performance is "often an unreliable index" of learning.

¹²³⁶Chapter 12 plant (n. 1178). Statistics on the transcript does not erase a three-year-through-Algebra-II floor those named pages tend to name.

¹²³⁷Chapter 3 §3.3–3.4: ban versus allow and workflows (a)–(f) are practice suggestions grounded in Bastani, Soderstrom and Bjork, and Oreopoulos and Low, not a homeschool RCT. No year-plus kitchen GPT-Tutor RCT in the literature opened 30 August 2026. No homeschool AP Statistics RCT. This section reuses the list on data-and-chance objects and still calls them practice suggestions.

or a statistics package if the tiny problem is a computation. The point is whether the explanation landed hard enough to hear a wrong design sentence at the table.¹²³⁸

(c) Extra isomorphic inference items with the parent-held key. Same structure, new numbers, new cover story. Two pages: student sheet and parent key. Five to eight items, not a completed copy of tonight’s assigned FRQ. Isomorphic here means a new one-sample proportion interval after last night’s one-sample proportion interval, or a new “which design is this?” after yesterday’s design item — not a packet that mixes every unit on day two of Unit 3. Mix later. The child never sees the key. Sample the set. Work one item before the child sees the sheet. Bastani’s GPT Base was fully correct only 51 percent of the time on high-school items, with 8 percent arithmetic errors even when the method was right. A fluent wrong key is worse than no extra practice. A known-correct key still beats a fluent wrong one.¹²³⁹ Models hallucinate statistics: invented p -values, swapped hypotheses, causal verbs on observational data, residual plots described that were never plotted, data sets whose columns do not add. The parent who can hear a wrong answer can also hear a fake interval.

(d) Hint script, not an answer key for the child. Level 1: a question only — “Is this a sample or a census?” “Were the treatments assigned at random?” Level 2: name the concept without numbers — “random assignment licenses a causal comparison inside these units; random sampling licenses generalization.” Level 3: the next step, still not the final interval or the final sentence. One likely mistake and what to say if it appears. Parent reads the hints aloud. Do not put the final answer anywhere the child might see.¹²⁴⁰

(f) Diagnose a child’s already-wrong work. Photograph the page *after* the attempt. Parent-only prompt: name the most likely misconception in one sentence; one question I should ask the child, not the correct steps; one isomorphic item to retest after we talk. The child answers the parent, not the model. The misconception this chapter exists to hit: “the correlation is 0.9 so X causes Y .” Also: “ p is small, so the null is false and the treatment worked.” Also: “we asked the group, so this is the population.” Also: “I simulated it, so it is true.”¹²⁴¹

Generate a dataset and a wrong conclusion for the child to attack. Practice suggestion, not an RCT. This is the statistics-specific job that Chapter 3’s six workflows did not have to name, because fractions do not hallucinate a causal verb as fluently as a paragraph about r . Parent account. Ask the model for a small table — twenty rows is enough — of two variables, plus a one-sentence

¹²³⁸Chapter 3 workflow (a): explain the concept to the parent first. Statistics objects: voluntary-response bias; random assignment vs random sampling; small p -value is not a cause; $r = 0.9$ is not causation; a simulation estimate is not a proof.

¹²³⁹Chapter 3 workflow (c): isomorphic items, parent-held key. Bastani author PDF: GPT Base fully correct 51 percent of the time when asked for the answer; 8 percent arithmetic errors even when the method was right. Models hallucinate statistics — invented p -values, swapped hypotheses, causal verbs on observational data — as a pedagogical risk analogous to that 51 percent, not as a new named hallucination benchmark. A known-correct key still beats a fluent wrong one.

¹²⁴⁰Chapter 3 workflow (d): hint script that withholds the solution, escalating from a question to a named concept to the next step. Bastani GPT Tutor hidden prompt is the analogue (withhold the full solution; ask what they have tried; point out a named mistake; escalate slowly). Practice suggestion, not an RCT.

¹²⁴¹Chapter 3 workflow (f): diagnose already-wrong work; child answers the parent, not the model. Load-bearing misconceptions for this chapter: “ $r = 0.9$ so X causes Y ”; “ p is small, so the null is false and the treatment worked”; “we asked the group, so this is the population”; “I simulated it, so it is true.”

conclusion that is *wrong on purpose*: a causal claim from an observational scatter; a generalization from a voluntary-response sample; a mean treated as “they are all about this”; a p -value treated as the probability the null is true. Put the wrong conclusion and the answer key — what is wrong, and a better sentence — on the parent page only. The child gets the table and a prompt: “What is wrong with this conclusion? Write a better one.” The child writes unaided. The parent holds the key. Then talk. Then one isomorphic attack. Do not ask the model to grade the child’s paragraph as the last word. Models hallucinate statistics. Sample the marking.¹²⁴²

(e) Kitchen surveys as a hands-on workflow. The model may help design a question that has variability. The child collects and represents. “Help me write three statistical questions we can answer with a week of seedling measurements, and three that we cannot.” The parent picks. The child measures, plots, writes a sentence about shape and spread. The model does not collect. The model does not plot. The model does not write the sentence. Finding 26 analogue still holds: the kitchen data are a second representation, not Unit 1.¹²⁴³

Ban, child-facing, during the attempt.

Letting the model compute a p -value the child has not set up. That is Bastani’s crutch with a statistics button. The child must name the parameter, the hypotheses, the conditions, and the procedure. A package or Wolfram may then check the arithmetic. A chat model that prints a fluent test from a pasted table has done the child’s problem.¹²⁴⁴

“Analyze this and write the AP paragraph.” Ban that prompt, child-facing and, as a substitute for the child’s writing, parent-facing too. The AP free-response question is unaided work. College Board still requires the student to produce unaided writing in the exam room. Practice suggestion, not an RCT: every FRQ the child will meet is written without a model, on paper, timed. The model may afterwards, on the parent account, mark against a rubric the parent pasted from an official FRQ — and the parent still samples the marking, because models hallucinate theorems, drop negative signs, and, in this chapter, invent significance.¹²⁴⁵ Paste-the-FRQ is Chapter 3’s paste-the-worksheet in a new font.

Photo-to-key on tonight’s statistics worksheet. Photomath, Mathway, and Symbolab list statistics among subjects. Vendor “designed for learning” is vendor. The mechanism is GPT Base.¹²⁴⁶

¹²⁴²Statistics-specific allow, practice suggestion, not an RCT: parent asks the model for a small dataset plus a wrong conclusion the child will attack; parent holds the key; child writes unaided. Ban the prompt “analyze this and write the AP paragraph” as a substitute for that writing. Models hallucinate statistics (n. 1239). Sample any later marking.

¹²⁴³Chapter 3 workflow (e): hands-on; the model is a materials generator. Kitchen surveys: the model may help design a question that has variability; the child collects and represents. Finding 26 analogue (n. 1219).

¹²⁴⁴Ban, child-facing, during the attempt: letting the model compute a p -value the child has not set up. Bastani crutch (n. 1235) with a statistics button. Procedure choice is the child’s; a package or Wolfram may check arithmetic after (n. 1234).

¹²⁴⁵Chapter 3 §3.7: AP-style free response is unaided, on paper, timed. Practice suggestion, not an RCT. The model may afterwards, parent account, mark against a rubric the parent pasted from an official FRQ; the parent still samples the marking. Paste-the-FRQ is paste-the-worksheet. Ban “analyze this and write the AP paragraph.”

¹²⁴⁶Photomath, <https://photomath.com/>; Mathway, <https://www.mathway.com/>; Symbolab, <https://www.symbolab.com/>; fetched 30 August 2026. Statistics listed among subjects. **Vendor** “designed for learning.” No product-specific unaided-transfer RCT opened. Mechanism: Bastani’s GPT Base (n. 1235). Photo-to-key is the ban.

Companion persona. Open-web agent in the child’s hands. Detectors as police on the child’s written conclusions. Chapter 3 §3.3 and §3.6. Liang, Yuksekgonul, Mao, Wu, and Zou 2023: average false-positive rate 61.22 percent (arXiv) / 61.3 percent (*Patterns*) on human TOEFL essays. A homeschooler who writes a careful paragraph about why a sample is biased is exactly the writer those detectors smear.¹²⁴⁷

Session shape, Chapter 3 §3.8, scaled to this chapter. Practice suggestion, not an RCT. Unaided warm-up: one “census or sample,” one correlation-versus-causation, one probability model or “which procedure.” Teach from the family’s book, parent already having used (a) and, if needed, (b) *before* sitting down. Struggle. Hint not key. Hands-on or a table the child draws if the miss was design or chance structure. Unaided exit ticket: a mixed three-item check without the model. “Finish your stats with ChatGPT so we can go to soccer” is GPT Base, the pattern with a large-sample causal harm estimate.¹²⁴⁸ If Khanmigo is in the house, expect the Tennessee pattern unless the parent requires the hint dialogue. Oreopoulos and Low, NBER w35620, August 2026: intention-to-treat 0.040 population SD per term; do not invent a standalone 0.06 as ITT. 96 percent tried it at least once; the median student messaged it on only a third of practice days and in 17 percent of mistake sessions; 14.5 percent of messages contained a mathematical question or a step of reasoning. Practice minutes predicted MAP; chat did not, descriptively. Working paper, not yet journal-peer-reviewed.¹²⁴⁹

A *hypothetical*, labelled as such, can put the ban in a room. A parent whose sixteen-year-old pastes a scatter plot into a chatbot with “write the AP paragraph,” copies a fluent residual analysis, and scores 100 percent on the page, then cannot say on tomorrow’s unaided three-item check whether $r = 0.9$ licenses a cause, is running GPT Base.¹²⁵⁰ The empirical pieces are Bastani’s crutch, Chapter 3’s FRQ-unaided rule, and Soderstrom and Bjork’s performance/learning split. The age, the scatter, and the 100 percent are hypothetical. The move is not: take the paste, put the table on paper, demand a sentence the child writes, and run the unaided check.

Hermes Agent, if used, lives on the parent machine. Model split this day (Nous Portal): Hermes 4 or any strong chat model for parent explanation and hint scripts; agentic examples — Claude

¹²⁴⁷Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, and James Zou, “GPT detectors are biased against non-native English writers,” *Patterns* 4, no. 7 (2023); arXiv 2304.02819 average false-positive rate 61.22 percent on human TOEFL essays; *Patterns* 61.3 percent. Do not use detectors to police a homeschooler’s written statistical conclusions. Companion-harm cases — *Garcia v. Character Technologies*, *Raine v. OpenAI* — are allegations, not verdicts, not a literature on mathematics learning. Chapter 3 §3.6. COPPA amended Rule published 22 April 2025, general compliance 22 April 2026. California A.B. 1159 is not law as of 30 August 2026 (Senate third reading).

¹²⁴⁸Chapter 3 §3.8 session shape, scaled: unaided warm-up, teach, struggle, hint not key, unaided exit ticket. Practice suggestion, not an RCT. “Finish your stats with ChatGPT so we can go to soccer” is GPT Base (n. 1235).

¹²⁴⁹Philip Oreopoulos and Nina Low, “One Click Away: AI Tutoring with Khanmigo in a Two-Year School Experiment,” NBER Working Paper 35620, August 2026. Two-year cluster RCT, eighteen Tennessee middle schools, assignment to Khan Academy plus Khanmigo. Table 5 pooled ITT, baseline-adjusted: 0.040 population SD per term; do not invent a standalone 0.06 as ITT. Abstract’s 0.06–0.08 is a school-year figure. 96 percent tried Khanmigo at least once; median student messaged it on only a third of practice days and in 17 percent of mistake sessions; 14.5 percent of messages contained a mathematical question or a step of reasoning. Practice minutes predicted MAP; chat did not, descriptively. Working paper, not yet journal-peer-reviewed. Gains resemble Khan Academy practice without AI. Not a homeschool study.

¹²⁵⁰Composite *hypothetical*, labelled as such. Empirical pieces: Bastani’s crutch (n. 1235), Chapter 3’s FRQ-unaided rule (n. 1245), Soderstrom and Bjork (n. 1235). The age, the scatter, and the 100 percent are hypothetical.

Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro — for the tool loop that turns a lesson note into a student sheet, a parent key, and a wrong-conclusion attack set. Hermes 4 is **not recommended inside Hermes Agent**. Hermes 4 ≠ Hermes Agent. Not a child-safe school SKU. Memory in `~/hermes/` is sensitive. Hard no: messaging gateway to a child’s Telegram or WhatsApp.¹²⁵¹ Grok Bot, if used, is included with SuperGrok and Cursor Pro/Teams as of the 26 August 2026 expansion post. Cloud storage required; Legacy Privacy Mode unsupported; shared computer per user; no education SKU and no FERPA claim on the pages opened. Grok ≠ Grok Bot ≠ El Salvador Grok-as-tutor. Hard no: the child’s hands; an IEP PDF on the shared computer; signing the Bot into a school portal.¹²⁵²

Facts and arithmetic checks are not an LLM job to memorize by chat. A z-score is arithmetic. Expected value is a sum of products. GPT Base was fully correct 51 percent of the time, with 8 percent arithmetic errors even when the method was right. Use a dumb calculator or a statistics package or Wolfram after the child has set up the sum. Use the model to generate the deck of “which procedure” prompts, not to sit between the child and the sum.¹²⁵³

The open questions this chapter leaves standing are the ones a sequel would fill with analogy. Home-school AP Statistics participation rate: **UNVERIFIED**. GAISE II page-level claims: not opened. Whether stats-as-substitute is actually chosen in kitchens: not in these sources. College-credit comparison of AP Statistics versus introductory-statistics dual enrollment: not a single national experiment in the opened literature; IES dual-enrollment outcomes study first report expected 2027.¹²⁵⁴ No homeschool AP Statistics RCT. No year-plus kitchen GPT-Tutor RCT. Hold those holes. Do not close them with a kitchen scene.

Everything Chapters 4 through 11 taught still has to leave the house as a record a stranger can read.

¹²⁵¹Nous Research, Hermes Agent, <https://hermes-agent.org/>, fetched 30 August 2026. Parent-machine tool; MIT license; memory in `~/hermes/`. Hermes 4 ≠ Hermes Agent; Hermes 4 not recommended inside Hermes Agent (Nous Portal docs). Portal agent-work examples this day: Claude Sonnet 4.6, GPT-5.5 Pro, Gemini 3 Pro Preview, DeepSeek V4 Pro. Not a child-safe school SKU. Hard no: messaging gateway to a child’s Telegram / WhatsApp. Practice suggestion for parent-side generation of a student sheet, parent key, and wrong-conclusion attack set, not an RCT.

¹²⁵²xAI, “Introducing Grok Bot,” 11 August 2026; “Grok Bot is now included with more plans,” 26 August 2026; Grok Bot FAQ, fetched 30 August 2026. Cloud storage required; Legacy Privacy Mode unsupported; one computer per user shared across Bots; no education SKU and no FERPA claim on the pages opened. Grok ≠ Grok Bot ≠ El Salvador Grok-as-tutor (xAI announcement 11 December 2025 is an announcement, not an RCT; independent evaluation **UNVERIFIED**). Hard no: the child’s hands; IEP/504 PDFs on the shared computer; signing the Bot into a school portal.

¹²⁵³Chapter 3 / 04-ai load-bearing: flash retrieval of arithmetic facts is not an LLM job. Bastani: GPT Base fully correct 51 percent; 8 percent arithmetic errors even when the method was right. A z-score and an expected-value sum are arithmetic. Use a calculator, a statistics package, or Wolfram after the child has set up the sum. Use the model to generate a “which procedure” deck, not to sit between the child and the sum.

¹²⁵⁴Open questions this chapter leaves standing. Homeschool AP Statistics participation rate **UNVERIFIED** (n. 1170). GAISE II page-level claims not opened (n. 1172; brief open question on GAISE). Whether stats-as-substitute is actually chosen in kitchens: not in these sources (n. 1179). College-credit comparison of AP Statistics versus intro-stats dual enrollment: not a single national experiment in the opened literature; IES/NCEE, Outcomes Associated with Dual Enrollment Programs, first report expected 2027, <https://ies.ed.gov/use-work/evaluations/outcomes-associated-dual-enrollment-programs>. No homeschool AP Statistics RCT. No year-plus kitchen GPT-Tutor RCT. NCES 2020-125 dual-enrollment supply is **public-school** supply (82 percent of public schools with grades 9–12 offered dual or concurrent enrollment, 2017–18); homeschool-specific dual-enrollment counts **not in these sources**. Hold those holes. Do not close them with a kitchen scene.

Course titles. Grades. Credits. An AP seat ordered through a school, or a dual-enrollment transcript from a college, or a CLEP score the receiving institution actually takes — different objects, overlapping destinations, no automatic credit. There is no national homeschool diploma. State compliance is not college evidence. Chapter 12 is that export layer. This chapter was the through-line for data and chance, from three categories in Grade 1 to a non-calculus college-equivalent that does not substitute for calculus. The child writes the paragraph. The parent holds the key. The agent does not take the test.

Chapter 12

Records, Transcripts, College-Ready

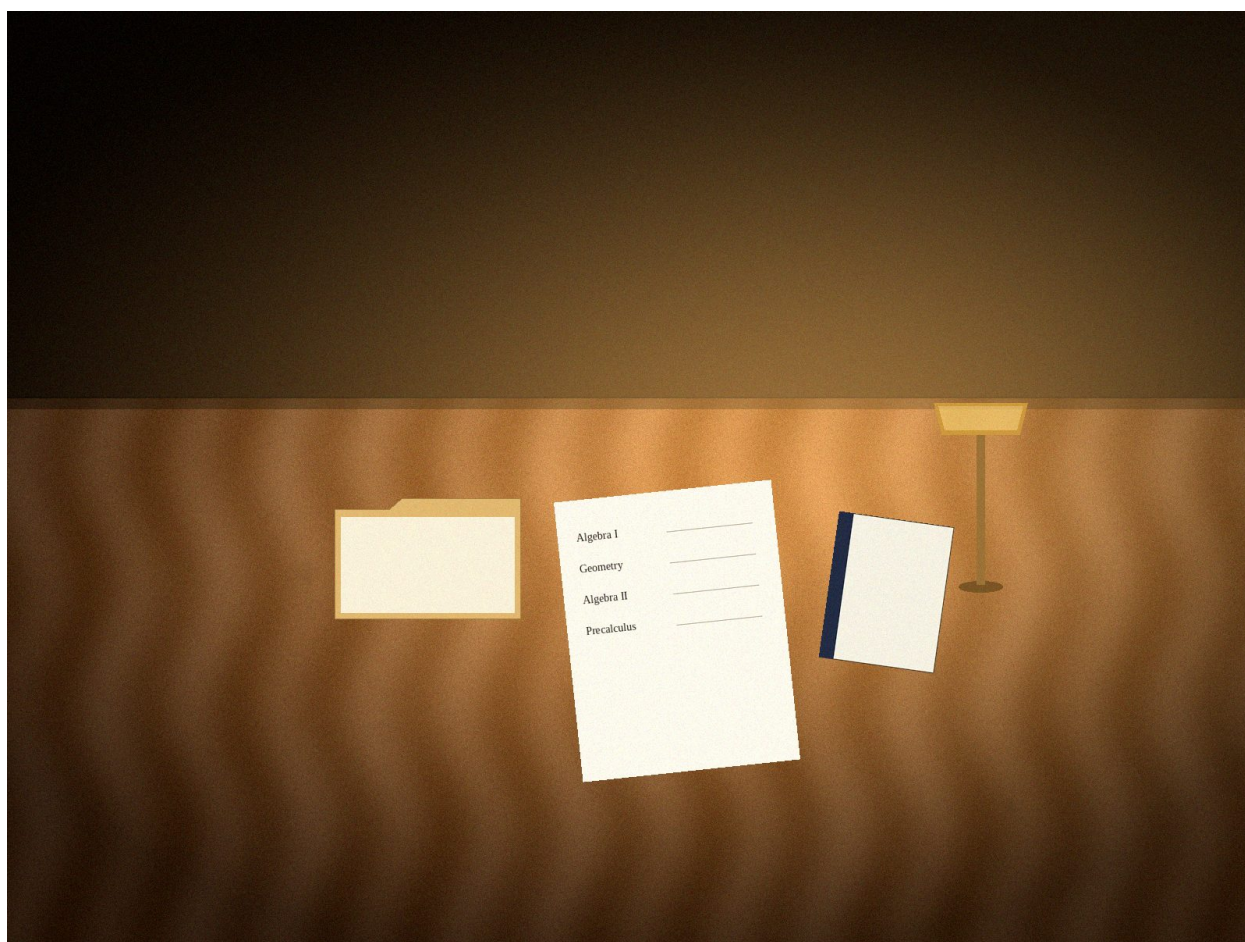


Figure 13: A manila folder, a transcript page with generic course titles, and a plain test booklet.

Two documents sit on the table this chapter is written from. Neither is a kitchen scene. Neither is a diploma. The first is a state card from the Home School Legal Defense Association. Texas, on the card this research opened: notification, no; assessment, no; subjects, yes; teacher qualifications, none. Mathematics is a named subject. A written curriculum is required under the *Leeper* private-

school reading the card cites. A portfolio is not. A standardized test is not. A high-school diploma issued by a national office does not exist on the card, because no such office exists.¹²⁵⁵ The card is advocacy, attorney-reviewed for members. It is not the Texas Education Code and it is not the *Leeper* opinion. Illinois and Idaho sit next to Texas on the same advocacy map as other no-notice examples. A parent in those rooms can finish a legal year of mathematics without producing a single sheet a registrar would file.

The second document is a National Collegiate Athletic Association Core-Course Worksheet, dated July 2025 on the Eligibility Center's student-resources page. It asks who taught the course, what texts were used, what the prerequisites were, how the grade was assigned, and what content was covered. The parent or administrator signs. The grade and the credit on the worksheet must match the transcript. A course titled something a stranger cannot map onto Algebra I or higher is not a mathematics core course. A CLEP pass is not a core course. Evaluation happens only after an NCAA school requests it.¹²⁵⁶ That packet is not a statute. It is a private association's eligibility rules. It binds athletic scholarships. It will not be satisfied by the Texas card's silence.

Those two documents are the split this chapter has to keep in view. Compliance is one job. Export is another. There is no national homeschool diploma. State compliance is not college evidence. A family can satisfy Texas, Illinois, or Idaho and still have nothing a university admissions office or an Eligibility Center evaluator can read. A family can satisfy New York's Individualized Home Instruction Plan, Pennsylvania's portfolio, or North Carolina's annual mathematics-inclusive test and still have a course titled "Saxon 8/7" or "Epsilon" where a registrar wanted Algebra I. Chapters 4 through 11 taught the mathematics. This chapter is the export layer. It does not rewrite the kitchen Chapter 1 already named. It does not reopen the outcomes fight Chapter 1 already printed. It does not dump a 50-state legal manual from listicles that disagreed with each other in search snippets and were not used as primaries.¹²⁵⁷ It translates work already done into titles, grades, credits, and external signals a stranger can map.

College-readiness in mathematics is a documentation problem as much as a teaching problem. The teaching problem was the rest of the book: fractions as the gate, concepts and procedures that iterate, the child who struggles before the tool, the parent who can hear a wrong answer. The documentation problem is this chapter's. A kitchen calculus course taught from a textbook can count if the worksheet is honest and the course is college-preparatory. A CLEP Calculus pass

¹²⁵⁵Home School Legal Defense Association, "Texas Homeschool Laws (at a glance)," <https://hsllda.org/legal/texas>, and "How to Comply with Texas's Homeschool Law," <https://hsllda.org/post/how-to-comply-with-texas-homeschool-law>. **Advocacy**, attorney-reviewed for members, not the Texas Education Code and not the *Leeper* opinion. Notification no; assessment no; subjects yes (mathematics among them); teacher qualifications none; written curriculum under the private-school reading. There is no national homeschool diploma on the card or in any federal statute this stream opened.

¹²⁵⁶NCAA Eligibility Center, *Homeschool Core-Course Worksheet* (July 2025), http://fs.ncaa.org/Docs/eligibility_center/Student_Resources/Core_Course_Worksheet.pdf. Teacher of record, texts, prerequisites, grading basis, content, parent or administrator signature; grade and credit must match the transcript. CLEP and credit-by-exam exclusion, and evaluation-after-request, are in the *Home School Toolkit 2025–26*, http://fs.ncaa.org/Docs/eligibility_center/Student_Resources/Home_School_Toolkit.pdf. Primary association documents, not blogs.

¹²⁵⁷HomeschoolLeap, HomeSchoolie, and similar 50-state listicles disagreed with each other in search snippets and were not used as primaries. See the book outline's manuscript-wide do-not-invent list and 03-homeschool.md §7. HSLDA cards are advocacy maps of variation, not statutes.

cannot substitute for an NCAA core course. An Advanced Placement seat in May is controlled by a school coordinator, not by a parent who self-studied. Dual enrollment is the clean external transcript. Named university pages this research opened — Pittsburgh, Southern Methodist, North Central, Southwest Baptist — want, as a pattern and not as a national rule, about three years of high-school mathematics through Algebra II as a floor. A comprehensive 2026 policy statement from the National Association for College Admission Counseling dedicated to homeschool mathematics was **not opened** as a PDF and remains **UNVERIFIED**. This stream used named university pages instead. That gap stays flagged.¹²⁵⁸

What this chapter is not. It is not a promise that a homeschooler will outscore a public-school student on a nationally representative mathematics exam. The National Center for Education Statistics has never published a nationally representative homeschool NAEP mathematics score, and Chapter 1 already refused to crown a convenience-sample percentile.¹²⁵⁹ It is not a pitch for a tutor, a co-op, or a community college. Those are release valves the 2016 Parent and Family Involvement tables already counted — any tutor, 23 percent; any local homeschool group or co-op, 31 percent; a small share of the 9–12 band also in college classes — and they become transcript facts when the parent is the ceiling.¹²⁶⁰ It is not a sequel to Chapter 3. An agent may generate a transcript template or a course description. The parent still signs. Fabricating hours or a diploma is a ban, not a workflow. None of the tools Chapter 3 named is FERPA-ready as a kitchen product. It is not the coda. The coda's job is what an honest book can promise. This chapter's job is the paper trail that promise would have to rest on.

The thesis still holds, and it is still not a causal claim about homeschool mathematics. A parent who is not a math specialist can take a child from first-grade number sense through high-school calculus if three things hold: the parent understands the concept well enough to hear a wrong answer; the child does the struggle before the tool; and any agent is a supplement hosted and constrained by the parent — not a substitute for the child's work and not a school-safe product by default. The sources do not prove that thesis. They do not contradict it. They constrain how the work, once done, can be shown to a stranger. Promise a path, not a percentile.

What follows is the export, named from documents. Section 12.1 splits the legal object from the college object so a parent cannot confuse a satisfied statute with a readable record. Section 12.2 installs the transcript as a translation document: titles a registrar recognizes, not publisher level names; placement and acceleration as record-keeping choices; the Common App parent-as-counselor path, flagged for confirmation in the year of publication. Section 12.3 prints the NCAA mathematics

¹²⁵⁸A comprehensive National Association for College Admission Counseling policy statement dedicated to homeschool mathematics was **not opened** as a 2026 PDF. Open question 24 in the book outline. This chapter uses named university pages (nn. 1312–1315) instead. Do not invent a 2026 NACAC PDF.

¹²⁵⁹National Center for Education Statistics has never published a nationally representative homeschool NAEP mathematics score. NHES asks subjects taught, not achievement. Chapter 1 §1.7 and n. 109 print the fight (Ray/NHERI convenience samples versus Lubienski, McCracken/ICHER, Kunzman and Gaither 2020, Cardus 2023) and refuse a crown. This chapter does not reopen it as a finding.

¹²⁶⁰Jiashan Cui and Rachel Hanson, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey*, NCES 2020-001 (Washington, DC: National Center for Education Statistics, 2019), <https://nces.ed.gov/pubs2020/2020001.pdf>, Table 3 (any tutor 23 percent; any local homeschool group or co-op 31 percent) and Table 4 (1 percent of homeschoolers also attended college, interpret with caution). Federal web tables. Last opened subject and provider tables; NHES:2023 First Look did not republish them.

core from NCAA pages, not from blogs: Algebra I or higher; three years for Division I and two for Division II among sixteen core courses; CLEP is not a core course. Section 12.4 keeps SAT, PSAT, ACT, AP, CLEP, and dual enrollment from collapsing into one “external signal.” The SAT homeschool high-school code is 970000. AP seats are school-controlled. CLEP is self-register. Dual enrollment is the clean college transcript. Section 12.5 prints what named university pages actually ask, as a pattern and not a rule, and leaves the unopened NACAC PDF standing. Section 12.6 reprints parent expectations and the Cardus bachelor’s-rate facts without reopening Chapter 1’s fight, and names what NCES does not show. Section 12.7 folds special-education and anxiety back to Chapters 1 and 2 as callouts, states the AI rule for records, and closes on the paper trail. Access date for the fetches under these claims remains 30 August 2026.

12.1 No national diploma; compliance is not college evidence

There is no national homeschool diploma. That sentence is load-bearing and easy to over-read. It does not mean a family cannot issue a diploma. Many do. A parent, an umbrella school, a church-school statute, or a private-school affidavit can put a graduation date on a transcript and a signature under it. It means there is no federal office, no national homeschool bureau, and no single piece of paper that converts a satisfied state statute into college evidence. Colleges and the NCAA Eligibility Center will still want a transcript. Some will want course descriptions. Some will want an external signal — an SAT or ACT score, an AP score, a CLEP transcript, a dual-enrollment transcript. None of those objects is produced by filing a notice, or by not filing one.

State compliance is not college evidence. The variation Chapter 1 already named as a legal-object problem is, in this chapter, an export problem. The teaching chapters could not assume a single paperwork load. This chapter cannot assume that the paperwork a state asked for is the paperwork a registrar will accept.

Pennsylvania, on the advocacy how-to this research opened, is the textbook portfolio state under option 1: an annual affidavit or unsworn declaration to the superintendent by 1 August; a portfolio that holds a log, work samples, test results, and an evaluator letter; standardized testing in mathematics and reading in grades 3, 5, and 8; secondary mathematics that must include general mathematics, algebra, and geometry, citing 24 P.S. § 13-1327.1.¹²⁶¹ That portfolio is a compliance object. It can be a useful archive when a parent later writes a course description. It is not, by itself, a high-school transcript a stranger can map onto Algebra I, Geometry, and Algebra II. A family that keeps a beautiful portfolio and titles the year’s work “Math 10” has satisfied one reader and confused the next.

New York mixes quarterly narratives with tests. On the HSLDA how-to this research opened — advocacy reading 8 NYCRR § 100.10, not the regulation itself — the parent files an Individualized Home Instruction Plan listing syllabi, curriculum materials, textbooks, or a plan of instruction; logs

¹²⁶¹Home School Legal Defense Association, “Pennsylvania Homeschool Affidavit and Declaration,” <https://hsllda.org/post/pennsylvania-homeschool-affidavit>, and “Pennsylvania Homeschool Laws (at a glance),” <https://hsllda.org/legal/pennsylvania>. **Advocacy** citing 24 P.S. § 13-1327.1: option 1 annual affidavit or unsworn declaration to the superintendent by 1 August; portfolio (log, work samples, test results, evaluator letter); standardized testing in mathematics and reading in grades 3, 5, and 8; secondary mathematics including general mathematics, algebra, and geometry. Confirm the statute. Parent may not administer the option 1 tests (advocacy how-to).

hours; files quarterly reports; and produces an annual assessment. Narrative is permitted in grades 1–3. Standardized testing is required at least every other year in grades 4–8 and every year in grades 9–12. Named tests include Iowa, the California Achievement Test, Stanford, CTBS, MAT (retired), a State Education Department test, and PASS. SAT and ACT do *not* qualify as the annual assessment. Composite above the 33rd percentile or one year of growth.¹²⁶² A New York family can therefore finish high school with a thick IHIP file and still need a separate transcript for college, and can sit the SAT for admissions without using that score to close the year’s compliance. Two objects. Two readers.

North Carolina: one-time notice to the Division of Non-Public Education, not to the local district; the person providing academic instruction must have a high-school diploma or equivalent; nine calendar months; an annual nationally standardized test that *must include mathematics*, plus English grammar, reading, and spelling; scores kept available one year.¹²⁶³ The HSLDA card lists state-mandated subjects as no. The test-record requirement is how mathematics enters the compliance list. A family that never taught mathematics would fail the test-record requirement, not a curriculum-naming requirement. A family that taught mathematics, sat the test, and kept the scores for a year has complied. A college that asks for Algebra II by title has asked for something the test does not name.

Texas, Illinois, and Idaho, on the cards this research opened, require none of that testing for compliance.¹²⁶⁴ Texas’s how-to, still advocacy, names mathematics among required subjects and a written curriculum under the *Leeper* private-school interpretation. A parent in Texas may legally teach calculus without notifying anyone and without a mathematics credential, provided a bona fide written curriculum includes mathematics. That parent will still meet a university page that wants three years through Algebra II and an Eligibility Center that wants a signed transcript plus a worksheet per core course. The legal silence is not a college silence. It is a documentation burden that falls entirely on the kitchen.

Must a mathematics curriculum be named for compliance? Often no. Texas requires *a* written curriculum covering mathematics, not a named title. New York’s plan requires a list of syllabi, curriculum materials, textbooks, or a plan of instruction. Pennsylvania’s affidavit requires an outline of proposed educational objectives by subject area, not a publisher name. North Carolina, on the

¹²⁶²Home School Legal Defense Association, “How to Comply with New York’s Homeschool Law,” <https://hsllda.org/post/how-to-comply-with-new-yorks-homeschool-law>. **Advocacy** reading of 8 NYCRR § 100.10. IHIP; hours; arithmetic grades 1–6; mathematics credits grades 7–8 and 9–12; quarterly reports; annual assessment (narrative OK in 1–3; standardized at least every other year in 4–8; every year in 9–12). Named tests: Iowa, CAT, Stanford, CTBS, MAT (retired), SED, PASS. SAT/ACT do not qualify. Composite above the 33rd percentile or one year of growth. Confirm against NYSED regulation before legal advice.

¹²⁶³Home School Legal Defense Association, “How to Comply with North Carolina’s Homeschool Law,” <https://hsllda.org/post/how-to-comply-with-north-carolinas-homeschool-law>, and “North Carolina Homeschool Laws (at a glance),” <https://hsllda.org/legal/north-carolina>. **Advocacy**. One-time notice to the Division of Non-Public Education; instructor high-school diploma or equivalent; nine calendar months; annual nationally standardized test that must include mathematics plus English grammar, reading, and spelling; scores kept available one year. Card lists state-mandated subjects as no. Confirm against the North Carolina statute.

¹²⁶⁴Home School Legal Defense Association, Texas, Illinois, and Idaho “at a glance” cards: <https://hsllda.org/legal/texas>; <https://hsllda.org/legal/illinois>; <https://hsllda.org/legal/idaho>. **Advocacy**. Notification no; assessment no; subjects yes; teacher qualifications none. Examples of no-notice compliance, not a 50-state census.

card, does not mandate subjects but mandates a test that includes mathematics. The legal object is coverage, a record, or a score — not Saxon, not Singapore, not Math-U-See. The college object is a course title a stranger can map. Confusing those two objects is how a parent treats a satisfied statute as a finished transcript.

This section is not a 50-state legal manual. HomeschoolLeap, HomeSchoolie, and similar listicles disagreed with each other in search snippets and were not used as primaries.¹²⁶⁵ HSLDA is advocacy, the way a parent finds the state's code. The code is the authority. Parents must read their own state's statute and regulation. Massachusetts, on the card: notification yes; assessment yes; subjects yes; teacher qualifications no. Local approval has historically been required. Legislative fights over approval versus notice were in flux on HSLDA's news feed in 2026. This book will not freeze a 2026 bill as law.¹²⁶⁶ West Virginia's how-to, opened as a testing example, describes an annual assessment option: a nationally normed test covering reading, language, mathematics, science, and social studies, with a fourth-stanine (23rd percentile) or improvement treated as acceptable progress.¹²⁶⁷ That is another compliance object. It is still not a transcript.

The tests homeschoolers actually sit split the same way. State-required batteries are compliance: North Carolina's annual nationally standardized test; New York's named list, with SAT and ACT excluded; Pennsylvania option 1's mathematics and reading tests in grades 3, 5, and 8, which the parent may not administer. Voluntary and college-facing tests are export: Iowa Assessments (ITBS/ITED), Stanford Achievement Test, California Achievement Test in the CAT/TerraNova lineage, the Classic Learning Test used by some Christian colleges — Treleaven 2022 analyzed CLT homeschool scores in *Home School Researcher*, an advocacy journal, and is named here as the existence of that literature, not as a finding this chapter will crown — plus PSAT/NMSQT, SAT, ACT, AP Calculus AB and BC, AP Statistics, AP Precalculus, and CLEP College Algebra, Precalculus, and Calculus.¹²⁶⁸ Sitting a compliance test does not produce a college-facing score unless the instrument is one a college already reads. Sitting SAT or ACT does not close New York's annual assessment. Two readers. Two calendars.

Umbrella schools, cottage schools, and some church-school statutes can issue transcripts. Chapter 1 already noted that they are a legal wrapper in some states — California's private-school option among them — and that they are not in the NCES curriculum-source table as a named mathematics pedagogy.¹²⁶⁹ They can put a school name and an administrator signature on the page. They do

¹²⁶⁵See n. 1257. Listicles were not primaries. Parents must read their own state's code.

¹²⁶⁶Home School Legal Defense Association, "Massachusetts Homeschool Laws (at a glance)," <https://hsllda.org/legal/massachusetts>. **Advocacy.** Notification yes; assessment yes; subjects yes; teacher qualifications no. Local approval historically required. 2026 legislative fights on approval versus notice were in flux on HSLDA's news feed. Do not freeze pending legislation as law. Open question 25 in the book outline.

¹²⁶⁷Home School Legal Defense Association, "How to Comply with West Virginia's Homeschool Law," <https://hsllda.org/post/how-to-comply-with-west-virginias-homeschool-law>. **Advocacy.** Opened as a testing example: nationally normed test covering reading, language, mathematics, science, social studies; fourth stanine (23rd percentile) or improvement treated as acceptable progress. Not a transcript rule.

¹²⁶⁸Voluntary and college-facing instruments as listed in 03-homeschool.md §3.1. Classic Learning Test: some Christian colleges accept it; Southwest Baptist University names CLT alongside ACT/SAT (n. 1315). Treleaven 2022 analyzed CLT homeschool scores in *Home School Researcher* (**advocacy journal**). Named as the existence of that literature, not as a finding this chapter crowns. PSAT/SAT/ACT/AP/CLEP logistics: n. 1292.

¹²⁶⁹Cui and Hanson, NCES 2020-001, Table 6 (curriculum sources) and related participation tables. Umbrella schools are a legal wrapper in some states (California private-school option among them). They can issue transcripts. They

not magically teach calculus, and they do not convert a publisher level name into Algebra I unless someone writes Algebra I. Full-time virtual public or private school is a different legal object again. Pew, restating NCES, noted 1.8 percent of K–12 students in full-time virtual public or private school whose parents did *not* consider them homeschooled; NCES 2023 also reports 0.7 percent full-time virtual *as homeschoolers*. Do not collapse those rooms.¹²⁷⁰ A child in a virtual public charter has a teacher of record and, often, a school transcript the registrar already knows how to file. That is not the parent-as-administrator packet this chapter is writing.

The design fact for every later section is blunt. In Texas a parent may legally finish high school without a portfolio, without a test, and without a notice, and still owe a stranger a transcript. In New York the same parent already has quarterly grades and a standardized battery and still owes the stranger titles, credits, and, if athletics are in view, a worksheet per course. In Pennsylvania option 1 the parent already has a portfolio and an evaluator letter and still owes the stranger a document that looks like the NCAA example transcript rather than a folder of work samples. There is no national diploma that collapses those rooms. State compliance is not college evidence.

What a stranger can read is titles, grades, and credits. That is the next section.

12.2 Course titles a stranger can map

There is no national homeschool transcript. Practice, from the NCAA Eligibility Center’s own example transcript and from the named college pages in section 12.5, is a translation document, not a scrapbook. The fields a stranger can use without a phone call are few and specific. Student name. Address. Ninth-grade start date, which NCAA requires as month/day/year. Course titles that a registrar can map onto Algebra I, Geometry, Algebra II, Precalculus, Calculus, Statistics — not “Math 10,” not “Saxon 8/7,” not “Epsilon,” not “fun with numbers,” unless a parenthetical does the mapping on the same line. Grades. Credit units. A grading scale. Academic year. Graduation date. Administrator signature.¹²⁷¹ Dual-enrollment courses listed on the homeschool transcript and backed by the official college transcript. The NCAA credit rule, because it is the strictest increment rule this research opened, is a useful default even for a family that will never file an Eligibility Center packet: increments of .25, .50, .75, or 1.0, or of .34 and .67; **no course more than 1.0 credit** on a single line.¹²⁷² A year of Algebra I is 1.0. A semester is 0.50. A course that tries to claim 1.5 on one line has left the example.

are not a named mathematics pedagogy in the NCES curriculum-source table. Chapter 1 §1.6.

¹²⁷⁰Ellena Sempeles and Jiashan Cui, *Parent and Family Involvement in Education: 2023*, NCES 2024-113 (Washington, DC: National Center for Education Statistics, September 2024), <https://nces.ed.gov/pubs2024/2024113.pdf>; Pew Research Center, “A look at homeschooling in the U.S.,” 20 February 2025, <https://www.pewresearch.org/short-reads/2025/02/20/a-look-at-homeschooling-in-the-us/>. 1.8 percent full-time virtual as public or private school students whose parents did *not* consider them homeschooled; 0.7 percent full-time virtual as homeschoolers. Do not collapse full-time public virtual into parent-directed homeschool.

¹²⁷¹NCAA Eligibility Center, homeschool transcript example, http://fs.ncaa.org/Docs/eligibility_center/Student_Resources/Home_Scho Fields illustrated: student name, address, ninth-grade start date, course titles (Algebra I, Geometry, Algebra II, Precalculus sequence on the example), grades, credits, grading scale, academic year, graduation date, administrator signature. Named college pages (nn. 1312–1315) ask for the same grain: titles, grades, credits.

¹²⁷²NCAA Eligibility Center, *Home School Toolkit 2025–26*. Credit increments of .25 / .50 / .75 / 1.0 or .34 / .67; **no course more than 1.0 credit**. Ninth-grade start date required as month/day/year. Primary association rule, useful as a default increment even for families that will never file.

Publisher level names are not titles. Saxon Math 8/7 is a skill-level book in a Houghton Mifflin Harcourt homeschool sequence. It is not “eighth grade” and it is not Algebra I. Math-U-See Epsilon is a fractions *level*, not “fifth grade” and not a high-school course. Singapore Dimensions Math 6, on the vendor FAQ Chapter 1 already quoted, “includes some Pre-algebra topics, but is not a Pre-algebra text in the traditional sense; it is a Grade 6 text.” Beast Academy levels are puzzle-hard elementary and middle-grades books; they are not Geometry.¹²⁷³ A transcript that says “Saxon 8/7, A” has told a registrar the publisher and the book and has not told the registrar the mathematics. A transcript that says “Pre-algebra (Saxon 8/7), A” has done the translation. A transcript that says “Algebra I, A” and keeps “Saxon Algebra 1, 4th edition” for the course-description page has done it twice, which is what SMU’s page will ask for when it wants textbook titles in the descriptions.

The title rule is not pedantry. It is how a kitchen course becomes a core course, a Carnegie unit, or a line a Common App counselor form can carry. NCAA’s mathematics core is Algebra I or higher. A course titled “Consumer Math,” “Fun with Numbers,” or “Saxon 8/7” without a parenthetical does not meet that definition on its face. Named university pages in section 12.5 ask for Algebra I, Geometry, and Algebra II by those names, or for three years of mathematics that a reader will try to map onto those names. NMAP’s Algebra II is not “mostly quadratics plus a little log.” Chapter 9 already said so. This chapter will not re-teach Algebra II. It will say: if the year’s work omitted trigonometry and logarithms, do not title it Algebra II on a document a stranger will treat as Algebra II. Honesty on the worksheet is an NCAA rule. Honesty on the title is the same rule applied to every reader.

Placement and acceleration are record-keeping choices, not only teaching choices. Chapter 1 already killed the calendar-grade as a legal object. Publishers themselves place by skill, not by birthday. Singapore Math Inc. ships untimed PDFs, about an hour, and treats starting at an earlier level as common and acceptable. Houghton Mifflin Harcourt’s Saxon middle-grades test uses score bands to place into Math 5/4, 6/5, 7/6, 8/7, Algebra 1/2, or Algebra 1, and the publisher says Saxon is skill-level rather than grade-level. Math-U-See’s placement tool is a yes/no mastery questionnaire; the scoring guide in the placement materials uses a roughly 80–89 percent mastery threshold to advance. Beast Academy uses unit pretests after an age-based first guess.¹²⁷⁴ A child of ten may place into Singapore 3A, Saxon 6/5, and Math-U-See Gamma on the same afternoon. That is not a contradiction. It is three grain-sizes of “grade.” The National Mathematics Advisory Panel’s Finding 15 is the research sentence under that practice: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; claims that children of a given age “cannot” learn con-

¹²⁷³Singapore Math Inc., Dimensions Math FAQ (**vendor**): Dimensions Math 6 “includes some Pre-algebra topics, but is not a Pre-algebra text in the traditional sense; it is a Grade 6 text.” Houghton Mifflin Harcourt / Saxon Homeschool, *Middle Grades Math Placement Test* and *Saxon Homeschool Math Scope and Sequence* (**vendor**): Math 5/4, 6/5, 7/6, 8/7 are skill-level names, not calendar grades. Math-U-See / Demme Learning (**vendor**): Epsilon is a fractions *level*. Art of Problem Solving, Beast Academy (**vendor**): levels 1–5, puzzle-hard elementary and middle grades; “generally recommend[s] that students start about a level ‘behind’” because of problem difficulty. Chapter 1 §1.5. Publisher names are not transcript titles.

¹²⁷⁴Singapore Math Inc., placement tests, <https://www.singaporemath.com/pages/placement-tests> (**vendor**): untimed PDFs, about an hour; starting at an earlier level common and acceptable. HMH / Saxon, *Middle Grades Math Placement Test* (**vendor**): score bands into 5/4 through Algebra 1; skill-level rather than grade-level. Math-U-See placement tool, <https://curriculumexpress.mathusee.com/> (**vendor**): yes/no mastery; scoring guide uses an ~80–89 percent mastery threshold to advance. Beast Academy unit pretests A–D per level (**vendor**). A child of ten may place into three different “grades” on one afternoon; that is three grain-sizes, not a contradiction.

tent because they are “too young,” “not in the appropriate stage,” or “not ready” have “consistently been shown to be wrong.” Brain-immaturity claims are not justified even when students possess the prerequisite knowledge. Finding 30: gifted students with motivation should be allowed to move faster.¹²⁷⁵ Accelerate a child who has the prerequisites. Do not skip fractions because a child is “good at math.” Delay is equally normal, and for the student with unfinished arithmetic it is often the correct professional judgment.

NCAA and colleges care about the high-school transcript window — ninth-grade start date; Algebra I or higher as core — not about whether Algebra I happened at twelve. Taking Algebra I in eighth grade *and* listing it as a high-school credit is a record-keeping choice with NCAA implications. The current Toolkit is the document to read for pre-ninth coursework rules. This chapter will not invent those rules from a blog.¹²⁷⁶ A family that finishes Algebra I at thirteen and Geometry at fourteen still needs three years of Algebra I or higher in the high-school window if Division I is the aim, or two if Division II is the aim. Acceleration does not erase the year-count. It changes which titles fill the years. A family that delays Algebra I until fifteen still needs a title a stranger can map, and still meets the same university floor in section 12.5. Delay is not a compliance failure in Texas. It is a college problem if the transcript then stops at Algebra I.

Unschooling is a named minority practice and a poor match for this document unless the parent translates. NCES 2016: 12 percent of homeschooled students were taught with “mostly or strictly informal learning.” The unschooling tradition associated with John Holt is real. This book’s thesis is closer to structured or mastery work than to pure unschooling; Chapter 1 said so without scorning the twelve percent.¹²⁷⁷ Lived mathematics — cooking, building, games, later self-directed study — does not arrive as “Algebra II, 1.0, B+.” New York’s IHIP, Pennsylvania’s portfolio, and an NCAA Core-Course Worksheet all require a title, a content description, and a grading basis after the fact. The translation is possible. It is work. A parent who refuses the translation has chosen a practice that will not export cleanly. That is a fact about documents, not a verdict on the practice.

Common App is the other translation surface most families will actually meet. Practice, from practitioner guides this research opened: the student marks homeschooled; invites the parent, or an umbrella administrator, as counselor; the counselor account uploads the School Report, the tran-

¹²⁷⁵National Mathematics Advisory Panel, *Foundations for Success: The Final Report of the National Mathematics Advisory Panel* (Washington, DC: U.S. Department of Education, 2008), Findings 15 and 30, ERIC ED500486, <https://files.eric.ed.gov/fulltext/ED500486.pdf>. Finding 15: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; claims that children of a given age “cannot” learn content because they are “too young,” “not in the appropriate stage,” or “not ready” have “consistently been shown to be wrong.” Brain-immaturity claims are not justified even when students possess the prerequisite knowledge. Finding 30: gifted students with motivation should be allowed to move faster. Accelerate a child who has the prerequisites; do not skip fractions because a child is “good at math.” Nation Empowered / Iowa Acceleration Scale literature: not opened this pass; effect sizes **UNVERIFIED**.

¹²⁷⁶Pre-ninth coursework rules: read the current NCAA *Home School Toolkit*. This chapter does not invent whether eighth-grade Algebra I may be listed as a high-school credit. Acceleration does not, by itself, create three years of high-school mathematics in the window the association counts.

¹²⁷⁷Cui and Hanson, NCES 2020-001, Table 6: 77 percent “mostly or strictly formal curriculum”; 12 percent “mostly or strictly informal learning.” Informal share higher among students whose parent reported a disability (20 percent informal / 62 percent formal). Unschooling associated with John Holt is a named minority practice. The 2023 First Look did not republish the teaching-style table. **UNVERIFIED** whether 2023 microdata still show ~12 percent informal. Chapter 1 §1.1.

script, a school profile, and course descriptions. The former student-facing “homeschool supplement” has been folded into counselor homeschool questions. Parent-as-counselor is standard, not a workaround.¹²⁷⁸ Confirm against Common App help text in the year of publication. The help article this stream opened, dated 7 May 2024, describes a recommender account that is invitation-only after the student assigns a counselor — the mechanism a homeschool parent uses.¹²⁷⁹ It does not, in the text opened, reprint the current counselor-form homeschool questions. That confirmation stays open. Do not treat a 2024 practitioner summary as 2026 help text.

A school profile, in this room, is a one-page description of the homeschool: who taught, what the grading scale is, whether the school is the kitchen or an umbrella, how many students, whether dual-enrollment and AP are used. It is not a marketing brochure. It is not a diploma. SMU’s page will want course descriptions that include textbook titles. Pittsburgh will want a transcript by year with grades and a Home School Supplemental Form. Those are named later. The operational sentence here is: the parent is the counselor unless an umbrella administrator is. Plan for that invitation. Do not wait for a district counselor who does not exist.

Artificial intelligence belongs on this page as a clerk, not as a signer. Chapter 3 already locked the kitchen condition: the agent does not take the test; the child struggles first; the parent hosts and constrains the tool; none of Hermes Agent, Grok Bot, consumer ChatGPT, or Khanmigo is a school-safe product by default; Grok Bot requires cloud storage, has no education SKU, and is not FERPA-ready.¹²⁸⁰ This chapter inherits that lock and adds a records rule. Allow: on the parent’s account, generate a transcript *template* with the NCAA example fields, or a course-description paragraph that lists the textbook, the major topics, the grading basis, and the credit. The parent checks every line against the actual work. The parent signs. Ban: asking a model to invent hours, to invent a diploma, to invent a grading scale the family never used, to back-fill a Core-Course Worksheet with content the course did not cover, or to put a student’s name, an IEP, or worksheet photographs on a shared cloud computer the parent does not treat as a records closet. Fabricating a record is not a documentation problem. It is a false statement on a document a college or an athletic association will treat as signed. An agent that drafts a template has not signed. An agent that fills hours the family did not keep has been asked to lie. Point back to Chapter 3 for the product distinctions. Do not sequel them here.

A *hypothetical* illustration, labelled as such, can put the title rule in a room without inventing a reported family. A parent who used Math-U-See Algebra 1, a co-op geometry class, and Teaching Textbooks Algebra 2 writes three lines: “Algebra I (Math-U-See Algebra 1), A, 1.0, 2023–24”;

¹²⁷⁸Practitioner guides (not Common App help text) describe the former student-facing “homeschool supplement” as folded into counselor homeschool questions, and parent-as-counselor as standard. **Confirm against Common App help text in the year of publication.** Do not treat a practitioner summary as 2026 help text.

¹²⁷⁹Common App Member Support, “How can I create a recommender account?,” 7 May 2024, <https://membersupport.commonapp.org/s/article/I-am-a-counselor-or-recommender-How-do-I-create-my-account>. Counselor account is invitation-only after the student assigns a recommender — the mechanism a homeschool parent uses as counselor. Dated 2024. Current counselor-form homeschool questions were not reprinted in the text opened. Confirmation stays open.

¹²⁸⁰Chapter 3, especially §§3.2 and 3.6. Grok Bot: persistent cloud computer; Legacy Privacy Mode unsupported; no education SKU and no FERPA claim on pages opened 30 August 2026. Hermes Agent: self-hosted open-web agent, not a child-safety school product. FRAME.md: no documented K–12 student-facing autonomous open-web agent as a school product; Grok Bot is not FERPA-ready. This chapter inherits that lock and adds a records rule (template and course description allowed; fabricating hours or a diploma banned).

“Geometry (co-op; Prentice Hall), A–, 1.0, 2024–25”; “Algebra II (Teaching Textbooks Algebra 2), B+, 1.0, 2025–26.” The course-description page names the texts and the teacher of record — parent, co-op instructor, parent again. The dual-enrollment College Algebra the student sat at the community college that spring is listed on the homeschool transcript *and* sent as an official college transcript.¹²⁸¹ The empirical pieces that composite combines are in the tables and the named pages: 77 percent formal curriculum; 31 percent any co-op; dual enrollment visible and small in NCES 2016; NCAA and SMU wanting titles and texts. The three titles, the three years, and the sitting at the community college are hypothetical. The rule is not.

NCAA is a private association’s rules that bind athletic scholarships. That is the next section.

12.3 NCAA math core: Algebra I or higher; CLEP is not a core course

The National Collegiate Athletic Association is a private association. Its initial-eligibility rules are not homeschool law and they are not a federal statute. They bind athletic scholarships at member schools. A family that will never file an Eligibility Center packet can still use the mathematics definition, because it is the cleanest operational sentence this research opened for what a high-school mathematics course is when a stranger has to read it: college-preparatory high-school work in mathematics at Algebra I or higher.¹²⁸² Everything below that line — arithmetic, consumer mathematics, a publisher’s middle-grades book without a parenthetical — is not a mathematics core course on NCAA’s page. Everything at or above that line can be, if the worksheet is honest.

This section is written from NCAA pages, not from blogs. The sources are the Division I initial-eligibility page, the Home School Toolkit PDF for 2025–26, the Core-Course Worksheet dated July 2025, the Home School Email Cover Sheet, and the Eligibility Center’s example transcript.¹²⁸³ Practitioner listicles that restate those rules without linking them are not used as primaries. If a later printing needs a rule this stream did not open — in particular, the current treatment of coursework completed before ninth grade — the Toolkit is the document to re-open. Do not invent that rule here.

A core course, in the Toolkit’s definition, is college-preparatory high-school work in English, **mathematics (Algebra I or higher)**, science, social science, world language, comparative religion, or philosophy. That list is the association’s list. It is not a suggestion that comparative religion is mathematics. It is a reminder that “core” is larger than mathematics and that mathematics, inside that larger object, has a floor. Algebra I or higher. Geometry counts. Algebra II counts. Precalculus counts. Calculus counts. Statistics at a college-preparatory high-school level can count if the

¹²⁸¹Labelled *hypothetical*. Empirical pieces: Cui and Hanson, NCES 2020-001 (77 percent formal curriculum; 31 percent any co-op; dual enrollment visible and small in Table 4); NCAA example transcript and SMU course-description ask (nn. 1271, 1313). The three titles, years, and community-college sitting are hypothetical.

¹²⁸²NCAA, Division I initial-eligibility requirements, <https://www.ncaa.org/eligibility-center/initial-eligibility-requirements/division-i/>; *Home School Toolkit 2025–26*. A core course is college-preparatory high-school work in English, mathematics (Algebra I or higher), science, social science, world language, comparative religion, or philosophy. Private association rules, not a statute.

¹²⁸³NCAA Eligibility Center: Division I initial-eligibility page (n. 1282); *Home School Toolkit 2025–26*; *Homeschool Core-Course Worksheet* (July 2025); Home School Email Cover Sheet, http://fs.ncaa.org/Docs/eligibility_center/Student_Resources/Cover_Sheet.pdf (updated January 2023; still linked from the 2025–26 Toolkit); homeschool transcript example (n. 1271). These five are the primaries. Blogs are not.

worksheet supports it. “Saxon 8/7” without a parenthetical does not. “Fun with numbers” does not.

Division I: sixteen core courses, including **three years of mathematics at Algebra I or higher**. Ten of those sixteen, including seven in English, mathematics, or science, must be completed before the seventh semester. Minimum **2.3** core GPA. Final transcript and proof of graduation.¹²⁸⁴

Division II: sixteen core courses, including **two years of mathematics at Algebra I or higher**. Minimum **2.2** core GPA.¹²⁸⁵ Those year-counts are among sixteen, not instead of sixteen. A student with three years of mathematics and a thin English record has not finished the Division I object. A student with two years of mathematics and a beautiful calculus course has not finished Division I’s mathematics count, and has finished Division II’s only if the two years are Algebra I or higher. Three and two are the mathematics floors. They are not a diploma.

The homeschool packet is specific. Transcript. Administrator and Accordance Statement — who taught, who graded, who awarded credit, and a statement that the program was conducted in accordance with state law. **One Core-Course Worksheet per core course**. Teacher of record. Texts. Prerequisites. Grading basis. Content. Parent or administrator signature. Grade and credit on the worksheet must match the transcript.¹²⁸⁶ The Cover Sheet, still linked from the 2025–26 Toolkit and updated January 2023 on the file this stream opened, is the submission checklist: transcript, administrator statement, worksheets, proof of graduation.¹²⁸⁷ Evaluation occurs only after an NCAA school requests it. A family that builds the packet in eleventh grade and waits for a coach’s request is inside the process. A family that treats a finished packet as an eligibility letter is not. The association does not pre-approve a kitchen.

Audited, CLEP, and credit-by-exam courses are **not** NCAA-approved core courses.¹²⁸⁸ That sentence is the one blogs get wrong, and it is the one this chapter will not get wrong. CLEP College Algebra, CLEP Precalculus, and CLEP Calculus are real College Board exams. Section 12.4 will say how a parent registers for them without a school. They can produce college credit at institutions that grant it. They cannot substitute for an NCAA mathematics core course. A student who sits CLEP Calculus instead of taking a year of high-school calculus, and who needs that year as one of three Division I mathematics cores, has bought a college-credit object and has not bought an NCAA core course. Do not promise NCAA eligibility from CLEP. Do not promise NCAA eligibility from a course titled “fun with numbers.”

¹²⁸⁴NCAA Division I initial-eligibility page and *Home School Toolkit 2025–26*. Sixteen core courses including **three years of mathematics (Algebra I or higher)**; ten core courses (seven in English, mathematics, or science) before the seventh semester; minimum **2.3** core GPA; final transcript and proof of graduation.

¹²⁸⁵NCAA Division II initial-eligibility requirements, as restated in the *Home School Toolkit 2025–26* and the Division I/II pages this stream opened. Sixteen core courses including **two years of mathematics (Algebra I or higher)**; minimum **2.2** core GPA.

¹²⁸⁶NCAA *Homeschool Core-Course Worksheet* (July 2025). One worksheet per core course. Teacher of record, texts, prerequisites, grading basis, content, parent or administrator signature. Grade and credit must match the transcript. Administrator and Accordance Statement (who taught, graded, awarded credit; conducted in accordance with state law) is a separate packet piece in the Toolkit.

¹²⁸⁷NCAA Home School Email Cover Sheet (January 2023 file, still linked from the 2025–26 Toolkit). Submission checklist: transcript, administrator statement, worksheets, proof of graduation. Evaluation occurs only after an NCAA school requests it (*Toolkit*).

¹²⁸⁸NCAA *Home School Toolkit 2025–26*: audited, CLEP, or credit-by-exam courses are **not** NCAA-approved core courses. A CLEP Calculus pass cannot substitute for a mathematics core course. Do not promise NCAA eligibility from CLEP.

A kitchen calculus course taught from a textbook can count if the worksheet is honest and the course is college-preparatory. That is the other direction of the same rule. The teacher of record can be the parent. The text can be a homeschool edition. The content description has to be calculus, not a wish. Dual-enrollment Calculus I from the community college is a college transcript, which NCAA treats differently — and, on the Toolkit’s own logic, often more cleanly — than a parent-graded course.¹²⁸⁹ Keep the college transcript. List the course on the homeschool record. Do not invent a homeschool dual-enrollment participation rate to make that path sound typical. NCES 2016 saw a small share of 9–12 also in college classes. Section 12.4 will print the public-school supply figure and refuse to launder it as a homeschool count.

Pre-ninth Algebra I is the record-keeping choice section 12.2 already named. NCAA cares about the high-school window and about Algebra I or higher as core. Whether an eighth-grade Algebra I may be listed as a high-school credit, and under what documentation, is a Toolkit question. This chapter will not invent the answer. The operational warning is enough: acceleration does not, by itself, create three years of high-school mathematics. A student who finishes Algebra I at thirteen still needs the Division I count to be filled with Algebra I or higher *in the window the association counts*. Read the current Toolkit. Do not trust a forum restatement.

Co-op mathematics classes create a teacher of record useful for the worksheet. That is a fact about a form, not a pitch for a co-op. NCES 2016: 31 percent of homeschoolers had any instruction from a local homeschooling group or co-op.¹²⁹⁰ A Tuesday geometry class with a named instructor, a named text, and a grade the instructor assigned is easier to describe on a worksheet than a year the parent is not sure how to narrate. The parent remains the homeschool administrator of record unless an umbrella school is. The worksheet still needs the parent or administrator signature. The co-op teacher does not replace the packet.

The mathematics implication for Chapters 7 through 11 is now an export implication. Transcript Algebra I as Algebra I. Transcript Geometry as Geometry. Transcript Algebra II as Algebra II only if the year’s work earned that title in NMAP’s sense. Transcript Precalculus, Calculus, Statistics by those names. Keep the publisher in a parenthetical or on the description page. Do not ask CLEP to do NCAA’s job. Do not ask a state test to do it either. New York already said SAT and ACT do not close the annual assessment. NCAA now says CLEP does not close a core course. Different associations. Same lesson: objects do not substitute because they are all “math.”

External signals — SAT, PSAT, ACT, AP, CLEP, dual enrollment — are different objects again. That is the next section.

12.4 SAT, AP, CLEP, dual enrollment — different objects

The popular story collapses four or five college-facing objects into one sentence: “get some outside validation.” The documents do not. The SAT is an admissions test a homeschooler registers for

¹²⁸⁹NCAA *Home School Toolkit 2025–26*: dual-enrollment college courses can count if documented from the college. A community-college transcript is a different object from a parent-graded kitchen course and is, on the Toolkit’s logic, often cleaner. Keep the official college transcript.

¹²⁹⁰Cui and Hanson, NCES 2020-001, Table 3: 31 percent any instruction from a local homeschooling group or co-op. A named co-op instructor is useful as teacher of record on a worksheet. Not a pitch for a co-op. The parent remains administrator of record unless an umbrella school is.

online. The PSAT is an “away student” sitting arranged at a school. An AP exam is a school-controlled seat in May, even if the student self-studied. CLEP is a credit-by-exam the student registers for without a school. Dual enrollment is a college course that produces a college transcript. ACT is a parallel admissions test. The Classic Learning Test is accepted at some, often Christian, colleges. Each receiving college sets policy. None of those objects is produced by a satisfied state statute. None of them is an NCAA core course except insofar as a dual-enrollment course is documented from the college. Secondary journalism that ranks AP above CLEP for selective colleges is labelled journalism. A single national experiment comparing AP, dual enrollment, and CLEP mathematics outcomes at scale was not found. An IES dual-enrollment outcomes study’s first report is expected in 2027.¹²⁹¹ This section is logistics without invented registration recipes.

College Board’s “Homeschoolers’ Guide to College Board Assessments,” by Jacob Mishook, dated 29 June 2026, is the primary this stream opened for SAT, PSAT, AP, and CLEP as they apply to homeschoolers.¹²⁹² Use it. Do not replace it with a blog.

SAT. Register online. The homeschool high-school code is **970000**. Fee waivers are available directly from College Board for eligible low-income homeschoolers. Free practice: Bluebook full-length tests; Official SAT Prep on Khan Academy.¹²⁹³ That code is a load-bearing digit string. A parent who invents a school code, or who leaves the field blank, has left the guide. The SAT is an admissions test. It is not New York’s annual assessment. It is not an NCAA core course. It is not a diploma. It is one external signal a named university page may read when the homeschool transcript is otherwise opaque.

PSAT 8/9, PSAT 10, PSAT/NMSQT. Homeschoolers sit as “away students.” Contact a participating school at least four weeks ahead. National Merit eligibility runs only through PSAT/NMSQT. Co-ops may become testing centers.¹²⁹⁴ The four-week lead is College Board’s, not this book’s invention. A family that decides in October that a junior should sit the National Merit paper has missed the arrangement window if the school’s deadline has passed. This chapter will not invent a list of host schools. The guide’s instruction is to contact a participating school. Homeschool-specific access mechanics beyond that sentence remain **UNVERIFIED** as a registration recipe.¹²⁹⁵

¹²⁹¹College-credit comparison study of AP versus dual enrollment versus CLEP mathematics outcomes at scale: not found as a single national experiment (brief open question 23). IES/NCEE, “Outcomes Associated with Dual Enrollment Programs,” project page, first report expected 2027, <https://ies.ed.gov/use-work/evaluations/outcomes-associated-dual-enrollment-programs>. Secondary journalism that ranks AP above CLEP for selective colleges is labelled journalism. Each receiving college sets policy.

¹²⁹²Jacob Mishook, “The Homeschoolers’ Guide to College Board Assessments,” College Board All Access, 29 June 2026, <https://allaccess.collegeboard.org/homeschoolers-guide-college-board-assessments>. Primary College Board policy communication for SAT code 970000, PSAT away-student rules, AP exam-only via school coordinator, CLEP self-registration, and fee waivers. Association document, not a college.

¹²⁹³Mishook, “Homeschoolers’ Guide,” 29 June 2026. SAT: register online; homeschool high-school code **970000**; fee waivers available directly from College Board for eligible low-income homeschoolers; free practice via Bluebook full-length tests and Official SAT Prep on Khan Academy.

¹²⁹⁴Mishook, “Homeschoolers’ Guide,” 29 June 2026. PSAT 8/9, PSAT 10, PSAT/NMSQT: “away students”; contact a participating school at least four weeks ahead; National Merit only via PSAT/NMSQT; co-ops may become testing centers.

¹²⁹⁵Homeschool-specific PSAT and AP exam-access mechanics beyond the College Board 29 June 2026 guide and the AP Students help article (n. 1298): **UNVERIFIED** as a registration recipe. Outline open question 16. Do not invent a host-school list or an ordering path.

ACT remains a parallel admissions test. Many homeschoolers sit it. This research did not open a homeschool-specific ACT participation rate and will not invent one. **CLT**, the Classic Learning Test, is accepted at some, often Christian, colleges. Southwest Baptist University names CLT alongside ACT and SAT on the homeschool page this stream opened.¹²⁹⁶ Treleven 2022, in *Home School Researcher*, analyzed CLT homeschool scores. That journal is an advocacy journal. The paper is named as the existence of a literature, not as a finding this chapter will crown.

AP. Forty-plus exams. A score of 3 or higher may yield credit; each college sets policy. Homeschoolers **may self-study**. A formal AP course is recommended, not required. Parents **cannot order exams**. Find a school via the AP Course Ledger. The AP coordinator enrolls the student in an “exam only” My AP section. Mid-November order deadline. Homeschool educators may submit courses to the AP Course Audit to become authorized providers.¹²⁹⁷ Those sentences are from the 29 June 2026 guide and from College Board’s own AP Students help article, “Do I need to enroll in a class section in My AP if I’m homeschooled?”¹²⁹⁸ They are the entire registration recipe this book is allowed to print. Do not invent a homeschool AP exam-ordering path beyond them. Do not promise that self-study yields a seat in May. Seats are school-controlled. A family that finishes a beautiful home AP Calculus AB course in April and has not found a coordinator has a course and does not have an exam. Chapter 10 already planted that fact. This chapter installs it as logistics.

AP Calculus AB is eight units; BC is those eight plus parametric, polar, and vector-valued functions and infinite sequences and series. AP Statistics, effective Fall 2026, is a one-semester, introductory, non-calculus-based college course; the published prerequisite is first-year algebra. College Board’s revised framework encourages students who have the background to take **both** AP Statistics and AP Calculus. It is not published as a substitute for calculus in a STEM calculus sequence.¹²⁹⁹ Chapter 11 already said so. The export rule is the title: Statistics or AP Statistics; Calculus or AP Calculus AB or AP Calculus BC. Whether homeschoolers actually sit AP Statistics at any measurable rate is **UNVERIFIED**.¹³⁰⁰ Do not invent a participation rate. Do not treat an AP score as automatic

¹²⁹⁶Southwest Baptist University, “Homeschool College Admission,” <https://sbuniv.edu/admissions/homeschool-students.php>. Names ACT, SAT, and CLT. Test-optional for admission; scores may affect scholarships. Homeschool ACT participation rate: not opened; not invented.

¹²⁹⁷Mishook, “Homeschoolers’ Guide,” 29 June 2026. AP: 40+ exams; score 3+ may yield credit (each college sets policy); homeschoolers may self-study; formal AP course recommended not required; parents **cannot order exams**; AP Course Ledger; coordinator enrolls “exam only”; mid-November order deadline; homeschool educators may submit courses to the AP Course Audit. Do not promise that self-study yields a May seat. Seats are school-controlled.

¹²⁹⁸College Board AP Students Help Center, “Do I need to enroll in a class section in My AP if I’m homeschooled?,” <https://apstudents.collegeboard.org/help-center/do-i-need-enroll-class-section-my-ap-if-im-homeschooled>. Homeschoolers cannot order AP Exams; find a school via the AP Course Ledger; exam-only My AP section; mid-November order deadline. Entire registration recipe this book is allowed to print.

¹²⁹⁹College Board, AP Calculus AB/BC course pages and AP Statistics Course and Exam Description, Effective Fall 2026 (partially opened 30 August 2026). AB: eight units. BC: those eight plus Units 9–10 (parametric/polar/vector; infinite sequences and series). AP Statistics: one-semester introductory **non-calculus-based** college course; published prerequisite a first-year algebra course; five units. Revised framework encourages both AP Statistics and AP Calculus when background allows; not published as a calculus substitute. Chapters 10 and 11. AP Calculus CED prerequisite paragraph now extracted (Chapter 10 n. 1084). Full CED topic-level tables below the eight and ten unit titles remain **UNVERIFIED**.

¹³⁰⁰Whether homeschoolers sit AP Statistics at any measurable rate: **UNVERIFIED**. Do not invent a participation rate. NCES 2016 probability ever-taught (24 percent of high-school-level homeschoolers) is not an AP Statistics completion

credit. Do not treat an AP score as an NCAA core course.

CLEP. Thirty-four subjects, including College Algebra, Precalculus, and Calculus. Register via My CLEP. Test centers or remote proctoring; remote, in the United States, for ages 13 and up on the pages this stream used. About 2,900 colleges grant credit — College Board’s figure, not an independent census. No school enrollment required. Modern States partnership: free prep and an exam voucher. Students eligible for an SAT fee waiver may be CLEP-fee-waiver-eligible.¹³⁰¹ CLEP Calculus, on College Board pages opened for Chapter 10, is a one-semester college-calculus exam; ACE’s recommended credit-granting score is 50 for four semester hours; each institution sets its own policy. A list price of \$97 appeared on the Calculus page as of the 30 August 2026 fetch; confirm the current fee at publication.¹³⁰² CLEP is the object a parent can actually order without a school. That is its operational virtue. It is also why it cannot do NCAA’s job and why it cannot be promised as automatic credit. Different object from AP. Different object from dual enrollment. Different object from SAT.

Dual enrollment is the cleanest external transcript the kitchen can buy when the parent is the ceiling. The student takes College Algebra, Precalculus, or Calculus I as a college student. The college sends an official transcript. NCAA can read that transcript. Admissions offices already know how to file it. Pittsburgh’s page says college courses should be designated and official college transcripts sent. SMU’s page names dual-credit, AP, IB, and historical SAT Subject scores as additional measures in core academic subjects including mathematics.¹³⁰³ Cost ranges from free, in some state programs, to ordinary community-college tuition. State dual-credit laws vary. Texas litigation over dual-credit access appeared on HSLDA’s Texas page in 2026; that is advocacy news, not a holding this chapter will cite as doctrine.¹³⁰⁴ Keep the syllabus. List the course on the homeschool transcript. Send the college transcript. Do not invent a homeschool dual-enrollment rate.

The supply figures this research opened are not that rate. NCES 2020-125, from the 2017–18 National Teacher and Principal Survey: 82 percent of public schools with any of grades 9–12

rate. Chapter 11.

¹³⁰¹Mishook, “Homeschoolers’ Guide,” 29 June 2026. CLEP: 34 subjects including College Algebra, Precalculus, and Calculus; register via My CLEP; test centers or remote proctoring (13+ in the United States on the pages used); ~2,900 colleges grant credit (College Board’s figure, not an independent census); no school enrollment required; Modern States partnership for free prep and an exam voucher; SAT-fee-waiver-eligible students may be CLEP-fee-waiver-eligible. Credit is not automatic.

¹³⁰²College Board, CLEP Calculus exam page, <https://clep.collegeboard.org/clep-exams/calculus>, accessed 30 August 2026. One-semester college calculus; about 60 percent limits and differential calculus and 40 percent integral calculus on one College Board statement (another statement on the same family of pages used ~10 percent / ~50 percent / ~40 percent — Chapter 10 already noted the wording split). ACE recommended credit-granting score 50 for 4 semester hours; each institution sets policy. List price \$97 on the Calculus page this access day; confirm current CLEP fee at publication.

¹³⁰³University of Pittsburgh, “Home Schooled Applicants,” <https://admissions.pitt.edu/home-school/>; Southern Methodist University, “Application Guide for Homeschool, Non-Traditional Students,” <https://www.smu.edu/admission/apply/undergraduate-admission/home-schooled-nontraditional-students>. Pitt: designate college courses and send official college transcripts. SMU: dual-credit, AP, IB, A-Level, historical SAT II as additional measures in core academic subjects including mathematics.

¹³⁰⁴Home School Legal Defense Association Texas page, 2026 news item on dual-credit access litigation. **Advocacy news, not a holding to cite as doctrine.** State dual-credit laws vary. Cost ranges from free (some state programs) to ordinary community-college tuition. Do not invent a typical homeschool dual-enrollment price or rate.

offered dual or concurrent enrollment.¹³⁰⁵ That is **public-school supply**, not a homeschool participation rate. IPEDS 2022–23: nearly 2.5 million dual-enrolled students, a figure this stream has as IES project-page and CCD journalism, not as a homeschool count.¹³⁰⁶ Homeschool-specific dual-enrollment counts are **not in these sources**. NCES 2016 Table 4: 18 percent of homeschoolers also attended a school — 15 percent public, 2 percent private (interpret with caution), 1 percent college (interpret with caution).¹³⁰⁷ Dual enrollment is visible and small in that table. High-school families using community-college algebra or calculus will not show up as a large national percentage and should not be written as the typical path. They are a release valve. They are a clean transcript. They are not “what homeschoolers do.”

When the parent is the ceiling, the release valves are already in the 2016 tables: any tutor, 23 percent; any co-op, 31 percent; a small share of 9–12 also in college classes.¹³⁰⁸ Those are facts, not a pitch. A co-op mathematics class creates a teacher of record useful for an NCAA worksheet. A weekly tutor does not, by itself, produce an external transcript; the parent remains the grader unless the tutor’s institution issues one. A community-college course produces an official transcript NCAA and admissions offices already know how to read. Public-school à la carte access is state-law dependent; HSLDA publishes “public school access” pages per state, not opened in full for this stream.¹³⁰⁹ Umbrella schools can issue transcripts; they do not magically teach calculus. A *hypothetical*, labelled as such: a parent who last passed Algebra I in 1998 hires a weekly tutor for Algebra II, or enrolls the student in College Algebra at the community college, while remaining the homeschool administrator of record.¹³¹⁰ That is a normal high-school plan, not a failure of the kitchen. Chapter 1 already used that composite. This chapter says how to put it on the record: tutor-supported Algebra II is still Algebra II, parent-graded, with the tutor named on a course description if the parent wants the help visible; dual-enrollment College Algebra is a college line plus a college transcript.

Live debate, printed as a debate and not as a ranking this book will settle: dual enrollment versus AP versus CLEP as the external signal. They are different objects. Each receiving college sets

¹³⁰⁵National Center for Education Statistics, *Dual or Concurrent Enrollment in Public Schools in the United States*, NCES 2020-125 (December 2020), <https://nces.ed.gov/pubs2020/2020125.pdf>. From the 2017–18 National Teacher and Principal Survey: **82 percent** of public schools with any of grades 9–12 offered dual or concurrent enrollment. **Public-school supply**, not a homeschool participation rate. NCES definition: enrollment in a class that offers both high-school and college credit.

¹³⁰⁶IES/NCEE project page and Community College Daily / AACC, “DataPoints: New data on dual enrollment,” August 2024, <https://www.ccdaily.com/2024/08/datapoints-new-data-on-dual-enrollment/>, reporting IPEDS 2022–23 counts: nearly 2.5 million dual-enrolled students. **Journalism** for the CCD article; IES page is a federal project description. Homeschool-specific counts **not in these sources**.

¹³⁰⁷Cui and Hanson, NCES 2020-001, Table 4: 18 percent of homeschoolers also attended a school (15 percent public; 2 percent private, interpret with caution; 1 percent college, interpret with caution). Dual enrollment visible and small. Not a typical path.

¹³⁰⁸Cui and Hanson, NCES 2020-001, Table 3: any tutor or private teacher 23 percent; any local homeschool group or co-op 31 percent; Table 4 college cell, caution. Release valves when the parent is the ceiling. Facts, not a pitch.

¹³⁰⁹HSLDA “public school access” pages exist per state and were not opened in full for this stream. Public-school à la carte access is **state-law dependent**. Do not write a 50-state access manual.

¹³¹⁰Labelled *hypothetical*. Same composite Chapter 1 §1.4 already used from NCES pieces (tutor 23 percent; college cell small and caution). A parent who last passed Algebra I in 1998 hiring a weekly tutor for Algebra II, or enrolling the student in College Algebra, while remaining administrator of record, is a normal high-school plan, not a failure of the kitchen. Not a reported family.

policy. AP is a high-school-course-plus-exam object with a school-controlled seat. CLEP is a credit-by-exam the student can order. Dual enrollment is a college course. Journalism that ranks AP above CLEP at selective colleges is journalism. College Board itself tells families to search each college's policy. This book will not invent a national preference order. It will say: if the destination is a named university page in section 12.5, read that page; if the destination is NCAA, do not ask CLEP to be a core course; if the parent is the ceiling, dual enrollment is the cleanest of the three as a transcript, because it arrives as a transcript.

Do not promise AP, CLEP, or dual-enrollment credit is automatic. Do not promise that AP self-study yields a May seat. Do not invent homeschool AP or dual-enrollment participation rates. Do not treat SAT code 970000 as optional. Do not treat PSAT as a walk-in. What named colleges actually print is the next section.

12.5 Named university pages: a pattern, not a rule

A comprehensive NACAC policy statement dedicated to homeschool mathematics was **not opened** as a 2026 PDF and remains **UNVERIFIED**. This stream used named university pages instead.¹³¹¹ That sentence stays in the body, not only in a note. Anyone who hands a parent a “NACAC says homeschoolers must...” paragraph that this research did not fetch is handing a document this book will not launder. The pattern below is four pages. It is not a census. It is not a rule.

University of Pittsburgh, on the Home Schooled Applicants page this stream opened: minimum **three years of mathematics**, plus four of English, three of science, three of social studies, two of arts or humanities. Home School Supplemental Form. Transcript by year with grades. Evaluator or supervisor progress evaluations if applicable. College courses designated and official college transcripts sent. Final transcript from an accredited homeschool organization **or** signed by the person responsible.¹³¹² Three years of mathematics is the floor on that page. The page does not, in the text opened, name Algebra I, Geometry, and Algebra II as the three. A parent who fills three years with “Math 9,” “Math 10,” and “Math 11” has met a year-count and has failed the title rule in section 12.2. A parent who fills three years with Algebra I, Geometry, and Algebra II has met both the year-count and the title a stranger can map.

Southern Methodist University, on the Application Guide for Homeschool and Non-Traditional Students: Home School Supplement; holistic review; **minimum suggested** four units of English, **three of mathematics (Algebra I, Geometry, Algebra II)**, three of science with two lab, three of social science, a two-year foreign-language sequence. Transcript with consistent grading and course descriptions including textbook titles. Dual-credit, AP, IB, A-Level, and historical SAT II scores as additional measures “in core academic subject areas such as English literature, math, and science.”¹³¹³ SMU is the page that names the three titles. It is also the page that wants the textbook

¹³¹¹See n. 1258. NACAC 2026 homeschool-mathematics statement **not opened**. Named university pages only. Pattern, not a rule.

¹³¹²University of Pittsburgh Office of Admissions, “Home Schooled Applicants,” <https://admissions.pitt.edu/homeschool/>. Minimum **3 years of mathematics** (plus 4 English, 3 science, 3 social studies, 2 arts/humanities); Home School Supplemental Form; transcript by year with grades; evaluator/supervisor progress evaluations if applicable; college courses designated and official college transcripts sent; final transcript from an accredited homeschool organization **or** signed by the person responsible. Named page, not NACAC policy.

¹³¹³Southern Methodist University Admission, “Application Guide for Homeschool, Non-Traditional Students,”

on the description. A kitchen that used Saxon Algebra 1, a co-op geometry text, and Teaching Textbooks Algebra 2 can satisfy that ask by writing the titles and the books. A kitchen that used “the app” and cannot name the book has a description problem SMU has already described.

North Central College, Homeschool Applicants: transcript with course names, grades, credit, and year; recommended **three years of mathematics including algebra and geometry, one course beyond geometry preferred**.¹³¹⁴ The “beyond geometry” clause is the first named preference in this set for a fourth course. It is a preference on a recommendation list, not a statute. Precalculus, Algebra II after a geometry year, or a dual-enrollment College Algebra can be that course. A family that stops at Algebra I and geometry has not met the preference.

Southwest Baptist University, Homeschool Students: three units of mathematics among recommended college-preparatory work; 2.50 GPA; parent or organization transcript; ACT, SAT, or CLT named; test-optional for admission, with scores that may affect scholarships.¹³¹⁵ SBU is the Christian-college specimen in the set. It names a year-count, not the three titles. It names CLT. It is test-optional for admission and not test-blind for money. A family that treats “test-optional” as “mathematics-optional” has misread the page.

Pattern, not a rule: three years of high-school mathematics through Algebra II is the floor most of these pages name. Pittsburgh says three years. SMU says three, and names Algebra I, Geometry, Algebra II. North Central says three including algebra and geometry, one beyond geometry preferred. Southwest Baptist says three units. Selective STEM programs will want precalculus or calculus **plus** an external signal — AP, dual enrollment, SAT mathematics. This research did not open a STEM-program census to prove that last clause as a rate. It is the operational reading of SMU’s “additional measures,” of Pittsburgh’s request for official college transcripts, and of every high-school chapter in this book that refused to treat Algebra I as a finished college-preparatory record.

The kitchen that stops at Algebra I because “the state doesn’t require it” has a college problem in every named example above. Texas does not require a test. Illinois does not. Idaho does not. North Carolina requires a mathematics-inclusive battery and does not require Algebra II by title. New York requires mathematics credits and a standardized composite, not Algebra II by name. Pennsylvania option 1 requires general mathematics, algebra, and geometry at the secondary level — closer to the university floor than Texas, and still not a Pittsburgh transcript. Compliance can be satisfied below the floor every named page prints. That is the split section 12.1 installed. This section names the cost of living in it.

<https://www.smu.edu/admission/apply/undergraduate-admission/home-schooled-nontraditional-students>. Holistic review; Home School Supplement; **minimum suggested** four units English, **three mathematics (Algebra I, Geometry, Algebra II)**, three science (two lab), three social science, two-year foreign-language sequence; transcript with consistent grading and course descriptions including textbook titles; dual-credit, AP, IB, A-Level, SAT II (historical) as additional measures “in core academic subject areas such as English literature, math, and science.”

¹³¹⁴North Central College Admissions, “Homeschool Applicants,” <https://northcentralcollege.edu/apply/homeschool-applicants>. Transcript with course names, grades, credit, year; recommended **3 years of mathematics including algebra and geometry, one course beyond geometry preferred**.

¹³¹⁵Southwest Baptist University Admissions, “Homeschool Students,” <https://sbuniv.edu/admissions/homeschool-students.php>. 3 units mathematics among recommended college-preparatory work; 2.50 GPA; parent or organization transcript; ACT/SAT/CLT named; test-optional for admission; scores may affect scholarships. Christian-college specimen in the set.

NCES 2016 already showed the coverage problem as a population fact, not as a scold. High-school-level homeschoolers, subjects ever taught during home instruction: Algebra I, 79 percent; geometry, 52 percent; Algebra II, 49 percent; calculus, 15 percent, interpret with caution. The 2012 report: 88 percent basic algebra, 54 percent geometry, about a third advanced algebra, calculus “not often.”¹³¹⁶ Algebra I is common. Algebra II and geometry are not universal. Calculus at home is a minority. A book that treats the SMU three as automatic has left those tables. A book that treats the SMU three as optional because “the state doesn’t require it” has left the named pages. Both errors are documentation errors as much as teaching errors. Chapters 7 through 9 exist because the titles have to be earned. This chapter exists because they also have to be written down.

What these pages do *not* say is as useful as what they say. None of them, in the text opened, offers a national homeschool diploma as a substitute for a transcript. None of them treats a state compliance letter as a transcript. None of them asks for Saxon, Singapore, or Math-U-See by franchise. SMU asks for textbook titles inside course descriptions — the opposite of a franchise demand. Pittsburgh will accept a final transcript signed by the person responsible. That person is, in most kitchens, the parent Chapter 1 already installed as the main provider for 78 percent of homeschooled students in 2016.¹³¹⁷ Parent-as-counselor on Common App and parent-as-signer on a Pittsburgh transcript are the same adult.

Do not upgrade four pages into a NACAC rule. Do not invent a 2026 NACAC PDF. Do not write “colleges require three years through Algebra II” as if a census had been opened. Write: the named pages this research opened want, as a pattern, about three years through Algebra II as a floor; selective STEM will want more plus an external signal; the kitchen that stops at Algebra I because the state is silent has a problem on every page in the set. Confirm each destination’s current page in the year of application. Pages move.

Parent expectations and adult outcomes are the next section. They are printed as facts already installed in Chapter 1, not as a second fight.

12.6 Expectations, Cardus, and what NCES does not show

This chapter is not the outcomes fight. Chapter 1 already printed it. Brian D. Ray and the National Home Education Research Institute publish convenience-sample composites that advocates quote — the 2009 specimen, about the 86th percentile overall, 84th in mathematics, 89th in reading, recruited through testing services and networks, disproportionately White, Christian, two-parent, higher-income, higher parental education. Christopher Lubienski, Tiffany Puckett, and T. Jameson Brewer, Chelsea McCracken at ICHER, Robert Kunzman and Milton Gaither’s 2020 survey of more than 2,000 texts, and the *Washington Post* of 11 December 2023 quoting Kunzman are the

¹³¹⁶Cui and Hanson, NCES 2020-001, Table 8: high-school-level homeschoolers, subjects ever taught during home instruction — Algebra I 79 percent; geometry 52; Algebra II 49; calculus 15 (interpret with caution). Jeremy Redford, Danielle Battle, and Stacey Bielick, *Homeschooling in the United States: 2012*, NCES 2016-096, <https://files.eric.ed.gov/fulltext/ED569947.pdf>: 88 percent basic algebra; 54 percent geometry; about a third advanced algebra; calculus “not often” (10 percent, interpret with caution). Do not narrate 88 → 79 as decline without checking standard errors and questionnaire differences. Both figures: Algebra I common; Algebra II and geometry not universal; calculus at home a minority. NHES:2023 did not republish these tables (open question 2).

¹³¹⁷Cui and Hanson, NCES 2020-001, Table 3: mother main provider for 78 percent of homeschooled students. Chapter 1 §1.1. Parent-as-counselor on Common App and parent-as-signer on a Pittsburgh transcript are the same adult.

critique. NCES has never published a nationally representative homeschool NAEP mathematics score. Do not crown a winner with a convenience sample. Do not reopen the 84th-percentile specimen here as “what homeschool mathematics does.”¹³¹⁸

What belongs in an export chapter is narrower: what parents said they expected, what one nationally representative adult survey found about bachelor’s completion, and what no opened source shows about college calculus or STEM majors. Those facts constrain the promise this chapter can make. They do not tell a parent whether to write Algebra II on a transcript. They tell a parent not to treat a finished transcript as a percentile.

NCES 2016 parent expectations, Table 10: 31 percent of homeschool parents expected a bachelor’s degree; 30 percent a graduate or professional degree; 13 percent high school only. Enrolled-school parents: 29 percent bachelor’s, 43 percent graduate.¹³¹⁹ Homeschool parents were *less* likely, not more, to expect graduate school. The table is not a finding that homeschool mathematics is less ambitious. It is a finding that the parent this book is written for is not, as a population, shopping for a gifted pipeline. Beast Academy and Art of Problem Solving exist for families who are. They are not the modal kitchen, and they do not appear on Pittsburgh’s page as a required franchise.

Cardus Education Survey 2023, reported by Albert Cheng and Angela Watson in 2024, is the adult-attainment document. Ipsos KnowledgePanel; 2,350 United States adults ages 24–39 who completed at least high school; about 8 percent, $n = 181$, ever homeschooled. Think tank, not a federal statistical agency. A reputable panel vendor. Short-term homeschoolers, one to two years: bachelor’s or higher at 45 percent versus 46 percent among the never-homeschooled. Medium-term, three to seven years: 29 percent. Long-term, eight or more: 27 percent. Demographic controls applied. Authors explicitly refuse a causal interpretation.¹³²⁰ Only 17 percent of ever-homeschooled adults in that sample were homeschooled all of K–12; 20 percent for only one year. Duration is itself a source of heterogeneity. Daniel Hamlin and Albert Cheng, in the *Journal of School Choice* in 2022, made that point as a design claim: treating “homeschoolers” as one treatment is a design error.¹³²¹

¹³¹⁸Chapter 1 §1.7 and notes 81–88. Brian D. Ray / NHERI *Homeschool Progress Report* (2009 specimen): about the 86th percentile overall, 84th in mathematics, 89th in reading; convenience sample; **advocacy**. Christopher Lubienski, Tiffany Puckett, and T. Jameson Brewer, “Does Homeschooling ‘Work’?,” *Peabody Journal of Education* 88, no. 3 (2013). Chelsea McCracken / ICHER, “How to Mislead with Data.” Robert Kunzman and Milton Gaither, “Homeschooling: An Updated Comprehensive Survey of the Research,” *Other Education* 9, no. 1 (2020): 253–336. Moriah Balingit, *Washington Post*, 11 December 2023. This chapter does not reopen the specimen as “what homeschool mathematics does.”

¹³¹⁹Cui and Hanson, NCES 2020-001, Table 10. Homeschool parents: 31 percent expected a bachelor’s; 30 percent a graduate or professional degree; 13 percent high school only. Enrolled-school parents: 29 percent bachelor’s; 43 percent graduate. Homeschool parents less likely, not more, to expect graduate school.

¹³²⁰Albert Cheng and Angela Watson, *Diverse Outcomes for a Diverse Population: Findings About Homeschooled Adults from the Cardus Education Survey* (Cardus, 2024), <https://www.cardus.ca/research/diverse-outcomes-for-a-diverse-population/>. CES fielded 10–22 October 2023; Ipsos KnowledgePanel; 2,350 U.S. adults ages 24–39 who completed at least high school; ~8 percent / $n = 181$ ever homeschooled. Bachelor’s or higher: short-term (1–2 years) 45 percent versus never-homeschooled 46 percent; medium (3–7) 29 percent; long-term (8+) 27 percent. Demographic controls; authors refuse a causal interpretation. Only 17 percent of ever-homeschooled adults were homeschooled all K–12; 20 percent for only one year. Think tank, not a federal statistical agency. Does not report SAT mathematics, calculus completion, or STEM-major rates. Do not cite the bachelor’s-rate gap as the effect of homeschool mathematics or as proof that homeschooling harms STEM.

¹³²¹Daniel Hamlin and Albert Cheng, “Homeschooling, Perceived Social Isolation, and Life Tra-

Cardus does not report SAT mathematics, calculus completion, or STEM-major rates. It does undermine any sentence of the form “homeschoolers go to college at higher rates.” Medium- and long-term homeschoolers in this representative adult sample completed bachelor’s degrees at **lower** rates. Whether that is a mathematics-preparation problem, a values problem — long-term homeschoolers were less likely to say money is “very important” — a selection problem, or all three is **open**. Do not cite the bachelor’s-rate gap as the effect of homeschool mathematics. Do not cite it as proof that homeschooling harms STEM. Chapter 1 already said both. This section says them once so a parent who skipped Chapter 1’s close cannot treat a finished Algebra II line as a Cardus rebuttal or as a Cardus confirmation.

What NCES does not show is the rest of the honest paragraph. No nationally representative home-school NAEP mathematics score. NHES asks subjects taught, not achievement. Any sentence that says homeschoolers outperform public-school students in mathematics on NAEP would be false.¹³²² No named nationally representative source opened for this project gives the share of homeschool graduates who complete a college calculus course or who major in STEM. NCES 2016’s 15 percent is high-school-level ever-taught calculus *during home instruction*, interpret with caution — not college calculus, not a completion rate, not a STEM-major rate.¹³²³ That open question was planted in Chapter 1 for this chapter. This chapter leaves it standing. AoPS and Beast Academy vendor materials serve a gifted pipeline. They are not a population rate.

Selection effects are the reason a percentile cannot be the promise. Families who homeschool, who persist, who buy Beast Academy, who sit the SAT, who volunteer scores to NHERI, and who answer Cardus are different filters. Two-parent one-earner households with a college-educated parent are overrepresented in the NHES homeschool population relative to all children; they are *more* overrepresented in convenience testing samples. Any “homeschool mathematics advantage” that survives a comparison with public-school children of the same parental education, income, and two-parent status has not been shown in a national probability sample this stream could open.¹³²⁴ The teaching chapters promised a path, not a percentile. The export chapter promises a document a stranger can read, not a rank.

The path, as a documentation sentence: write the titles; keep the descriptions; sit an external signal the destination actually reads; when the parent is the ceiling, buy a transcript a college already

jectories: An Analysis of Formerly Homeschooled Adults,” *Journal of School Choice* (2022), <https://doi.org/10.1080/15582159.2022.2028338>. Duration of homeschooling as a heterogeneity dimension; cited by Cardus 2024 for the short/medium/long split. Treating “homeschoolers” as one treatment is a design error. Paywalled DOI.

¹³²²NCES has never published a nationally representative homeschool NAEP mathematics score. See n. 1259. Any sentence that says homeschoolers outperform public-school students in mathematics on NAEP would be false.

¹³²³Cui and Hanson, NCES 2020-001, Table 8: 15 percent of high-school-level homeschoolers had ever been taught calculus during home instruction (interpret with caution). Not college calculus. Not a completion rate. Not a STEM-major rate. No named nationally representative source opened for this stream gives those last two figures. Outline open question 5. AoPS / Beast Academy are **vendor** materials for a gifted pipeline, not a population rate.

¹³²⁴Selection paragraph: 03-homeschool.md §4.6. Families who homeschool, persist, buy Beast Academy, sit the SAT, volunteer scores to NHERI, and answer Cardus are different filters. Two-parent one-earner college-educated households are overrepresented in NHES relative to all children and more overrepresented in convenience testing samples. A homeschool mathematics advantage that survives comparison with public-school children of the same parental education, income, and two-parent status has not been shown in a national probability sample this stream could open. Promise a path, not a percentile.

knows how to file; do not treat a state compliance letter as that file; do not treat a convenience-sample percentile as a substitute for it. Special education and anxiety stay callouts, not a ghetto. The paper trail is the close.

12.7 Callouts back to Chapters 1–2; the paper trail is the close

Special education at home is not an Individualized Education Program with a Free Appropriate Public Education duty. Chapter 1 already installed that fact from 34 C.F.R. § 300.137(a) and from OSEP’s Questions and Answers 22-01 (February 2022). Children parentally placed in private schools have no individual right to FAPE and no individual right to the IEP they would have in public school. Local educational agencies must conduct child find, consult, and spend a proportionate share of IDEA Part B funds on equitable services documented in a *services plan* for those children designated to receive them. Whether homeschoolers are in that bucket depends on whether state law treats homeschools as private elementary or secondary schools.¹³²⁵ This chapter will not rewrite FAPE. It will not dump the Pennsylvania approval letter or the WestEd proportionate-share fork again. It will say, as a records callout: if a services plan exists, keep it in the file that is not the transcript; do not title a year of unfinished arithmetic “Algebra I” because a plan exists; do not ask an agent to generate an IEP the regulation does not create. Request an evaluation — child find exists even if services do not. If the parent is the ceiling for a learner with a mathematics disability, buy expertise rather than an AI workaround. Chapter 2 said the last of those as a teaching rule. This chapter says it as a file rule.

Maloney, Ramirez, Gunderson, Levine, and Beilock (2015) remains a school-homework study: first- and second-graders of more math-anxious parents learned less mathematics over the year and were more math-anxious at year’s end, *but only if those parents reported frequent homework help*. It is not a parent-as-sole-teacher randomized trial. Replication status is open. Chapter 1 installed the design user; Chapter 2 locked the method — prepare so you can hear a wrong answer; then do not take the pencil.¹³²⁶ Video curricula, a co-op teacher, a tutor, and dual enrollment are release valves when the parent is the ceiling, not a finding that the parent should stop hearing wrong answers. Teaching Textbooks’ “does 100% of the teaching” is a **vendor** claim. It is relevant to Maloney. It is not a finding that the administrator of record can stop being the administrator of record.¹³²⁷ The

¹³²⁵34 C.F.R. § 300.137(a): parentally placed private-school children with disabilities have no individual right to FAPE or to the IEP they would have in public school. eCFR, <https://www.ecfr.gov/current/title-34/subtitle-B/chapter-III/part-300/subpart-B/subject-group-ECFR3556f7ac2fe0a92/section-300.137>. U.S. Department of Education, OSERS/OSEP, *Questions and Answers on Serving Children with Disabilities Placed by Their Parents in Private Schools* (Q&A 22-01), revised February 2022, <https://sites.ed.gov/idea/idea-files/q-and-a-children-with-disabilities-private-schools-parentally-placed/>. National Center for Systemic Improvement (WestEd), 2025 proportionate-share quick reference: whether homeschoolers are in the bucket depends on whether state law treats homeschools as private schools. Chapter 1 §1.2 already installed this. This chapter does not rewrite FAPE.

¹³²⁶Erin A. Maloney, Gerardo Ramirez, Elizabeth A. Gunderson, Susan C. Levine, and Sian L. Beilock, “Intergenerational Effects of Parents’ Math Anxiety on Children’s Math Achievement and Anxiety,” *Psychological Science* 26, no. 9 (2015): 1480–1488, <https://doi.org/10.1177/0956797615592630>. School-homework study of first- and second-graders; effect only when math-anxious parents reported frequent homework help; math-specific, not reading. Not a parent-as-sole-teacher RCT. Replication status open (Chapter 1 n. 69). Chapter 2 locked the method: prepare so you can hear a wrong answer; then do not take the pencil.

¹³²⁷Teaching Textbooks homepage, <https://www.teachingtextbooks.com/>, accessed 30 August 2026. **Vendor**. “Does 100% of the teaching” and “100% of the grading” is vendor copy, not an effectiveness trial, and not a finding that

transcript still needs a signer. NCAA still needs a worksheet. Pittsburgh still wants a final transcript signed by the person responsible.

When the parent is the ceiling, the facts are already in the tables. Any tutor, 23 percent. Any co-op, 31 percent. A small share of 9–12 also in college classes. Dual enrollment produces the clean external transcript. A co-op class produces a teacher of record useful for a worksheet. A tutor produces help and, unless the tutor’s institution issues a transcript, not an external credit. Those are facts, not a directory and not a pitch. Public-school à la carte access is state-law dependent and was not opened state-by-state for this stream. Umbrella schools issue transcripts; they do not teach the derivative. Cost of dual enrollment ranges from free to ordinary community-college tuition, by state program; this book will not invent a typical homeschool price.¹³²⁸

Artificial intelligence, on a records page, is a clerk. Chapter 3 remains the method chapter. Grok Bot is not FERPA-ready. Hermes Agent is not a child-safety school product. Consumer ChatGPT is Bastani’s GPT Base if the child holds it. Khanmigo is a constrained tutor over Khan content, parent-gated, “never gives the answer” as **vendor** copy.¹³²⁹ Allow, on the parent’s account: a transcript template with the fields section 12.2 named; a course-description draft the parent checks against the book that was actually used; a school-profile outline that does not invent a faculty or a CEEB code the kitchen does not have. The parent still signs. Ban: fabricating hours; fabricating a diploma; fabricating a grading scale; back-filling a Core-Course Worksheet with content the course did not cover; putting a student’s disability file on a shared cloud computer; asking a model to “make this look like a school transcript” by inventing a school that does not exist. A false record is not an AI-use problem first. It is a signature problem. The tool makes the false record faster. Point to Chapter 3. Do not sequel it.

Placement remains the first teaching job and the last records job. Publishers place by skill. NMAP Finding 15 refuses an age lock when prerequisites are present; Finding 30 allows motivated gifted students to move faster.¹³³⁰ Acceleration and delay are normal. They change which titles fill the high-school window. They do not replace titles. They do not replace the NCAA year-count. They do not replace SMU’s three names. A ten-year-old in Algebra I is not a transcript problem. A sixteen-year-old whose transcript says “Saxon 8/7” when the work was pre-algebra is a transcript problem. A sixteen-year-old whose transcript says Algebra II when the work omitted trigonometry and logarithms is a honesty problem Chapter 9 already named.

The paper trail this chapter was written to install is now a list a parent can keep without a district the administrator of record can stop signing the transcript. List price not displayed on the homepage this access day (**UNVERIFIED**).

¹³²⁸Dual-enrollment cost: \$0 to full community-college tuition, by state program (03-homeschool.md §5.3). No typical homeschool price invented. Tutor / co-op / college shares: n. 1308. Public-school à la carte: n. 1309. Umbrella schools issue transcripts; they do not teach calculus.

¹³²⁹Chapter 3 §§3.1–3.2. Hamsa Bastani et al., “Generative AI without guardrails can harm learning,” *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122 — GPT Base practice +48 percent / unaided exam –17 percent. Khanmigo pricing, <https://khanmigo.ai/pricing>, accessed 30 August 2026: \$4/month or \$44/year; “Unlike ChatGPT, Khanmigo never gives you the answer” is **vendor**. Grok Bot not FERPA-ready (n. 1280). Allow on the parent’s account: transcript template, course-description draft. Ban: fabricating hours or a diploma; back-filling a worksheet; putting a disability file on a shared cloud computer.

¹³³⁰NMAP 2008, Findings 15 and 30. See n. 1275. Placement tests: n. 1274. Acceleration and delay change which titles fill the high-school window; they do not replace titles, the NCAA year-count, or SMU’s three names.

office.

1. There is no national homeschool diploma. State compliance is not college evidence. Texas, Illinois, and Idaho can be satisfied in silence. New York, Pennsylvania, and North Carolina can be satisfied with a file a registrar will still not treat as a transcript. Read the state's code. Do not treat an HSLDA card as the code.
2. Write course titles a stranger can map: Algebra I, Geometry, Algebra II, Precalculus, Calculus, Statistics. Publisher level names belong in a parenthetical or on a description page. Credits in NCAA increments; no course more than 1.0 on a line. Ninth-grade start date as month/day/year if athletics are in view. Administrator signature.
3. NCAA mathematics core is Algebra I or higher. Division I: three years among sixteen core courses, 2.3 core GPA. Division II: two years among sixteen, 2.2 core GPA. One worksheet per core course. CLEP is not a core course. Evaluation after a school requests it. From NCAA, not from blogs.
4. SAT homeschool code 970000. PSAT as an away student, four weeks' lead. ACT as a parallel test. CLT at some, often Christian, colleges. AP seats are school-controlled; parents cannot order exams; self-study does not guarantee a May seat. CLEP is self-register; about 2,900 colleges grant credit on College Board's figure; credit is not automatic. Dual enrollment is the clean college transcript. Homeschool dual-enrollment and AP Statistics participation rates were not in the sources opened.
5. Named university pages want, as a pattern and not a rule, about three years through Algebra II as a floor. Pittsburgh: three years. SMU: Algebra I, Geometry, Algebra II, suggested, plus textbook titles in descriptions. North Central: three including algebra and geometry, one beyond geometry preferred. Southwest Baptist: three units. A 2026 NACAC homeschool-mathematics PDF was not opened. Confirm each destination's current page.
6. Parent expectations in 2016 were more modest than a franchise brochure. Cardus 2023 found lower bachelor's rates among medium- and long-term homeschoolers, noncausal, and does not report a mathematics score. NCES publishes no homeschool NAEP mathematics score and no STEM-major rate this stream could open. Promise a path, not a percentile.
7. IDEA at home is not an IEP with FAPE. Point to Chapter 1. If the parent is the ceiling, tutor, co-op, or community college — facts, not a pitch. An agent may draft a template. The parent signs. Do not fabricate hours or a diploma. Tools are not FERPA-ready. Point to Chapter 3.

That list is the export layer for everything Chapters 4 through 11 taught. It is not a diploma. It is not a percentile. It is not the coda.

The coda's job is what an honest book can promise. This chapter does not write it. The promise, if the coda is honest, will have to live inside constraints the sources already named: a parent who can hear a wrong answer; a child who struggles before the tool; an agent that is parent-hosted; fractions treated as the gate; concepts and procedures iterated; unaided exit tickets; course titles a stranger can map. It will not be able to promise a 2026 national headcount, a homeschool-curriculum winner, a national NAEP mathematics advantage, a parent-as-sole-teacher randomized trial, automatic AP or CLEP or dual-enrollment credit, an IEP with FAPE, that cooking is algebra, or that the chatbot

takes the test. The paper trail is what the kitchen can actually put in a stranger's hand. The honest book's remaining job is to say what that trail can and cannot mean.

Coda

A Path, Not a Percentile

A parent who is not a math specialist can take a child from first-grade number sense through high-school calculus *if* three things hold: the parent understands the concept well enough to hear a wrong answer; the child does the struggle before the tool; and any agent is a supplement hosted and constrained by the parent — not a substitute for the child’s work and not a school-safe product by default.¹³³¹ Promise, here, does not mean a prediction. It does not mean a guarantee. It means a claim the chapters already stood on, with named documents, that survives a second reading after the open questions have been left standing. The sources do not prove that thesis as a causal claim about homeschool mathematics. They do not contradict it. They constrain how it can be attempted. A close that converted the constraint into a guarantee would be a different, less honest book.

The three conditions are locks, not a personality. Hearing a wrong answer is the parent knowing the day’s concept well enough to stay out of the child’s working memory — to hear 15 on $3 + 4 + 5 = 3 + \square$ as an operational equal sign, and $2/5$ on $1/2 + 1/3$ as a magnitude failure. Chapters 1 and 2 installed that lock against Maloney and colleagues 2015, which is a school-homework interaction, not a parent-as-sole-teacher RCT, and against Ashcraft and Kirk’s working-memory finding, which is math-specific. The child doing the struggle before the tool is Soderstrom and Bjork before Bastani: current performance is often an unreliable index of learning; a practice score that rose because a solver finished the page is not the unaided exam. Chapter 3 locked the AI lane so that Chapters 4 through 11 could reuse ban versus allow without becoming a sequel. The agent as parent-hosted supplement is a constraint on the tool, not a product. Hermes Agent and Grok Bot do not get a victory lap on this page. Neither does Saxon, Singapore, Beast Academy, Math-U-See, Teaching Textbooks, or Khanmigo. Those were named and compared. None of them was sold. None of them is the thesis.

Chapters 1 through 3 are the locks. Chapters 4 through 11 are the mathematics those locks make teachable: number and the equal sign; the fractions gate; ratio, signed numbers, and early functions; Algebra I; Geometry; Algebra II and precalculus; calculus as AB, BC, and CLEP kept from collapsing; data and chance as a through-line that is not stealth calculus. Chapter 12 is the export. The coda’s job is not to tour them. It is to say, in one place, what the book can promise and what it cannot, so a later reader cannot treat a named open question as a solved chapter. A hedge keeps

¹³³¹Working thesis, locked in FRAME.md (30 August 2026) and restated from the preface. The sources constrain the attempt; they do not prove a causal claim about homeschool mathematics. Access date for all URLs in these notes: 30 August 2026.

the claim and adds a qualifier. An open question is a claim the book refuses to make. The popular story fills every item below with an analogy, a vendor page, or a 2030 kitchen. This coda will not. It will not say the future is bright if we are thoughtful. The future is not a finding in the record. Brightness is a tone the conjunction uses when it has run out of documents. An honest book stops.

What an honest book can promise

It can promise a parent who can hear a wrong answer. Not a mathematics degree. Not a district credential. The 2016 tables already showed the mother as main provider for 78 percent, a formal curriculum in 77 percent, and no mathematics degree required in most of the opened states. The constructive response Maloney actually supports is preparation, then generation, then silence about worry — not “help less often” as a daily strategy for the person who is the teacher.

It can promise a child who struggles before the tool. Worked examples faded for novices; incorrect examples explained; a short blocked introduction then mixed retrieval; schemas, not keywords; the number line as the representation that scales; brief timed retrieval of already-known facts, distinguished from a high-stakes early speed test. Iterate conceptual and procedural knowledge. Do not wait for full understanding before any practice, and do not run pages of a new algorithm with no representation and no why. Five strands, interwoven; ranking them is misguided. NMAP Finding 27 and Kapur 2014 remain different comparisons.

It can promise an agent that is parent-hosted. Ban paste-the-worksheet. Allow parent-first explanation, isomorphic practice with a parent-held key, a hint script, a hands-on activity, diagnosis of work the child already produced. Bastani: GPT Base +48 percent practice / -17 percent unaided exam; GPT Tutor +127 percent / indistinguishable from control. Oreopoulos and Low: Khanmigo ITT 0.040 population SD per term; gains resemble Khan without AI; do not invent standalone 0.06 as ITT.¹³³² One AI chapter, then a lane. Not five chapters. Not a sequel. The model may prepare the adult and the next problem. It may not do the child’s problem.

It can promise fractions treated as the gate. NMAP: the most important foundational skill not currently developed. Siegler and colleagues 2012: age-10 fractions uniquely predicted high-school algebra, 0.15 SD (UK) / 0.17 SD (US) per 1 SD fractions after controls — prediction, not a curriculum RCT. “Multiply makes bigger” is whole-number bias, not cute folklore. Diagnose fractions, signed numbers, and the equal sign before buying Algebra I. Cooking can be a second representation. It is not the gate.

It can promise concepts and procedures iterated, unaided exit tickets, and course titles a stranger can map. Algebra I, Geometry, Algebra II — or Integrated I–III — as titles a registrar recognizes, not “Saxon 8/7” or “Epsilon” without a parenthetical. NCAA mathematics core is Algebra I or higher;

¹³³²Load-bearing numbers reused here without new claims: NCES 2024-113 (3.4 percent / ~1,765,000, 2022–23); NCES 2020-001 (mother 78 / formal 77 / calculus 15 caution; Algebra I in 6–8 at 41 percent that year); Siegler et al. 2012 (UK 0.15 / US 0.17 SD); Rohrer et al. 2014 (72% vs. 38%, $d = 1.05$, $n = 140$, two-week delay) and 2020 (61% vs. 38%, $d = 0.83$, 54 classes); Jitendra et al. 2012 ($g = 1.24 / 1.27$ / null transfer); Bastani et al. 2025 (+48% / -17%; GPT Tutor +127% / -0.004 NS); Oreopoulos and Low, NBER w35620 (0.040 population SD per term); Liang et al. 2023 (61.22 / 61.3); Maloney et al. 2015 (interaction with frequent help; school-homework study); Cardus 2023 / Cheng and Watson 2024 (29% / 27% vs. 46%, noncausal); 34 C.F.R. § 300.137(a) and OSEP Q&A 22-01 (no individual FAPE at home). Full citations live in the notes that first spent them.

CLEP is not a core course. AP Statistics is a different college-equivalent. AB is eight units; BC is those eight plus two; CLEP Calculus is a third object. State compliance is not college evidence. There is no national homeschool diploma.

What the can-list adds up to is not a program a catalog could sell. It is a set of constraints a program would have to survive. If it promises a percentile, it fails Chapter 1. If it promises that a “grade 6” book is a legal grade, it fails Correction 3. If it promises that cooking is algebra, it fails NMAP Finding 26. If it promises that unguarded fluency is learning, it fails Bastani. If it promises an IEP with FAPE at the kitchen table, it fails 34 C.F.R. § 300.137(a). If it promises that the chatbot takes the test, it fails Chapter 3. A vendor that wanted a victory lap would find this list unsatisfying. That is a feature of an honest book, not a defect to be repaired in a coda.

What it cannot promise

It cannot promise a 2026 national homeschool headcount. NHES:2023 — 3.4 percent / approximately 1,765,000 in 2022–23 — is the latest federal PFI opened. Pulse and NHERI 2024–25 figures are different instruments. Do not fill with Pulse. Do not treat 3.408 million as NCES.

It cannot promise a homeschool-curriculum winner. No RCT of Saxon versus Singapore versus Math-U-See versus Teaching Textbooks versus Beast Academy *in homeschool* was opened. School-side ESSA ratings are vendor and school evidence. Cathy Duffy and Rainbow Resource are catalogue. Kunzman and Gaither already said the curriculum research is thin.

It cannot promise a nationally representative homeschool NAEP mathematics advantage, or a causal college-attainment effect of homeschool mathematics. Ray/NHERI convenience samples are a fight. Cardus 2023’s lower bachelor’s rates among medium- and long-term homeschoolers are noncausal. Only 17 percent of ever-homeschooled adults in that sample were homeschooled all K–12. STEM-major and college-calculus completion rates for homeschool graduates in a probability sample: no named source opened. NCES 2016’s 15 percent is high-school-level ever-taught calculus during home instruction, interpret with caution, not college calculus.

It cannot promise a parent-as-sole-teacher RCT. Maloney 2015 is homework help. Replication status of the interaction is open. It cannot promise a national figure for daily math minutes. **UNVERIFIED / no study found.** It cannot promise NHES:2023 microdata on teaching style and math subjects; the First Look did not republish the 2016 tables.

It cannot promise a year-plus kitchen GPT-Tutor RCT. Bastani is Turkish high school. Oreopoulos and Low is Tennessee middle school with a teacher present. Whether a parent sitting next to the child and requiring a hint dialogue is a different intervention than the Tennessee ITT is a design hypothesis, not a fitted result. Photomath-specific unaided-transfer RCT: **UNVERIFIED.** A 2026 Chinese staggered-adoption study was not opened as a PDF; do not print 18/20.

It cannot promise automatic AP, CLEP, or dual-enrollment credit. Each institution sets policy. AP seats are school-controlled. CLEP is self-register and is not an NCAA core course. Dual enrollment is the clean external transcript and is not a national entitlement. IES dual-enrollment outcomes report expected 2027. NACAC 2026 guidance specific to homeschool mathematics was not opened as a PDF.

It cannot promise an IEP with FAPE at home. It cannot promise that cooking is algebra. It cannot promise that the chatbot takes the test. It cannot promise that detectors police written mathematics (Liang 61.22 / 61.3). It cannot promise that A.B. 1159 is law — Senate third reading, 30 August 2026. It cannot promise GSM8K saturation as geometry proof. It cannot freeze Massachusetts 2026 notice-versus-approval fights as statute. It cannot promise a 2026 state-by-state CCSSM census. It cannot quote Fuchs 2013/2014 coefficients, Brunmair and Richter’s exact math-task g , Levine/Mix number-talk coefficients, or Beilock 2010 teacher-anxiety coefficients; those primaries were not extracted or not opened. It cannot promise an isolated timer effect. It cannot promise interleaving d from grade 7 as a finding about grade 2 or about proof. It cannot promise optimal dosage of concepts versus procedures for whole topics. It cannot promise measured transfer from informal activity to formal items.

The cannot-list is not a confession that the book found nothing. It is a specification of what the findings are not. Siegler’s 0.15 / 0.17 is a finding. It is not a fractions-curriculum RCT. Rohrer’s 72 percent versus 38 percent, $d = 1.05$, is a finding. It is not Dunlosky “high utility,” and it is not the 2020 $d = 0.83$. Bastani’s -17 percent is a finding. It is not a homeschool year. Oreopoulos and Low’s 0.040 SD per term is a finding. It is not Bloom’s two-sigma tutor. Cardus’s 29 / 27 versus 46 is a finding. It is not the effect of homeschool math. A coda that converted those “nots” into a program for 2030 would have rejoined the popular story on the last page.

Standing open questions

Copy, condensed, so no chapter can be misread as having filled them. No 2026 national headcount. NHES:2023 microdata on teaching style and math subjects. Homeschool-curriculum RCT. Daily math minutes. STEM-major and college-calculus rates. Parent-as-sole-teacher gap. Year-plus kitchen GPT-Tutor RCT. Photomath-specific unaided-transfer RCT. Khanmigo durability past the term. AP Calculus CED topic-level tables below the eight and ten unit titles; AP Precalculus topic lists below the four official unit titles (unit names and MC weights now taken from the course page; official AP Calculus prerequisite paragraph now extracted). NMAP Table 2 geometry/measurement remainder. NCTM Focal Points official book. Named vendor primaries still closed. Integrated I–III topic-by-topic. 2026 CCSSM state census. Homeschool dual-enrollment and AP participation rates. Fuchs 2013/2014 numerical effect sizes. Isolated timer; interleaving outside grade 7. Concepts-versus-procedures dosage for whole topics. Informal-to-formal transfer, measured. Proof, modeling, and the limit concept; named 2026 ChatGPT two-column-proof eval. A.B. 1159 enactment; El Salvador evaluation; Setzer/Garcia docket terms; student-facing open-web school agent census. AP versus dual enrollment versus CLEP outcomes at scale. NACAC 2026 PDF. Unschooling math outcomes beyond the 12 percent share; 2026 homeschool AI-usage surveys; Massachusetts 2026 legislative outcome; and the remaining closed primaries the brief already listed.

Do not fill any of those with analogy.¹³³³ Hold the thesis unless a primary contradicts it.

¹³³³Open questions 1–25 are copied, condensed, from *Mathematics for Homeschooling: Book Brief* (30 August 2026). They are gaps in the memos, not invitations to invent. A.B. 1159 remained on Senate third reading as of 30 August 2026, not law. NHERI 3.408 million remains advocacy, not NCES.

Four lanes, one last time

Every mathematics chapter of this book owed the parent four lanes. They were book architecture, not a finding, and they were not four identical boilerplate headings.

Explain the mathematics to the parent. Not a franchise scope-and-sequence. Not a child's worksheet. The adult who stopped at algebra has to know what a unit fraction is, what a ratio is not, what the equal sign is, what a function is, what a limit is not, and what AP Statistics is not, well enough to hear the wrong answer.

Teaching guidance. Placement before purchase. Iterate. Number line. Schema, not keyword. Explicit modeling for a student already behind, without using Kapur to cancel NMAP 27. Accelerate a child who has the prerequisites; do not skip fractions because a child is "good at math." When the parent is the ceiling, a tutor, a co-op, or dual enrollment is a normal high-school plan, not a failure of the kitchen.

Tips that survive evidence. Retrieve known facts, sometimes against a clock, after they are known. Interleave after a short blocked introduction, with the grade-7 objects attached to the *ds*. Do not mash 1.05 with 0.83 with 1.24 with 1.57 with 0.37 with 0.60 with +48/-17 with 0.040. Kitchen-math is a second representation, not a curriculum.

Parent-hosted AI. Child does the work. Parent holds the key. Ban paste-the-worksheet, photo-to-key, companion, open web in the child's hands. Allow the six parent workflows Chapter 3 named, as practice suggestions, not as a homeschool RCT. Facts are not an LLM job. AP free response is unaided. Wolfram is a check. Detectors fail. COPPA compliance is 22 April 2026. A.B. 1159 is not law. Not a school-safe product by default.

Not a franchise. Not a sequel. Not a 180-day plan. Grade labels remain sequence anchors, not prison bars. CCSSM remains a map, not a homeschool mandate. Access date for the fetches under these claims remains 30 August 2026. **UNVERIFIED** items stay flagged.

The kitchen table is not a district. Chapter 1 opened on that sentence, and it still holds. A district has a scope-and-sequence committee, a teacher of record with a credential, a special-education team that writes an IEP, a transcript office, and an external examiner who is not the parent. The kitchen has one adult, usually the mother, usually a purchased book, legally allowed in most of the country to teach Algebra II without a mathematics degree, and a legal object that is not "a math program." Texas may require a written curriculum covering mathematics and no notice. New York requires an Individualized Home Instruction Plan, hours, quarterly reports, and a standardized mathematics-inclusive battery in high school. Those are not the same paperwork load, and neither of them is a registrar's file.

The record still has to leave the house. A satisfied statute is not a transcript. A publisher level is not a course title. A CLEP pass is not an NCAA core course. An AP seat in May is not produced by self-study alone. Dual enrollment is the clean external transcript when the family can get it. Named college pages this research opened wanted, as a pattern and not a rule, about three years through Algebra II as a floor. Chapter 12 translated the work of Chapters 4 through 11 into titles, grades, credits, and external signals a stranger can map. That translation is not optional. The thesis, if it is attempted, ends on paper a stranger can read. There is no national homeschool diploma. There is

no 2030 kitchen in this close. There is no instruction to buy the video curriculum and stop hearing wrong answers, and no instruction to let the agent tutor. Those sentences reinstall the conjunction in one paragraph.

Do not close with a percentile. Close with the constraint the documents actually support: hear a wrong answer, let the child struggle before the tool, host the agent if you use one, treat fractions as the gate, iterate, and export the record. The next year of mathematics will require knowledge the parent may not yet have. A first-grader can already write 15 on a non-canonical equation. A calculus student can still treat the equal sign as “compute now.” A fluent model can still finish the page. The work remains. The open questions remain. The path is not a percentile. The kitchen table is not a district. The record still has to leave the house.

That is not a failure of the book. It is the attempt the book named.

A Note on Sources

This book is sourced to documents opened on 30 August 2026, or to the intermediary that was actually read when a full text could not be opened. It does not invent titles, publishers, years, page numbers, author names, coefficients, prices, or quotations. Where a bibliographic element could not be verified from a catalog, a journal landing page, a fetched PDF, or a manuscript note, the element is omitted. An UNVERIFIED flag in a note or in the bibliography means the full text was not read, or a named element (a coefficient, a price, a page) was not extracted. Unfetched is not a finding. Gaps are not filled by analogy.

A fact-check pass corrected identities and wording that an earlier draft had wrong. Those corrections are locked and are not to be reverted. The seventh author of Siegler, Duncan, Davis-Kean, Duckworth, Claessens, Engel, Susperreguy, and Chen 2012 is Maria Ines Susperreguy, as printed on the author PDF; the accents were not in the primary. *Adding It Up* page 116 carries the five strands, “interwoven and interdependent”; the extreme-positions sentence is page 409. National Mathematics Advisory Panel Finding 15 is two sentences: what is developmentally appropriate is “largely contingent on prior opportunities to learn”; claims that children of a given age cannot learn certain content because they are “too young,” “not in the appropriate stage,” or “not ready” have “consistently been shown to be wrong.” Brain-immaturity claims are not justified even when prerequisite knowledge is present. The two sentences are not to be mashed into “wrong if prerequisite knowledge is present.” Finding 10 says the capabilities are “mutually supportive,” not “mutually reinforcing.” NCTM process-standard order is Problem Solving; Reasoning and Proof; Communication; Connections; Representation. Booth, Lange, Koedinger, and Newton 2013: the second author is Karin E. Lange, not Kenneth. Rohrer, Dedrick, Hartwig, and Cheung 2020: the fourth author is Chi-Ngai Cheung, not Alexander. NCES 2020-001 is Cui and Hanson, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey*, not the 2016 First Look. School GAISE II is 2020; 2016 is the college GAISE report. AlphaGeometry2’s arXiv title is “Gold-medalist Performance in Solving Olympiad Geometry with AlphaGeometry2.” Oreopoulos and Low, NBER Working Paper 35620: the second author is Nina Low, not Nathan. The official AP Calculus prerequisite paragraph has now been extracted from the Course and Exam Description front matter; unit-guide topic rows remain UNVERIFIED. AP Precalculus unit names and multiple-choice weights are now from the official course page; topic lists below those titles remain UNVERIFIED. Brunmair and Richter’s interleaving meta-analysis was not opened; this book does not quote a mathematical-task g .

Full texts and coefficients that remain unread are flagged, not filled. Among them: the official NCTM *Curriculum Focal Points* book (paywalled; grade lists from a third-party posting); Greer

1994 “multiply makes bigger” data (cite Ni and Zhou, Siegler, and 5.NF.5 instead); Fuchs et al. 2013/2014 fractions-intervention numerical effect sizes (papers located; coefficients not copied); Brunmair and Richter’s exact math-task g ; a Photomath-named unaided-transfer RCT; a named 2026 evaluation of consumer ChatGPT on high-school two-column proofs; Beilock et al. 2010 teacher-anxiety coefficients (fetched secondary; not quoted as read); Carbonneau, Marley, and Selig 2013 full PDF (overall $d \approx 0.37$ remains approximate); Rittle-Johnson and Star 2007/2009 comparison-of-methods exact d ; Uttal / DeLoache dual-representation papers; the virtual-versus-physical manipulatives literature; Barnett and Ceci 2002 transfer-taxonomy exact dimensions; Project Follow Through original Abt / Stebbins evaluation tables; English-learner-specific mathematics RCT effect sizes; Mix / Levine / Gunderson number-talk coefficients; Wai, Lubinski, and Benbow 2009 spatial-STEM coefficients; Hyde et al. 2006 scaffolding paper; Ashcraft and Krause later reviews (Ashcraft and Kirk 2001 $r = -.40$ is used as already in the memos); Tall and Vinner 1981 full PDF (named; quotations UNVERIFIED); Hermes 4 MATH / GSM8K technical-report metrics (arXiv HTML failed); GAISE / GAISE II page-level claims (ASA primary not opened in full); Woodward et al., *Improving Mathematical Problem Solving in Grades 4 Through 8* (NCEE 2012-4055), existence confirmed, full text UNVERIFIED this pass; ACT and The Education Trust, *On Course for Success* (2005), cited via CCSSM Appendix A; Adelman 1999 as NMAP’s citation; NRC *Mathematics Learning in Early Childhood* (2009) full book this pass; McCallum Progressions (2023) compiled PDF not line-read; Gelman and Gallistel 1978 volume not re-opened (five principles via later quotations); Siegler and Chen 2008 / Siegler and Jenkins 1989 tables; Hiebert and Grouws 2007 full handbook chapter. A further list of named papers cited in the notes but not independently opened this pass — among them Sweller 1985/1988, Renkl 2003, Kalyuga 2003, Cepeda 2006, Rohrer and Taylor 2006/2007, Taylor and Rohrer 2010, Knuth 2006, Star 2005, Rittle-Johnson 1999/2001/2006, Schneider 2011, Hecht and Vagi 2010, Fuson and Briars 1990, Baroody and Ginsburg 1983, Kieran 1981, Vosniadou 2008, Duncan 2007, Ma 1999 — stays flagged where the bibliography so marks them.

Author lines that could not be locked are omitted rather than guessed. Remaining co-authors after the named first authors of Fuchs et al. 2021, Fuchs et al. 2013, Jitendra et al. 2013, Duncan et al. 2007, Peltier and colleagues 2025, Phan et al. / *Humanity’s Last Exam in Nature*, and Chervonyi et al. / *AlphaGeometry2* were not locked from a title-page refetch. Ray 2009 *Homeschool Progress Report* is an HSLDA-hosted specimen; publisher, place, and page details were not independently catalogued. Assouline / *Nation Empowered* / Iowa Acceleration Scale remaining authors and effect sizes stay UNVERIFIED. Hyde 2006 remaining authors and full journal citation were not locked. The Saxon Homeschool scope-and-sequence PDF year was not on the title this pass. The mathed.net CCSSM matrix year was not on the page; the fetch timed out.

Vendor, price, and Course-and-Exam-Description holes are left open. Teaching Textbooks list price on the publisher page, 30 August 2026, was not displayed; the official scope-and-sequence PDF was not opened. Beast Academy Online list price did not render. RightStart, Math Mammoth, and Life of Fred official scopes were not opened on vendor primaries. Wolfram|Alpha Pro current price, an IXL live homeschool invoice, the DreamBox underlying ESSA study PDF, MATHia direct family purchase as of 30 August 2026, Claude Projects current UI and file limits, standalone Microsoft Math Solver app status, and Gemini for Education feature list / homeschool eligibility remain UNVERIFIED. AP Calculus CED unit-guide topic rows below the eight and ten titles were not extracted (front matter was). AP Precalculus topic lists below the four official unit titles were

not opened. Appendix A Integrated Mathematics I–III topic-by-topic (pp. 49–90) was only partly extracted. NMAP Table 2 geometry/measurement remainder was truncated in the ERIC fetch. The exact final-report locus of the memo’s fractions sentence — “knowledge of fractions is the most important foundational skill not currently developed among American students” — was not relocated in the ERIC PDF this pass; Chapter 4 of that report uses “The most important foundational skill not presently developed appears to be proficiency with fractions.” The memo-locked sentence is left standing, not rewritten, and not replaced from journalism.

Homeschool, law, and outcomes holes are likewise left standing. There is no 2026 national homeschool headcount in these sources; NHES:2023 is the latest federal Parent and Family Involvement survey opened. NHES:2023 microdata on teaching style, curriculum sources, and math subjects were not republished in the First Look. No homeschool-curriculum RCT was found. No national study of typical daily homeschool math minutes was found (NCES 2016 is weekly-hour bands, not subject minutes). No nationally representative homeschool NAEP mathematics score was opened. STEM-major and college-calculus completion rates for homeschool graduates in a probability sample: no named source opened. Maloney et al. 2015 is a school-homework study, not a parent-as-sole-teacher experiment. No year-plus kitchen GPT-Tutor RCT was found. Homeschool AP and dual-enrollment participation rates are not in these sources. The NACAC 2026 homeschool-mathematics PDF was not opened. Common App counselor-form homeschool questions in the year of publication: confirm; 2024 help text was opened for the invitation-only counselor account only. NHERI’s North Carolina 2024–25 register 165,243 versus DNPE’s own latest PDF: UNVERIFIED. Official 2026 England elective-home-education registration count was left out as journalism. The Johns Hopkins Homeschool Hub Pulse page fetch was blocked; 2025–26 figures are not invented. 2026 AI-usage surveys among homeschoolers were not opened. Massachusetts 2026 legislative outcome on approval versus notice is not frozen as law. California A.B. 1159 is not law as of 30 August 2026 (Senate third reading). Setzer / Character.AI settlement terms UNVERIFIED. El Salvador Grok-as-tutor independent evaluation UNVERIFIED. An EU AI Act × 2026 homeschool practical guide was not opened. Non-AP homeschool calculus tables of contents (Saxon Calculus 2nd ed.; Art of Problem Solving Calculus; a community-college Calculus I catalog sample) were not opened. Slope/intercept mixup prevalence: no named prevalence study opened. Area-as-chant-without-unit prevalence: no single named study opened. Isolated timer effect: Peltier 2025 is labelled preprint; IES 2021 studies bundled timing. A 2026 state-by-state CCSSM census is not asserted; this book will not say “most states.” Humanity’s Last Exam math-subset percentage was not broken out on lastexam.ai. FrontierMath Tiers 1–3 live accuracy percentages: hub opened, plot not extracted. Exact GSM8K percentages for named 2026 models were not extracted. The IES dual-enrollment outcomes report is expected 2027.

Liang, Yuksekgonul, Mao, Wu, and Zou 2023: 61.22 percent is the authors’ average false-positive rate on 91 human TOEFL essays (arXiv); *Patterns* HTML rounds the same average to 61.3 percent. The book keeps 61.22 in the body and reports both. It does not use 61.22 as the only number, and it does not invent a third rate.

The book does not revive cut or barred claims: Brunmair and Richter $g = 0.34$; NMAP Finding 15 mashed as “wrong if prerequisite knowledge is present”; Finding 10 quoted as “mutually reinforcing”; *Adding It Up* extreme-positions sentence attributed to page 116; NCTM process order with Representation before Connections; Siegler seventh author with accents not in the primary; Booth

second author “Kenneth E. Lange”; Rohrer 2020 fourth author “Alexander Cheung”; NCES 2020-001 as the 2016 First Look or as “Jinghong Cui and Chelsea Hanson”; GAISE II school report dated 2016; an invented AlphaGeometry2 title; Oreopoulos second author “Nathan Low”; invented AP Precalculus unit weights or an invented AP Calculus prerequisite; NHERI 3.408 million as NCES, or Pulse as comparable to NHES; Ray/NHERI 84th-percentile math as “what homeschool mathematics does”; Cardus 29/27 versus 46 as the causal effect of homeschool math; Maloney 2015 as a parent-as-sole-teacher RCT or as “anxious parents should not teach”; Khanmigo standalone 0.06 as intention-to-treat (that band is school-year, not Table 5 ITT); Bastani percentages mashed with Oreopoulos standard deviations or treated as Cohen’s d ; Liang 61.22 as the only number; A.B. 1159 as enacted law; COPPA school-authorization amendments as finalized; Hermes 4 as Hermes Agent; Grok Bot as the El Salvador tutor; adaptive practice as an agent; Humanity’s Last Exam scores updated past the 30 August 2026 lastexam.ai snapshot; GSM8K saturation as a geometry-proof score; AlphaGeometry2 as a kitchen chatbot; detectors as catching cheaters or as homeschool evaluation; CLEP as an NCAA core course; AP Statistics as a STEM-calculus substitute; a “grade 6” book as a legal grade; cooking as fractions or algebra (NMAP Finding 26); an IEP with FAPE at the kitchen table (34 C.F.R. § 300.137(a)); a 2026 national homeschool headcount; a homeschool-curriculum RCT; a homeschool NAEP mathematics score; invented prices, invented CED topic rows, or invented prevalence percentages. Effect sizes are not averaged. Open questions numbered in the research brief remain standing.

Bibliography

Mathematics for Homeschooling

Michael Gannotti

Chicago Manual of Style, 17th edition, notes-bibliography system

This list is a working apparatus for a university-press or serious trade book. It is compiled from the four research source files of 30 August 2026 (01-sources.json through 04-sources.json) and from every work actually named in the manuscript notes (manuscript/00-preface.md, 00-introduction.md, ch01.md through ch12.md, coda.md). It does not invent titles, publishers, years, DOIs, URLs, or page numbers. Where a bibliographic element could not be verified from a catalog, landing page, fetched PDF, or manuscript note, it is omitted and recorded in GAPS.md. Errata (factcheck/front-errata.md, mid1-errata.md, mid2-errata.md, end-errata.md) were read so that cut claims are not revived and so that authorship and wording corrections (Siegler seventh author Maria Ines Susperreguy; NRC *Adding It Up* p. 116 versus p. 409; NMAP Finding 15 two sentences; Finding 10 “mutually supportive”; NCTM process order Connections then Representation; Booth second author Karin E. Lange; Rohrer 2020 fourth author Chi-Ngai Cheung; NCES 2020-001 Cui and Hanson; GAISE II school report 2020 not 2016; AlphaGeometry2 actual arXiv title; Oreopoulos second author Nina Low; AP Calculus official prerequisite now extracted; AP Precalculus unit names and MC weights from the official course page) are the versions listed here.

Status tags (after each entry, in square brackets):

- **Cited** — named, quoted, or used as load-bearing authority in the present manuscript notes.
- **Consulted** — used in the research JSONs; not named on the manuscript page.
- **Vendor** — product documentation or vendor claim, labelled as such.
- **Preprint / working paper** — not a journal article as of the access date.
- **Advocacy / Journalism / catalogue** — labelled as such.
- **UNVERIFIED** — full text not read, or a named element (coefficient, price, page) not extracted.

Digital sources: access date **30 August 2026**, unless a fact-check fetch the same day is noted.

Hanging-indent ready: first line flush; subsequent lines indented three spaces.

1. Standards, sequences, and college-credit maps

Achieve in partnership with the Common Core State Standards writing team. *Appendix A: Designing High School Mathematics Courses Based on the Common Core State Standards*. 2010. Oregon Department of Education hosting copy, <https://www.oregon.gov/ode/educator-resources/standards/mathematics/Documents/math-appendix-a-model-course-pathways.pdf>. Accessed 30 August 2026. Pathways (Traditional, Integrated, compacted) are “models, not mandates.” Geometry six critical areas; G.CO.9–11 instructional note quoted in full against this copy, 30 August 2026. Integrated Mathematics I–III topic-by-topic (pp. 49–90) only partly extracted. [Cited]

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